AWS Certified Solutions Architect Professiona

By Stéphane Maarek

COURSE →

EXTRA PRACTICE EXAMS →

Disclaimer: These slides are copyrighted and strictly for personal use only

- This document is reserved for people enrolled into the Ultimate AWS Certified Solutions Architect Professional course
- **Please do not share this document**, it is intended for personal use and exam preparation only, thank you.
- If you've obtained these slides for free on a website that is not the course's website, please reach out to piracy@datacumulus.com. Thanks!
- **Best of luck for the exam and happy learning!**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Table of Contents

- Identity & Federation
- Security
- Compute & Load Balancing
- Storage
- Caching
- Databases
- Service Communication
- Data Engineering
- Monitoring
- Deployment & Instance Management

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Table of Contents

- Cost Control
- Migrations
- Amazon VPC
- Machine Learning
- Other Services
- Exam Preparation
- Congratulations

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Certified Solutions Architect Professional Course

SAP-C02

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Setting the right expectations for this course

- This course is all slides based
  - I'm assuming you have experience using AWS
  - No hands-on will come with the course. You should know the basics
  - It's fast paced. Your time is valuable. Feel free to slow me down to 0.75x
- If you just passed the AWS Certified Solutions Architect Associate cert
  - I recommend you go through AWS Certified Developer, SysOps & DevOps
  - I know you are eager to get the SAP certification, but take your time
- The AWS knowledge needed for the SA Pro exam
  - Is extremely similar to the knowledge for SAA
  - The questions are more complex, and knowing details is very important
  - It's possible that multiple answers are correct, but one is the most appropriate

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

The AWS Certified Solutions Architect Professional Exam

- Is HARD
- Tests real AWS experience
- Will test you on some very subtle service features

- I have included quizzes for every single section BUT…
  - The quizzes are not "scenario based" / "exam-like"
  - They only help you extract some important notions out of what you're learning
  - This is my optimal way of teaching you about specific topics
  - Please trust my teaching process

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Practice Exams

- This course does not come with practice exams
  - Practice Exams come separately
  - I really want to focus this course on the knowledge needed

- Warning:
  - This course is on the NEW CERTIFICATION (SAP-C02)
  - You may see outdated content in other practice exams, other courses, etc…
  - This course is not incomplete, it's more targeted towards the knowledge you actually need to know to pass the exam

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Identity & Federation

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM – What should you know by now

- Users: long term credentials
- Groups
- Roles: short-term credentials, uses STS
  - EC2 Instance Roles: uses the **EC2 metadata** service. One role at a time per instance
  - Service Roles: API Gateway, CodeDeploy, etc…
  - Cross Account roles
- Policies
  - AWS Managed
  - Customer Managed
  - Inline Policies
- Resource Based Policies (S3 bucket, SQS queue, etc…)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM Policies Deep Dive

- Anatomy of a policy: JSON doc with Effect, Action, Resource, Conditions, Policy Variables
- Explicit DENY has precedence over ALLOW
- Best practice: use least privilege for maximum security
  - Access Advisor: See permissions granted and when last accessed
  - Access Analyzer: Analyze resources that are shared with external entity
- Navigate Examples at: https://docs.aws.amazon.com/IAM/latest/UserGuide/access_policies_examples.html

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "ec2:AttachVolume",
                "ec2:DetachVolume"
            ],
            "Resource": "arn:aws:ec2:*:*:instance/*",
            "Condition": {
                "StringEquals": {"ec2:ResourceTag/Department": "Development"}
            }
        },
        {
            "Effect": "Allow",
            "Action": [
                "ec2:AttachVolume",
                "ec2:DetachVolume"
            ],
            "Resource": "arn:aws:ec2:*:*:volume/*",
            "Condition": {
                "StringEquals": {"ec2:ResourceTag/VolumeUser": "${aws:username}"}
            }
        }
    ]
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM AWS Managed Policies

AdministratorAccess

```
{
   "Version": "2012-10-17",
   "Statement": [
      {
         "Effect": "Allow",
         "Action": "*",
         "Resource": "*"
      }
   ]
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM AWS Managed Policies

PowerUserAccess

```
{
   "Version": "2012-10-17",
   "Statement": [
      {
         "Effect": "Allow",
         "NotAction": [
            "iam:*",
            "organizations:*",
            "account:*"
         ],
         "Resource": "*"
      },…
```

```
      …{
         "Effect": "Allow",
         "Action": [
            "iam:CreateServiceLinkedRole",
            "iam:DeleteServiceLinkedRole",
            "iam:ListRoles",
            "organizations:DescribeOrganization”,
            "account:ListRegions"
         ],
         "Resource": "*"
      }
   ]
}
```

Note how ”NotAction” is used instead of Deny

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM Policies Conditions

"Condition" : { "{condition-operator}" : { "{condition-key}" : "{condition-value}" }}

Operators:

- String (StringEquals, StringNotEquals, StringLike…)
  - "Condition": {"StringEquals": {"aws:PrincipalTag/job-category": "iamuser-admin"}}
  - "Condition": {"StringLike": {"s3:prefix": ["", "home/", "home/${aws:username}/"]}}
- Numeric (NumericEquals, NumericNotEquals, NumericLessThan…)
- Date (DateEquals, DateNotEquals, DateLessThan…)
- Boolean (Bool):
  - “Condition": {"Bool": {"aws:SecureTransport": "true"}}
  - "Condition": {"Bool": {"aws:MultiFactorAuthPresent": "true"}}
- (Not)IpAddress:
  - "Condition": {"IpAddress": {"aws:SourceIp": "203.0.113.0/24"}}
- ArnEquals, ArnLike
- Null: "Condition":{"Null":{"aws:TokenIssueTime":"true"}}

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM Policies Variables and Tags

Example: ${aws:username}

- "Resource": ["arn:aws:s3:::mybucket/${aws:username}/*"]

AWS Specific:

- aws:CurrentTime, aws:TokenIssueTime, aws:principaltype, aws:SecureTransport, aws:SourceIp, aws:userid, ec2:SourceInstanceARN

Service Specific:

- s3:prefix, s3:max-keys, s3:x-amz-acl, sns:Endpoint, sns:Protocol…

Tag Based:

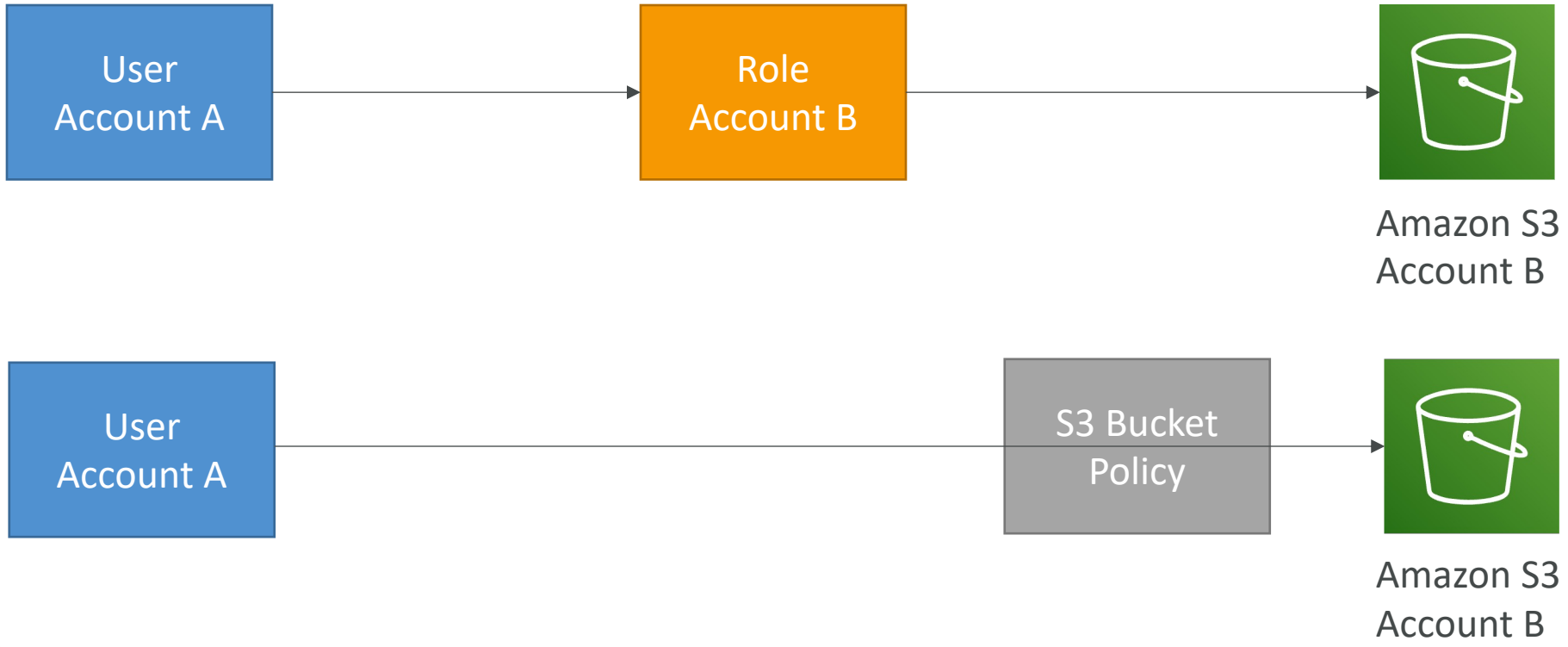
- iam:ResourceTag/key-name, aws:PrincipalTag/key-name…

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM Roles vs Resource Based Policies

- Attach a policy to a resource (example: S3 bucket policy) versus attaching of a using a role as a proxy



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM Roles vs Resource Based Policies

- When you assume a role (user, application or service), you give up your original permissions and take the permissions assigned to the role
- When using a resource-based policy, the principal doesn't have to give up any permissions
- Example: User in account A needs to scan a DynamoDB table in Account A and dump it in an S3 bucket in Account B.
- Supported by: Amazon S3 buckets, SNS topics, SQS queues, Lambda functions, ECR, Backup, EFS, Glacier, AWS Artifact, Secrets Manager, ACM, KMS, CloudWatch Logs, API Gateway, EventBridge etc…

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM Permission Boundaries

- IAM Permission Boundaries are supported for users and roles (not groups)
- Advanced feature to use a managed policy to set the maximum permissions an IAM entity can get.

Example:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "s3:*",
                "cloudwatch:*",
                "ec2:*"
            ],
            "Resource": "*"
        }
    ]
}
```

IAM Permission Boundary

+

```
{
  "Version": "2012-10-17",
  "Statement": {
    "Effect": "Allow",
    "Action": "iam:CreateUser",
    "Resource": "*"
  }
}
```

IAM Permissions Through IAM Policy

=

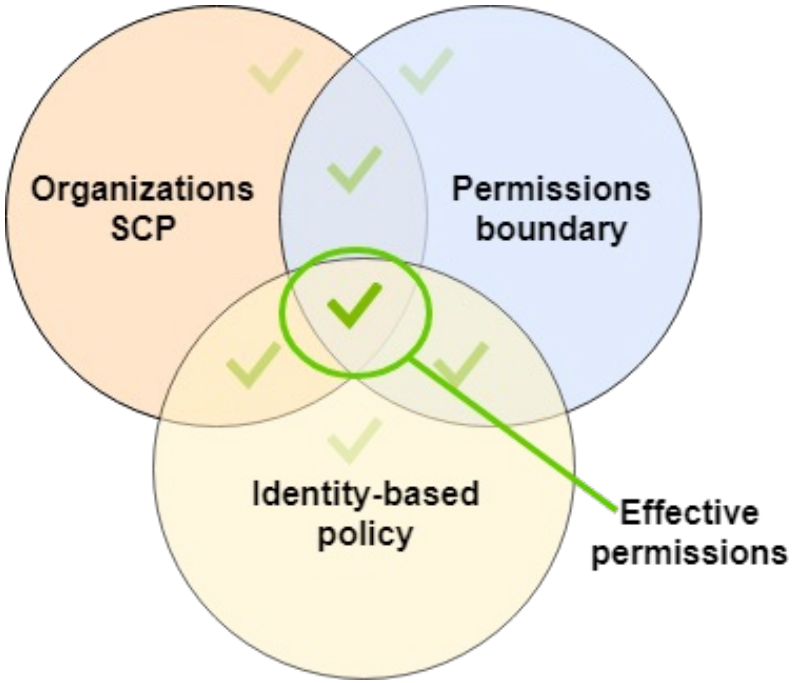
No Permissions

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM Permission Boundaries

- Can be used in combinations of AWS Organizations SCP



https://docs.aws.amazon.com/IAM/latest/UserGuide/access_policies_boundaries.html

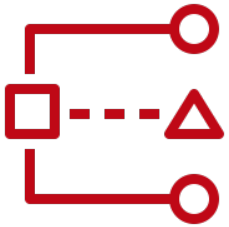
Use cases

- Delegate responsibilities to non administrators within their permission boundaries, for example create new IAM users
- Allow developers to self-assign policies and manage their own permissions, while making sure they can't "escalate" their privileges (= make themselves admin)
- Useful to restrict one specific user (instead of a whole account using Organizations & SCP)

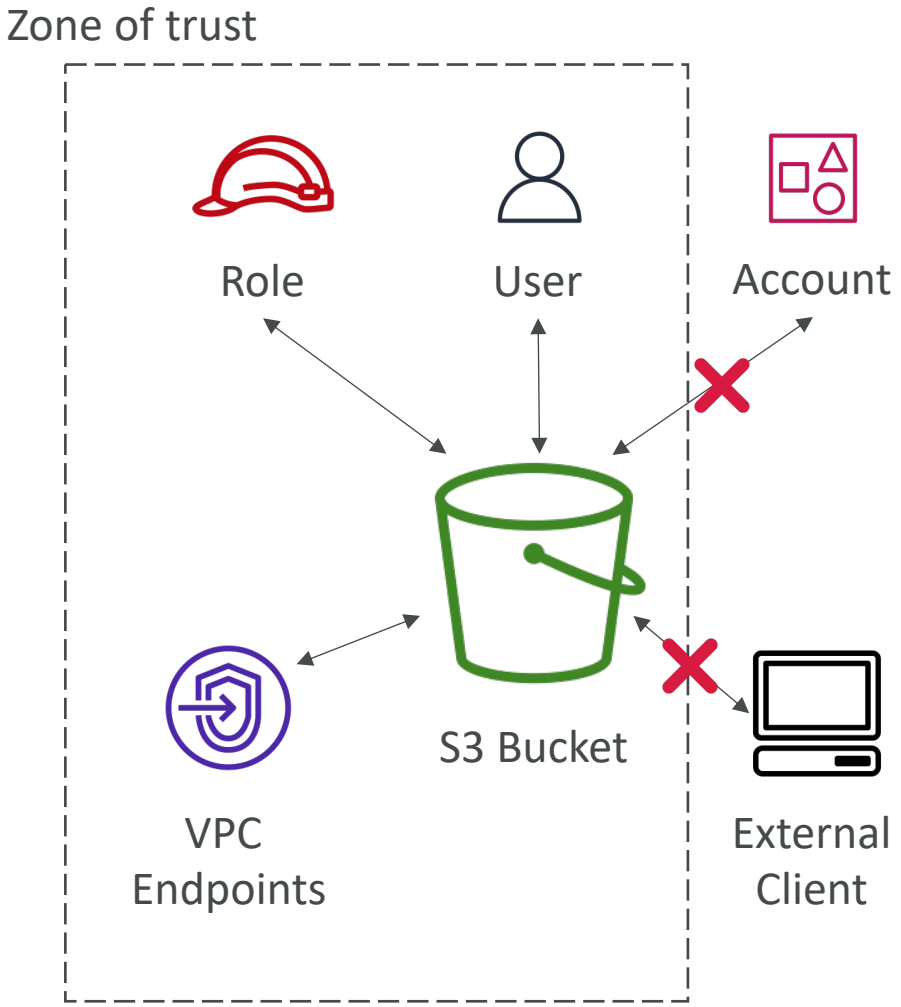
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM Access Analyzer



- Find out which resources are shared externally
  - S3 Buckets
  - IAM Roles
  - KMS Keys
  - Lambda Functions and Layers
  - SQS queues
  - Secrets Manager Secrets
- Define **Zone of Trust** = AWS Account or AWS Organization
- Access outside zone of trusts => findings



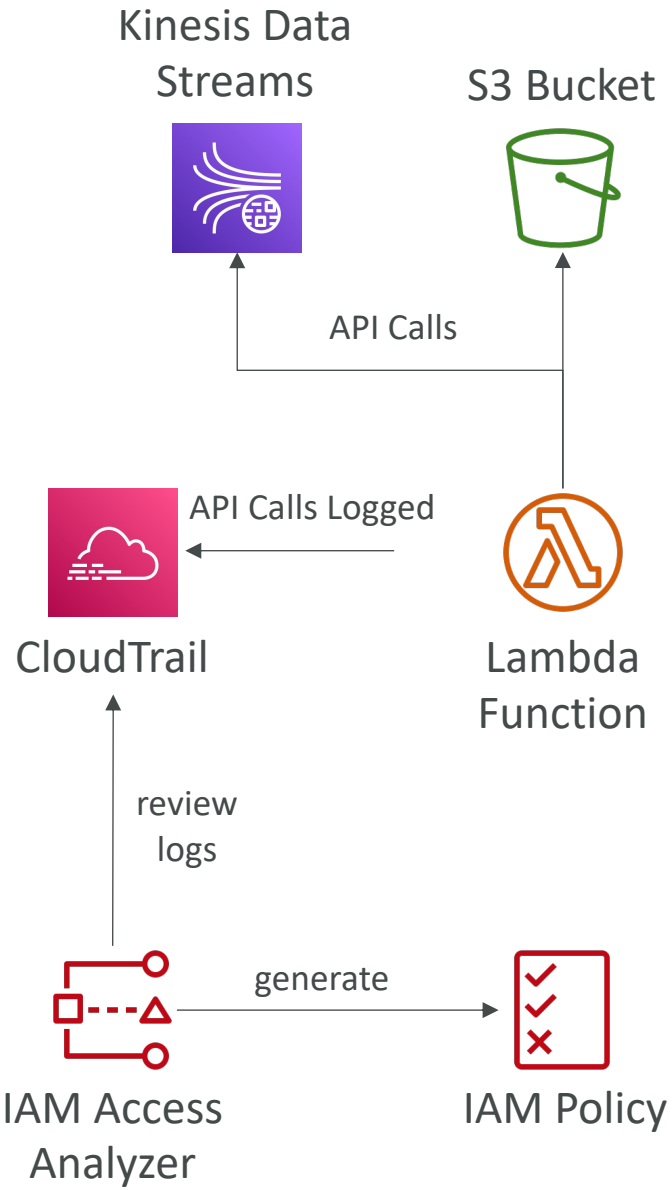
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM Access Analyzer

- IAM Access Analyzer Policy Validation
  - Validates your policy against IAM policy grammar and best practices
  - General warnings, security warnings, errors, suggestions
  - Provides actionable recommendations

- IAM Access Analyzer Policy Generation
  - Generates IAM policy based on access activity
  - CloudTrail logs is reviewed to generate the policy with the **fine-grained permissions and the appropriate Actions and Services**
  - Reviews CloudTrail logs for up to 90 days

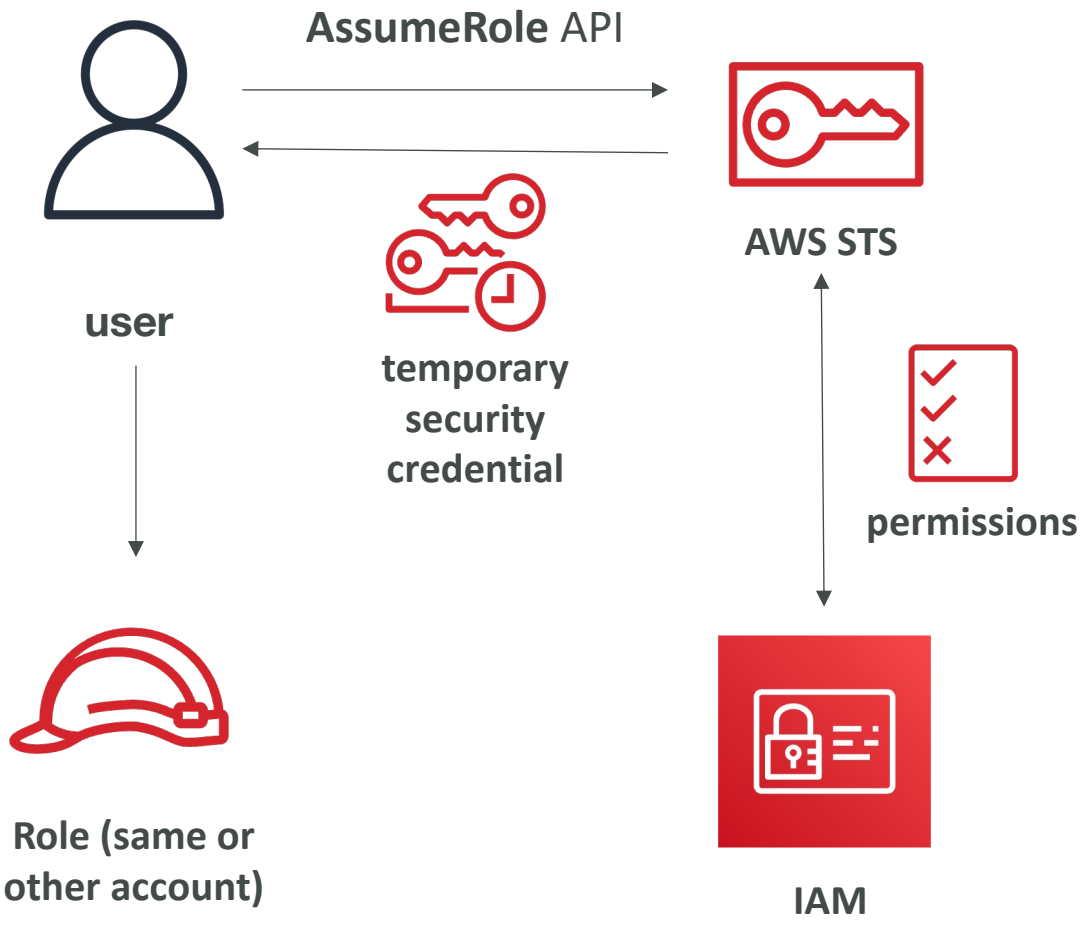


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Using STS to Assume a Role

- Define an IAM Role within your account or cross-account
- Define which principals can access this IAM Role
- Use AWS STS (Security Token Service) to retrieve credentials and impersonate the IAM Role you have access to (**AssumeRole API**)
- Temporary credentials can be valid between 15 minutes to 12 hour



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Assuming a Role with STS

- Provide access for an IAM user in one AWS account that you own to access resources in another account that you own
- Provide access to IAM users in AWS accounts owned by third parties
- Provide access for services offered by AWS to AWS resources
- Provide access for externally authenticated users (identity federation)

- Ability to revoke active sessions and credentials for a role (by adding a policy using a time statement – AWSRevokeOlderSessions)

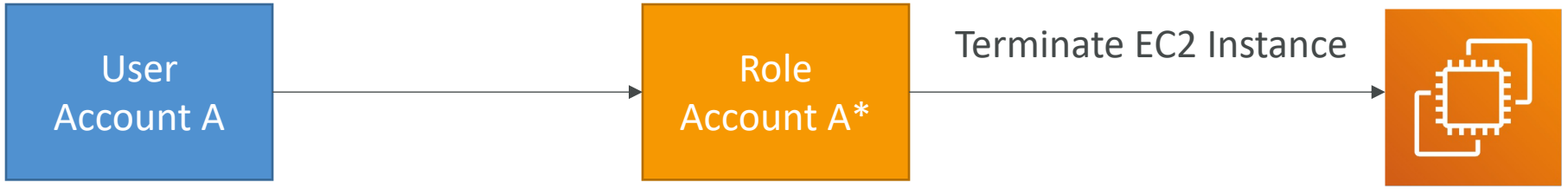
When you assume a role (user, application or service), you give up your original permissions and take the permissions assigned to the role

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Providing Access to an IAM User in Your or Another AWS Account That You Own

- You can grant your IAM users permission to switch to roles within your AWS account or to roles defined in other AWS accounts **that you own**.

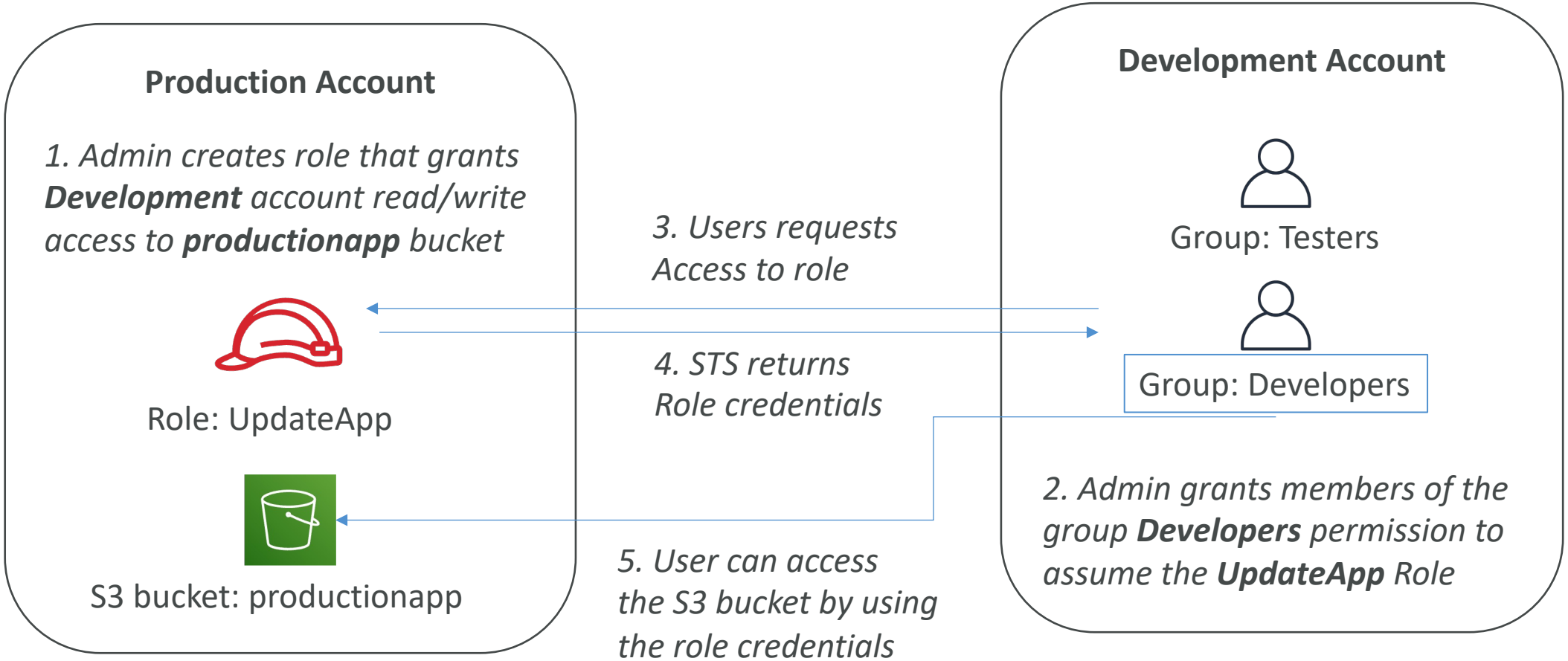


- Benefits:
  - You must explicitly grant your users permission to assume the role.
  - Your users must actively switch to the role using the AWS Management Console or assume the role using the AWS CLI or AWS API
  - **You can add multi-factor authentication (MFA) protection to the role** so that only users who sign in with an MFA device can assume the role
  - Least privilege + auditing using CloudTrail

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cross account access with STS



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

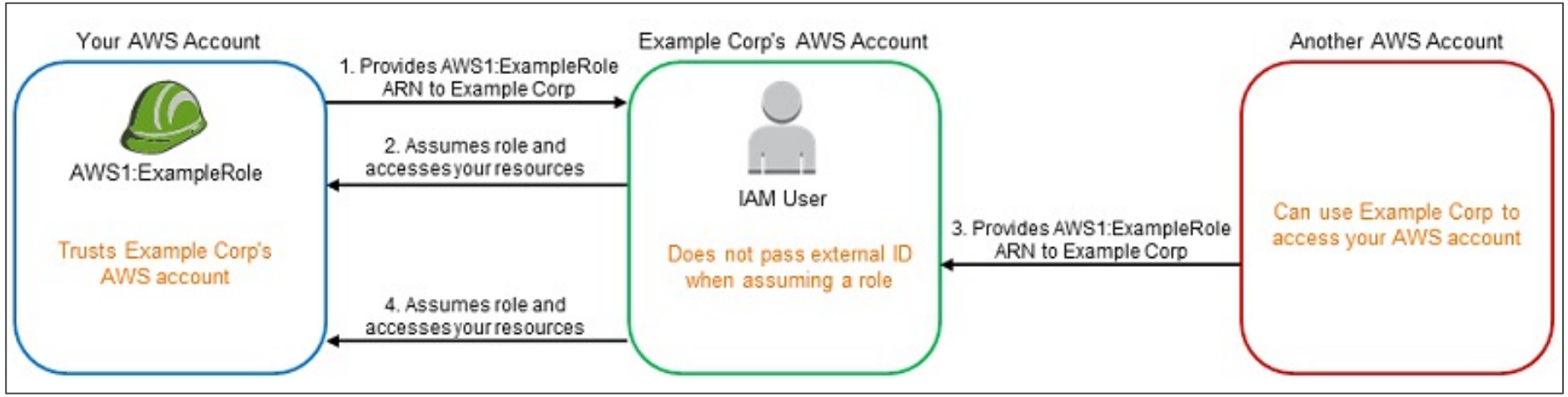
Providing Access to AWS Accounts Owned by Third Parties

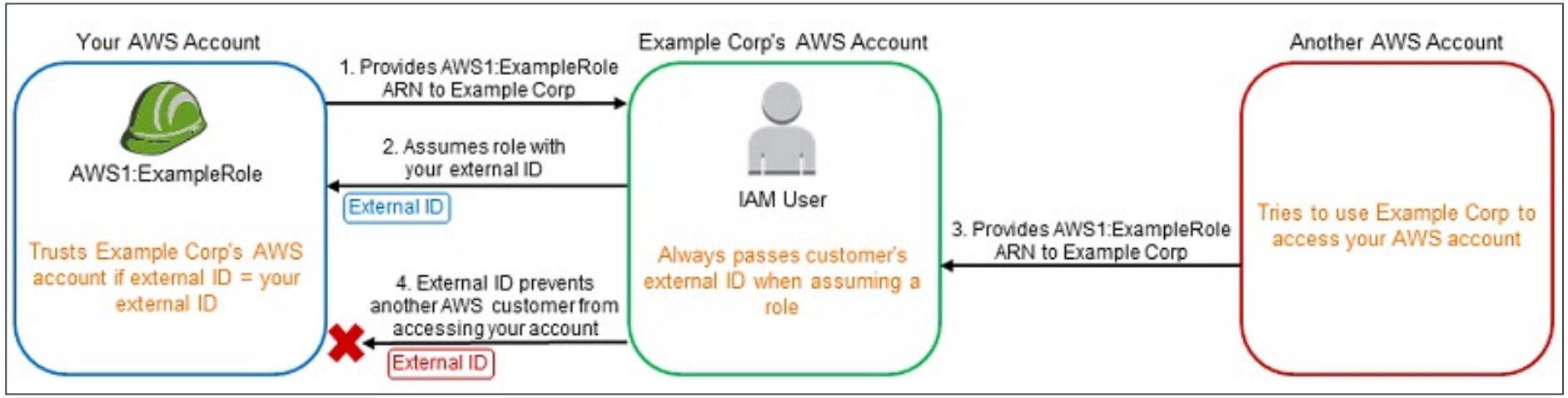
- Zone of trust = accounts, organizations that you own
- Outside Zone of Trust = 3rd parties
- Use IAM Access Analyzer to find out which resources are exposed
- For granting access to a 3rd party:
  - The 3rd party AWS account ID
  - **An External ID** (secret between you and the 3rd party)
    - To uniquely associate with the role between you and 3rd party
    - Must be provided when defining the trust and when assuming the role
    - Must be chosen by the 3rd party
  - Define permissions in the IAM policy

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

The confused deputy



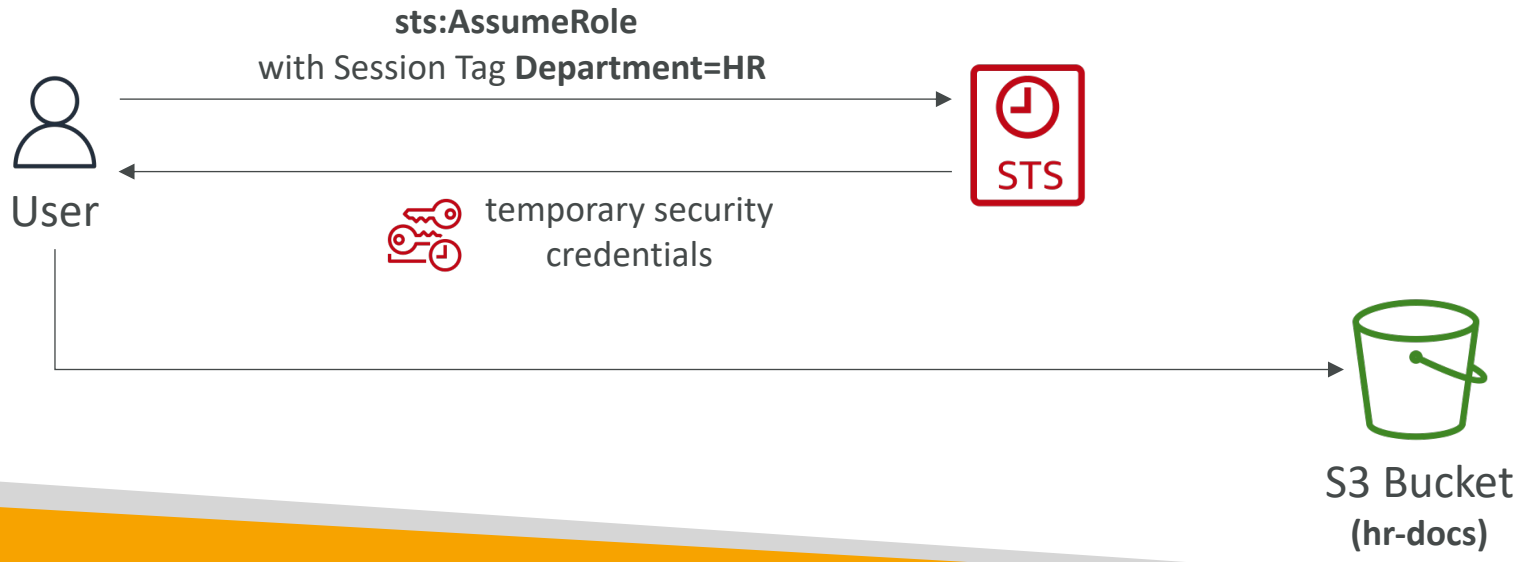


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Session Tags in STS

- Tags that you pass when you assume an IAM Role or federate user in STS
- aws:PrincipalTag Condition
  - Compares the tags attached to the principal making the request with the tag you specified in the policy
  - Example: allow a principal to pass session tags only if the principal making the request has the specified tags




```
{
    "Version": "2012-10-17",
    "Statement": [
      {
        "Effect": "Allow",
        "Action": "s3:GetObject",
        "Resource": "arn:aws:s3:::hr-docs/*",
        "Condition": {
          "StringEquals": {
            "aws:PrincipalTag/Department": "HR"
          }
        }
      }
    ]
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

STS Important APIs

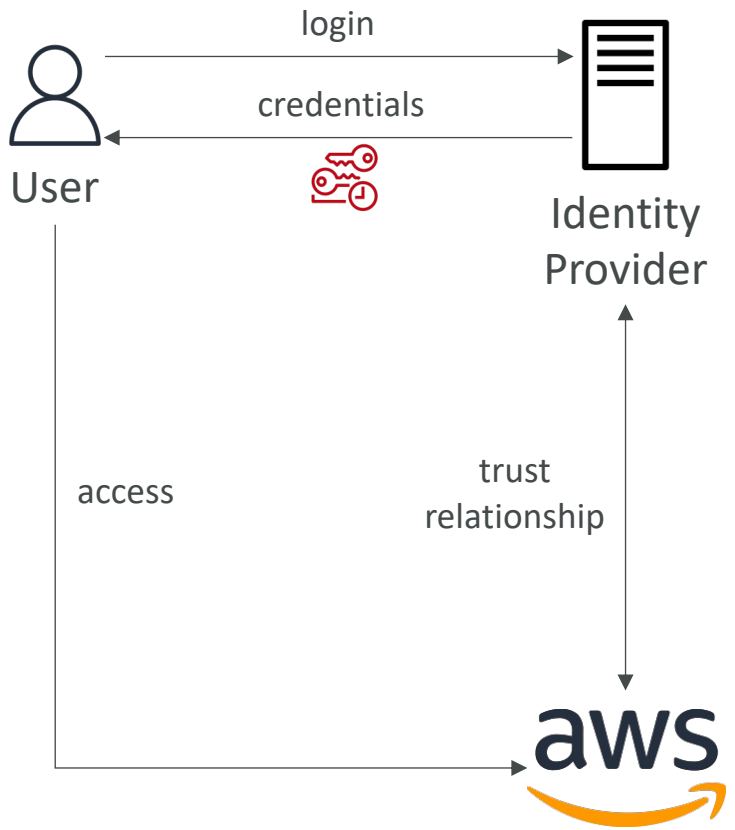
- **AssumeRole**: access a role within your account or cross-account
- **AssumeRoleWithSAML**: return credentials for users logged with SAML
- **AssumeRoleWithWebIdentity**: return creds for users logged with an IdP
  - Example providers include Amazon Cognito, Login with Amazon, Facebook, Google, or any OpenID Connect-compatible identity provider
  - **AWS recommends using Cognito instead**
- **GetSessionToken**: for MFA, from a user or AWS account root user
- **GetFederationToken**: obtain temporary creds for a federated user, usually a proxy app that will give the creds to a distributed app inside a corporate network

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Identity Federation in AWS

- Give users outside of AWS permissions to access AWS resources in your account
- **You don't need to create IAM Users (user management is outside AWS)**
- Use cases:
  - A corporate has its own identity system (e.g., Active Directory)
  - Web/Mobile application that needs access to AWS resources
- Identity Federation can have many flavors:
  - SAML 2.0
  - Custom Identity Broker
  - Web Identity Federation With(out) Amazon Cognito
  - IAM Identity Center



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SAML 2.0 Federation

- Security Assertion Markup Language 2.0 (SAML 2.0)
- Open standard used by many identity providers (e.g., ADFS)
  - Supports integration with Microsoft Active Directory Federations Services (ADFS)
  - Or any SAML 2.0–compatible IdPs with AWS
- Access to AWS Console, AWS CLI, or AWS API using temporary credentials
  - No need to create IAM Users for each of your employees
  - Need to setup a trust between AWS IAM and SAML 2.0 Identity Provider (both ways)

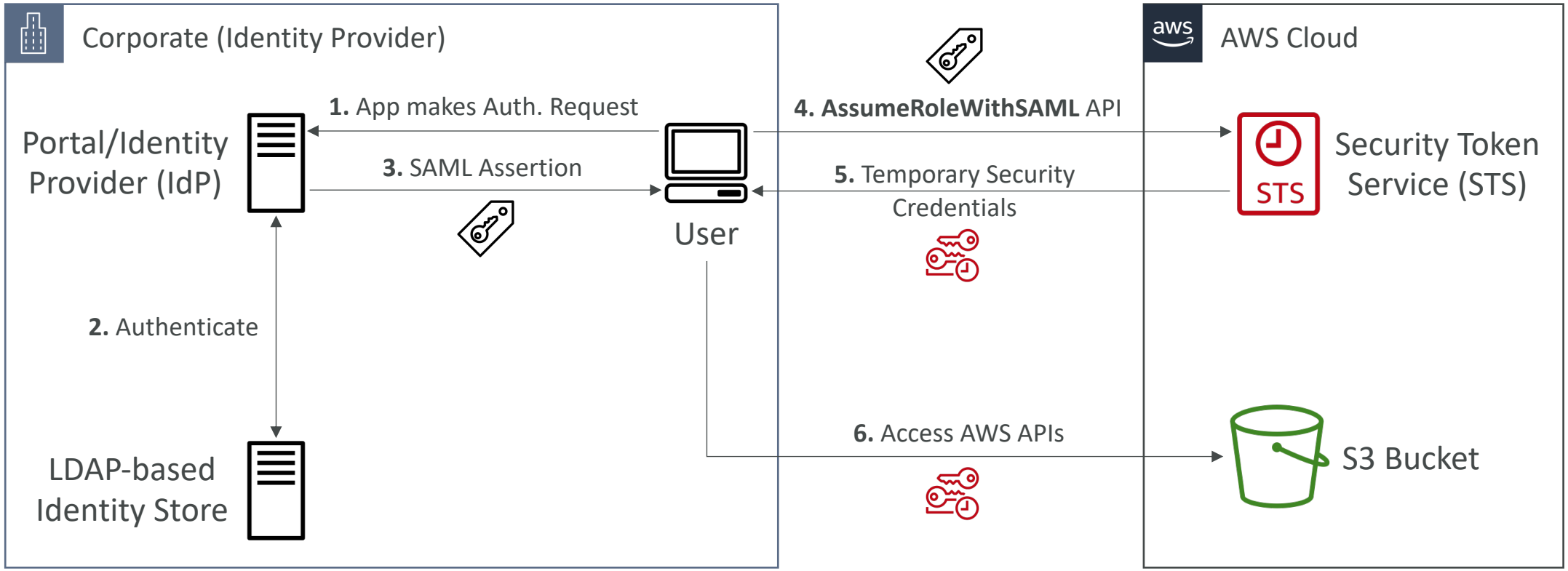
- Under-the-hood: Uses the STS API **AssumeRoleWithSAML**

- SAML 2.0 Federation is the "old way", **IAM Identity Center** Federation is the new managed and simpler way
  - https://docs.aws.amazon.com/singlesignon/latest/userguide/what-is.html

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

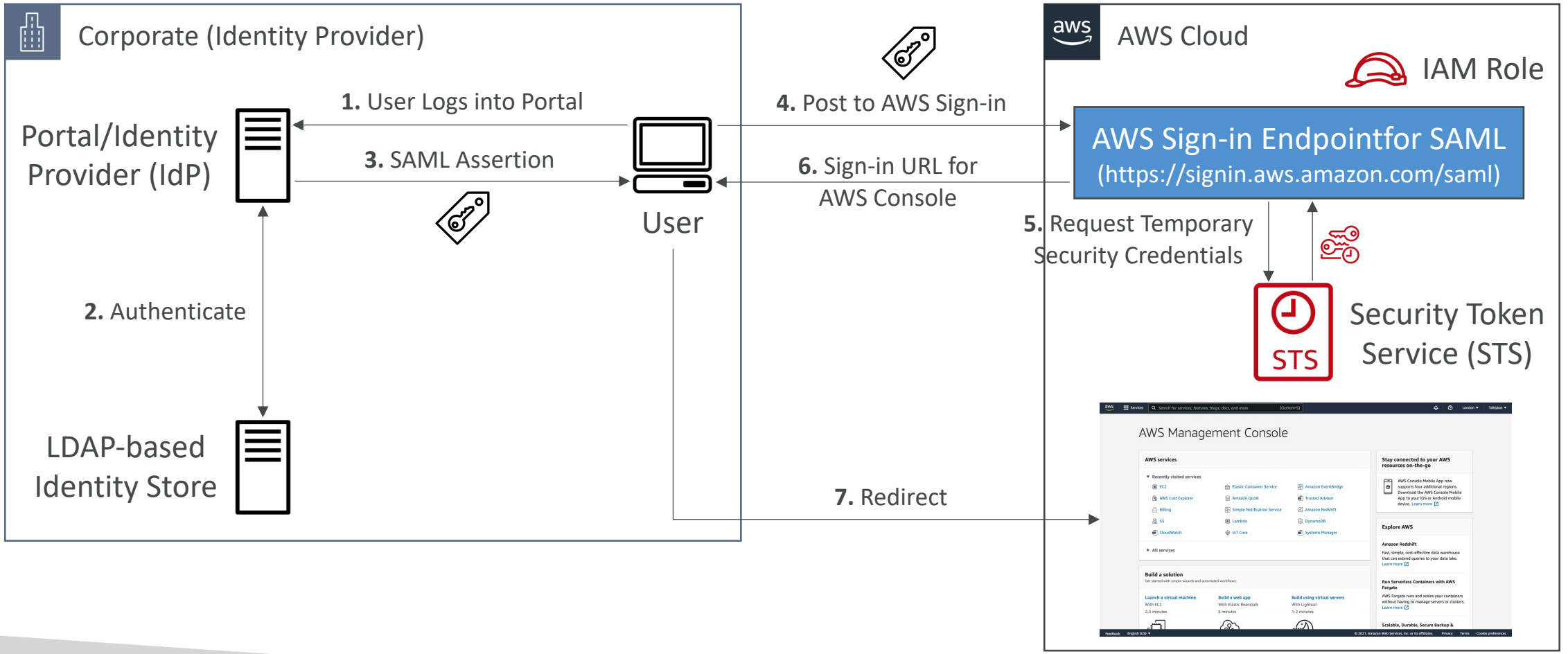
SAML 2.0 Federation – AWS API Access



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

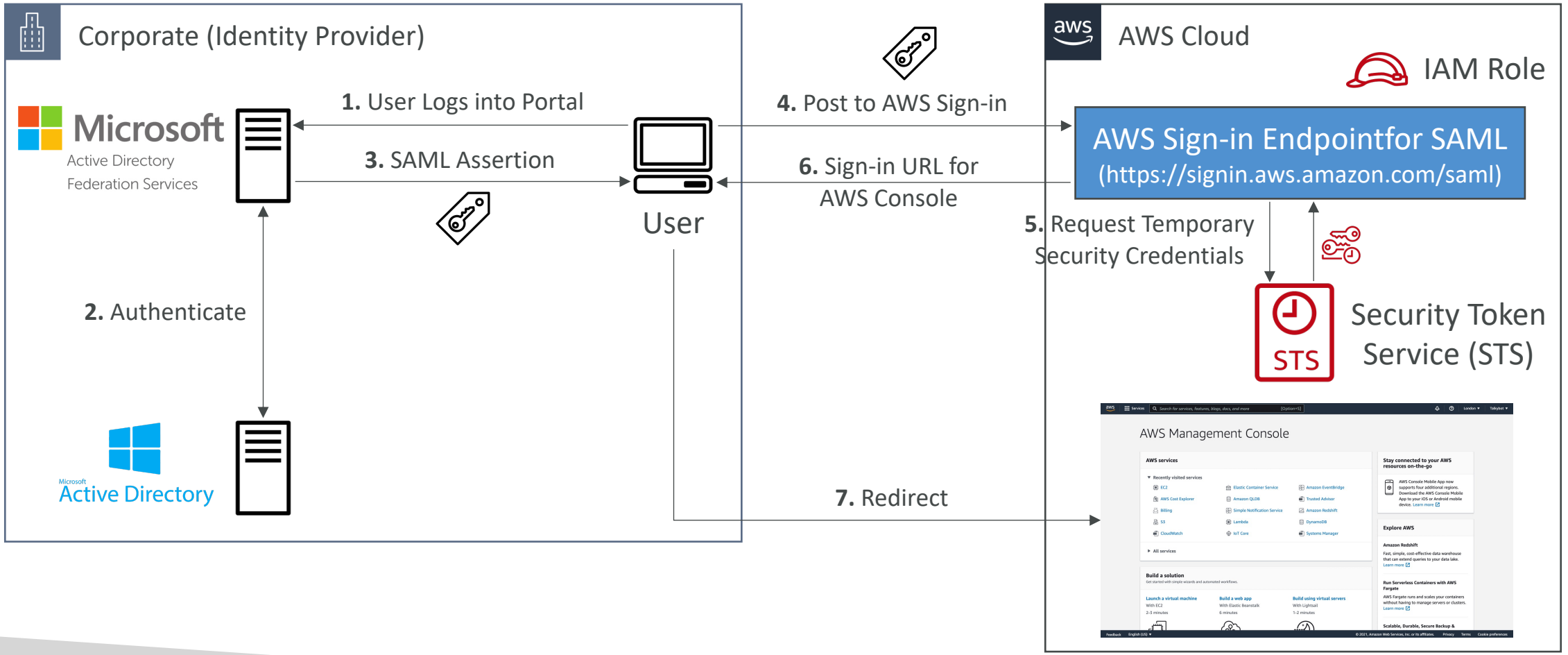
SAML 2.0 Federation – AWS Console Access



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SAML 2.0 Federation – Active Directory FS (ADFS)

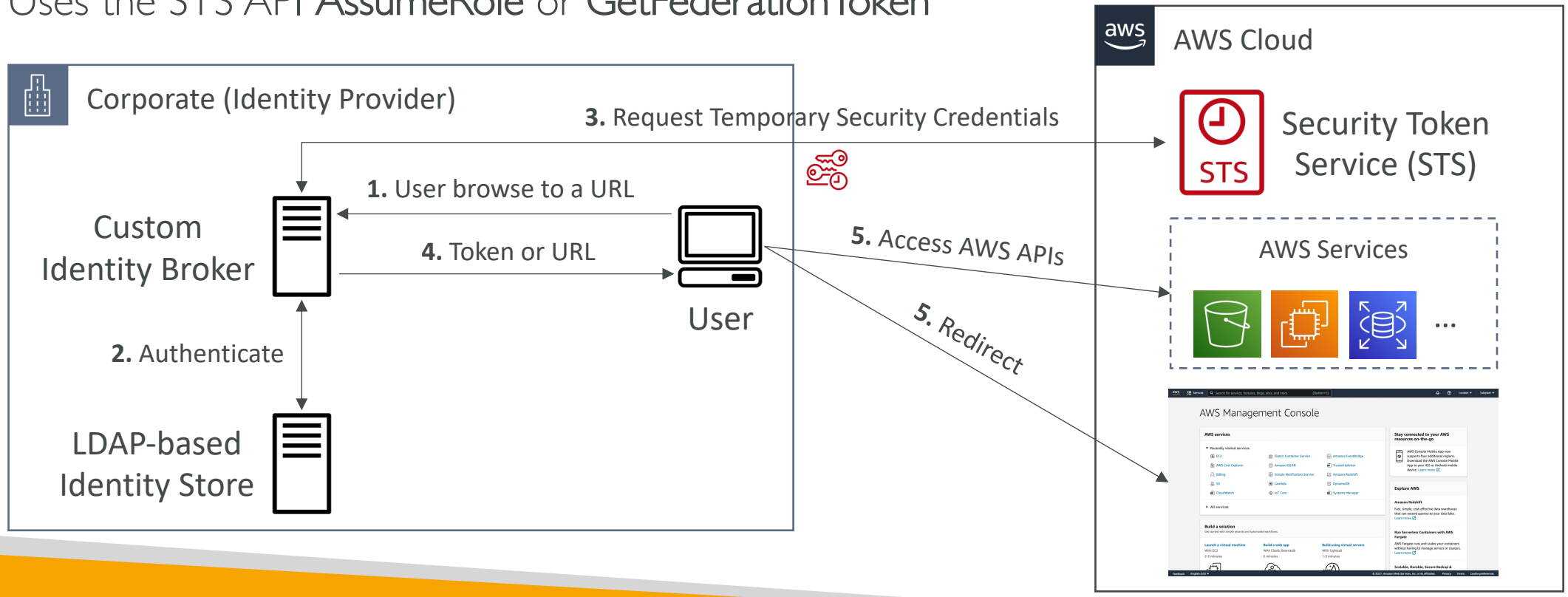


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Custom Identity Broker Application

- Use only if Identity Provider is **NOT** compatible with SAML 2.0
- The Identity Broker Authenticates users & requests temporary credentials from AWS
- **The Identity Broker must determine the appropriate IAM Role**
- Uses the STS API **AssumeRole** or **GetFederationToken**

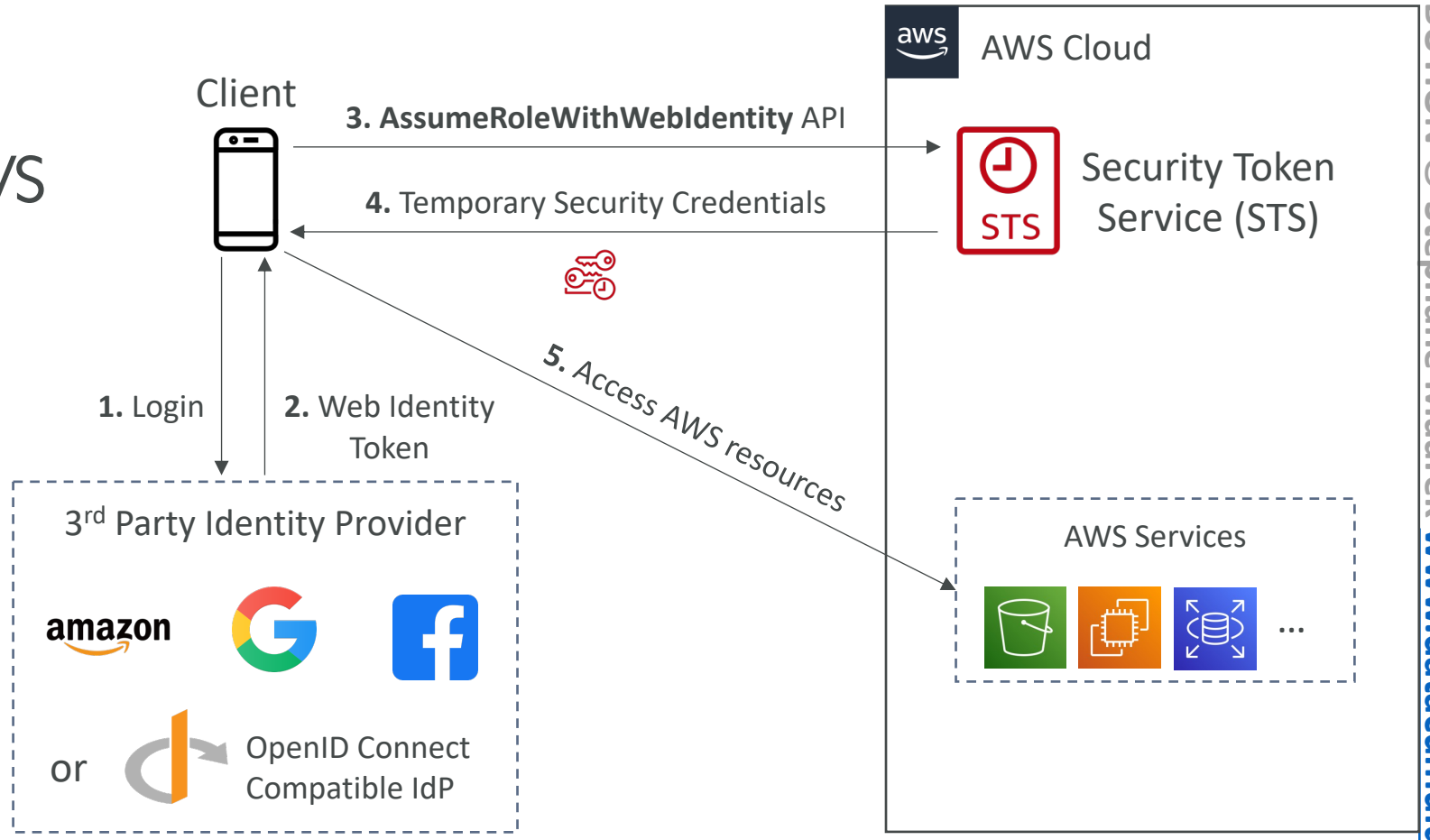


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Web Identity Federation – Without Cognito

- Not recommended by AWS – use Cognito instead

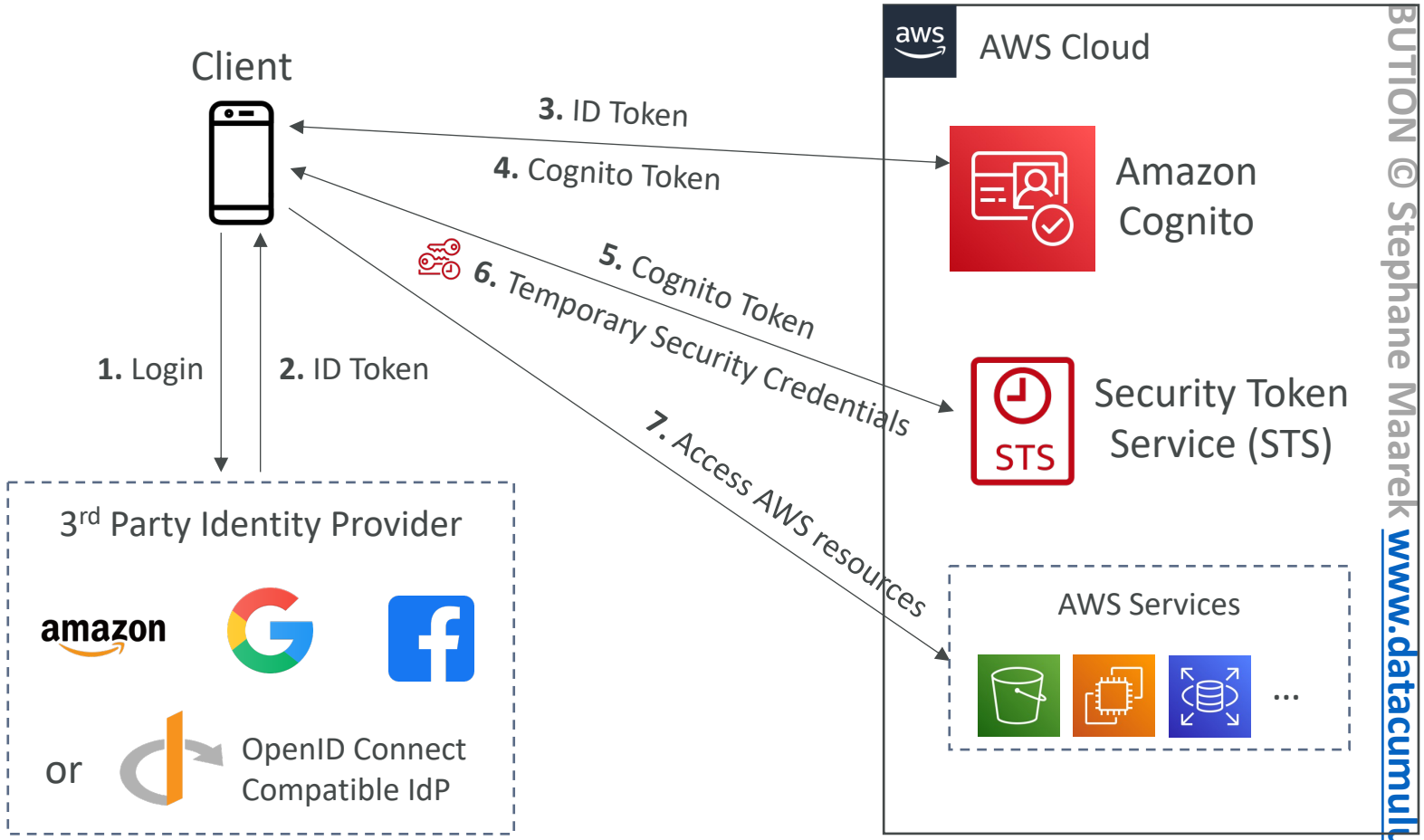


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Web Identity Federation – With Cognito

- Preferred over for Web Identity Federation
  - Create IAM Roles using Cognito with the least privilege needed
  - Build trust between the OIDC IdP and AWS
- Cognito benefits:
  - Supports anonymous users
  - Supports MFA
  - Data Synchronization
- Cognito replaces a Token Vending Machine (TVM)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Web Identity Federation – IAM Policy

- After being authenticated with Web Identity Federation, you can identify the user with an IAM policy variable

- Examples:
  - cognito-identity.amazonaws.com:sub
  - www.amazon.com:user_id
  - graph.facebook.com:id
  - accounts.google.com:sub

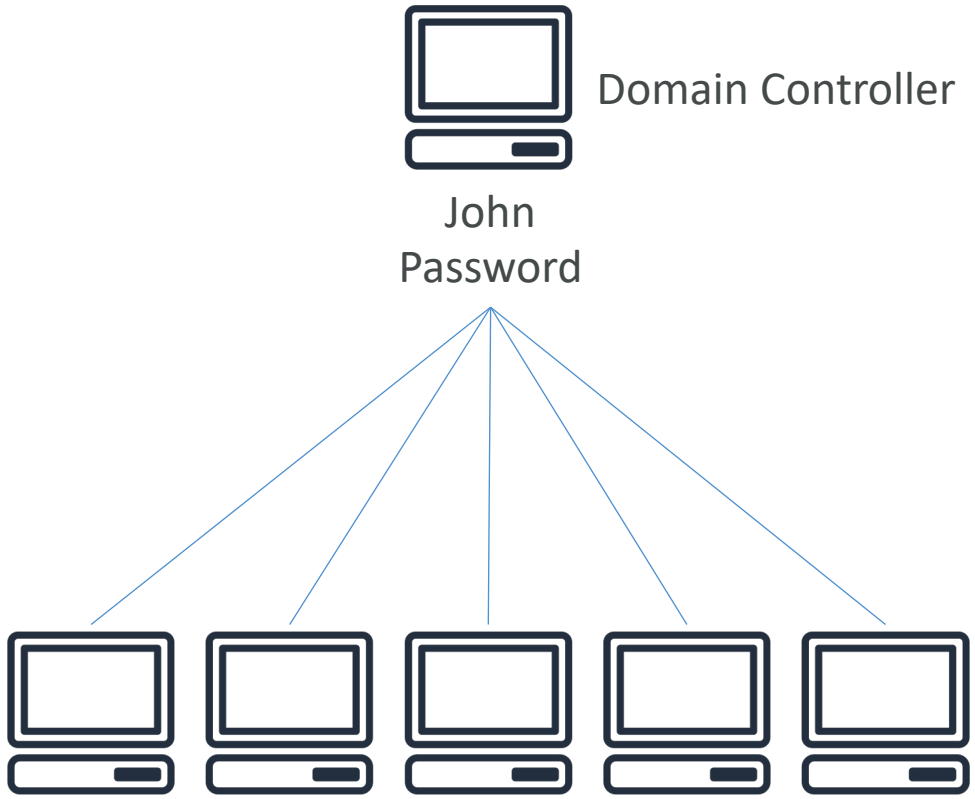
```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "s3:ListBucket",
            "Resource": "arn:aws:s3:::myBucket",
            "Condition": {
                "StringLike": {
                    "s3:prefix": "Amazon/mynumbersgame/${www.amazon.com:user_id}/*"
                }
            }
        }, {
            "Effect": "Allow",
            "Action": ["s3:GetObject", "s3:PutObject", "s3:DeleteObject"],
            "Resource": [
                "arn:aws:s3:::myBucket/amazon/mynumbersgame/${www.amazon.com:user_id}",
                "arn:aws:s3:::myBucket/amazon/mynumbersgame/${www.amazon.com:user_id}/*"
            ]
        }
    ]
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

What is Microsoft Active Directory (AD)?

- Found on any Windows Server with AD Domain Services
- Database of **objects**: User Accounts, Computers, Printers, File Shares, Security Groups
- Centralized security management, create account, assign permissions
- Objects are organized in **trees**
- A group of trees is a **forest**

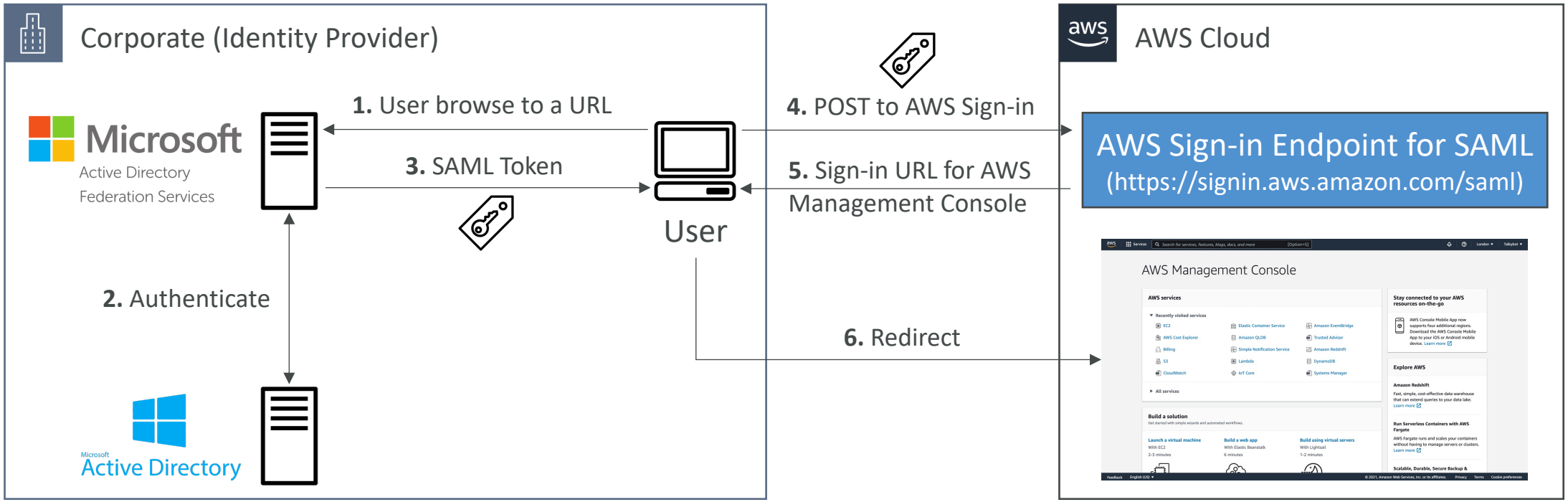


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

What is ADFS (AD Federation Services)?

- ADFS provides Single Sign-On across applications
- SAML across 3rd party: AWS Console, Dropbox, Office365, etc…

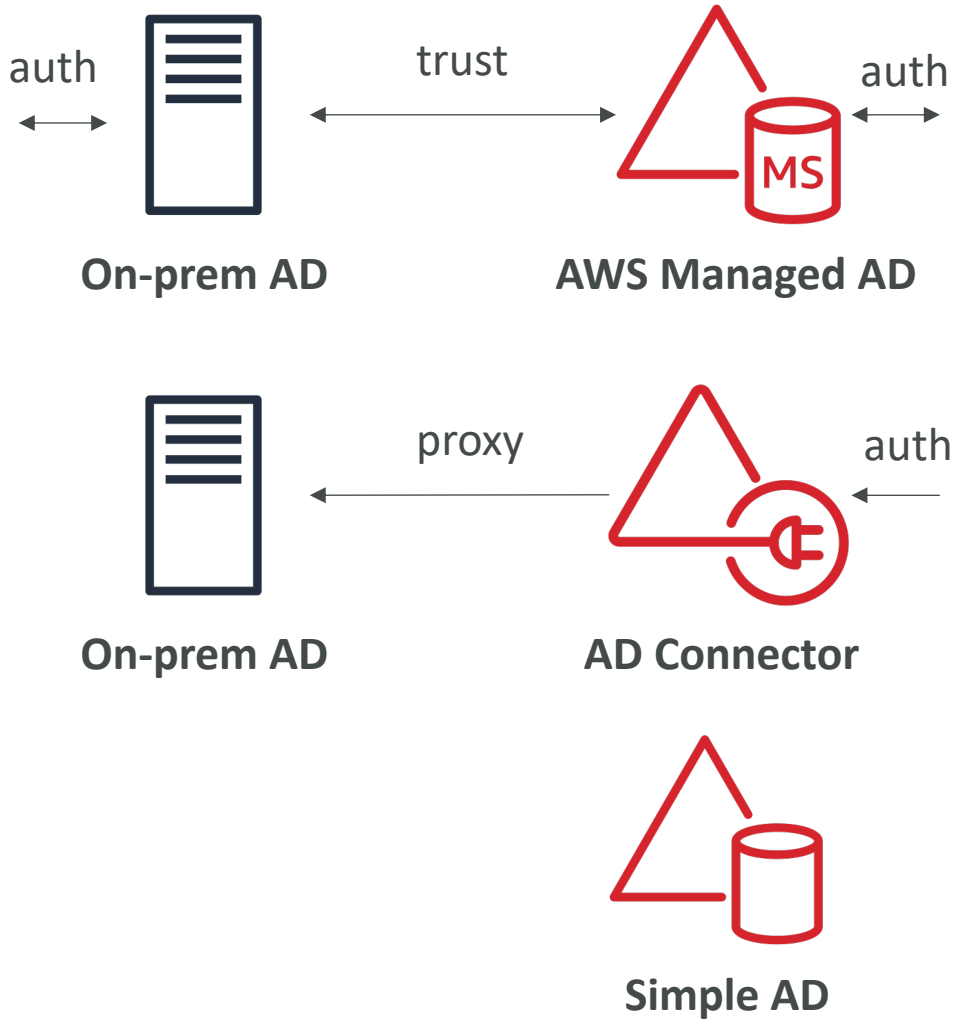


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Directory Services

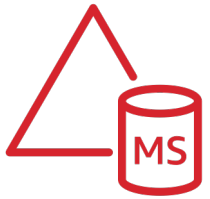
- AWS Managed Microsoft AD
  - Create your own AD in AWS, manage users locally, supports MFA
  - Establish "trust" connections with your on-premises AD
- AD Connector
  - Directory Gateway (proxy) to redirect to on-premises AD, supports MFA
  - Users are managed on the on-premises AD
- Simple AD
  - AD-compatible managed directory on AWS
  - Cannot be joined with on-premises AD



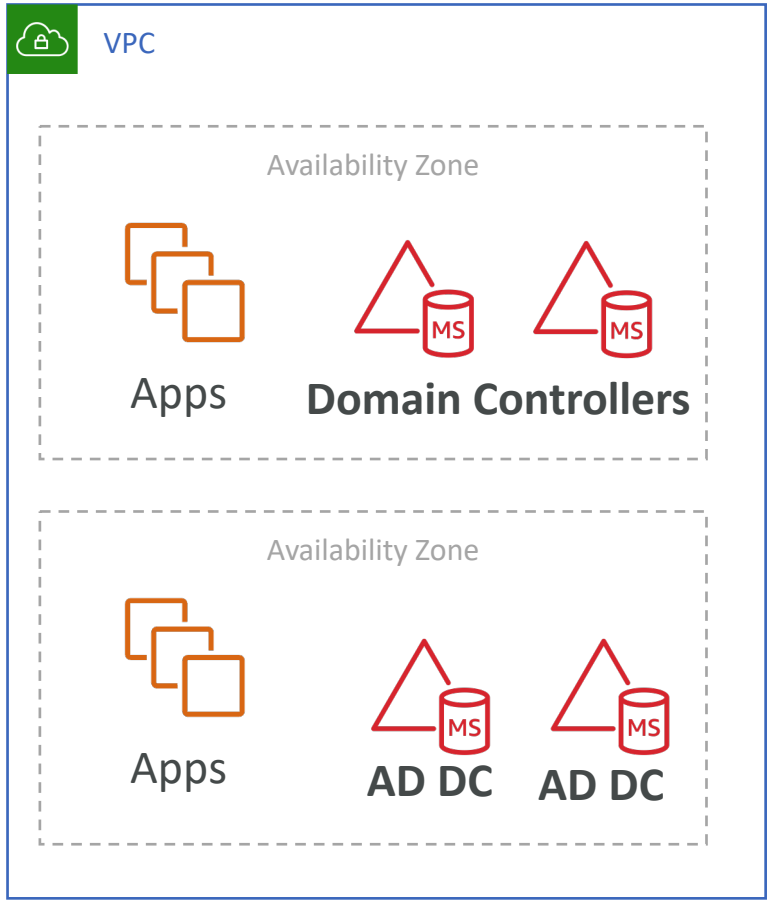
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Directory Services
AWS Managed Microsoft AD



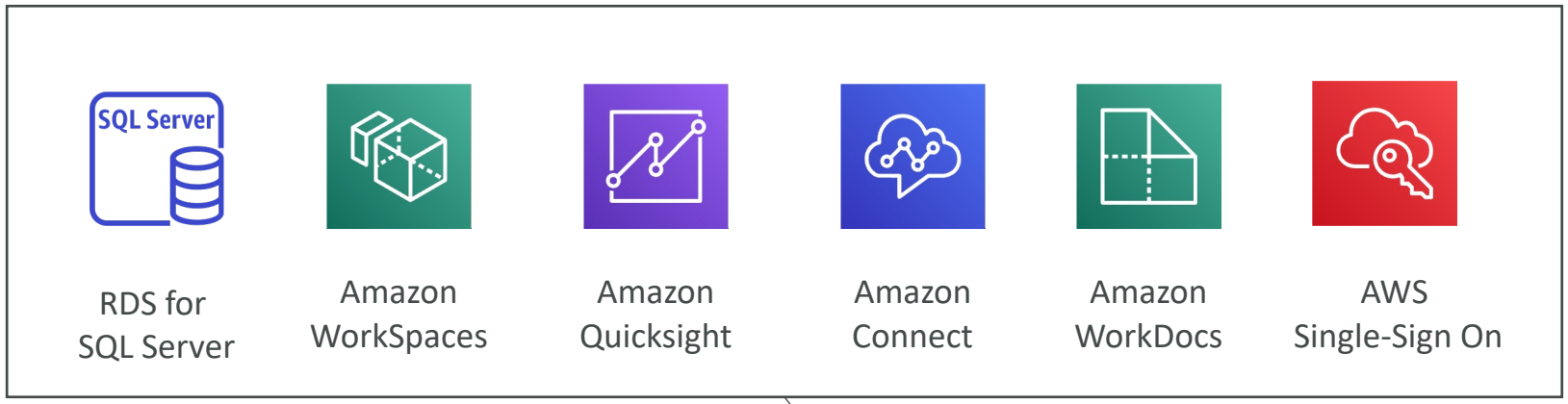
- Managed Service: Microsoft AD in your AWS VPC
- EC2 Windows Instances:
  - EC2 Windows instances can join the domain and run traditional AD applications (sharepoint, etc)
  - Seamlessly Domain Join Amazon EC2 Instances from Multiple Accounts & VPCs
- Integrations:
  - RDS for SQL Server, AWS Workspaces, Quicksight…
  - AWS SSO to provide access to 3rd party applications
- Standalone repository in AWS or joined to on-premises AD
- Multi AZ deployment of AD in 2 AZ, # of DC (Domain Controllers) can be increased for scaling
- Automated backups
- Automated Multi-Region replication of your directory

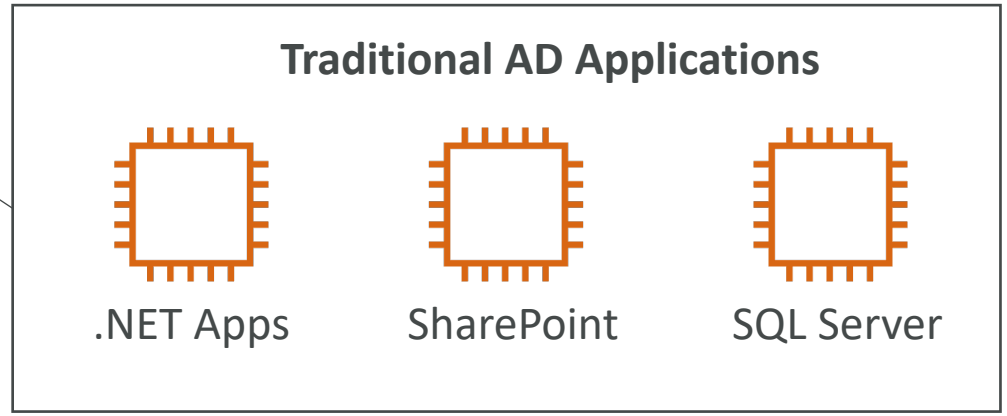


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Microsoft Managed AD - Integrations



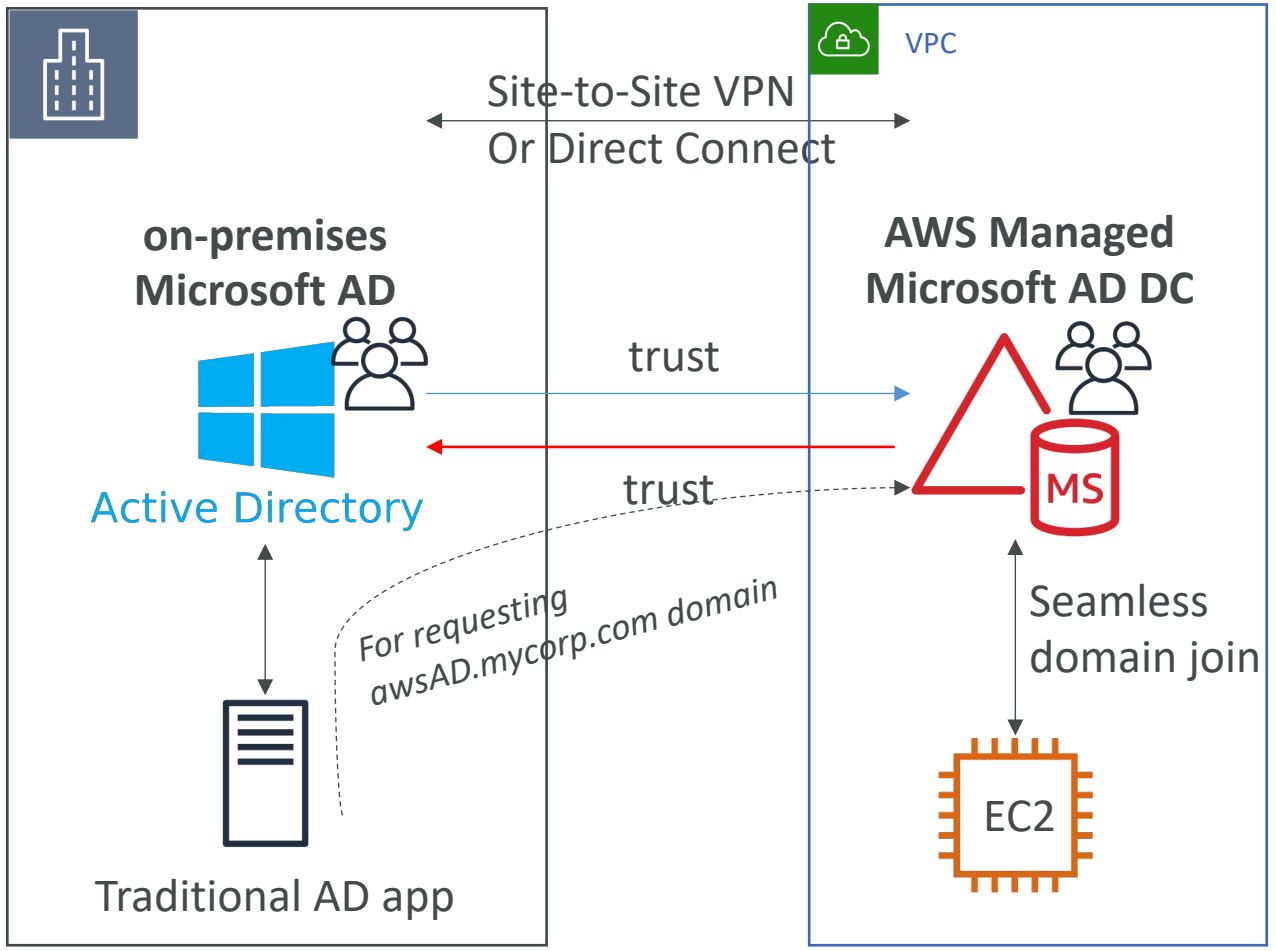


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Connect to on-premises AD

- Ability to connect your on-premises Active Directory to AWS Managed Microsoft AD
- Must establish a Direct Connect (DX) or VPN connection
- Can setup three kinds of forest trust:
  - **One-way trust:** AWS => on-premises
  - **One-way trust:** on-premises => AWS
  - **Two-way forest trust:** AWS ⇔ on-premises
- Forest trust is different than synchronization

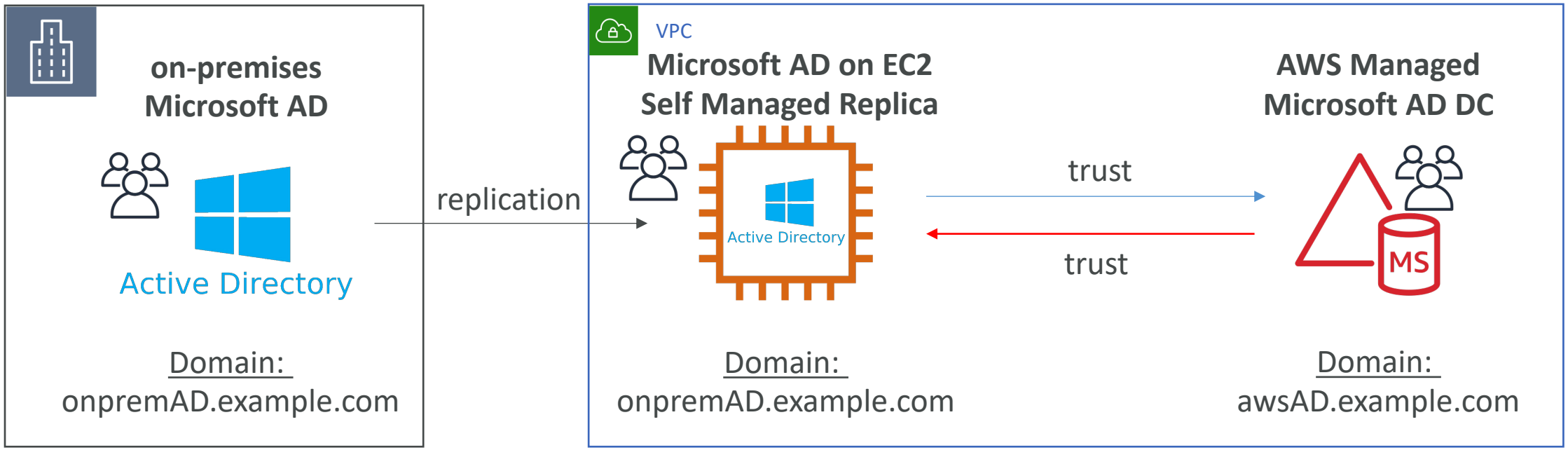


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Solution Architecture: Active Directory Replication

- You may want to create a replica of your AD on EC2 in the cloud to minimize latency of in case DX or VPN goes down
- Establish trust between the AWS Managed Microsoft AD and EC2

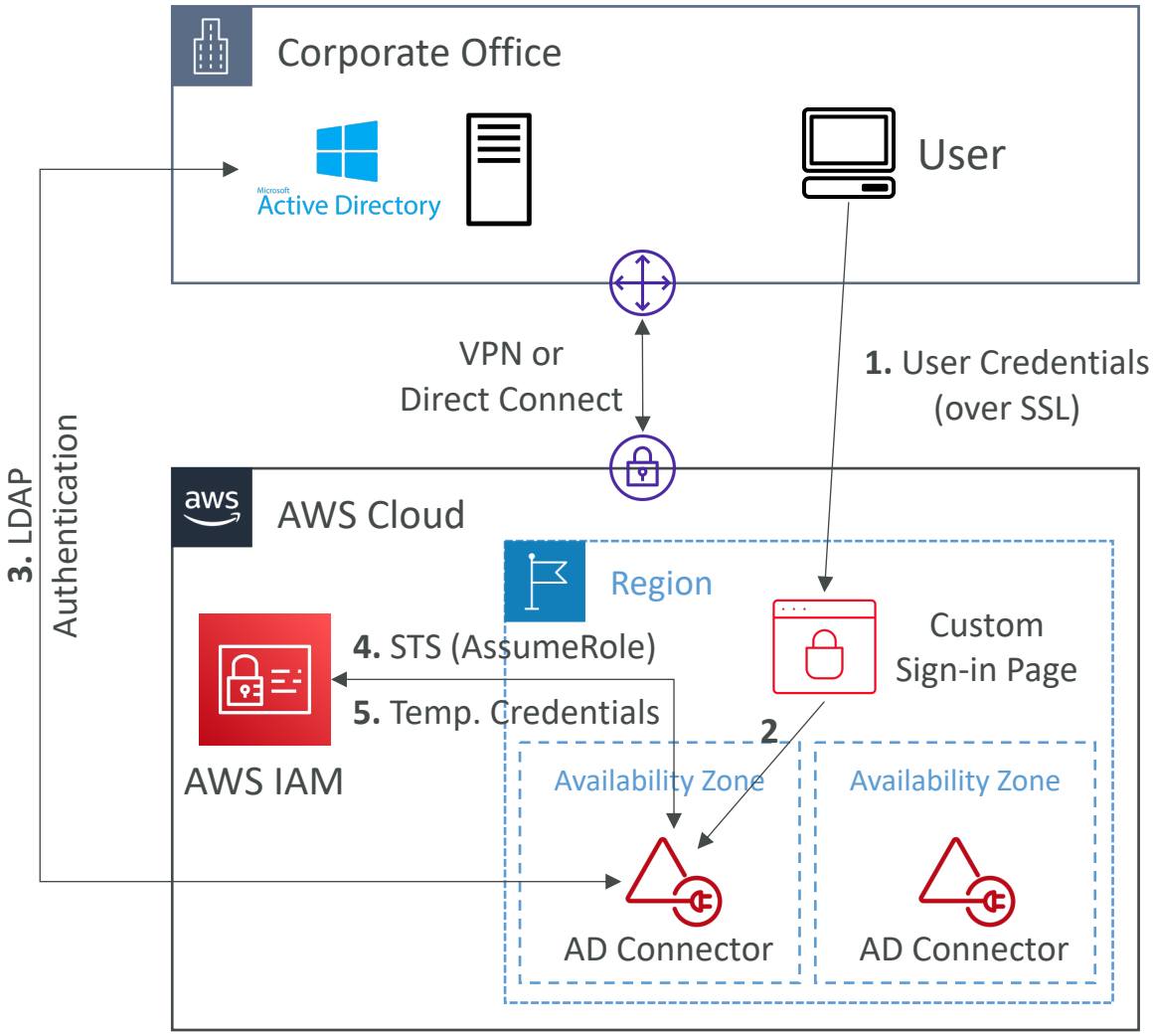


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Directory Services
AD Connector

- **AD Connector** is a directory gateway to redirect directory requests to your on-premises Microsoft Active Directory
- No caching capability
- Manage users solely on-premises, no possibility of setting up a trust
- VPN or Direct Connect
- Doesn't work with SQL Server, doesn't do seamless joining, can't share directory



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

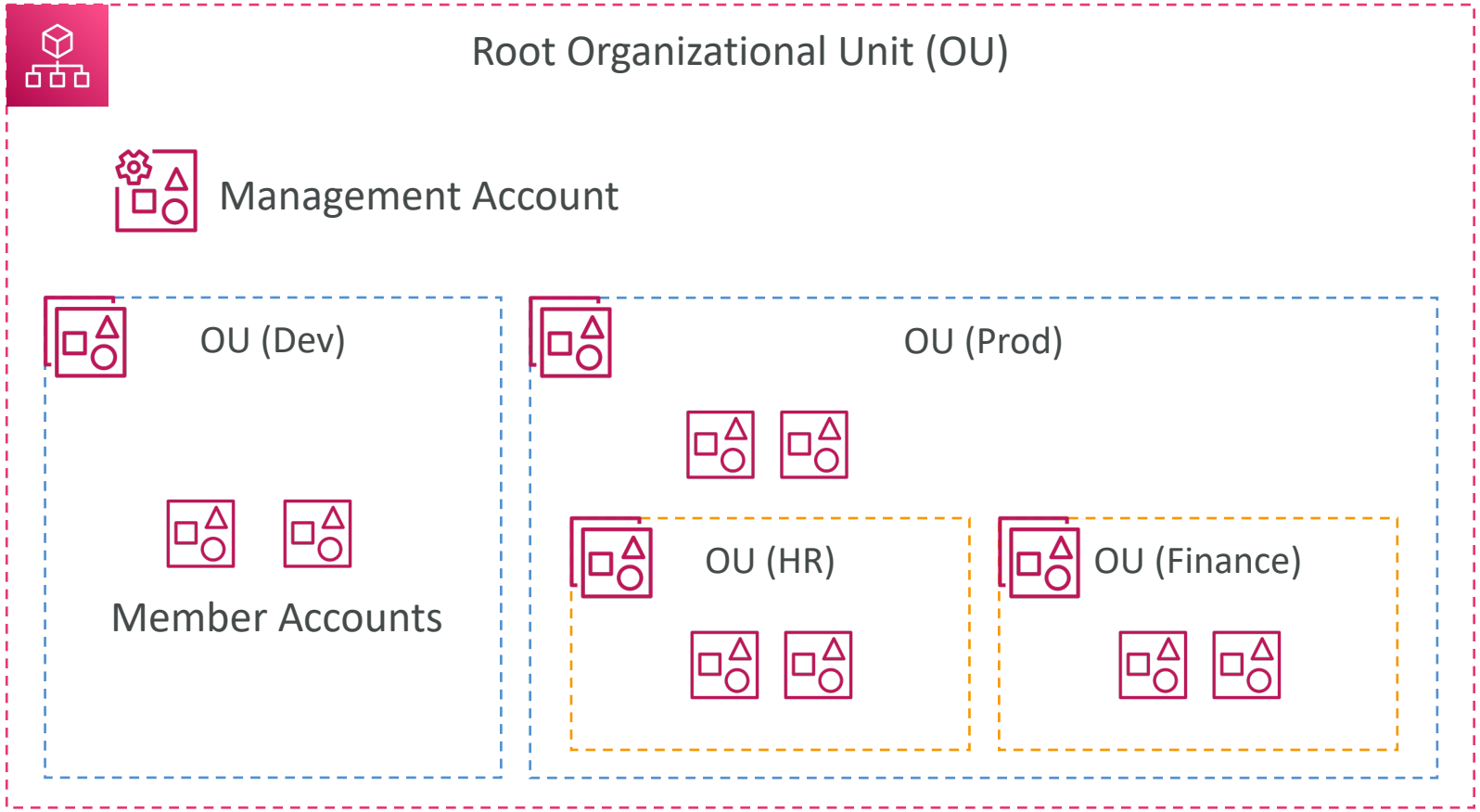
AWS Directory Services
Simple AD

- **Simple AD** is an inexpensive Active Directory–compatible service with the common directory features.
- Supports joining EC2 instances, manage users and groups
- Does not support MFA, RDS SQL server, AWS SSO
- Small: 500 users, large: 5000 users
- Powered by Samba 4, compatible with Microsoft AD
- lower cost, low scale, basic AD compatible, or LDAP compatibility
- No trust relationship

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Organizations

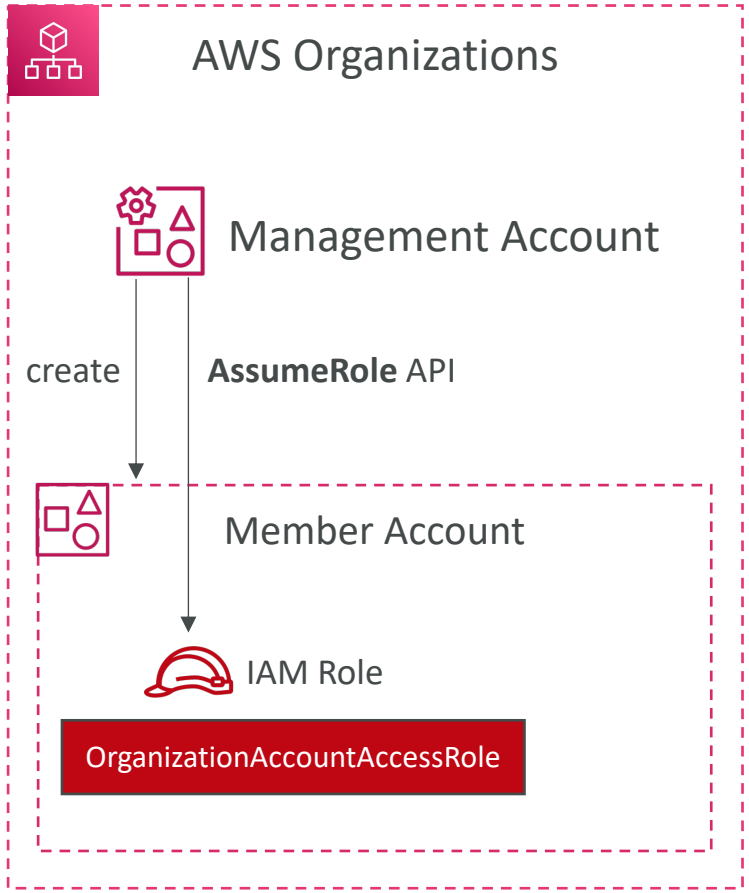


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Organizations - OrganizationAccountAccessRole

- IAM role which grants full administrator permissions in the Member account to the Management account
- Used to perform admin tasks in the Member accounts (e.g., creating IAM users)
- Could be assumed by IAM users in the Management account
- Automatically added to all new Member accounts created with AWS Organizations
- Must be created manually if you invite an existing Member account



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Multi Account Strategies

- Create accounts per **department**, per **cost center,** per **dev / test / prod**, based on **regulatory restrictions** (using SCP), for **better resource isolation** (ex: VPC), to have **separate per-account service limits**, isolated account for **logging**,

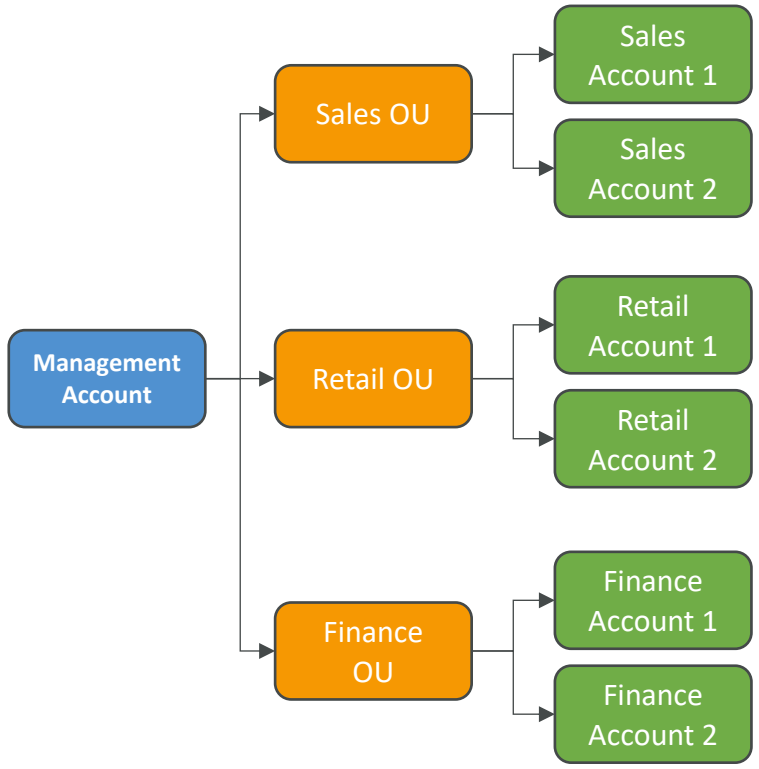
- Multi Account vs. One Account Multi VPC
- Use tagging standards for billing purposes
- Enable CloudTrail on all accounts, send logs to central S3 account
- Send CloudWatch Logs to central logging account
- Strategy to create an account for security

© Stephane Maarek

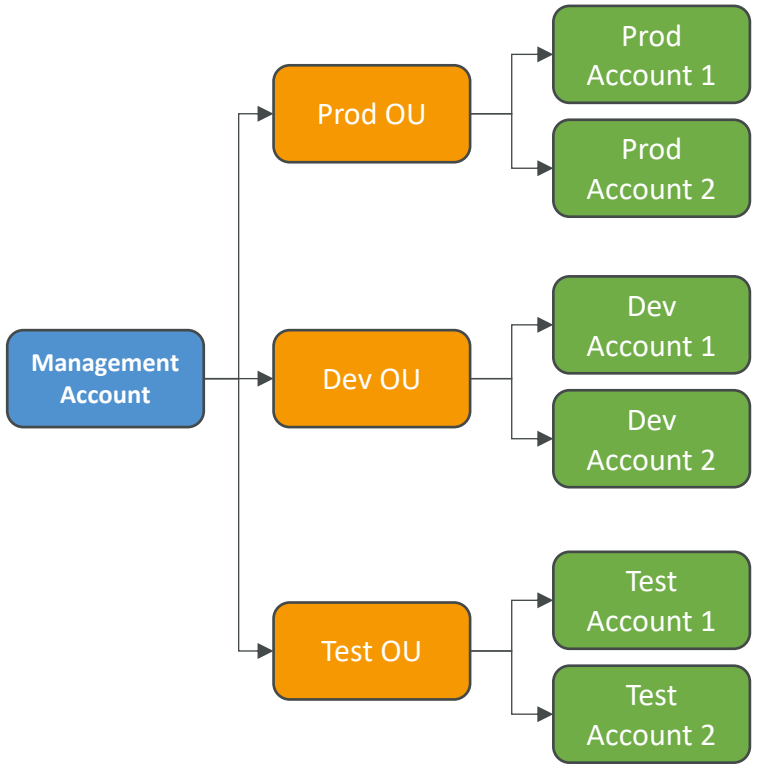
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Organizational Units (OU) - Examples

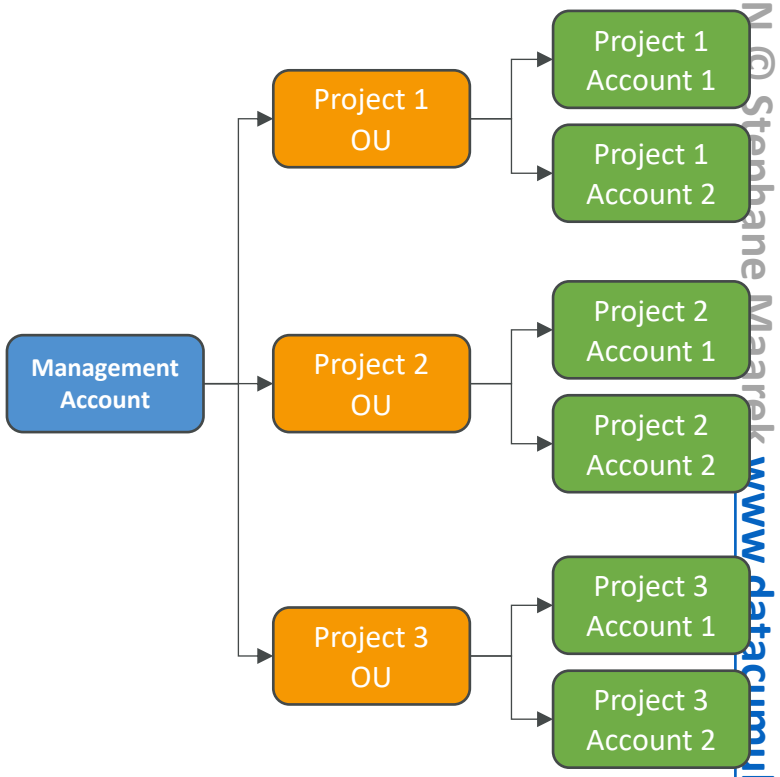
Business Unit



Environmental Lifecycle



Project-Based



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Organization - Feature Modes

- Consolidated billing features:
  - Consolidated Billing across all accounts - single payment method
  - Pricing benefits from aggregated usage (volume discount for EC2, S3…)

- All Features (Default):
  - Includes consolidated billing features, SCP
  - Invited accounts must approve enabling all features
  - Ability to apply an SCP to prevent member accounts from leaving the org
  - Can't switch back to Consolidated Billing Features only

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Organizations – Reserved Instances

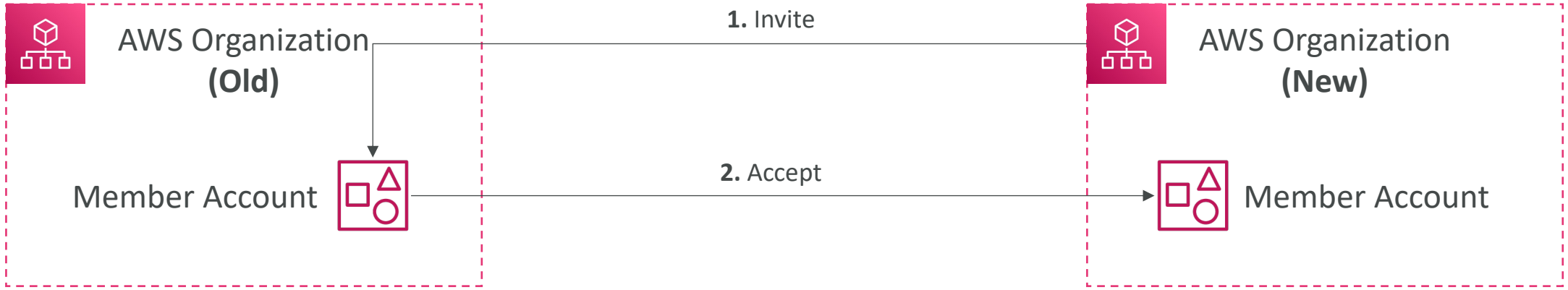
- For billing purposes, the consolidated billing feature of AWS Organizations treats all the accounts in the organization as one account.
- This means that **all accounts** in the organization can receive the hourly cost benefit of Reserved Instances that are purchased **by any other account.**
- **The payer account (Management account) of an organization** can turn off Reserved Instance (RI) discount and Savings Plans discount sharing for any accounts in that organization, including the payer account
- This means that RIs and Savings Plans discounts aren't shared between any accounts that have sharing turned off.
- To share an RI or Savings Plans discount with an account, both accounts must have sharing turned on

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Organizations – Moving Accounts

1. Send an invite to the member account from the AWS Organization
2. Accept the invite to the new Organization from the member account



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

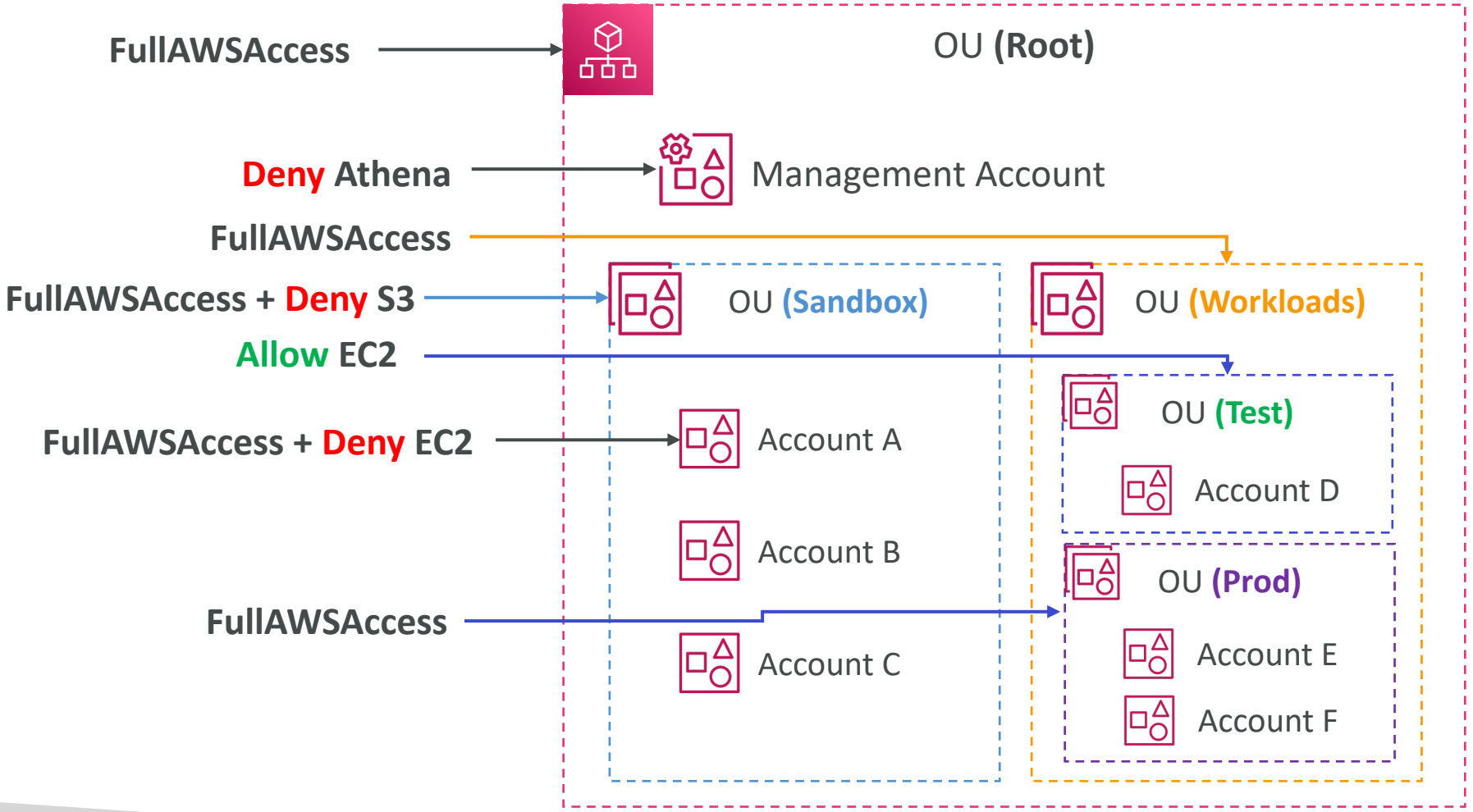
Service Control Policies (SCP)

- Define allowlist or blocklist IAM actions
- Applied at the **OU** or **Account** level
- Does not apply to the Management Account
- SCP is applied to all the **Users and Roles** in the account, including Root user
- The SCP does not affect Service-linked roles
  - Service-linked roles enable other AWS services to integrate with AWS Organizations and can't be restricted by SCPs.
- SCP must have an explicit Allow from the root a each OU in the direct path to the target account (does not allow anything by default)
- Use cases:
  - Restrict access to certain services (for example: can't use EMR)
  - Enforce PCI compliance by explicitly disabling services

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SCP Hierarchy



- Management Account
  - Can do anything (no SCP apply)
- Account A
  - Can do anything
  - EXCEPT S3 (explicit Deny from Sandbox OU)
  - EXCEPT EC2 (explicit Deny)
- Account B & C
  - Can do anything
  - EXCEPT S3 (explicit Deny from Sandbox OU)
- Account D
  - Can access EC2
- Prod OU & Account E & F
  - Can do anything

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

SCP Examples
Blocklist and Allowlist strategies

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "AllowsAllActions",
            "Effect": "Allow",
            "Action": "*",
            "Resource": "*"
        },
        {
            "Sid": "DenyDynamoDB",
            "Effect": "Deny",
            "Action": "dynamodb:*",
            "Resource": "*"
        }
    ]
}
```

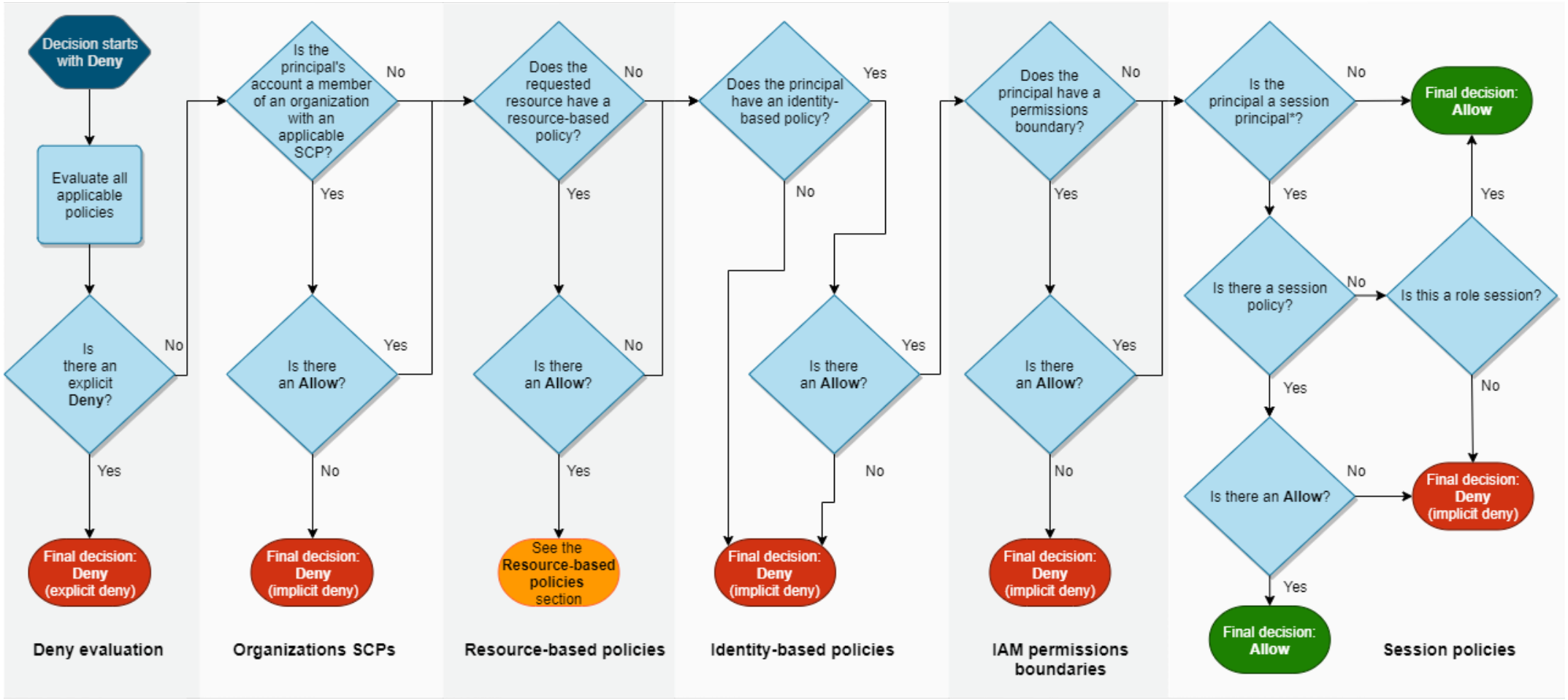
```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "ec2:*",
                "cloudwatch:*"
            ],
            "Resource": "*"
        }
    ]
}
```

More examples: https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_example-scps.html

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM Policy Evaluation Logic



*A session principal is either a role session or an IAM federated user session.

https://docs.aws.amazon.com/IAM/latest/UserGuide/reference_policies_evaluation-logic.html

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Restricting Tags with IAM Policies

- You can restrict specific Tags on AWS resources
- Using the **aws:TagKeys** Condition Key
  - Validate the Tag Keys attached to a resource against the Tag Keys in the IAM Policy
- Example: allow IAM users to create EBS Volumes only if it has the "Env" and "CostCenter" Tags
- Use either **ForAllValues** (must have all keys) or **ForAnyValue** (must have any of these keys at a minimum)

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "ec2:CreateVolume",
            "Resource": "arn:aws:ec2:*:*:volume/*",
            "Condition": {
                "ForAllValues:StringEquals": {
                    "aws:TagKeys": ["Env", "CostCenter"]
                }
            }
        }
    ]
}
```

Match **All** Keys

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "ec2:CreateVolume",
            "Resource": "arn:aws:ec2:*:*:volume/*",
            "Condition": {
                "ForAnyValue:StringEquals": {
                    "aws:TagKeys": ["Env", "CostCenter"]
                }
            }
        }
    ]
}
```

Match **Any** Keys

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Using SCP to Deny a Region aws:RequestedRegion

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Deny",
            "Action": ["ec2:*", "rds:*", "dynamodb:*"],
            "Resource": "*",
            "Condition": {
                "StringEquals": {
                    "aws:RequestedRegion": ["eu-central-1", "eu-west-1"]
                },
                "ArnNotLike": {
                    "aws:PrincipalARN": [
                        "arn:aws:iam::*:role/Role1AllowedToBypassThisSCP",
                        "arn:aws:iam::*:role/Role2AllowedToBypassThisSCP"
                    ]
                }
            }
        }
    ]
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Using SCP to Restrict Creating Resources without appropriate Tags

- Prevent IAM Users/Roles in the affected Member accounts from creating resources if they don't have a specific Tags

- Example: restrict launching an EC2 instance if it doesn't have the "Project" and "CostCenter" Tags

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "DenyRunInstanceWithNoProjectTag",
            "Effect": "Deny",
            "Action": "ec2:RunInstances",
            "Resource": ["arn:aws:ec2:*:*:instance/*", "arn:aws:ec2:*:*:volume/*"],
            "Condition": {
                "Null": {
                    "aws:RequestTag/Project": "true"
                }
            }
        },
        {
            "Sid": "DenyRunInstanceWithNoCostCenterTag",
            "Effect": "Deny",
            "Action": "ec2:RunInstances",
            "Resource": ["arn:aws:ec2:*:*:instance/*", "arn:aws:ec2:*:*:volume/*"],
            "Condition": {
                "Null": {
                    "aws:RequestTag/CostCenter": "true"
                }
            }
        }
    ]
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION

AWS Organizations – Tag Policies

- Helps you standardize tags across resources in an AWS Organization
- Ensure consistent tags, audit tagged resources, maintain proper resources categorization, …
- You define Tag keys and their allowed values
- Helps with AWS Cost Allocation Tags and Attribute-based Access Control
- Prevent any non-compliant tagging operations on specified services and resources
- Generate a report that lists all tagged/non-compliant resources
- Use Amazon EventBridge to monitor non-compliant tags

```
{
    "tags": {
        "costcenter": {
            "tag_key": {
                "@@assign": "CostCenter"
            },
            "tag_value": {
                "@@assign": ["100", "200"]
            },
            "enforced_for": {
                "@@assign": ["secretsmanager:*"]
            }
        }
    }
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Organizations – AI Services Opt-out Policies

- Certain AWS AI services may use your content for continuous improvement of Amazon AI/ML services
- Example: Amazon Lex, Amazon Comprehend, Amazon Polly, …

- You can opt-out of having your content stored or used by AWS AI services
- Create an Opt-out Policy that enforces this setting across all Member accounts and AWS Regions
- You can opt-out all AI services or selected services
- Can be attached to Organization Root, specific OU, or individual Member account

```
{
    "services": {
        "default": {
            "opt_out_policy": {
                "@@assign": "optOut"
            }
        }
    }
}
```

All Services

```
{
    "services": {
        "rekognition": {
            "opt_out_policy": {
                "@@assign": "optOut"
            }
        },
        "lex": {
            "opt_out_policy": {
                "@@assign": "optOut"
            }
        }
    }
}
```

ONLY Rekognition & Lex

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Organizations – Backup Policies

- AWS Backup enables you to create Backup Plans that define how to backup your AWS resources

- JSON documents that define Backup Plans across an AWS Organization
- Gives you granular control over backing up your resources (e.g., backup frequency, time window, backup region, …)
- Can be attached to Organization Root, specific OU, or individual Member account
- **Immutable** Backup Plans appear in Member accounts (view ONLY)

```
{
    "plans": {
        "PII_Backup_Plan": {
            "regions": {
                "@@assign": ["us-east-1"]
            },
            "rules": {
                "My_Hourly_Rule": {
                    "schedule_expression": { "@@assign": "cron(0 5 ? * * *)" },
                    "start_backup_window_minutes": { "@@assign": "60" },
                    "complete_backup_window_minutes": { "@@assign": "604800" },
                    "enable_continuous_backup": { "@@assign": false },
                    "target_backup_vault_name": { "@@assign": "My_Backup_Vault" },
                    "lifecycle": {
                        "move_to_cold_storage_after_days": { "@@assign": "180" },
                        "delete_after_days": { "@@assign": "270" }
                    }
                }
            }
        }
    }
}
```

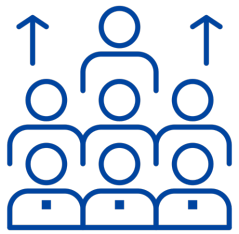
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS IAM Identity Center (successor to AWS Single Sign-On)

- One login (single sign-on) for all your
  - **AWS accounts in AWS Organizations**
  - Business cloud applications (e.g., Salesforce, Box, Microsoft 365, …)
  - SAML2.0-enabled applications
  - EC2 Windows Instances

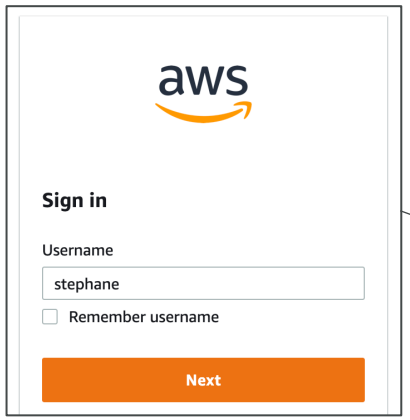
- Identity providers
  - Built-in identity store in IAM Identity Center
  - 3rd party: Active Directory (AD), OneLogin, Okta…



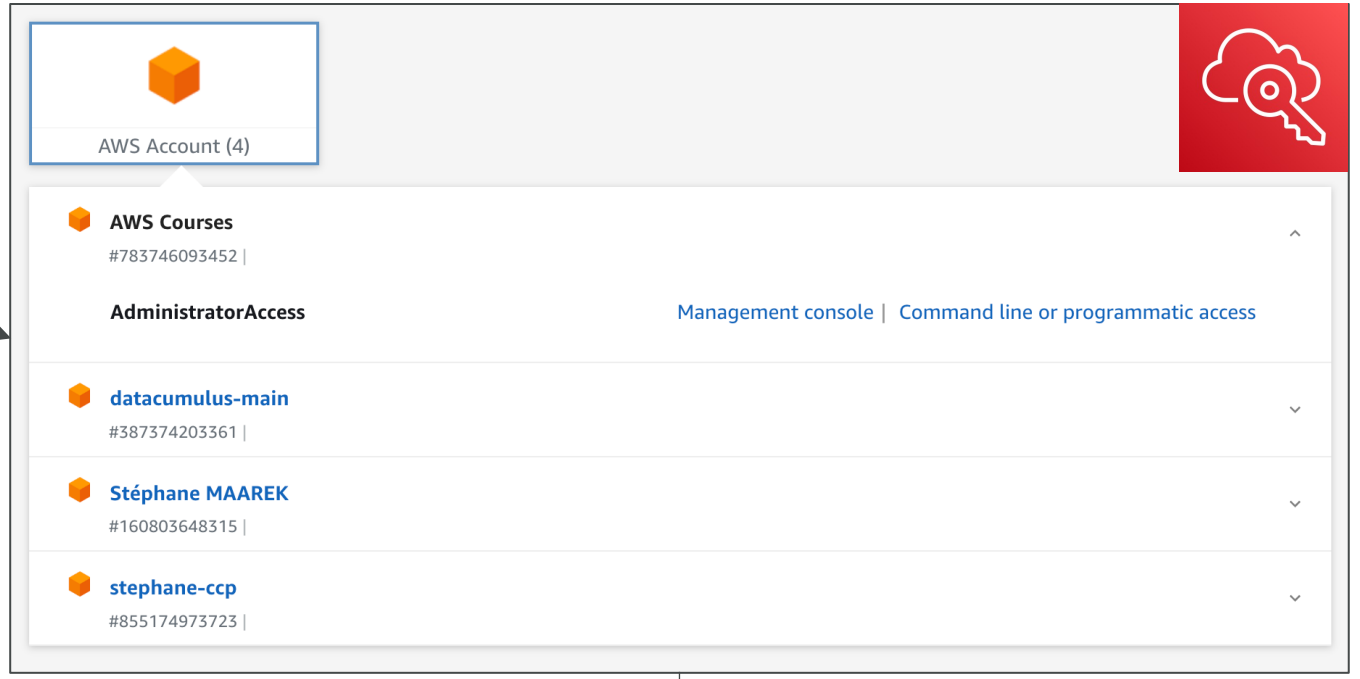
© Stephane Maarek

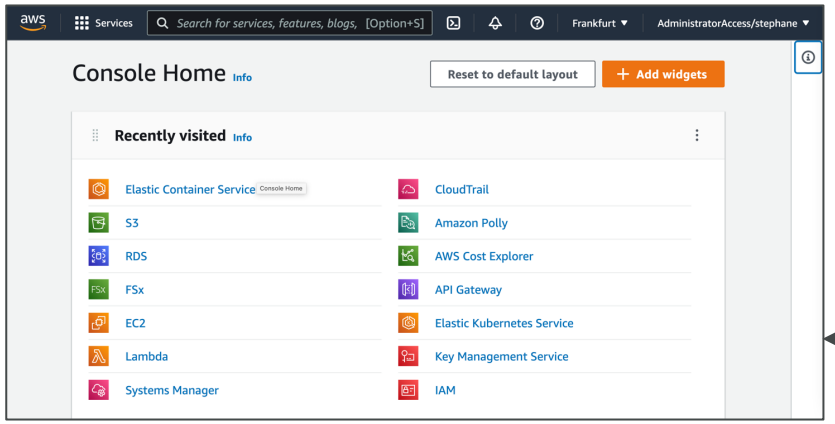
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS IAM Identity Center – Login Flow



AWS IAM Identity Center

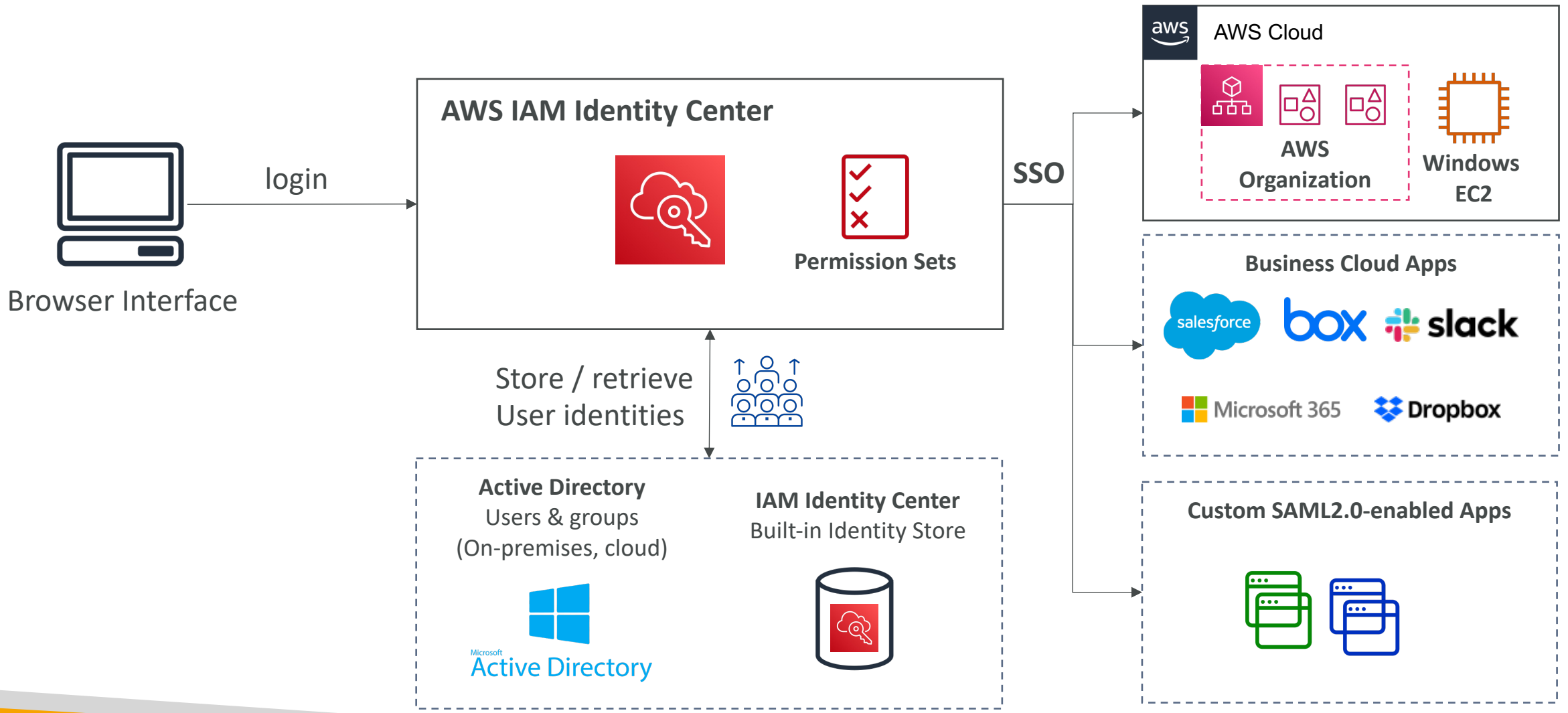




NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

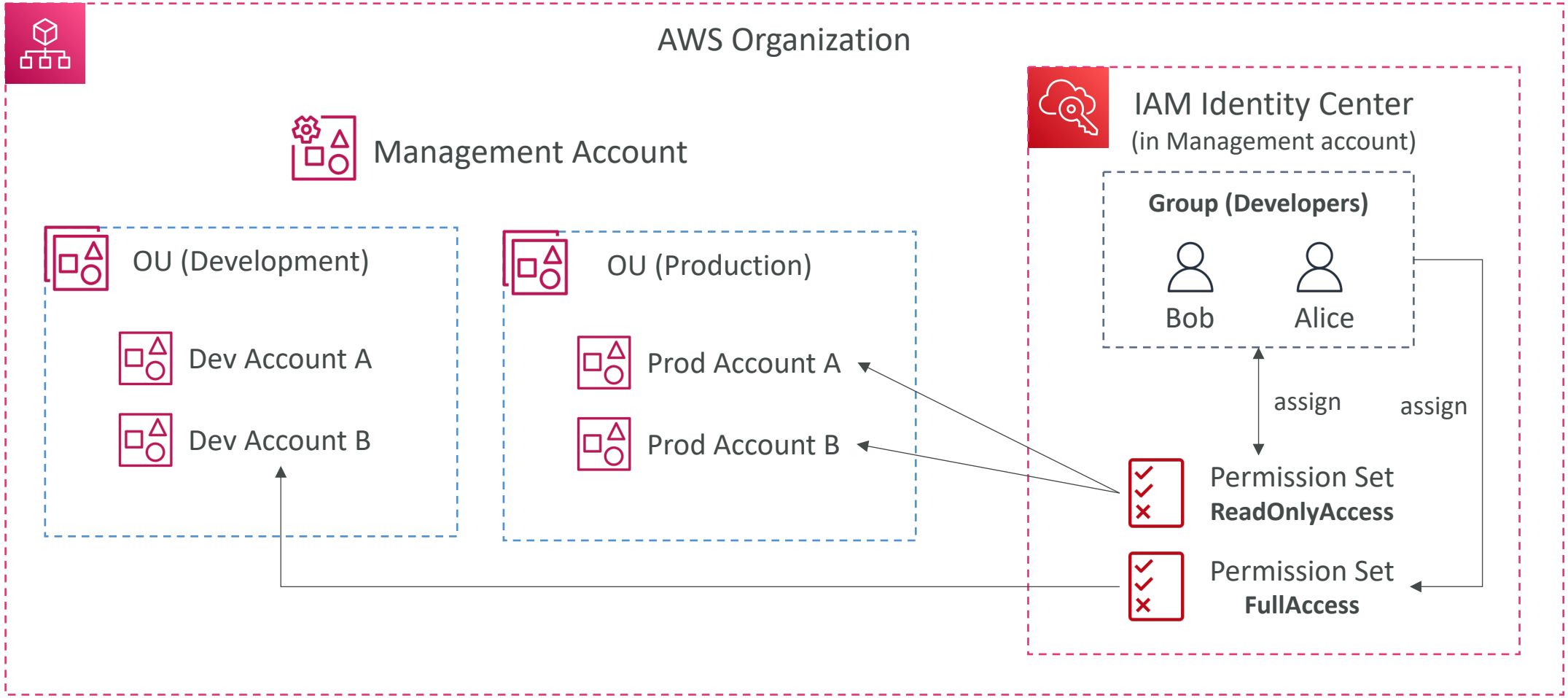
AWS IAM Identity Center



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM Identity Center

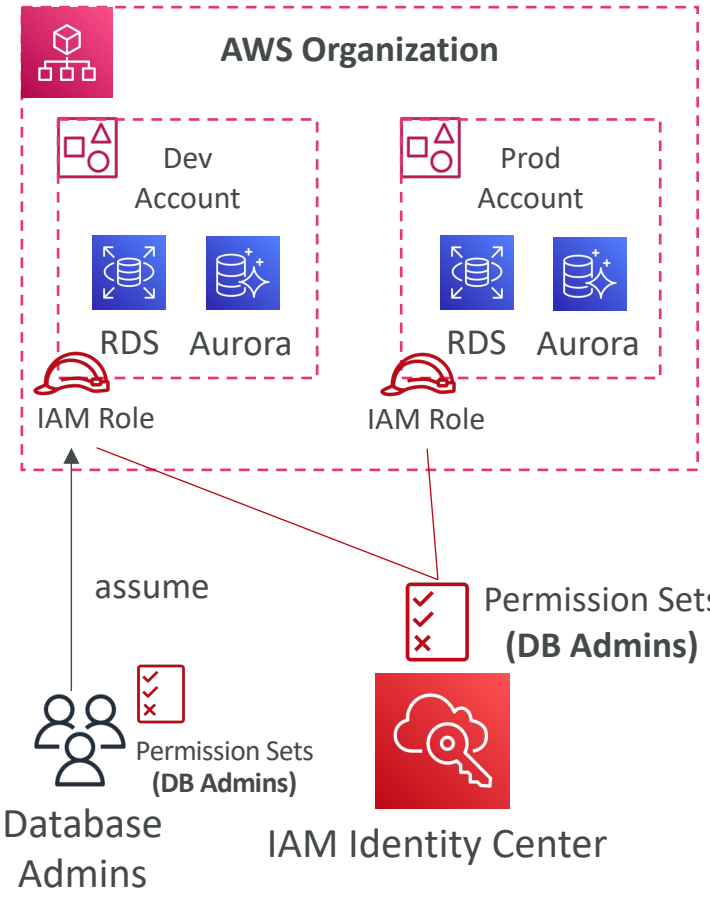


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS IAM Identity Center
Fine-grained Permissions and Assignments

- **Multi-Account Permissions**
  - Manage access across AWS accounts in your AWS Organization
  - Permission Sets – a collection of one or more IAM Policies assigned to users and groups to define AWS access
- **Application Assignments**
  - SSO access to many SAML 2.0 business applications (Salesforce, Box, Microsoft 365, …)
  - Provide required URLs, certificates, and metadata
- **Attribute-Based Access Control (ABAC)**
  - Fine-grained permissions based on users' attributes stored in IAM Identity Center Identity Store
  - Example: cost center, title, locale, …
  - Use case: Define permissions once, then modify AWS access by changing the attributes



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Control Tower

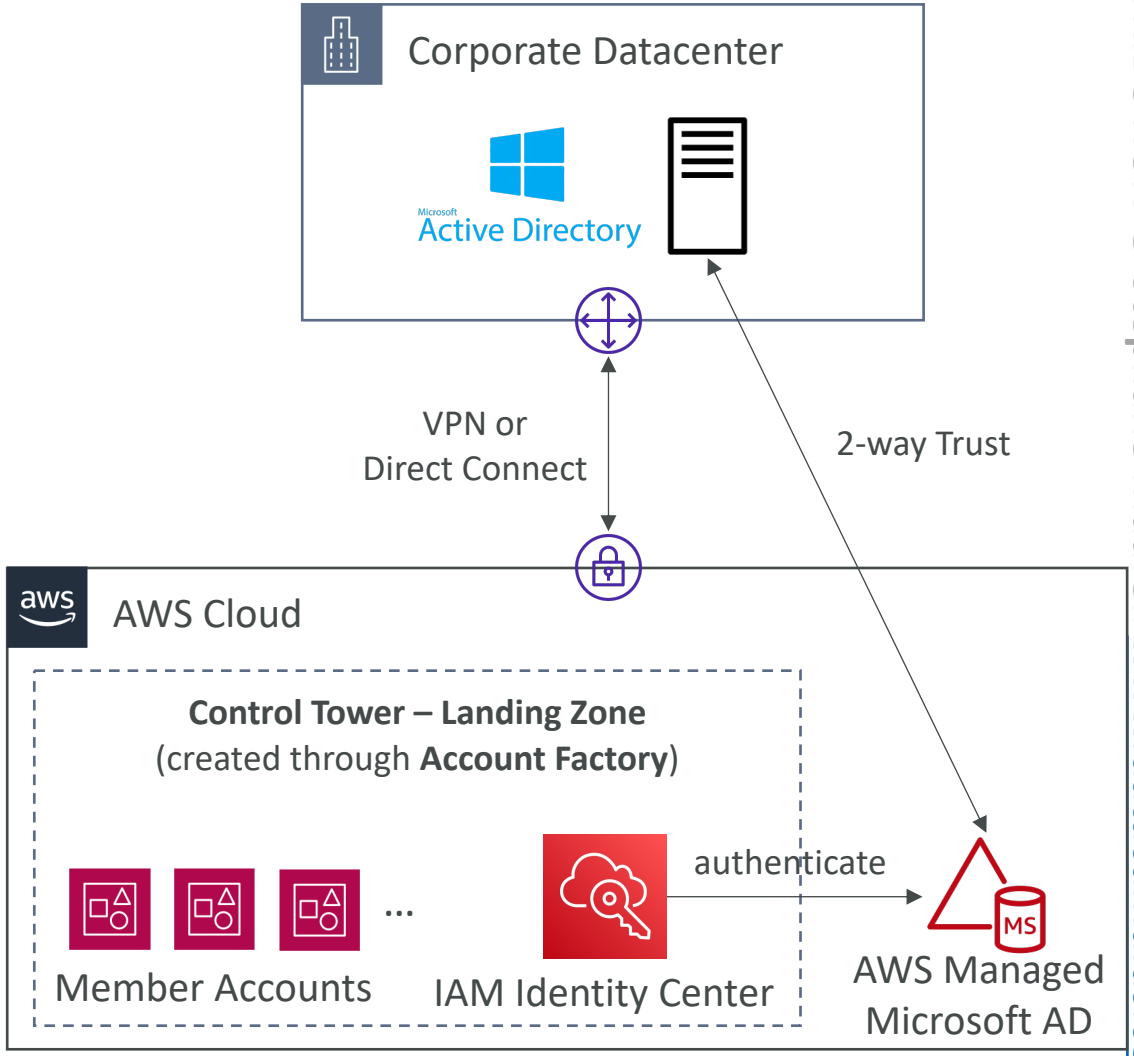
- Easy way to **set up and govern a secure and compliant multi-account AWS environment** based on best practices
- Benefits:
  - Automate the set up of your environment in a few clicks
  - Automate ongoing policy management using guardrails
  - Detect policy violations and remediate them
  - Monitor compliance through an interactive dashboard
- AWS Control Tower runs on top of AWS Organizations:
  - It automatically sets up AWS Organizations to organize accounts and implement SCPs (Service Control Policies)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Control Tower – Account Factory

- Automates account provisioning and deployments
- Enables you to create pre-approved baselines and configuration options for AWS accounts in your organization (e.g., VPC default configuration, subnets, region, …)
- Uses AWS Service Catalog to provision new AWS accounts

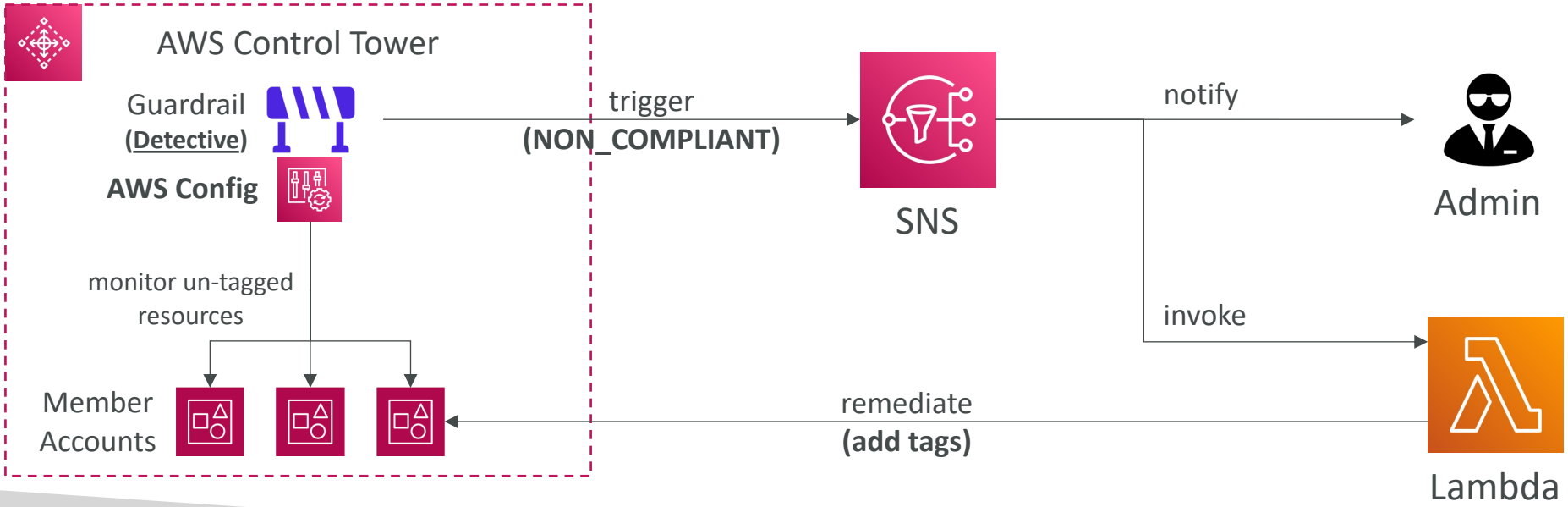


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Control Tower – Detect and Remediate Policy Violations

- Guardrail
  - Provides ongoing governance for your Control Tower environment (AWS Accounts)
  - **Preventive – using SCPs** (e.g., Disallow Creation of Access Keys for the Root User)
  - **Detective – using AWS Config** (e.g., Detect Whether MFA for the Root User is Enabled)
  - Example: identify non-compliant resources (e.g., untagged resources)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Control Tower – Guardrails Levels

- Mandatory
  - Automatically enabled and enforced by AWS Control Tower
  - Example: Disallow public Read access to the Log Archive account
- Strongly Recommended
  - Based on AWS best practices (optional)
  - Example: Enable encryption for EBS volumes attached to EC2 instances
- Elective
  - Commonly used by enterprises (optional)
  - Example: Disallow delete actions without MFA in S3 buckets

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Resource Access Manager (RAM)

- Share AWS resources that you own with other AWS accounts
- Share with any account or within your Organization
- Avoid resource duplication!
- VPC Subnets
  - Allow to have all the resources launched in the same subnets
  - Must be from the same AWS Organizations.
  - Cannot share default security groups and default VPC
  - Participants can manage their own resources in there
  - Participants can't view, modify, delete resources that belong to other participants or the owner
- AWS Transit Gateway
- Route 53 (Resolver Rules, DNS Firewall Rule Groups)
- License Manager Configurations

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

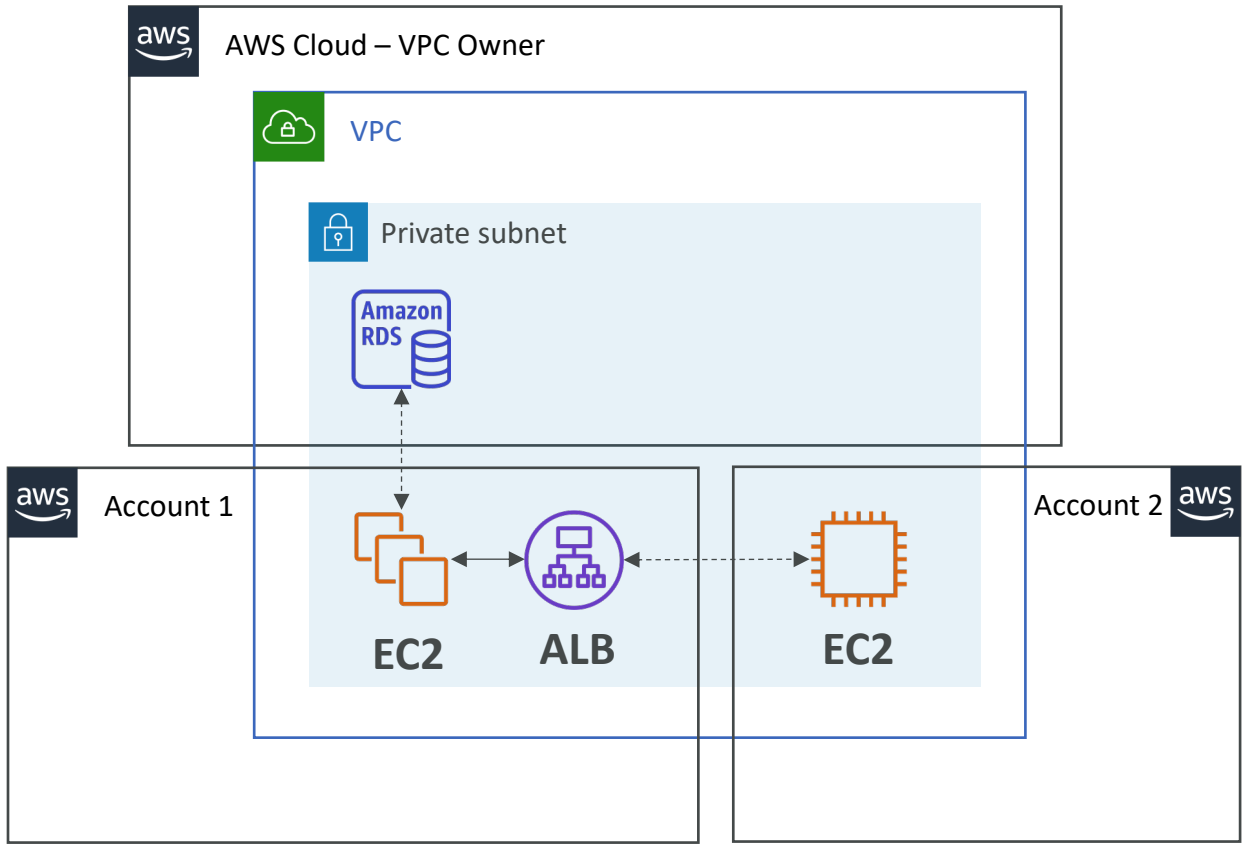
AWS Resource Access Manager (RAM)

- Aurora DB Clusters
- ACM Private Certificate Authority
- CodeBuild Project
- EC2 (Dedicated Hosts, Capacity Reservation)
- AWS Glue (Catalog, Database, Table)
- AWS Network Firewall Policies
- AWS Resource Groups
- Systems Manager Incident Manager (Contacts, Response Plans)
- AWS Outposts (Outpost, Site)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Resource Access Manager – VPC example



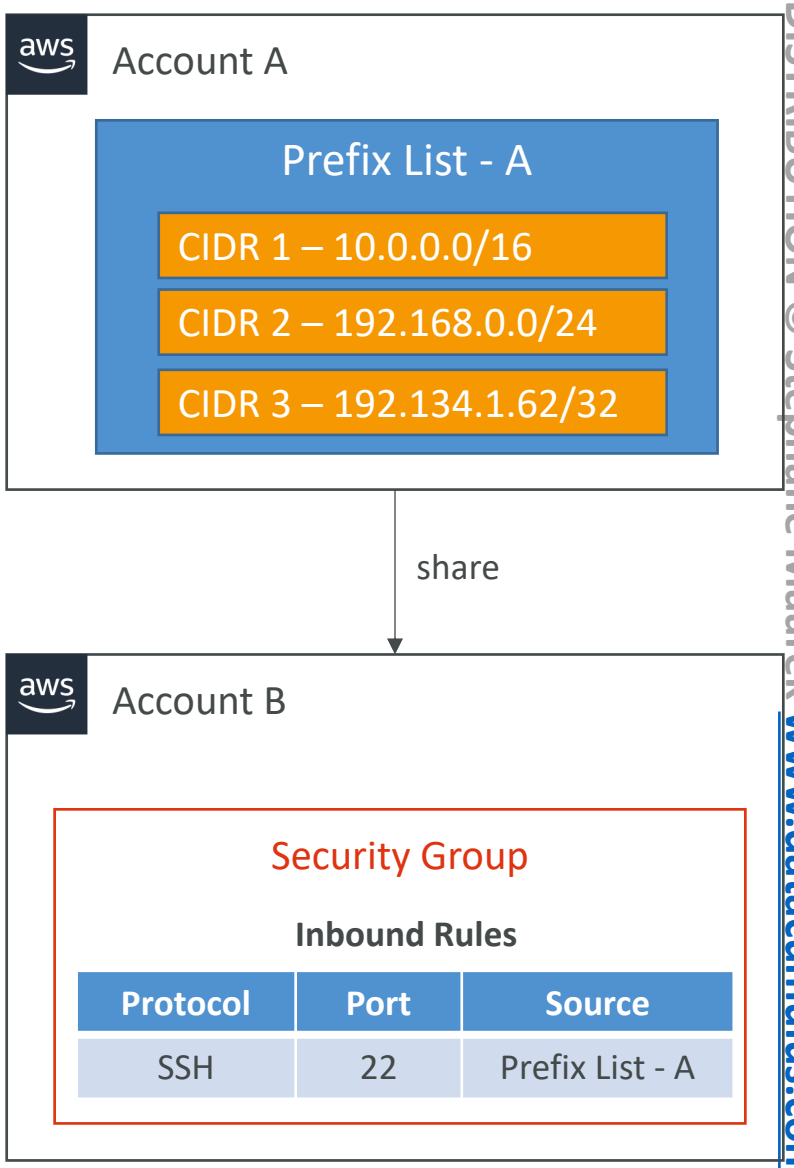
- Each account…
  - is responsible for its own resources
  - cannot view, modify or delete other resources in other accounts
- Network is shared so…
  - Anything deployed in the VPC can talk to other resources in the VPC
  - Applications are accessed easily across accounts, using private IP!
  - Security groups from other accounts can be referenced for maximum security
- Use cases
  - Applications within the same trust boundaries
  - Applications with a high degree of interconnectivity

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Resource Access Manager Managed Prefix List

- A set of one or more CIDR blocks
- Makes it easier to configure and maintain Security Groups and Route Tables
- Customer-Managed Prefix List
  - Set of CIDRs that you define and manage by you
  - Can be shared with other AWS accounts or AWS Organization
  - Modify to update many security groups at once
- AWS-Managed Prefix List
  - Set of CIDRs for AWS services
  - You can't create, modify, share, or delete them

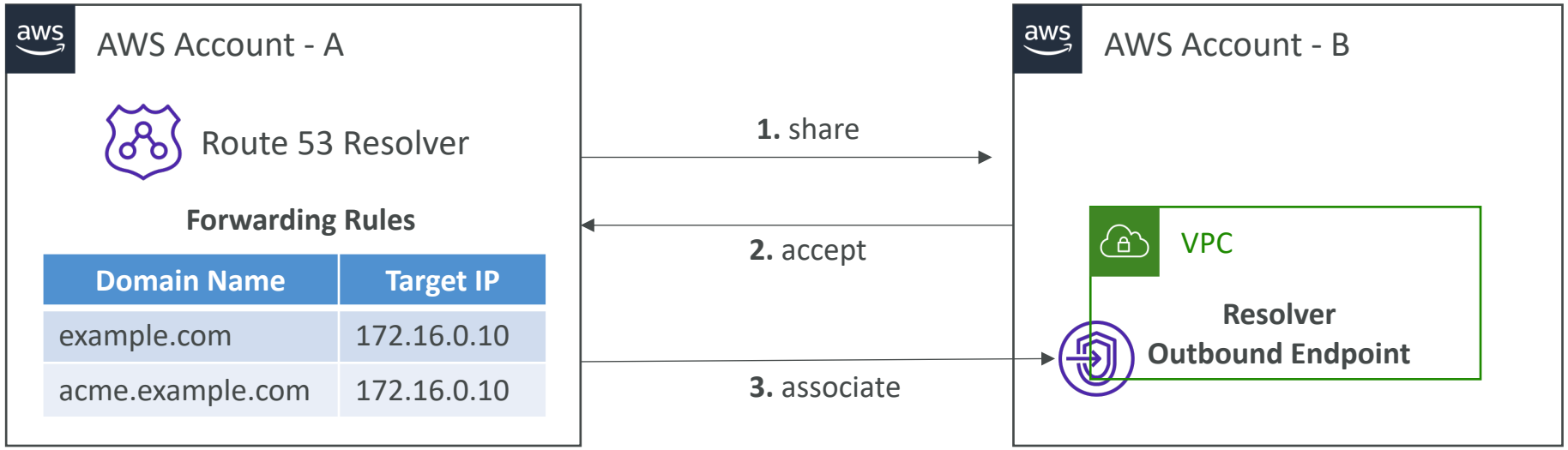


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Resource Access Manager
Route 53 Outbound Resolver

- Helps you scale forwarding rules to your DNS in case you have multiple accounts and VPC



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Summary of Identity & Federation

- Users and Accounts all in AWS
- AWS Organizations
- AWS Control Tower to setup secure & complaint multi-account AWS environment (best practices)
- Federation with SAML
- Federation without SAML with a custom IdP (**GetFederationToken**)
- IAM Identity Center to connect to multiple AWS Accounts (Organization) and SAML apps
- Web Identity Federation (not recommended)
- Cognito for most web and mobile applications (has anonymous mode, MFA)
- AWS Directory Service:
  - **Managed Microsoft AD** – standalone or setup trust AD with on-premises, has MFA, seamless join, RDS integration
  - **AD Connector** – proxy requests to on-premises
  - **Simple AD** – standalone & cheap AD-compatible with no MFA, no advanced capabilities
- AWS RAM to share resources (example VPC subnets)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Security

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

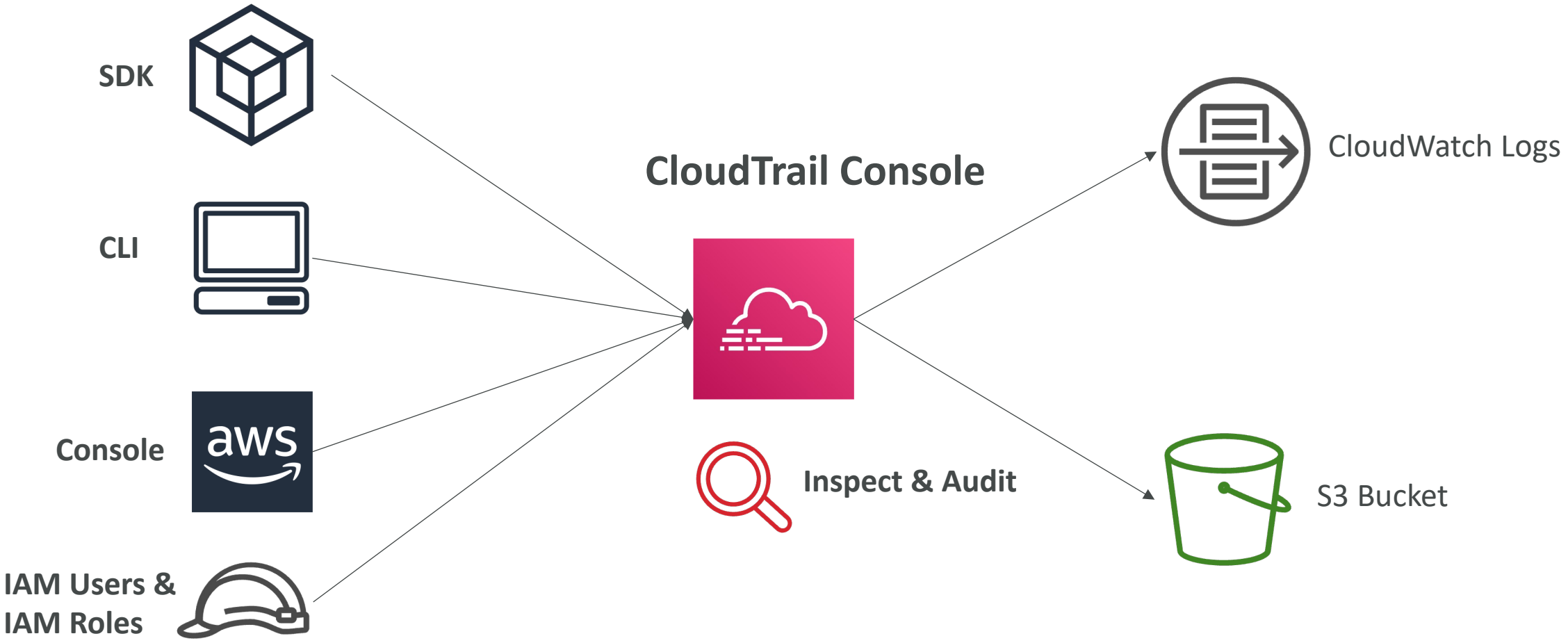
AWS CloudTrail

- Provides governance, compliance and audit for your AWS Account
- CloudTrail is enabled by default!
- Get an history of events / API calls made within your AWS Account by:
  - Console
  - SDK
  - CLI
  - AWS Services
- Can put logs from CloudTrail into CloudWatch Logs or S3
- A trail can be applied to All Regions (default) or a single Region.
- If a resource is deleted in AWS, investigate CloudTrail first!

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudTrail Diagram



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudTrail Events

- Management Events:
  - Operations that are performed on resources in your AWS account
  - Examples:
    - Configuring security (IAM **AttachRolePolicy**)
    - Configuring rules for routing data (Amazon EC2 **CreateSubnet**)
    - Setting up logging (AWS CloudTrail **CreateTrail**)
  - By default, trails are configured to log management events.
  - Can separate **Read Events** (that don't modify resources) from **Write Events** (that may modify resources)

- Data Events:
  - By default, data events are not logged (because high volume operations)
  - Amazon S3 object-level activity (ex: **GetObject**, **DeleteObject**, **PutObject**): can separate Read and Write Events
  - AWS Lambda function execution activity (the **Invoke** API)

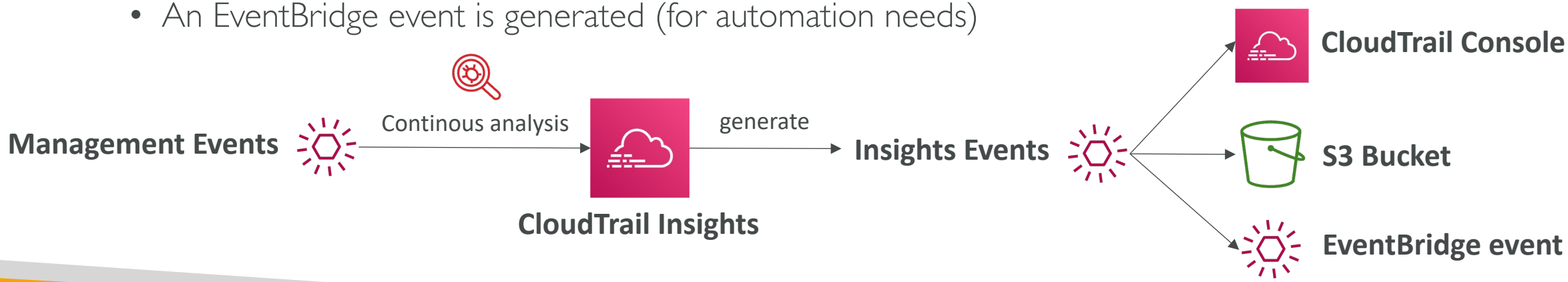
- CloudTrail Insights Events:
  - See next slide ☺

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudTrail Insights

- Enable **CloudTrail Insights to detect unusual activity** in your account:
  - inaccurate resource provisioning
  - hitting service limits
  - Bursts of AWS IAM actions
  - Gaps in periodic maintenance activity
- CloudTrail Insights analyzes normal management events to create a baseline
- And then **continuously analyzes write events to detect unusual patterns**
  - Anomalies appear in the CloudTrail console
  - Event is sent to Amazon S3
  - An EventBridge event is generated (for automation needs)

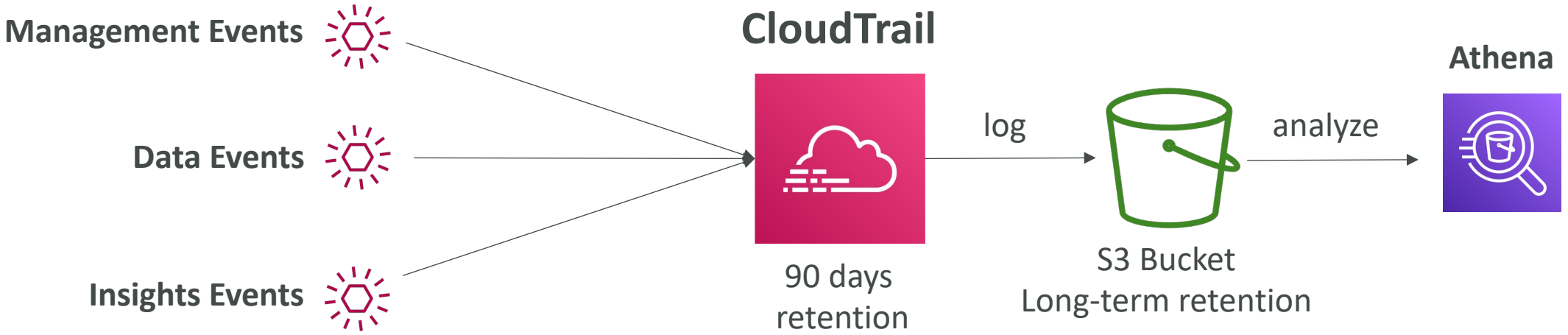


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudTrail Events Retention

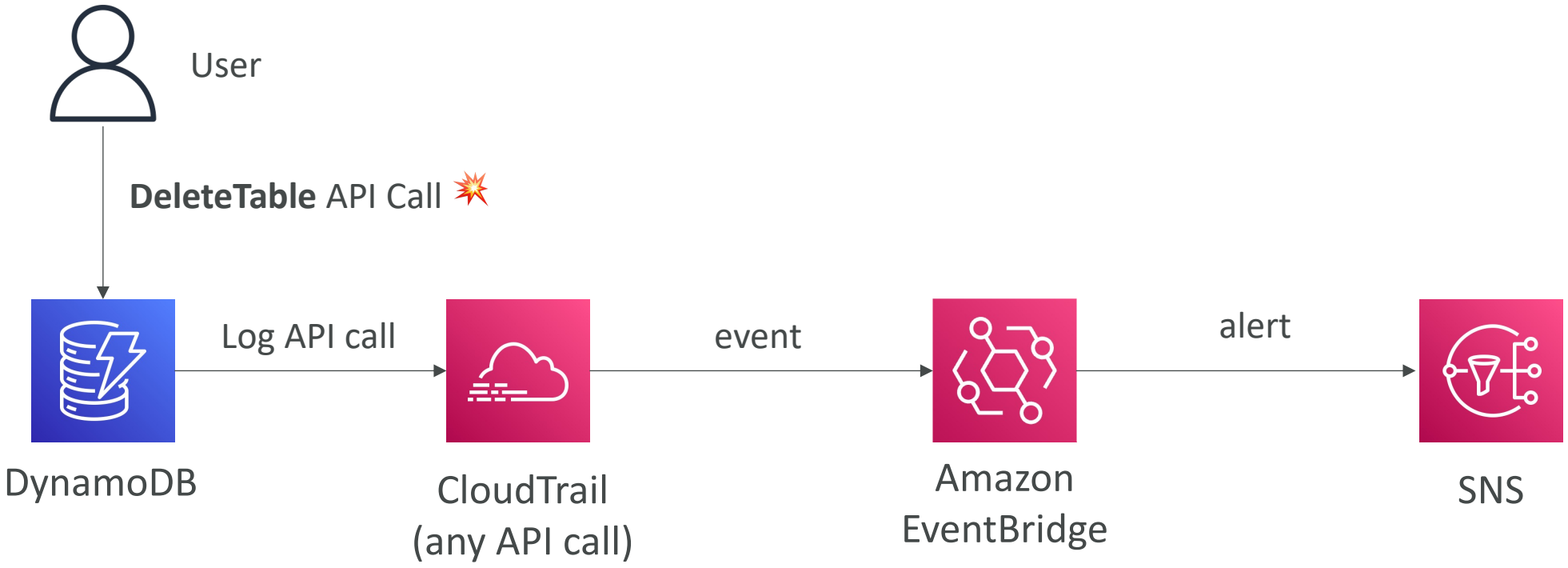
- Events are stored for 90 days in CloudTrail
- To keep events beyond this period, log them to S3 and use Athena



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

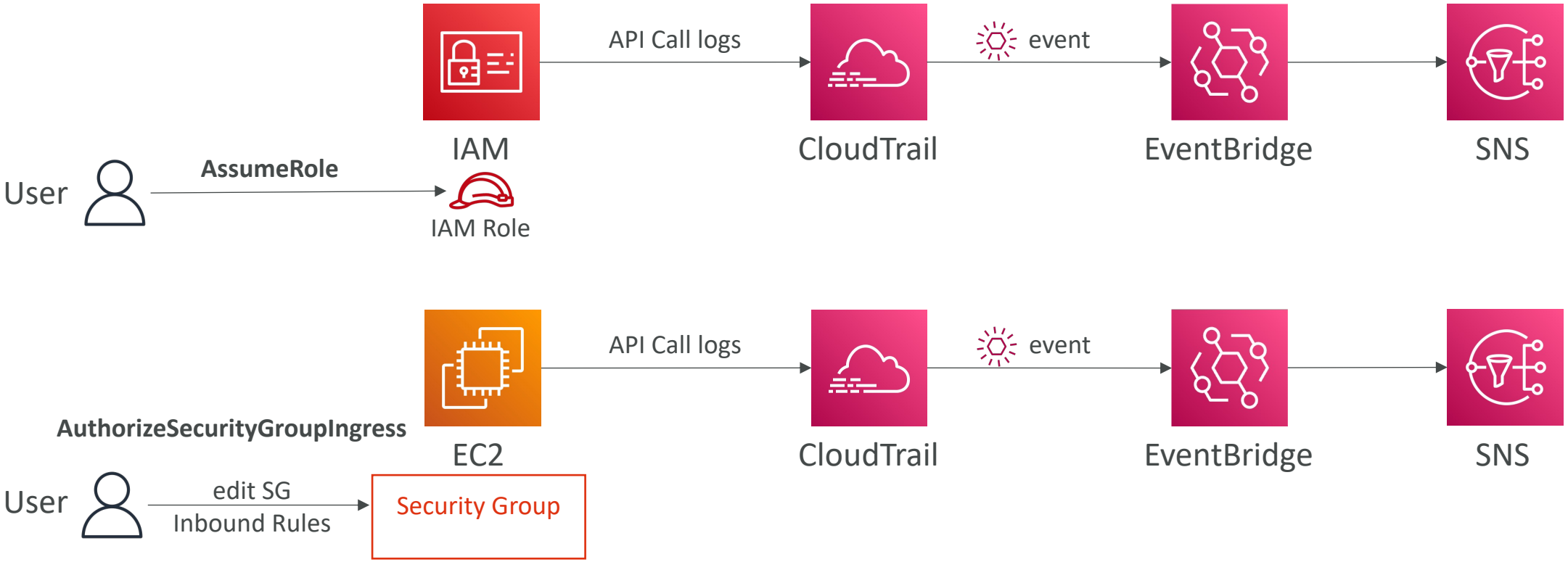
Amazon EventBridge – Intercept API Calls



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

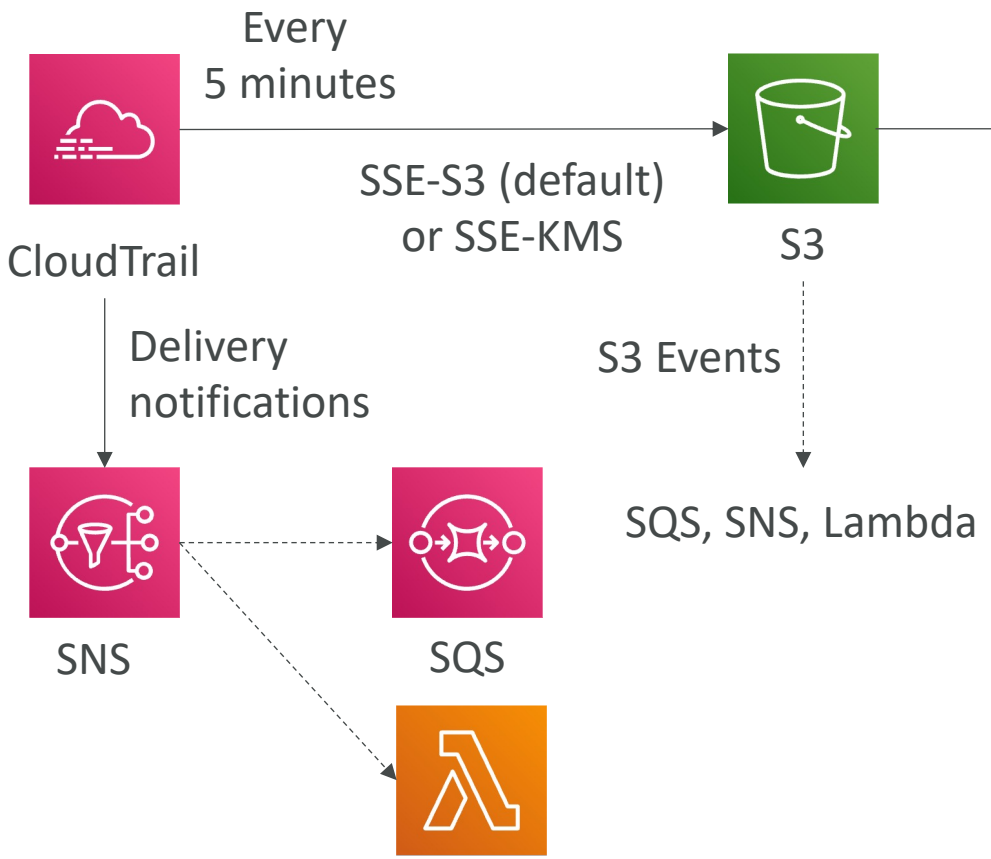
Amazon EventBridge + CloudTrail



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudTrail – Solution Architecture: Delivery to S3



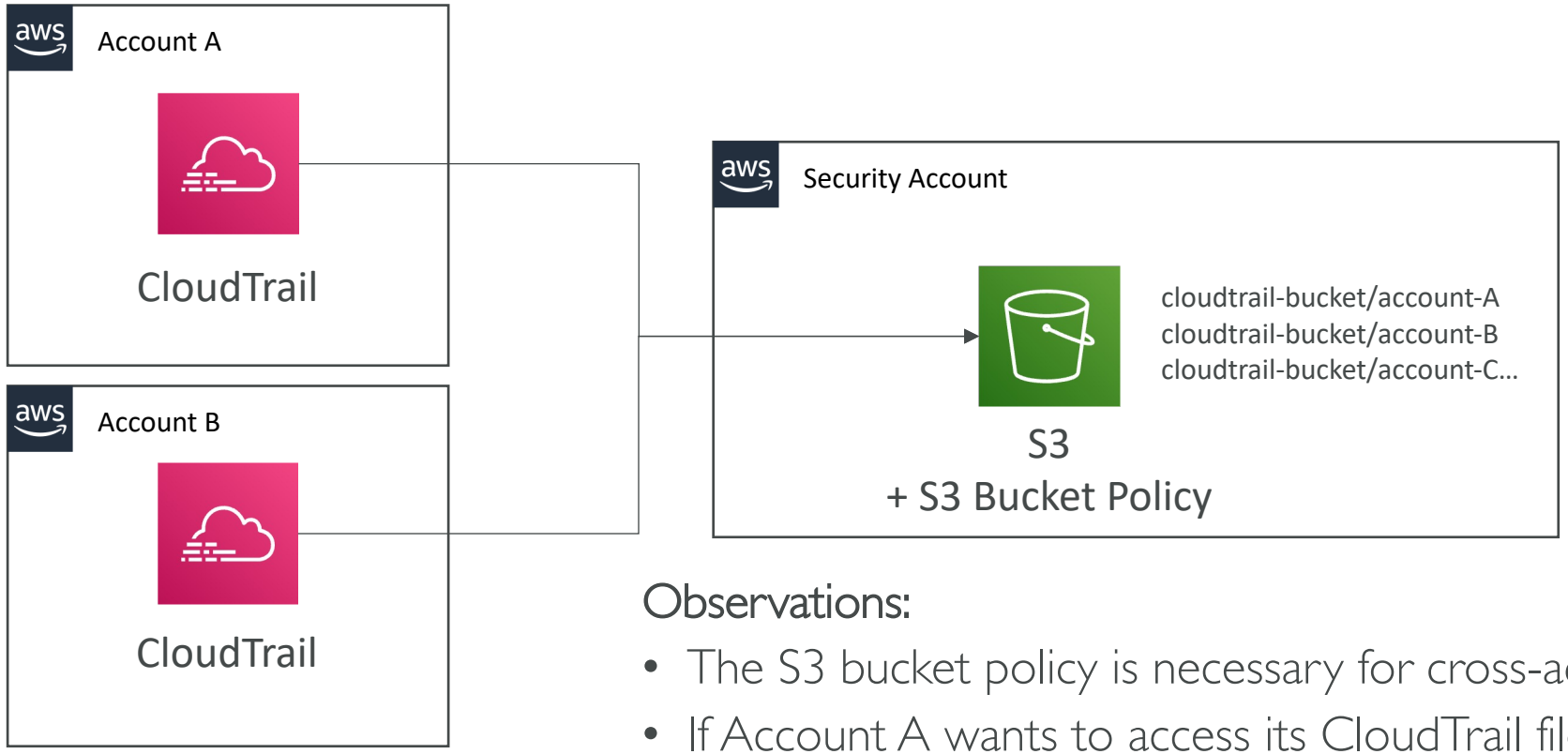
S3 Enhancements:

- Enable Versioning
- MFA Delete Protection
- S3 Lifecycle Policy (S3 IA, Glacier…)
- S3 Object Lock
- SSE-S3 or SSE-KMS encryption
- Feature to perform CloudTrail Log File Integrity validation
(SHA-256 for hashing and signing)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudTrail - Solution Architecture: Multi Account, Multi Region Logging



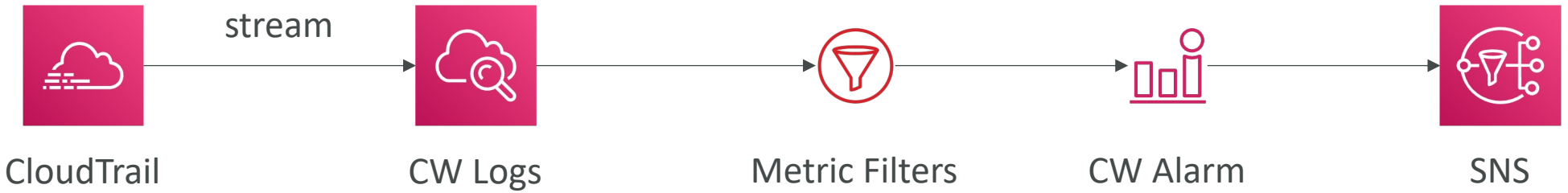
Observations:

- The S3 bucket policy is necessary for cross-account delivery
- If Account A wants to access its CloudTrail files:
  - Option 1: create a cross-account role and assume the role
  - Option 2: edit the bucket policy

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudTrail - Solution Architecture: Alert for API calls



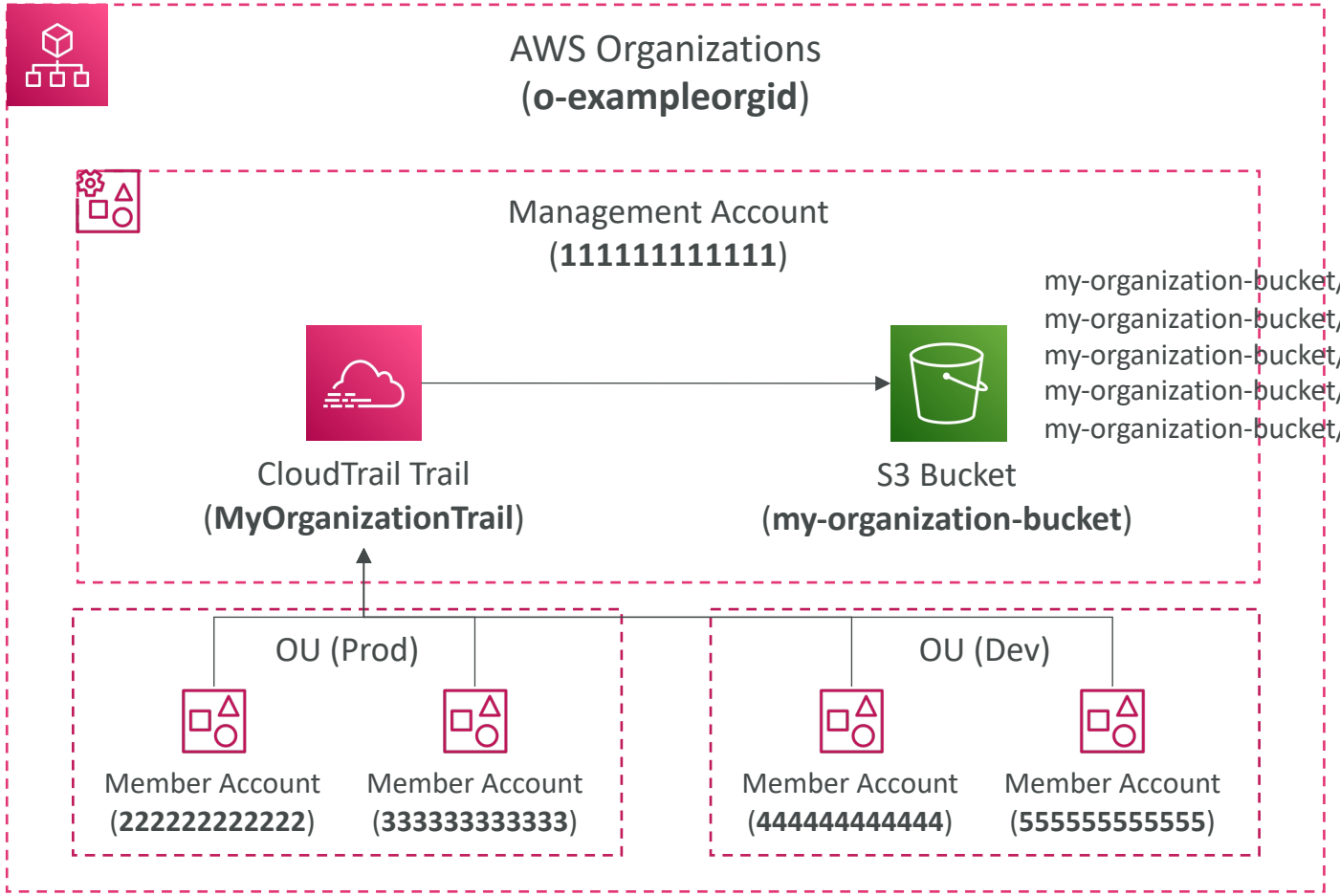
- Log filter metrics can be used to detect a high level of API happening
- Ex: Count occurrences of EC2 **TerminateInstances** API
- Ex: Count of API calls per user
- Ex: Detect high level of Denied API calls

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudTrail – Solution Architecture: Organizational Trail

The Organizational Trail is created in the management account.



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudTrail: How to react to events the fastest?

Overall, CloudTrail may take up to 15 minutes to deliver events

- EventBridge:
  - Can be triggered for any API call in CloudTrail
  - The fastest, most reactive way
- CloudTrail Delivery in CloudWatch Logs:
  - Events are streamed
  - Can perform a metric filter to analyze occurrences and detect anomalies
- CloudTrail Delivery in S3:
  - Events are delivered every 5 minutes
  - Possibility of analyzing logs integrity, deliver cross account, long-term storage

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS KMS (Key Management Service)

- Anytime you hear "encryption" for an AWS service, it's most likely KMS
- Easy way to control access to your data, AWS manages keys for us
- Fully integrated with IAM for authorization
- Seamlessly integrated into:
  - Amazon EBS: encrypt volumes
  - Amazon S3: Server-side encryption of objects
  - Amazon Redshift: encryption of data
  - Amazon RDS: encryption of data
  - Amazon SSM: Parameter store
  - Etc…
- KMS Keys are regional, and can only used in the region they're created in

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

KMS – KMS Key Types

- Symmetric (AES-256 keys)
  - First offering of KMS, single encryption key that is used to Encrypt and Decrypt
  - AWS services that are integrated with KMS use Symmetric KMS keys
  - Necessary for envelope encryption
  - You never get access to the KMS key unencrypted (must call KMS API to use)
- Asymmetric (RSA & ECC key pairs)
  - Public (Encrypt) and Private Key (Decrypt) pair
  - Used for Encrypt/Decrypt, or Sign/Verify operations
  - The public key is downloadable, but you can't access the Private Key unencrypted
  - Use case: encryption outside of AWS by users who can't call the KMS API

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Types of KMS Keys

- Customer Managed Keys
  - Create, manage and use, can enable or disable
  - Possibility of rotation policy (new key generated every year, old key preserved)
  - Can add a Key Policy (resource policy) & audit in CloudTrail
  - Leverage for envelope encryption
- AWS Managed Keys
  - Used by AWS service (aws/s3, aws/ebs, aws/redshift)
  - Managed by AWS (automatically rotated every 1 year)
  - View Key Policy & audit in CloudTrail
- AWS Owned Keys
  - Created and managed by AWS, use by some AWS services to protect your resources
  - Used in multiple AWS accounts, but they are not in your AWS account
  - You can't view, use, track, or audit

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Types of KMS Keys

KMS Key	Customer Managed Key	AWS Managed Key	AWS Owned Key
Can view metadata?	✓	✓	✗
Can manage?	✓	✗	✗
Used only for my AWS account?	✓	✓	✗
Automatic Rotation	90 – 2560 days (365 default)	Required (every 1 year)	Varies
On-demand Rotation	✓	✗	✗

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

KMS Key Material Origin

- Identifies the source of the key material in the KMS key
- Can't be changed after creation

- KMS (AWS_KMS) – default
  - AWS KMS creates and manages the key material in its own key store

- External (EXTERNAL)
  - You import the key material into the KMS key
  - You're responsible for securing and managing this key material outside of AWS

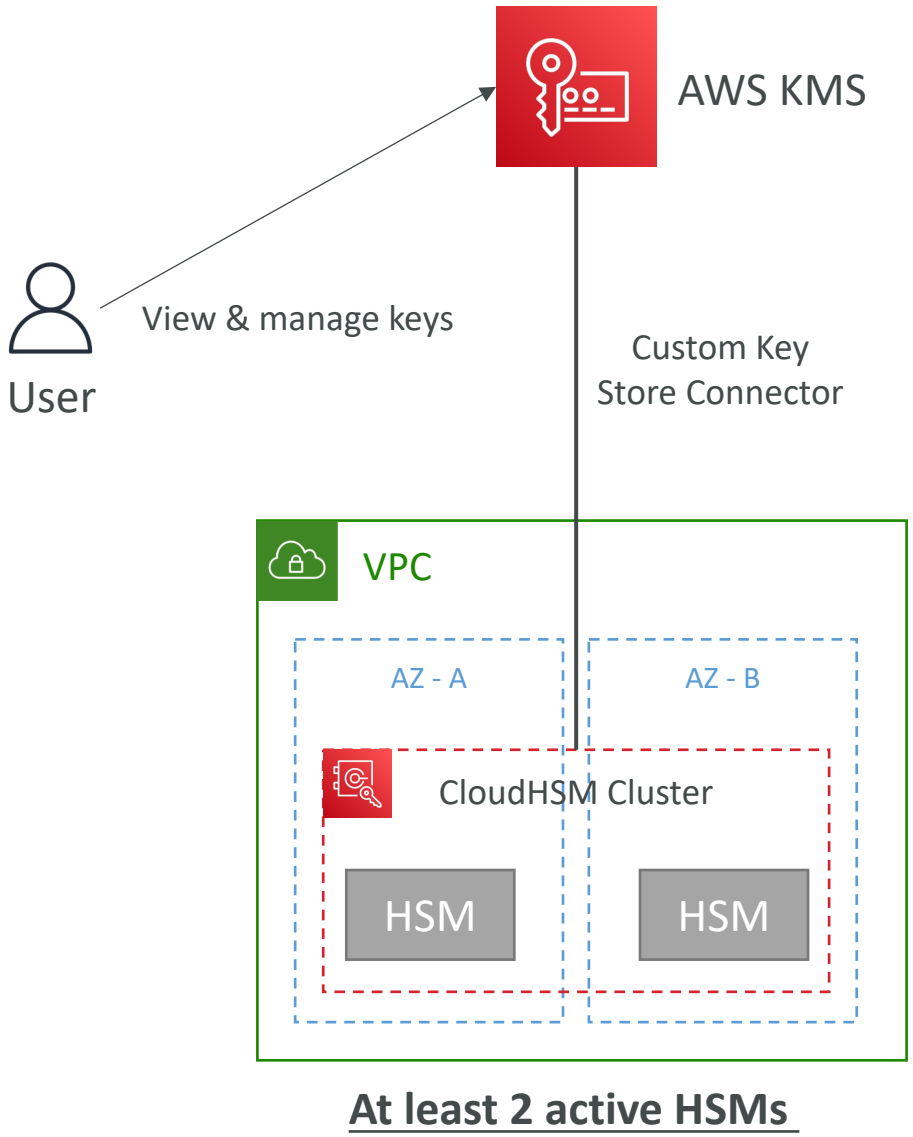
- Custom Key Store (AWS_CLOUDHSM)
  - AWS KMS creates the key material in a custom key store (CloudHSM Cluster)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

KMS Key Source – Custom Key Store (CloudHSM)

- Integrate KMS with CloudHSM cluster as a Custom Key Store
- Key materials are stored in a CloudHSM cluster that you own and manage
- The cryptographic operations are performed in the HSMs
- Use cases:
  - You need direct control over the HSMs
  - KMS keys needs to be stored in a dedicated HSMs

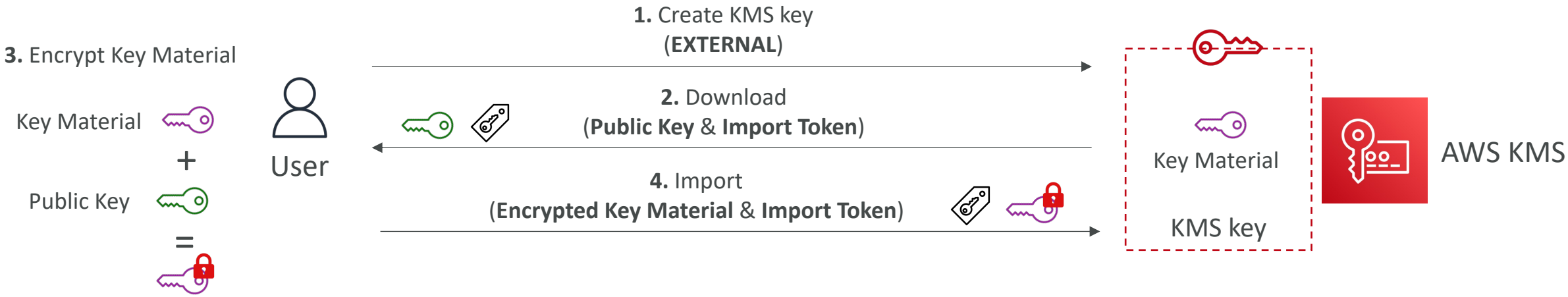


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

KMS Key Source - External

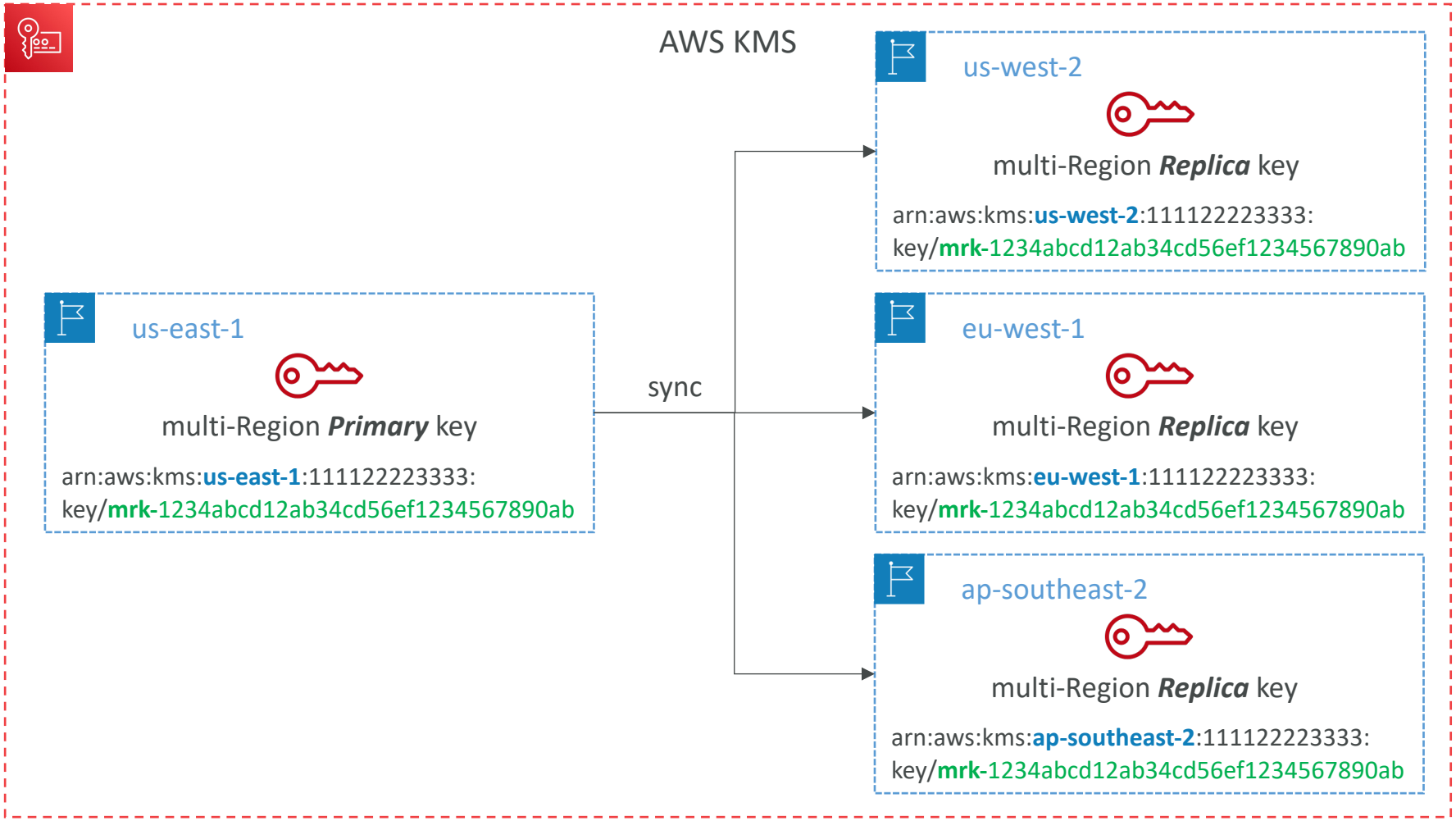
- Import your own key material into KMS key, Bring Your Own Key (BYOK)
- **You're responsible for key material's security, availability, and durability outside of AWS**
- Supports both **Symmetric and Asymmetric** KMS keys
- Can't be used with Custom Key Store (CloudHSM)
- Manually rotate your KMS key (Automatic Key Rotation is NOT supported)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

KMS Multi-Region Keys



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

KMS Multi-Region Keys

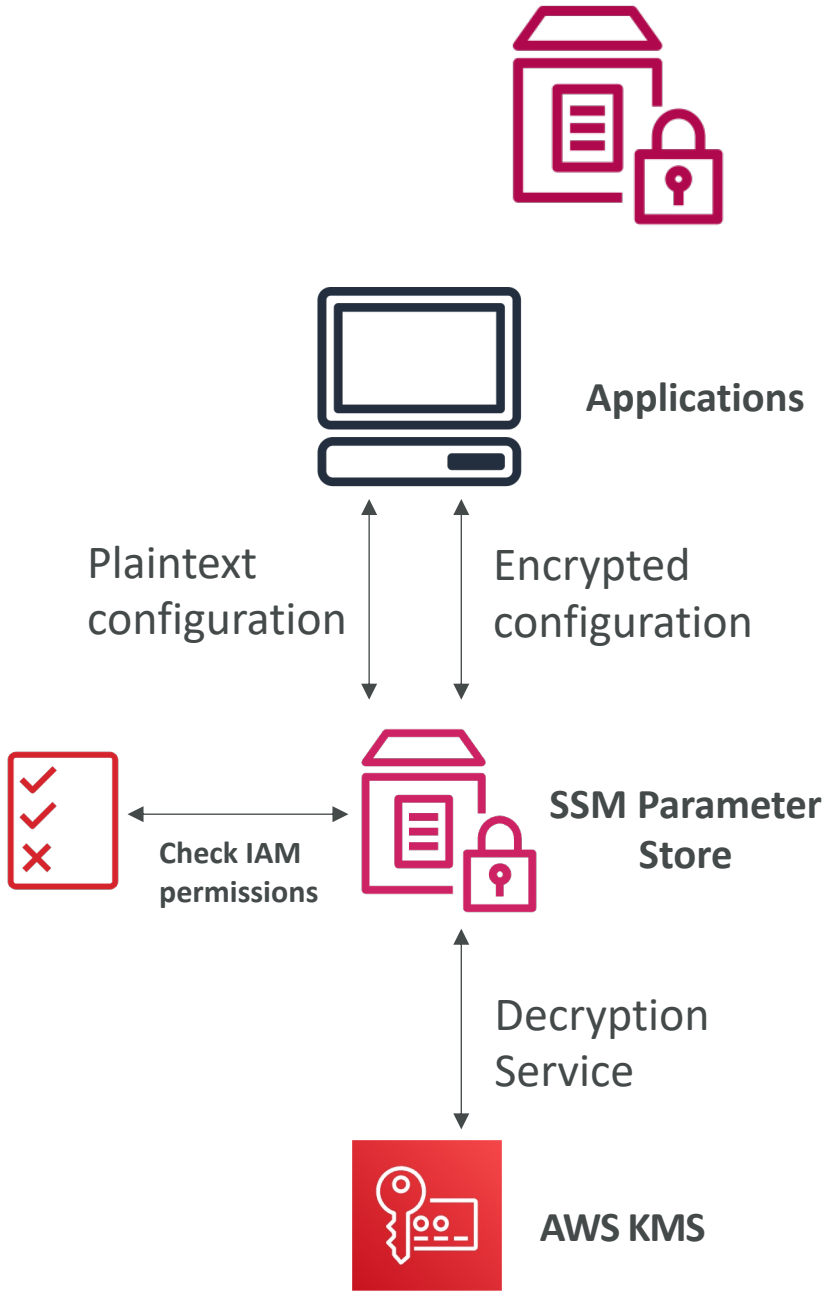
- A set of identical KMS keys in different AWS Regions that can be used interchangeably (~ same KMS key in multiple Regions)
- Encrypt in one Region and decrypt in other Regions (No need to re-encrypt or making cross-Region API calls)
- Multi-Region keys have the same key ID, key material, automatic rotation, …
- KMS Multi-Region are NOT global (Primary + Replicas)
- Each Multi-Region key is managed **independently**
- Only one primary key at a time, can promote replicas into their own primary
- Use cases: Disaster Recovery, Global Data Management (e.g., DynamoDB Global Tables), Active-Active Applications that span multiple Regions, Distributed Signing applications, …

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SSM Parameter Store

- Secure storage for configuration and secrets
- Optional Seamless Encryption using KMS
- Serverless, scalable, durable, easy SDK
- Version tracking of configurations / secrets
- Security through IAM
- Notifications with Amazon EventBridge
- Integration with CloudFormation

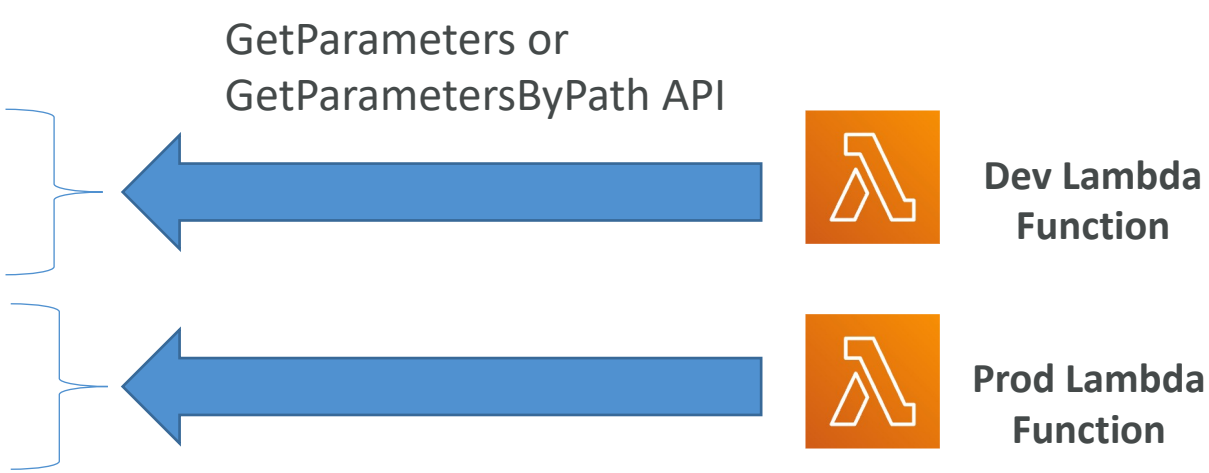


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SSM Parameter Store Hierarchy

- /my-department/
  - my-app/
    - dev/
      - db-url
      - db-password
    - prod/
      - db-url
      - db-password
  - other-app/
- /other-department/
- */aws/reference/secretsmanager/secret_ID_in_Secrets_Manager*
- */aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2 (public)*



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Standard and advanced parameter tiers

	Standard	Advanced
Total number of parameters allowed (per AWS account and Region)	10,000	100,000
Maximum size of a parameter value	4 KB	8 KB
Parameter policies available	No	Yes
Cost	No additional charge	Charges apply
Storage Pricing	Free	$0.05 per advanced parameter per month

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Parameters Policies (for advanced parameters)

- Allow to assign a TTL to a parameter (expiration date) to force updating or deleting sensitive data such as passwords
- Can assign multiple policies at a time

Expiration (to delete a parameter)

```
{
    "Type":"Expiration",
    "Version":"1.0",
    "Attributes":{
       "Timestamp":"2020-12-02T21:34:33.000Z"
    }
}
```

ExpirationNotification (EventBridge)

```
{
    "Type":"ExpirationNotification",
    "Version":"1.0",
    "Attributes":{
       "Before":"15",
       "Unit":"Days"
    }
}
```

NoChangeNotification (EventBridge)

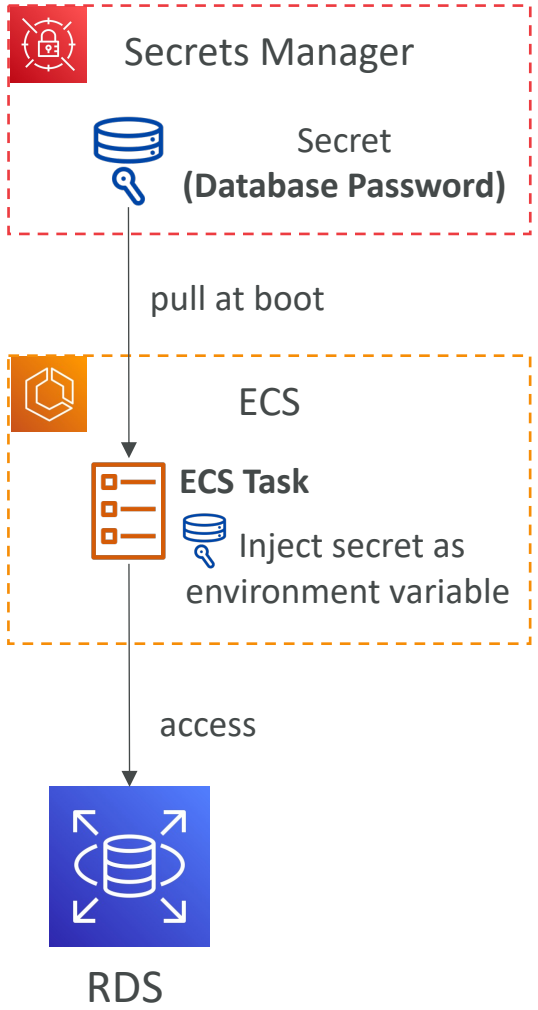
```
{
    "Type":"NoChangeNotification",
    "Version":"1.0",
    "Attributes":{
       "After":"20",
       "Unit":"Days"
    }
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Secrets Manager

- Meant for storing secrets (e.g., passwords, API keys, …)
- Capability to force **rotation of secrets** every X days
  - Automate generation of secrets on rotation (uses Lambda)
  - Natively supports Amazon RDS (all supported DB engines), Redshift, DocumentDB
  - Support other databases and services (custom Lambda function)
- Control access to secrets using Resource-based Policy
- Integration with other AWS services to natively pull secrets from Secrets Manager: *CloudFormation, CodeBuild, ECS, EMR, Fargate, EKS, Parameter Store…*



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Secrets Manager – with CloudFormation

```yaml
Resources:
  # Secret resource with a randomly generated password in its SecureString JSON
  MyRDSDBInstanceRotationSecret:
    Type: AWS::SecretsManager::Secret
    Properties:
      GenerateSecretString:
        SecretStringTemplate: '{"username": "admin"}'
        GenerateStringKey: password
        PasswordLength: 16
        ExcludeCharacters: "\"@/\\"

  # RDS Instance resource. Its master username and password use dynamic references
  # to resolve values from Secrets Manager
  MyRDSDBInstance:
    Type: AWS::RDS::DBInstance
    Properties:
      DBInstanceClass: db.t2.micro
      Engine: mysql
      MasterUsername: !Sub "{{resolve:secretsmanager:${MyRDSDBInstanceRotationSecret}::username}}"
      MasterUserPassword: !Sub "{{resolve:secretsmanager:${MyRDSDBInstanceRotationSecret}::password}}"

  # SecretTargetAttachment resource which updates the referenced Secret with properties
  # about the referenced RDS instance
  SecretRDSDBInstanceAttachment:
    Type: AWS::SecretsManager::SecretTargetAttachment
    Properties:
      TargetType: AWS::RDS::DBInstance
      SecretId: !Ref MyRDSDBInstanceRotationSecret
      TargetId: !Ref MyRDSDBInstance
```

secret is generated

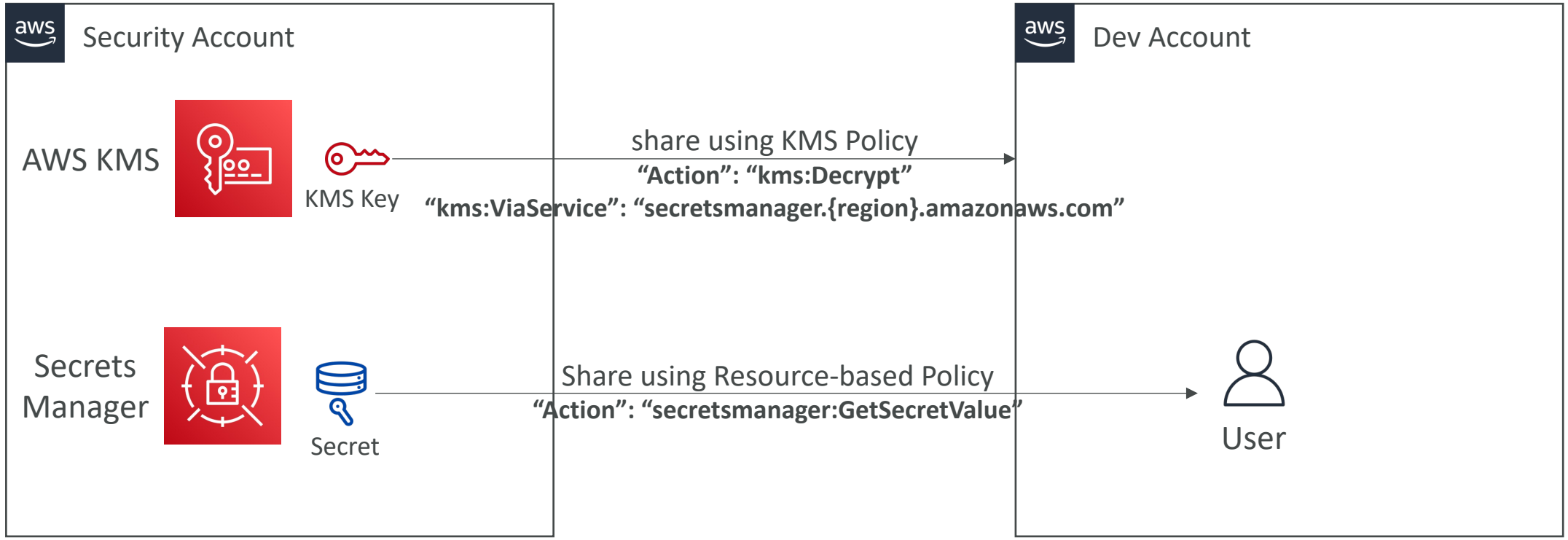
reference secret in RDS DB instance

link the secret to RDS DB instance

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Secrets Manager – Sharing Across Accounts



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SSM Parameter Store vs Secrets Manager

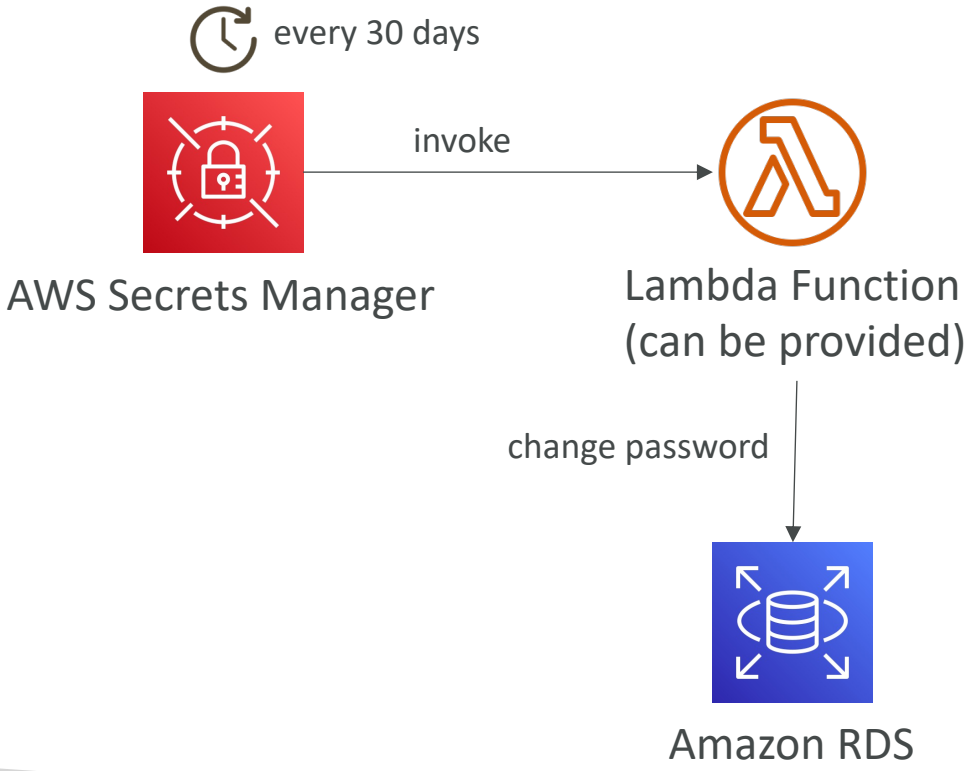
- Secrets Manager ($$$):
  - Automatic rotation of secrets with AWS Lambda
  - Lambda function is provided for RDS, Redshift, DocumentDB
  - KMS encryption is mandatory
  - Can integration with CloudFormation
- SSM Parameter Store ($):
  - Simple API
  - No secret rotation (can enable rotation using Lambda triggered by EventBridge)
  - KMS encryption is optional
  - Can integration with CloudFormation
  - Can pull a Secrets Manager secret using the SSM Parameter Store API

© Stephane Maarek

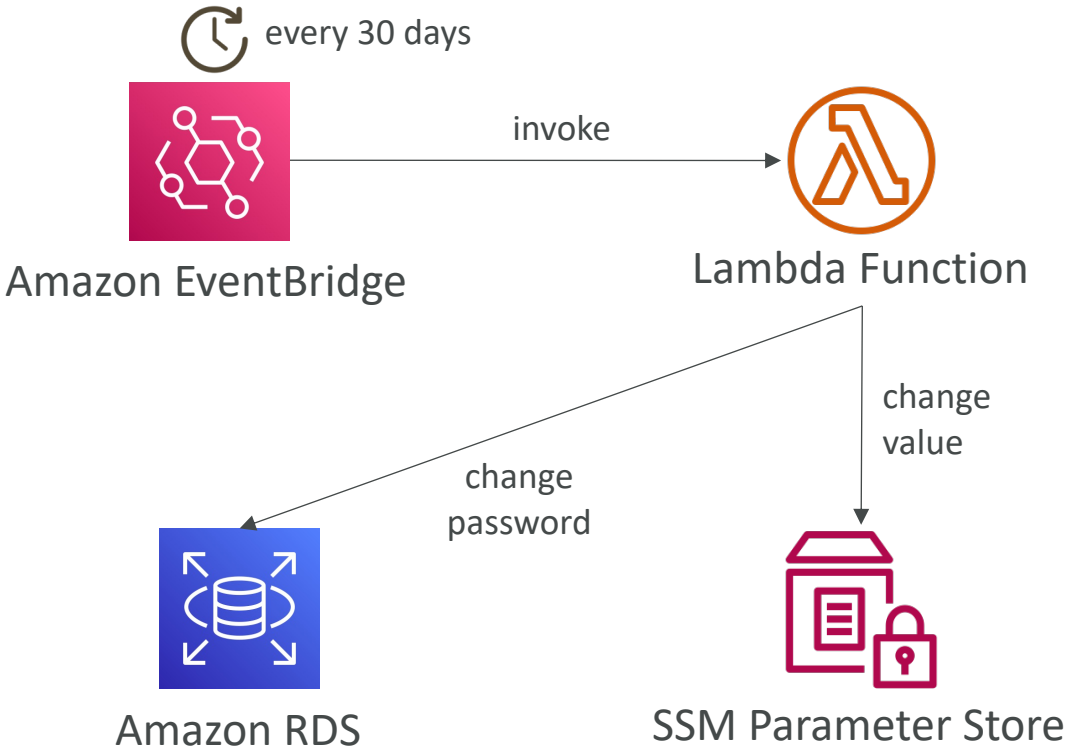
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SSM Parameter Store vs. Secrets Manager Rotation

AWS Secrets Manager



SSM Parameter Store



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

RDS - Security

- KMS encryption at rest for underlying EBS volumes / snapshots
- Transparent Data Encryption (TDE) for Oracle and SQL Server
- SSL encryption to RDS is possible for all DB (in-flight)
- IAM authentication for MySQL, PostgreSQL and MariaDB
- Authorization still happens within RDS (not in IAM)
- Can copy an un-encrypted RDS snapshot into an encrypted one
- CloudTrail cannot be used to track queries made within RDS

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SSL/TLS - Basics

- **SSL** refers to Secure Sockets Layer, used to encrypt connections
- **TLS** refers to Transport Layer Security, which is a newer version
- Nowadays, **TLS certificates are mainly used**, but people still refer as SSL

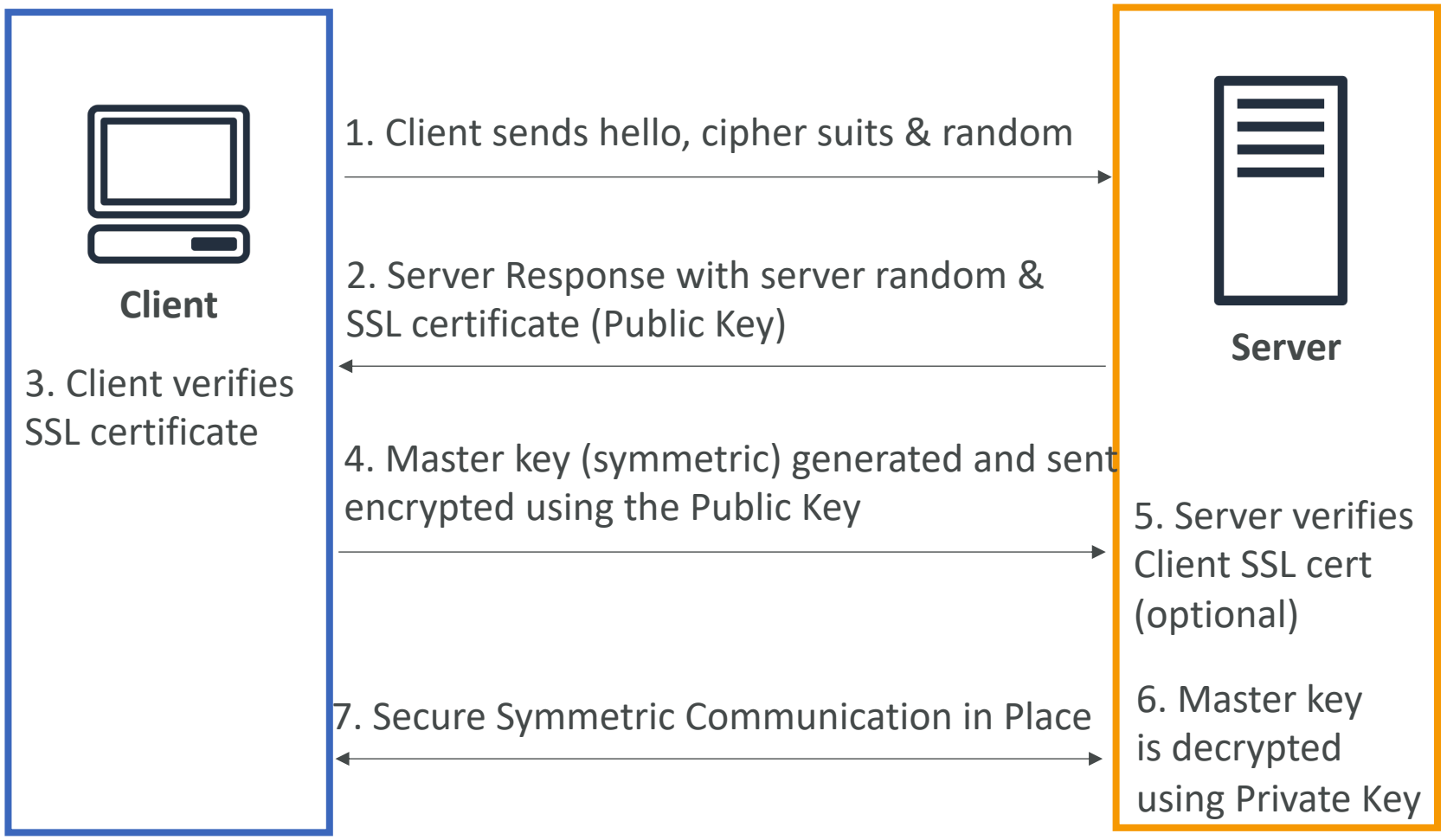
- Public SSL certificates are issued by Certificate Authorities (CA)
- Comodo, Symantec, GoDaddy, GlobalSign, Digicert, Letsencrypt, etc…

- SSL certificates have an expiration date (you set) and must be renewed

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SSL Encryption – How it works



- Asymmetric Encryption is expensive (SSL)
- Symmetric encryption is cheaper
- Asymmetric handshake is used to exchange a per-client random symmetric key
- Possibility of client sending an SSL certificate as well (two-way certificate)

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

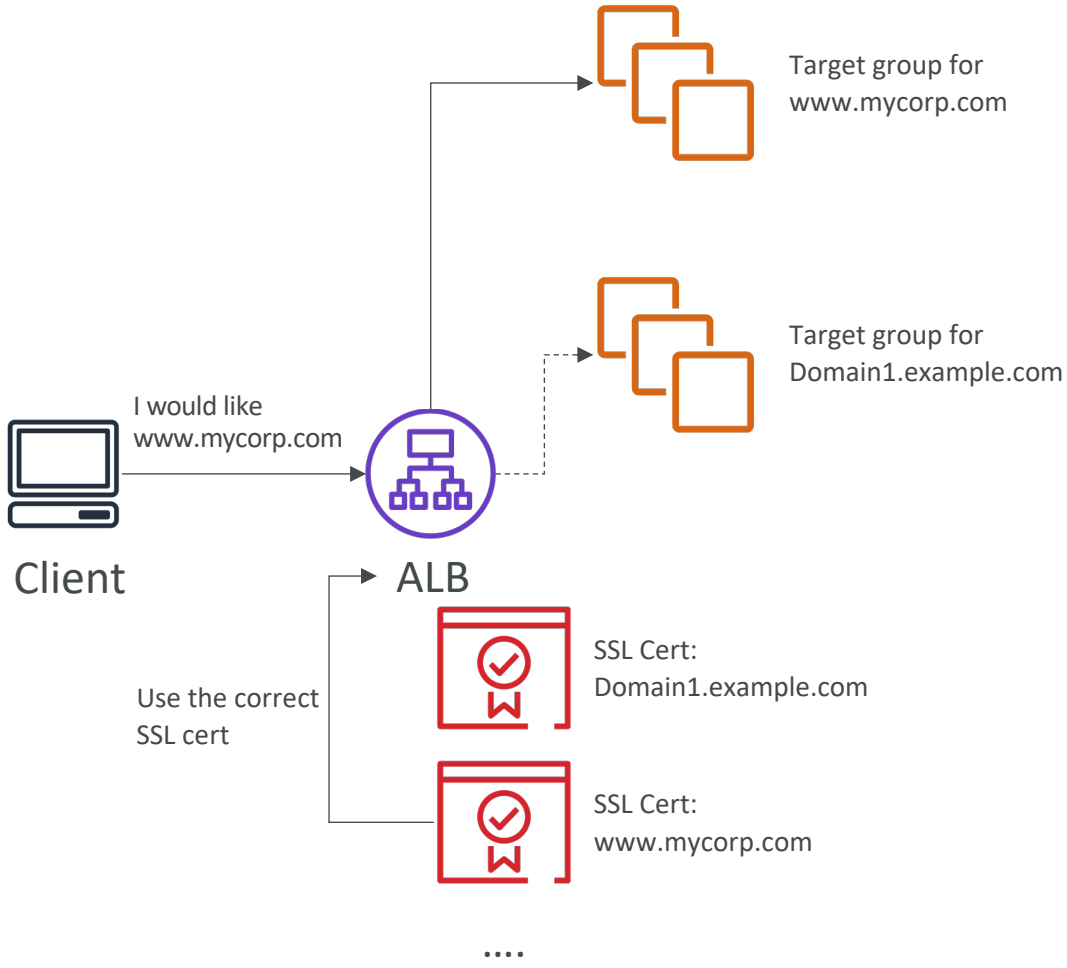
© Stephane Maarek

SSL – Server Name Indication (SNI)

- SNI solves the problem of loading **multiple SSL certificates onto one web server** (to serve multiple websites)
- It's a "newer" protocol, and requires the client to **indicate** the hostname of the target server in the initial SSL handshake
- The server will then find the correct certificate, or return the default one

Note:

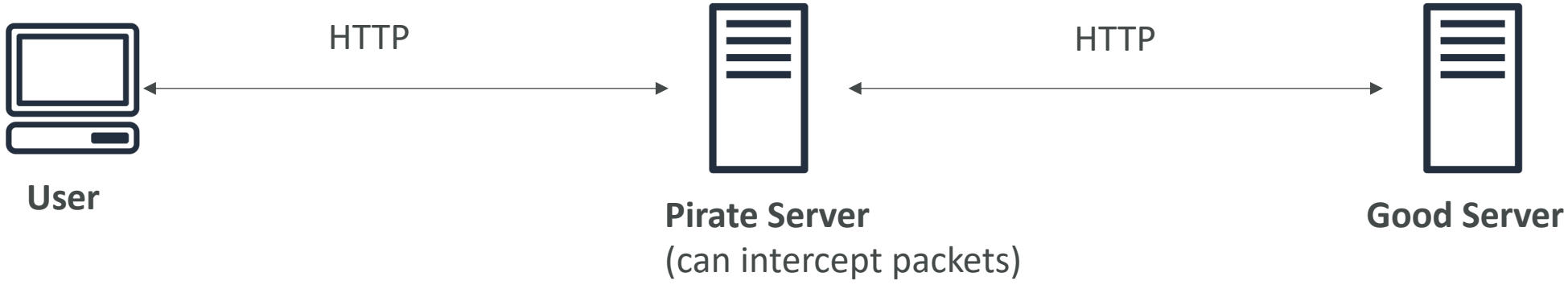
- Only works for ALB & NLB (newer generation), CloudFront
- Does not work for CLB (older gen)

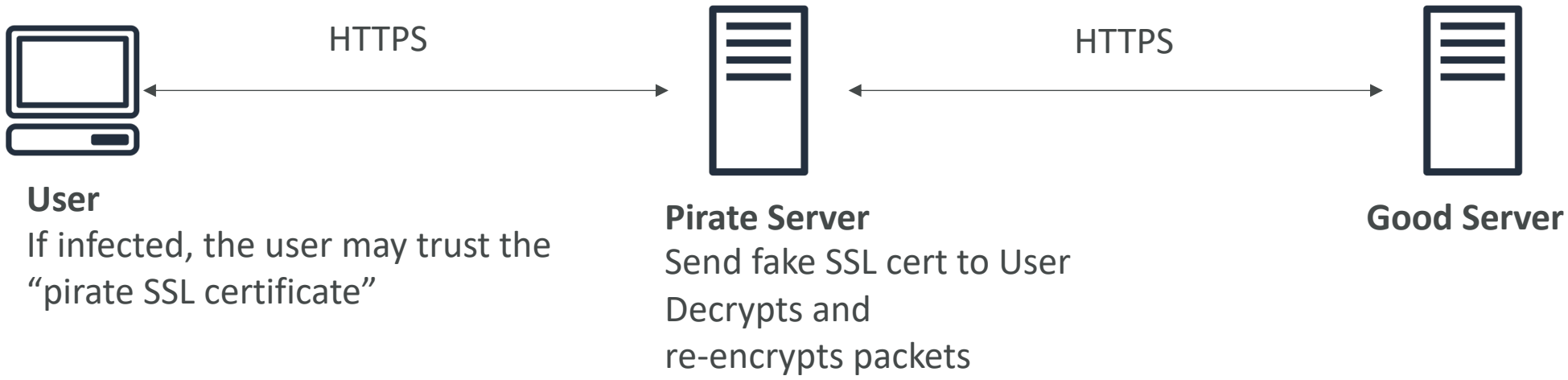


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SSL – Man in the Middle Attacks





© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SSL – Man in the Middle Attack
How to prevent

1. Don't use public-facing HTTP, use HTTPS (meaning, use SSL/TLS certificates)
2. Use a DNS that has DNSSEC
   - To send a client to a pirate server, a DNS response needs to be "forged" by a server which intercepts them
   - It is possible to protect your domain name by configuring DNSSEC
   - **Amazon Route 53 supports DNSSEC for domain registration.**
   - **Route 53 supports DNSSEC for DNS service as of December 2020 (using KMS)**
   - You could also run a custom DNS server on Amazon EC2 for example (**Bind** is the most popular, dnsmasq, KnotDNS, PowerDNS).

© Stephane Maarek

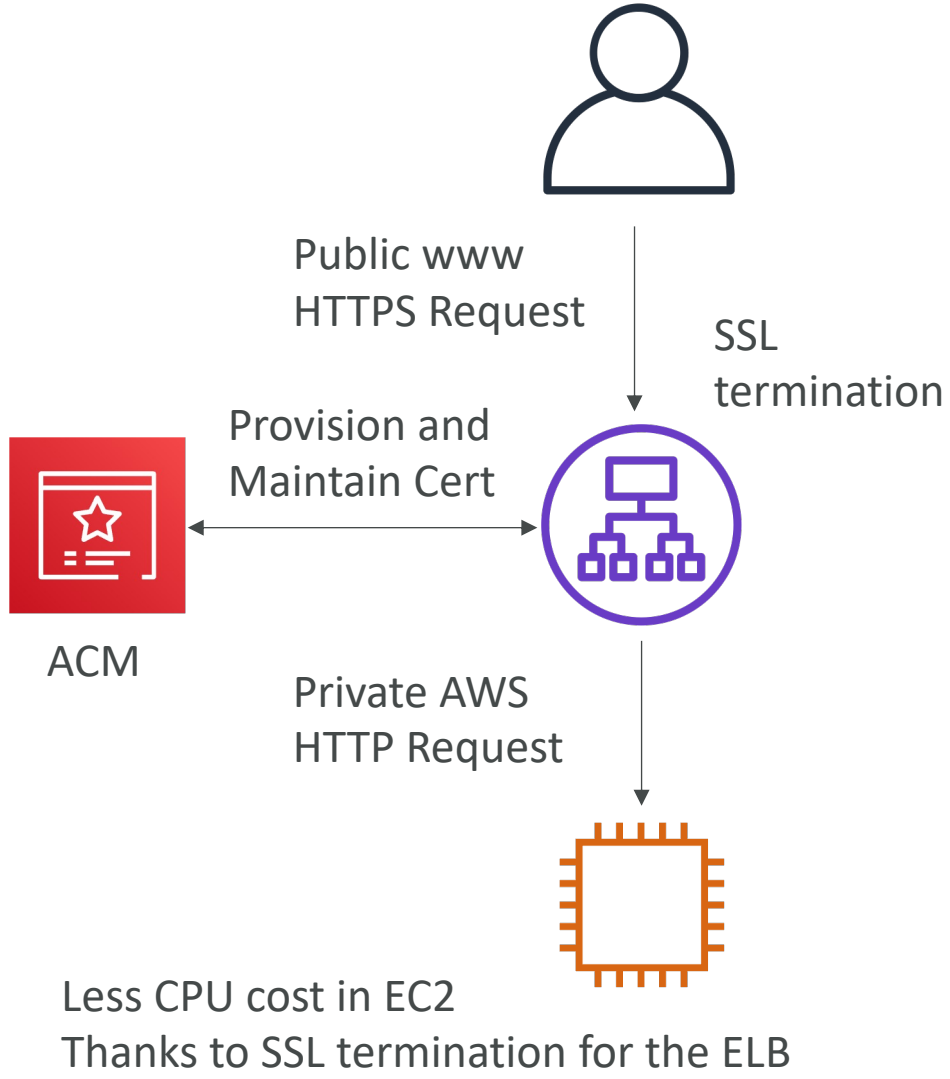
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Certificate Manager (ACM)

- To host public SSL certificates in AWS, you can:
  - Buy your own and upload them using the CLI
  - Have ACM provision and renew public SSL certificates for you (free of cost)

- ACM loads SSL certificates on the following integrations:
  - Load Balancers (including the ones created by EB)
  - CloudFront distributions
  - APIs on API Gateways

- SSL certificates is overall a pain to manually manage, so ACM is great to leverage in your AWS infrastructure!



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

ACM – Good to know

- Possibility of creating public certificates
  - Must verify public DNS
  - Must be issued by a trusted public certificate authority (CA)
- Possibility of creating private certificates
  - For your internal applications
  - You create your own private CA
  - Your applications must trust your private CA
- Certificate renewal:
  - Automatically done if generated provisioned by ACM
  - Any manually uploaded certificates must be renewed manually and re-uploaded
- ACM is a **regional** service
  - To use with a global application (multiple ALB for example), you need to issue an SSL certificate in each region where you application is deployed.
  - You cannot copy certs across regions

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

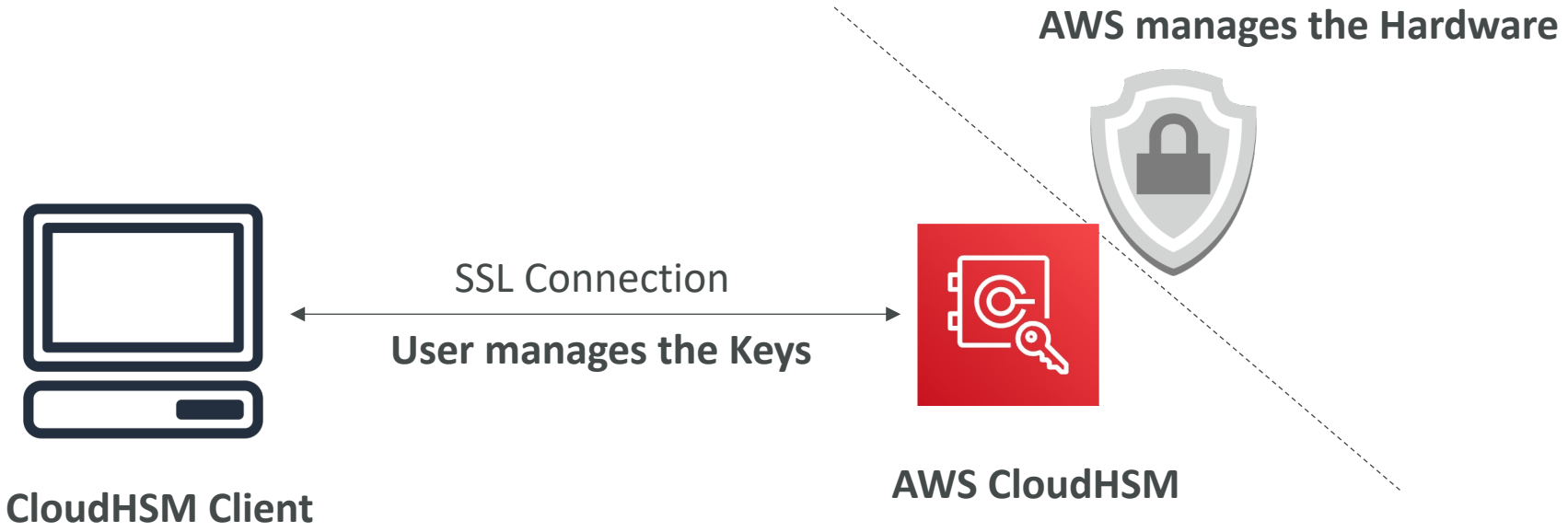
CloudHSM

- KMS => AWS manages the software for encryption
- CloudHSM => AWS provisions encryption **hardware**
- Dedicated Hardware (HSM = Hardware Security Module)
- You manage your own encryption keys entirely (not AWS)
- HSM device is tamper resistant, FIPS 140-2 Level 3 compliance
- Supports both symmetric and **asymmetric** encryption (SSL/TLS keys)
- No free tier available
- Must use the CloudHSM Client Software
- Redshift supports CloudHSM for database encryption and key management
- **Good option to use with SSE-C encryption**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudHSM Diagram



IAM permissions:

- CRUD an HSM Cluster

CloudHSM Software:

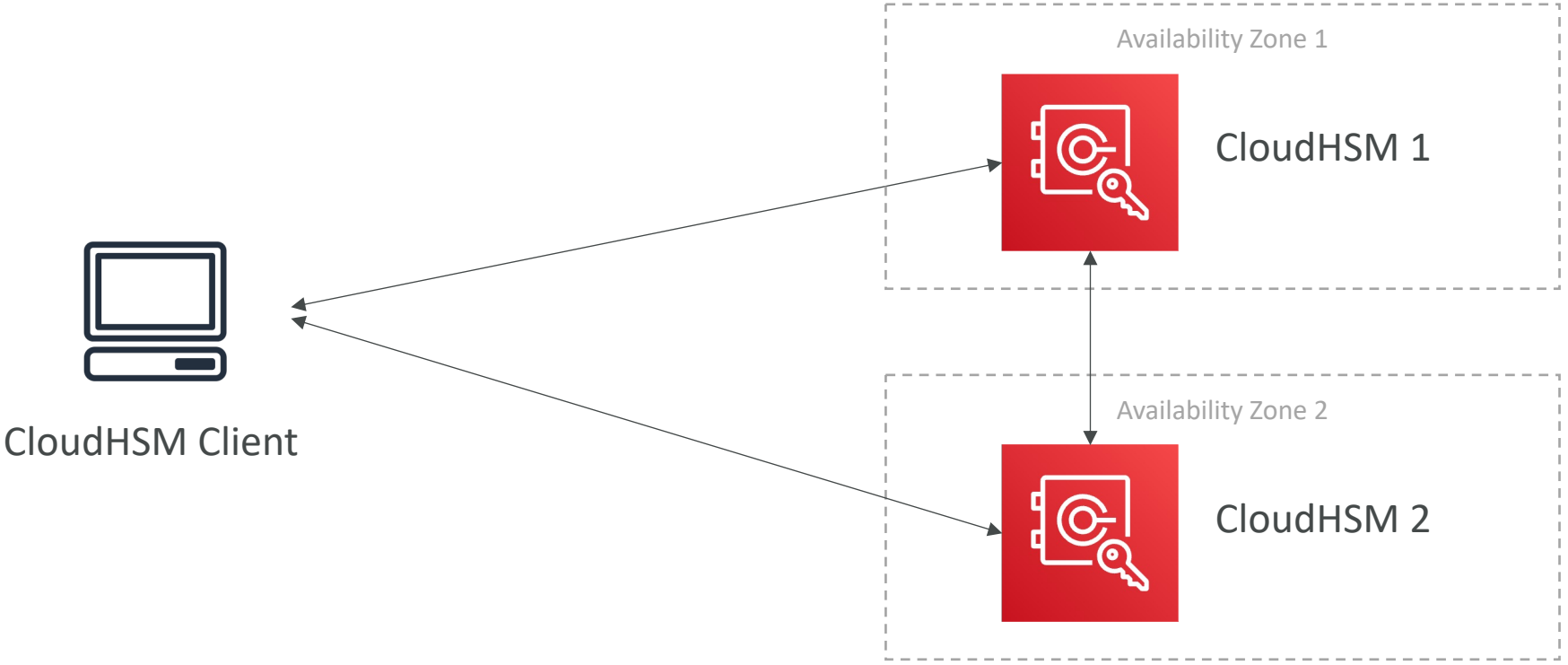
- Manage the Keys
- Manage the Users

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudHSM – High Availability

- CloudHSM clusters are spread across Multi AZ (HA)
- Great for availability and durability



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudHSM vs. KMS

Feature	AWS KMS	AWS CloudHSM
Tenancy	Multi-Tenant	Single-Tenant
Standard	FIPS 140-2 Level 3	FIPS 140-2 Level 3
Master Keys	• AWS Owned Keys • AWS Managed Keys • Customer Managed KMS Keys	Customer Managed CMK
Key Types	• Symmetric • Asymmetric • Digital Signing	• Symmetric • Asymmetric • Digital Signing & Hashing
Key Accessibility	Accessible in multiple AWS regions KMS Key Replication	• Deployed and managed in a VPC • Can be shared across VPCs (VPC Peering)
Cryptographic Acceleration	None	• SSL/TLS Acceleration • Oracle TDE Acceleration
Access & Authentication	AWS IAM	You create users and manage their permissions

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

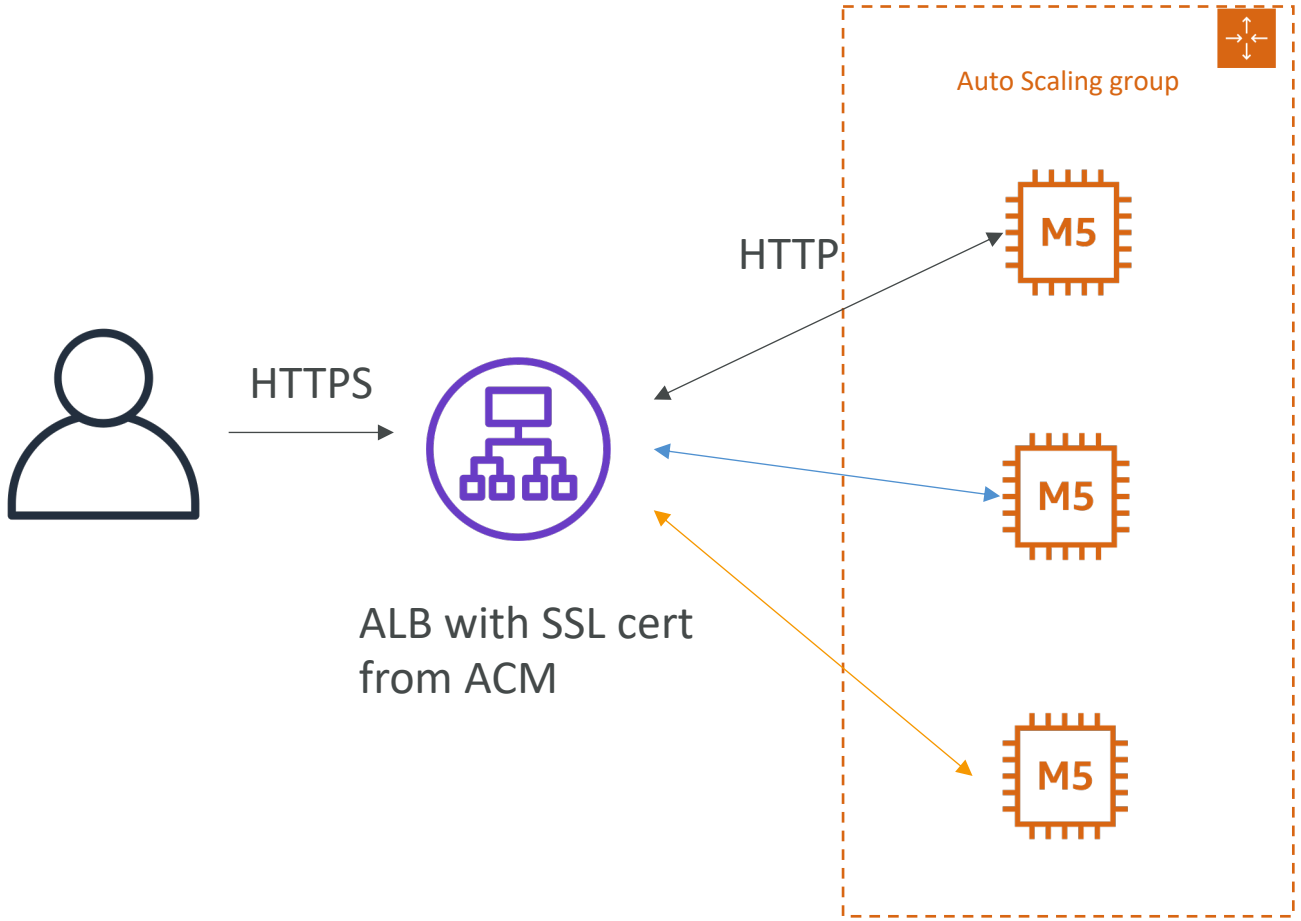
CloudHSM vs. KMS

Feature	AWS KMS	AWS CloudHSM
High Availability	AWS Managed Service	Add multiple HSMs over different AZs
Audit Capability	• CloudTrail • CloudWatch	• CloudTrail • CloudWatch • MFA support
Free Tier	Yes	No

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

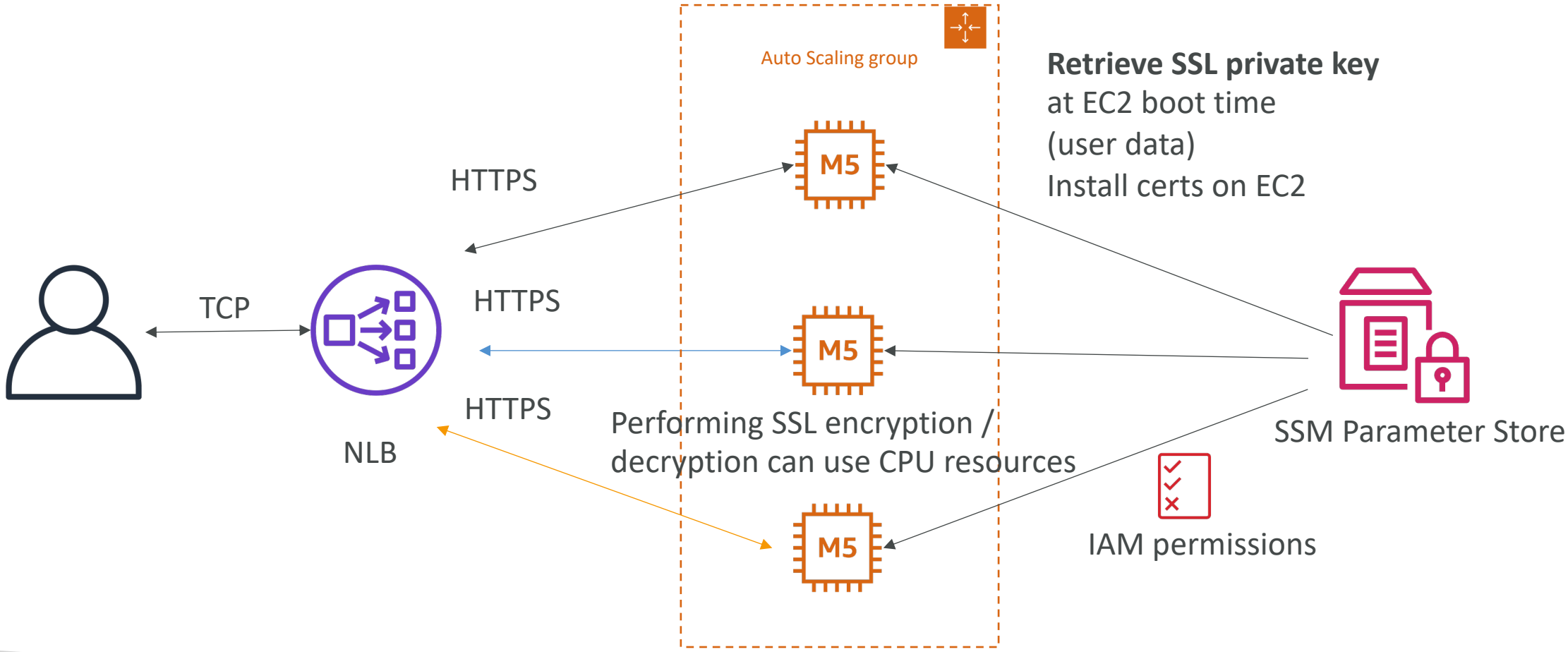
Solution Architecture: SSL on ALB



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Solution Architecture:
SSL on web server EC2 instances

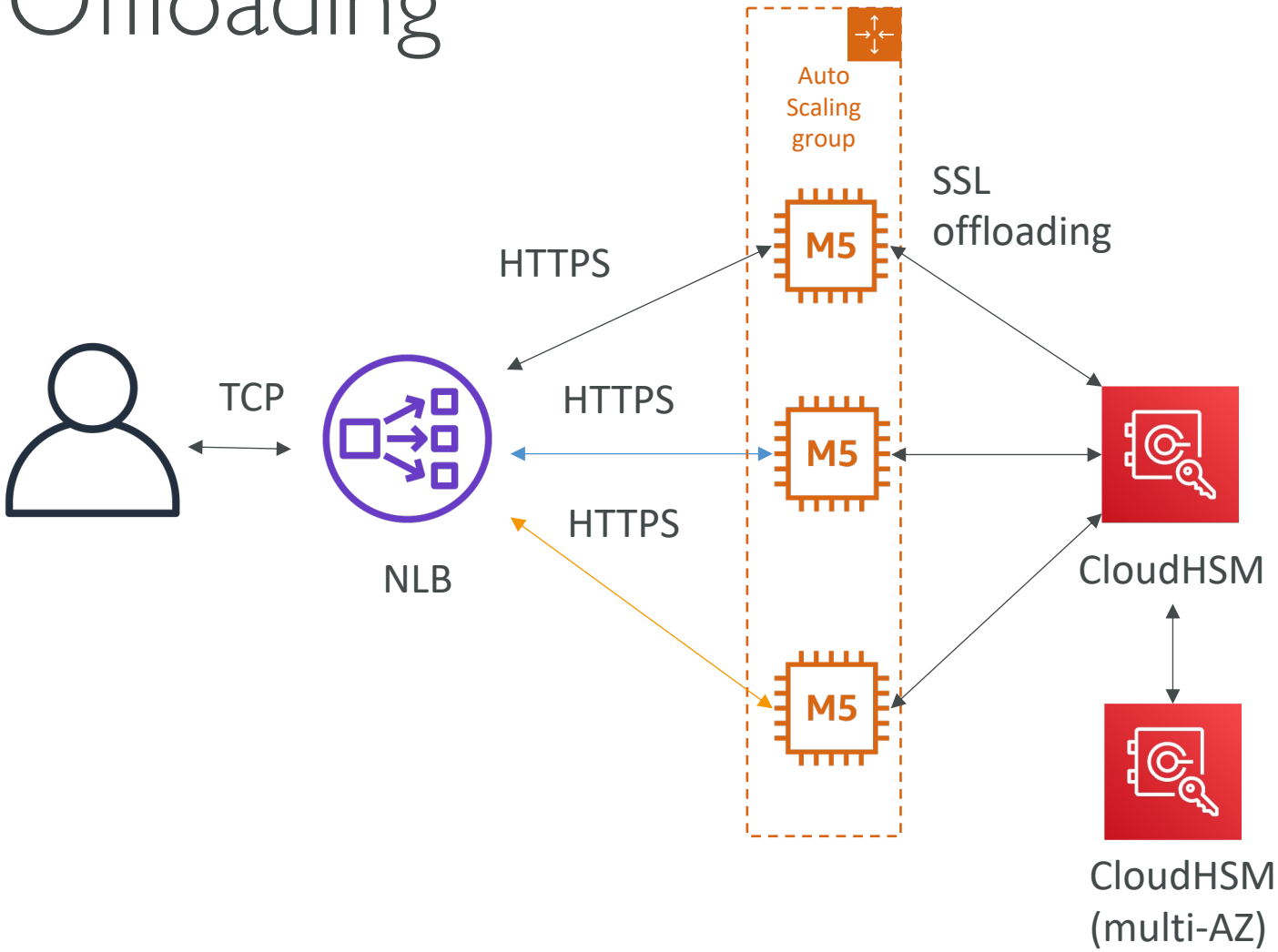


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Solution Architecture: CloudHSM – SSL Offloading

- You can offload SSL to CloudHSM (SSL Acceleration)
- Supported by NGINX, Apache Web servers and IIS for Windows Server
- Extra security: the SSL private key never leaves the HSM device
- Must setup a cryptographic user (CU) on the CloudHSM device



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Encryption for Objects

- **SSE-S3:** encrypts S3 objects using keys handled & managed by AWS
- **SSE-KMS**: leverage KMS to manage encryption keys
  - Key usage appears in CloudTrail
  - objects made public can never be read
  - On *s3:PutObject*, make the permission *kms:GenerateDataKey* is allowed
- **SSE-C**: when you want to manage your own encryption keys
- Client-Side Encryption

- **Glacier:** all data is AES-256 encrypted, key under AWS control

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Encryption in transit (SSL / TLS)

- Amazon S3 exposes:
  - HTTP endpoint: non encrypted
  - HTTPS endpoint: encryption in flight

- You're free to use the endpoint you want, but HTTPS is recommended
- HTTPS is mandatory for SSE-C

- To enforce HTTPS, use a Bucket Policy with aws:SecureTransport

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Events in S3 Buckets

- S3 Access Logs:
  - Detailed records for the requests that are made to a bucket
  - Might take hours to deliver
  - Might be incomplete (best effort)
- S3 Events Notifications:
  - Receive notifications when certain events happen in your bucket
  - E.g.: new objects created, object removal, restore objects, replication events
  - Destinations: SNS, SQS queue, Lambda
  - Typically delivered in seconds but can take minutes, notification for every object if versioning is enabled, else risk of one notification for two same object write done simultaneously
- Trusted Advisor:
  - Check the bucket permission (is the bucket public?)
- Amazon EventBridge:
  - Need to enable CloudTrail object level logging on S3 first
  - Target can be Lambda, SQS, SNS, etc…

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Security

- User based
  - IAM policies - which API calls should be allowed for a specific user from IAM console
- Resource Based
  - Bucket Policies - bucket wide rules from the S3 console - allows cross account
  - Object Access Control List (ACL) – finer grain
  - Bucket Access Control List (ACL) – less common

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Bucket Policies

- Use S3 bucket for policy to:
  - Grant public access to the bucket
  - Force objects to be encrypted at upload
  - **Grant access to another account (Cross Account)**
- Optional Conditions on:
  - **SourceIp**: Public IP or Elastic IP | **VpcSourceIp**: Private IP (through VPC Endpoint)
  - Source VPC or Source VPC Endpoint – only works with VPC Endpoints
  - CloudFront Origin Identity
  - MFA
- Examples here: https://docs.aws.amazon.com/AmazonS3/latest/dev/example-bucket-policies.html

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 pre-signed URLs

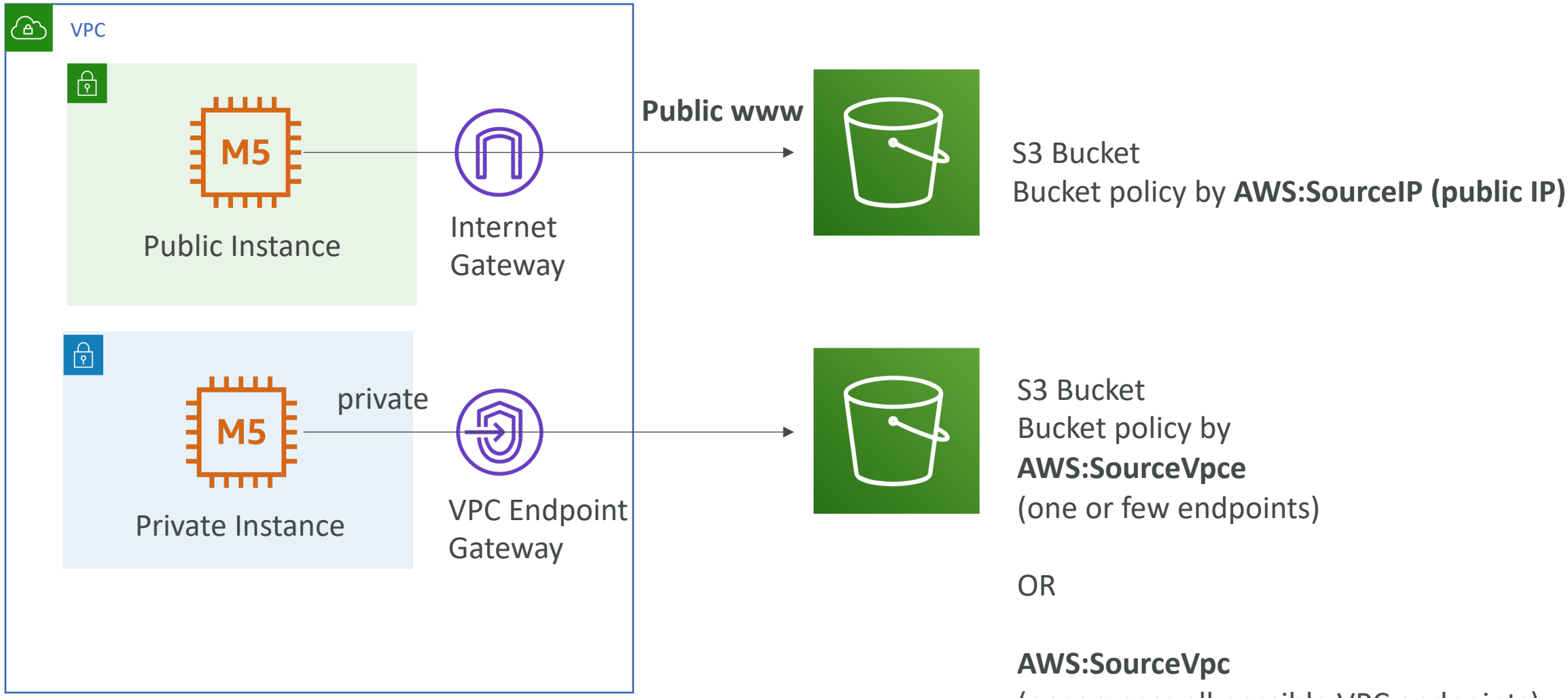
- Can generate pre-signed URLs using SDK or CLI
  - For downloads (easy, can use the CLI)
  - For uploads (harder, must use the SDK)
- Valid for a default of 3600 seconds, can change timeout with --expires-in [TIME_BY_SECONDS] argument
- Users given a pre-signed URL inherit the permissions of the person who generated the URL for GET / PUT

- Examples :
  - Allow only logged-in users to download a premium video on your S3 bucket
  - Allow an ever changing list of users to download files by generating URLs dynamically
  - Allow temporarily a user to upload a file to a precise location in our bucket

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Endpoint Gateway for S3



S3 Bucket
Bucket policy by **AWS:SourceIP (public IP)**

S3 Bucket
Bucket policy by
AWS:SourceVpce
(one or few endpoints)

OR

AWS:SourceVpc
(encompass all possible VPC endpoints)

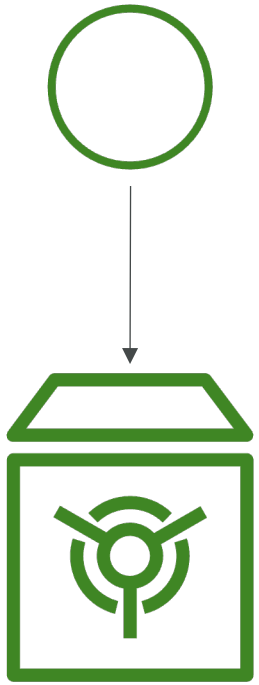
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Object Lock & Glacier Vault Lock

- S3 Object Lock
  - Adopt a WORM (Write Once Read Many) model
  - Block an object version deletion for a specified amount of time

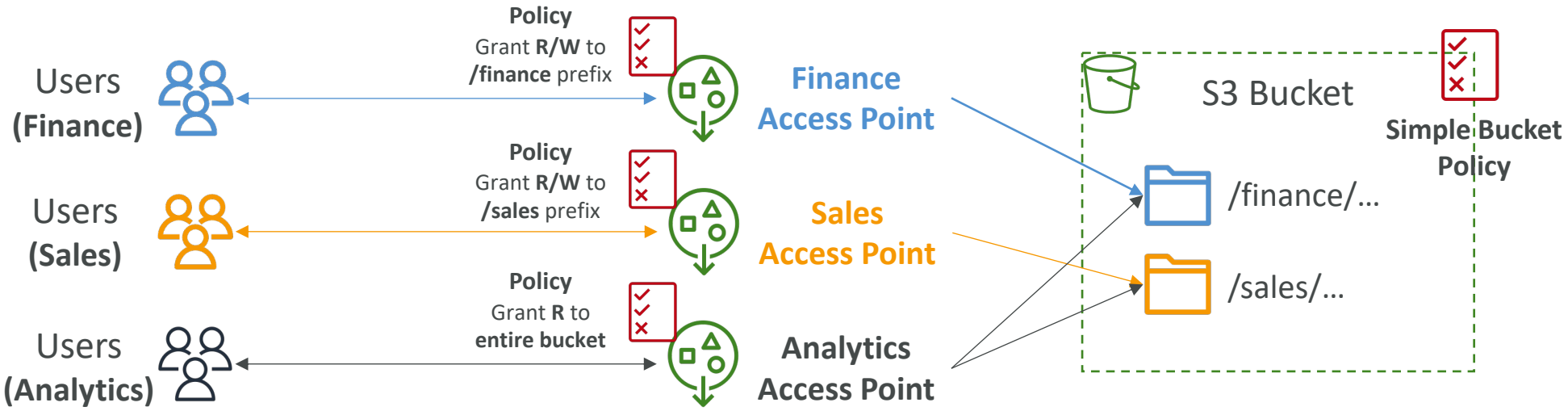
- Glacier Vault Lock
  - Adopt a WORM (Write Once Read Many) model
  - Lock the policy for future edits (can no longer be changed)
  - Helpful for compliance and data retention



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 – Access Points



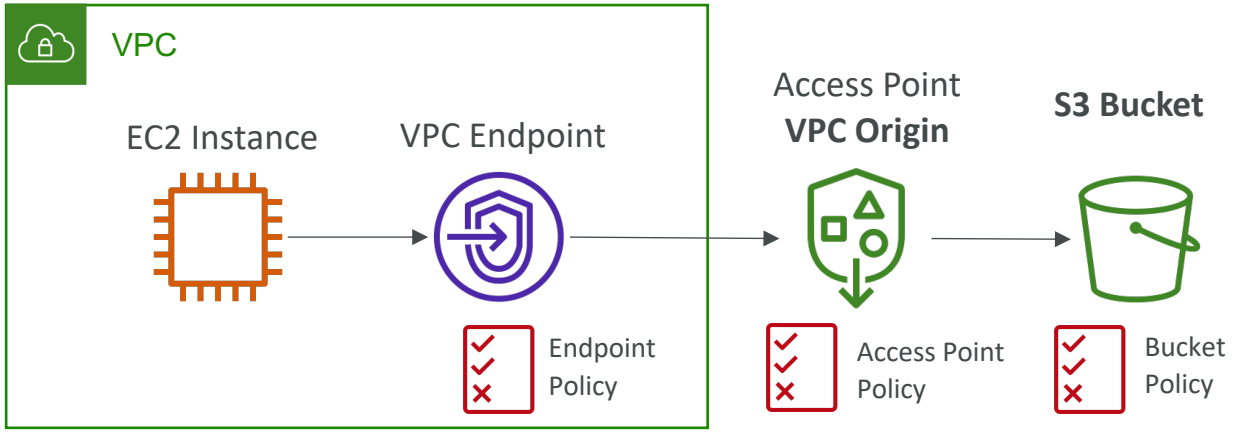
- Access Points simplify security management for S3 Buckets
- Each Access Point has:
  - its own DNS name (Internet Origin or VPC Origin)
  - an access point policy (similar to bucket policy) – manage security at scale

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 – Access Points – VPC Origin

- We can define the access point to be accessible only from within the VPC
- You must create a VPC Endpoint to access the Access Point (Gateway or Interface Endpoint)
- The VPC Endpoint Policy must allow access to the target bucket and Access Point



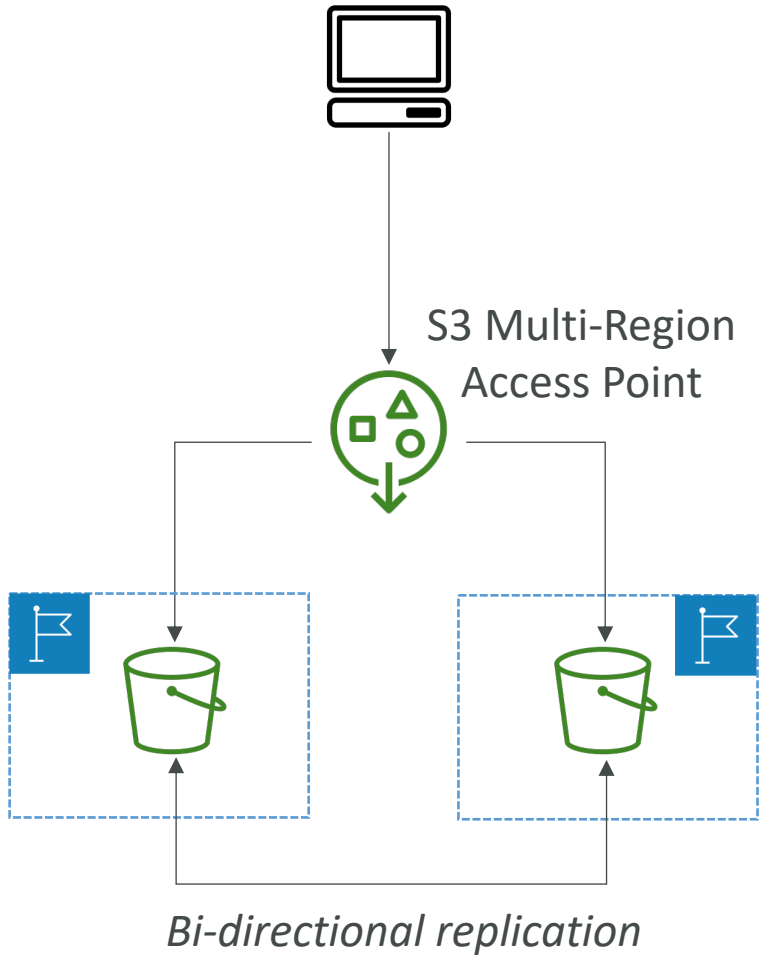

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Principal": "*",
            "Action": [
                "s3:GetObject"
            ],
            "Effect": "Allow",
            "Resource": [
                "arn:aws:s3:::awsexamplebucket1/*",
                "arn:aws:s3:us-west-2:123456789012:accesspoint/example-vpc-ap/object/*"
            ]
        }
    ]
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 – Multi-Region Access Points

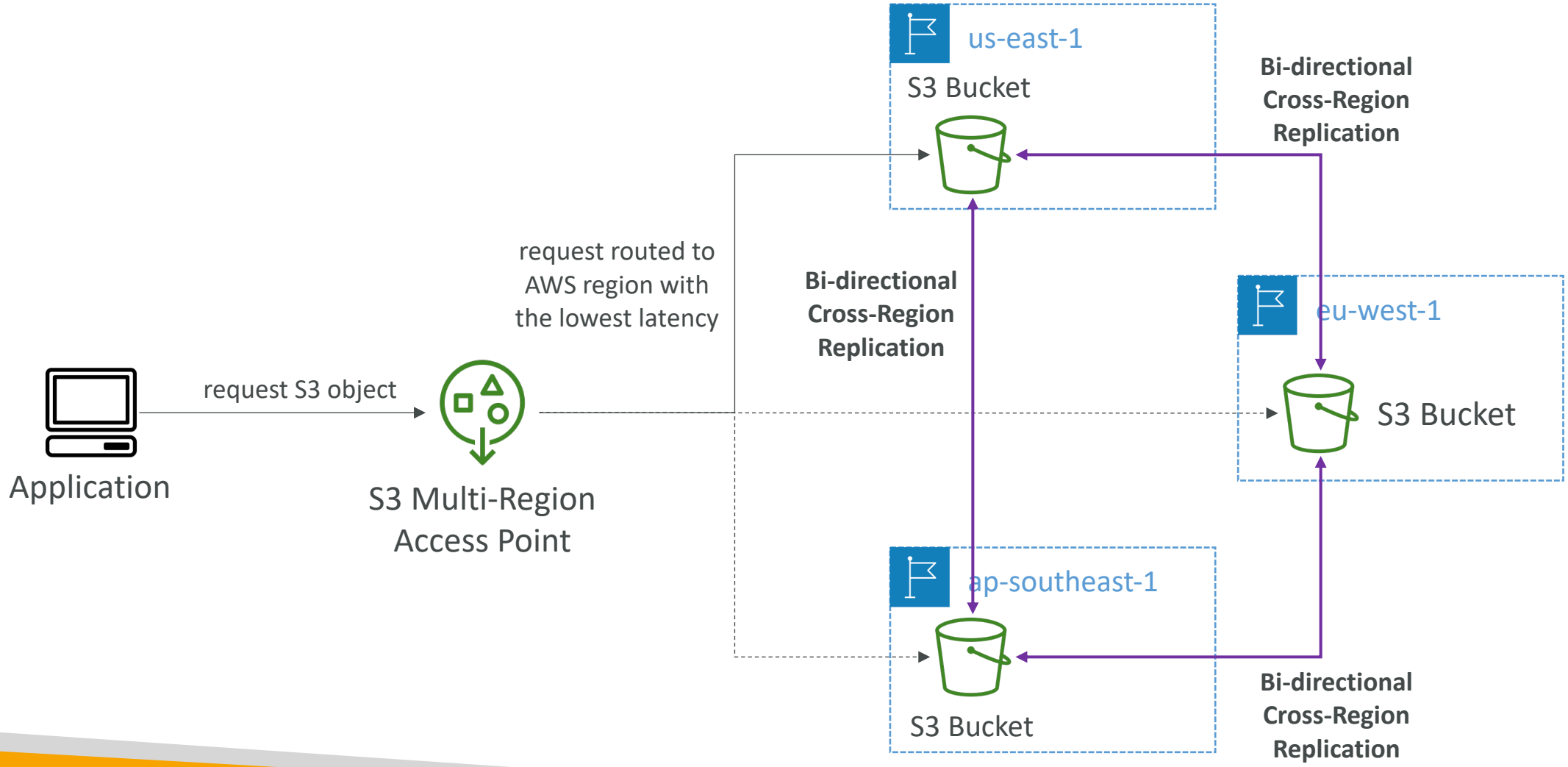
- Provide a global endpoint that span S3 buckets in multiple AWS regions
- Dynamically route requests to the nearest S3 bucket (lowest latency)
- Bi-directional S3 bucket replication rules are created to keep data in sync across regions
- **Failover Controls** – allows you to shift requests across S3 buckets in different AWS regions within minutes (Active-Active or Active-Passive)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

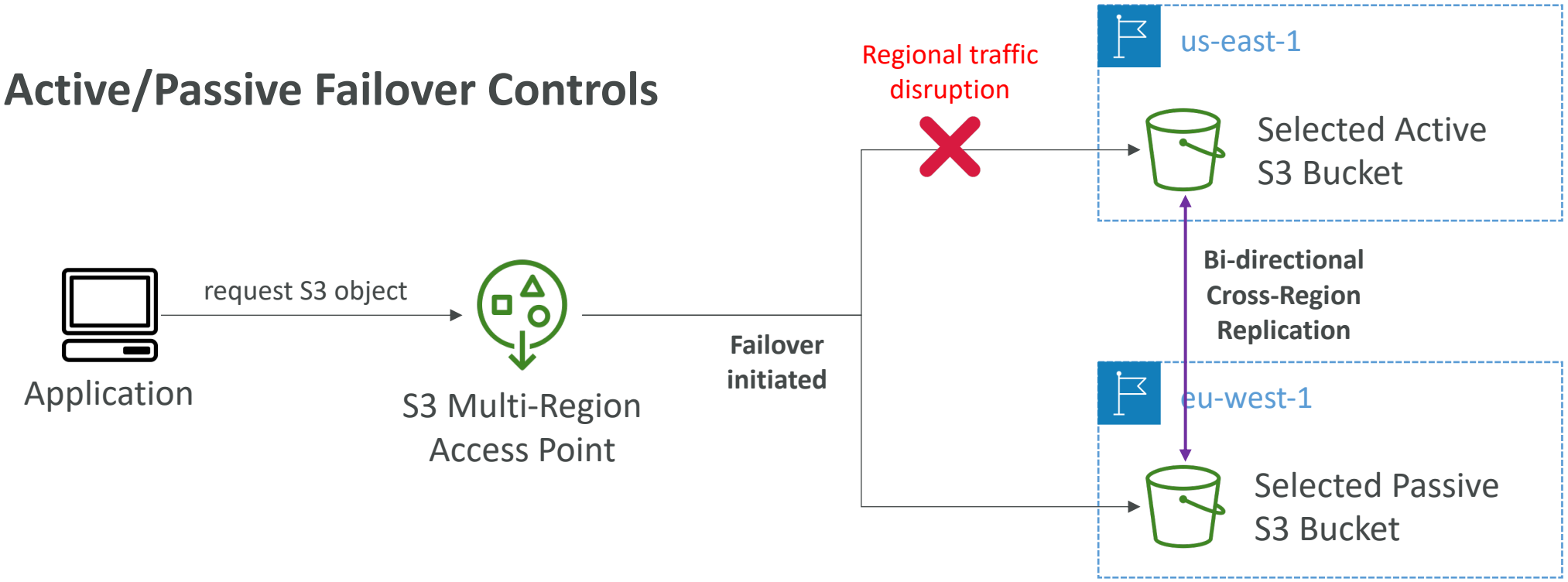
S3 – Multi-Region Access Points



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Multi-Region Access Points – Failover Controls



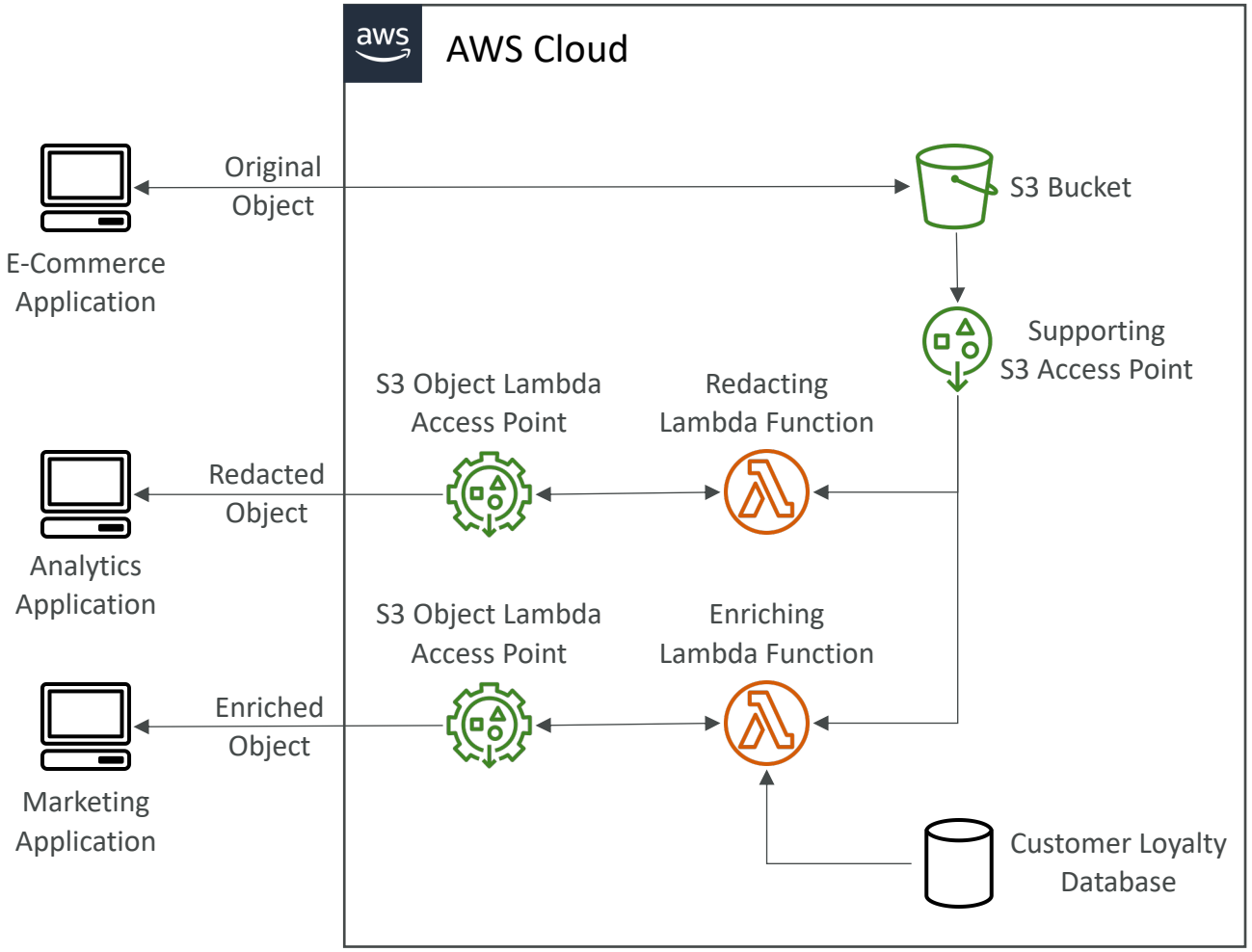
Works with active/active or active/passive setups

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Object Lambda

- Use AWS Lambda Functions to change the object before it is retrieved by the caller application
- Only one S3 bucket is needed, on top of which we create **S3 Access Point** and **S3 Object Lambda Access Points.**
- Use Cases:
  - Redacting personally identifiable information for analytics or non-production environments.
  - Converting across data formats, such as converting XML to JSON.
  - Resizing and watermarking images on the fly using caller-specific details, such as the user who requested the object.

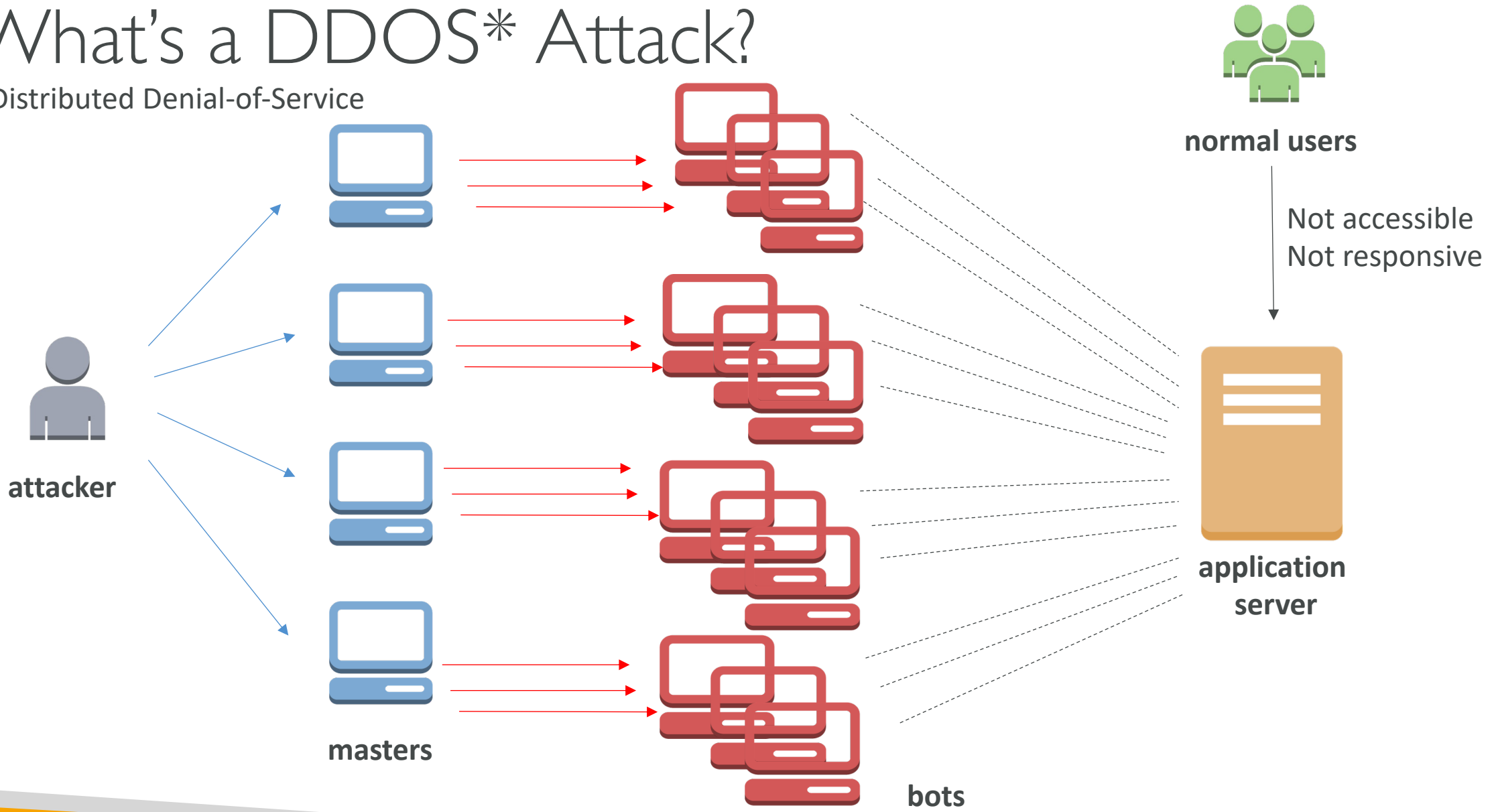


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

What's a DDOS* Attack?

*Distributed Denial-of-Service



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Type of Attacks on your infrastructure

- Distributed Denial of Service (DDoS):
  - When your service is unavailable because it's receiving too many requests
  - SYN Flood (Layer 4): send too many TCP connection requests
  - UDP Reflection (Layer 4): get other servers to send many big UDP requests
  - DNS flood attack: overwhelm the DNS so legitimate users can't find the site
  - Slow Loris attack: a lot of HTTP connections are opened and maintained

- Application level attacks:
  - more complex, more specific (HTTP level)
  - Cache bursting strategies: overload the backend database by invalidating cache

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

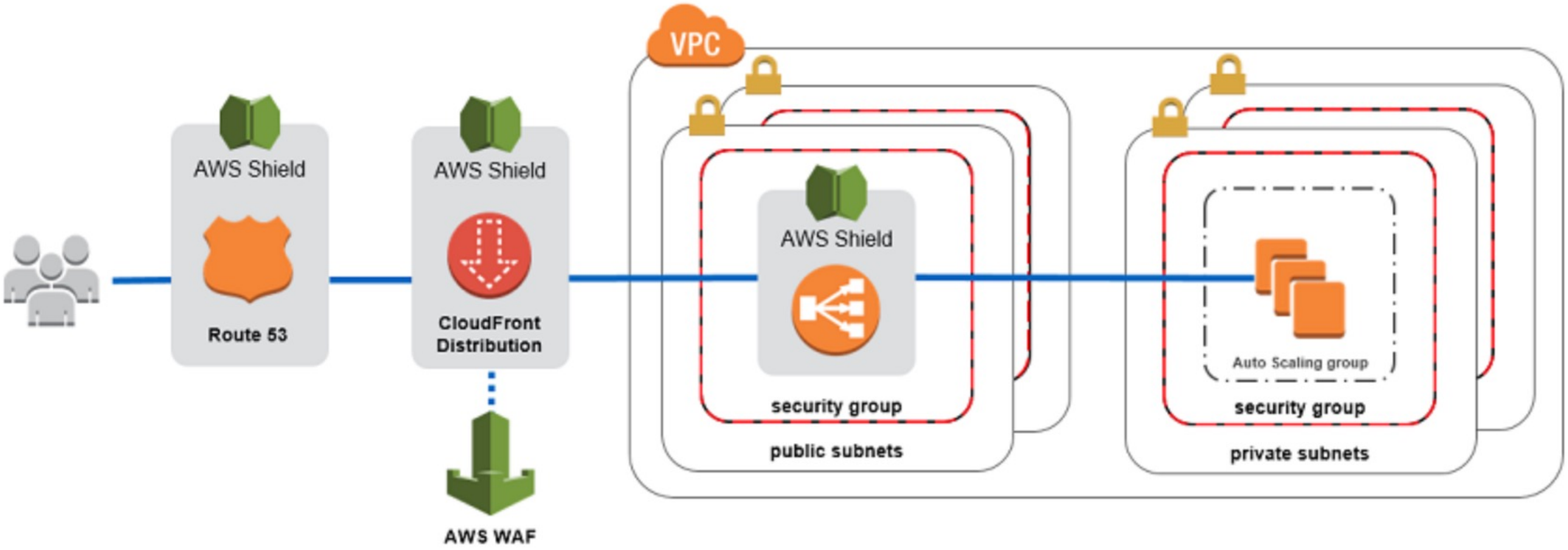
DDoS Protection on AWS

- **AWS Shield Standard:** protects against DDoS attack for your website and applications, for all customers at no additional costs
- **AWS Shield Advanced:** 24/7 premium DDoS protection
- **AWS WAF:** Filter specific requests based on rules
- **CloudFront and Route 53:**
  - Availability protection using global edge network
  - Combined with AWS Shield, provides DDoS attack mitigation at the edge
- Be ready to scale – leverage AWS Auto Scaling
- Separate static resources (S3 / CloudFront) from dynamic ones (EC2 / ALB)
- Read the whitepaper for details: https://d1.awsstatic.com/whitepapers/Security/DDoS_White_Paper.pdf

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Sample Reference Architecture



https://aws.amazon.com/answers/networking/aws-ddos-attack-mitigation/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Shield

- **AWS Shield Standard:**
  - Free service that is activated for every AWS customer
  - Provides protection from attacks such as SYN/UDP Floods, Reflection attacks and other layer 3/layer 4 attacks
- **AWS Shield Advanced:**
  - Optional DDoS mitigation service ($3,000 per month per organization)
  - Protect against more sophisticated attack on Amazon EC2, Elastic Load Balancing (ELB), Amazon CloudFront, AWS Global Accelerator, Route 53
  - 24/7 access to AWS DDoS response team (DRP)
  - Protect against higher fees during usage spikes due to DDoS

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS WAF – Web Application Firewall

- Protects your web applications from common web exploits (Layer 7)
- Deploy on **Application Load Balancer** (localized rules)
- Deploy on **API Gateway** (rules running at the regional or edge level)
- Deploy on **CloudFront** (rules globally on edge locations)
  - Used to front other solutions: CLB, EC2 instances, custom origins, S3 websites
- Deploy on AppSync (protect your GraphQL APIs)
- WAF is not for DDoS protection
- Define Web ACL (Web Access Control List):
  - Rules can include **IP addresses,** HTTP headers, HTTP body, or URI strings
  - Protects from common attack - **SQL injection** and Cross-Site Scripting (XSS)
  - Size constraints, Geo match
  - Rate-based rules (to count occurrences of events)
- Rule Actions: Count | Allow | Block | CAPTCHA | Challenge

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS WAF – Managed Rules

- Library of over 190 managed rules
- Ready-to-use rules that are managed by AWS and AWS Marketplace Sellers

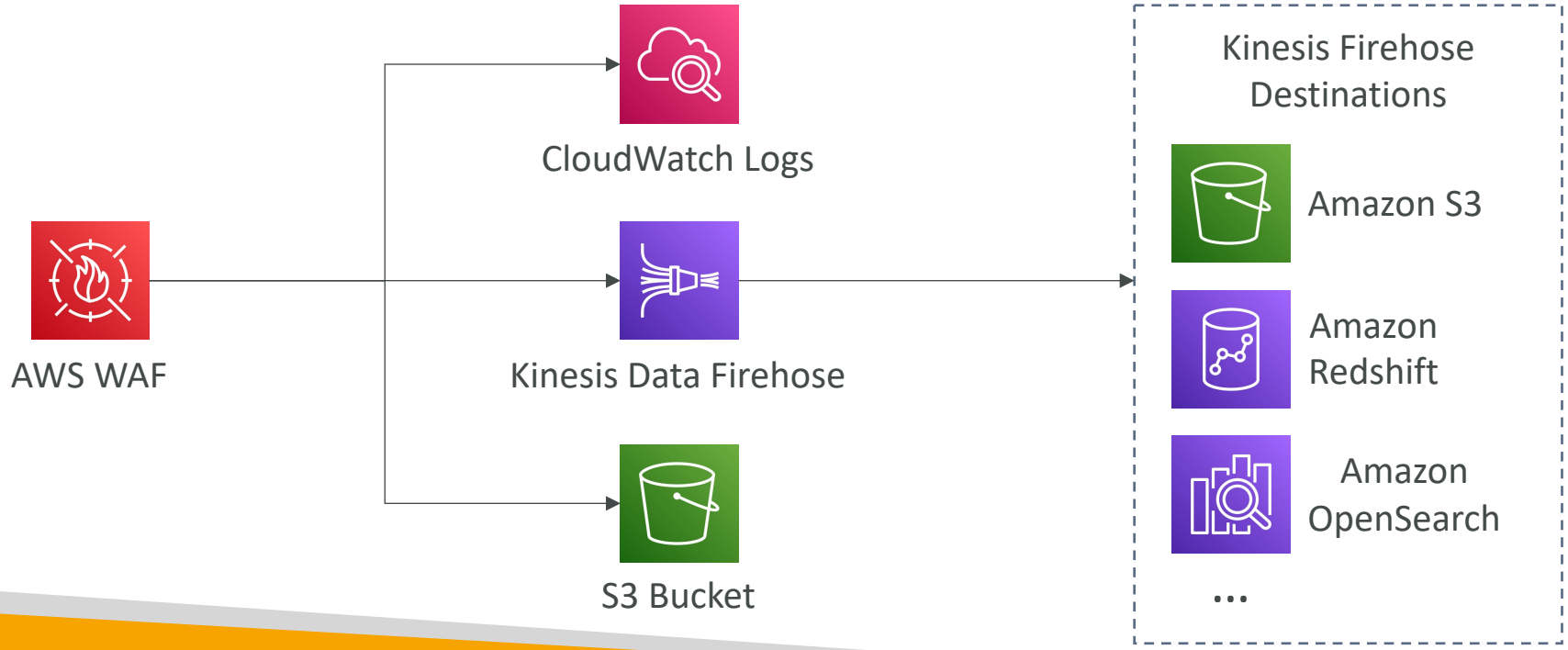
- **Baseline Rule Groups** – general protection from common threats
  - AWSManagedRulesCommonRuleSet, AWSManagedRulesAdminProtectionRuleSet, …
- **Use-case Specific Rule Groups** – protection for many AWS WAF use cases
  - AWSManagedRulesSQLiRuleSet, AWSManagedRulesWindowsRuleSet, AWSManagedRulesPHPRuleSet, AWSManagedRulesWordPressRuleSet, …
- **IP Reputation Rule Groups** – block requests based on source (e.g., malicious IPs)
  - AWSManagedRulesAmazonIpReputationList, AWSManagedRulesAnonymousIpList
- **Bot Control Managed Rule Group** – block and manage requests from bots
  - AWSManagedRulesBotControlRuleSet

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

WAF - Web ACL – Logging

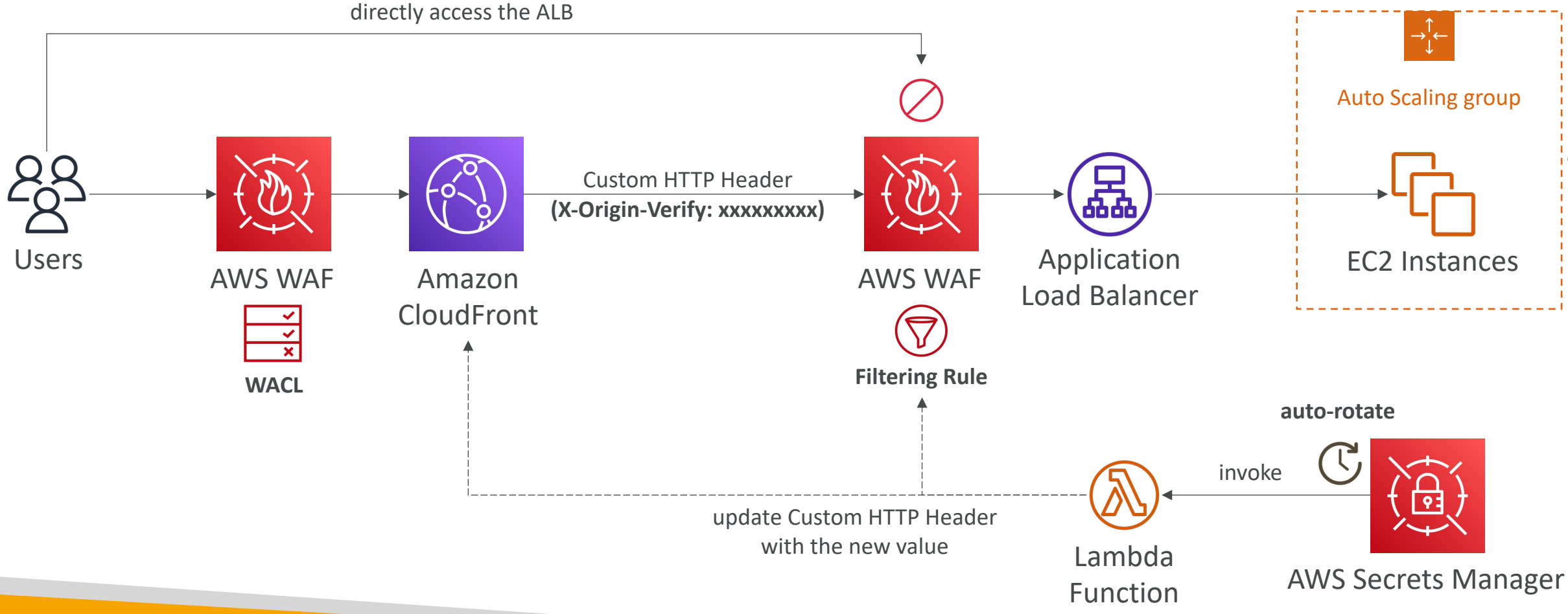
- You can send your logs to an:
  - Amazon CloudWatch Logs log group – 5 MB per second
  - Amazon Simple Storage Service (Amazon S3) bucket – 5 minutes interval
  - Amazon Kinesis Data Firehose – limited by Firehose quotas



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Solution Architecture – Enahnce CloudFront Origin Security with AWS WAF & AWS Secrets Manager



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Firewall Manager

- Manage rules in all accounts of an AWS Organization

- Security policy: common set of security rules
  - WAF rules (Application Load Balancer, API Gateways, CloudFront)
  - AWS Shield Advanced (ALB, CLB, NLB, Elastic IP, CloudFront)
  - Security Groups for EC2, Application Load Balancer and ENI resources in VPC
  - AWS Network Firewall (VPC Level)
  - Amazon Route 53 Resolver DNS Firewall
  - Policies are created at the region level

- Rules are applied to new resources as they are created (good for compliance) across all and future accounts in your Organization

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

WAF vs. Firewall Manager vs. Shield

AWS WAF

AWS Firewall Manager

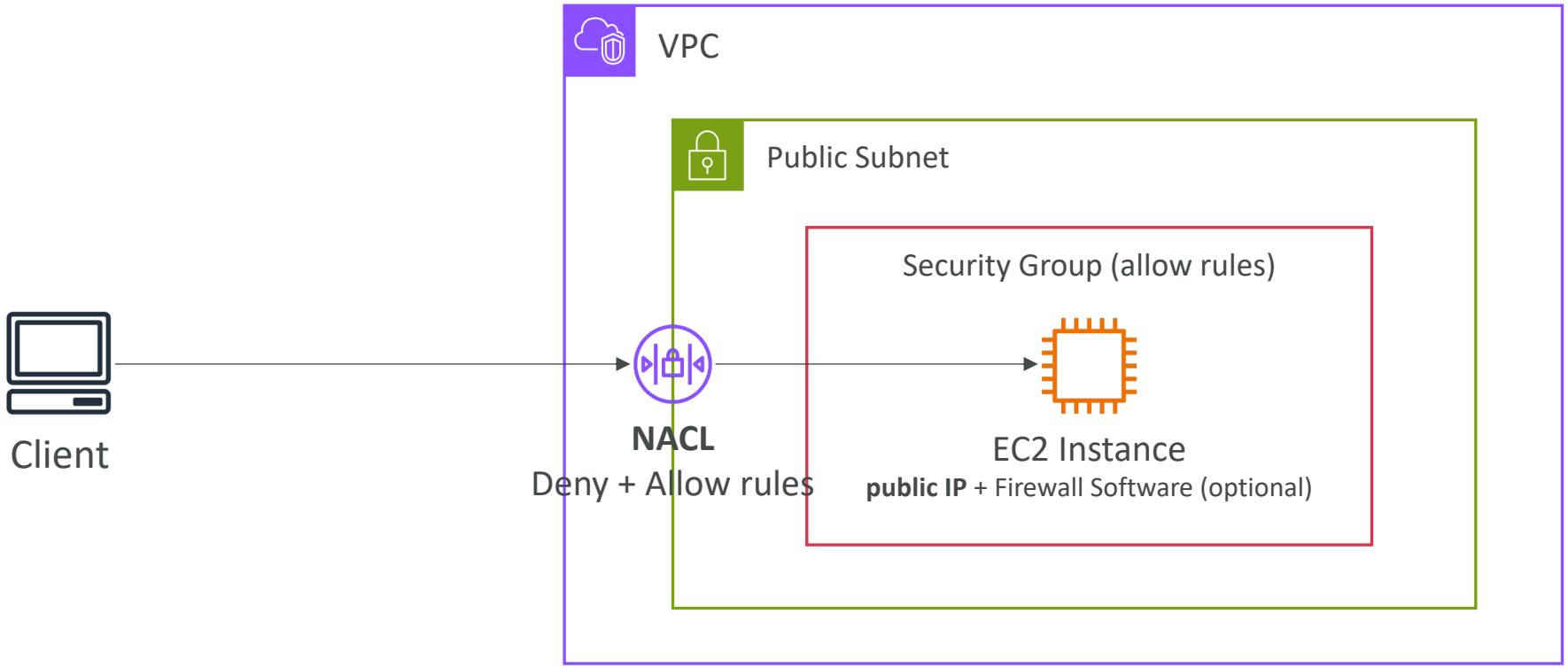
AWS Shield

- WAF, Shield and Firewall Manager are used together for comprehensive protection
- Define your Web ACL rules in WAF
- For granular protection of your resources, WAF alone is the correct choice
- If you want to use AWS WAF across accounts, accelerate WAF configuration, automate the protection of new resources, use Firewall Manager with AWS WAF
- Shield Advanced adds additional features on top of AWS WAF, such as dedicated support from the Shield Response Team (SRT) and advanced reporting.
- If you're prone to frequent DDoS attacks, consider purchasing Shield Advanced

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

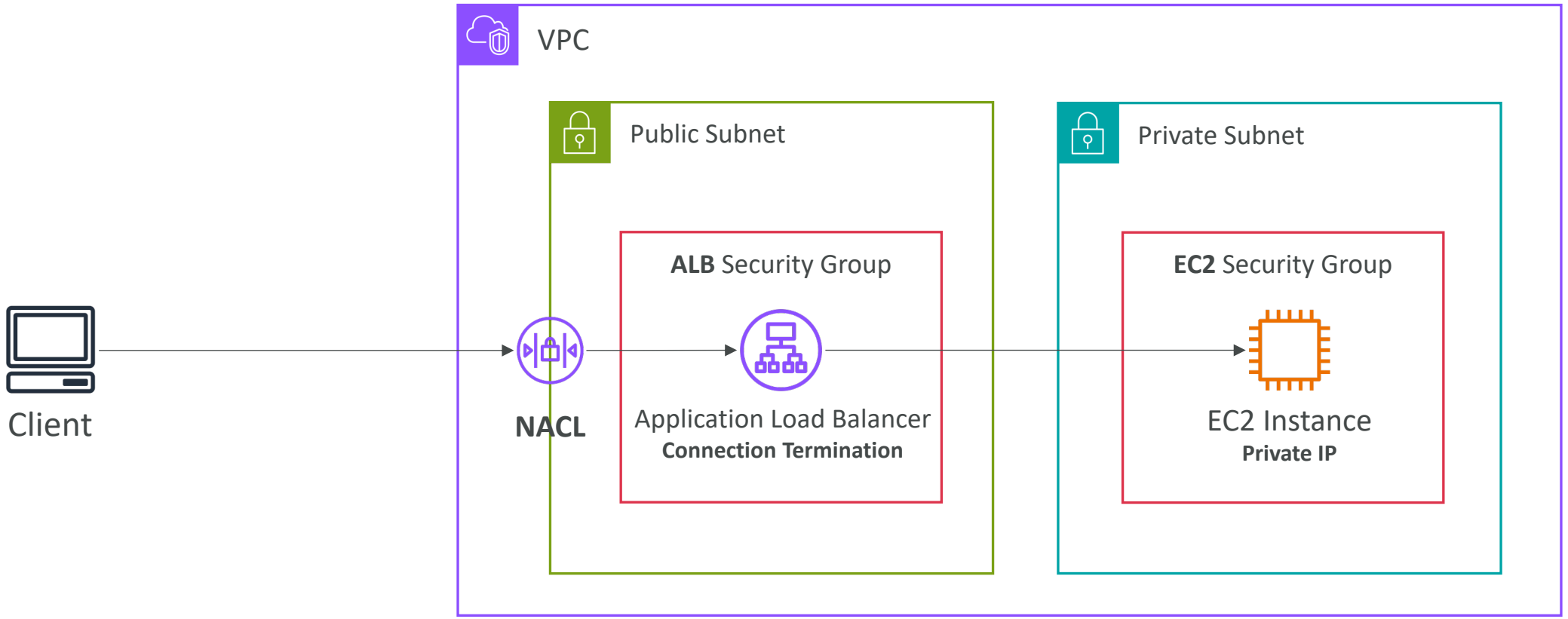
Blocking an IP address



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

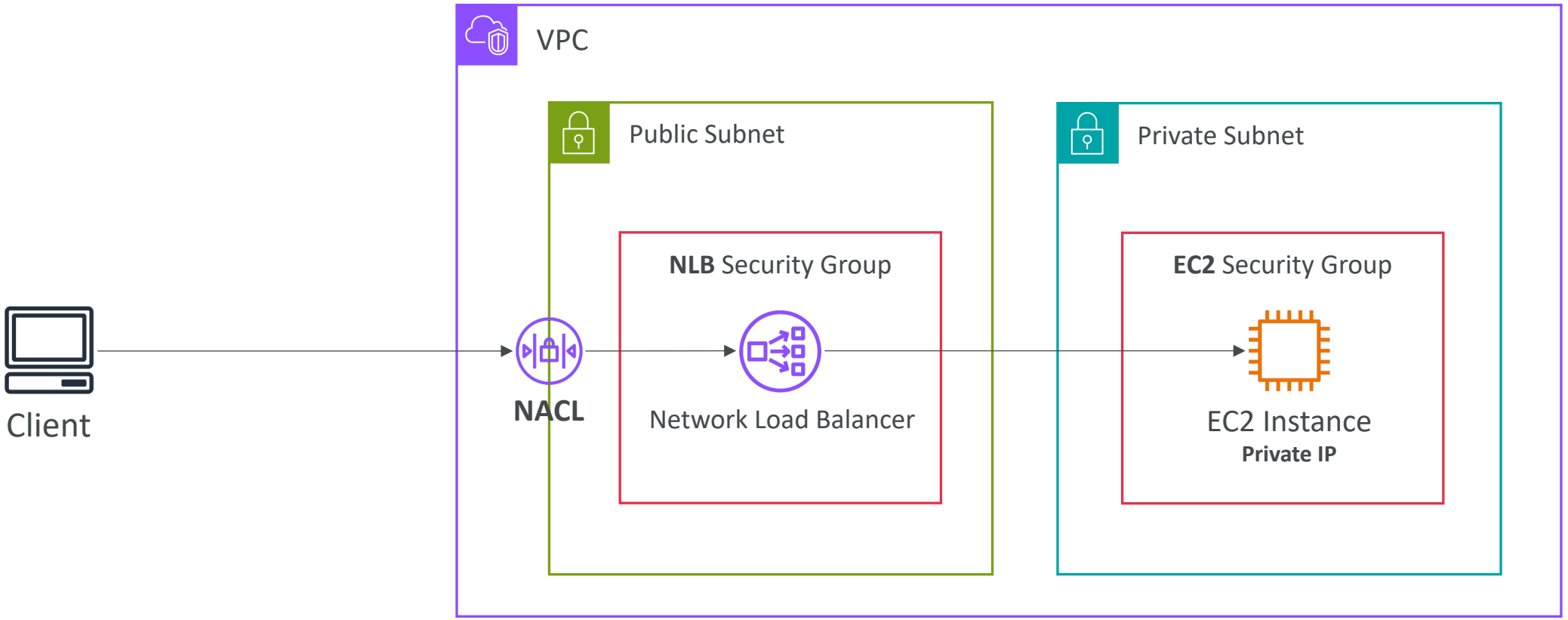
Blocking an IP address – with an ALB



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

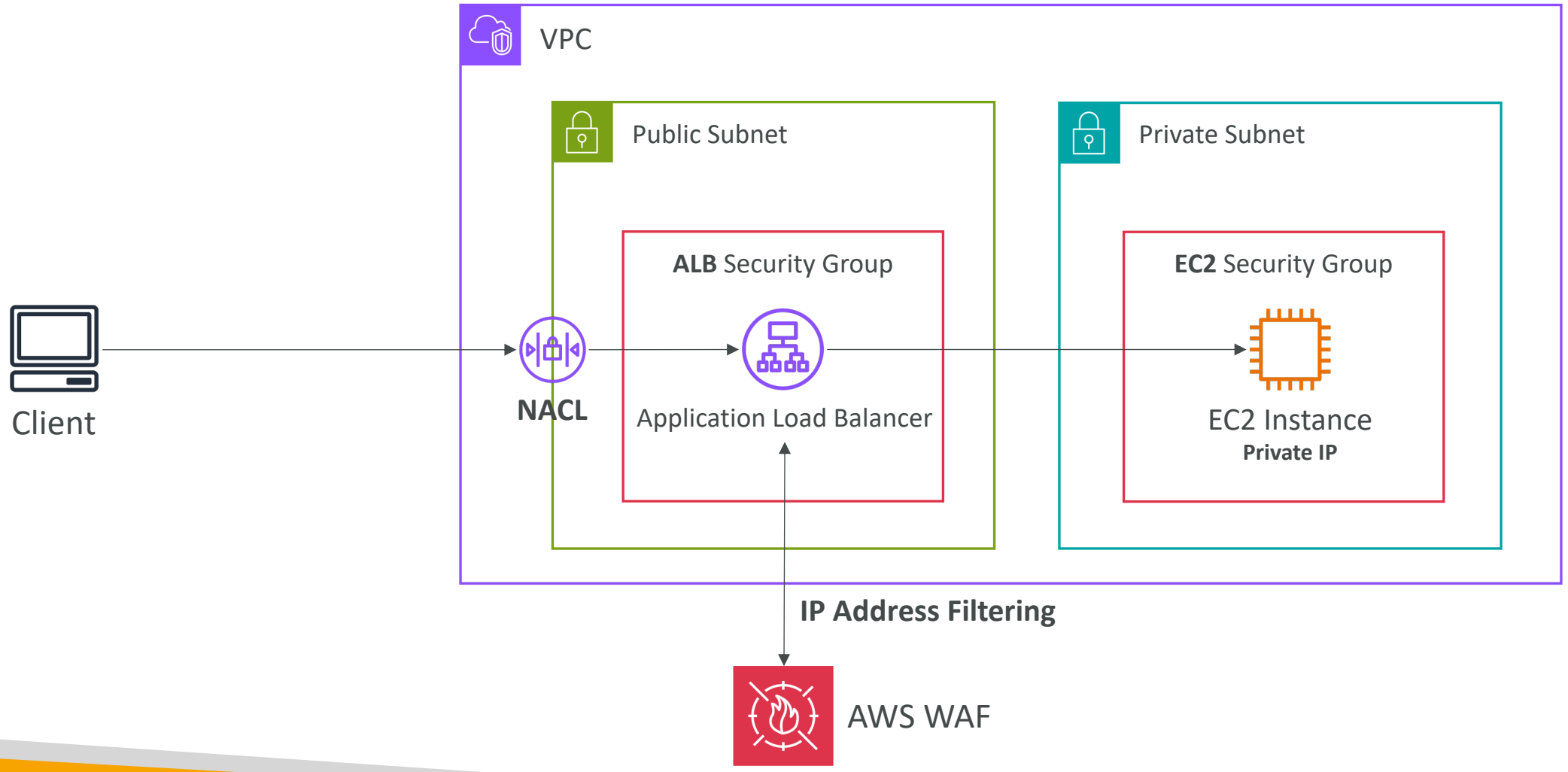
Blocking an IP address – with an NLB



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

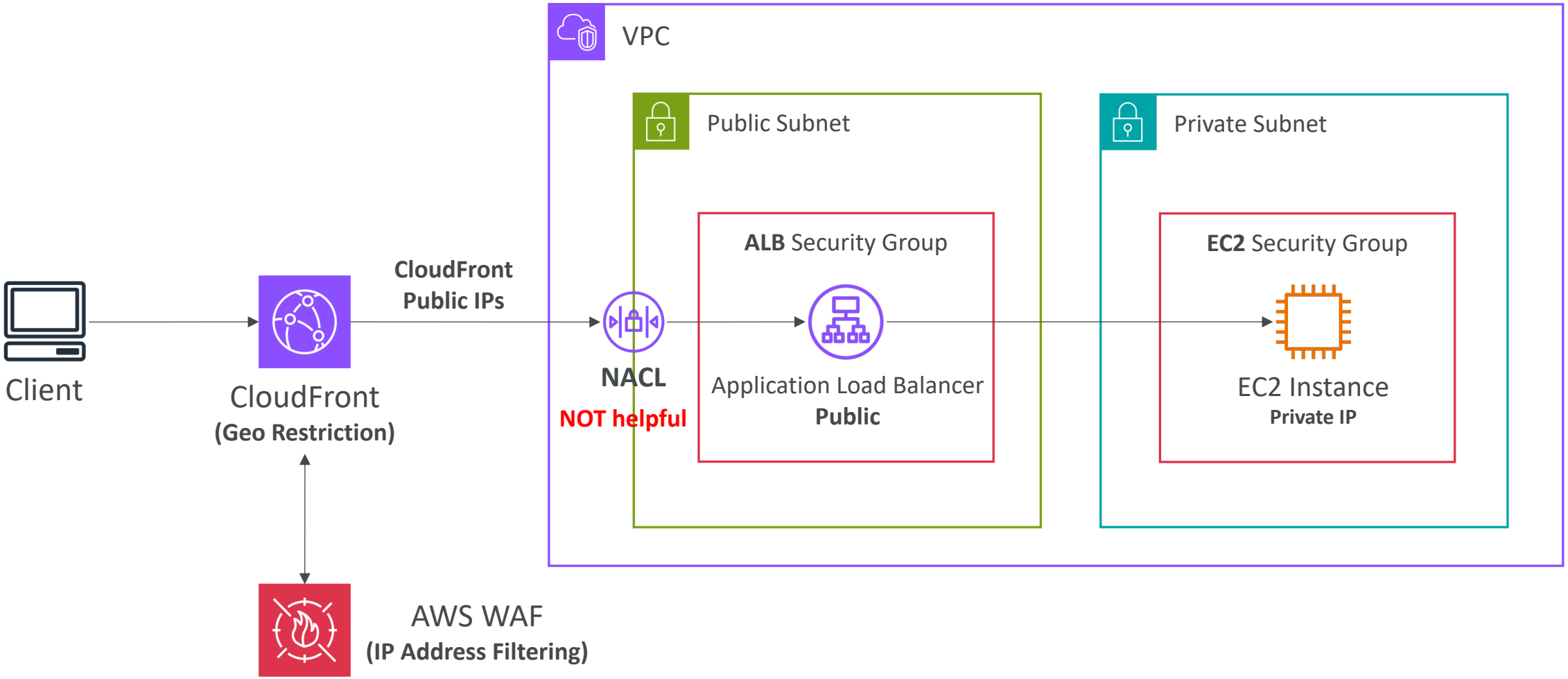
Blocking an IP address – ALB + WAF



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Blocking an IP address – ALB, CloudFront & WAF



© Stephane Maarek

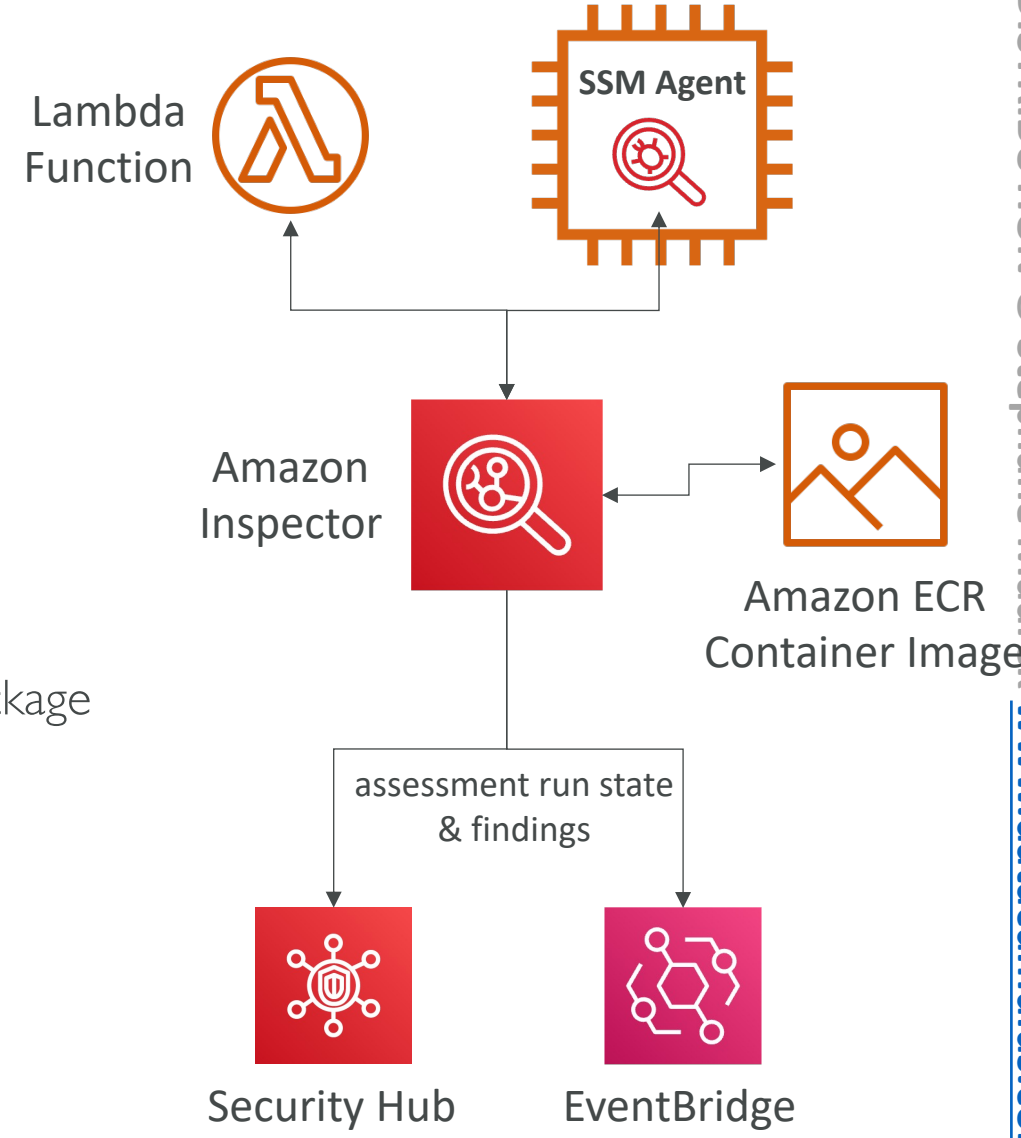
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Inspector

- Automated Security Assessments

- For EC2 instances
  - Leveraging the AWS System Manager (SSM) agent
  - Analyze against unintended network accessibility
  - Analyze the running OS against known vulnerabilities
- For Container Images push to Amazon ECR
  - Assessment of Container Images as they are pushed
- For Lambda Functions
  - Identifies software vulnerabilities in function code and package dependencies
  - Assessment of functions as they are deployed

- Reporting & integration with AWS Security Hub
- Send findings to Amazon Event Bridge



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

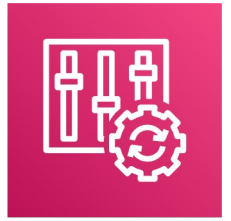
What does Amazon Inspector evaluate?

- **Remember: only for EC2 instances, Container Images & Lambda functions**
- Continuous scanning of the infrastructure, only when needed
- Package vulnerabilities (EC2, ECR & Lambda) – database of CVE
- Network reachability (EC2)
- A risk score is associated with all vulnerabilities for prioritization

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Config



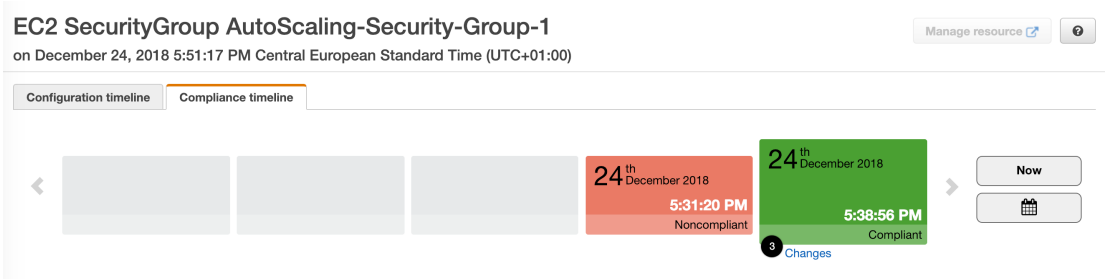
- Helps with auditing and recording **compliance** of your AWS resources
- Helps record configurations and changes over time
- **AWS Config Rules does not prevent actions from happening (no deny)**
- Questions that can be solved by AWS Config:
  - Is there unrestricted SSH access to my security groups?
  - Do my buckets have any public access?
  - How has my ALB configuration changed over time?
- You can receive alerts (SNS notifications) for any changes
- AWS Config is a per-region service
- Can be aggregated across regions and accounts

© Stephane Maarek

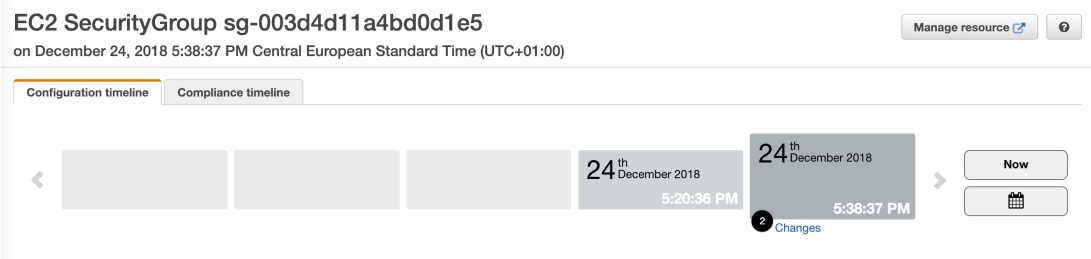
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Config Resource

- View compliance of a resource over time



- View configuration of a resource over time



- View CloudTrail API calls if enabled

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Config Rules

- Can use AWS managed config rules (over 75)
- Can make custom config rules (must be defined in AWS Lambda)
  - Evaluate if each EBS disk is of type gp2
  - Evaluate if each EC2 instance is t2.micro
- Rules can be evaluated / triggered:
  - For each config change
  - And / or: at regular time intervals
- Trigger Amazon EventBridge if the rule is non-compliant (chain with Lambda)
- Rules can have auto remediations through **SSM Automations**
  - If a resource is not compliant, you can trigger an auto remediation
  - Ex: remediate security group rules, stop instances with non-approved tags

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Managed Logs

- Load Balancer Access Logs (ALB, NLB, CLB) => to S3
  - Access logs for your Load Balancers
- CloudTrail Logs => to S3 and CloudWatch Logs
  - Logs for API calls made within your account
- VPC Flow Logs => to S3, CloudWatch Logs, Kinesis Data Firehose
  - Information about IP traffic going to and from network interfaces in your VPC
- Route 53 Access Logs => to CloudWatch Logs
  - Log information about the queries that Route 53 receives
- S3 Access Logs => to S3
  - Server access logging provides detailed records for the requests that are made to a bucket
- CloudFront Access Logs => to S3
  - Detailed information about every user request that CloudFront receives
- AWS Config => to S3

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

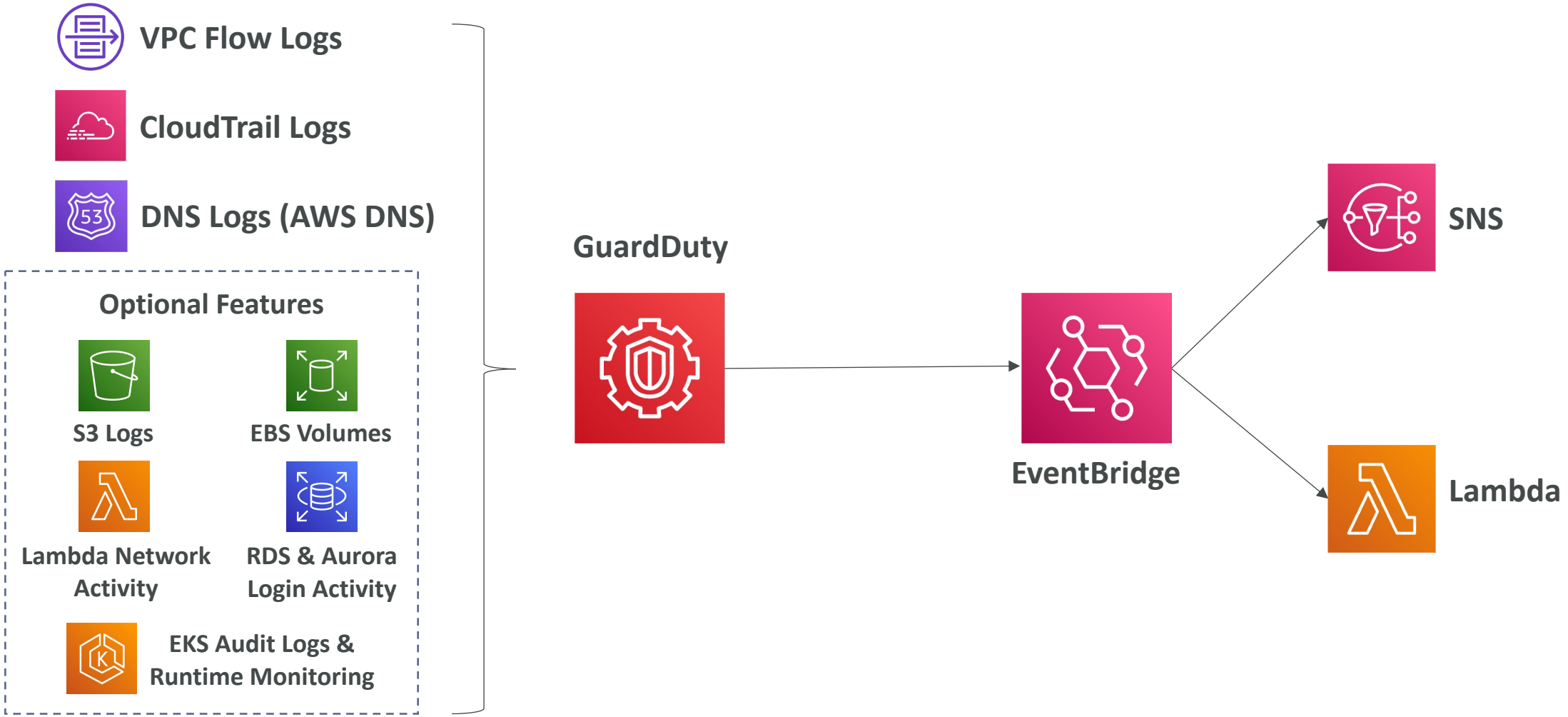
Amazon GuardDuty

- Intelligent Threat discovery to protect your AWS Account
- Uses Machine Learning algorithms, anomaly detection, 3rd party data
- One click to enable (30 days trial), no need to install software
- Input data includes:
  - **CloudTrail Events Logs** – unusual API calls, unauthorized deployments
    - **CloudTrail Management Events** – create VPC subnet, create trail, …
    - **CloudTrail S3 Data Events** – get object, list objects, delete object, …
  - **VPC Flow Logs** – unusual internal traffic, unusual IP address
  - **DNS Logs** – compromised EC2 instances sending encoded data within DNS queries
  - **Optional Feature** – EKS Audit Logs, RDS & Aurora, EBS, Lambda, S3 Data Events…
- Can setup **EventBridge rules** to be notified in case of findings
- EventBridge rules can target AWS Lambda or SNS
- **Can protect against CryptoCurrency attacks (has a dedicated "finding" for it)**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon GuardDuty

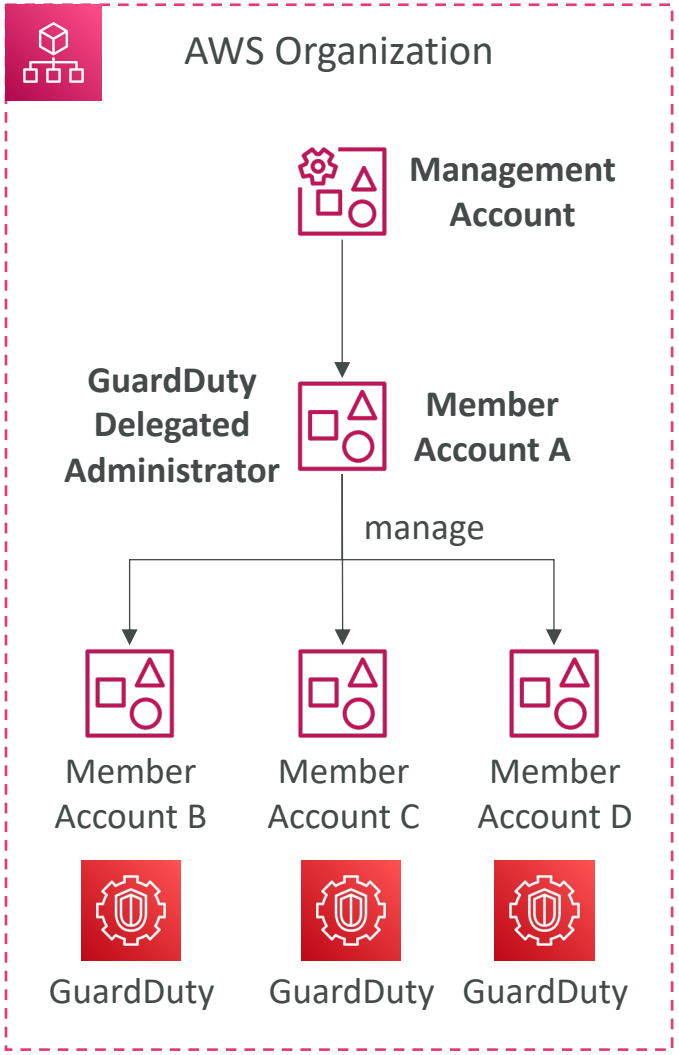


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

GuardDuty – Delegated Administrator

- AWS Organization member accounts can be designated to be a **GuardDuty Delegated Administrator**
- Have full permissions to enable and manage GuardDuty for all accounts in the Organization
- Can be done only using the Organization Management Account



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IAM Conditions

aws:SourceIp
restrict the client IP **from**
which the API calls are being made

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Deny",
            "Action": "*",
            "Resource": "*",
            "Condition": {
                "NotIpAddress": {
                    "aws:SourceIp": ["192.0.2.0/24", "203.0.113.0/2
                }
            }
        }
    ]
}
```

aws:RequestedRegion
restrict the region the
API calls are made **to**

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Deny",
            "Action": ["ec2:*", "rds:*", "dynamodb:*"],
            "Resource": "*",
            "Condition": {
                "StringEquals": {
                    "aws:RequestedRegion": ["eu-central-1", "eu-west-1"]
                }
            }
        }
    ]
}
```

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

IAM Conditions

ec2:ResourceTag
restrict based on **tags**

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": ["ec2:startInstances", "ec2:StopInstances"],
            "Resource": "arn:aws:ec2:us-east-1:123456789012:instance/*",
            "Condition": {
                "StringEquals": {
                    "ec2:ResourceTag/Project": "DataAnalytics",
                    "aws:PrincipalTag/Department": "Data"
                }
            }
        }
    ]
}
```

aws:MultiFactorAuthPresent
to force **MFA**

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "ec2:*",
            "Resource": "*"
        },
        {
            "Effect": "Deny",
            "Action": ["ec2:StopInstances", "ec2:TerminateInstances"],
            "Resource": "*",
            "Condition": {
                "BoolIfExists": {
                    "aws:MultiFactorAuthPresent": false
                }
            }
        }
    ]
}
```

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

IAM for S3

- s3:ListBucket permission applies to
  arn:aws:s3:::test
- => bucket level permission

- s3:GetObject, s3:PutObject, s3:DeleteObject applies to
  arn:awn:s3:::test/*
- => object level permission

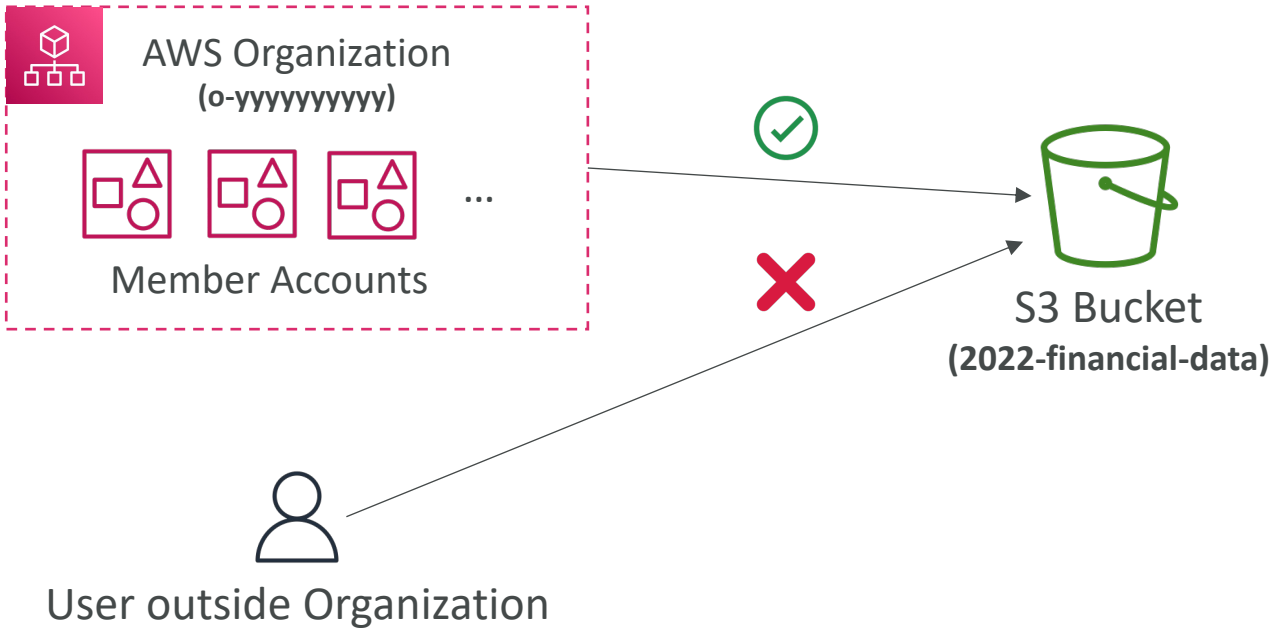
```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": ["s3:ListBucket"],
            "Resource": "arn:aws:s3:::test"
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3:PutObject",
                "s3:GetObject",
                "s3:DeleteObject"
            ],
            "Resource": "arn:aws:s3:::test/*"
        }
    ]
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Resource Policies & aws:PrincipalOrgID

- **aws:PrincipalOrgID** can be used in any resource policies to restrict access to accounts that are member of an AWS Organization

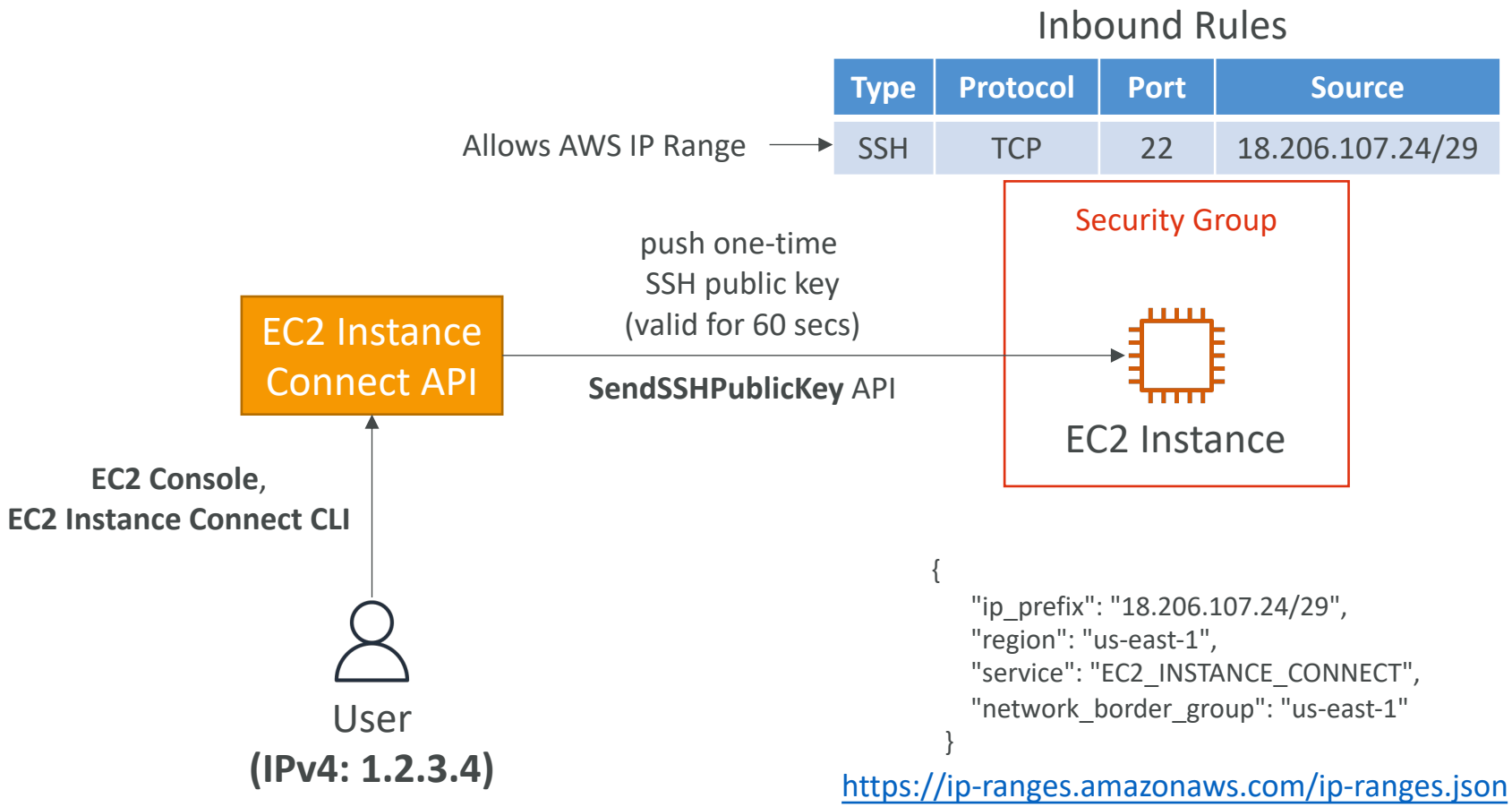



```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": ["s3:PutObject", "s3:GetObject"],
            "Resource": "arn:aws:s3:::2022-financial-data/*",
            "Condition": {
                "StringEquals": {
                    "aws:PrincipalOrgID": ["o-yyyyyyyyyy"]
                }
            }
        }
    ]
}
```

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

EC2 Instance Connect (SendSSHPublicKey API)



Note: All connections are logged in CloudTrail

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

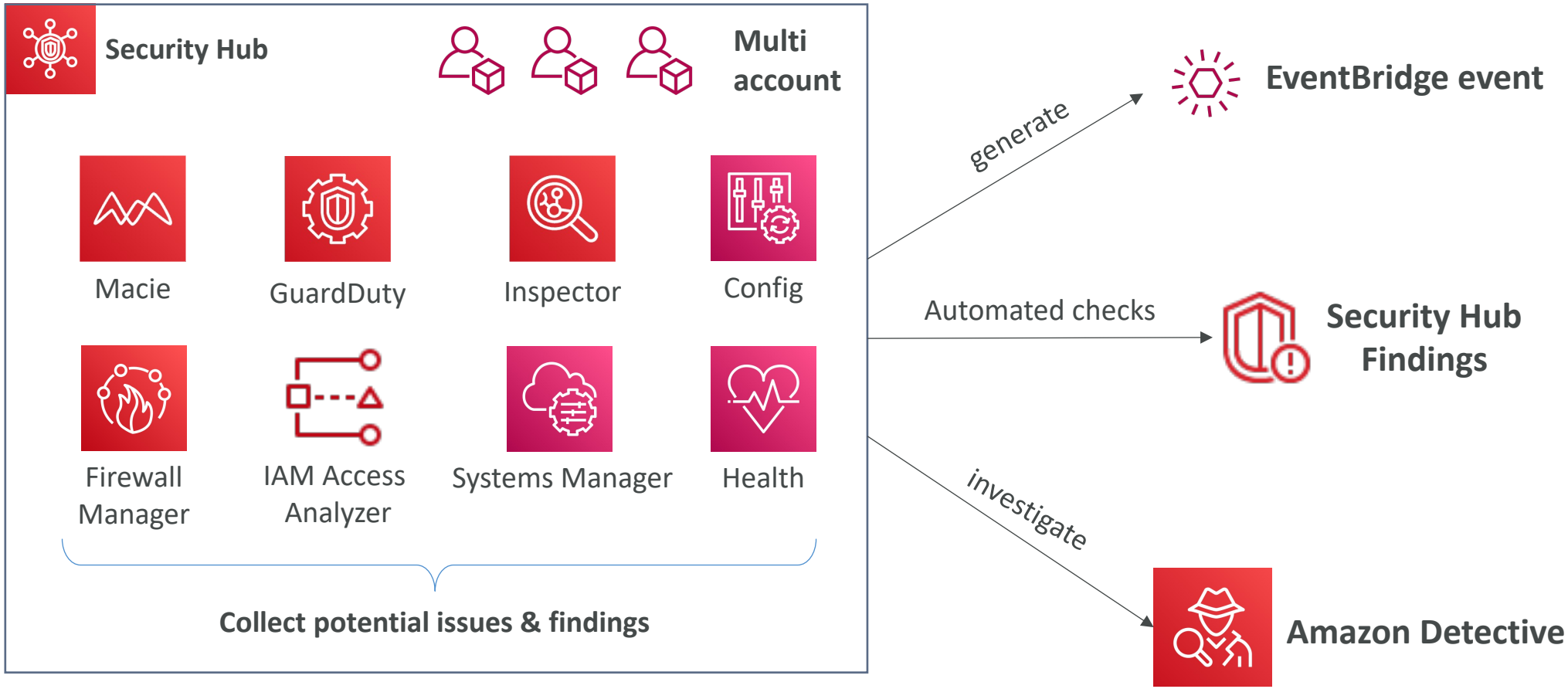
AWS Security Hub

- **Central security tool** to manage security **across several AWS accounts** and **automate security checks**
- Integrated dashboards showing current security and compliance status to quickly take actions
- Automatically aggregates alerts in predefined or personal findings formats from various AWS services & AWS partner tools:
  - Config
  - GuardDuty
  - Inspector
  - Macie
  - IAM Access Analyzer
  - AWS Systems Manager
  - AWS Firewall Manager
  - AWS Health
  - AWS Partner Network Solutions
- Must first enable the AWS Config Service

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Security Hub



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Detective

- GuardDuty, Macie, and Security Hub are used to identify potential security issues, or findings
- Sometimes security findings require deeper analysis to isolate the root cause and take action – it's a complex process
- Amazon Detective **analyzes, investigates, and quickly identifies the root cause of security issues or suspicious activities (using ML and graphs)**
- **Automatically collects and processes events** from VPC Flow Logs, CloudTrail, GuardDuty and create a unified view
- Produces visualizations with details and context to get to the root cause

© Stephane Maarek

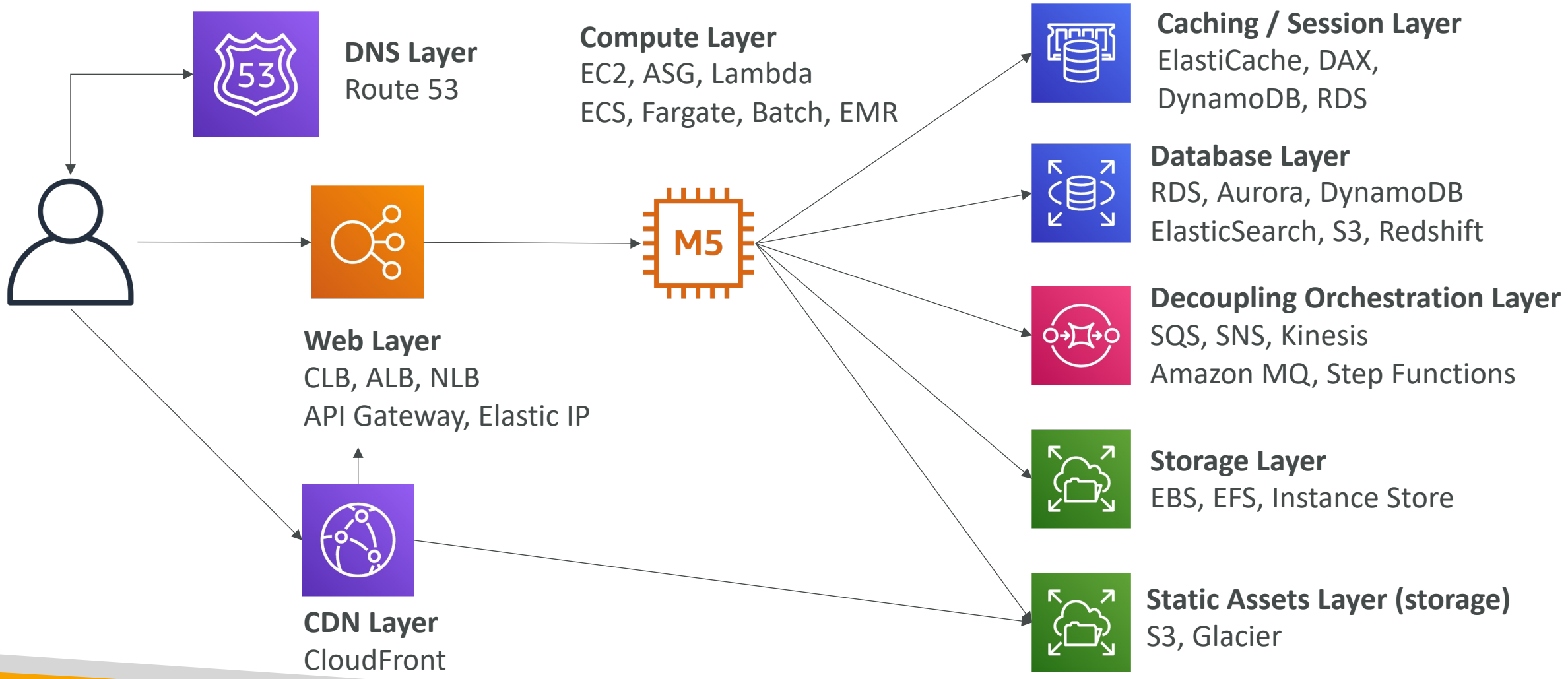
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Compute & Load Balancing

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Solution Architecture on AWS



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EC2 Instance Types – Main ones

- R: applications that needs a lot of RAM – in-memory caches
- C: applications that needs good CPU – compute / databases
- M: applications that are balanced (think "medium") – general / web app
- I: applications that need good local I/O (instance storage) – databases
- G: applications that need a GPU – video rendering / machine learning

- T2 / T3: burstable instances (up to a capacity)
- T2 / T3 - unlimited: unlimited burst

- Real-world tip: use https://www.ec2instances.info

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

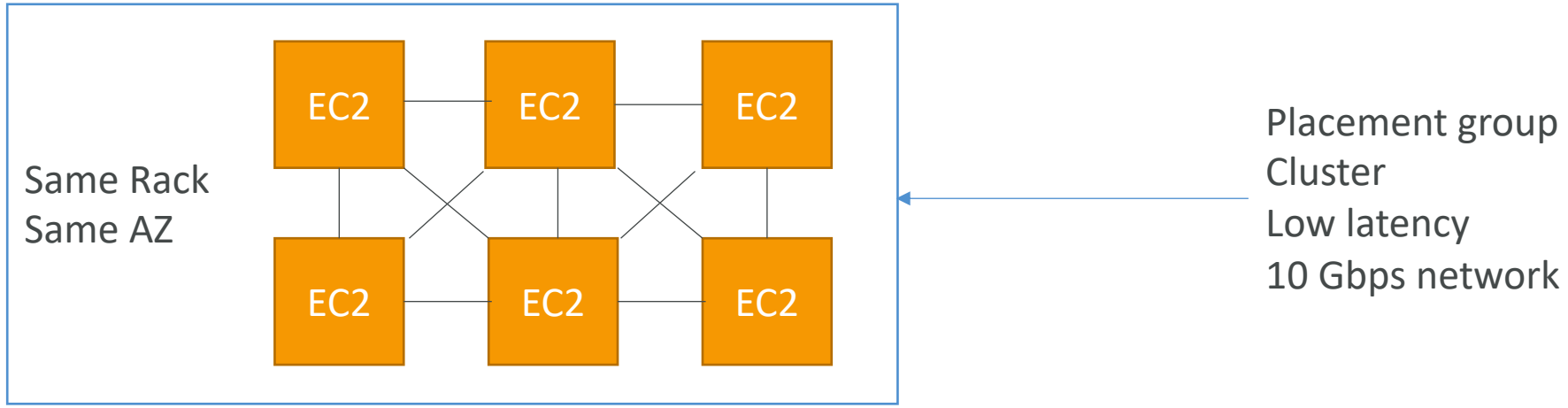
EC2 - Placement Groups

- Control the EC2 Instance placement strategy using placement groups
- Group Strategies:
  - *Cluster*—clusters instances into a low-latency group in a single Availability Zone
  - *Spread*—spreads instances across underlying hardware (max 7 instances per group per AZ) – critical applications
  - *Partition*—spreads instances across many different partitions (which rely on different sets of racks) within an AZ. Scales to 100s of EC2 instances per group (Hadoop, Cassandra, Kafka)
- **You can move** an instance into or out of a placement group
  - Your first need to **stop** it
  - You then need to use the CLI (modify-instance-placement)
  - You can then **start** your instance

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Placement Groups
Cluster

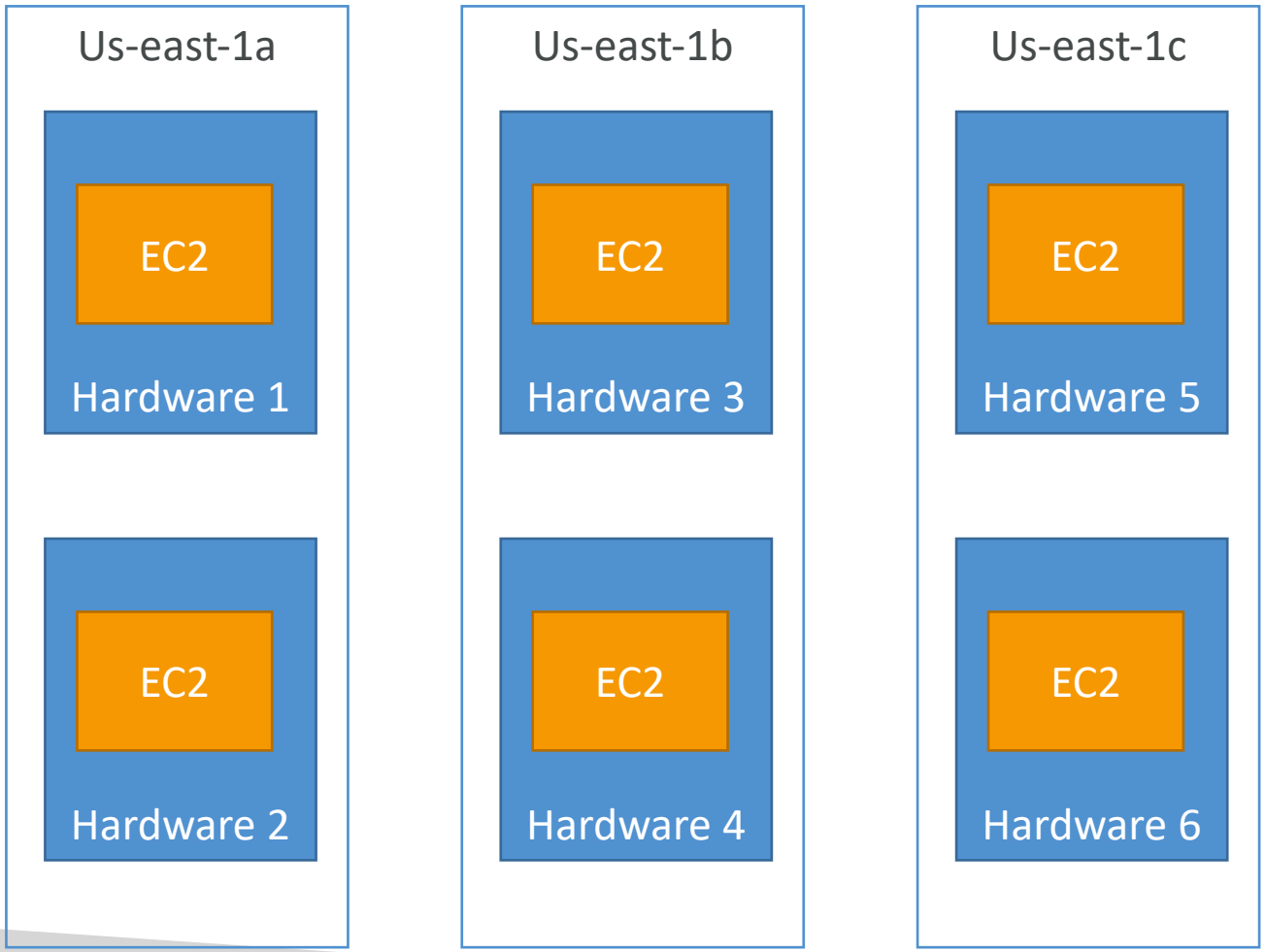


- Pros: Great network (10 Gbps bandwidth between instances with Enhanced Networking enabled - recommended)
- Cons: If the rack fails, all instances fails at the same time
- Use case:
  - Big Data job that needs to complete fast
  - Application that needs extremely low latency and high network throughput

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Placement Groups Spread

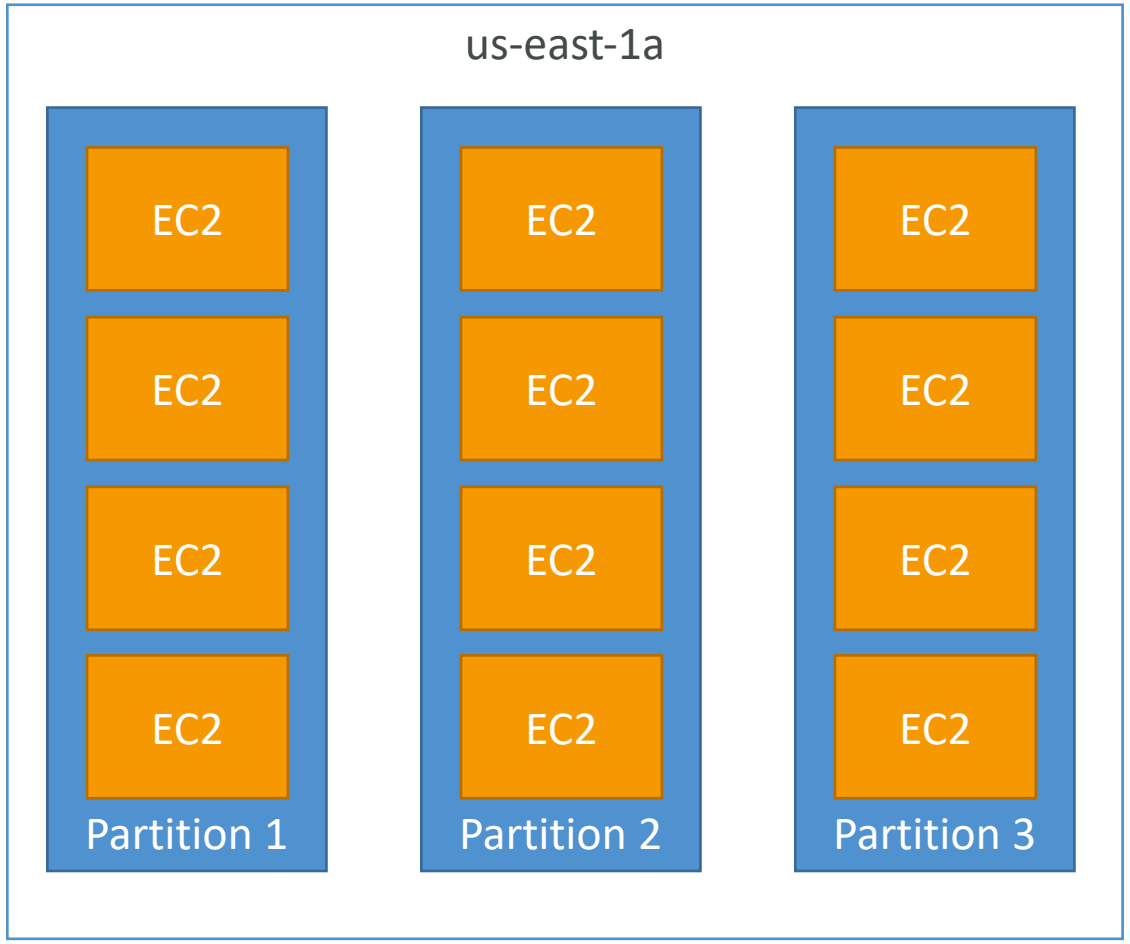


- Pros:
  - Can span across Availability Zones (AZ)
  - Reduced risk is simultaneous failure
  - EC2 Instances are on different physical hardware
- Cons:
  - Limited to 7 instances per AZ per placement group
- Use case:
  - Application that needs to maximize high availability
  - Critical Applications where each instance must be isolated from failure from each other

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Placements Groups Partition



- Up to 7 partitions per AZ
- Up to 100s of EC2 instances
- The instances in a partition do not share racks with the instances in the other partitions
- A partition failure can affect many EC2 but won't affect other partitions
- EC2 instances get access to the partition information as metadata
- Use cases: HDFS, HBase, Cassandra, Kafka

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EC2 Instance Launch Types

- **On Demand Instances:** short workload, predictable pricing, reliable
- **Spot Instances:** short workloads, for cheap, can lose instances (not reliable)
- **Reserved:** (MINIMUM 1 year)
  - **Reserved Instances:** long workloads
  - **Convertible Reserved Instances:** long workloads with flexible instances
  - **Highest to lowest discount:** All Upfront payment, Partial Upfront payment, no Upfront
- **Dedicated Instances:** no other customers will share your hardware
- **Dedicated Hosts:** book an entire physical server, control instance placement
  - Great for software licenses that operate at the core, or CPU socket level
  - Can define **host affinity** so that instance reboots are kept on the same host

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EC2 Graviton

- AWS Graviton Processors deliver the best price performance
- Supports many Linux OS, Amazon Linux 2, RedHat, SUSE, Ubuntu
- Not available for Windows instances

- **Graviton2** – 40% better price performance over comparable 5th generation x86-based instances
- **Graviton3** – Up to 3x better performance compared to Graviton2
- Use cases: app servers, microservices, HPC, CPU-based ML, video encoding, gaming, in-memory caches, …

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EC2 included metrics

- CPU: CPU Utilization + Credit Usage / Balance
- Network: Network In / Out
- Status Check:
  - Instance status = check the EC2 VM
  - System status = check the underlying hardware
- Disk: Read / Write for Ops / Bytes (only for instance store)

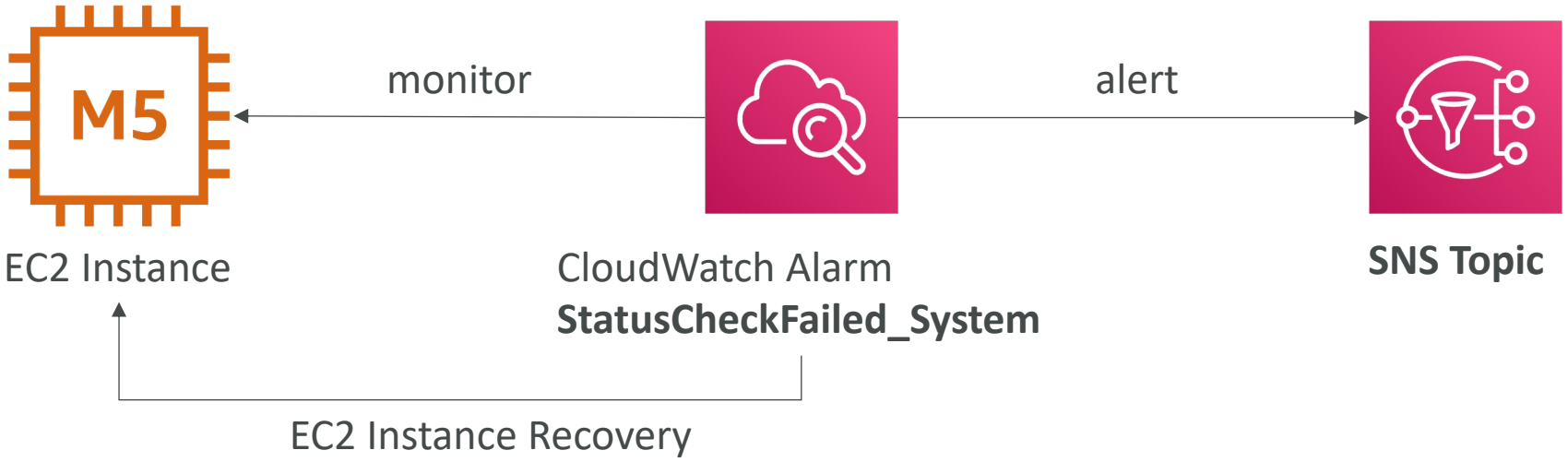
- RAM is NOT included in the AWS EC2 metrics

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EC2 Instance Recovery

- Status Check:
  - Instance status = check the EC2 VM
  - System status = check the underlying hardware



- **Recovery:** Same Private, Public, Elastic IP, metadata, placement group

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

High Performance Computing (HPC)

- The cloud is the perfect place to perform HPC
- You can create a very high number of resources in no time
- You can speed up time to results by adding more resources
- You can pay only for the systems you have used

- Perform genomics, computational chemistry, financial risk modeling, weather prediction, machine learning, deep learning, autonomous driving

- Which services help perform HPC?

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Data Management & Transfer

- AWS Direct Connect:
  - Move GB/s of data to the cloud, over a private secure network
- Snowball
  - Move PB of data to the cloud
- AWS DataSync
  - Move large amount of data between on-premise and S3, EFS, FSx for Windows

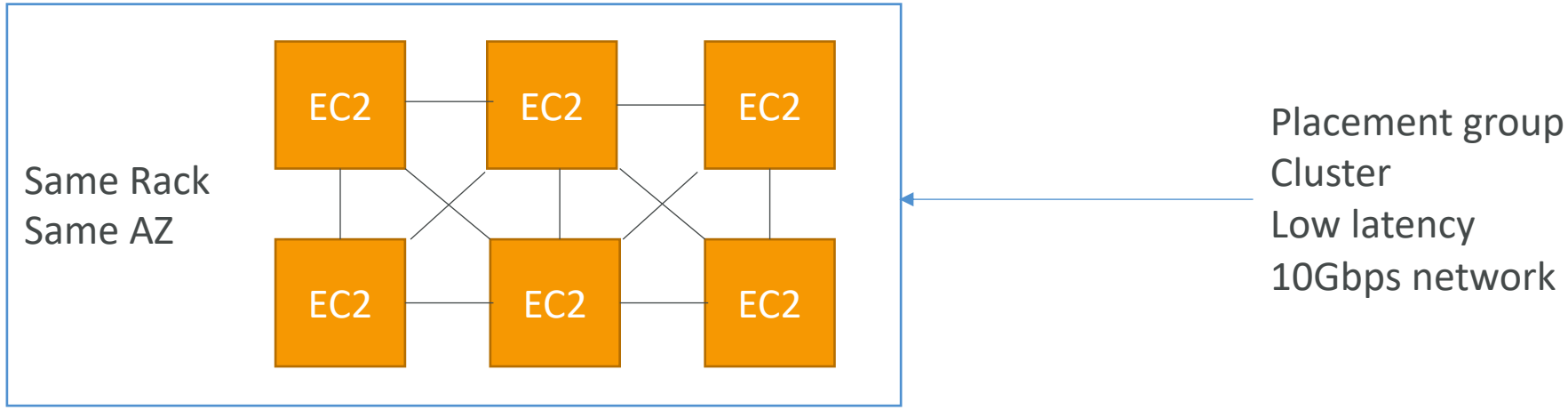
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Compute and Networking

- EC2 Instances:
  - CPU optimized, GPU optimized
  - Spot Instances / Spot Fleets for cost savings + Auto Scaling

- EC2 Placement Groups: **Cluster** for good network performance



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Compute and Networking

- EC2 Enhanced Networking (SR-IOV)
  - Higher bandwidth, higher PPS (packet per second), lower latency
  - Option 1: **Elastic Network Adapter (ENA)** up to 100 Gbps
  - Option 2: Intel 82599 VF up to 10 Gbps – LEGACY

- **Elastic Fabric Adapter (EFA)**
  - Improved ENA for **HPC**, only works for **Linux**
  - Great for inter-node communications, **tightly coupled workloads**
  - Leverages Message Passing Interface (MPI) standard
  - Bypasses the underlying Linux OS to provide low-latency, reliable transport

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Storage

- Instance-attached storage:
  - **EBS:** scale up to 256,000 IOPS with io2 Block Express
  - **Instance Store:** scale to millions of IOPS, linked to EC2 instance, low latency

- Network storage:
  - **Amazon S3:** large blob, not a file system
  - **Amazon EFS:** scale IOPS based on total size, or use provisioned IOPS
  - **Amazon FSx for Lustre:**
    - HPC optimized distributed file system, millions of IOPS
    - Backed by S3

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Automation and Orchestration

- AWS Batch
  - **AWS Batch** supports multi-node parallel jobs, which enables you to run single jobs that span multiple **EC2** instances.
  - Easily schedule jobs and launch EC2 instances accordingly

- AWS ParallelCluster
  - Open source cluster management tool to deploy HPC on AWS
  - Configure with text files
  - Automate creation of VPC, Subnet, cluster type and instance types

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Auto Scaling Groups – Dynamic Scaling Policies

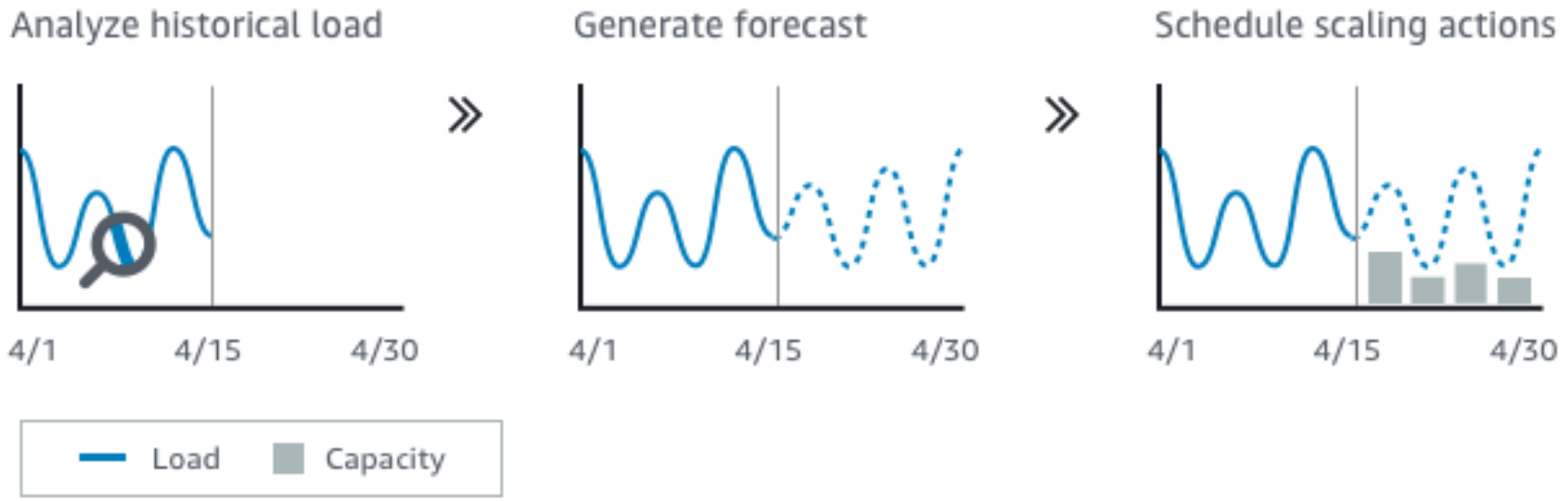
- Target Tracking Scaling
  - Most simple and easy to set-up
  - Example: I want the average ASG CPU to stay at around 40%
- Simple / Step Scaling
  - When a CloudWatch alarm is triggered (example CPU > 70%), then add 2 units
  - When a CloudWatch alarm is triggered (example CPU < 30%), then remove 1
- Scheduled Actions
  - Anticipate a scaling based on known usage patterns
  - Example: increase the min capacity to 10 at 5 pm on Fridays

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Auto Scaling Groups – Predictive Scaling

- **Predictive scaling:** continuously forecast load and schedule scaling ahead

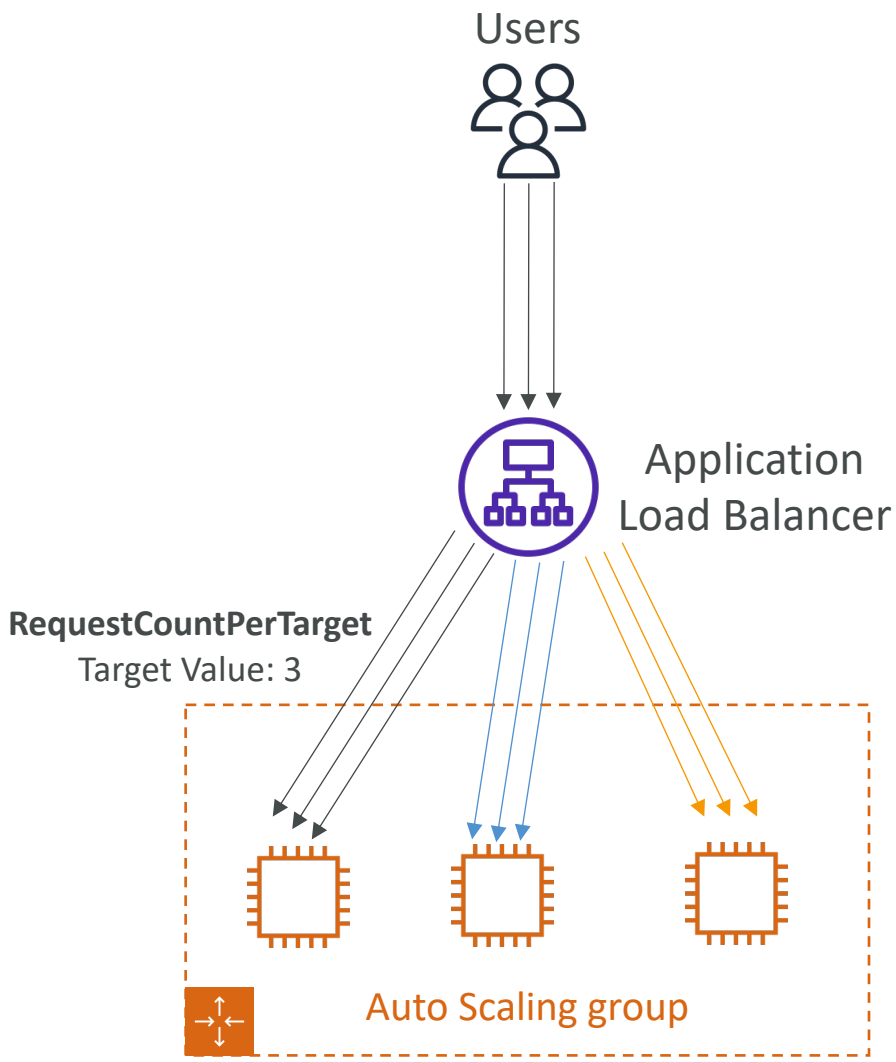


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Good metrics to scale on

- **CPUUtilization**: Average CPU utilization across your instances
- **RequestCountPerTarget**: to make sure the number of requests per EC2 instances is stable
- **Average Network In / Out** (if you're application is network bound)
- **Any custom metric** (that you push using CloudWatch)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Auto Scaling – Good to know

- **Spot Fleet support** (mix on Spot and On-Demand instances)

- **Lifecycle Hooks**:
  - Perform actions before an instance is in service, or before it is terminated
  - Examples: cleanup, log extraction, special health checks

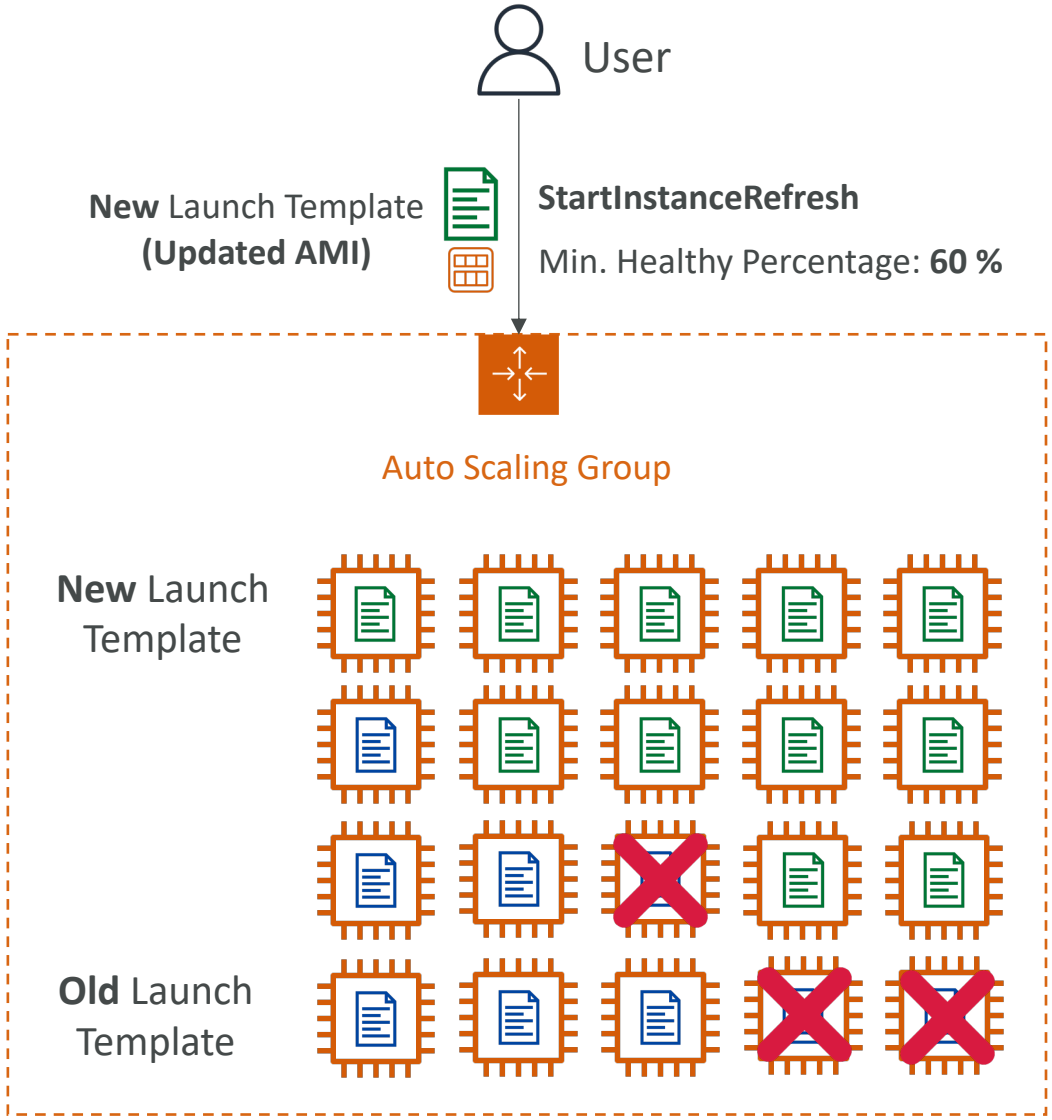
- To upgrade an AMI, must update the **launch configuration / template**
  - Then terminate instances manually (CloudFormation can help)
  - Or use EC2 Instance Refresh for Auto Scaling

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Auto Scaling – Instance Refresh

- Goal: update launch template and then re-creating all EC2 instances
- For this we can use the native feature of Instance Refresh
- Setting of minimum healthy percentage
- Specify warm-up time (how long until the instance is ready to use)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Auto Scaling – Scaling Processes

- **Launch:** Add a new EC2 to the group, increasing the capacity
- **Terminate:** Removes an EC2 instance from the group, decreasing its capacity.
- **HealthCheck:** Checks the health of the instances
- **ReplaceUnhealthy:** Terminate unhealthy instances and re-create them
- **AZRebalance:** Balancer the number of EC2 instances across AZ
- **AlarmNotification:** Accept notification from CloudWatch
- **ScheduledActions:** Performs scheduled actions that you create.
- **AddToLoadBalancer:** Adds instances to the load balancer or target group
- **InstanceRefresh**: Perform an instance refresh

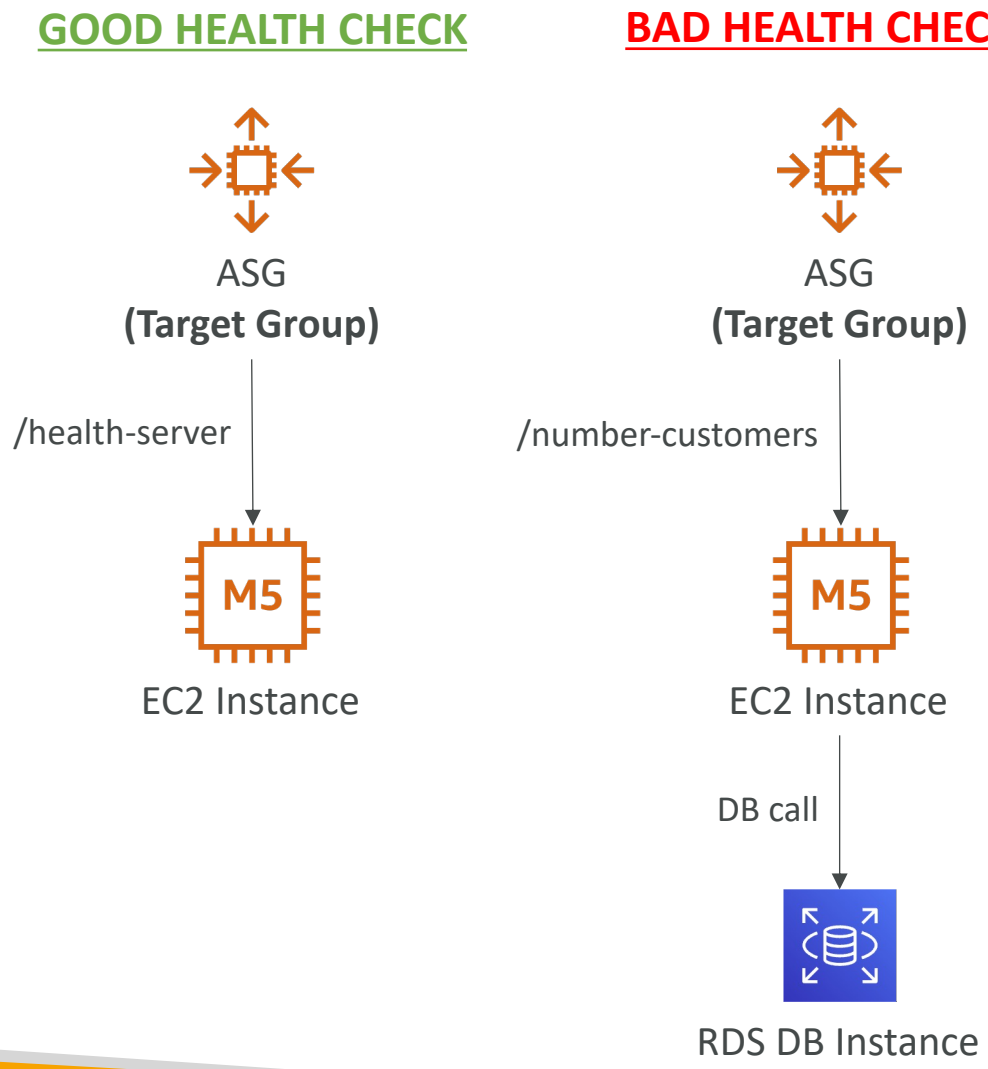
- We can suspend these processes!

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Auto Scaling – Health Checks

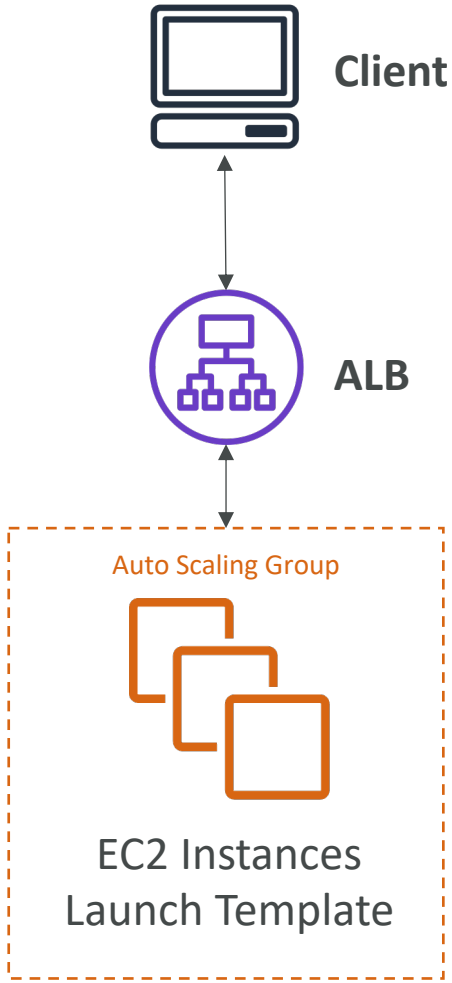
- Health checks available:
  - **EC2 Status Checks**
  - **ELB Health Checks (HTTP)**
  - **Custom Health Checks** – send instance's health to an ASG using AWS CLI or AWS SDK (**set-instance-health**)
- ASG will launch a new instance after terminating an unhealthy one
- Make sure the health check is simple and checks the correct thing



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

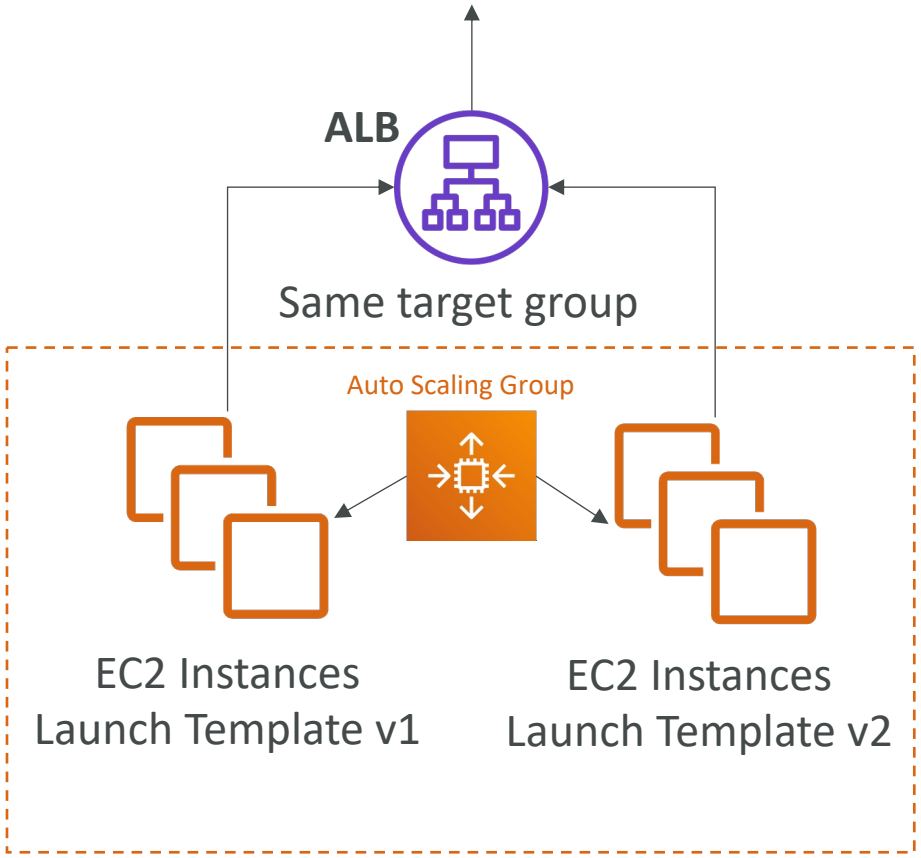
Auto Scaling – Updating an application

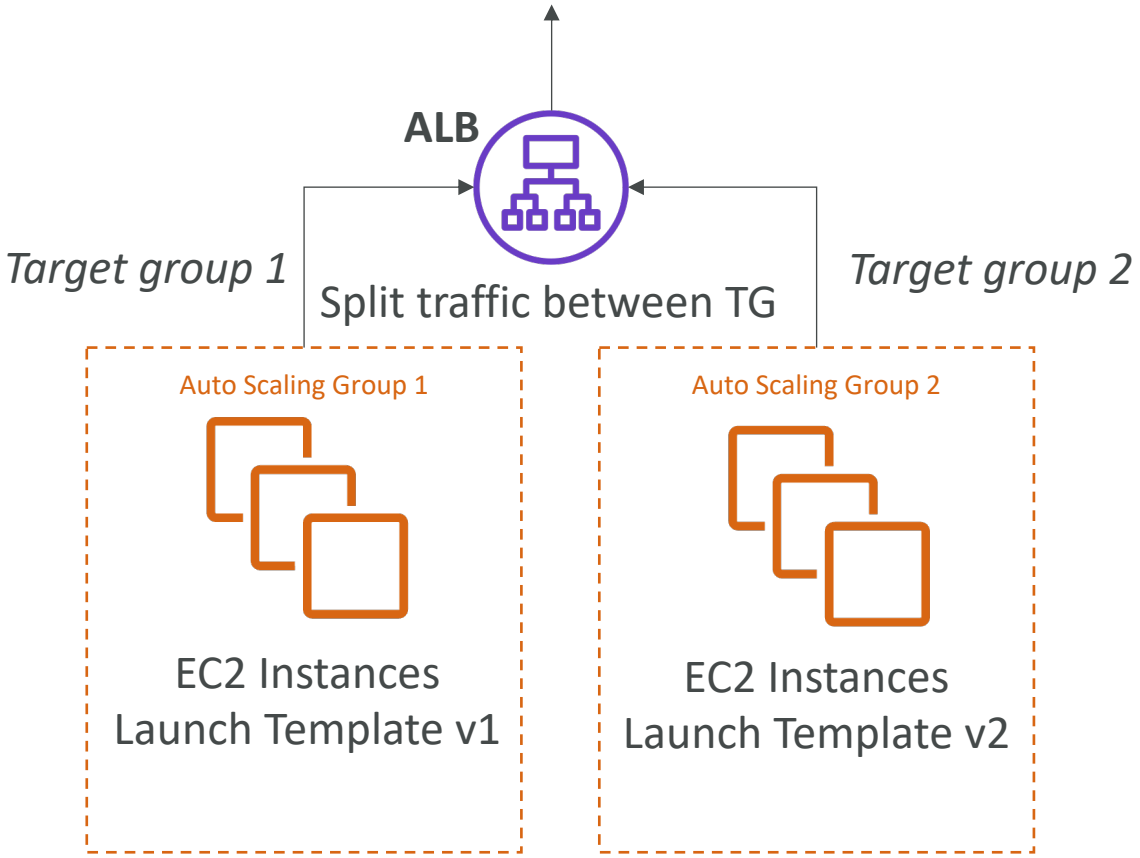


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Auto Scaling – Solution Architecture

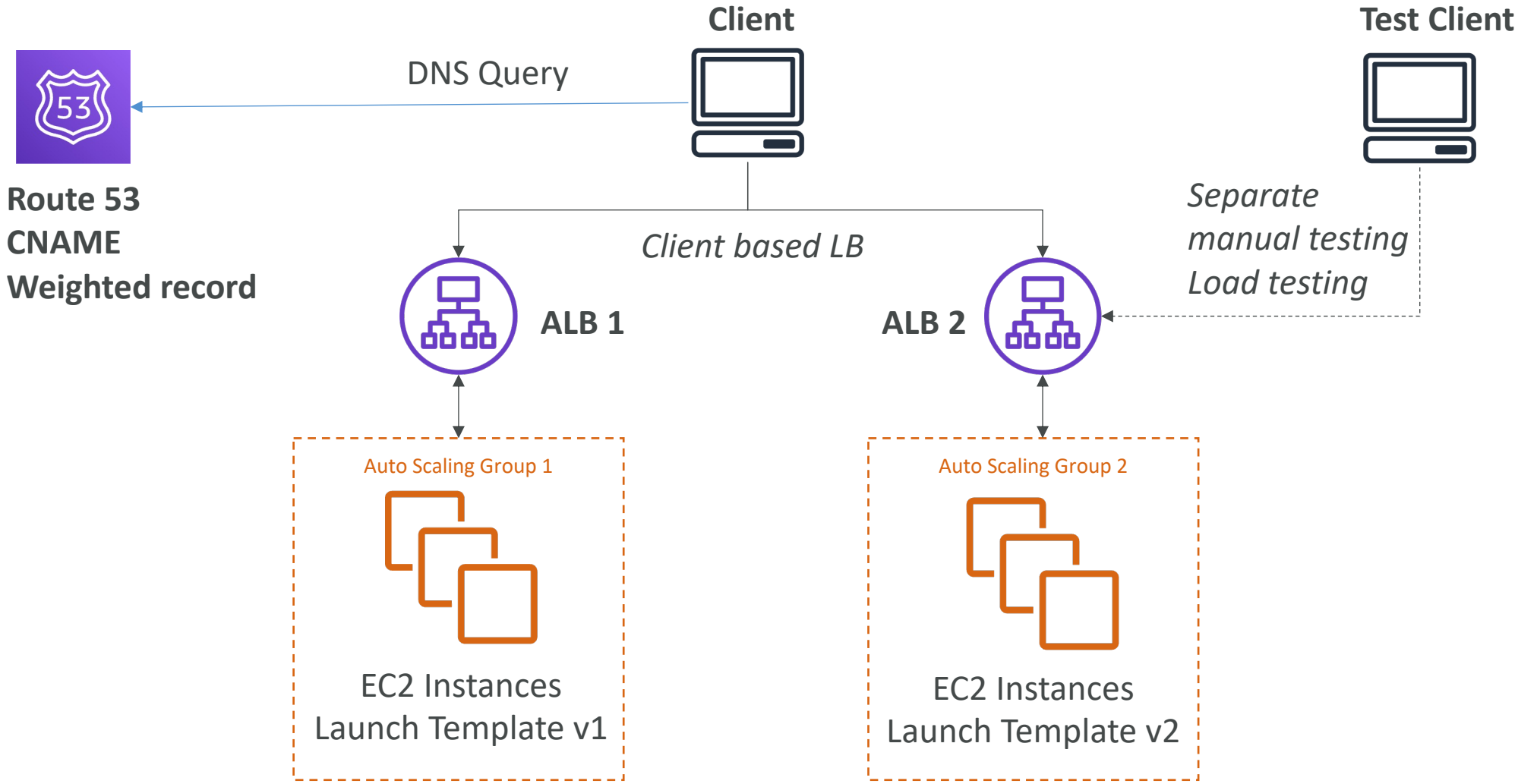




NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Auto Scaling – Solution Architecture



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EC2 Spot Instances

- Can get a discount of up to 90% compared to On-Demand
- Define **max spot price** and get the instance while **current spot price < max**
  - The hourly spot price varies based on offer and capacity
  - If the current spot price > your max price you can choose to **stop** or **terminate** your instance with a 2 minutes grace period.

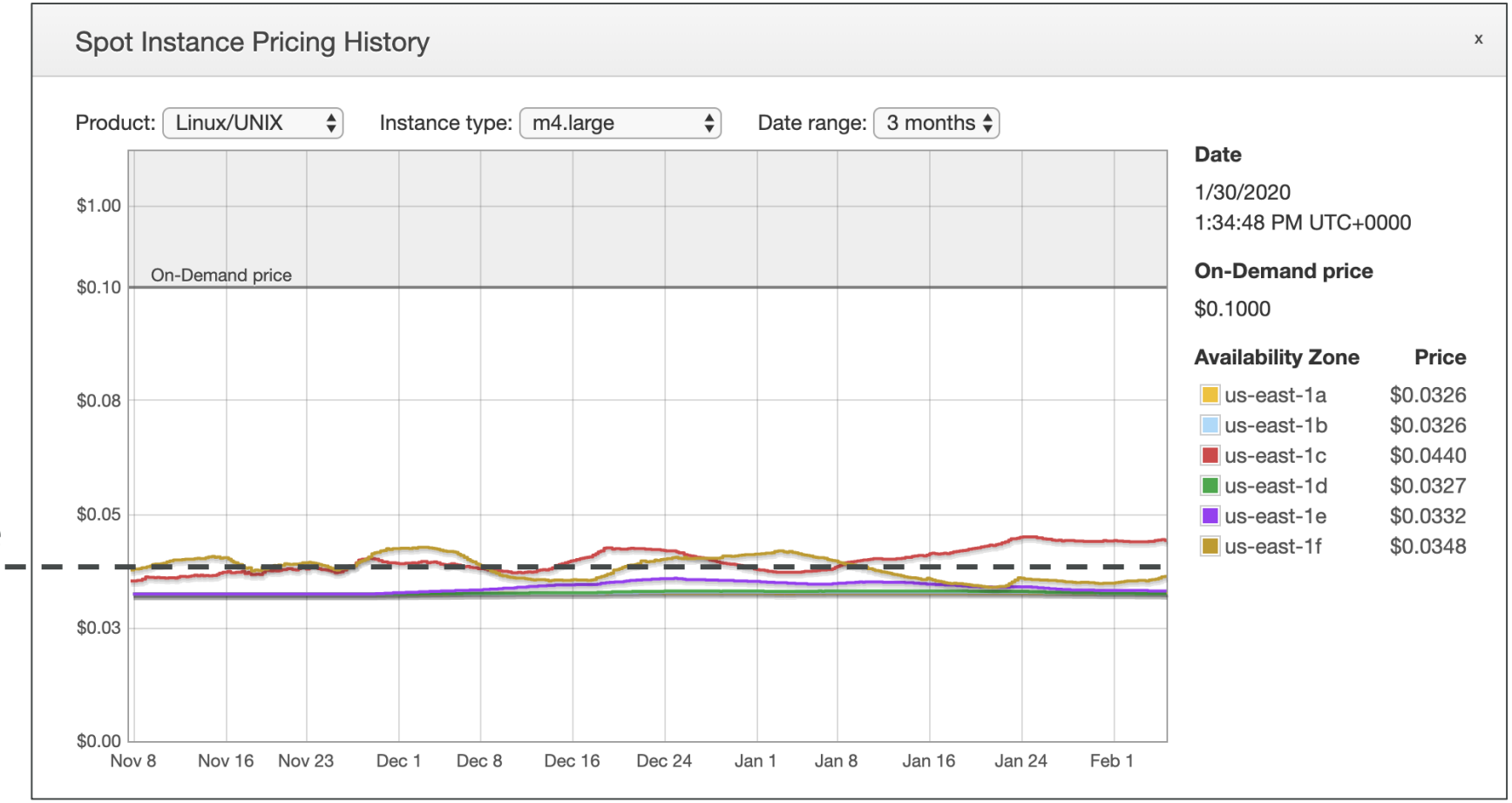
- Used for batch jobs, data analysis, or workloads that are resilient to failures.
- Not great for critical jobs or databases

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EC2 Spot Instances

User-defined max price



https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#SpotInstances:

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

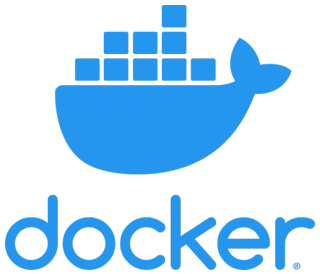
Spot Fleets

- Spot Fleets = set of Spot Instances + (optional) On-Demand Instances
- The Spot Fleet will try to meet the target capacity with price constraints
  - Define possible launch pools: instance type (m5.large), OS, Availability Zone
  - Can have multiple launch pools, so that the fleet can choose
  - Spot Fleet stops launching instances when reaching capacity or max cost
- Strategies to allocate Spot Instances:
  - **lowestPrice:** from the pool with the lowest price (cost optimization, short workload)
  - **diversified:** distributed across all pools (great for availability, long workloads)
  - **capacityOptimized:** pool with the optimal capacity for the number of instances
  - **priceCapacityOptimized (recommended):** pools with highest capacity available, then select the pool with the lowest price (best choice for most workloads)
- Spot Fleets allow us to automatically request Spot Instances with the lowest price

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

What is Docker?



- Docker is a software development platform to deploy apps
- Apps are packaged in **containers** that can be run on any OS
- Apps run the same, regardless of where they're run
  - Any machine (no compatibility issues, predictable behavior)
  - Less work
  - Easier to maintain and deploy
  - Works with any language, any OS, any technology
- Control how much memory / CPU is allocated to your container
- Scale containers up and down very quickly (seconds)
- More efficient than Virtual machines

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Docker Containers Management on AWS

- To manage containers, we need a container management platform

- Amazon Elastic Container Service (Amazon ECS)
  - Amazon's own container platform

Amazon ECS

- Amazon Elastic Kubernetes Service (Amazon EKS)
  - Amazon's managed Kubernetes (open source)

Amazon EKS

- AWS Fargate
  - Amazon's own Serverless container platform
  - Works with ECS and with EKS

AWS Fargate

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECS – Use cases

- Run Microservices
  - Run multiple Docker containers on the same machine
  - Easy Service Discovery features to enhance communication
  - Direct integration with Application Load Balancer and Network Load Balancer
  - Auto Scaling capability
- Run Batch Processing / Scheduled Tasks
  - Schedule ECS tasks to run on On-demand / Reserved / Spot instances
- Migrate Applications to the Cloud
  - Dockerize legacy applications running on-premises
  - Move Docker containers to run on Amazon ECS

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

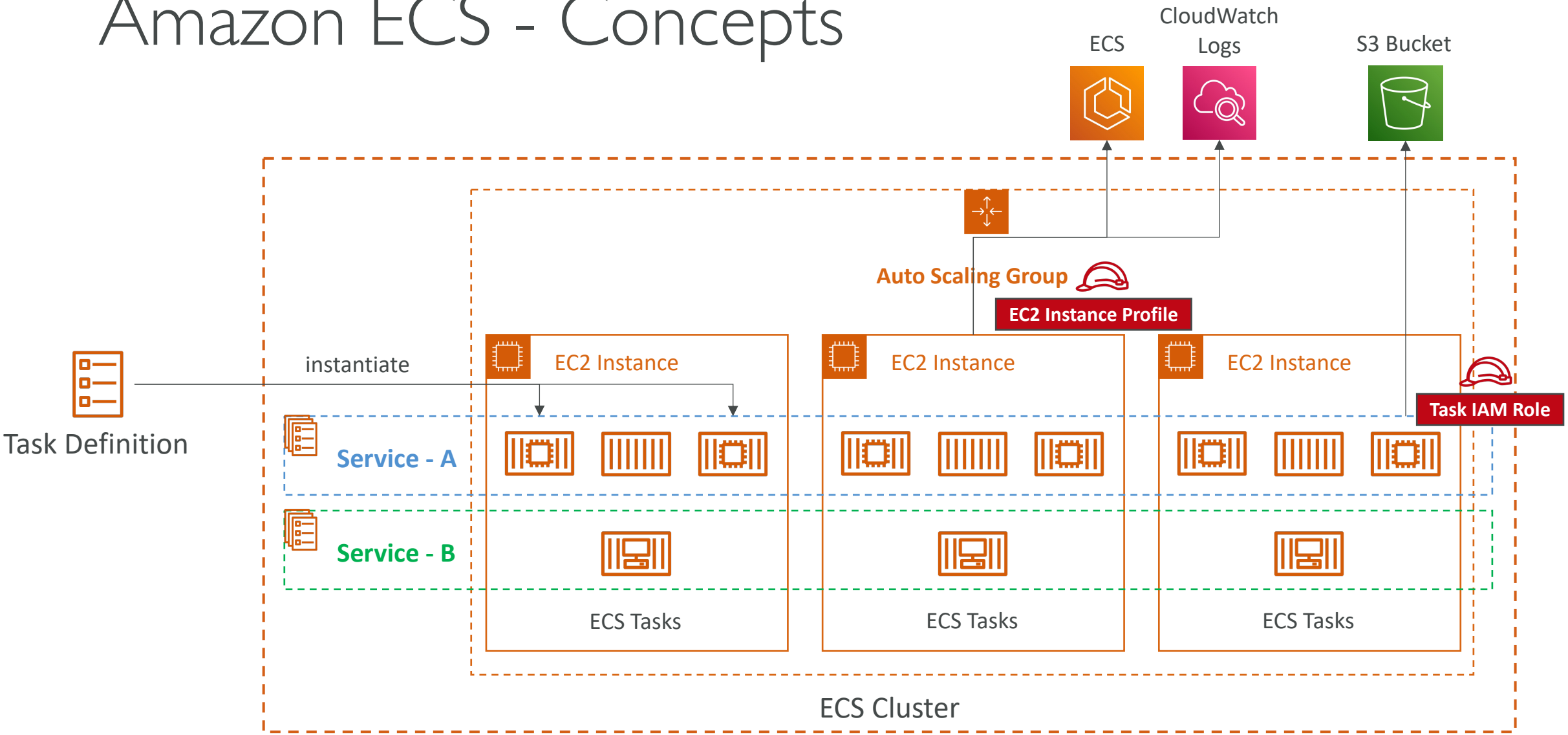
Amazon ECS – Concepts

- **ECS Cluster** – logical grouping of EC2 instances
- **ECS Service** – defines how many tasks should run and how they should be run
- **Task Definitions** – metadata in **JSON form** to tell ECS how to run a Docker container (image name, CPU, RAM, …)
- **ECS Task** – an instance of a Task Definition, a running Docker container(s)
- **ECS IAM Roles**
  - **EC2 Instance Profile** – used by the EC2 instance (e.g., make API calls to ECS, send logs, …)
  - **ECS Task IAM Role** – allow each task to have a specific role (e.g., make API calls to S3, DynamoDB, …)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECS - Concepts

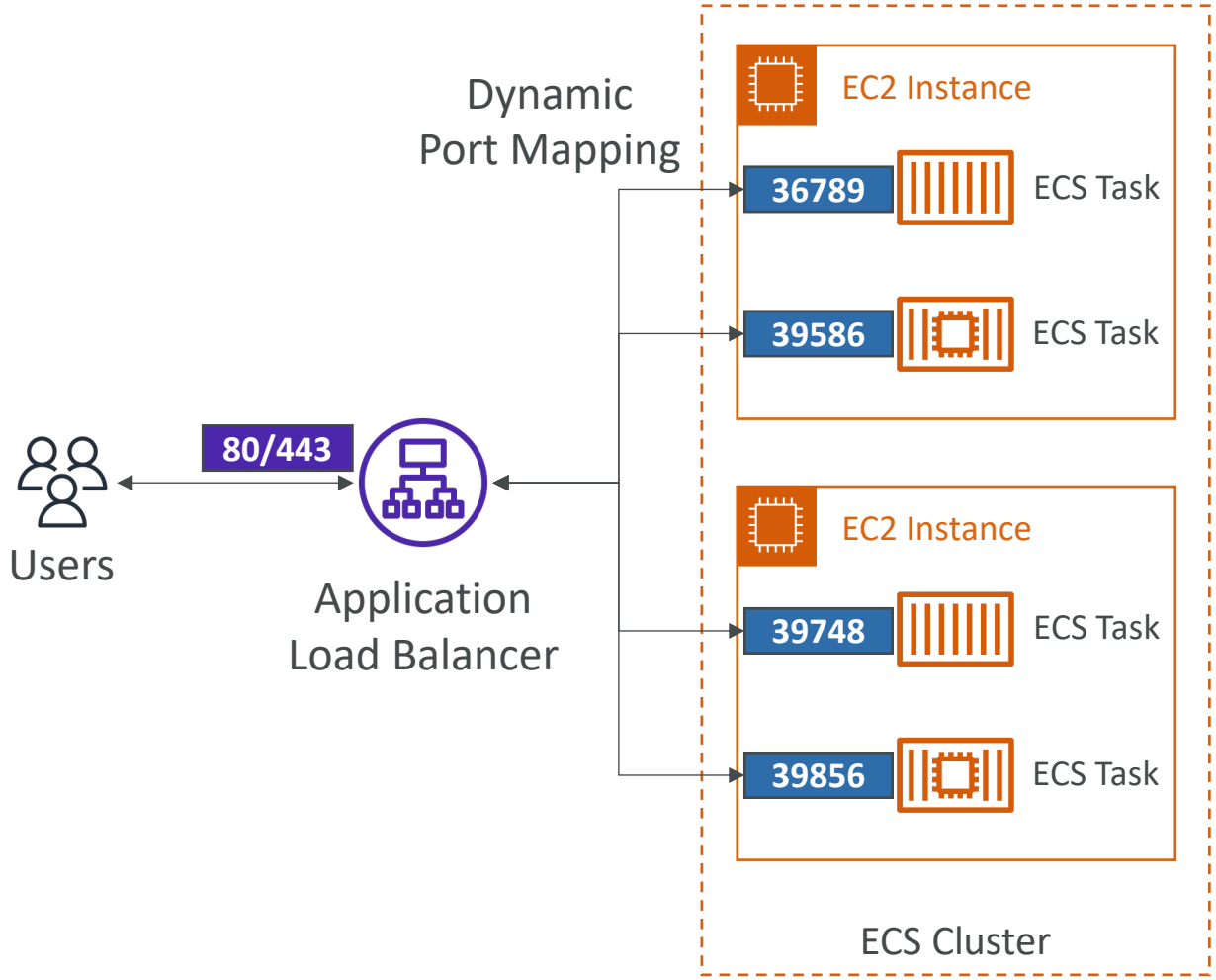


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECS – ALB Integration

- We get **Dynamic Port Mapping**
- Allows you to run multiple instances of the same application on the same EC2 instance
- The ALB finds the right port on your EC2 Instances
- Use cases:
  - Increased resiliency even if running on one EC2 instance
  - Maximize utilization of CPU / cores
  - Ability to perform rolling upgrades without impacting app uptime

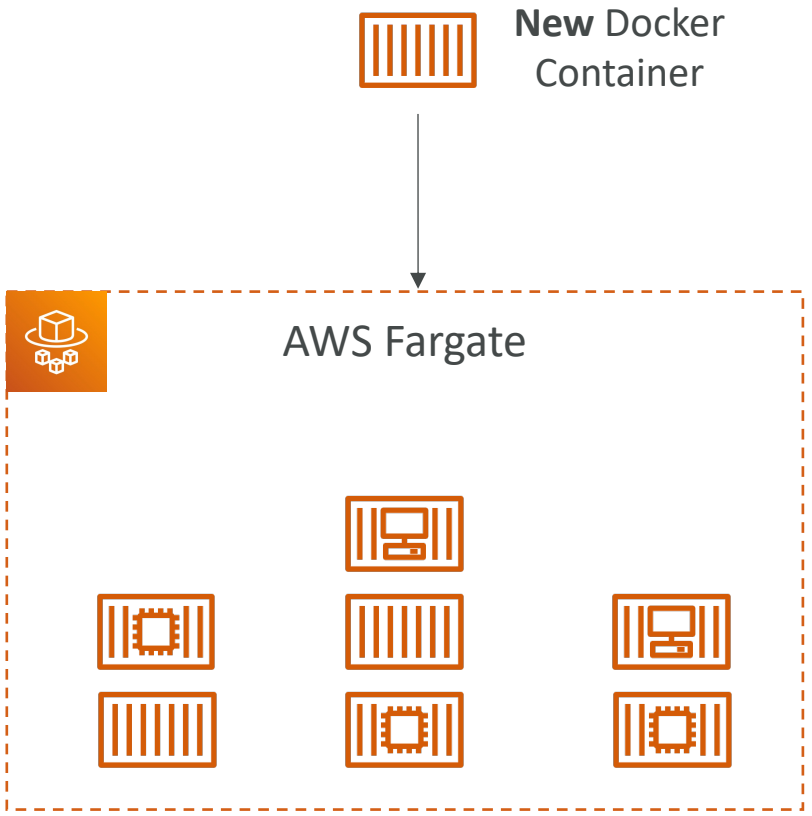


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

AWS Fargate

- Launch Docker containers on AWS
- **You do not provision the infrastructure (no EC2 instances to manage)**
- **It's all serverless!**
- You create task definitions
- AWS runs containers for you based on the CPU / RAM you need
- To scale, just increase the number of tasks. Simple! No more EC2 instances ☺



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECS – Security & Networking

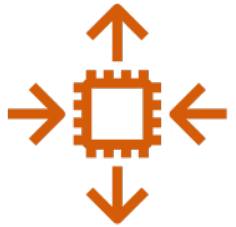
- You can inject secrets and configurations as Environment Variables into running Docker containers
  - Integration with **SSM Parameter Store** and **Secrets Manager**

- ECS Tasks Networking
  - **none** – no network connectivity, no port mappings
  - **bridge** – uses Docker's virtual container-based network
  - **host** – bypass Docker's network, uses the underlying host network interface
  - **awsvpc**
    - Every tasks launched on the instance gets its own ENI and a private IP address
    - Simplified networking, enhanced security, Security Groups, monitoring, VPC Flow Logs
    - Default mode for Fargate tasks

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECS – Service Auto Scaling



- Automatically increase/decrease the desired number of tasks
- Amazon ECS leverages **AWS Application Auto Scaling**
- CPU and RAM is tracked in CloudWatch at the ECS Service level

- **Target Tracking** – scale based on target value for a specific CloudWatch metric
- **Step Scaling** – scale based on a specified CloudWatch Alarm
- **Scheduled Scaling** – scale based on a specified date/time (predictable changes)

- ECS Service Auto Scaling (task level) ≠ EC2 Auto Scaling (EC2 instance level)
- Fargate Auto Scaling is much easier to setup (because **Serverless**)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECS – Spot Instances

- ECS Classic (EC2 Launch Type)
  - Can have the underlying EC2 instances as Spot Instances (managed by an ASG)
  - Instances may go into draining mode to remove running tasks
  - Good for cost savings, but will impact reliability

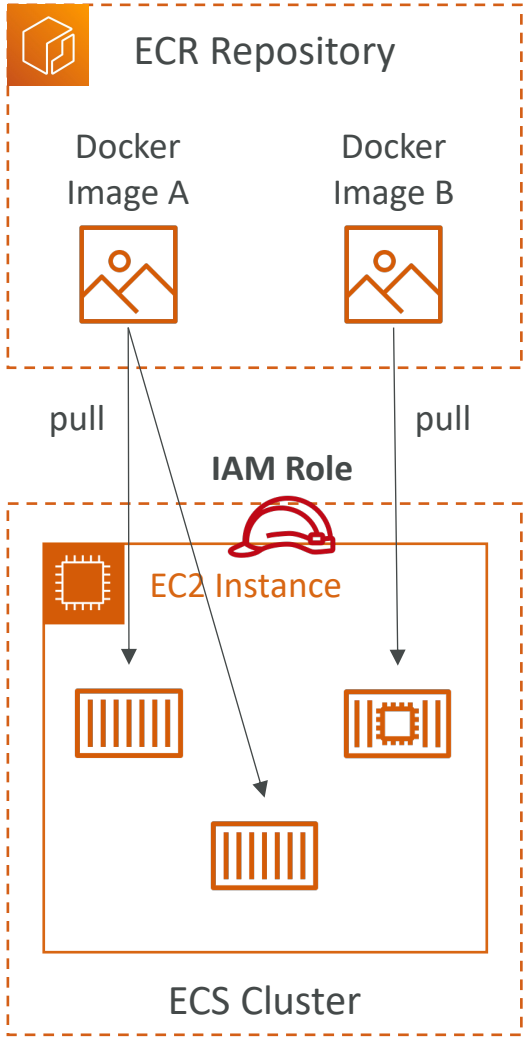
- AWS Fargate
  - Specify minimum of tasks for on-demand baseline workload
  - Add tasks running on FARGATE_SPOT for cost-savings (can be reclaimed by AWS)
  - Regardless of On-demand or Spot, Fargate scales well based on load

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECR - Elastic Container Registry

- Store and manage Docker images on AWS
- **Private** and **Public** repository (**Amazon ECR Public Gallery** https://gallery.ecr.aws)
- Fully integrated with ECS
- Access is controlled through IAM (permission errors => check policy)
- Supports image vulnerability scanning, versioning, image tags, image lifecycle, …

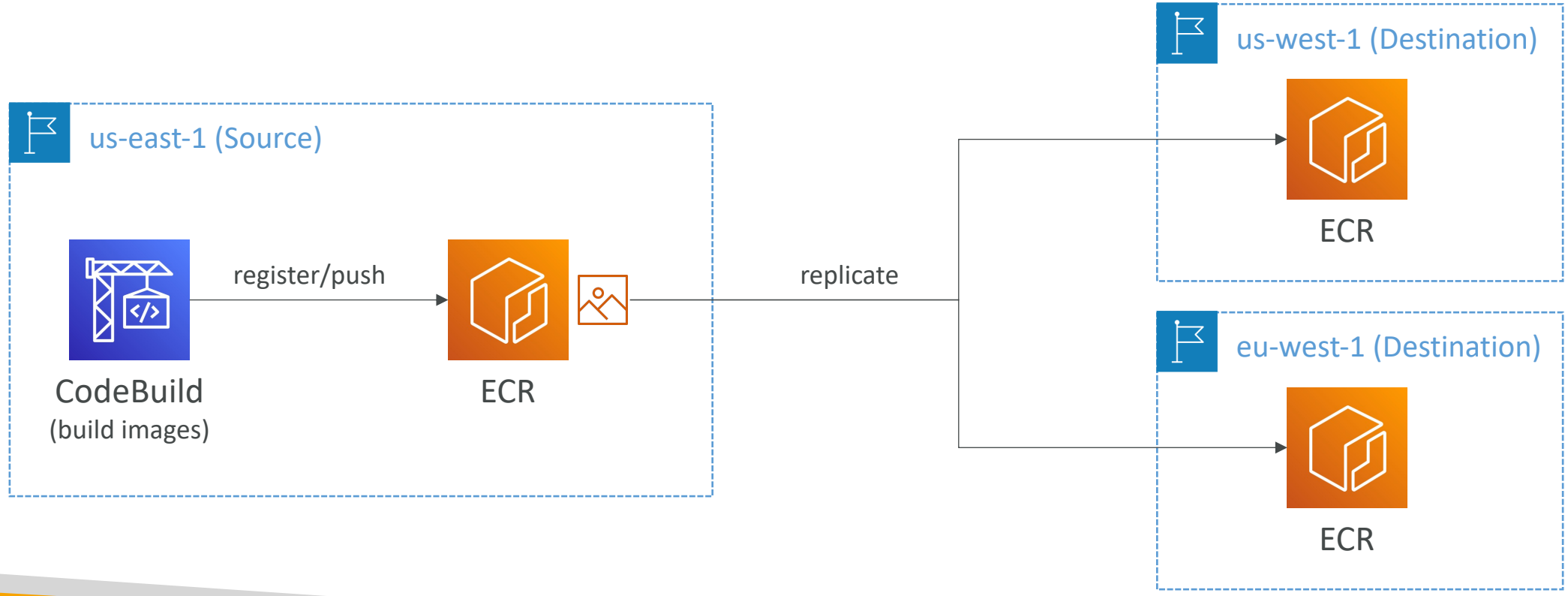


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECR – Cross Region Replication

- ECR private registry supports both **cross-Region** and **cross-account** replication

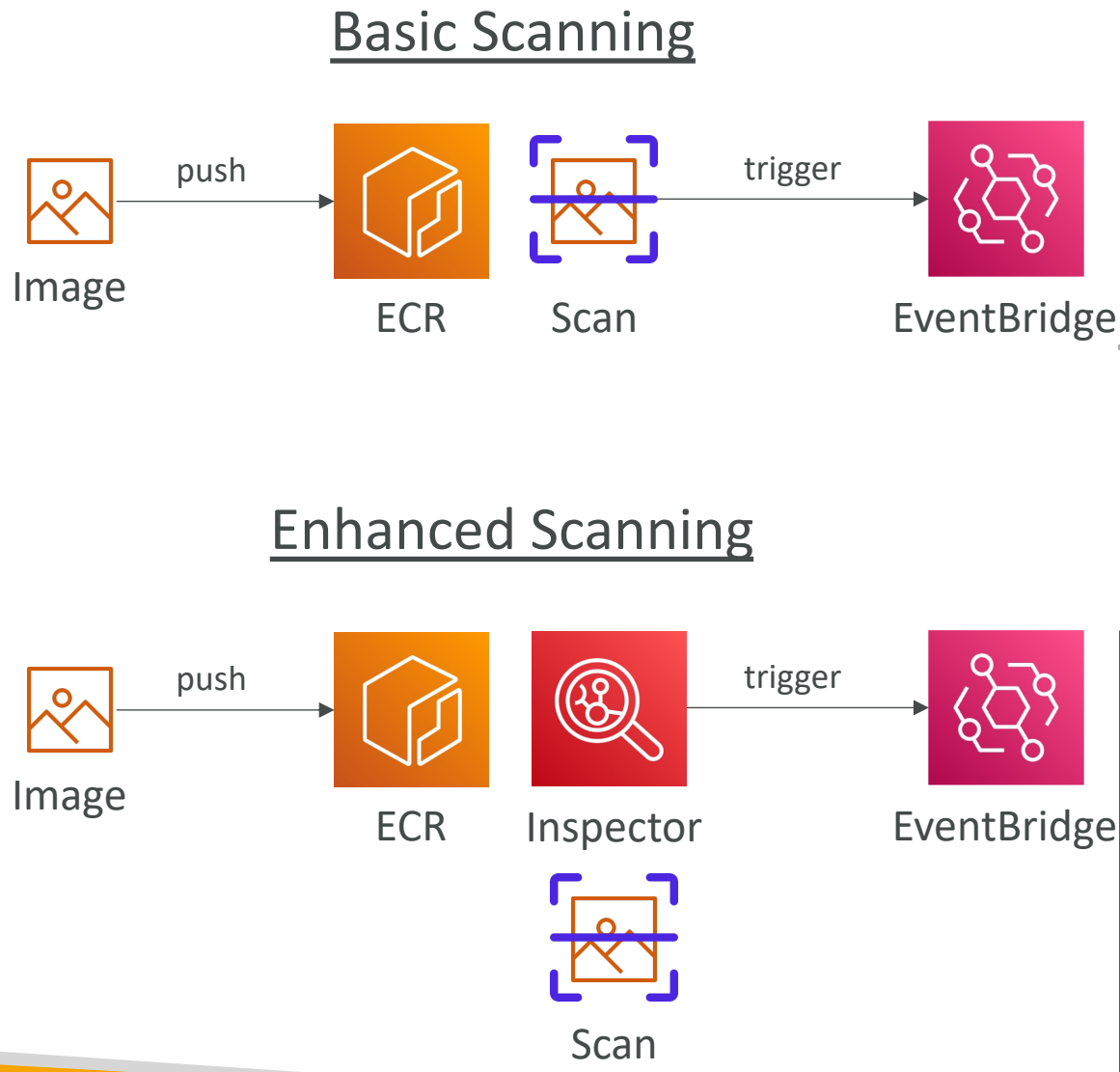


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECR – Image Scanning

- Manual Scan or Scan on Push
- Basic Scanning – Common CVE
- Enhanced Scanning – Leverages Amazon Inspector (OS & Programming Language vulnerabilities)
- Scan results can be retrieved from within the AWS console



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

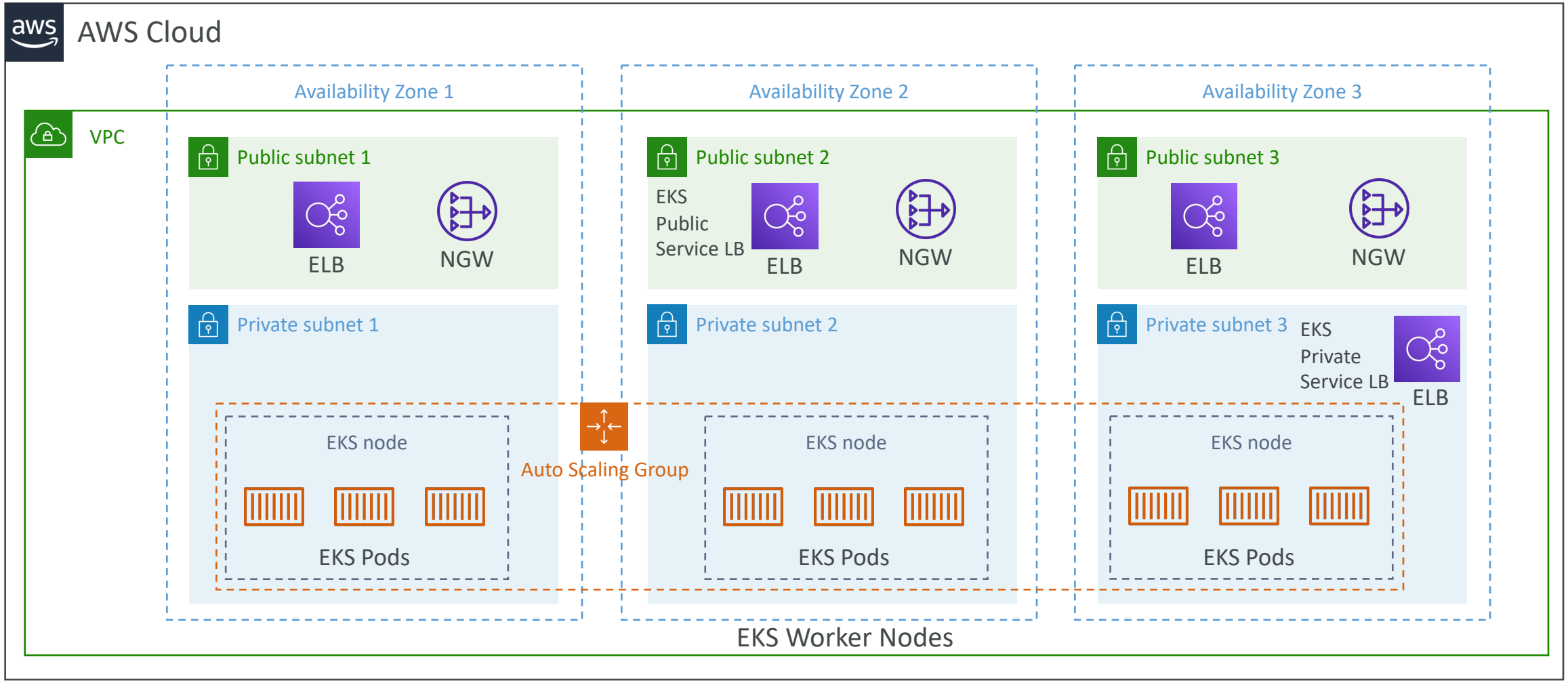
Amazon EKS Overview

- Amazon EKS = Amazon Elastic **Kubernetes** Service
- It is a way to launch **managed Kubernetes clusters on AWS**
- Kubernetes is an **open-source system** for automatic deployment, scaling and management of containerized (usually Docker) application
- It's an alternative to ECS, similar goal but different API
- EKS supports **EC2** if you want to deploy worker nodes or **Fargate** to deploy serverless containers
- **Use case:** if your company is already using Kubernetes on-premises or in another cloud, and wants to migrate to AWS using Kubernetes
- **Kubernetes is cloud-agnostic** (can be used in any cloud – Azure, GCP…)
- For multiple regions, deploy one EKS cluster per region
- Collect logs and metrics using **CloudWatch Container Insights**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon EKS - Diagram



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon EKS – Node Types

- Managed Node Groups
  - Creates and manages Nodes (EC2 instances) for you
  - Nodes are part of an ASG managed by EKS
  - Supports On-Demand or Spot Instances
- Self-Managed Nodes
  - Nodes created by you and registered to the EKS cluster and managed by an ASG
  - You can use prebuilt AMI - Amazon EKS Optimized AMI
  - Supports On-Demand or Spot Instances
- AWS Fargate
  - No maintenance required; no nodes managed

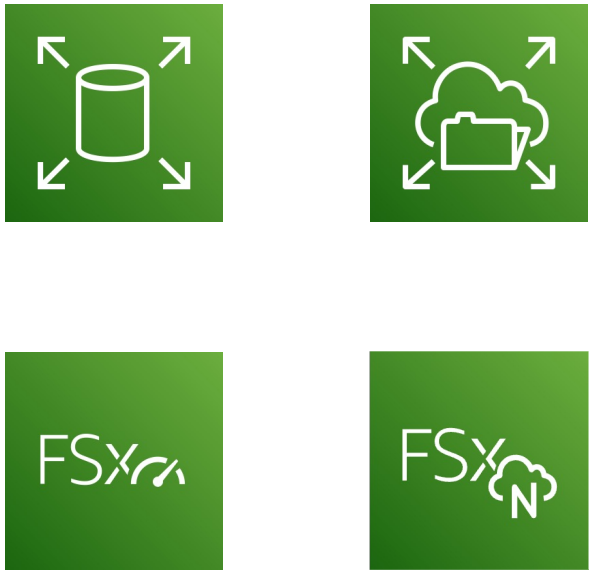
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon EKS – Data Volumes

- Need to specify **StorageClass** manifest on your EKS cluster
- Leverages a **Container Storage Interface (CSI)** compliant driver

- Support for…
- Amazon EBS
- Amazon EFS (works with Fargate)
- Amazon FSx for Lustre
- Amazon FSx for NetApp ONTAP

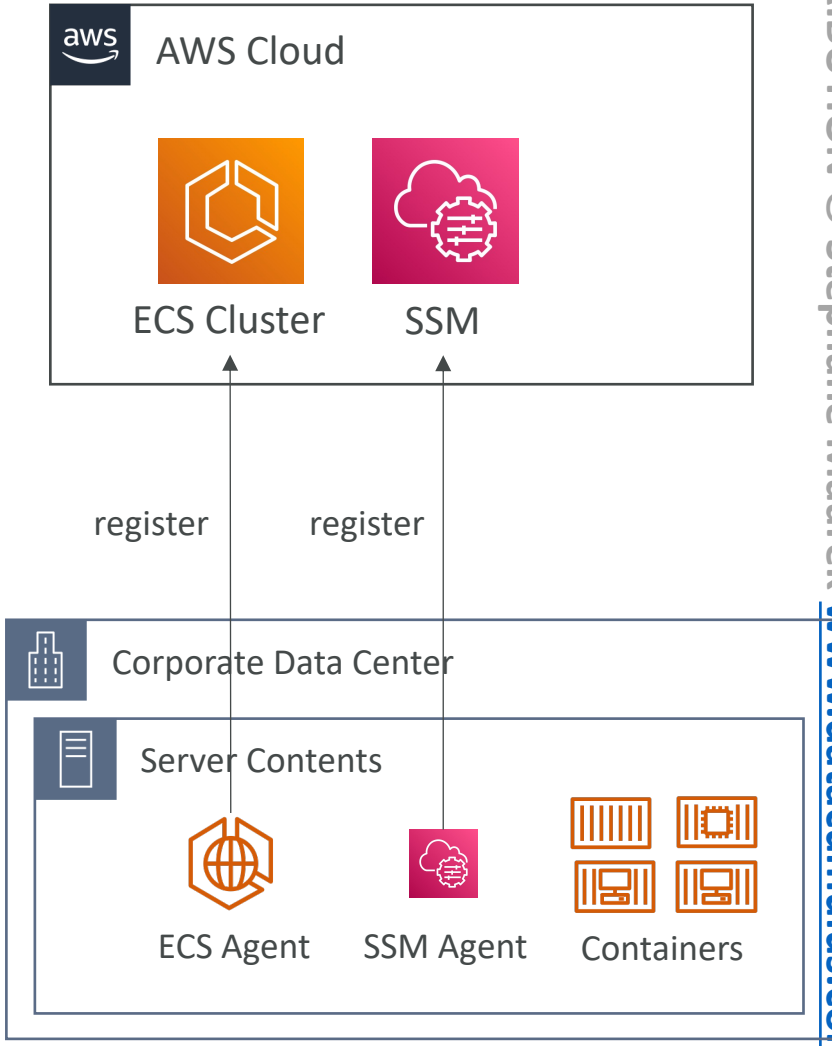


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECS Anywhere

- Easily run containers on Customer-managed infrastructure (on-premises, VMs, …)
- Allows customers to deploy native Amazon ECS tasks in any environment
- Fully-managed Amazon ECS Control Plane
- **ECS Container Agent** and **SSM Agent** needs to be installed
- **"EXTERNAL"** Launch Type
- Must have a stable connection to the AWS Region
- Use cases:
  - Meet compliance, regulatory, and latency requirements
  - Run apps outside AWS Regions and closer to their other services
  - On-premises ML, video processing, data processing, …



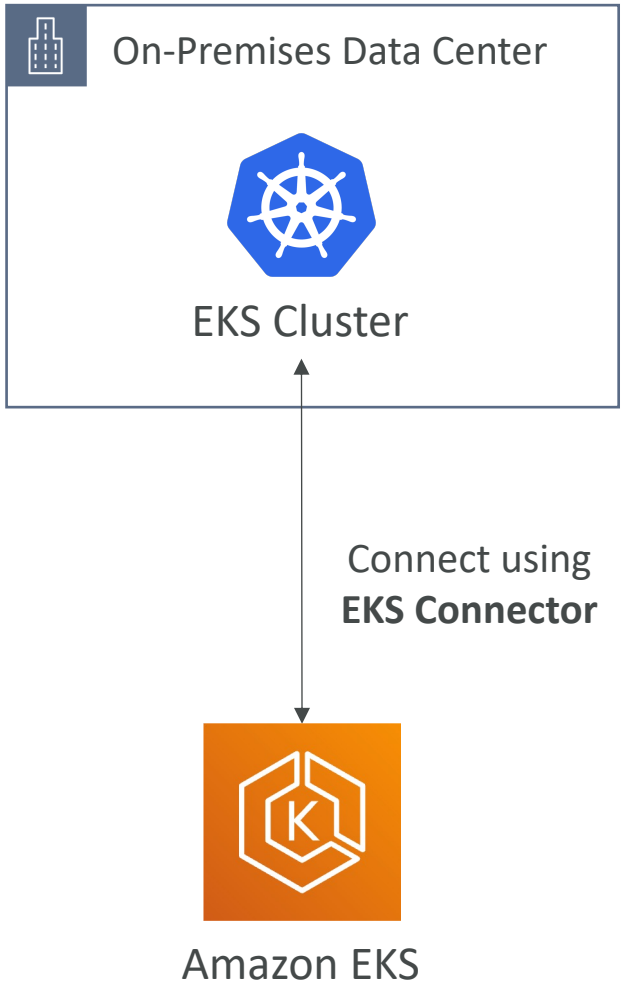
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Amazon EKS Anywhere

- Create and operate Kubernetes clusters created outside AWS
- Leverage the **Amazon EKS Distro** (AWS' bundled release of Kubernetes)
- Reduce support costs and avoid maintaing redundant 3rd party tools

- Install using the **EKS Anywhere Installer**
- Optionally use the **EKS Connector** to connect the EKS Anywhere clusters to AWS
  - Fully Connected & Partially Disconnected: you can connect to Amazon EKS Anywhere clusters to AWS, and leverage the EKS console
  - Fully Disconnected: must install the EKS Distro and leverage open-source tools to manage your clusters



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Lambda Integrations
Main ones

API Gateway

Kinesis

DynamoDB

AWS S3 –
Simple Storage Service

AWS IoT
Internet of Things

Amazon EventBridge

CloudWatch Logs

AWS SNS

AWS Cognito

Amazon
SQS

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Example: Serverless Thumbnail creation

trigger

push

New thumbnail in S3

push

Image name
Image size
Creation date
etc…

New image in S3

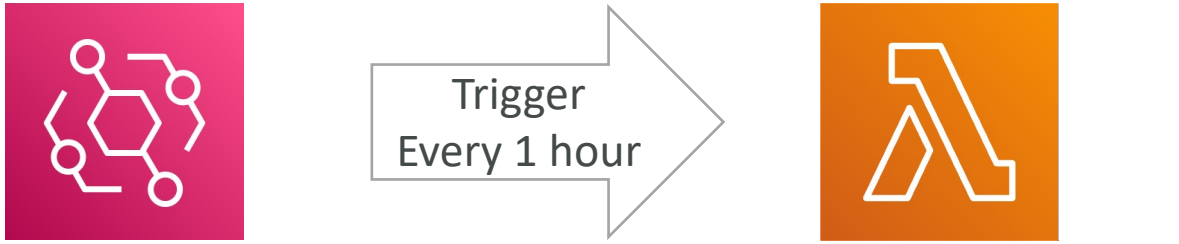
AWS Lambda Function Creates a Thumbnail

Metadata in DynamoDB

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Example: Serverless CRON Job



Amazon EventBridge

AWS Lambda Function
Perform a task

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Lambda Language Support (runtimes)

- Node.js (JavaScript)
- Python
- Java
- C# (.NET Core) / Powershell
- Ruby
- Custom Runtime API (community supported, example Rust or Golang)

- Lambda Container Image
  - The container image must implement the Lambda Runtime API
  - ECS / Fargate is preferred for running arbitrary Docker images

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Lambda – Limits to know

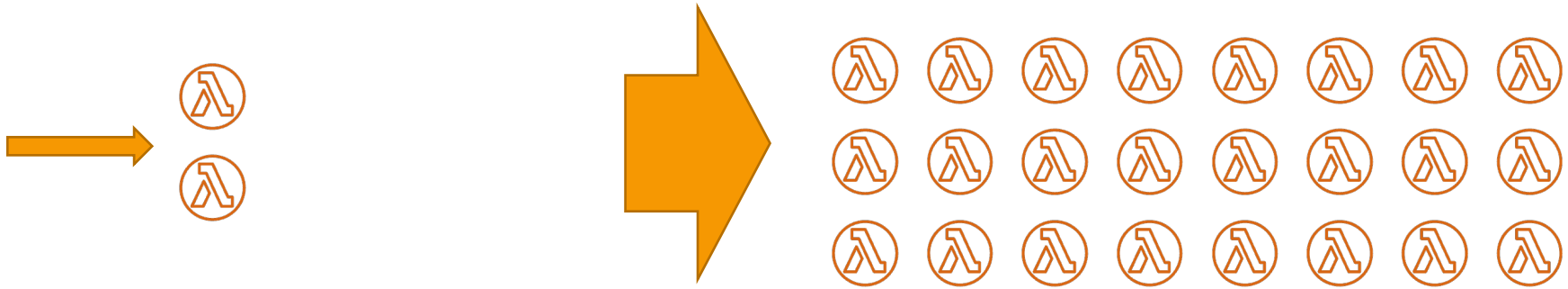
- **RAM** – 128 MB to 10,240 MB (10 GB)
- **CPU** – is linked to RAM (cannot be set manually)
  - 2 vCPUs are allocated at 1,769 MB of RAM
  - 6 vCPUs are allocated at 10,240 MB of RAM
- **Timeout** – up to 15 minutes
- **/tmp Storage** – 10,240 MB
- **Deployment Package** – 50 MB (zipped) , 250 MB (unzipped) including layers
- **Concurrent Executions** – 1000 (soft limit that can be increased)
- **Container Image Size** – 10 GB
- **Invocation Payload (request/response)** – 6 MB (sync), 256 KB (async)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Lambda Concurrency and Throttling

- Concurrency limit: up to 1000 concurrent executions



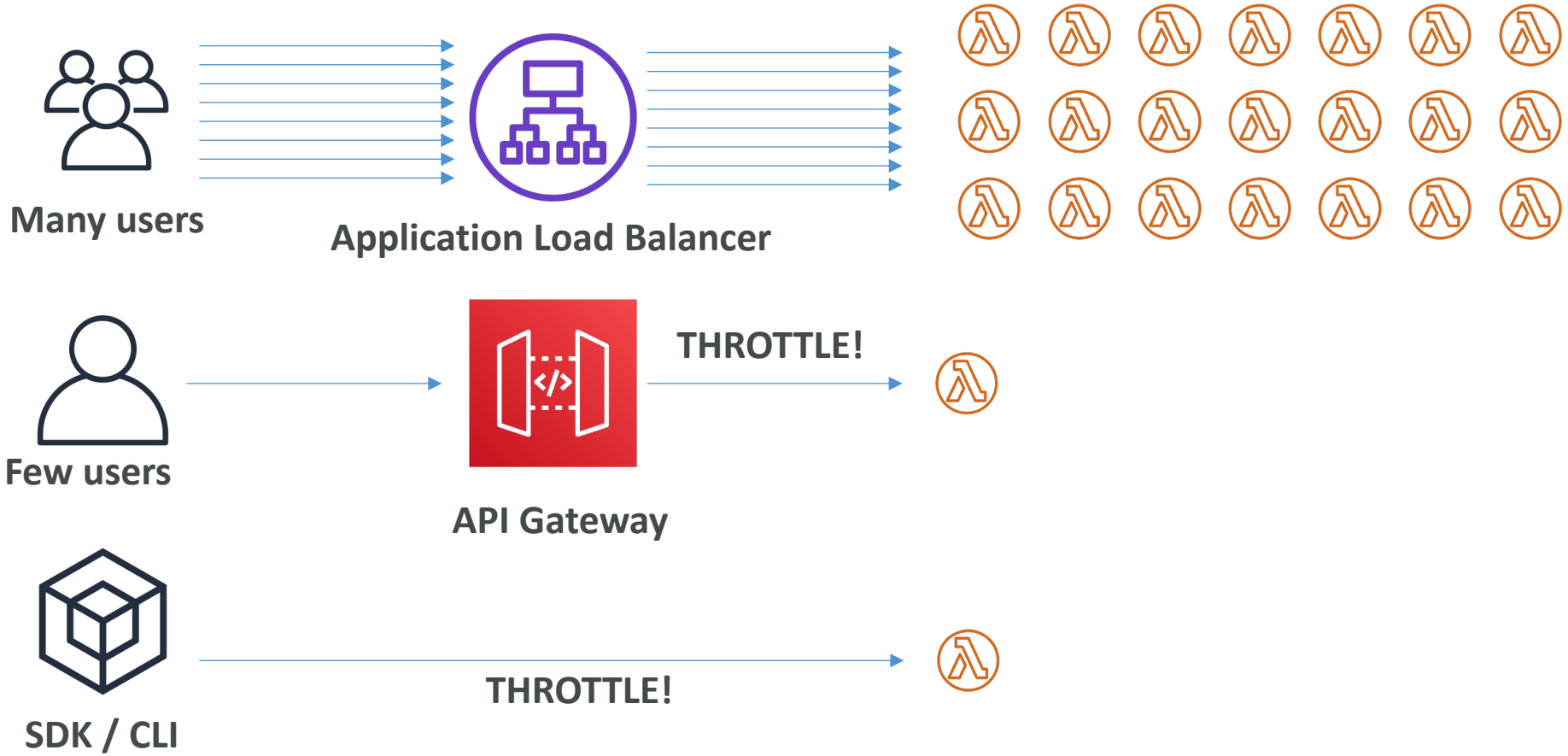
- Can set a "**reserved concurrency**" at the function level (=limit)
- Each invocation over the concurrency limit will trigger a "Throttle"
- Can request a quota increase in AWS Service Quotas

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Lambda Concurrency Issue

- If you don't reserve (=limit) concurrency, the following can happen:

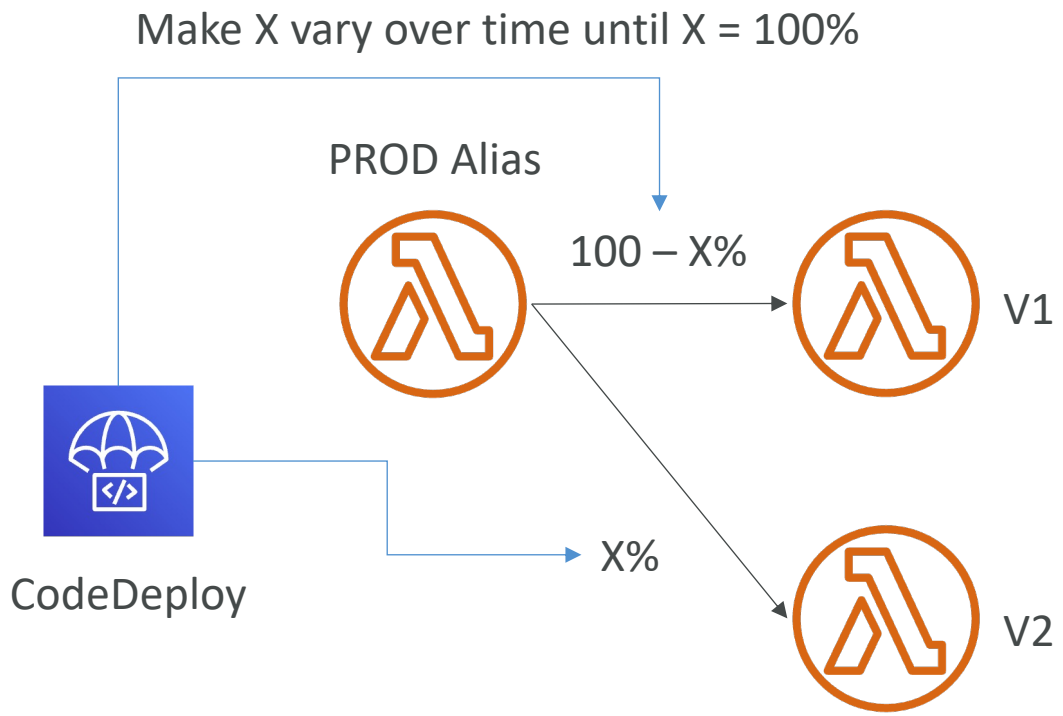


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Lambda & CodeDeploy

- **CodeDeploy** can help you automate traffic shift for Lambda aliases
- Feature is integrated within the SAM framework
- **Linear:** grow traffic every N minutes until 100%
  - Linear10PercentEvery3Minutes
  - Linear10PercentEvery10Minutes
- **Canary:** try X percent then 100%
  - Canary10Percent5Minutes
  - Canary10Percent30Minutes
- **AllAtOnce:** immediate
- Can create Pre & Post Traffic hooks to check the health of the Lambda function



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

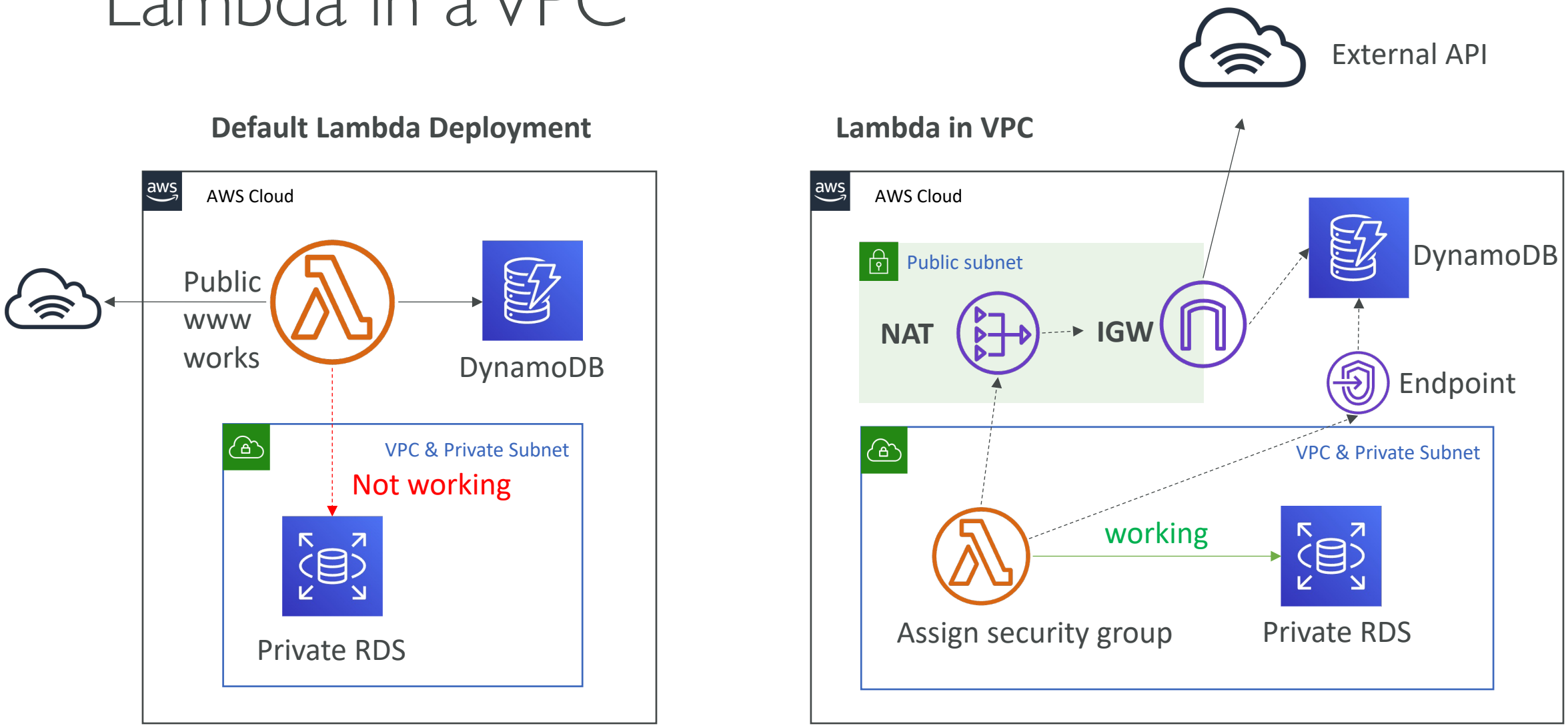
AWS Lambda Logging, Monitoring and Tracing

- CloudWatch:
  - AWS Lambda execution logs are stored in AWS CloudWatch Logs
  - AWS Lambda metrics are displayed in AWS CloudWatch Metrics (successful invocations, error rates, latency, timeouts, etc…)
  - **Make sure your AWS Lambda function has an execution role with an IAM policy that authorizes writes to CloudWatch Logs**
- X-Ray:
  - It's possible to trace Lambda with X-Ray
  - Enable in Lambda configuration (runs the X-Ray daemon for you)
  - Use AWS SDK in Code
  - **Ensure Lambda Function has correct IAM Execution Role**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Lambda in a VPC

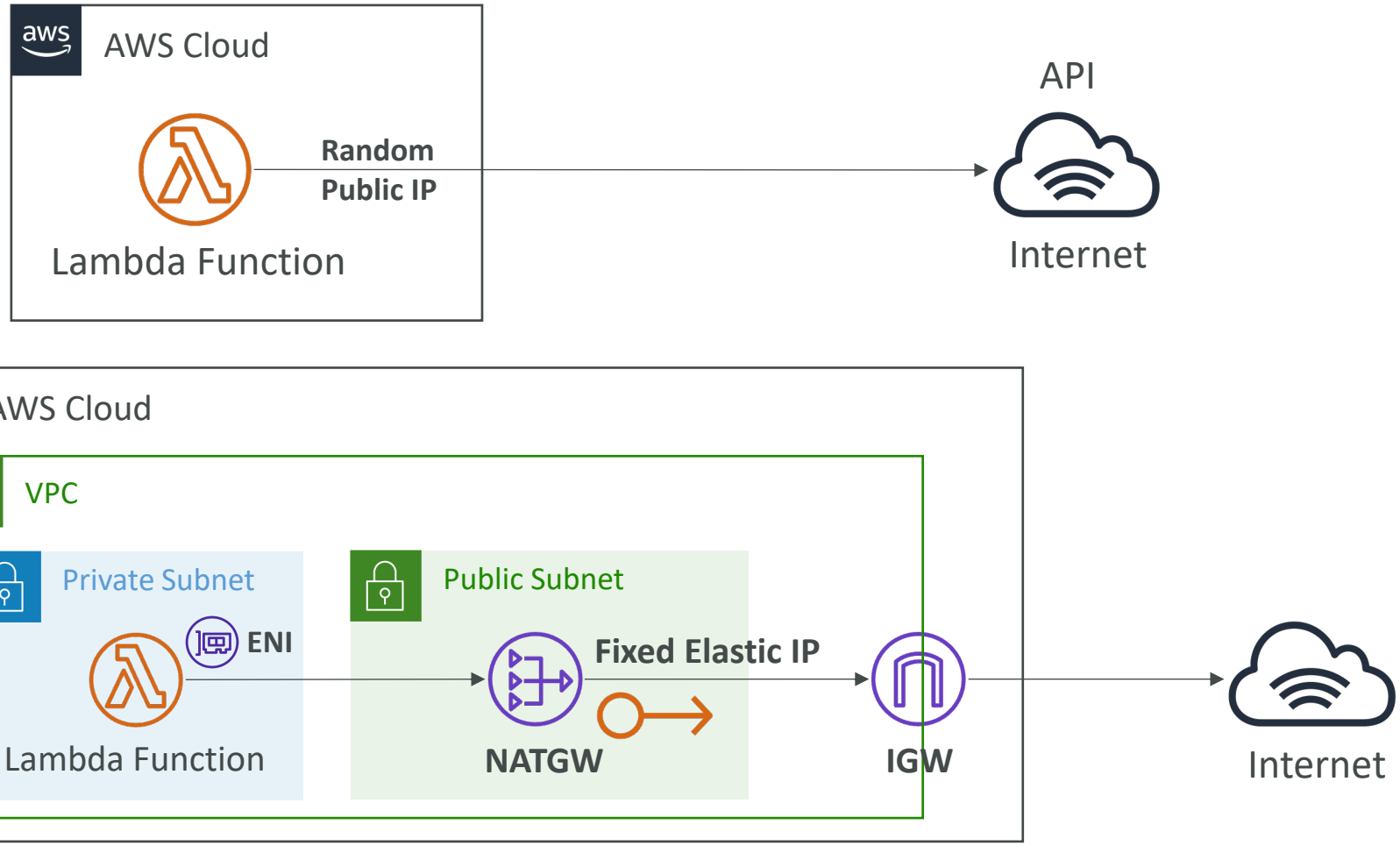


Note: Lambda - CloudWatch Logs works even without endpoint or NAT Gateway

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Lambda – Fixed Public IP for external comms

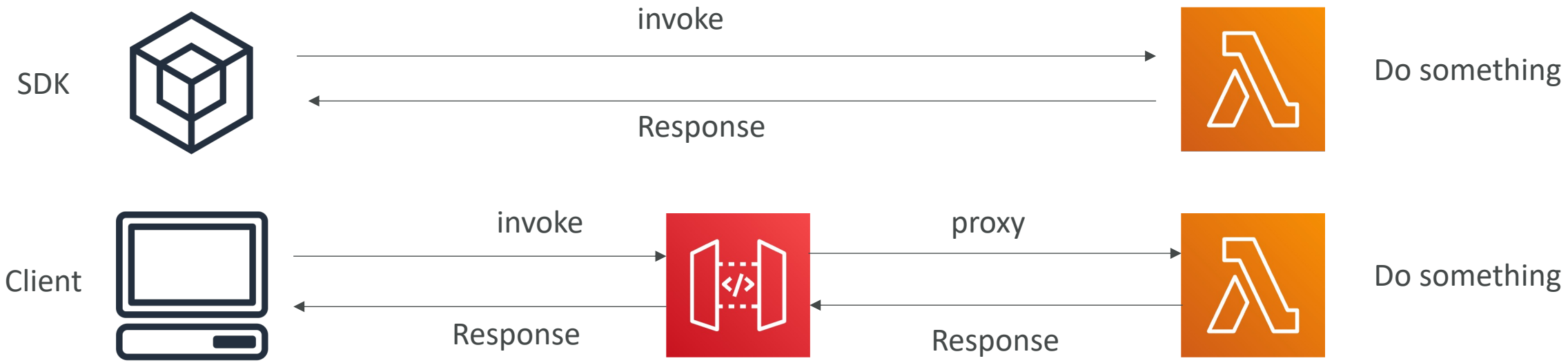


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Lambda – Synchronous Invocations

- Synchronous: CLI, SDK, API Gateway
  - Results is returned right away
  - Error handling must happen client side (retries, exponential backoff, etc…)



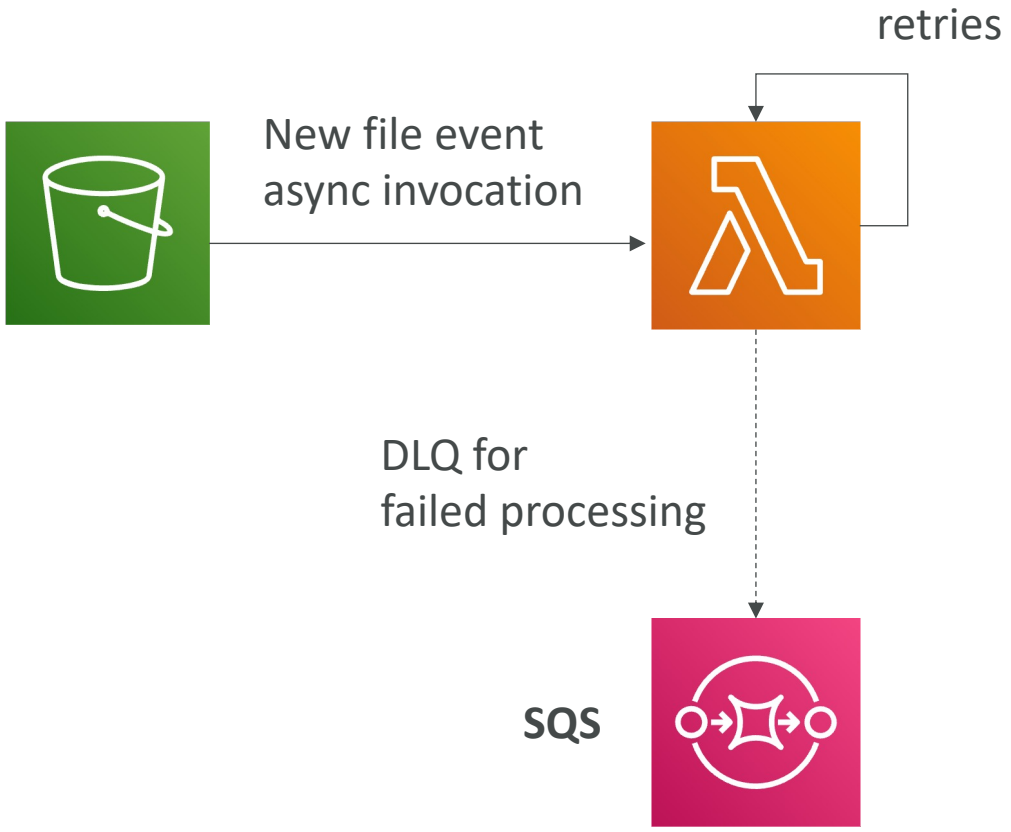
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Lambda – Asynchronous Invocation

- S3, SNS, Amazon EventBridge…
- Lambda attempts to retry on errors (3 tries total)
- Make sure the processing is **idempotent** (in case of retries)

- Can define a DLQ (dead-letter queue) – **SNS or SQS** – for failed processing

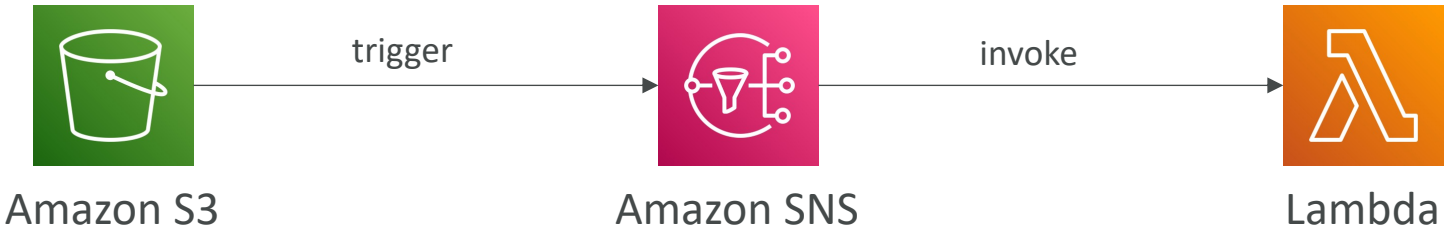


© Stephane Maarek

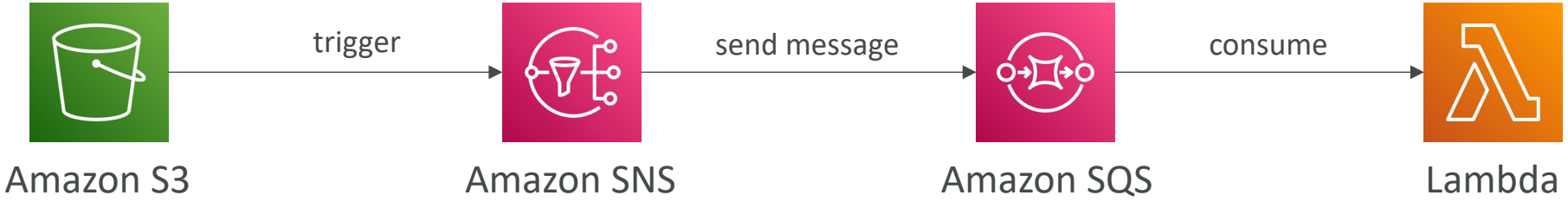
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Lambda – Architecture Discussion

Starts immediately
Parallel executions



Batched execution
Delay



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Types of load balancer on AWS

- AWS has 4 kinds of managed Load Balancers
- **Classic Load Balancer** (v1 - old generation) – 2009 – CLB
  - HTTP, HTTPS, TCP, SSL (secure TCP)
- **Application Load Balancer** (v2 - new generation) – 2016 – ALB
  - HTTP, HTTPS, WebSocket
- **Network Load Balancer** (v2 - new generation) – 2017 – NLB
  - TCP, TLS (secure TCP), UDP
- **Gateway Load Balancer** – 2020 – GWLB
  - Operates at layer 3 (Network layer) – IP Protocol

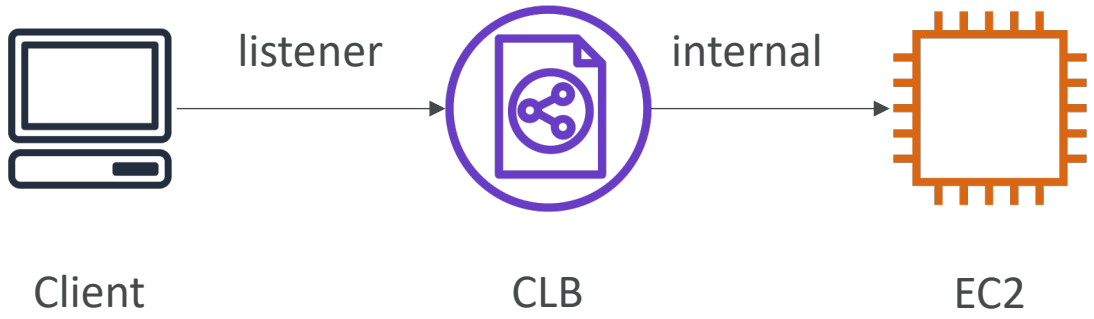
- Overall, it is recommended to use the newer generation load balancers as they provide more features
- Some load balancers can be setup as internal (private) or external (public) ELBs

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Classic Load Balancers (v1)

- Health Checks can be HTTP (L7) or TCP (L4) based including with SSL
- Supports only one SSL certificate
  - The SSL certificate can have many SAN (Subject Alternate Name), but the SSL certificate must be changed anytime a SAN is added / edited / removed
  - Better to use ALB with SNI (Server Name Indication) if possible
  - Can use multiple CLB if you want distinct SSL certificates
- TCP => TCP passes all the traffic to the EC2 instance
  - Only way to use 2-way SSL authentication



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Application Load Balancer (v2)

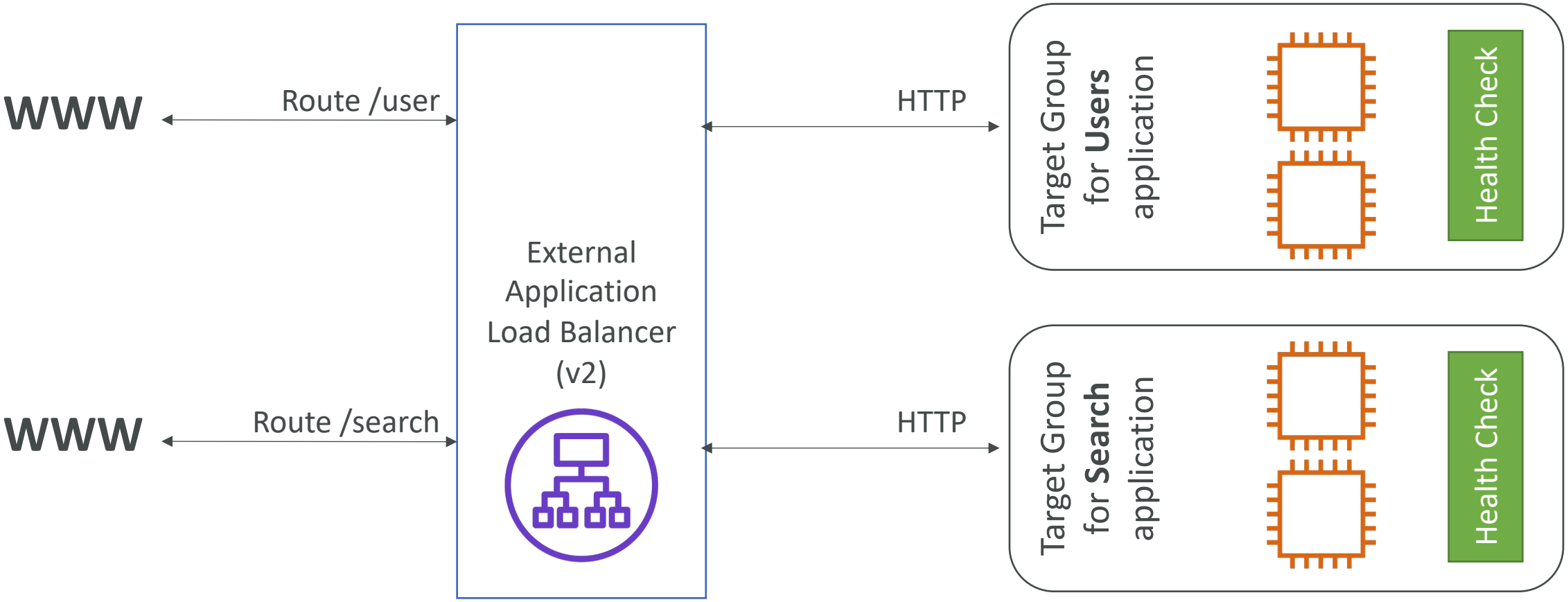
- Application load balancers is Layer 7 (HTTP)

- Load balancing to multiple HTTP applications across machines (target groups)
- Load balancing to multiple applications on the same machine (ex: containers) – great fit with ECS, has dynamic port mapping
- Support for HTTP/2 and WebSocket
- Support redirects (from HTTP to HTTPS for example)
- Routing Rules for path, headers, query string

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Application Load Balancer (v2)
HTTP Based Traffic



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Application Load Balancer (v2) Target Groups

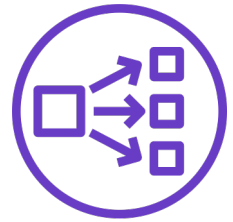
- **EC2 instances** (can be managed by an Auto Scaling Group) – HTTP
- **ECS tasks** (managed by ECS itself) – HTTP
- **Lambda functions** – HTTP request is translated into a JSON event
- **IP Addresses** – must be private IPs

- ALB can route to multiple target groups
- Health checks are at the target group level

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Network Load Balancer (v2)



- Network load balancers (Layer 4) allow to:
  - **Forward TCP & UDP traffic to your instances**
  - Handle millions of request per seconds
  - Less latency ~100 ms (vs 400 ms for ALB)

- **NLB has one static IP per AZ, and supports assigning Elastic IP** (helpful for whitelisting specific IP)

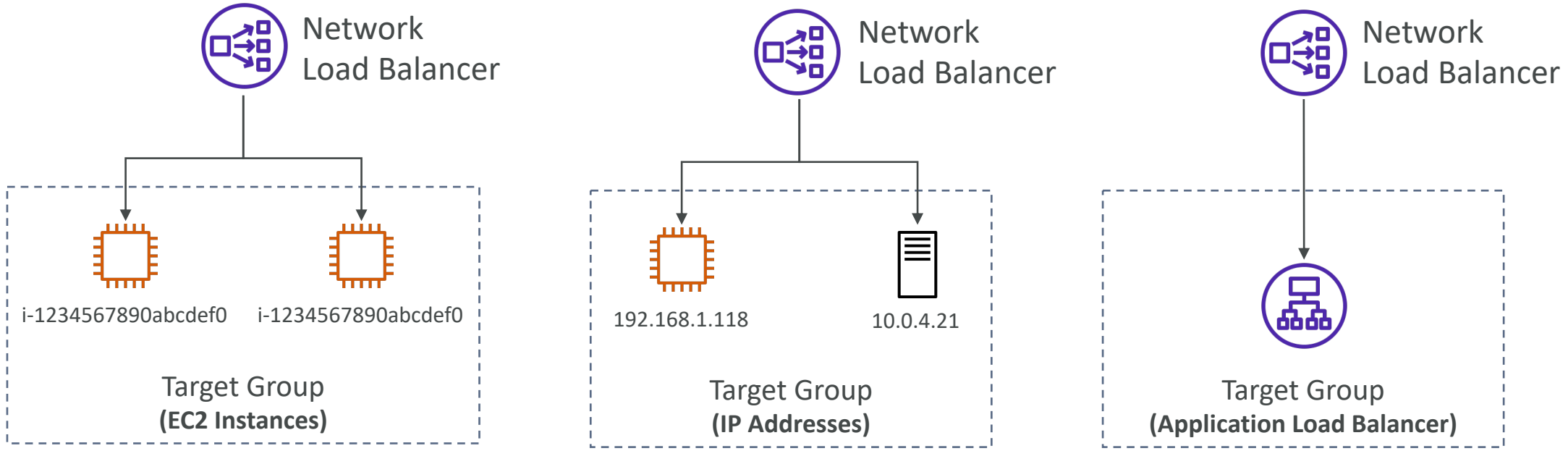
- NLB are used for extreme performance, TCP or UDP traffic
- Not included in the AWS free tier

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Network Load Balancer – Target Groups

- EC2 instances
- IP Addresses – must be private IPs
- Application Load Balancer

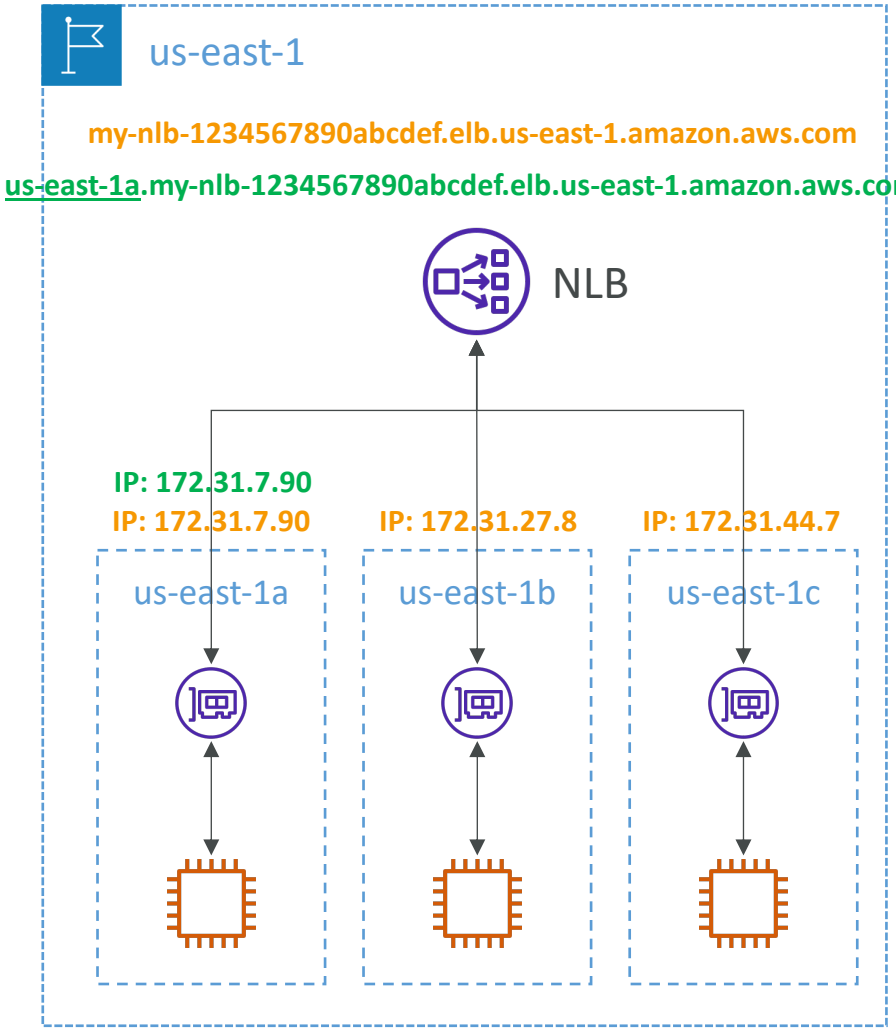


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Network Load Balancer – Zonal DNS Name

- Resolving **Regional NLB DNS** name returns the IP addresses for all NLB nodes in all enabled AZs
  - my-nlb-1234567890abcdef.elb.us-east-1.amazon.aws.com
- **Zonal DNS Name**
  - NLB has DNS names for each of its nodes
  - Use to determine the IP address of each node
  - us-east-1a.my-nlb-1234567890abcdef.elb.us-east-1.amazon.aws.com
  - Used to minimize latency and data transfer costs
  - You need to implement app specific logic



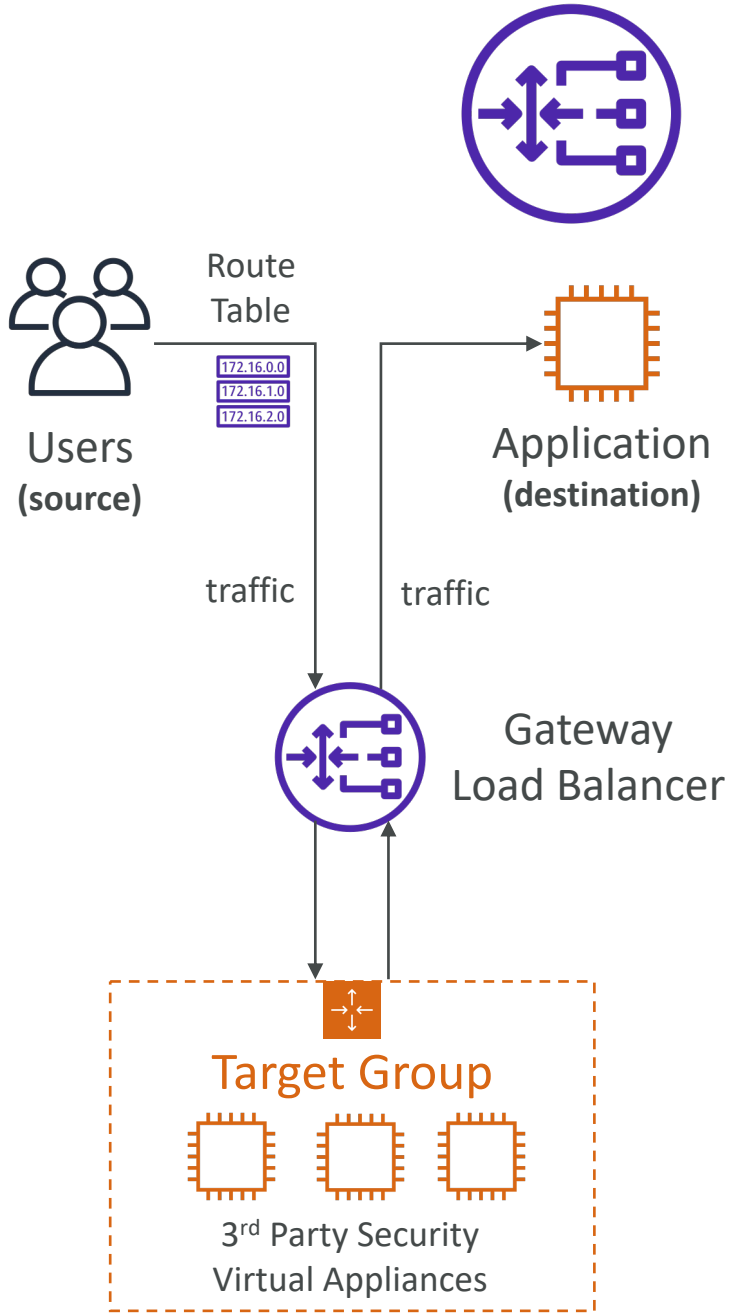
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Gateway Load Balancer

- Deploy, scale, and manage a fleet of 3rd party network virtual appliances in AWS
- Example: Firewalls, Intrusion Detection and Prevention Systems, Deep Packet Inspection Systems, payload manipulation, …

- Operates at Layer 3 (Network Layer) – IP Packets
- Combines the following functions:
  - **Transparent Network Gateway** – single entry/exit for all traffic
  - **Load Balancer** – distributes traffic to your virtual appliances
- Uses the **GENEVE** protocol on port **6081**

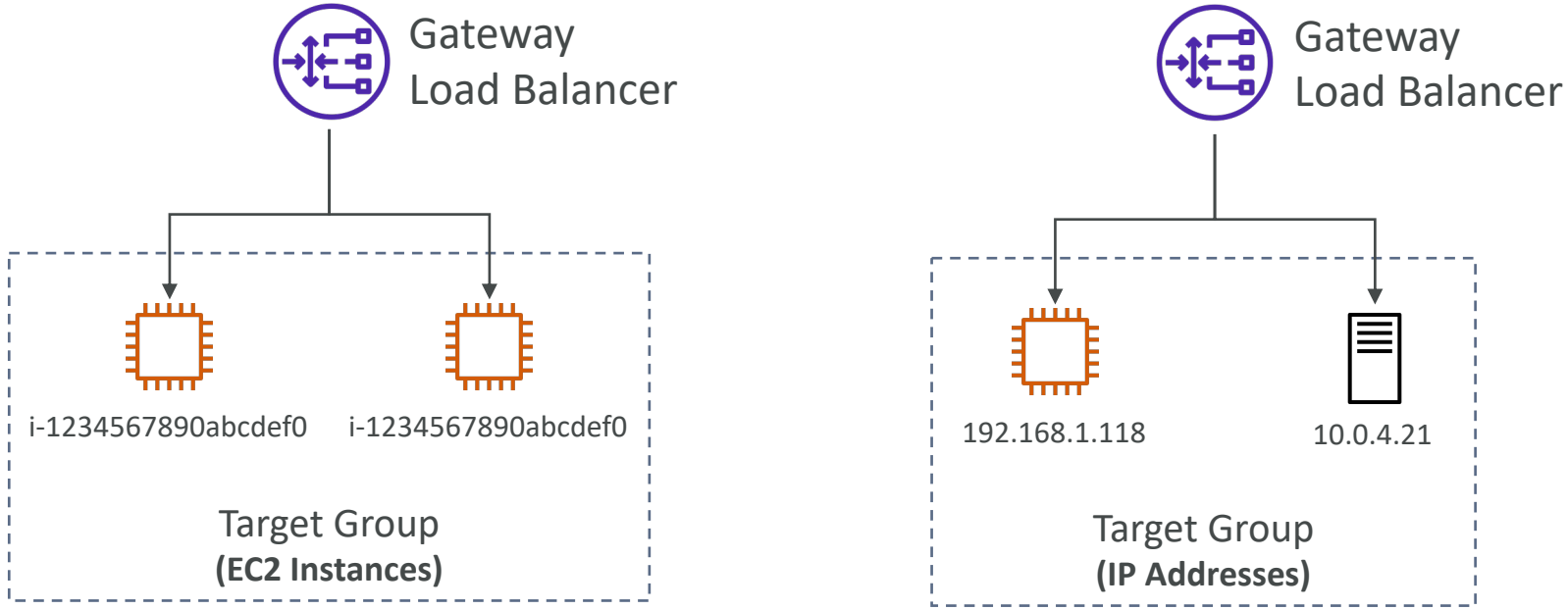


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Gateway Load Balancer – Target Groups

- EC2 instances
- IP Addresses – must be private IPs



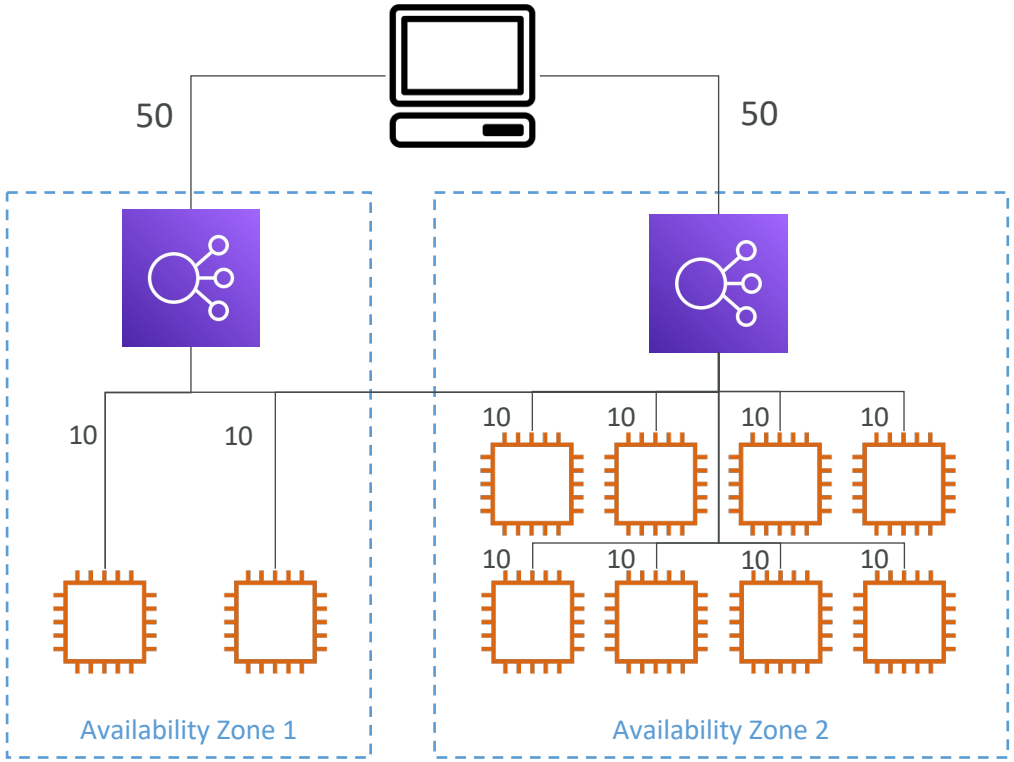
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cross-Zone Load Balancing

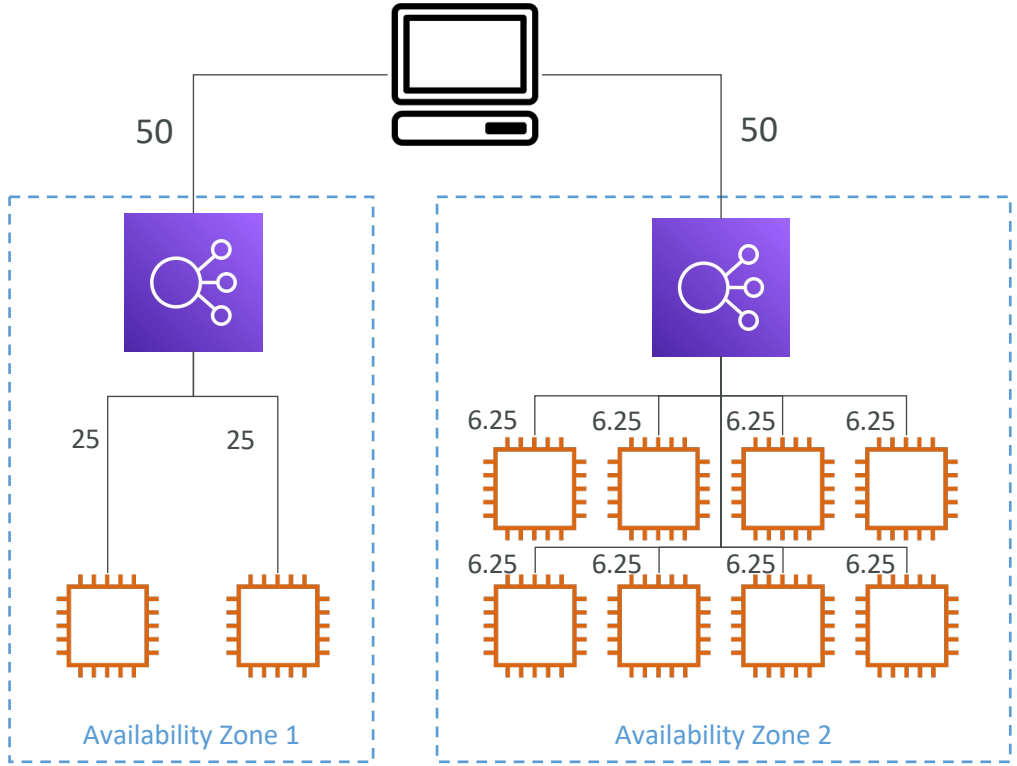
With Cross Zone Load Balancing:

each load balancer instance distributes evenly across all registered instances in all AZ



Without Cross Zone Load Balancing:

Requests are distributed in the instances of the node of the Elastic Load Balancer



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cross-Zone Load Balancing

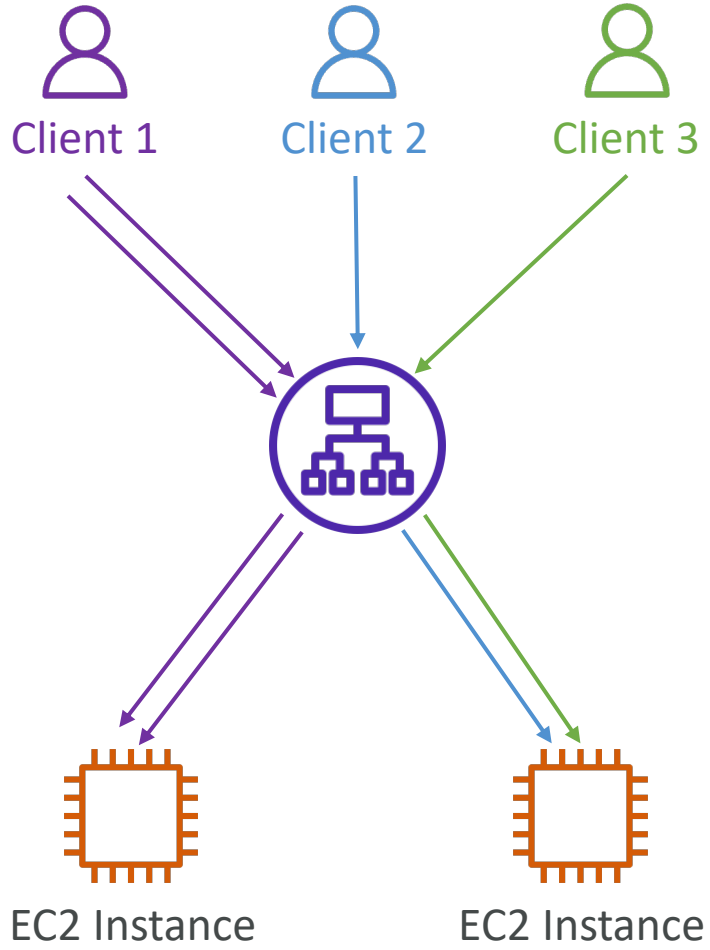
- Classic Load Balancer
  - Disabled by default
  - No charges for inter AZ data if enabled
- Application Load Balancer
  - Always on (can't be disabled)
  - No charges for inter AZ data
- Network Load Balancer
  - Disabled by default
  - You pay charges ($) for inter AZ data if enabled
- Gateway Load Balancer
  - Disabled by default
  - You pay charges ($) for inter AZ data if enabled

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Sticky Sessions (Session Affinity)

- It is possible to implement stickiness so that the same client is always redirected to the same instance behind a load balancer
- This works for Classic Load Balancers & Application Load Balancers
- The "cookie" used for stickiness has an expiration date you control
- Use case: make sure the user doesn't lose his session data
- Enabling stickiness may bring imbalance to the load over the backend EC2 instances

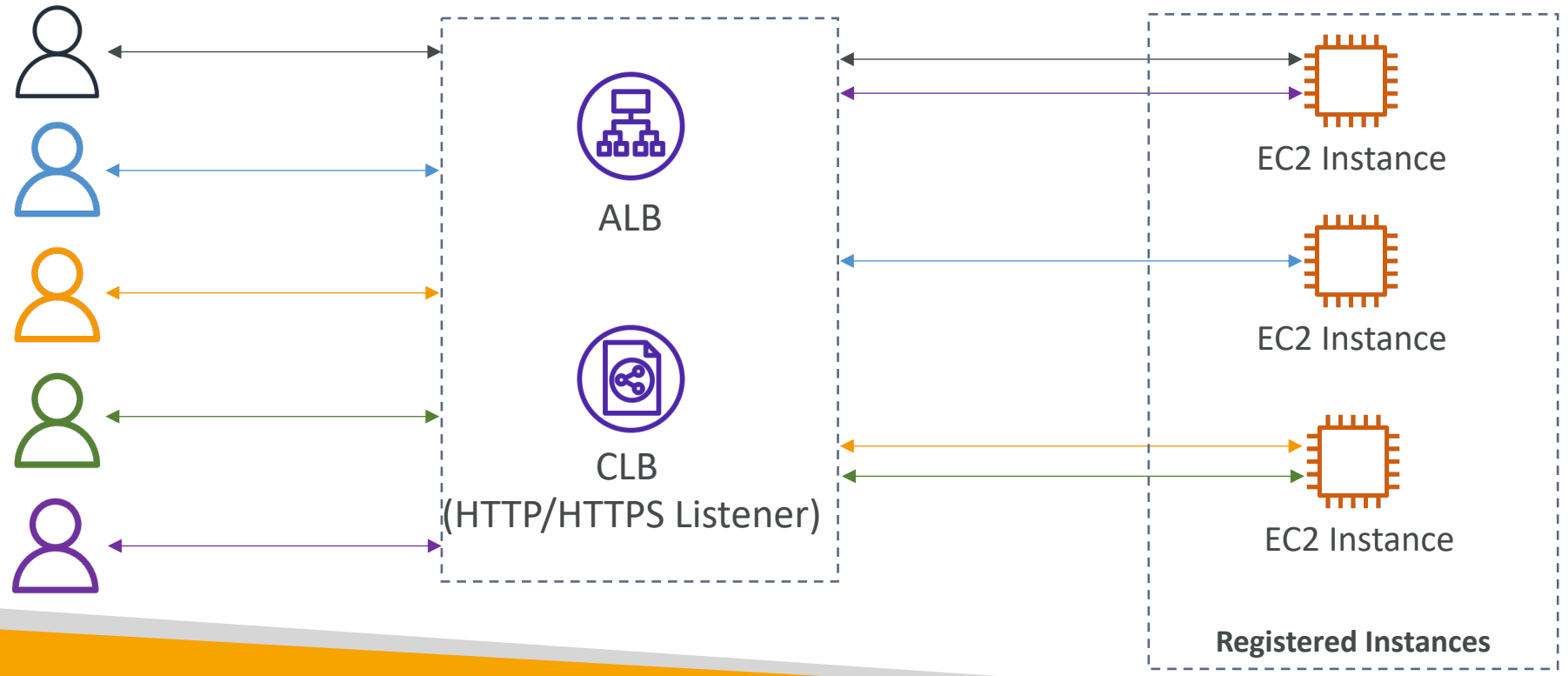


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Request Routing Algorithms – Least Outstanding Requests

- The next instance to receive the request is the instance that has the lowest number of pending/unfinished requests
- Works with **Application Load Balancer** and **Classic Load Balancer (HTTP/HTTPS)**

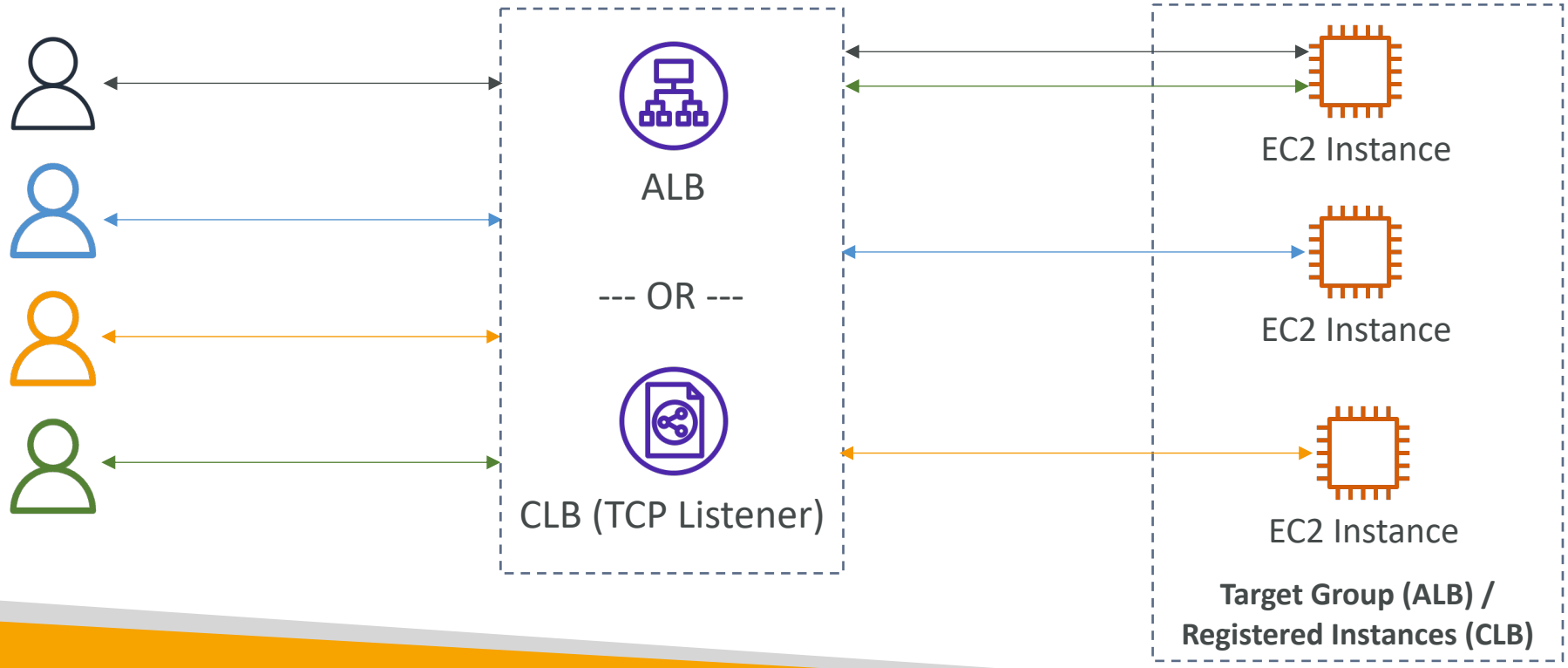


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Request Routing Algorithms – Round Robin

- Equally choose the targets from the target group
- Works with **Application Load Balancer** and **Classic Load Balancer (TCP)**

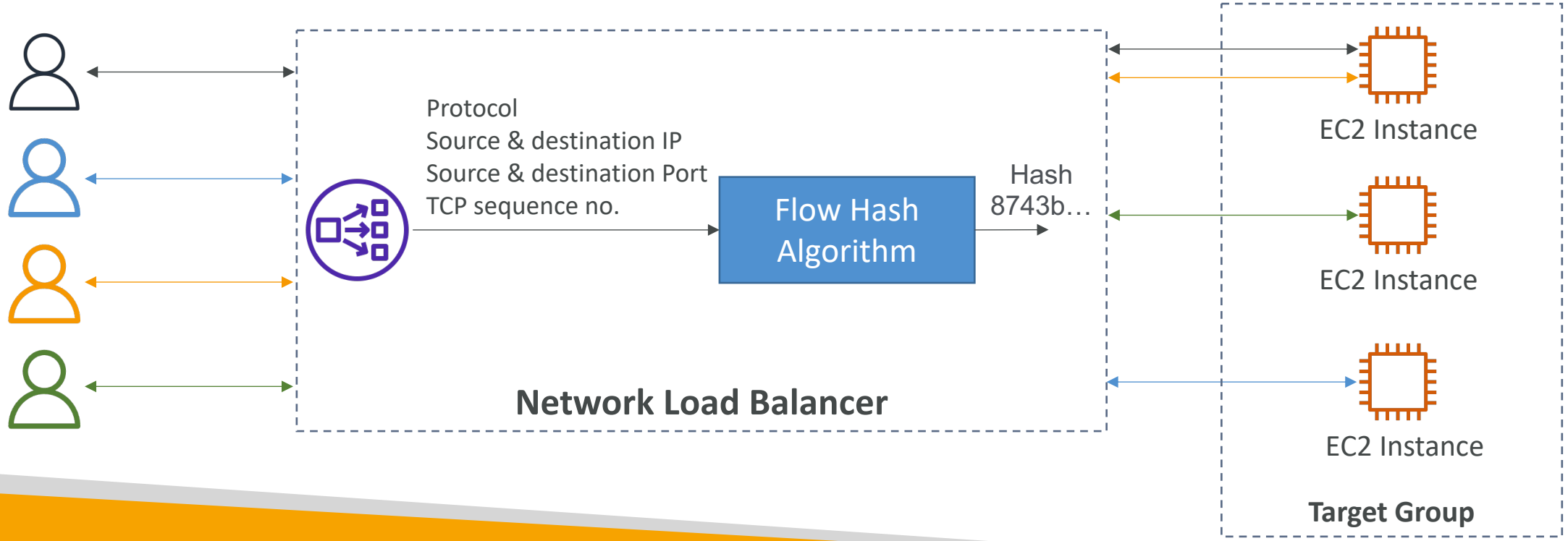


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Request Routing Algorithms – Flow Hash

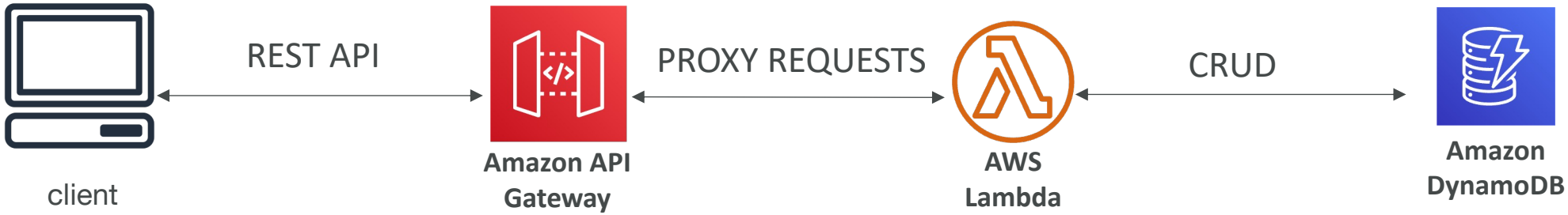
- Selects a target based on the protocol, source/destination IP address, source/destination port, and TCP sequence number
- Each TCP/UDP connection is routed to a single target for the life of the connection
- Works with **Network Load Balancer**



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway – Overview



- Helps expose Lambda, HTTP & AWS Services as an API
- API versioning, authorization, traffic management (API keys, throttles), huge scale, serverless, req/resp transformations, OpenAPI spec, CORS

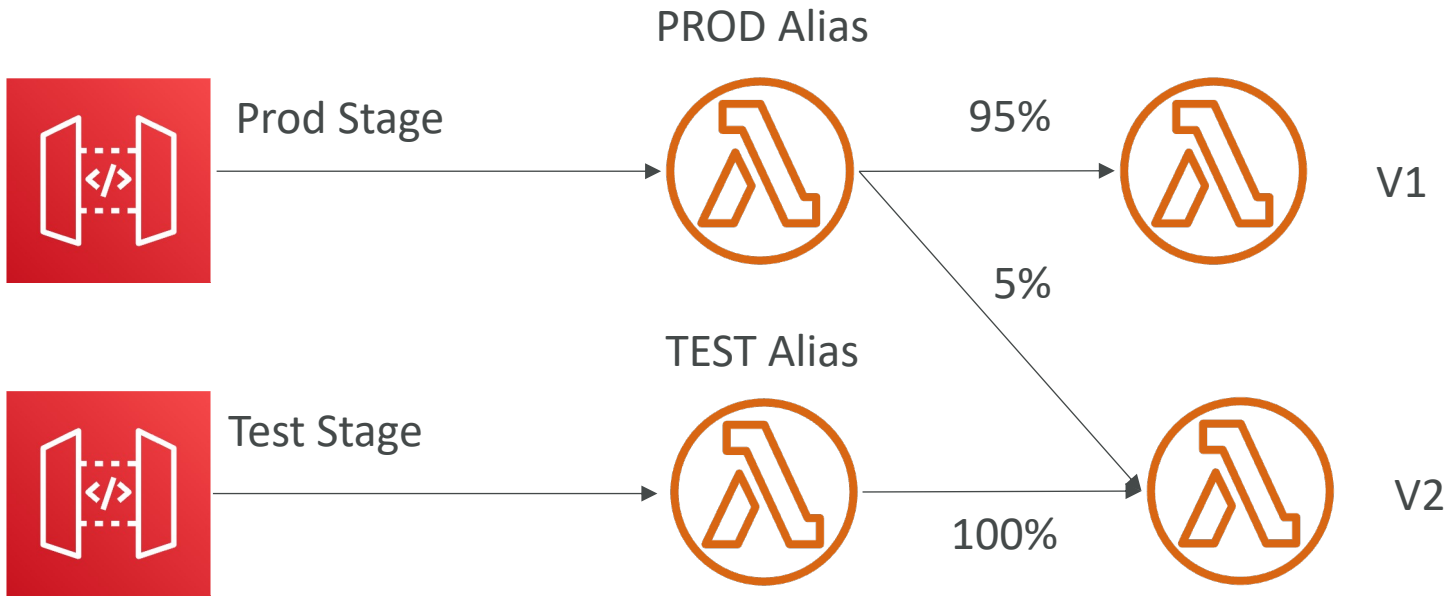
- Limits to know:
  - 29 seconds timeout
  - 10 MB max payload size

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway – Deployment Stages

- API changes are deployed to "Stages" (as many as you want)
- Use the naming you like for stages (dev, test, prod)
- Stages can be rolled back as a history of deployments is kept



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway – Integrations

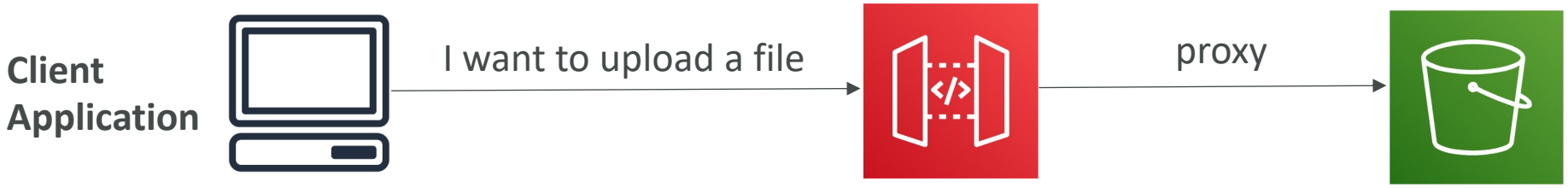
- HTTP
  - Expose HTTP endpoints in the backend
  - Example: internal HTTP API on premise, Application Load Balancer…
  - Why? Add rate limiting, caching, user authentications, API keys, etc…
- Lambda Function
  - Invoke Lambda function
  - Easy way to expose REST API backed by AWS Lambda
- AWS Service
  - Expose any AWS API through the API Gateway?
  - Example: start an AWS Step Function workflow, post a message to SQS
  - Why? Add authentication, deploy publicly, rate control…

© Stephane Maarek

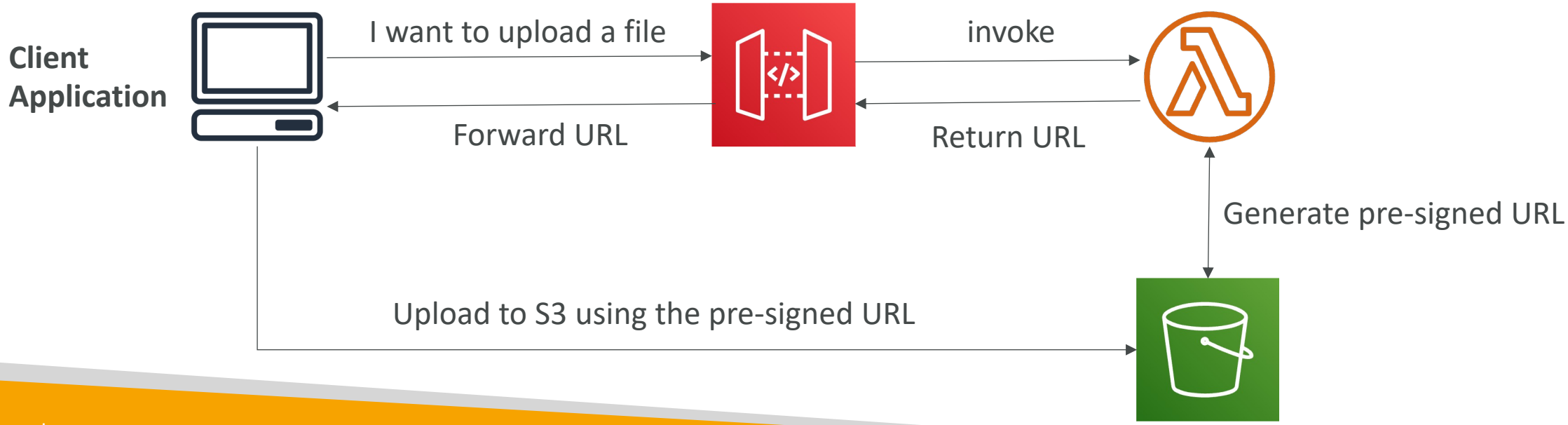
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Solution Architecture Discussion: API Gateway in front of S3

- You will be impacted by the **10 MB payload size limit**



- Better architecture:



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway - Endpoint Types

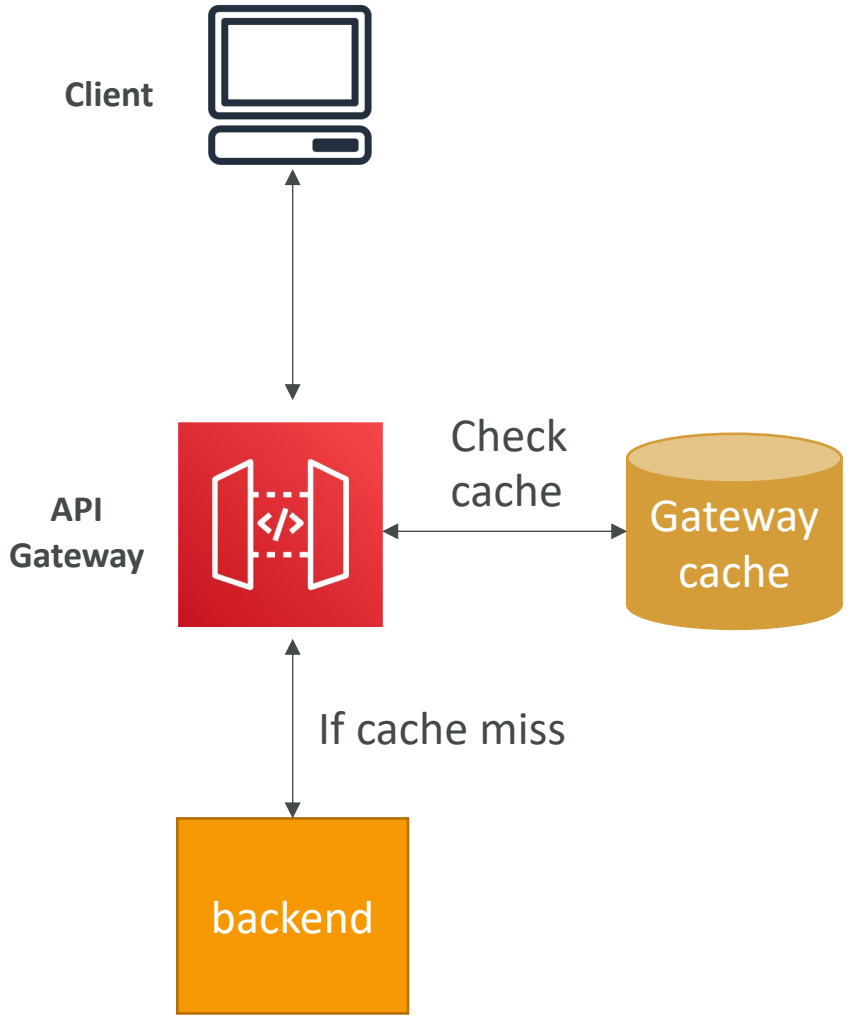
- **Edge-Optimized (default):** For global clients
  - Requests are routed through the CloudFront Edge locations (improves latency)
  - The API Gateway still lives in only one region
- **Regional:**
  - For clients within the same region
  - Could manually combine with CloudFront (more control over the caching strategies and the distribution)
- **Private:**
  - Can only be accessed from your VPC using an interface VPC endpoint (ENI)
  - Use a resource policy to define access

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Caching API responses

- Caching reduces the number of calls made to the backend
- Default TTL (time to live) is 300 seconds (min: 0s, max: 3600s)
- Caches are defined **per stage**
- Possible to override cache settings **per method**
- Clients can invalidate the cache with header: **Cache-Control: max-age=0** (with proper IAM authorization)
- Able to flush the entire cache (invalidate it) immediately
- Cache encryption option
- Cache capacity between 0.5GB to 237GB



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway - Errors

- 4xx means Client errors
  - 400: Bad Request
  - 403: Access Denied, WAF filtered
  - 429: Quota exceeded, Throttle

- 5xx means Server errors
  - 502: Bad Gateway Exception, usually for an incompatible output returned from a Lambda proxy integration backend and occasionally for out-of-order invocations due to heavy loads.
  - 503: Service Unavailable Exception
  - 504: Integration Failure – ex Endpoint Request Timed-out Exception
    API Gateway requests time out after 29 second maximum

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway – Security

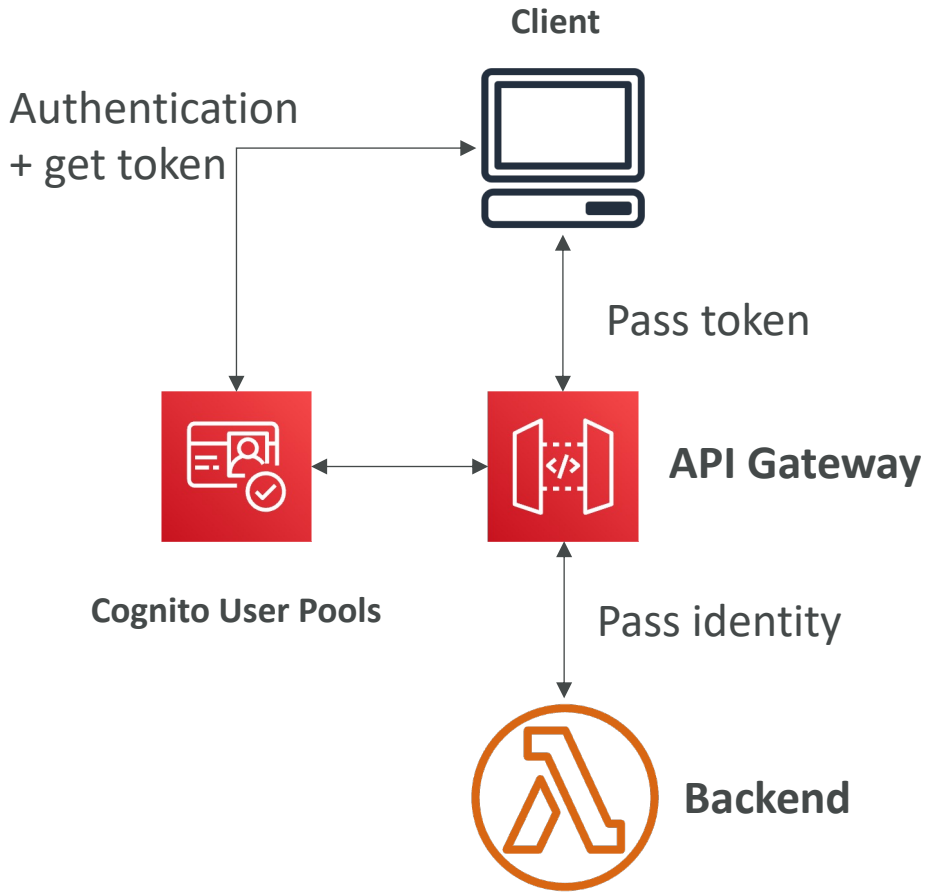
- Load SSL certificates and use Route53 to define a CNAME
- Resource Policy (~S3 Bucket Policy):
  - control who can access the API
  - Users from AWS accounts, IP or CIDR blocks, VPC or VPC Endpoints
- IAM Execution Roles for API Gateway at the API level
  - To invoke a Lambda Function, an AWS service…
- CORS (Cross-origin resource sharing):
  - Browser based security
  - Control which domains can call your API

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway – Authentication

- IAM based access (AWS_IAM)
  - Good for providing access within your infrastructure
  - Pass IAM credentials in headers through Sig V4
- Lambda Authorizer (formerly Custom Authorizer)
  - Use Lambda to verify a custom OAuth / SAML / 3rd party authentication
- Cognito User Pools
  - Client authenticates with Cognito
  - Client passes the token to API Gateway
  - API Gateway knows out-of-the-box how to verify to token



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway – Logging, Monitoring, Tracing

- CloudWatch Logs:
  - Enable CloudWatch logging at the Stage level (with Log Level – ERROR, INFO)
  - Can log full requests / responses data
  - Can send API Gateway Access Logs (customizable)
  - Can send logs directly into Kinesis Data Firehose (as an alternative to CW logs)
- CloudWatch Metrics:
  - Metrics are by stage, possibility to enable detailed metrics
  - *IntegrationLatency*, *Latency*, *CacheHitCount*, *CacheMissCount*
- X-Ray:
  - Enable tracing to get extra information about requests in API Gateway
  - X-Ray API Gateway + AWS Lambda gives you the full picture

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway – Usage Plans & API Keys

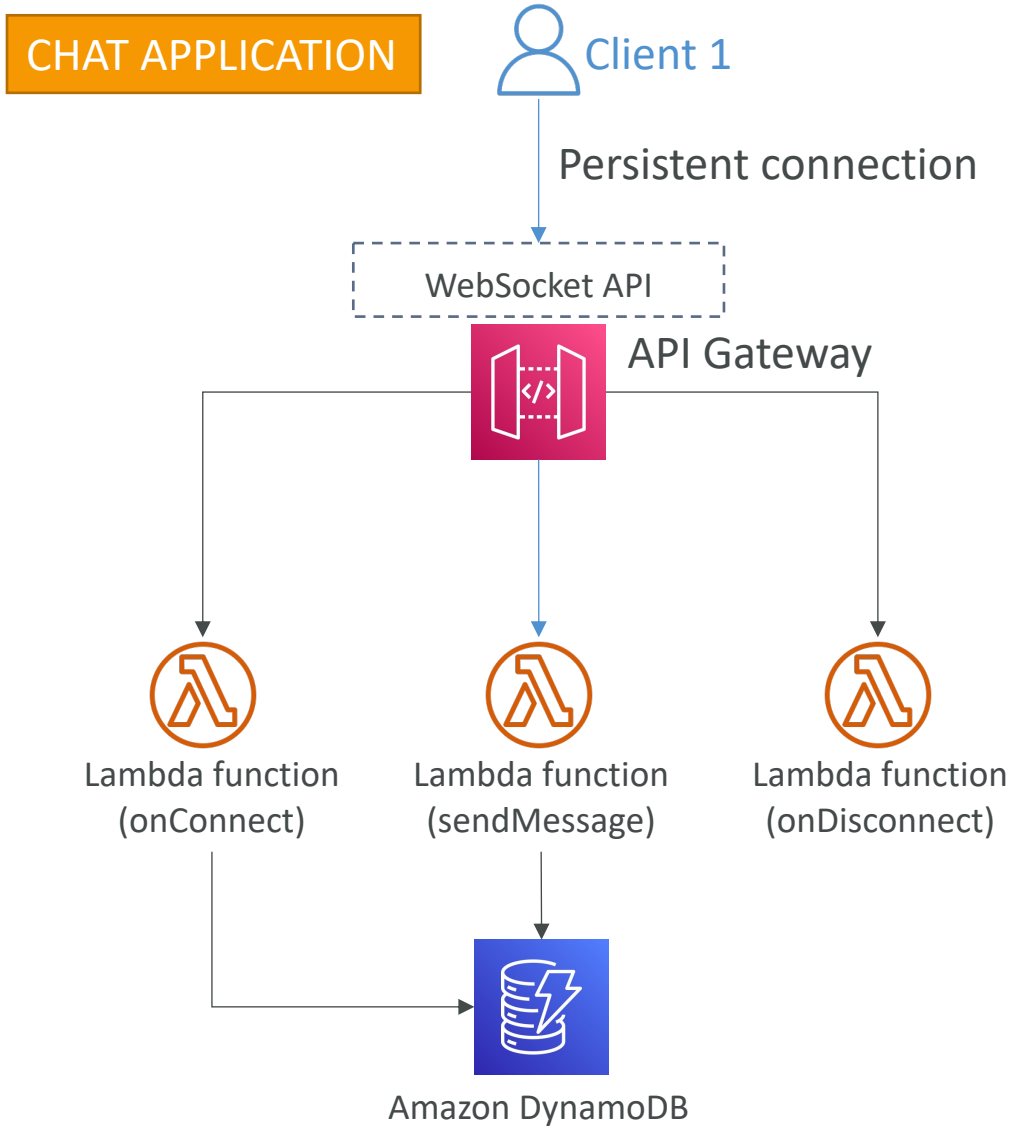
- If you want to make an API available as an offering ($) to your customers
- **Usage Plan:**
  - who can access one or more deployed API stages and methods
  - how much and how fast they can access them
  - uses API keys to identify API clients and meter access
  - configure throttling limits and quota limits that are enforced on individual client
- **API Keys:**
  - alphanumeric string values to distribute to your customers
  - Ex: WBjHxNtoAb4WPKBC7cGm64CBibIb24b4jt8jJHo9
  - Can use with usage plans to control access
  - Throttling limits are applied to the API keys
  - Quotas limits is the overall number of maximum requests
- **429 Too Many Requests:**
  - Account level throttling across all APIs in a region
  - Clients must implement retry mechanisms

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway – WebSocket API – Overview

- What's WebSocket?
  - Two-way interactive communication between a user's browser and a server
  - Server can push information to the client
  - This enables **stateful** application use cases
- WebSocket APIs are often used in **real-time applications** such as chat applications, collaboration platforms, multiplayer games, and financial trading platforms.
- Works with AWS Services (Lambda, DynamoDB) or HTTP endpoints

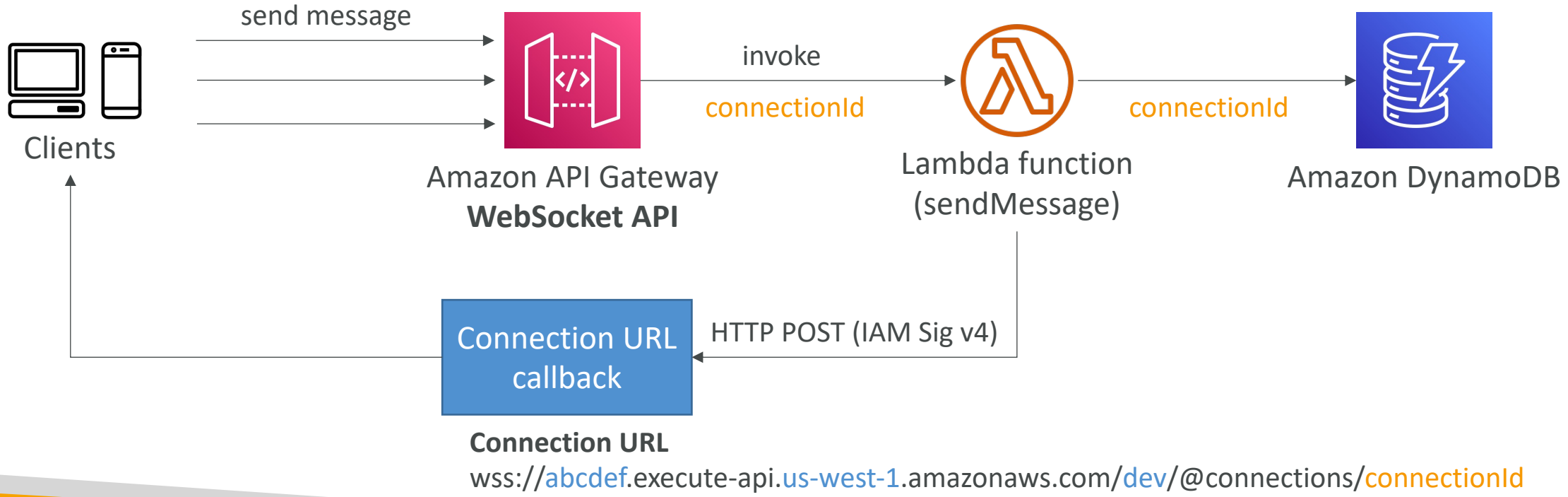


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Server to Client Messaging
@connections used for replies to clients

WebSocket URL
wss://abcdef.execute-api.us-west-1.amazonaws.com/dev



Connection URL
wss://abcdef.execute-api.us-west-1.amazonaws.com/dev/@connections/connectionId

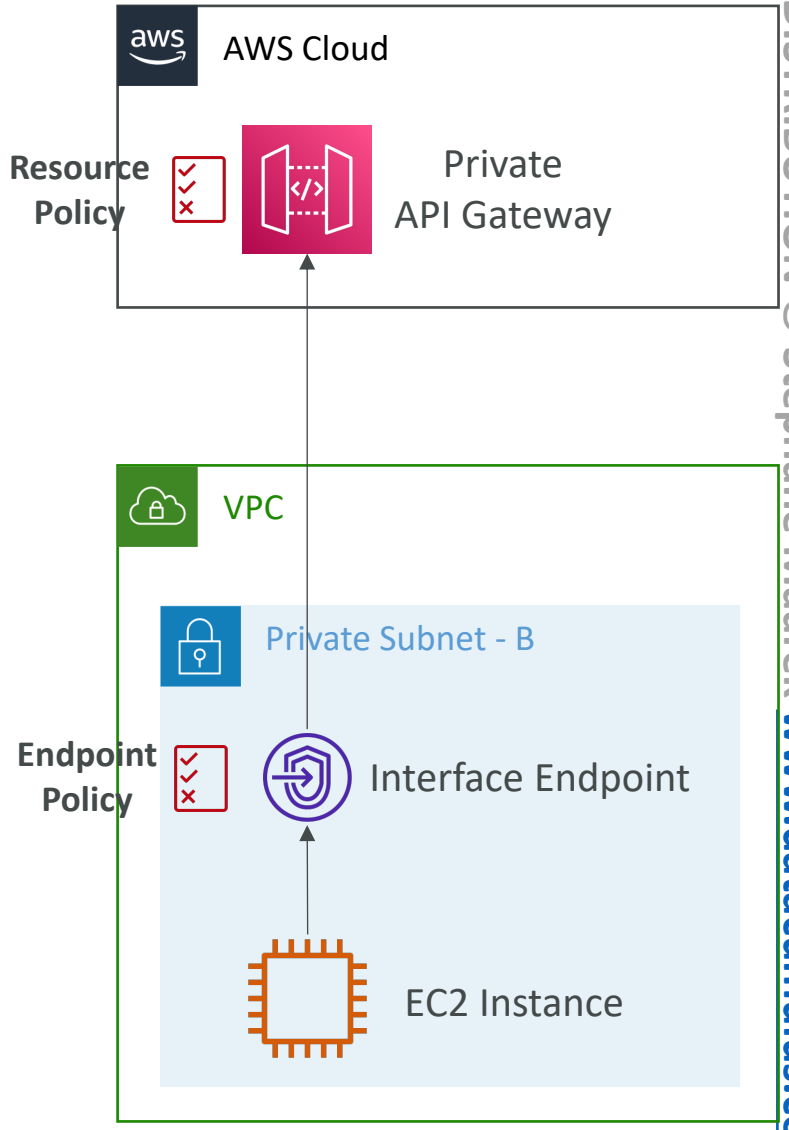
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway – Private APIs

- Can only be accessed from your VPC by using an VPC Interface Endpoint
- Each VPC Interface Endpoint can be used to access multiple Private APIs

- API Gateway Resource Policy
  - Allow or deny access to API from selected VPCs and VPC Endpoints, including across AWS accounts
  - **aws:SourceVpc** and **aws:SourceVpce**



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

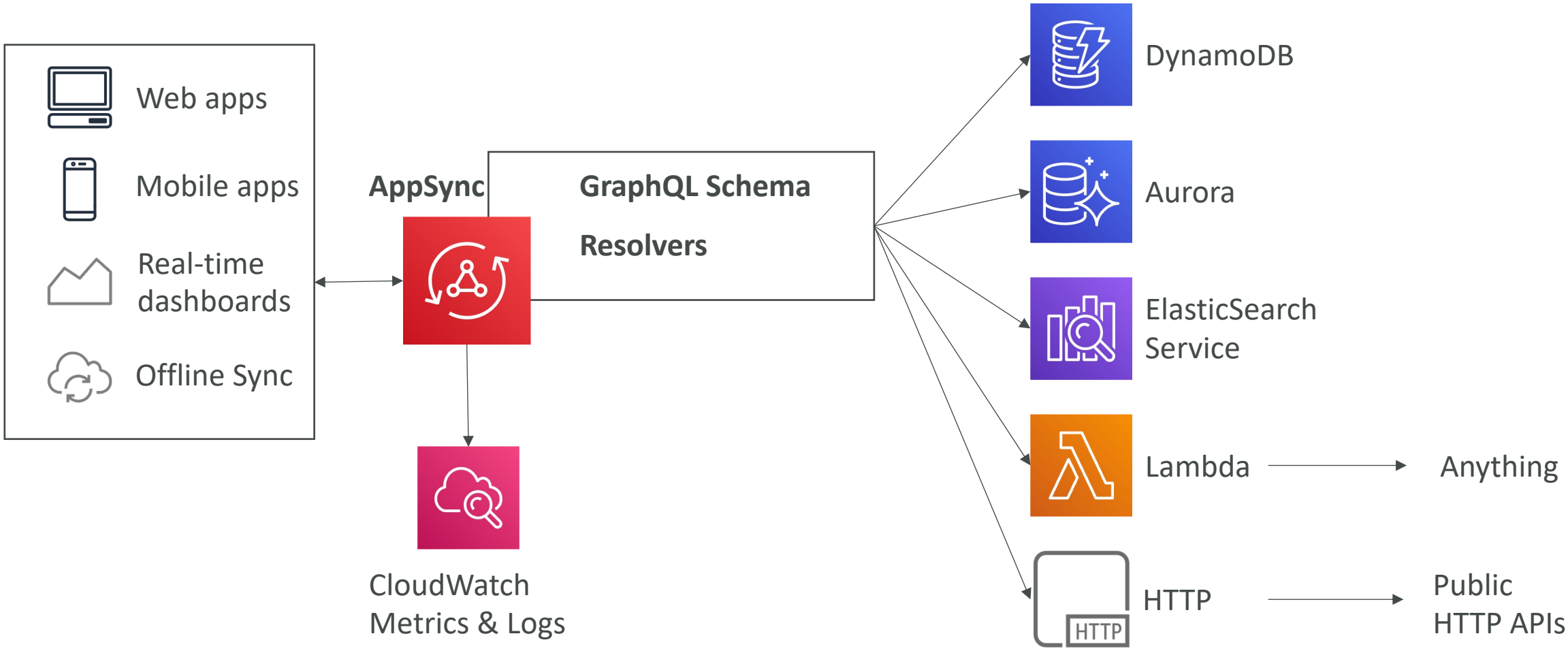
AWS AppSync - Overview

- **AppSync** is a managed service that uses **GraphQL**
- **GraphQL** makes it easy for applications to get exactly the data they need.
- This includes combining data from **one or more sources**
  - NoSQL data stores, Relational databases, HTTP APIs…
  - Integrates with DynamoDB, Aurora, Elasticsearch & others
  - Custom sources with AWS Lambda
- Retrieve data in **real-time with WebSocket or MQTT on WebSocket**
- For mobile apps: local data access & data synchronization
- It all starts with uploading one **GraphQL schema**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AppSync Diagram



© Stephane Maarek

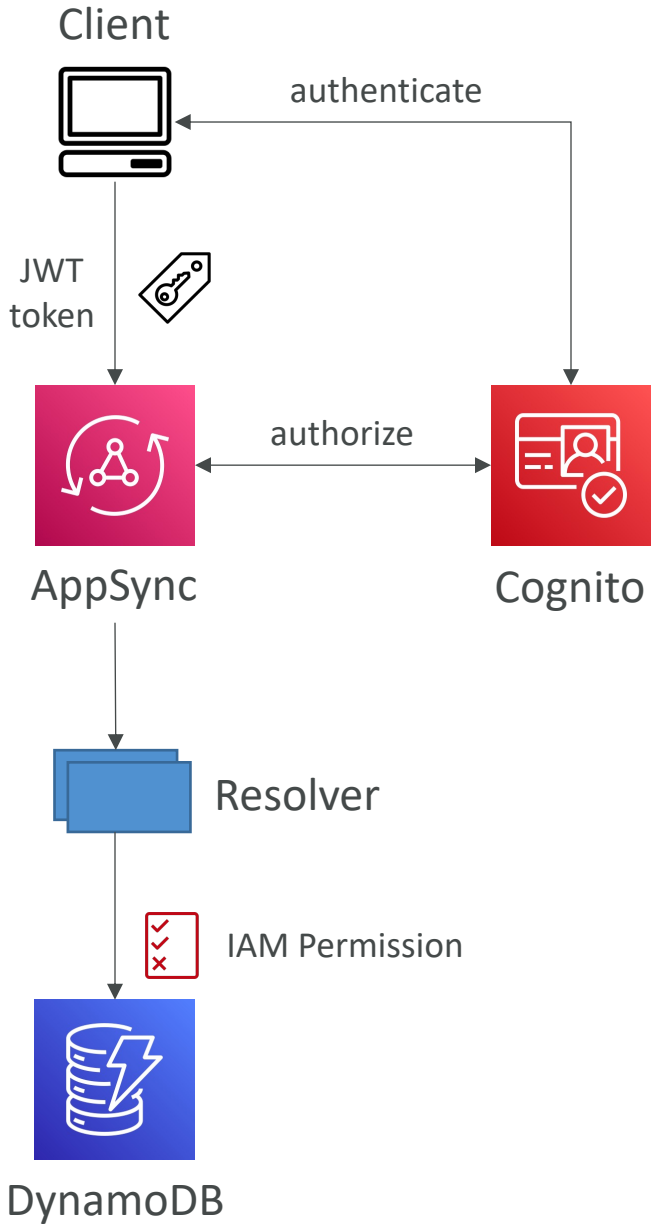
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AppSync – Cognito Integration

- Perform authorization on Cognito users based on the groups they belong to
- In the GraphQL schema, you can specify the security for Cognito groups

```
type Query {
    posts: [Post!]!
    @aws_auth(cognito_groups: ["Bloggers", "Readers"])
}

Type Mutation {
    addPost(id: ID!, title: String!): Post!
    @aws_auth(cognito_groups: ["Bloggers"])
}
```



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

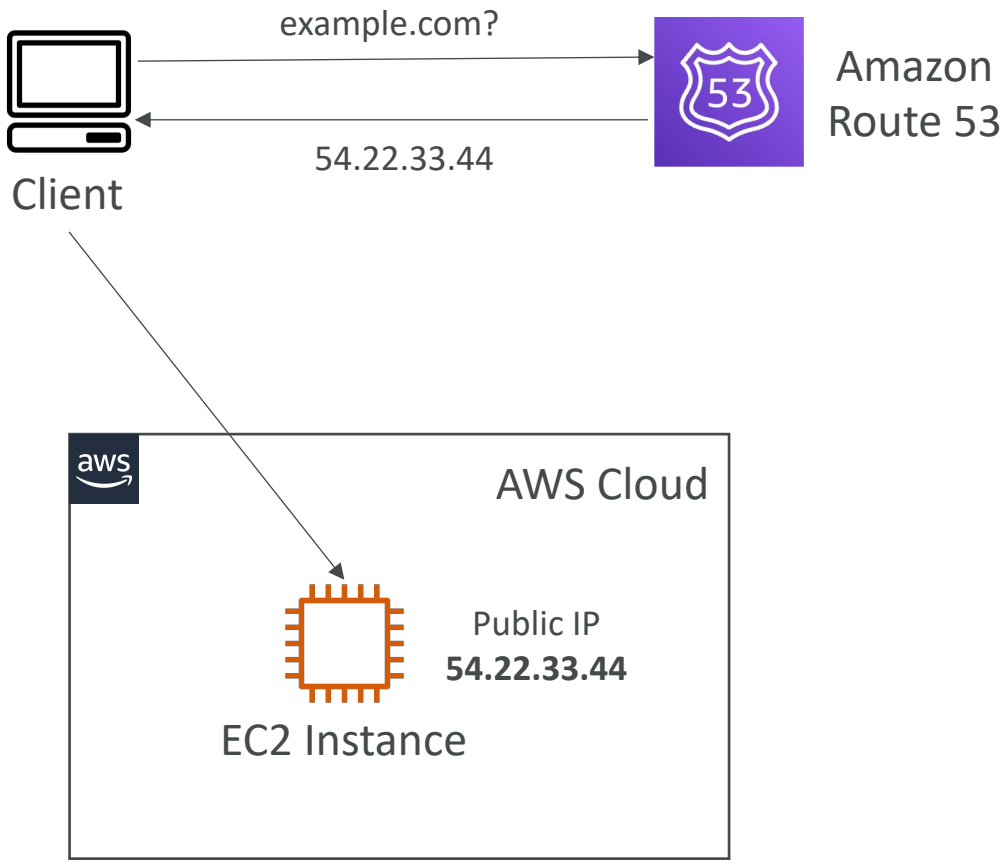
Route 53 – Record Types

- A – maps a hostname to IPv4
- AAAA – maps a hostname to IPv6
- CNAME – maps a hostname to another hostname
  - The target is a domain name which must have an A or AAAA record
  - Can't create a CNAME record for the top node of a DNS namespace (Zone Apex)
  - Example: you can't create for example.com, but you can create for www.example.com
- NS – Name Servers for the Hosted Zone
  - Control how traffic is routed for a domain

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route 53 – Diagram for A record



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route 53 – CNAME vs. Alias

- AWS Resources (Load Balancer, CloudFront...) expose an AWS hostname:
  - lb1-1234.us-east-2.elb.amazonaws.com and you want myapp.mydomain.com

- CNAME:
  - Points a hostname to any other hostname. (app.mydomain.com => blabla.anything.com)
  - ONLY FOR NON ROOT DOMAIN (aka. something.mydomain.com)
- Alias:
  - Points a hostname to an AWS Resource (app.mydomain.com => blabla.amazonaws.com)
  - Works for ROOT DOMAIN and NON ROOT DOMAIN (aka mydomain.com)
  - Free of charge
  - Native health check

© Stephane Maarek

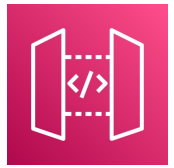
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route 53 – Alias Records Targets

- Elastic Load Balancers
- CloudFront Distributions
- API Gateway
- Elastic Beanstalk environments
- S3 Websites
- VPC Interface Endpoints
- Global Accelerator accelerator
- Route 53 record in the same hosted zone

Elastic
Load Balancer

Amazon
CloudFront



Amazon
API Gateway

Elastic Beanstalk

S3 Websites

VPC Interface
Endpoints

Global Accelerator

Route 53 Record
(same Hosted Zone)

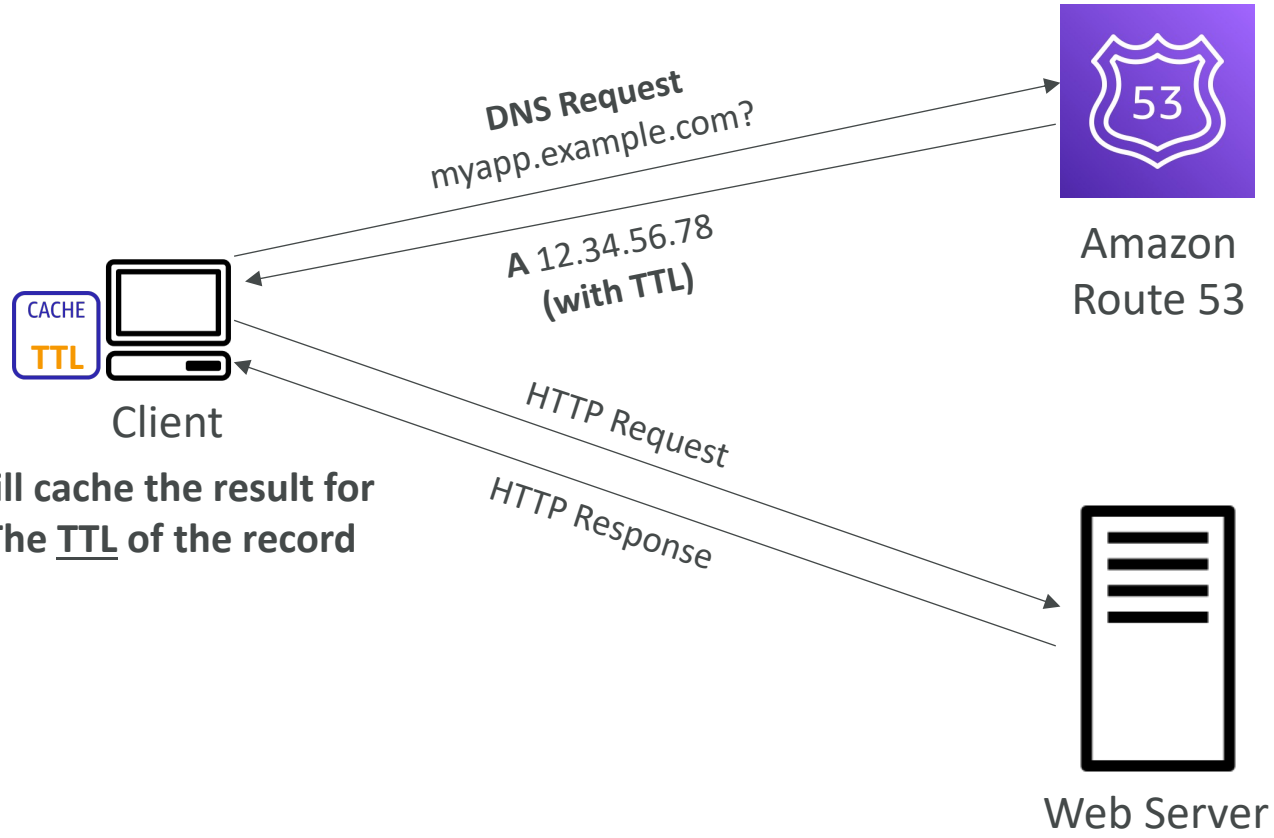
- You cannot set an ALIAS record for an EC2 DNS name

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route 53 – Records TTL (Time To Live)

- High TTL – e.g., 24 hr
  - Less traffic on Route 53
  - Possibly outdated records
- Low TTL – e.g., 60 sec.
  - More traffic on Route 53 ($$)
  - Records are outdated for less time
  - Easy to change records
- Except for Alias records, TTL is mandatory for each DNS record



© Stephane Maarek

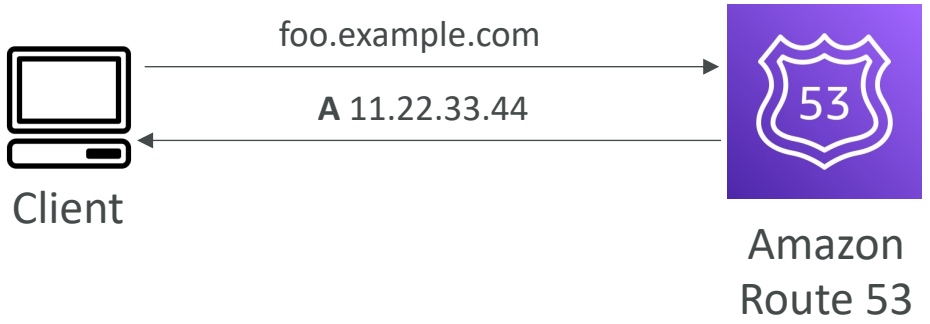
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Routing Policies – Simple

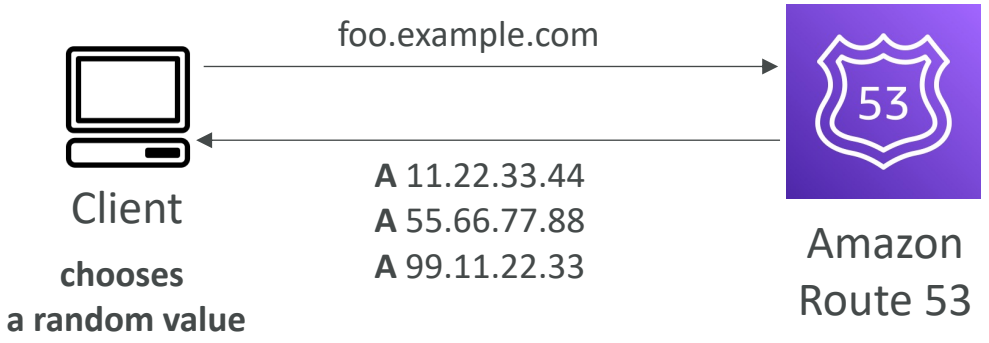
- Typically, route traffic to a single resource
- Can't be associated with Health Checks

- Can specify multiple values in the same record
- If multiple values are returned, a random one is chosen by the client

Single Value



Multiple Value

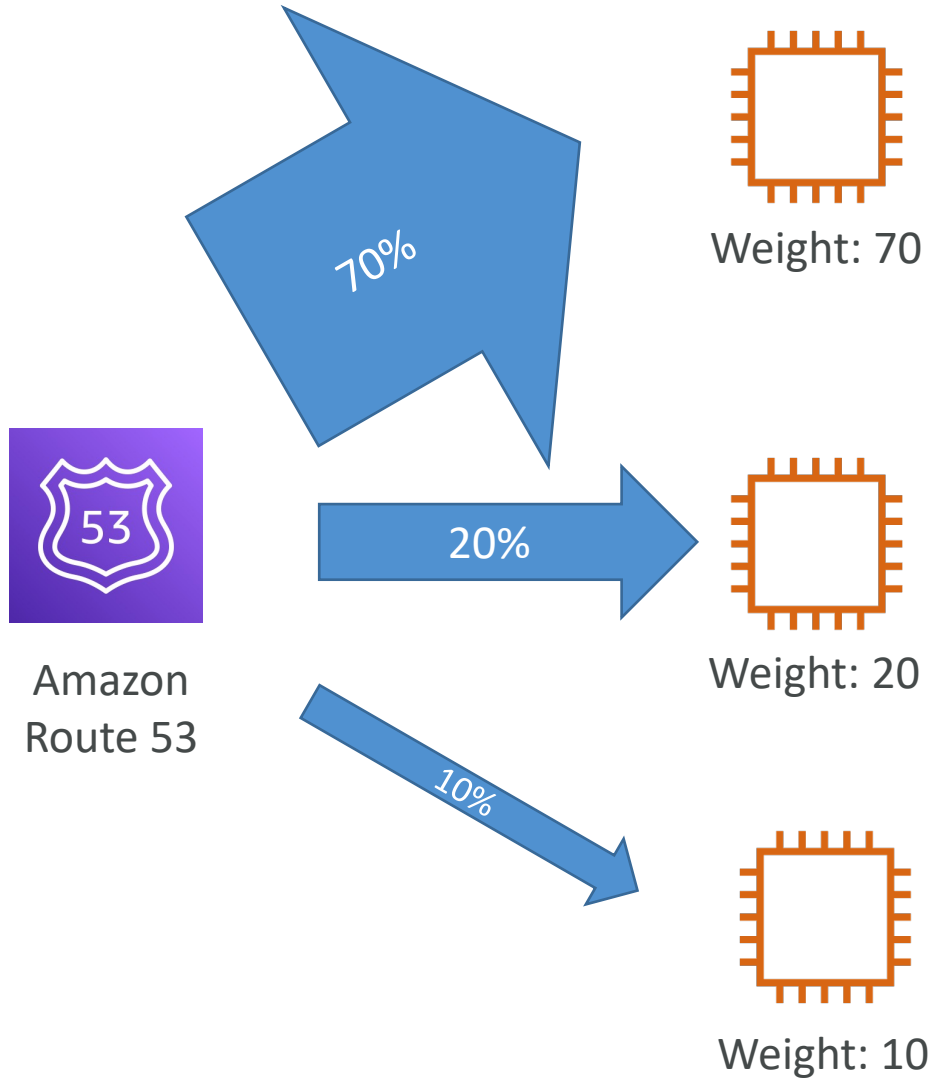


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Routing Policies – Weighted

- Control the % of the requests that go to each specific resource
- Can be associated with Health Checks
- Use cases: load balancing between regions, testing new application versions…

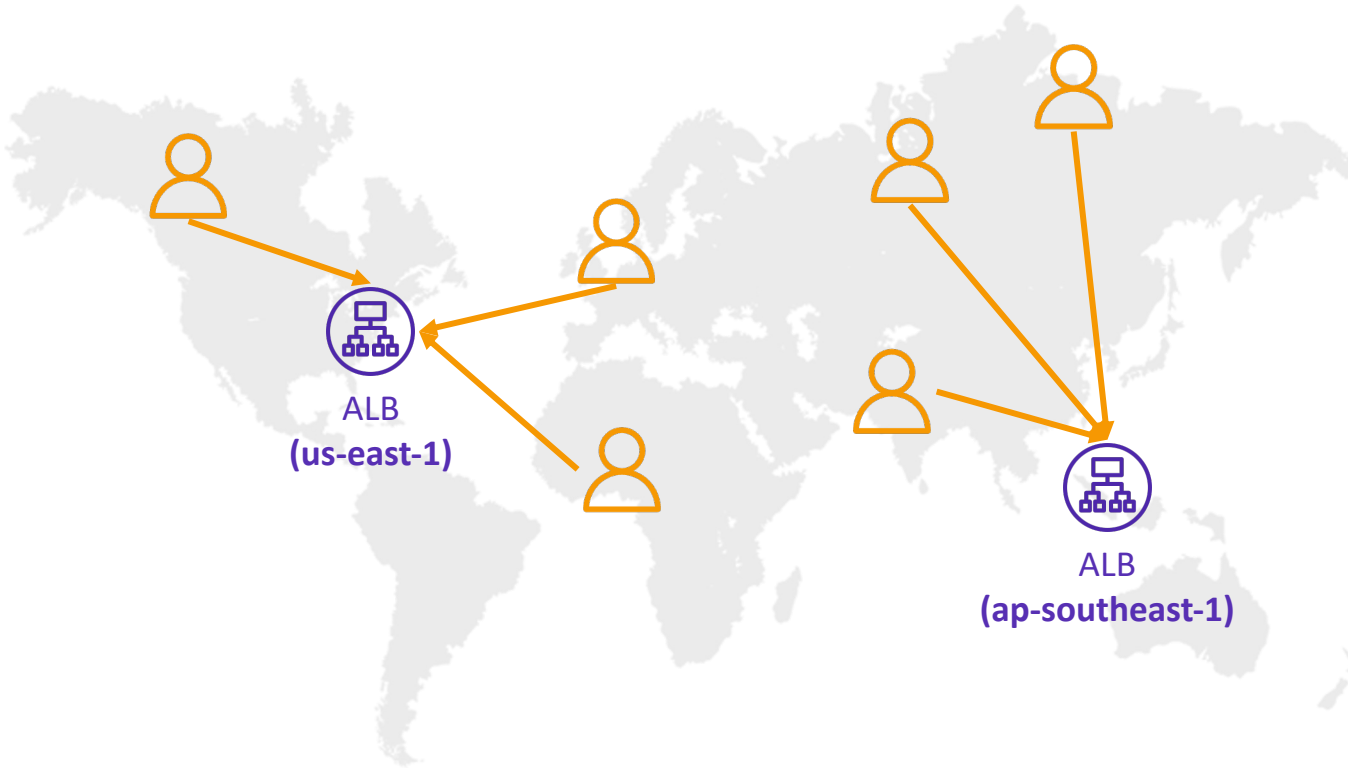


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Routing Policies – Latency-based

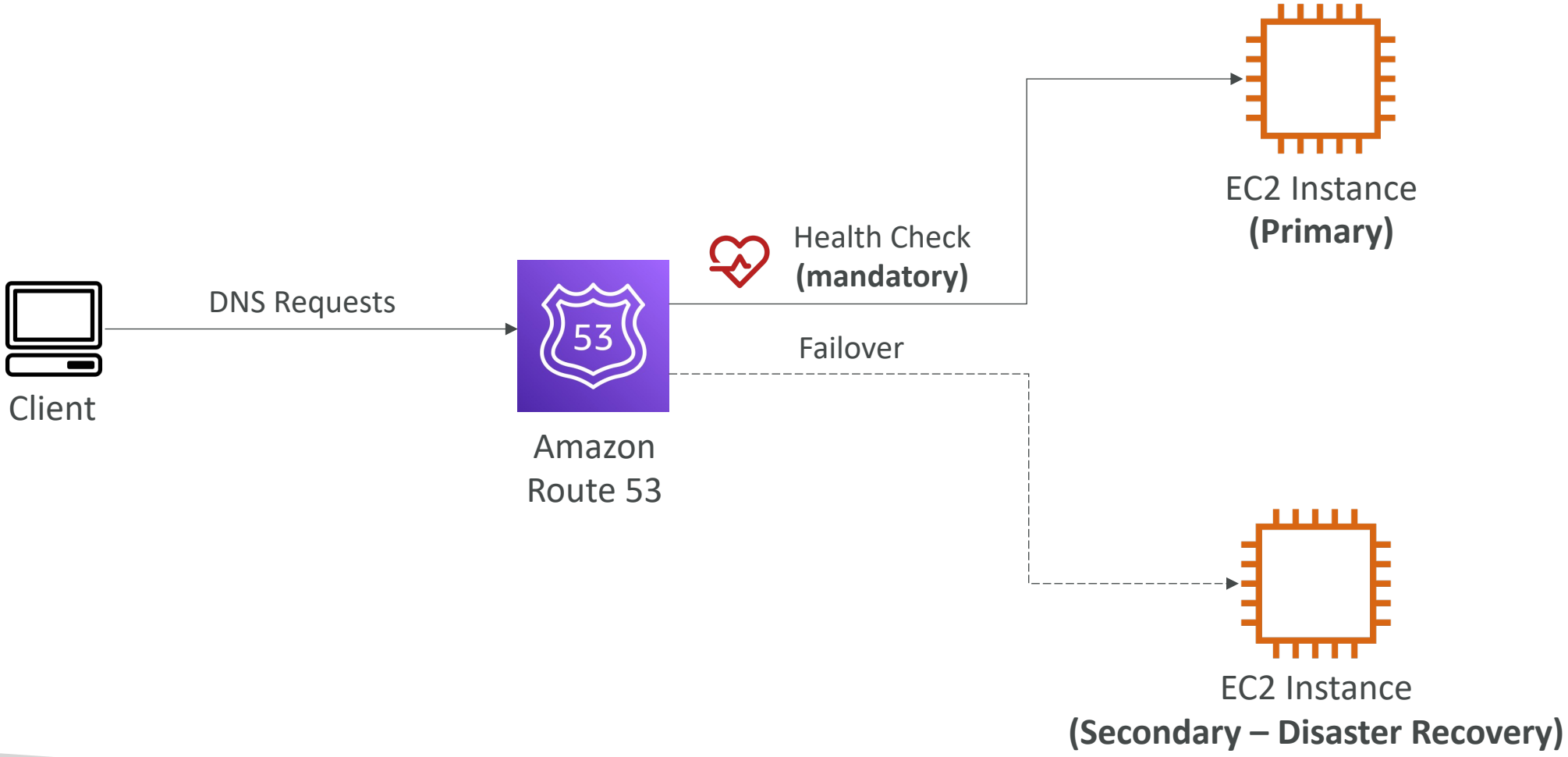
- Redirect to the resource that has the least latency close to us
- Super helpful when latency for users is a priority
- **Latency is based on traffic between users and AWS Regions**
- Germany users may be directed to the US (if that's the lowest latency)
- Can be associated with Health Checks (has a failover capability)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Routing Policies – Failover (Active-Passive)

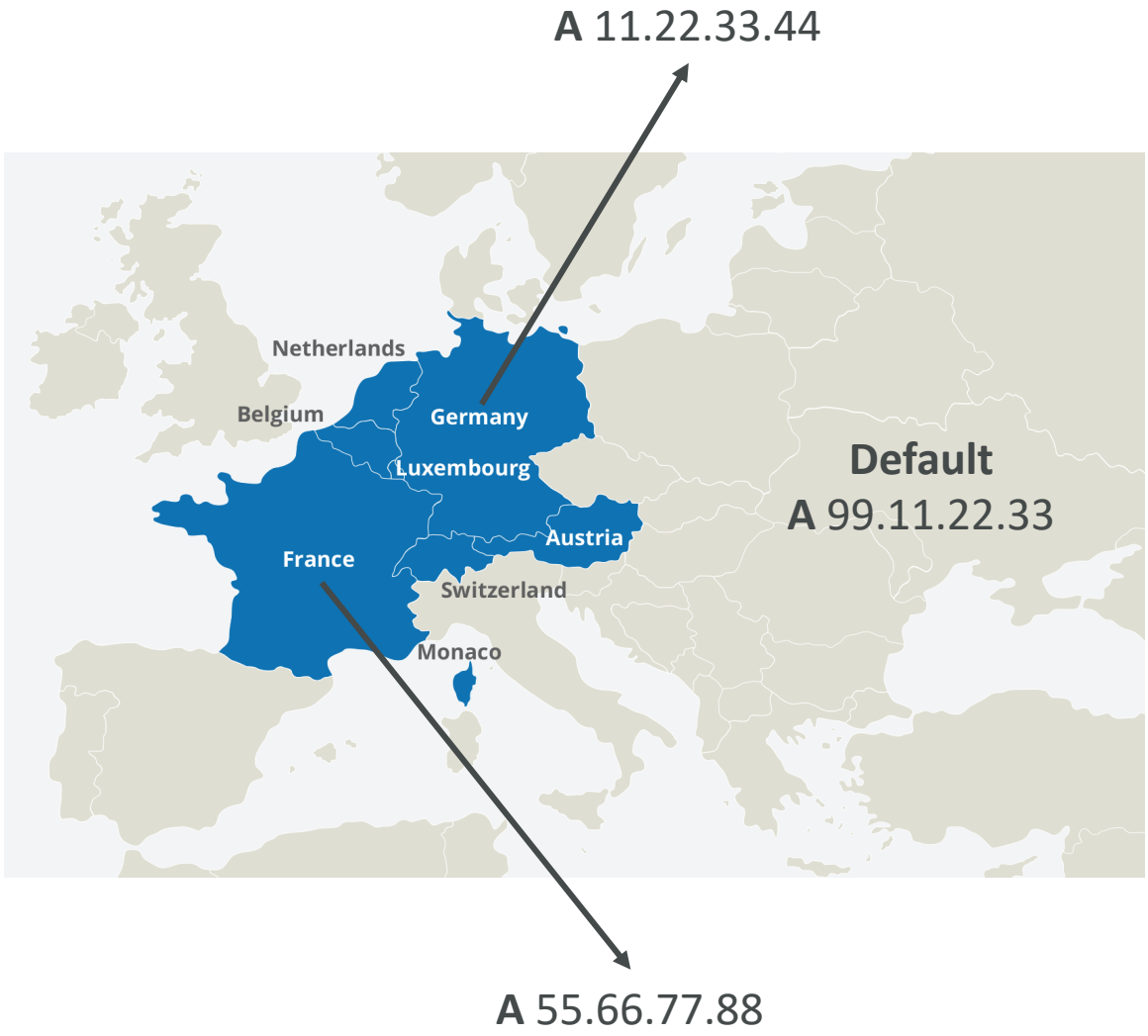


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Routing Policies – Geolocation

- Different from Latency-based!
- **This routing is based on user location**
- Specify location by Continent, Country or by US State (if there's overlapping, most precise location selected)
- Should create a **"Default"** record (in case there's no match on location)
- Use cases: website localization, restrict content distribution, load balancing, …
- Can be associated with Health Checks



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Routing Policies – Geoproximity

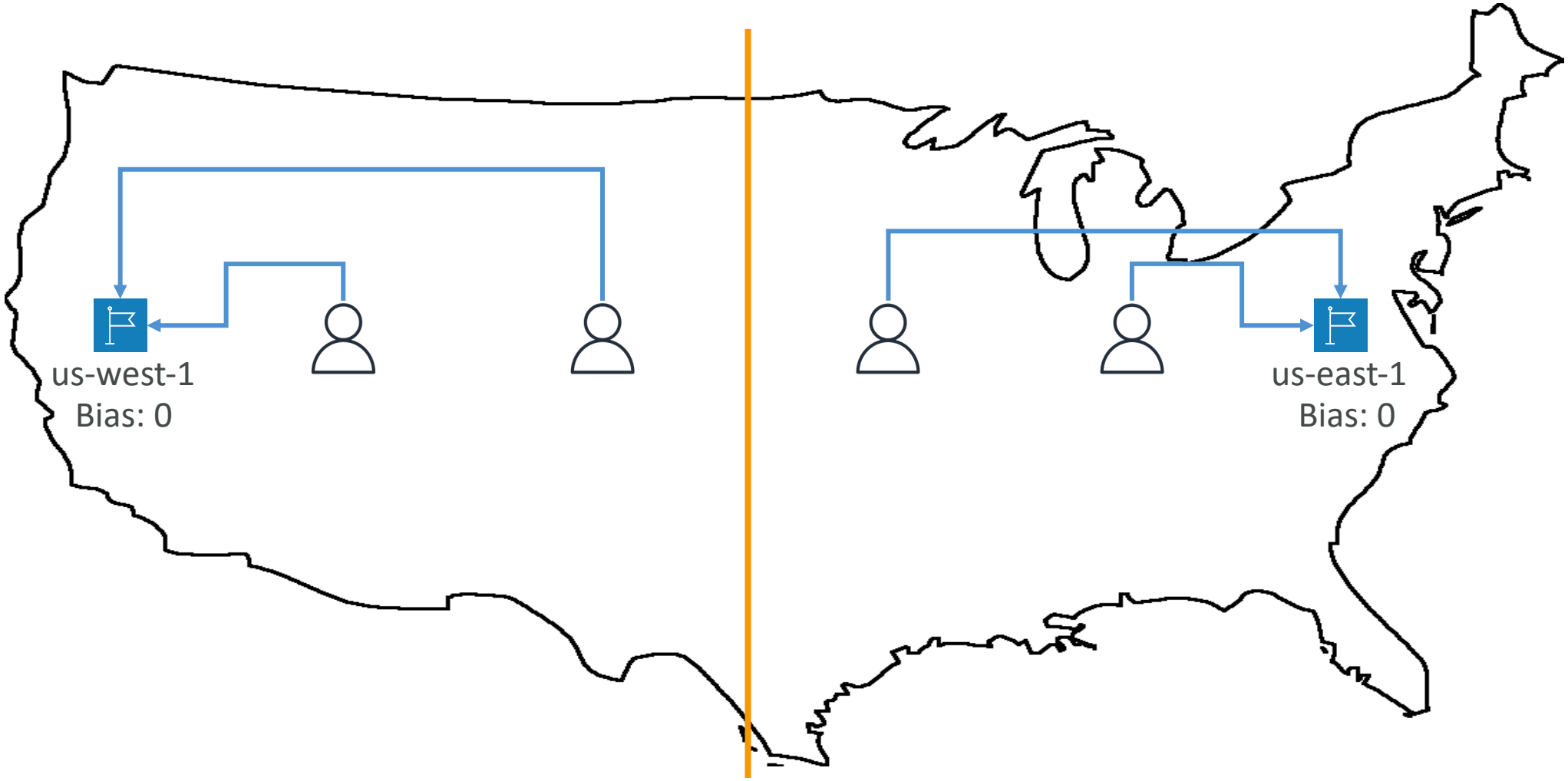
- Route traffic to your resources based on the geographic location of users and resources
- Ability **to shift more traffic to resources based** on the defined **bias**
- To change the size of the geographic region, specify **bias** values:
  - To expand (1 to 99) – more traffic to the resource
  - To shrink (-1 to -99) – less traffic to the resource

- Resources can be:
  - AWS resources (specify AWS region)
  - Non-AWS resources (specify Latitude and Longitude)
- You must use Route 53 **Traffic Flow** to use this feature

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

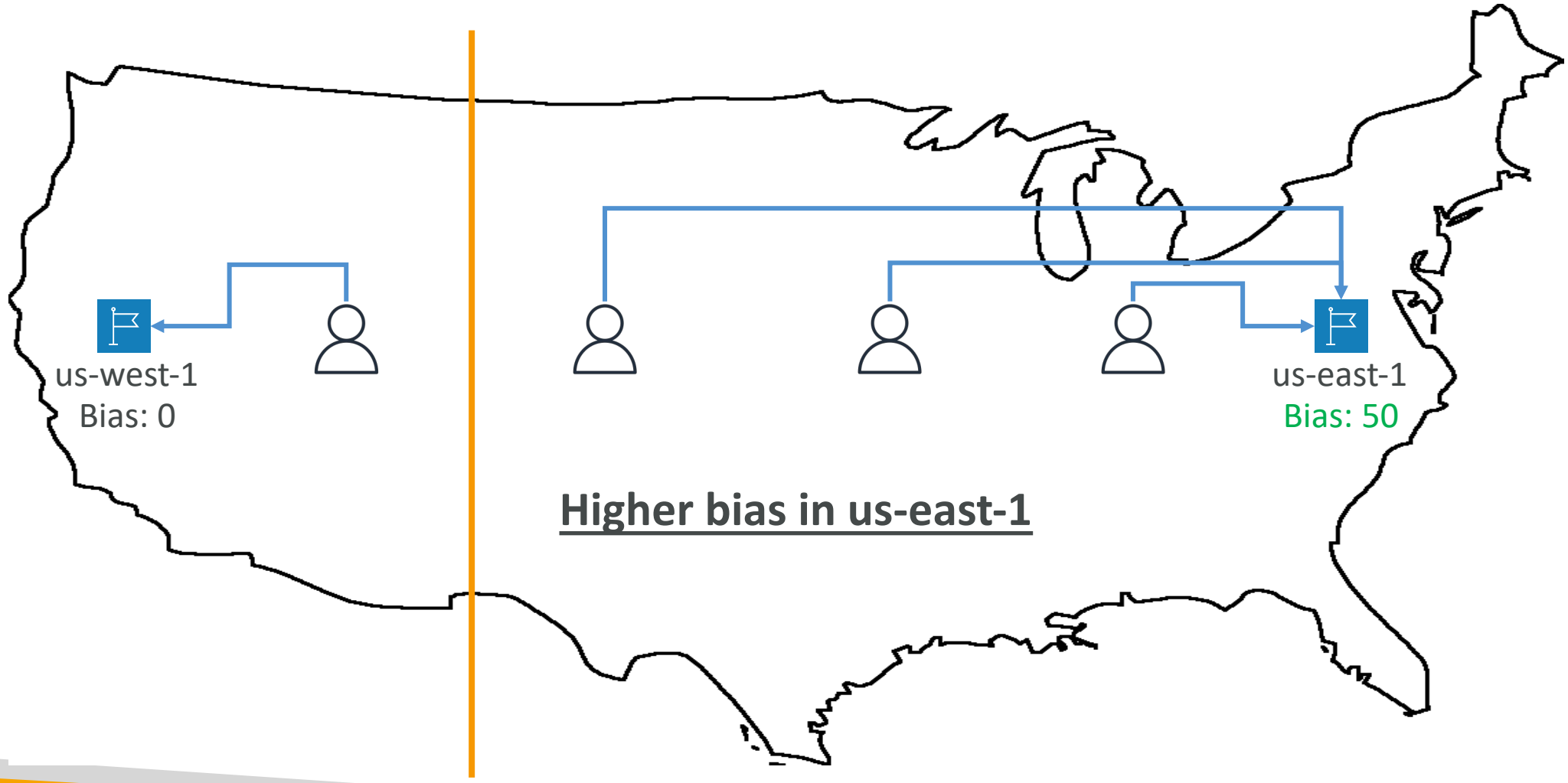
Routing Policies – Geoproximity



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Routing Policies – Geoproximity

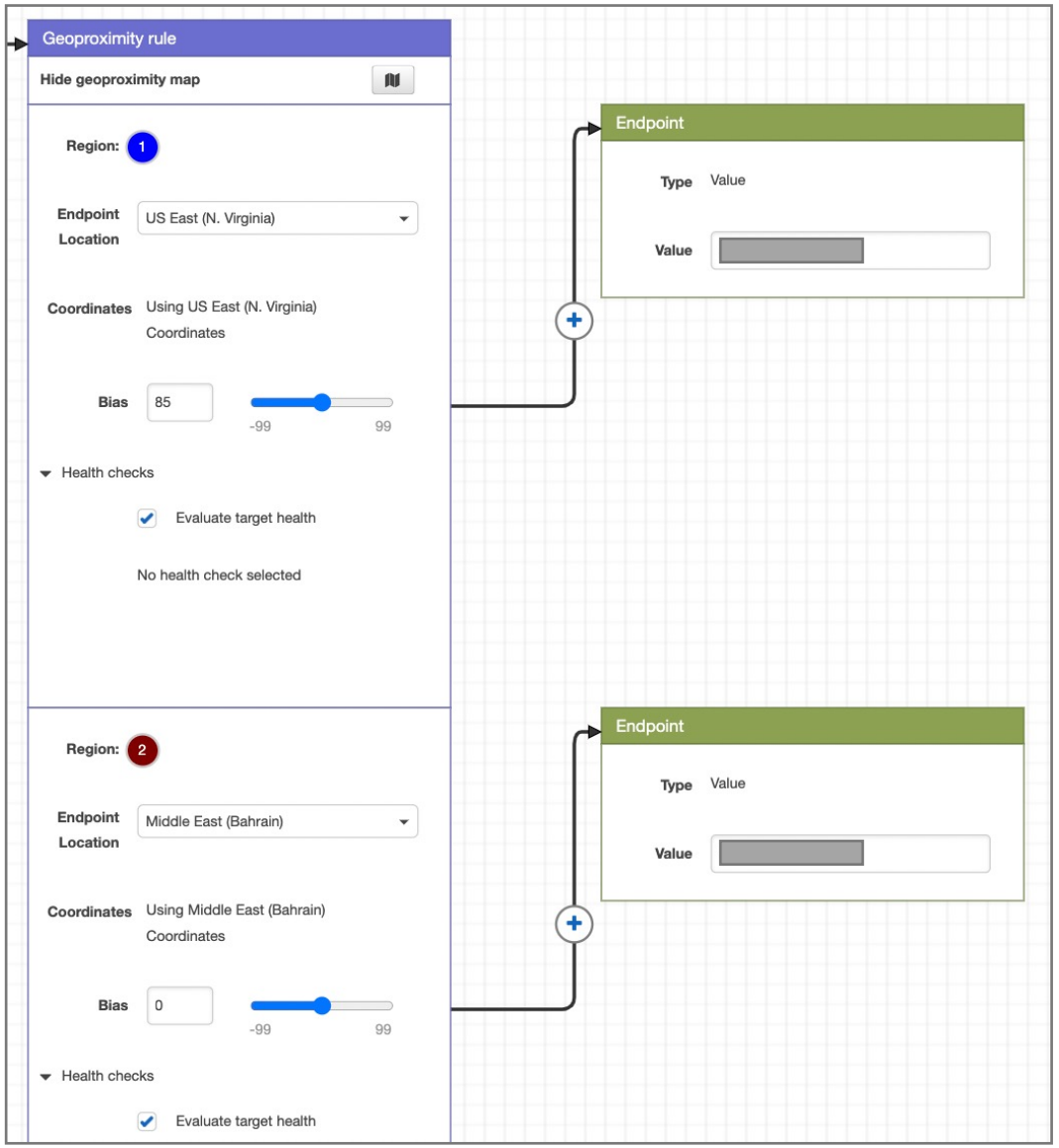


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route 53 – Traffic flow

- Simplify the process of creating and maintaining records in large and complex configurations
- Visual editor to manage complex routing decision trees
- Configurations can be saved as **Traffic Flow Policy**
  - Can be applied to different Route 53 Hosted Zones (different domain names)
  - Supports versioning



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Routing Policies – Multi-Value

- Use when routing traffic to multiple resources
- Route 53 return multiple values/resources
- Can be associated with Health Checks (return only values for healthy resources)
- Up to 8 healthy records are returned for each Multi-Value query
- Multi-Value is not a substitute for having an ELB

Name	Type	Value	TTL	Set ID	Health Check
www.example.com	A Record	192.0.2.2	60	Web1	A
www.example.com	A Record	198.51.100.2	60	Web2	B
www.example.com	A Record	203.0.113.2	60	Web3	C

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

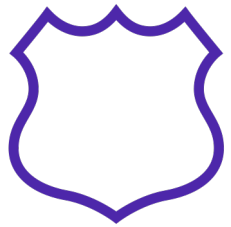
Routing Policies – IP-based Routing

- Routing is based on clients' IP addresses
- You provide a list of CIDRs for your clients and the corresponding endpoints/locations (user-IP-to-endpoint mappings)
- Use cases: Optimize performance, reduce network costs…
- Example: route end users from a particular ISP to a specific endpoint

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Route 53 – Hosted Zones



- A container for records that define how to route traffic to a domain and its subdomains

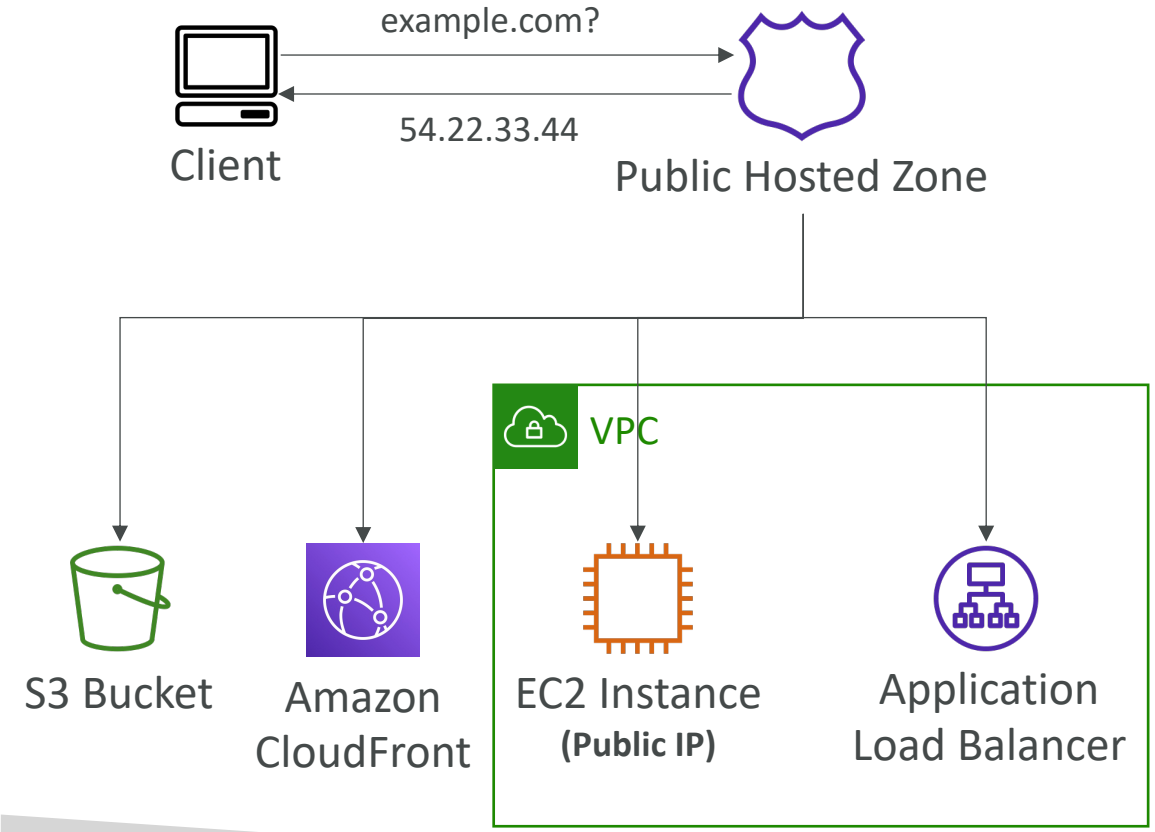
- **Public Hosted Zones** – contains records that specify how to route traffic on the Internet (public domain names)
  application1.mypublicdomain.com
- **Private Hosted Zones** – contain records that specify how you route traffic within one or more VPCs (private domain names)
  application1.company.internal

© Stephane Maarek

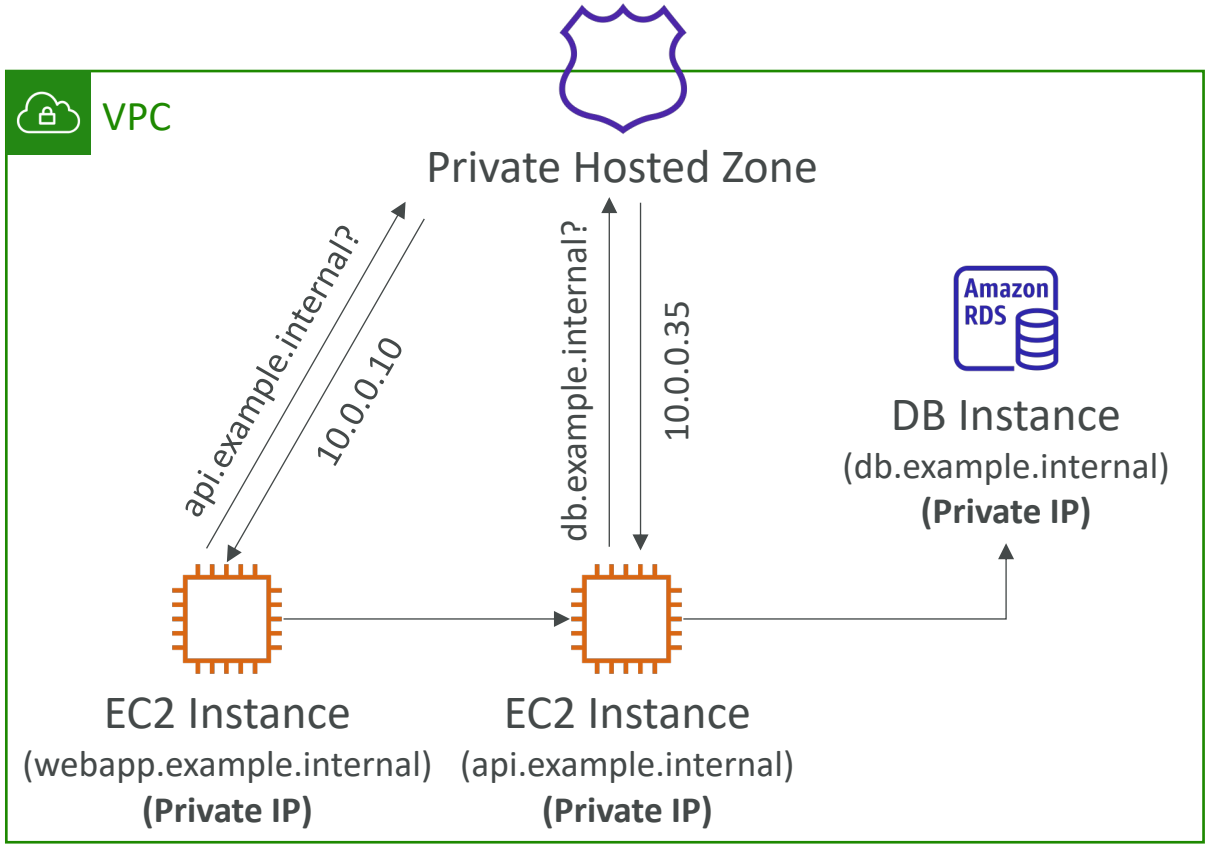
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route 53 – Public vs. Private Hosted Zones

Public Hosted Zone



Private Hosted Zone



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route 53 – Good to Know

- For internal private DNS (Private Hosted Zone), you must enable the VPC settings **enableDnsHostnames** and **enableDnsSupport**

- **DNS Security Extensions (DNSSEC)**
  - A protocol for securing DNS traffic, verifies DNS data integrity and origin
  - Protects against Man in the Middle (MITM) attacks
  - Route 53 supports both DNSSEC for Domain Registeration and DNSSEC Signing
  - Works only with **Public Hosted Zones**

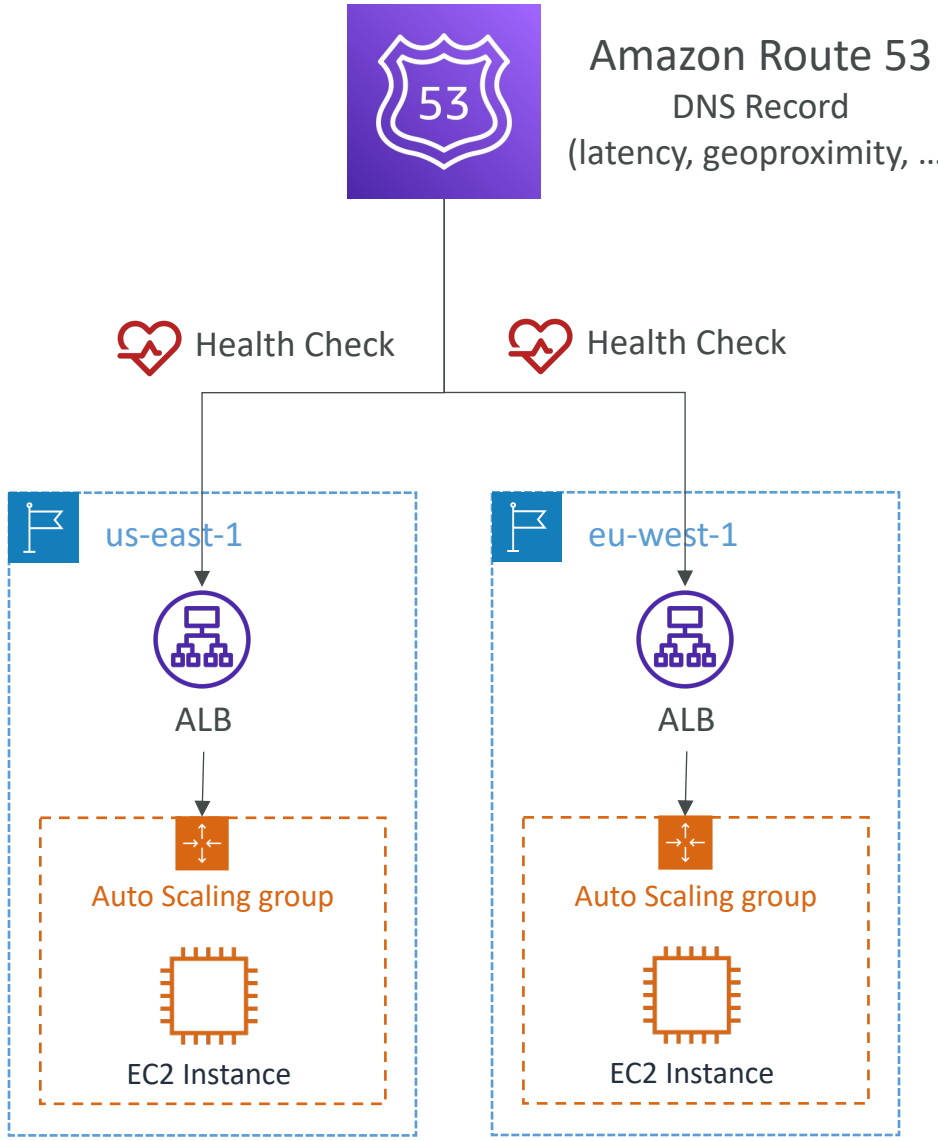
- **Route 53 with 3rd Registrar**
  - You can buy the domain out of AWS and use Route 53 as the DNS provider
  - Update the NS records on the 3rd party Registrar

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route 53 – Health Checks

- HTTP Health Checks are only for **public resources**
- Health Check => Automated DNS Failover:
  1. Health checks that monitor an endpoint (application, server, other AWS resource)
  2. Health checks that monitor other health checks (Calculated Health Checks)
  3. Health checks that monitor CloudWatch Alarms (full control !!) – e.g., throttles of DynamoDB, alarms on RDS, custom metrics, … (helpful for private resources)
- Health Checks are integrated with CW metrics

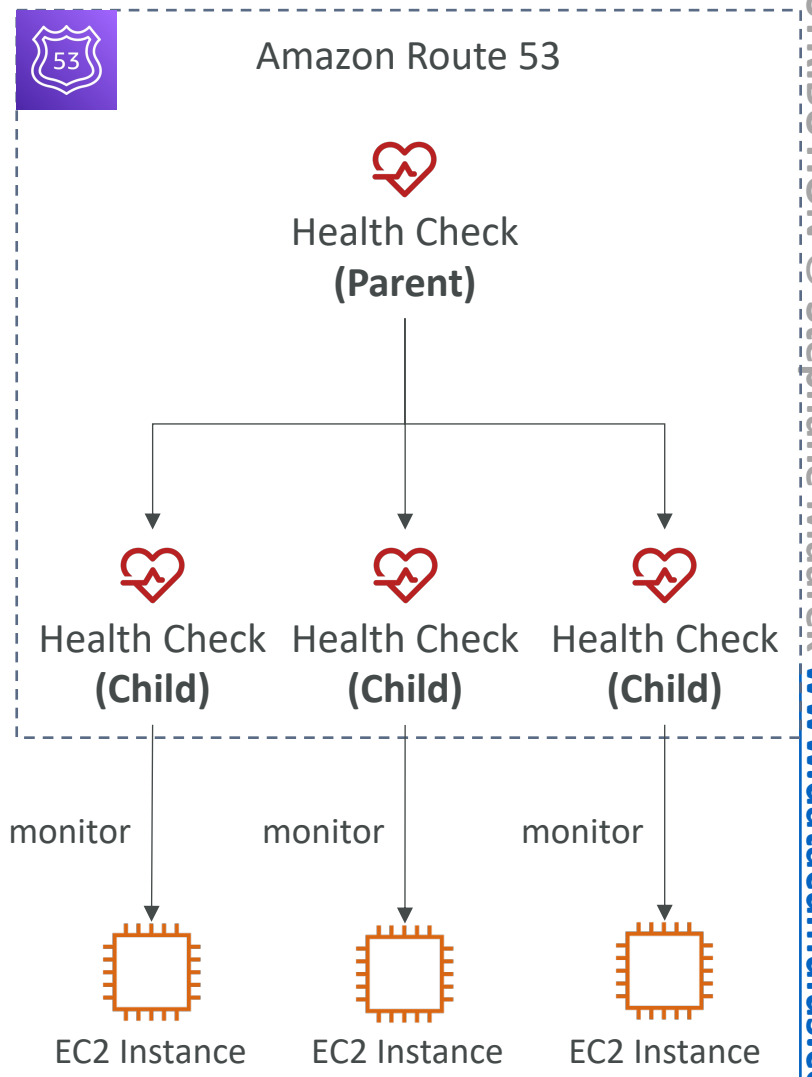


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route 53 – Calculated Health Checks

- Combine the results of multiple Health Checks into a single Health Check
- You can use **OR**, **AND**, or **NOT**
- Can monitor up to 256 Child Health Checks
- Specify how many of the health checks need to pass to make the parent pass
- Usage: perform maintenance to your website without causing all health checks to fail

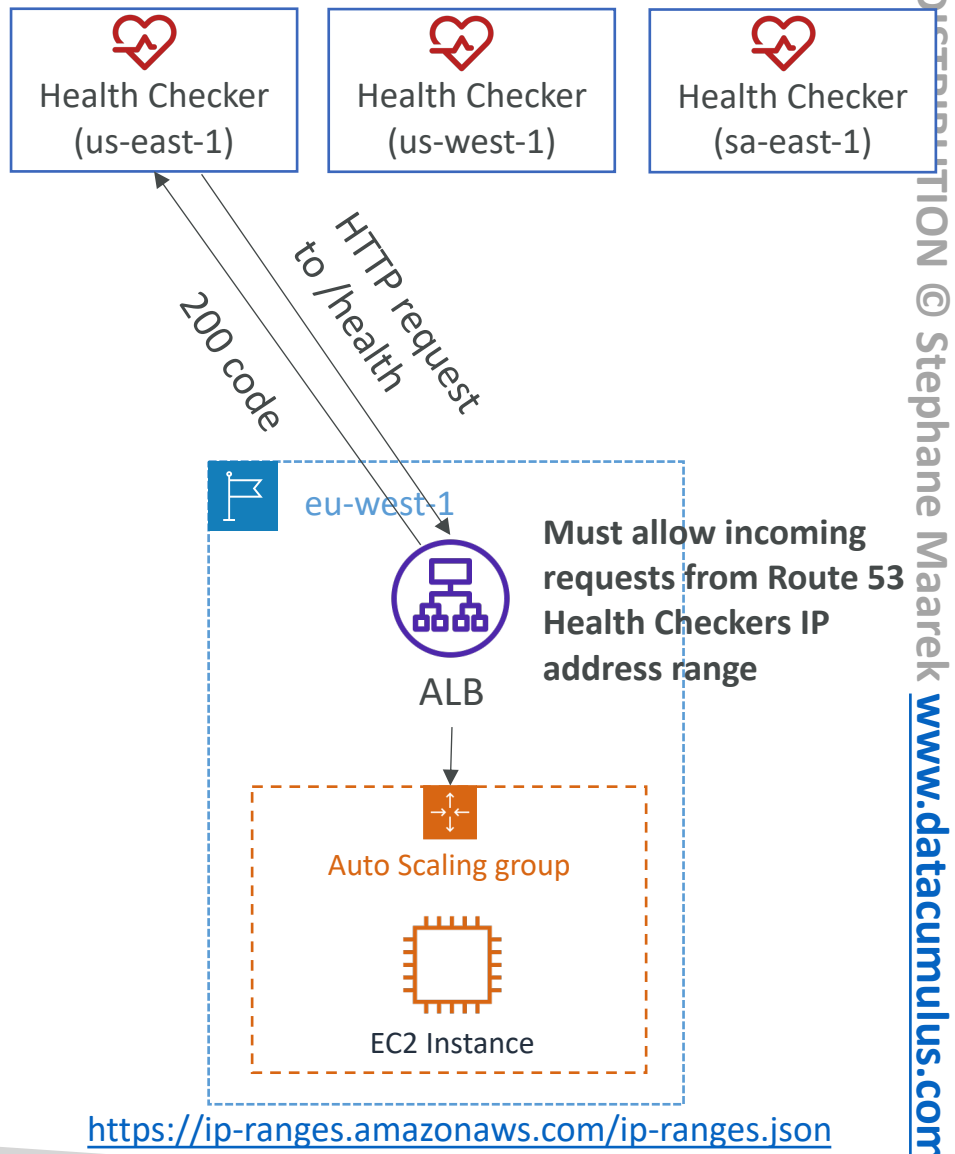


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Health Checks – Monitor an Endpoint

- About 15 global health checkers will check the endpoint health
- Health Checks pass only when the endpoint responds with the 2xx and 3xx status codes
- Health Checks can be setup to pass / fail based on the text in the first 5120 bytes of the response

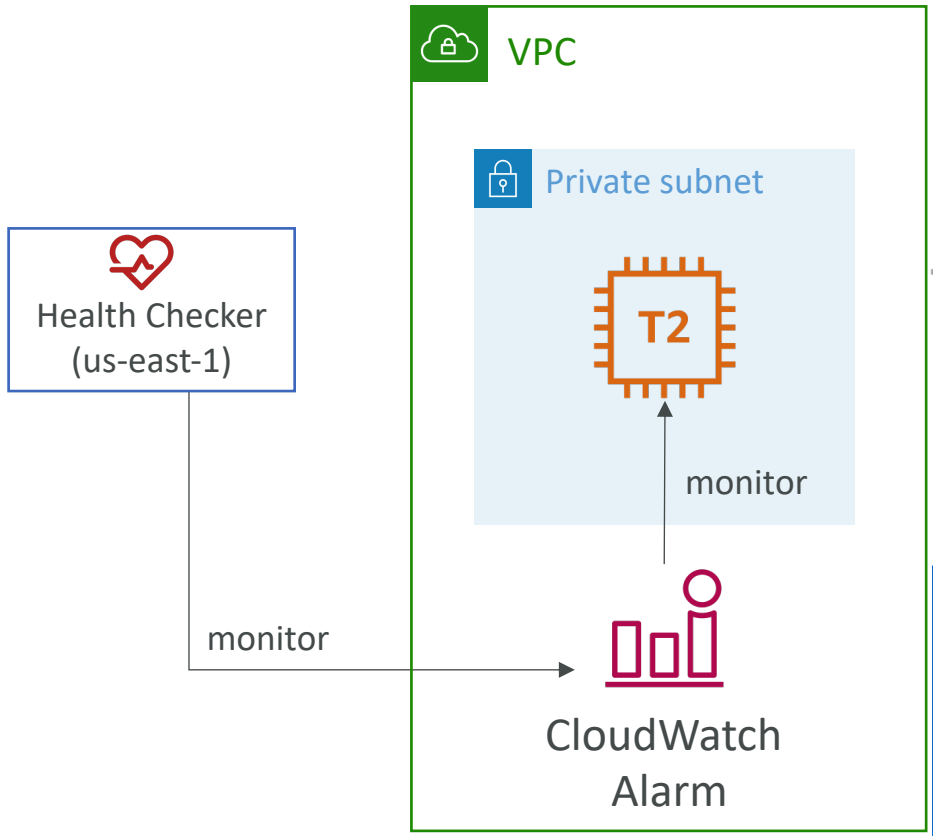


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Health Checks – Private Hosted Zones

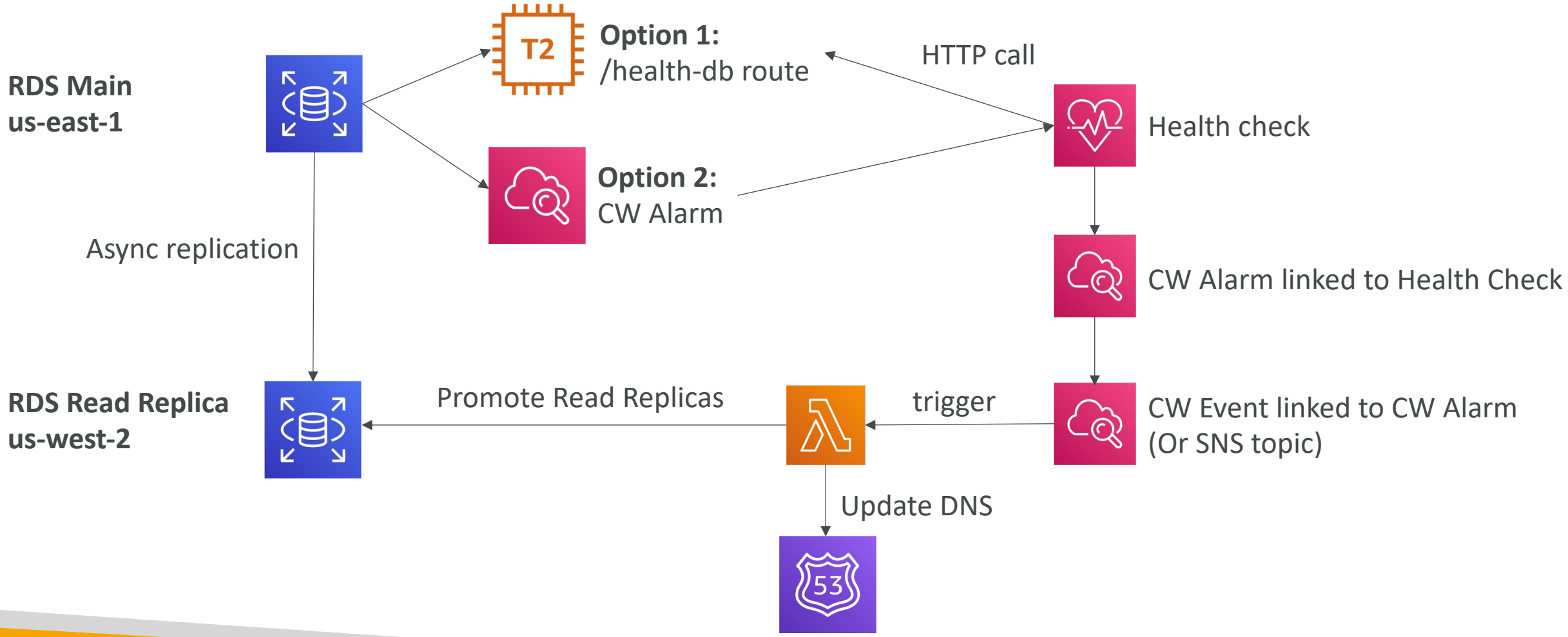
- Route 53 health checkers are outside the VPC
- They can't access **private** endpoints (private VPC or on-premises resource)
- You can create a **CloudWatch Metric** and associate a **CloudWatch Alarm**, then create a Health Check that checks the alarm itself



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Health Checks Solution Architecture
RDS multi-region failover

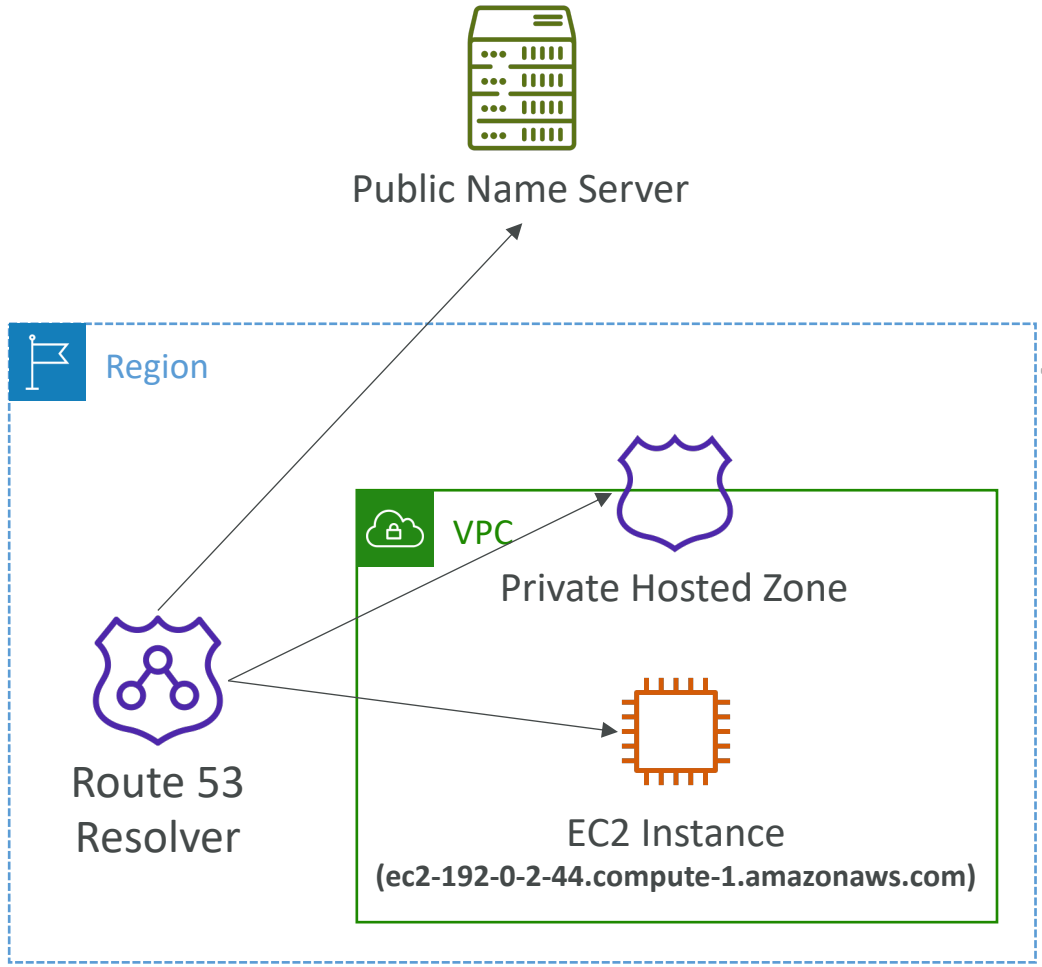


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route 53 – Hybrid DNS

- By default, Route 53 Resolver automatically answers DNS queries for:
  - Local domain names for EC2 instances
  - Records in Private Hosted Zones
  - Records in public Name Servers
- **Hybrid DNS** – resolving DNS queries between VPC (Route 53 Resolver) and your networks (other DNS Resolvers)
- Networks can be:
  - VPC itself / Peered VPC
  - On-premises Network (connected through Direct Connect or AWS VPN)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

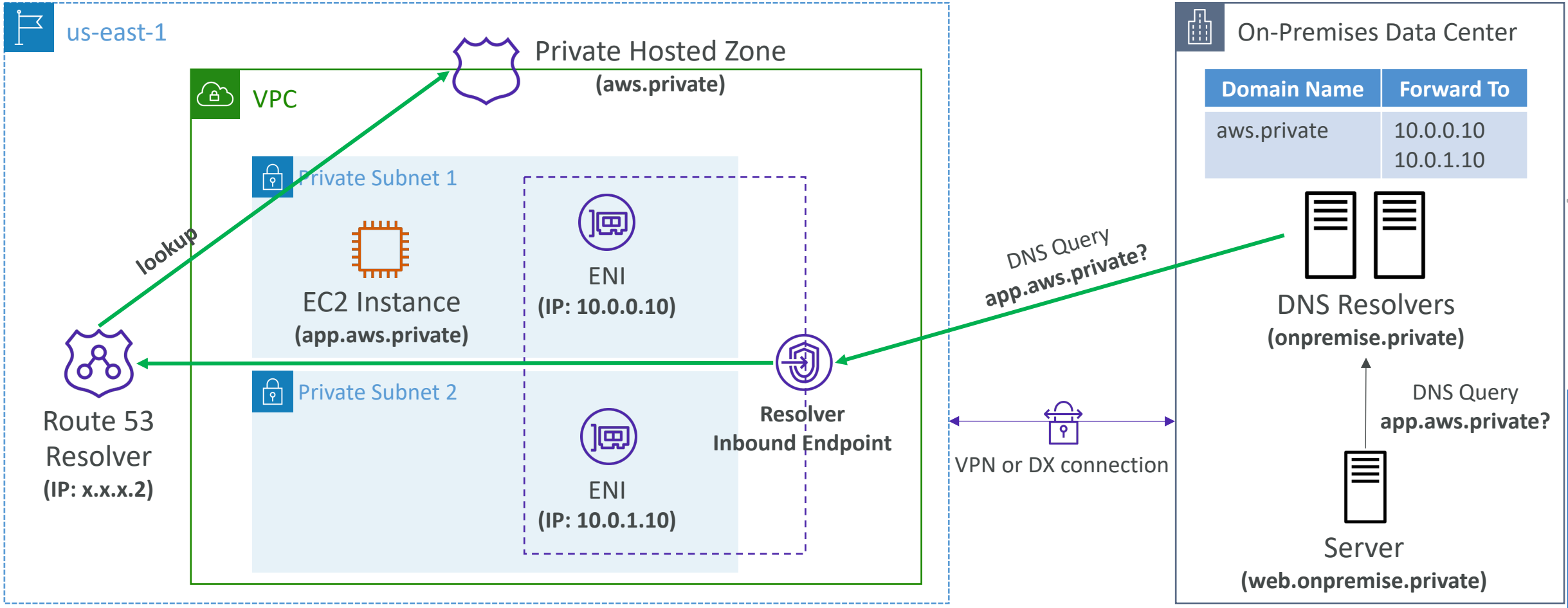
Route 53 – Resolver Endpoints

- Inbound Endpoint
  - DNS Resolvers on your network can forward DNS queries to Route 53 Resolver
  - Allows your DNS Resolvers to resolve domain names for AWS resources (e.g., EC2 instances) and records in Route 53 Private Hosted Zones
- Outbound Endpoint
  - Route 53 Resolver conditionally forwards DNS queries to your DNS Resolvers
  - Use **Resolver Rules** to forward DNS queries to your DNS Resolvers
- Associated with one or more VPCs in the same AWS Region
- Create in two AZs for high availability
- Each Endpoint supports 10,000 queries per second per IP address

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

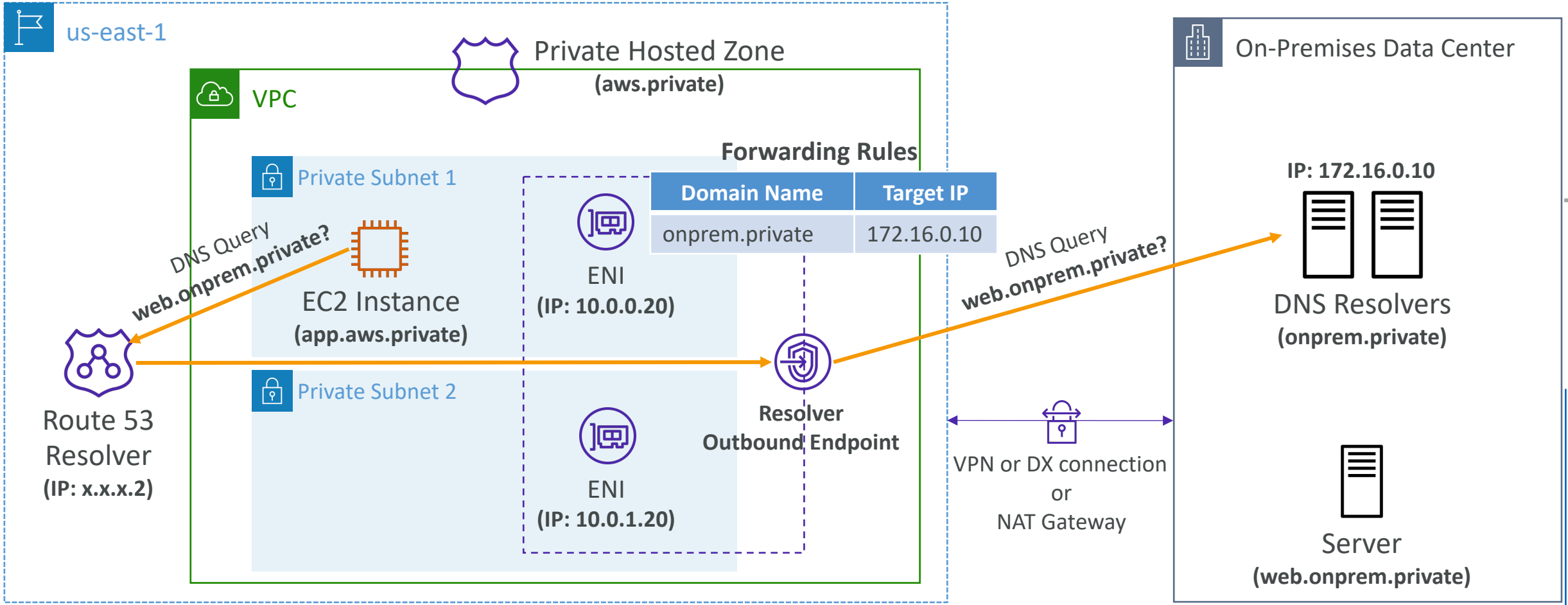
Route 53 – Resolver Inbound Endpoints



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route 53 – Resolver Outbound Endpoints

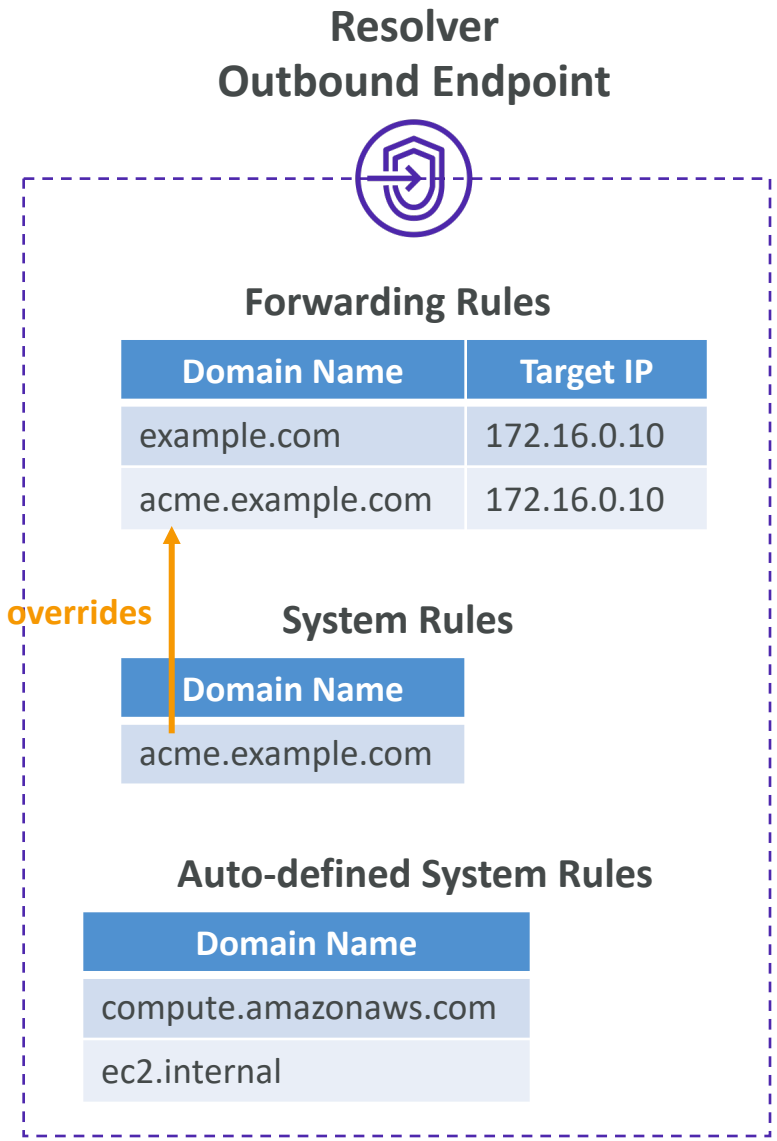


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route 53 – Resolver Rules

- Control which DNS queries are forwarded to DNS Resolvers on your network
- Conditional Forwarding Rules (Forwarding Rules)
  - Forward DNS queries for a specified domain and all its subdomains **to target IP addresses**
- System Rules
  - Selectively overriding the behavior defined in Forwarding Rules (e.g., don't forward DNS queries for a subdomain acme.example.com)
- Auto-defined System Rules
  - Defines how DNS queries for selected domains are resolved (e.g., AWS internal domain names, Privated Hosted Zones)
- If multiple rules matched, Route 53 Resolver chooses the most specific match
- Resolver Rules can be shared across accounts using AWS RAM
  - Manage them centrally in one account
  - Send DNS queries from multiple VPC to the target IP defined in the rule

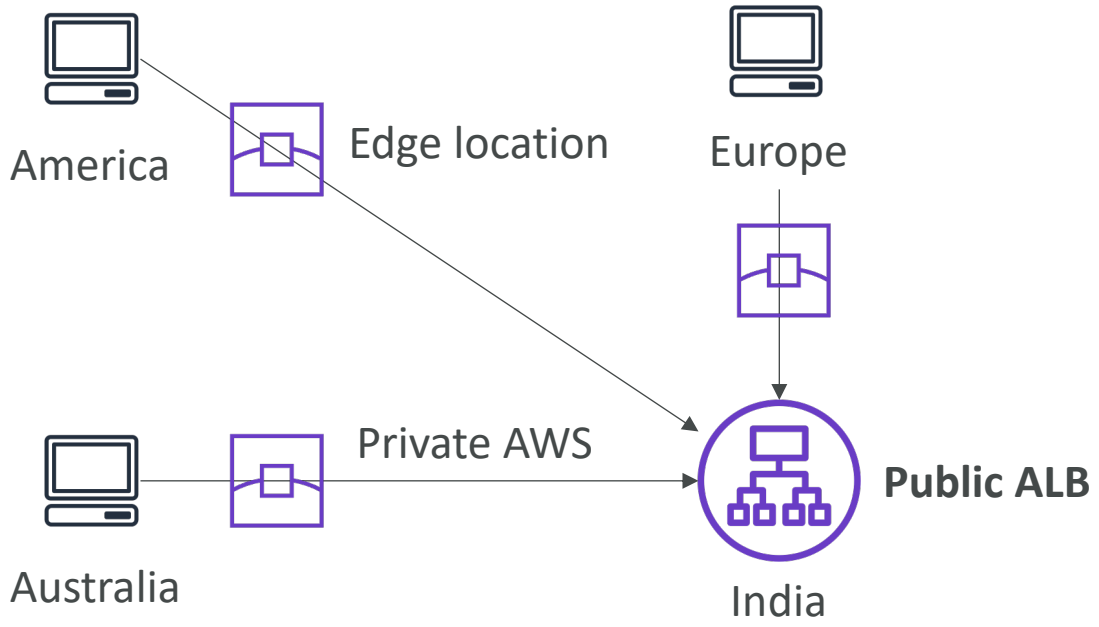


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Global Accelerator

- Leverage the AWS internal network to route to your application
- 2 Anycast IP are created for your application
- The Anycast IP send traffic directly to Edge Locations
- The Edge locations send the traffic to your application



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Global Accelerator

- Works with Elastic IP, EC2 instances, ALB, NLB, public or private
- Supports Client IP Address Preservation **except EIPs endpoints**
- Consistent Performance
  - Intelligent routing to lowest latency and fast regional failover
  - No issue with client cache (because the IP doesn't change)
  - Internal AWS network
- Health Checks
  - Global Accelerator performs a health check of your applications
  - Helps make your application global (failover less than 1 minute for unhealthy)
  - Great for disaster recovery (thanks to the health checks)
- Security
  - only 2 external IP need to be whitelisted
  - DDoS protection thanks to AWS Shield

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Global Accelerator vs CloudFront

- They both use the AWS global network and its edge locations around the world
- Both services integrate with AWS Shield for DDoS protection.

- CloudFront
  - Improves performance for both cacheable content (such as images and videos)
  - Dynamic content (such as API acceleration and dynamic site delivery)
  - Content is served at the edge

- Global Accelerator
  - Improves performance for a wide range of applications over TCP or UDP
  - Proxying packets at the edge to applications running in one or more AWS Regions.
  - Good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP
  - Good for HTTP use cases that require static IP addresses
  - Good for HTTP use cases that required deterministic, fast regional failover

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

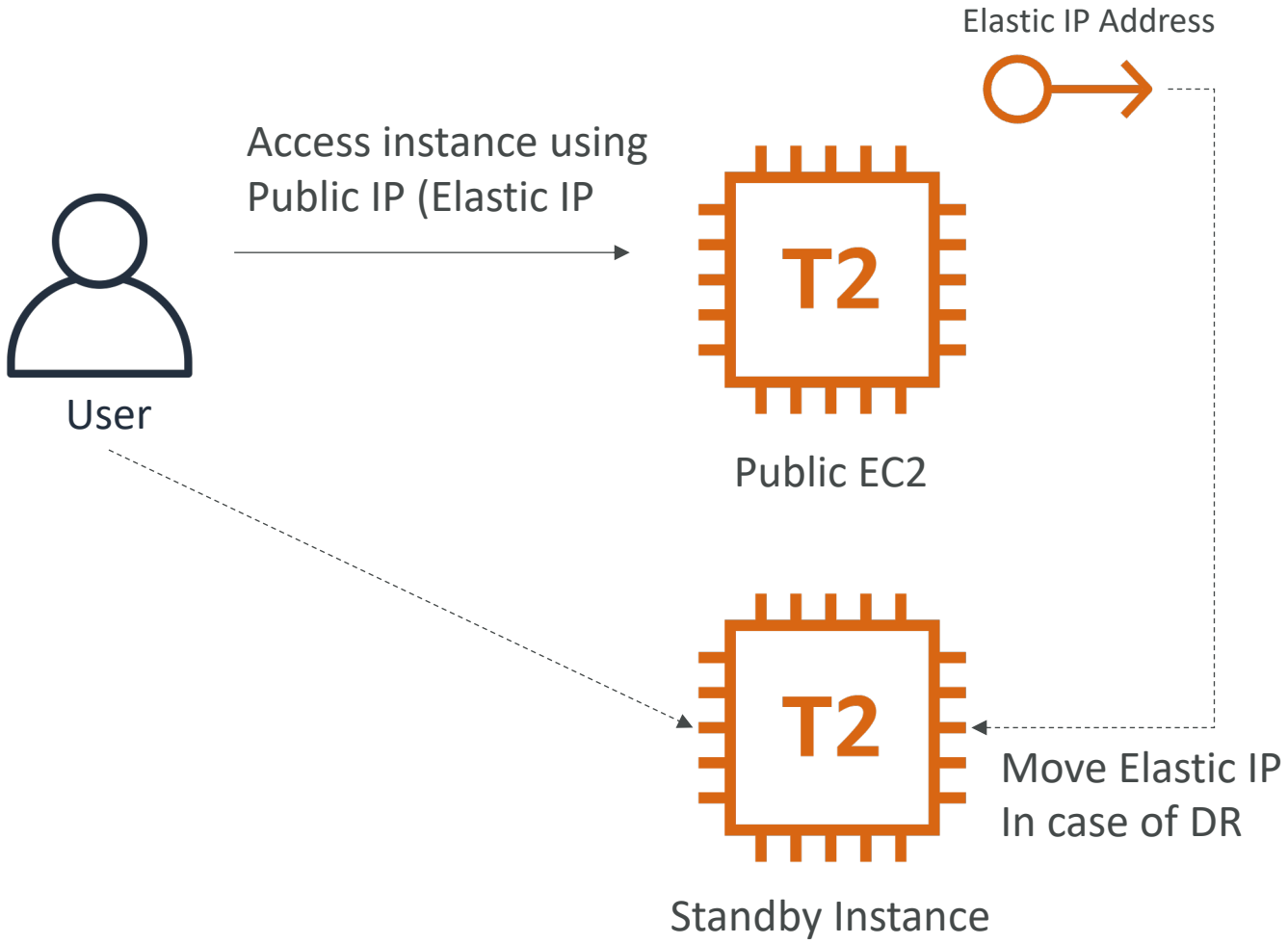
Solution Architecture Comparisons

- EC2 on its own with Elastic IP
- EC2 with Route53
- ALB + ASG
- ALB + ECS on EC2
- ALB + ECS on Fargate
- ALB + Lambda
- API Gateway + Lambda
- API Gateway + AWS Service
- API Gateway + HTTP backend (ex: ALB)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EC2 with Elastic IP

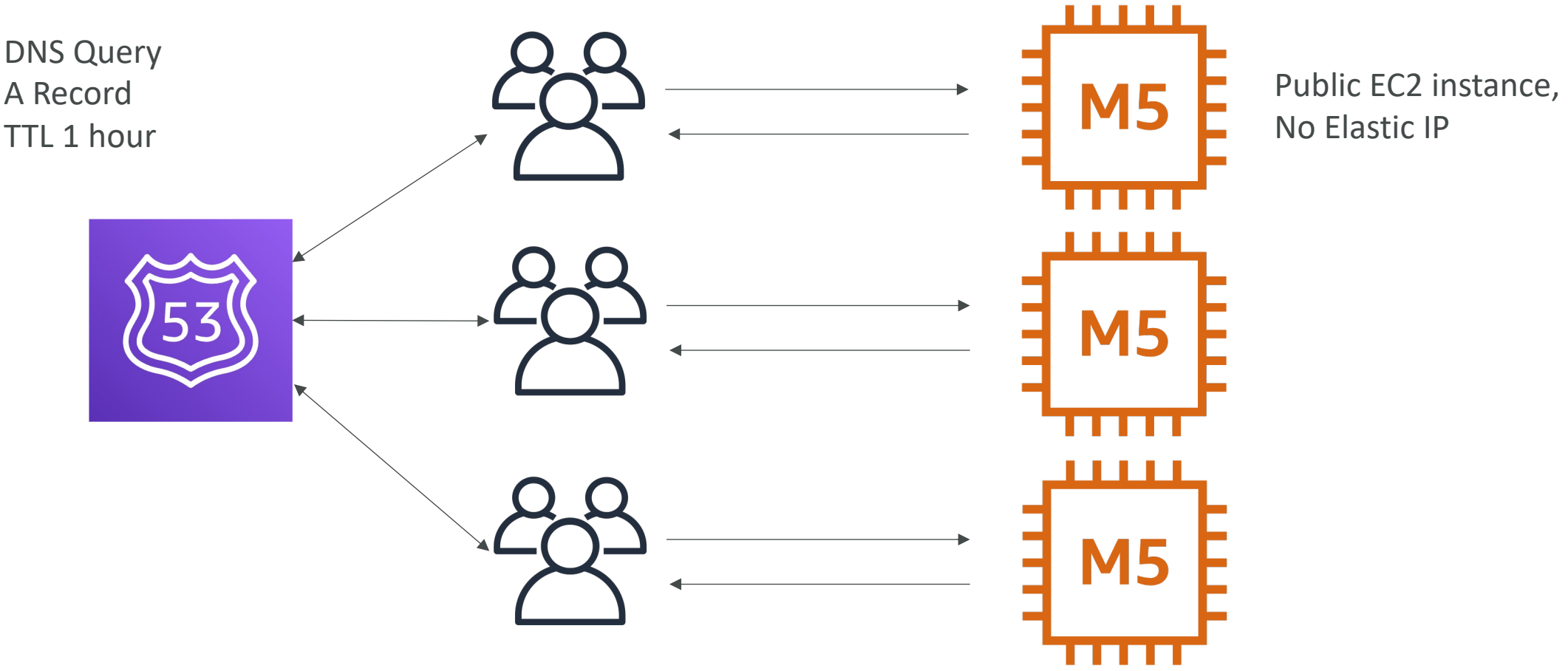


- Quick failover
- The client should not see the change happen
- Helpful if the client needs to resolve by static Public IP address
- Does not scale
- Cheap

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

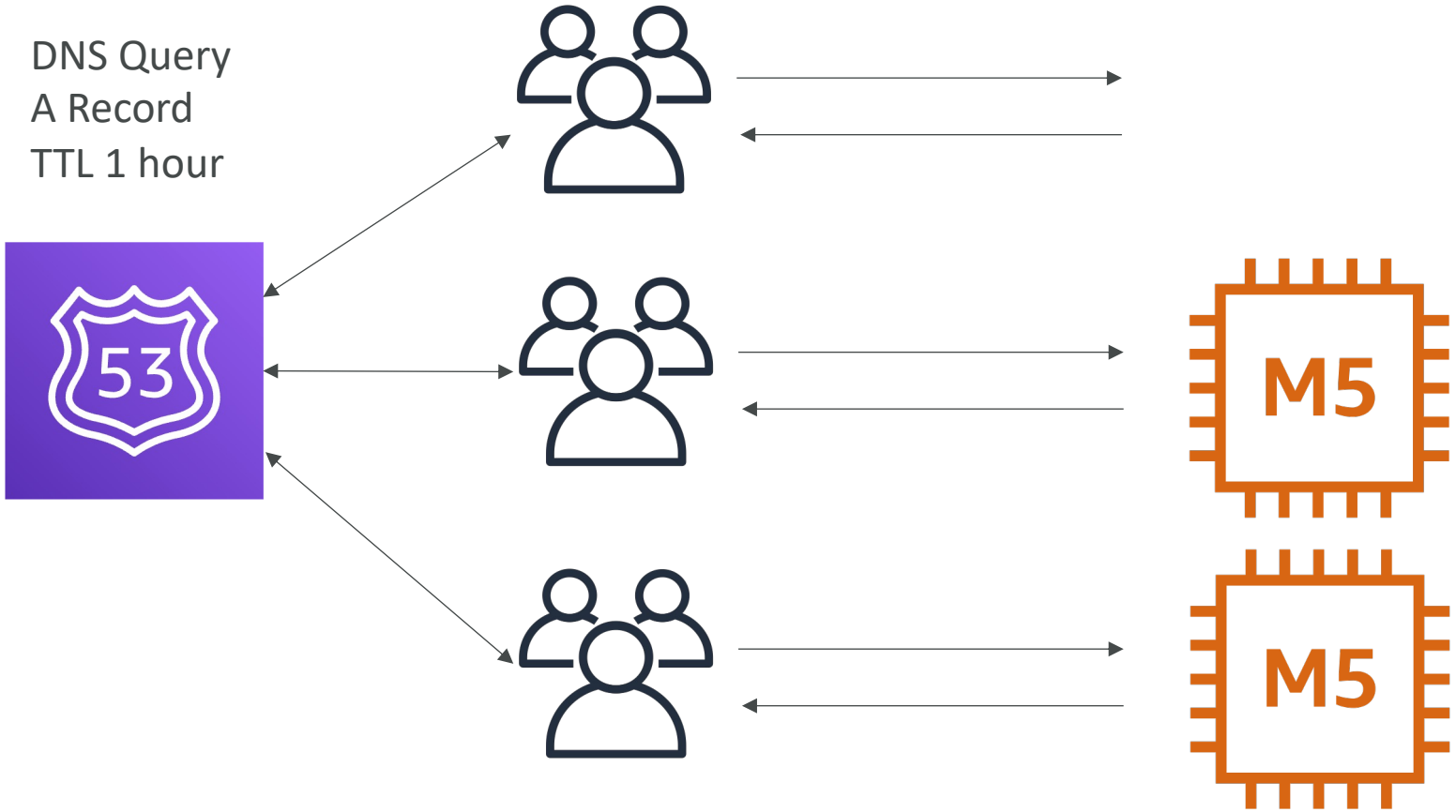
Stateless web app - scaling horizontally



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Stateless web app - scaling horizontally

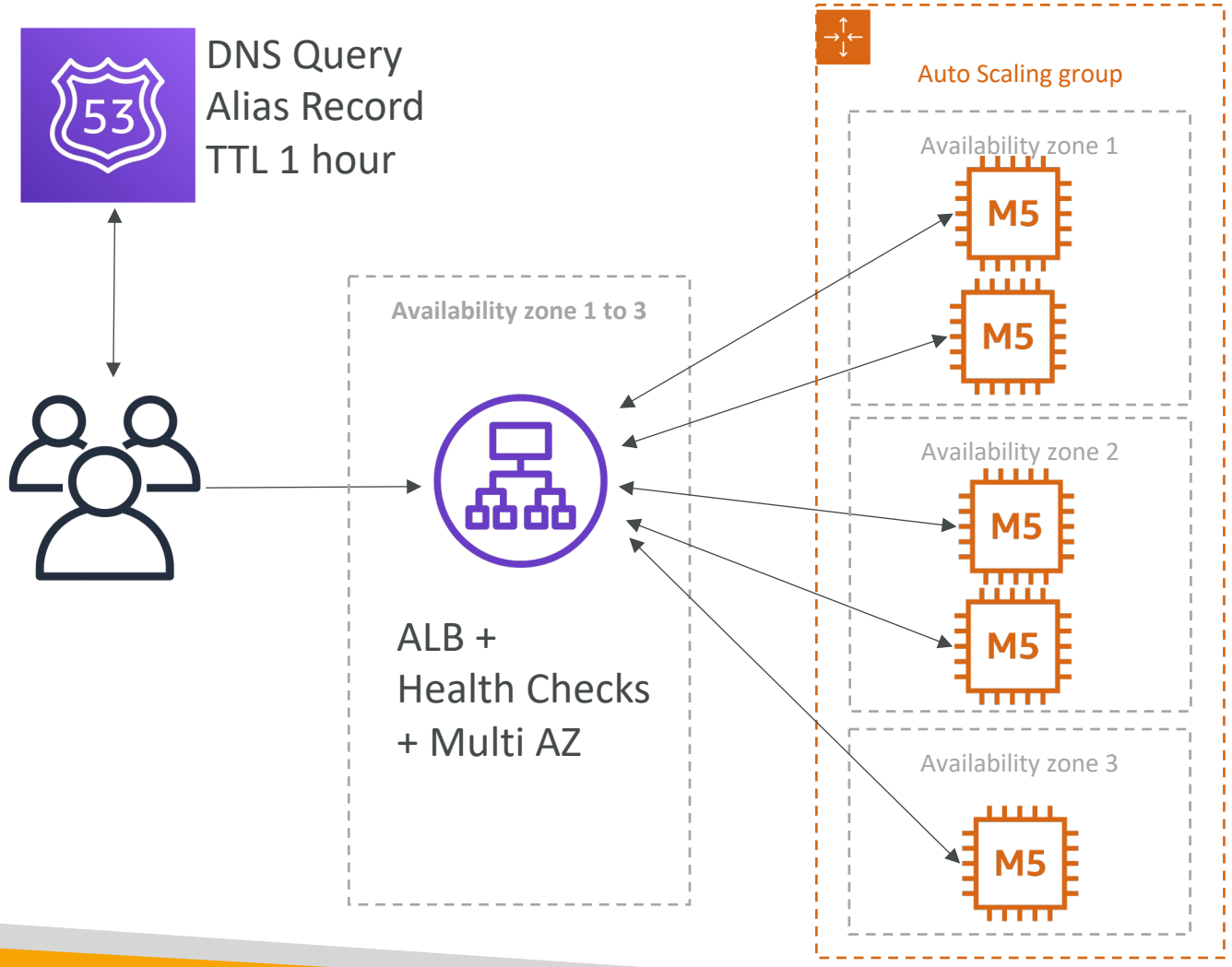


- "DNS-based load balancing"
- Ability to use multiple instances
- Route53 TTL implies client may get outdated information
- Clients must have logic to deal with hostname resolution failures
- Adding an instance may not receive full traffic right away due to DNS TTL

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

ALB + ASG



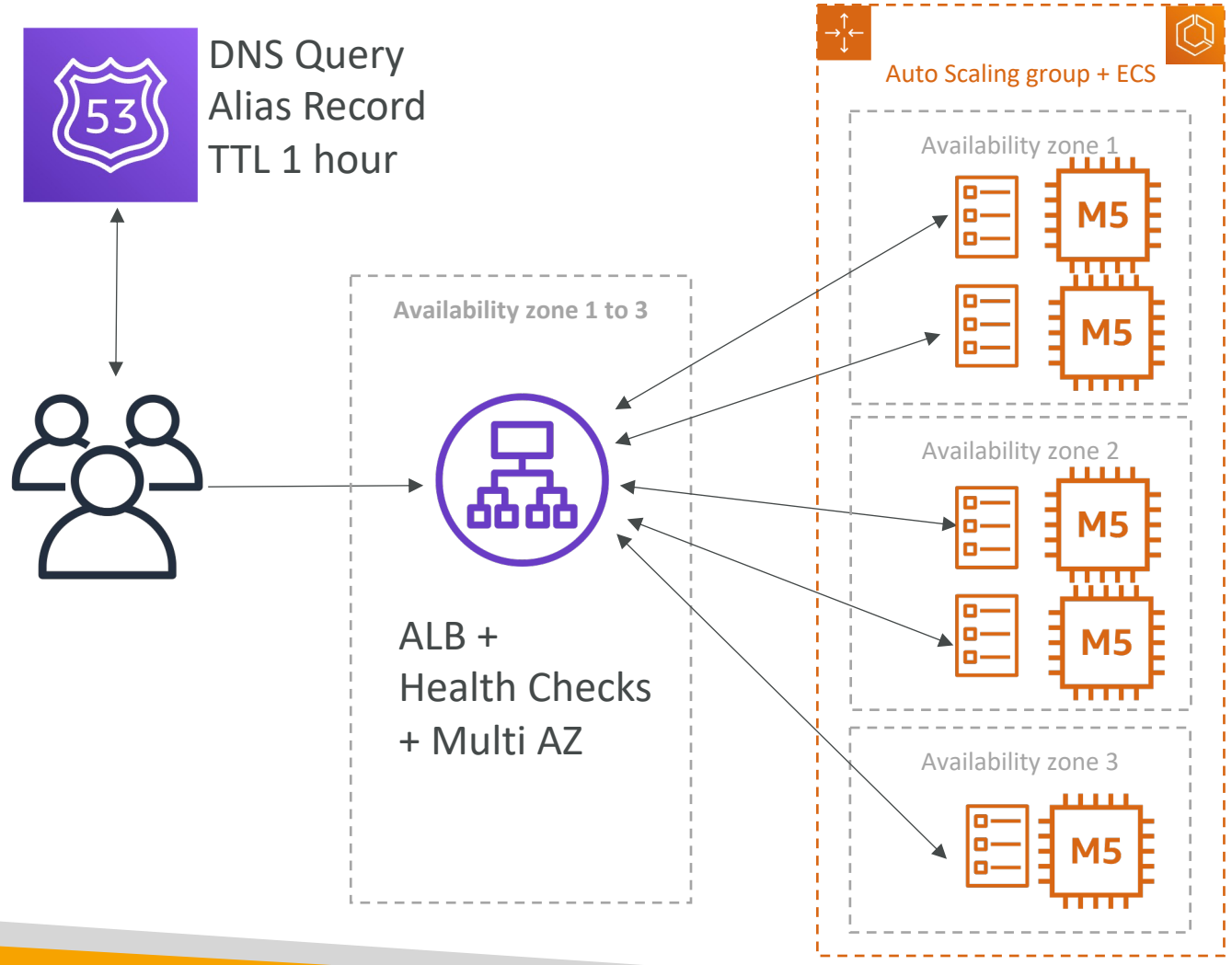
- Scales well, classic architecture
- New instances are in service right away.
- Users are not sent to instances that are out-of-service
- Time to scale is slow (EC2 instance startup + bootstrap) – AMI can help
- ALB is elastic but can't handle sudden, huge peak of demand (pre-warm)
- Could lose a few requests if instances are overloaded
- CloudWatch used for scaling
- Cross-Zone balancing for even traffic distribution

- Target utilization should be between 40% and 70%

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

ALB + ECS on EC2 (backed by ASG)

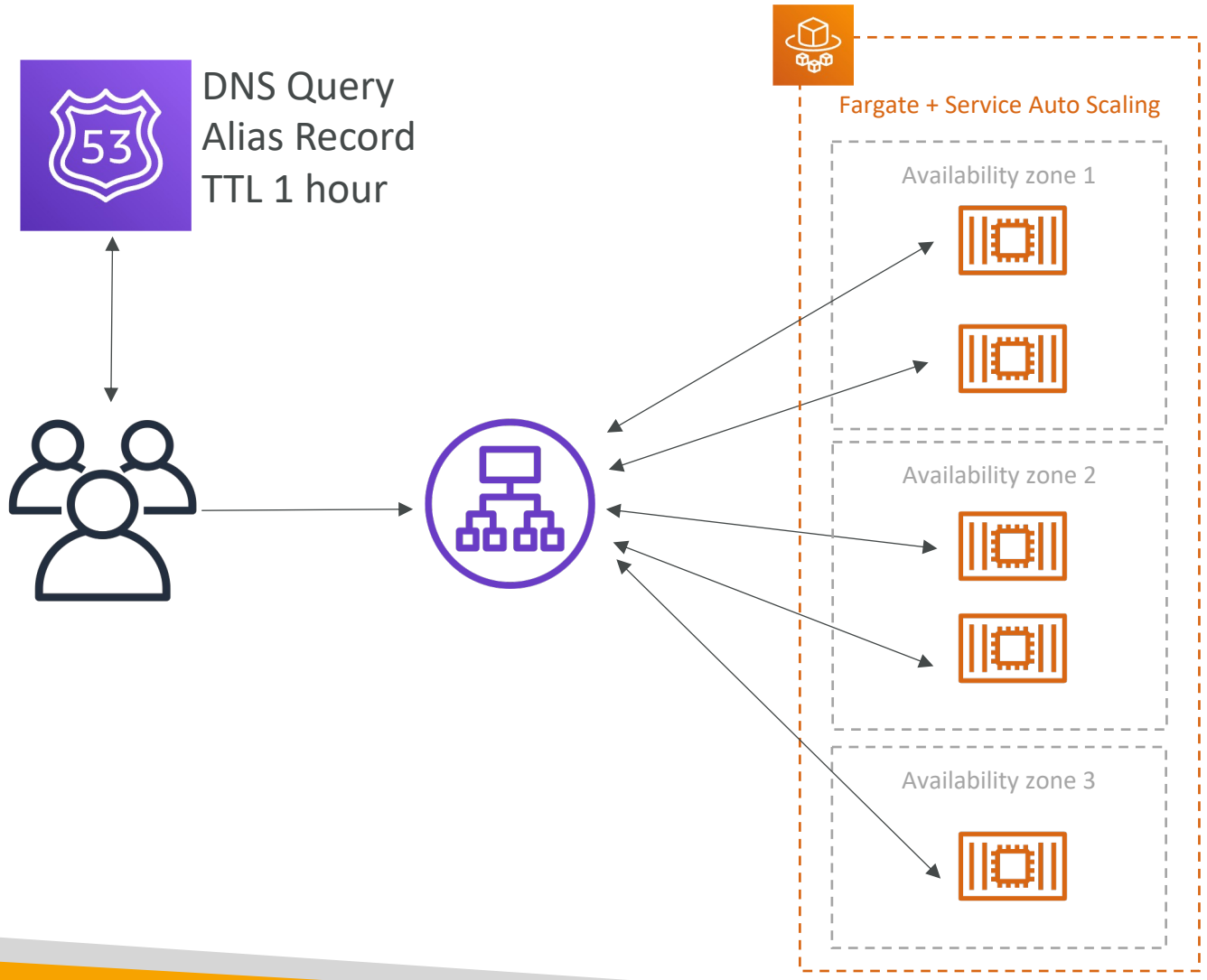


- Same properties as ALB + ASG
- Application is run on Docker
- ASG + ECS allows to have **dynamic port mappings**
- Tough to orchestrate ECS service auto-scaling + ASG auto-scaling

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

ALB + ECS on Fargate

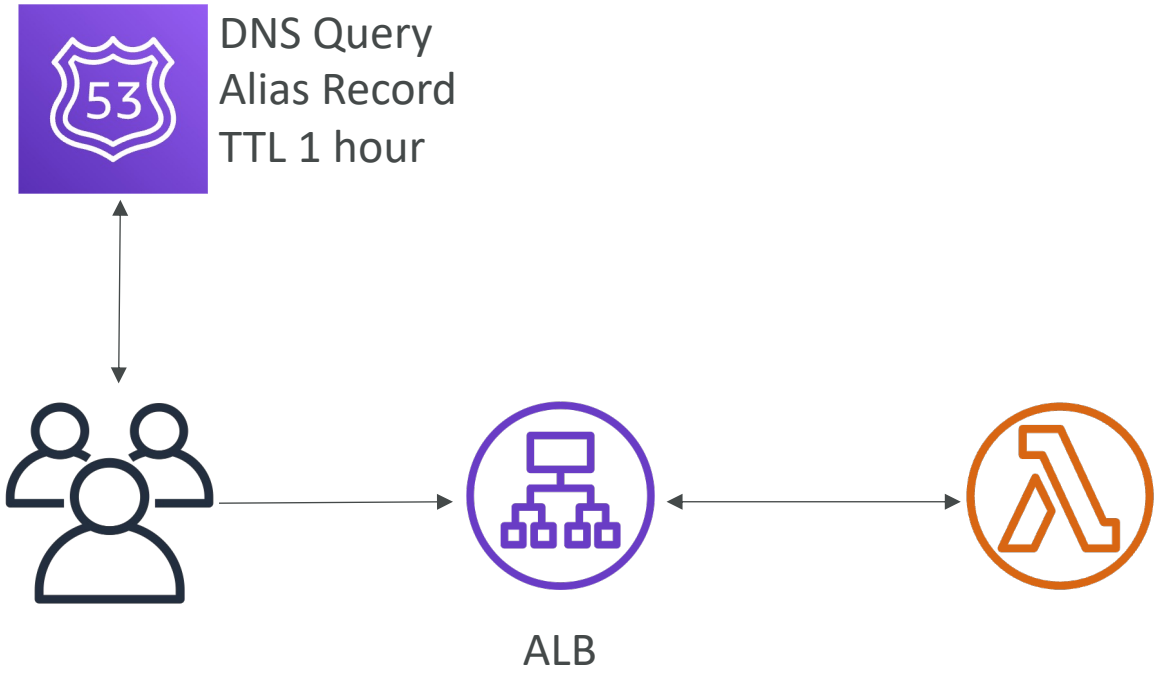


- Application is run on Docker
- Service Auto Scaling is easy
- Time to be in-service is quick (no need to launch an EC2 instance in advance)
- Still limited by the ALB in case of sudden peaks
- "serverless" application tier
- "managed" load balancer

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

ALB + Lambda

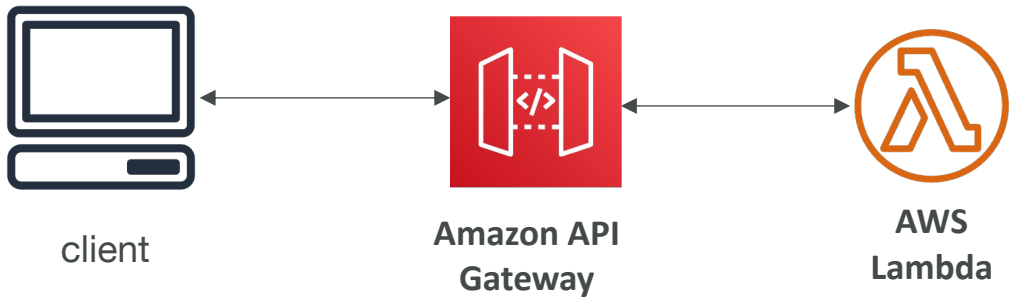


- Limited to Lambda's runtimes
- Seamless scaling thanks to Lambda
- Simple way to expose Lambda functions as HTTP/S without all the features from API Gateway
- Can combine with WAF (Web Application Firewall)
- Good for hybrid microservices
- Example: use ECS for some requests, use Lambda for others

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway + Lambda



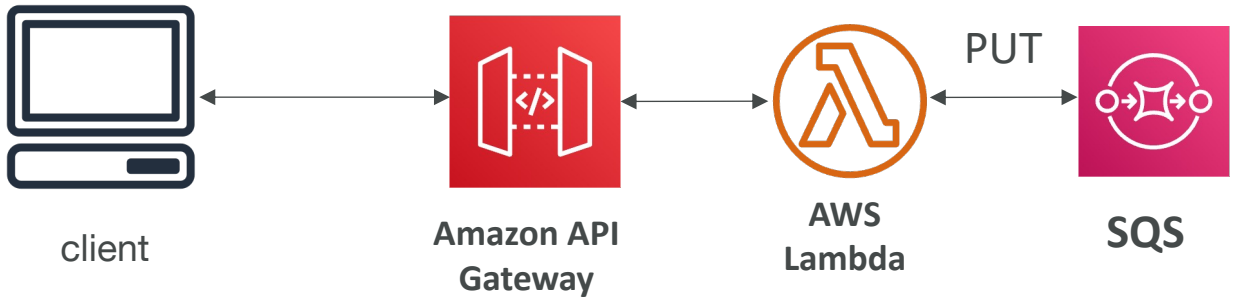
- Pay per request, seamless scaling, fully serverless
- Soft limits: 10000/s API Gateway, 1000 concurrent Lambda
- API Gateway features: authentication, rate limiting, caching, etc…
- Lambda Cold Start time may increase latency for some requests
- Fully integrated with X-Ray

© Stephane Maarek

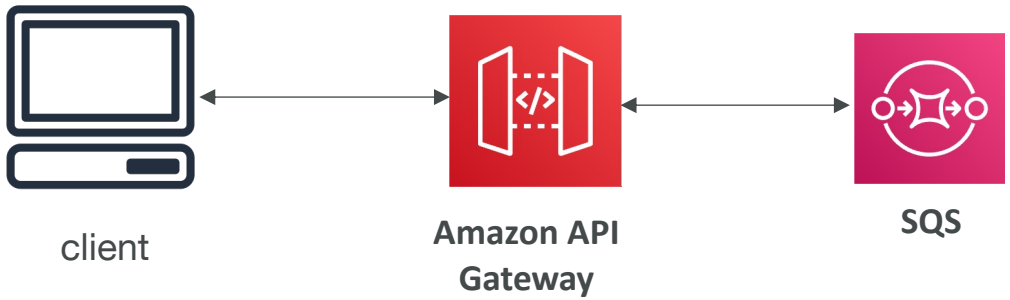
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway + AWS Service (as a proxy)

OK



BETTER

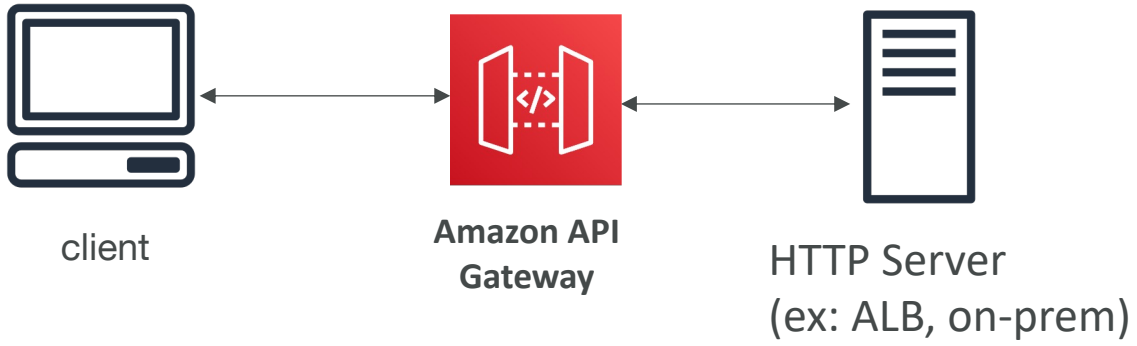


- Lower latency, cheaper
- Not using Lambda concurrent capacity, no custom code
- Expose AWS APIs securely through API Gateway
- SQS, SNS, Step Functions…
- Remember API Gateway has a payload limit of 10 MB (can be a problem for S3 proxy)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway + HTTP backend (ex: ALB)



- Use API Gateway features on top of custom HTTP backend (authentication, rate control, API keys, caching…)

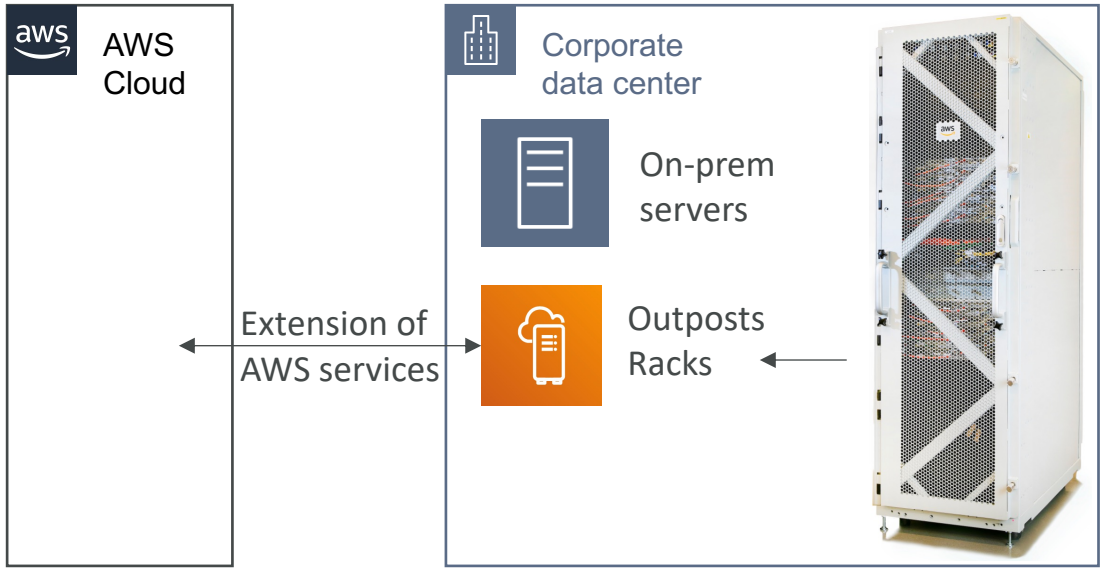
- Can connect to…
- on-premises service
- Application Load Balancer
- 3rd party HTTP service

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Outposts

- **Hybrid Cloud**: businesses that keep an on-premises infrastructure alongside a cloud infrastructure
- Therefore, two ways of dealing with IT systems:
  - One for the AWS cloud (using the AWS console, CLI, and AWS APIs)
  - One for their on-premises infrastructure
- **AWS Outposts are "server racks"** that offers the same AWS infrastructure, services, APIs & tools to build your own applications on-premises just as in the cloud
- **AWS will setup and manage "Outposts Racks"** within your on-premises infrastructure and you can start leveraging AWS services on-premises
- You are responsible for the Outposts Rack physical security

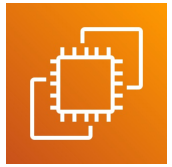


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Outposts

- Benefits:
  - Low-latency access to on-premises systems
  - Local data processing
  - Data residency
  - Easier migration from on-premises to the cloud
  - Fully managed service
- Some services that work on Outposts:


Amazon EC2

Amazon EBS

Amazon S3

Amazon EKS

Amazon ECS

Amazon RDS

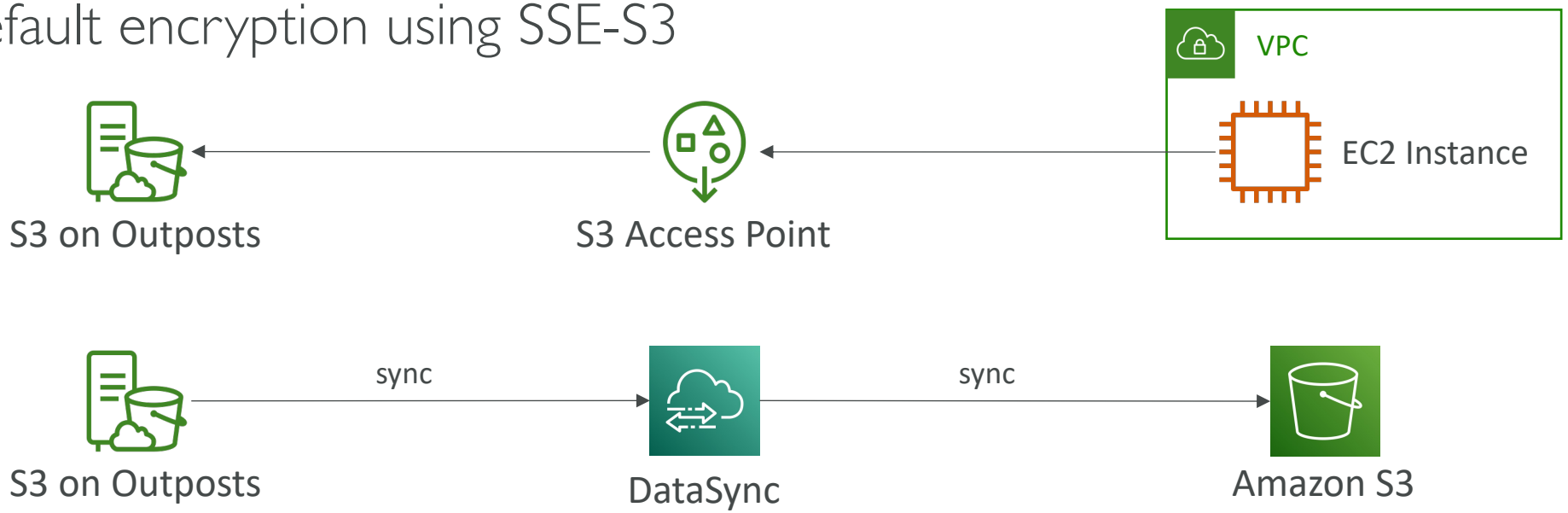
Amazon EMR

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 on AWS Outposts

- Use S3 APIs to store and retrieve data *locally* on AWS Outposts
- Keeping data close to on-premises applications
- Reduce data transfers to AWS Regions
- S3 Storage Class named **S3 Outposts**
- Default encryption using SSE-S3

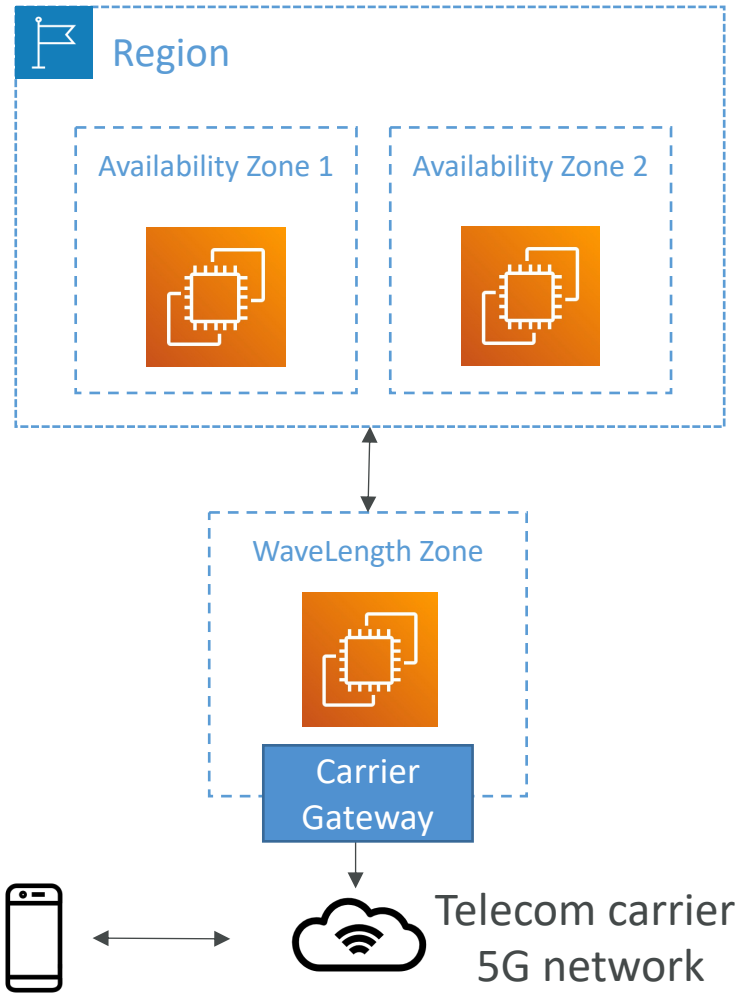


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS WaveLength

- **WaveLength Zones** are infrastructure deployments embedded within the telecommunications providers' datacenters at the edge of the 5G networks
- Brings AWS services to the edge of the 5G networks
- Example: EC2, EBS, VPC…
- Ultra-low latency applications through 5G networks
- Traffic doesn't leave the Communication Service Provider's (CSP) network
- High-bandwidth and secure connection to the parent AWS Region
- No additional charges or service agreements
- Use cases: Smart Cities, ML-assisted diagnostics, Connected Vehicles, Interactive Live Video Streams, AR/VR, Real-time Gaming, …

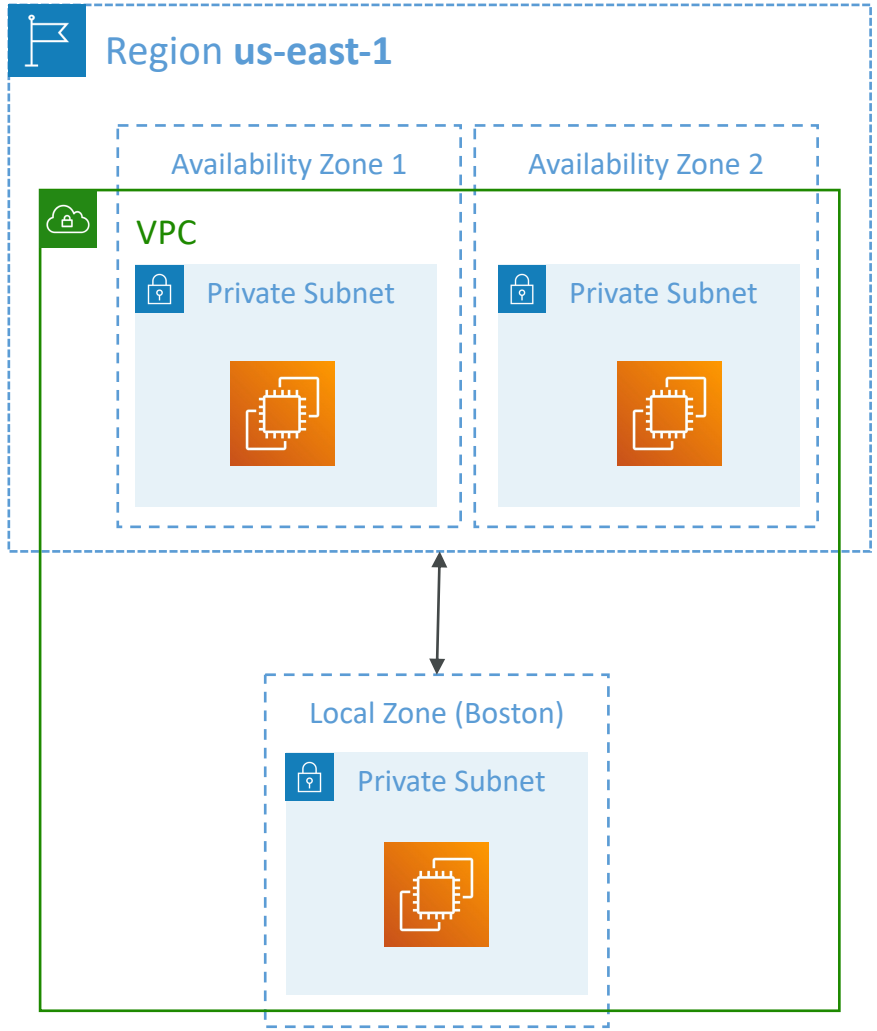


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Local Zones

- Places AWS compute, storage, database, and other selected AWS services **closer to end users to run latency-sensitive applications**
- Extend your VPC to more locations – "**Extension of an AWS Region**"
- Compatible with EC2, RDS, ECS, EBS, ElastiCache, Direct Connect …
- Example:
  - **AWS Region:** N. Virginia (us-east-1)
  - **AWS Local Zones:** Boston, Chicago, Dallas, Houston, Miami, …



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Storage

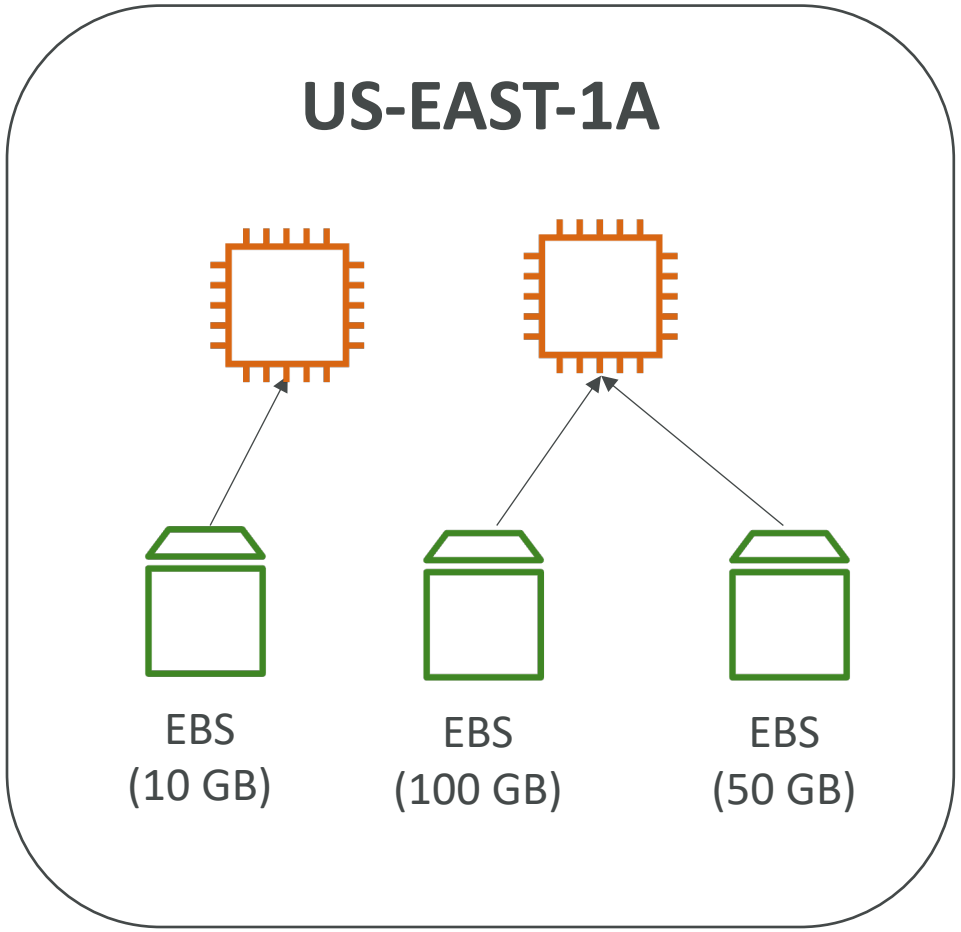
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EBS

- **Network** drive you attach to ONE instance only
- Linked to a specific availability zone (transfer: snapshot => restore)
- Volumes can be resized

- Make sure you choose an instance type that is EBS optimized to enjoy maximum throughput



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EBS Volume Types

- EBS Volumes come in 6 types
  - gp2 / gp3 (SSD): General purpose SSD volume that balances price and performance for a wide variety of workloads
  - io1 / io2 Block Express: Highest-performance SSD volume for mission-critical low-latency or high-throughput workloads
  - st1 (HDD): Low cost HDD volume designed for frequently accessed, throughput-intensive workloads
  - sc1 (HDD): Lowest cost HDD volume designed for less frequently accessed workloads
- EBS Volumes are characterized in Size | Throughput | IOPS (I/O Ops Per Sec)
- When in doubt always consult the AWS documentation – it's good!
- Only gp2/gp3 and io1/io2 can be used as boot volumes

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EBS Snapshots

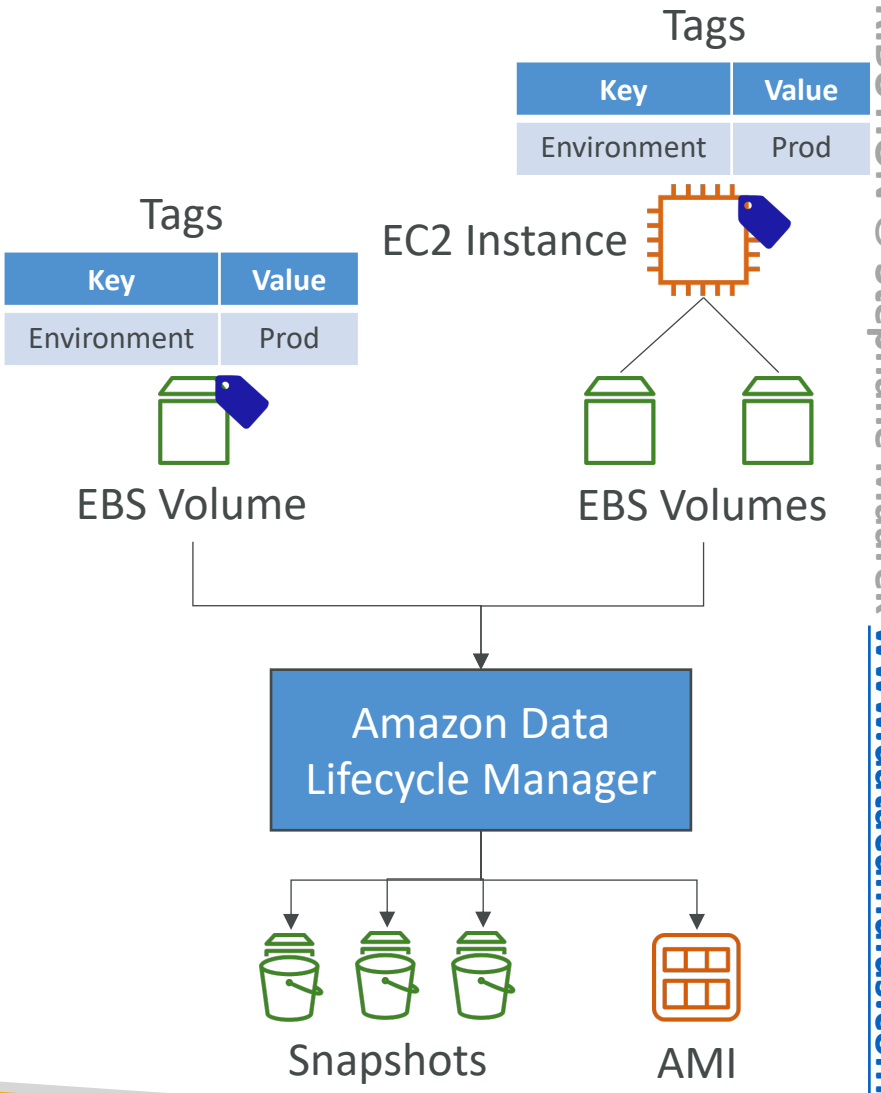
- Incremental – only backup changed blocks
- EBS backups use IO, and you shouldn't run them while your application is handling a lot of traffic
- Snapshots will be stored in S3 (but you won't directly see them)
- Not necessary to detach volume to do snapshot, but recommended
- Can copy snapshots across region (for DR)
- Can make Image (AMI) from Snapshot
- EBS volumes restored by snapshots need to be pre-warmed (use the Fast Snapshot Restore FSR feature or fio/dd command to read the entire volume)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Data Lifecycle Manager

- Automate the creation, retention, and deletion of EBS snapshots and EBS-backed AMIs
- Schedule backups, cross-account snapshot copies, delete outdated backups, …
- Uses resource tags to identify the resources (EC2 instances, EBS volumes)
- Can't be used to manage snapshots/AMIs created outside DLM
- Can't be used to manage instance-store backed AMIs



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Data Lifecycle Manager vs. AWS Backup

- Use Data Lifecycle Manager
  - when you want to automate the creation, retention, and deletion of EBS Snapshots

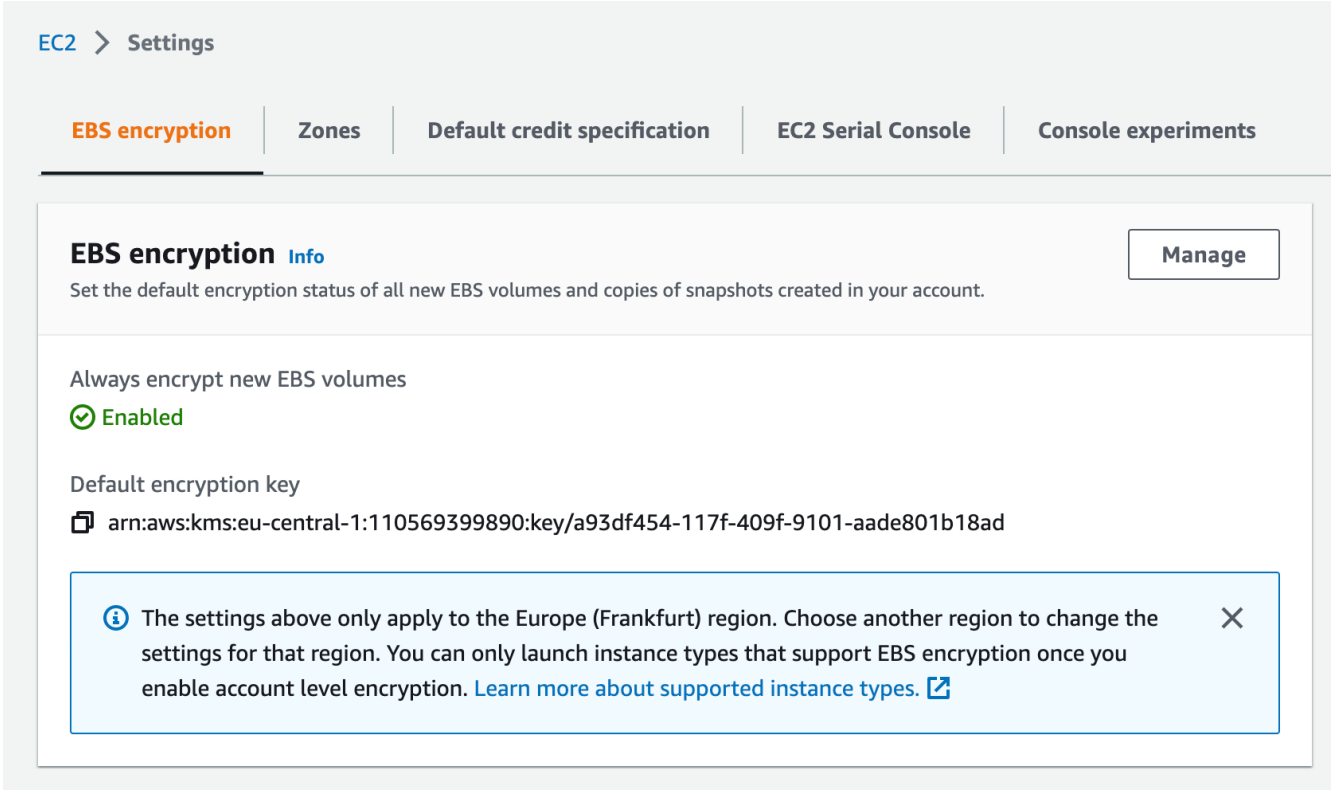
- Use AWS Backup
  - to manage and monitor backups across the AWS services you use, including EBS volumes, from a single place

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EBS Encryption – Account level setting

- New Amazon EBS volumes aren't encrypted by default
- There's an account-level setting to encrypt automatically new EBS volumes and Snapshots
- This setting needs to be enabled on a per-region basis

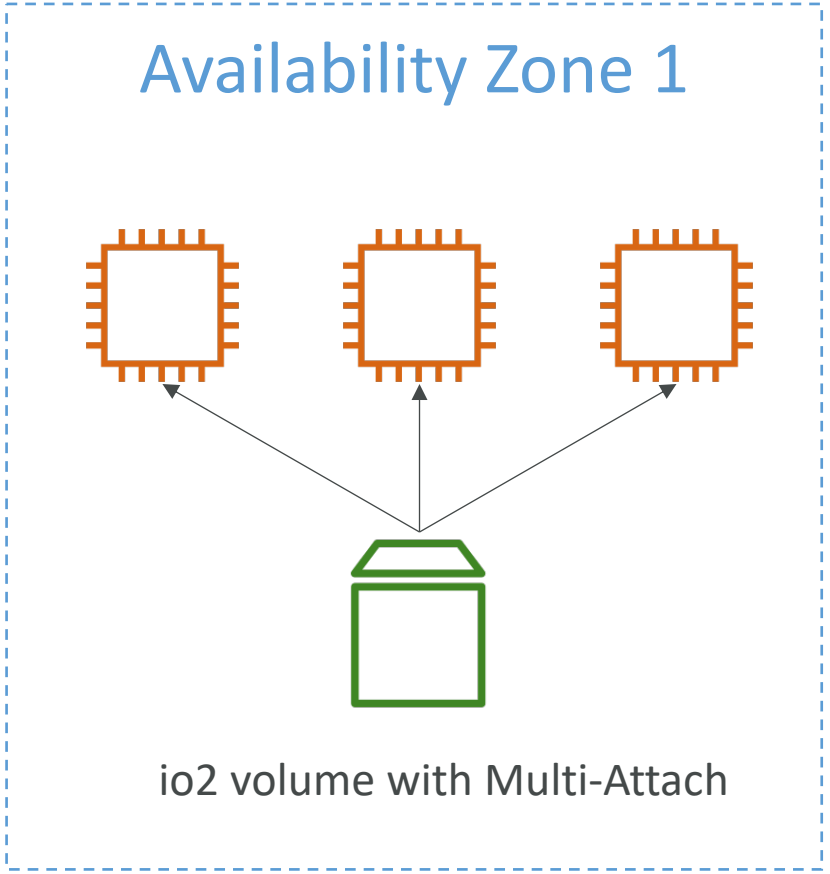


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EBS Multi-Attach – io1/io2 family

- Attach the same EBS volume to multiple EC2 instances in the same AZ
- Each instance has full read & write permissions to the volume
- Use case:
  - Achieve **higher application availability** in clustered Linux applications (ex: Teradata)
  - Applications must manage concurrent write operations
- Must use a file system that's cluster-aware (not XFS, EXT4, etc…)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Local EC2 Instance Store

- Physical disk attached to the physical server where your EC2 is
- Very High IOPS (because physical)
- Disks up to 7.5 TiB (can change over time), stripped to reach 60 TiB (can change over time…)
- Block Storage (just like EBS)
- Cannot be increased in size
- Risk of data loss if hardware fails

Very high IOPS

Instance Size	100% Random Read IOPS	Write IOPS
i3.large *	100,125	35,000
i3.xlarge *	206,250	70,000
i3.2xlarge	412,500	180,000
i3.4xlarge	825,000	360,000
i3.8xlarge	1.65 million	720,000
i3.16xlarge	3.3 million	1.4 million
i3.metal	3.3 million	1.4 million
i3en.large *	42,500	32,500
i3en.xlarge *	85,000	65,000
i3en.2xlarge *	170,000	130,000
i3en.3xlarge	250,000	200,000
i3en.6xlarge	500,000	400,000
i3en.12xlarge	1 million	800,000
i3en.24xlarge	2 million	1.6 million
i3en.metal	2 million	1.6 million

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Instance Store vs EBS

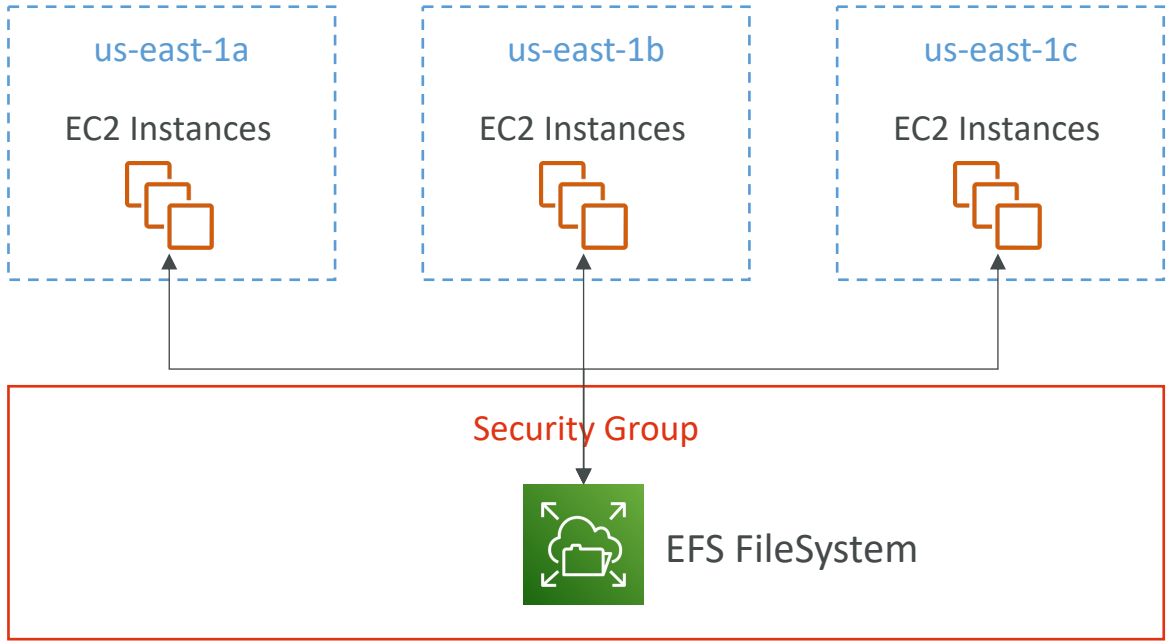
- Instance store is physically attached to the machine (ephemeral storage)
- EBS is a network drive (persistent)
- Pros:
  - Better I/O performance (EBS gp2 has a max IOPS of 16000, io1 of 64000, io2 Block Express of 256000)
  - Good for buffer / cache / scratch data / temporary content
  - Data survives reboots
- Cons:
  - On stop or termination, the instance store is lost
  - You can't resize the instance store
  - Backups must be operated by the user

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EFS – Elastic File System

- Managed NFS (network file system) that can be mounted on many EC2
- EFS works with EC2 instances in multi-AZ, & on–premises (DX & VPN)
- Highly available, scalable, expensive (3x gp2), pay per GB used



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EFS – Elastic File System

- Use cases: content management, web serving, data sharing, WordPress
- Compatible with Linux based AMI (not Windows), POSIX-compliant
- Uses NFSv4.1 protocol
- Uses security group to control access to EFS
- Encryption at rest using KMS

- POSIX file system (~Linux) that has a standard file API
- File system scales automatically, pay-per-use, no capacity planning!

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EFS – Performance & Storage Classes

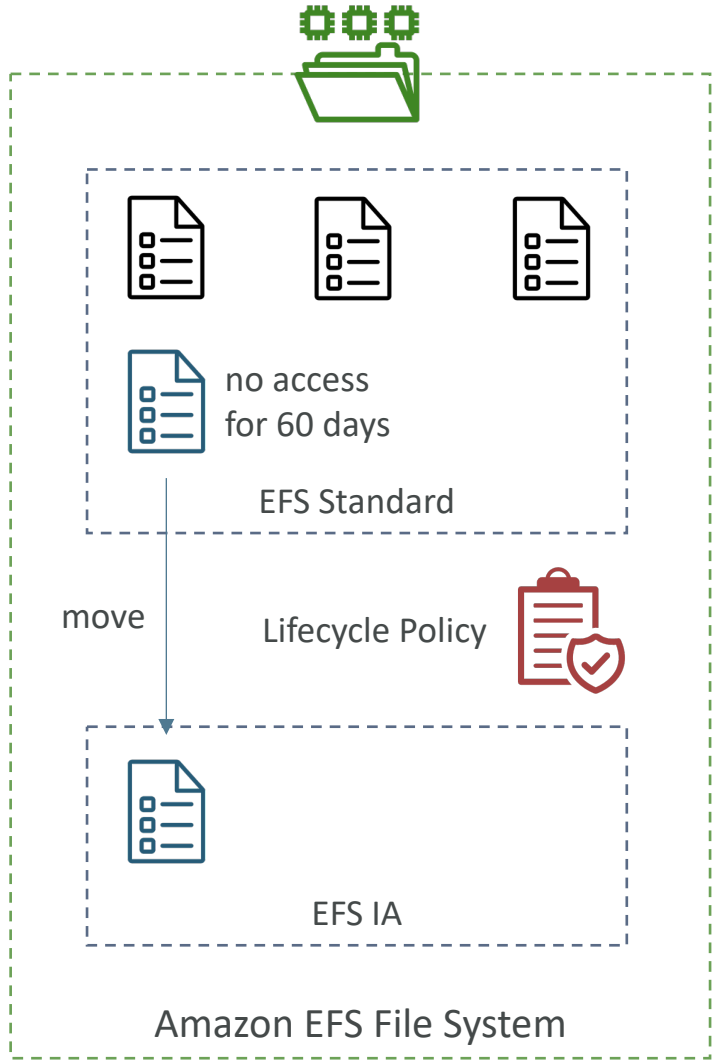
- EFS Scale
  - 1000s of concurrent NFS clients, 10 GB+ /s throughput
  - Grow to Petabyte-scale network file system, automatically
- Performance Mode (set at EFS creation time)
  - General Purpose (default) – latency-sensitive use cases (web server, CMS, etc…)
  - Max I/O – higher latency, throughput, highly parallel (big data, media processing)
- Throughput Mode
  - Bursting – 1 TB = 50MiB/s + burst of up to 100MiB/s
  - Provisioned – set your throughput regardless of storage size, ex: 1 GiB/s for 1 TB storage
  - Elastic – automatically scales throughput up or down based on your workloads
    - Up to 3GiB/s for reads and 1GiB/s for writes
    - Used for unpredictable workloads

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EFS – Storage Classes

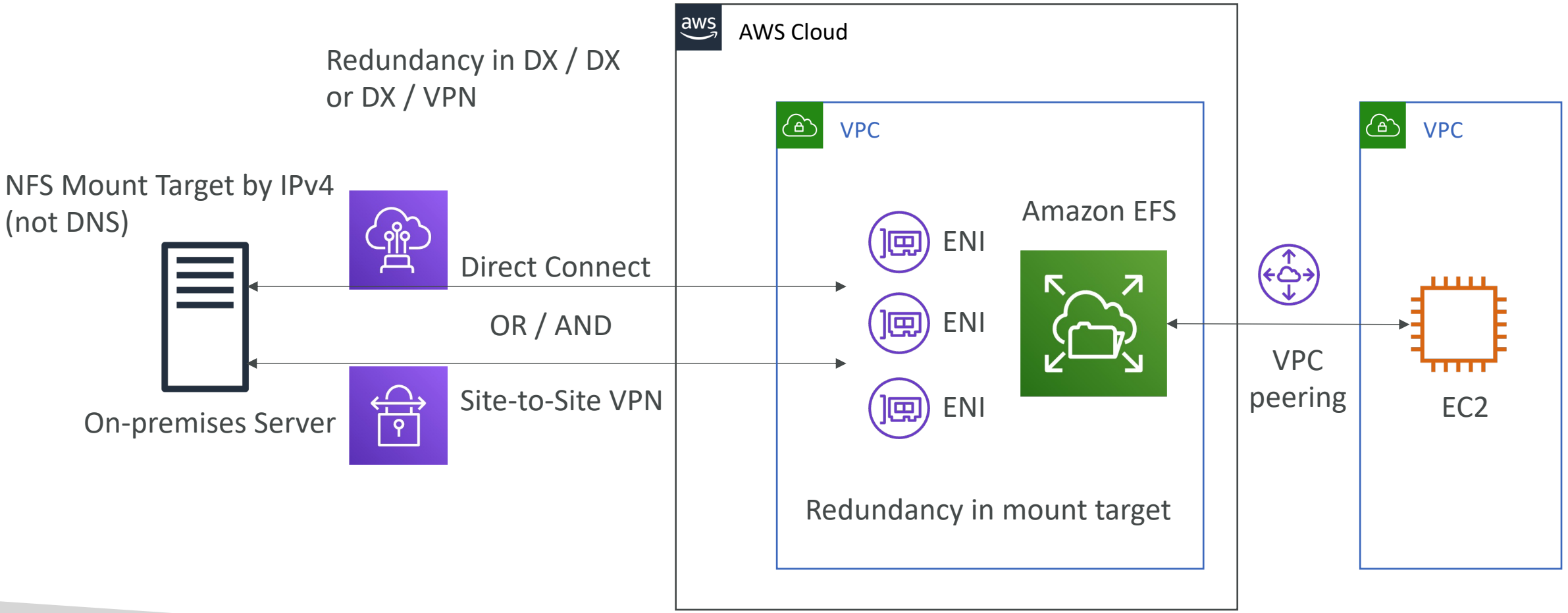
- Storage Tiers (lifecycle management feature – move file after N days)
  - **Standard**: for frequently accessed files
  - **Infrequent access (EFS-IA):** cost to retrieve files, lower price to store.
  - **Archive:** rarely accessed data (few times each year), 50% cheaper
  - Implement **lifecycle policies** to move files between storage tiers
- Availability and durability
  - Standard: Multi-AZ, great for prod
  - One Zone: One AZ, great for dev, backup enabled by default, compatible with IA (EFS One Zone-IA)
- Over 90% in cost savings



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EFS - On-premises & VPC Peering

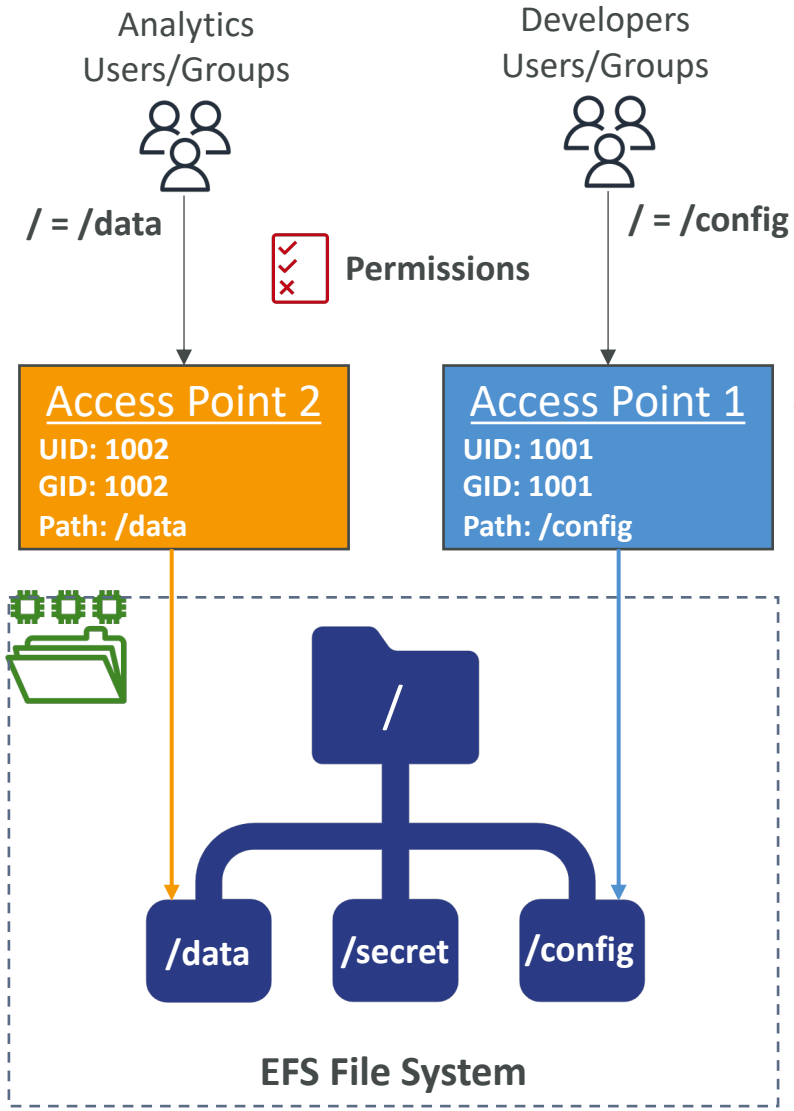


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EFS – Access Points

- Easily manage applications access to NFS environments
- Enforce a POSIX user and group to use when accessing the file system
- Restrict access to a directory within the file system and optionally specify a different root directory
- Can restrict access from NFS clients using IAM policies



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

EFS – File System Policies

- Resource-based policy to control access to EFS File Systems (same as S3 bucket policy)
- By default, it grants full access to all clients

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": { "AWS": "arn:aws:iam::123456789012:user/Stephane" },
            "Action": [
                "elasticfilesystem:ClientMount",
                "elasticfilesystem:ClientWrite"
            ],
            "Condition": {
                "Bool": { "aws:SecureTransport": "true" }
            }
        }
    ]
}
```

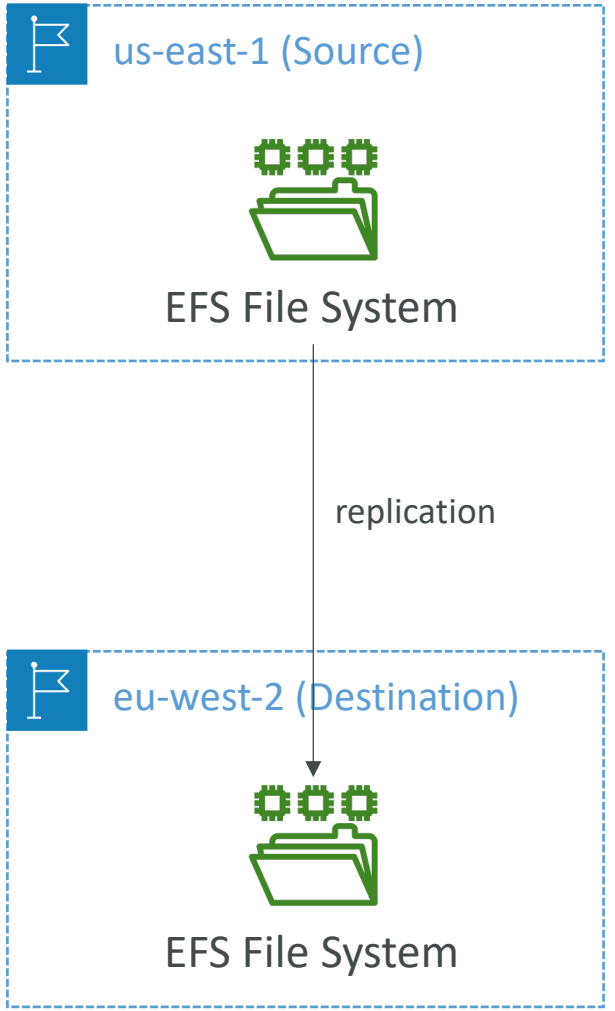
Grant Read & Write Access to A specific IAM User

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EFS – Cross-Region Replication

- Replicate objects in an EFS file system to another AWS Region
- Setup for new or existing EFS file systems
- Provides RPO and RTO of minutes
- Doesn't affect the provisioned throughput of the EFS file system
- Use cases: meet your compliance and business continuity goals



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

S3 – Overview

- Object storage, serverless, unlimited storage, pay-as-you-go
- Good to store static content (image, video files)
- Access objects by key, no indexing facility
- Not a filesystem, cannot be mounted natively on EC2

- Anti patterns:
  - Lots of small files
  - POSIX file system (use EFS instead), file locks
  - Search features, queries, rapidly changing data
  - Website with dynamic content

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Storage Classes Comparison

	Standard	Intelligent-Tiering	Standard-IA	One Zone-IA	Glacier Instant Retrieval	Glacier Flexible Retrieval	Glacier Deep Archive
Durability	99.999999999% == (11 9's)						
Availability	99.99%	99.9%	99.9%	99.5%	99.9%	99.99%	99.99%
Availability Zones	>= 3	>= 3	>= 3	1	>= 3	>= 3	>= 3
Min. Storage Duration Charge	None	None	30 Days	30 Days	90 Days	90 Days	180 Days
Min. Billable Object Size	None	None	128 KB	128 KB	128 KB	40 KB	40 KB
Retrieval Fee	None	None	Per GB retrieved	Per GB retrieved	Per GB retrieved	Per GB retrieved	Per GB retrieved

- You can transition objects between tiers (or delete) using S3 Lifecycle Policies

https://aws.amazon.com/s3/storage-classes/

© Stephane Maarek

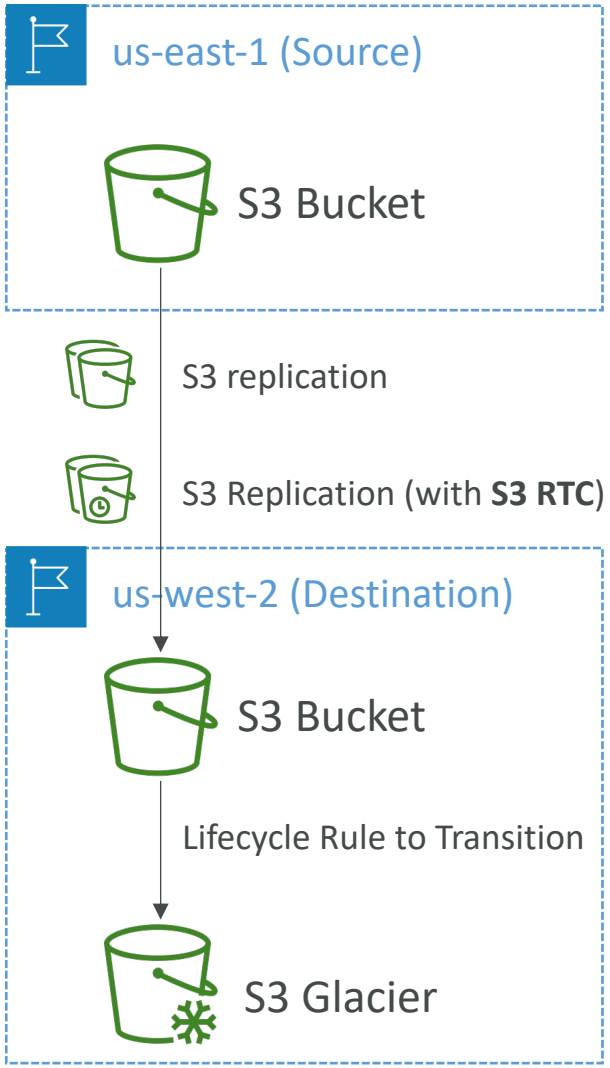
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 – Replication (Versioning enabled)

- Cross Region Replication (CRR)
- Same Region Replication (SRR)
- Combine with Lifecycle Rules

- Helpful to reduce latency, disaster recovery, security

- S3 Replication Time Control (S3 RTC)
  - Replicates most objects that you upload to Amazon S3 in seconds, and 99.99% of those objects within 15 minutes
  - Helpful for compliance, DR, etc..



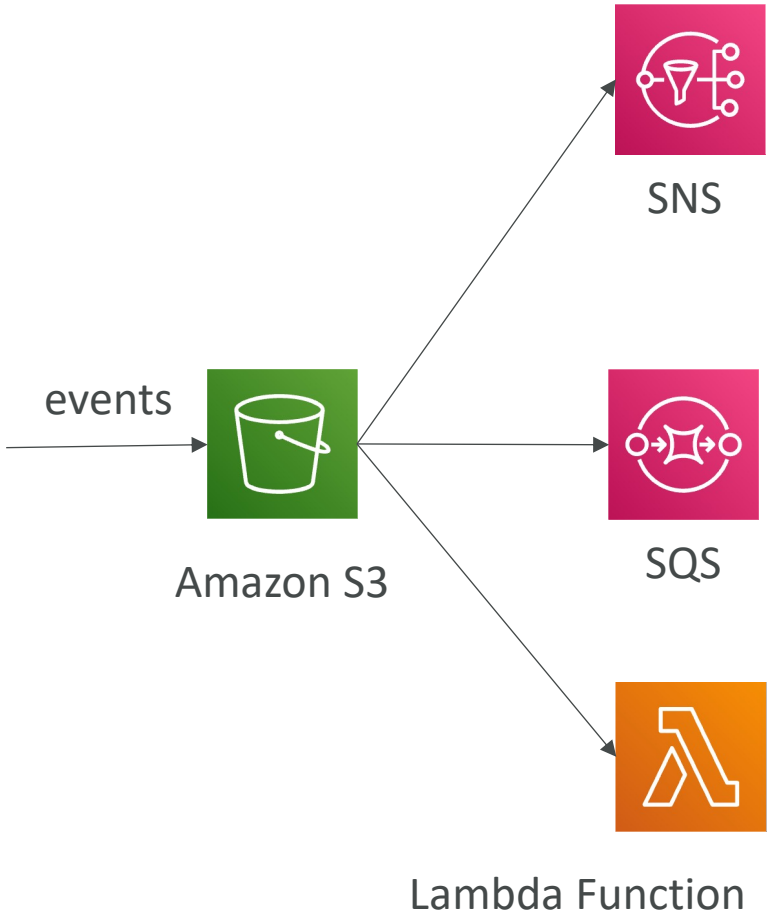
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Event Notifications

- S3:ObjectCreated, S3:ObjectRemoved, S3:ObjectRestore, S3:Replication…
- Object name filtering possible (*.jpg)
- Use case: generate thumbnails of images uploaded to S3
- **Can create as many "S3 events" as desired**

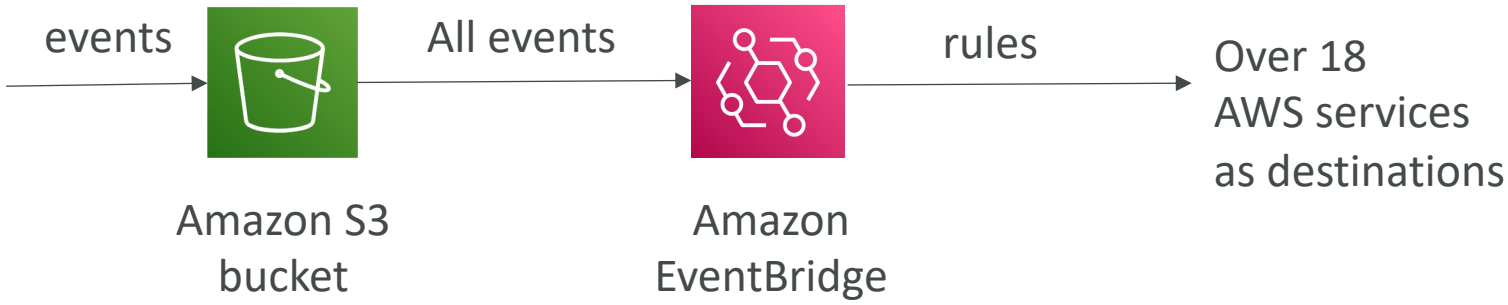
- S3 event notifications typically deliver events in seconds but can sometimes take a minute or longer



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Event Notifications with Amazon EventBridge



- **Advanced filtering** options with JSON rules (metadata, object size, name...)
- **Multiple Destinations** – ex Step Functions, Kinesis Streams / Firehose…
- **EventBridge Capabilities** – Archive, Replay Events, Reliable delivery

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 – Baseline Performance

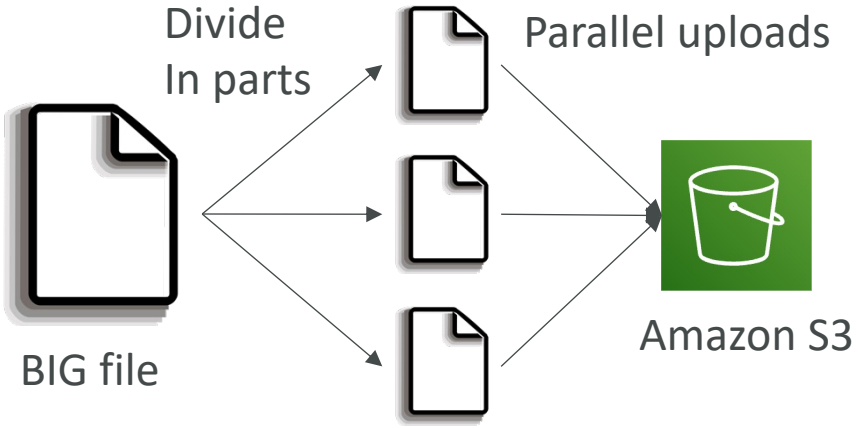
- Amazon S3 automatically scales to high request rates, latency 100-200 ms
- Your application can achieve at least **3,500 PUT/COPY/POST/DELETE or 5,500 GET/HEAD requests per second per prefix in a bucket.**
- There are no limits to the number of prefixes in a bucket.
- Example (object path => prefix):
  - bucket/folder1/sub1/file => /folder1/sub1/
  - bucket/folder1/sub2/file => /folder1/sub2/
  - bucket/1/file => /1/
  - bucket/2/file => /2/
- If you spread reads across all four prefixes evenly, you can achieve 22,000 requests per second for GET and HEAD

© Stephane Maarek

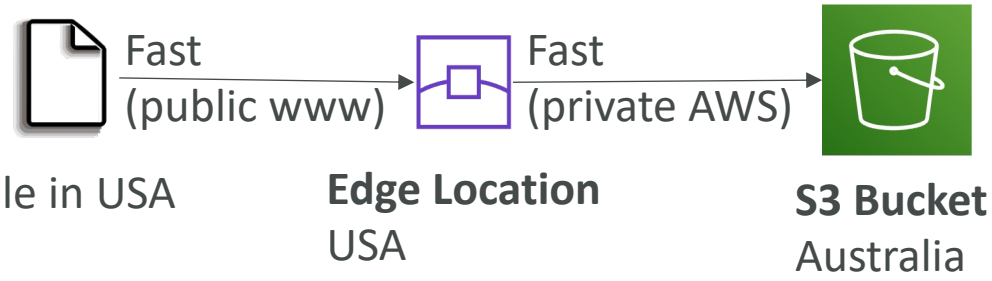
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Performance

- Multi-Part upload:
  - recommended for files > 100MB, must use for files > 5GB
  - Can help parallelize uploads (speed up transfers)



- S3 Transfer Acceleration
  - Increase transfer speed by transferring file to an AWS edge location which will forward the data to the S3 bucket in the target region
  - Compatible with multi-part upload



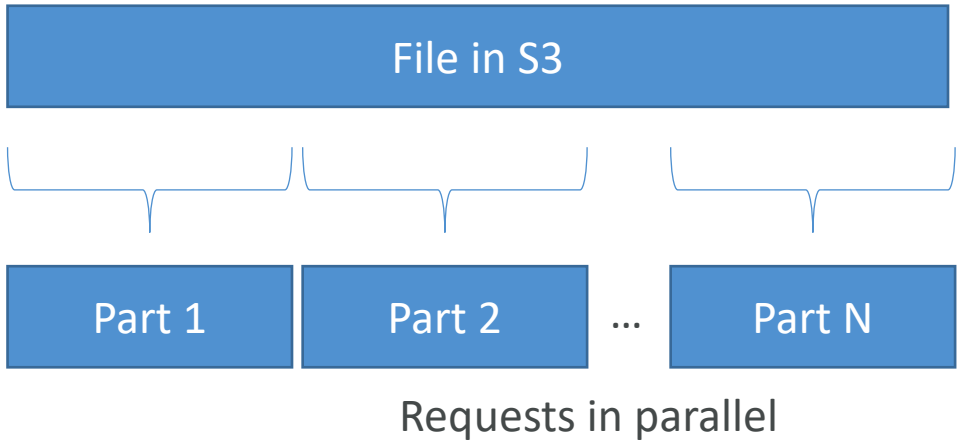
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

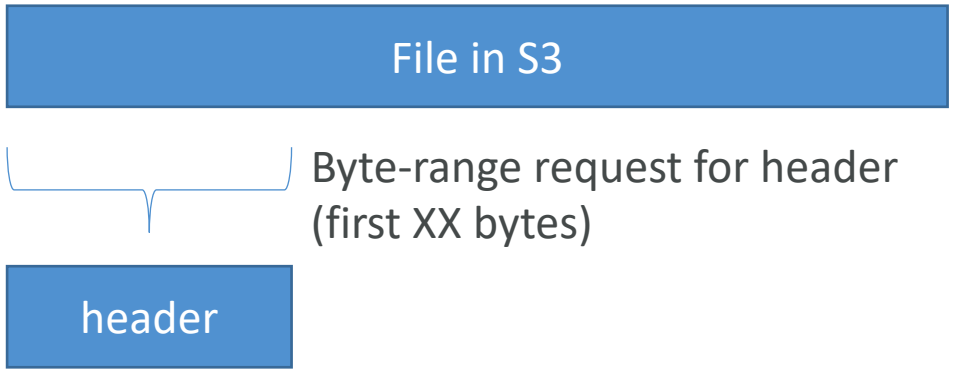
S3 Performance – S3 Byte-Range Fetches

- Parallelize GETs by requesting specific byte ranges
- Better resilience in case of failures

Can be used to speed up downloads



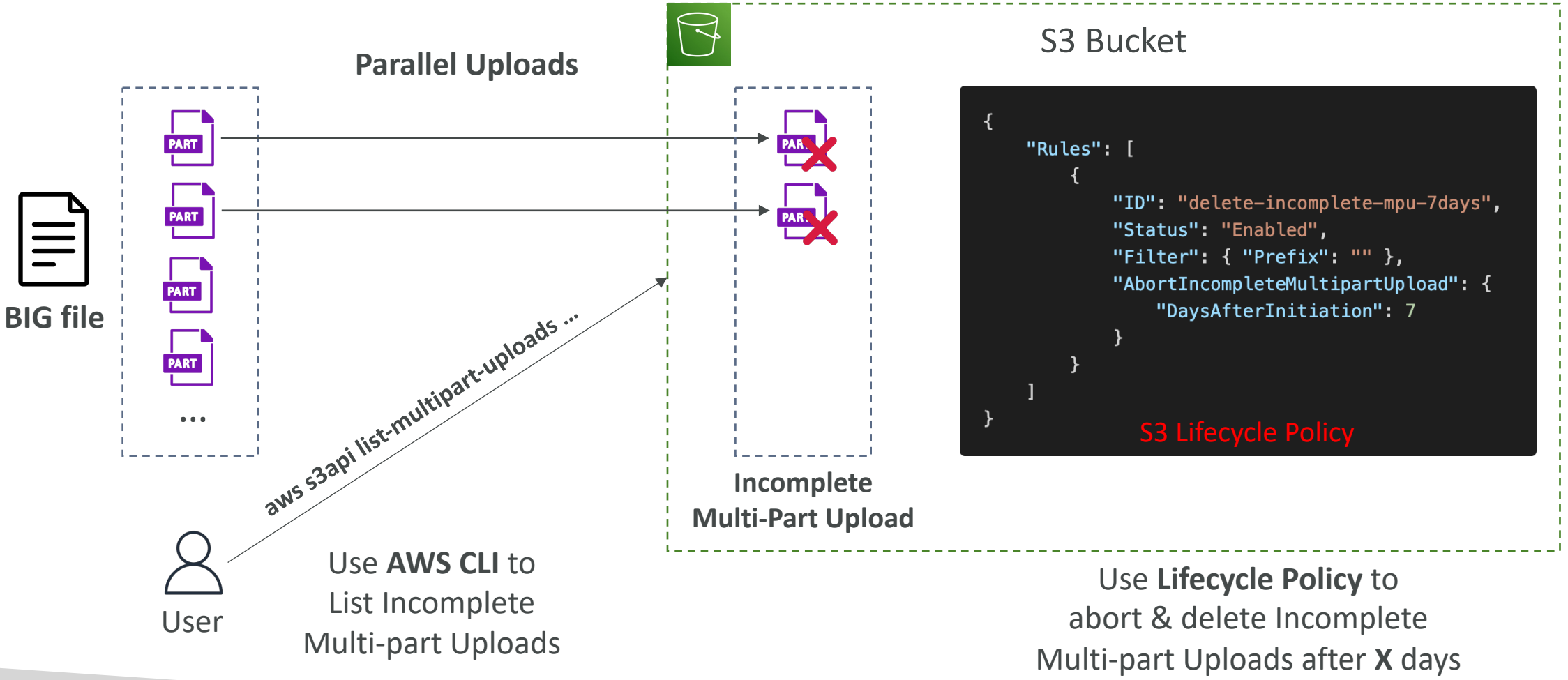
Can be used to retrieve only partial data (for example the head of a file)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Multi-Part Upload – Remove Incomplete Parts



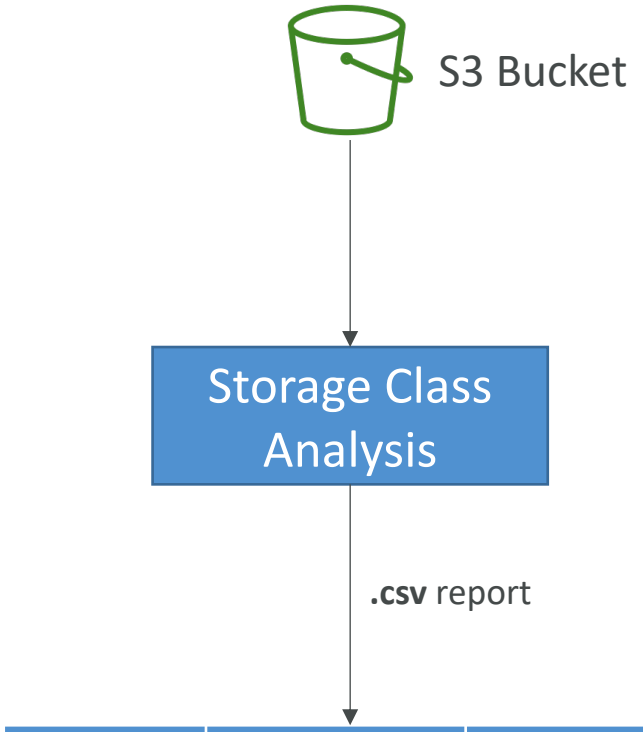
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Analytics – Storage Class Analysis

- May be seen as "Storage Class Analysis" at the exam
- Help you decide when to transition objects to the right storage class
- Recommendations for **Standard** and **Standard IA**
  - Does NOT work for One-Zone IA or Glacier
- Report is updated daily
- 24 to 48 hours to start seeing data analysis
- Visualize data in Amazon QuickSight

- Good first step to put together Lifecycle Rules (or improve them)!



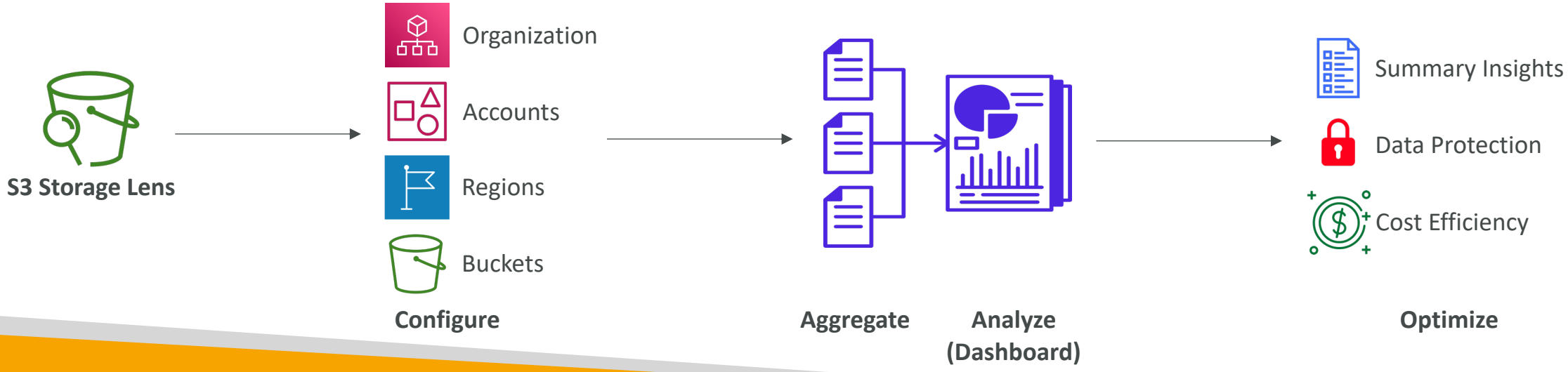
Date	StorageClass	ObjectAge
8/22/2022	STANDARD	000-014
8/25/2022	STANDARD	030-044
9/6/2022	STANDARD	120-149

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 – Storage Lens

- Understand, analyze, and optimize storage across entire AWS Organization
- Discover anomalies, identify cost efficiencies, and apply data protection best practices across entire AWS Organization (30 days usage & activity metrics)
- Aggregate data for Organization, specific accounts, regions, buckets, or prefixes
- Default dashboard or create your own dashboards
- Can be configured to export metrics daily to an S3 bucket (CSV, Parquet)

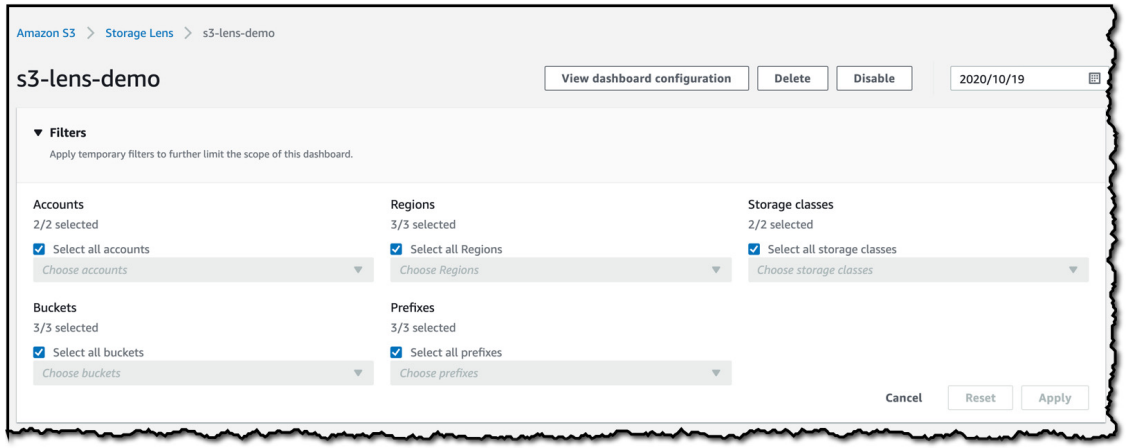


© Stephane Maarek

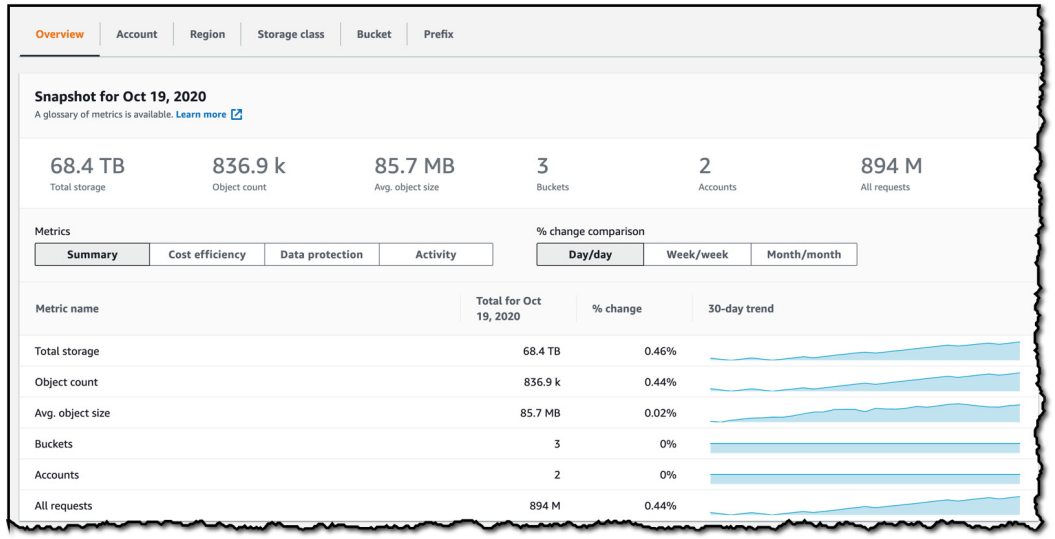
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Storage Lens – Default Dashboard

- Visualize summarized insights and trends for both free and advanced metrics
- Default dashboard shows Multi-Region and Multi-Account data
- Preconfigured by Amazon S3
- Can't be deleted, but can be disabled



https://aws.amazon.com/blogs/aws/s3-storage-lens/



https://aws.amazon.com/blogs/aws/s3-storage-lens/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Storage Lens – Metrics

- Summary Metrics
  - General insights about your S3 storage
  - StorageBytes, ObjectCount…
  - Use cases: identify the fastest-growing (or not used) buckets and prefixes

- Cost-Optimization Metrics
  - Provide insights to manage and optimize your storage costs
  - NonCurrentVersionStorageBytes, IncompleteMultipartUploadStorageBytes…
  - Use cases: identify buckets with incomplete multipart uploaded older than 7 days, Identify which objects could be transitioned to lower-cost storage class

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Storage Lens – Metrics

- Data-Protection Metrics
  - Provide insights for data protection features
  - VersioningEnabledBucketCount, MFADeleteEnabledBucketCount, SSEKMSEnabledBucketCount, CrossRegionReplicationRuleCount…
  - Use cases: identify buckets that aren't following data-protection best practices
- Access-management Metrics
  - Provide insights for S3 Object Ownership
  - ObjectOwnershipBucketOwnerEnforcedBucketCount…
  - Use cases: identify which Object Ownership settings your buckets use
- Event Metrics
  - Provide insights for S3 Event Notifications
  - EventNotificationEnabledBucketCount (identify which buckets have S3 Event Notifications configured)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Storage Lens – Metrics

- Performance Metrics
  - Provide insights for S3 Transfer Acceleration
  - TransferAccelerationEnabledBucketCount (identify which buckets have S3 Transfer Acceleration enabled)
- Activity Metrics
  - Provide insights about how your storage is requested
  - AllRequests, GetRequests, PutRequests, ListRequests, BytesDownloaded…
- Detailed Status Code Metrics
  - Provide insights for HTTP status codes
  - 200OKStatusCount, 403ForbiddenErrorCount, 404NotFoundErrorCount…

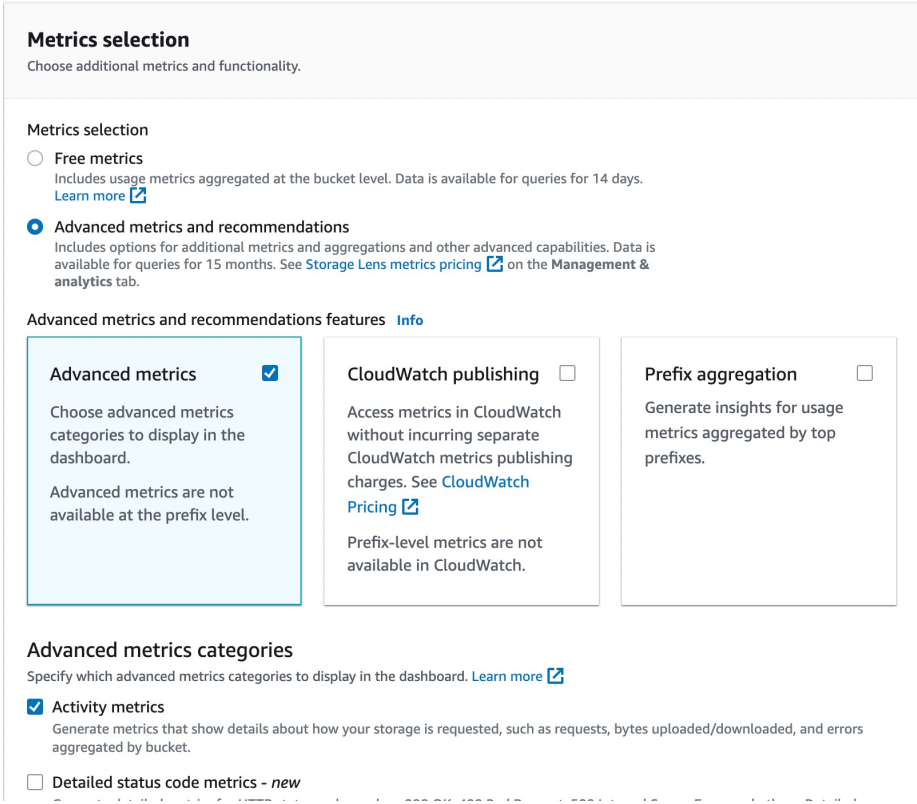
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Storage Lens – Free vs. Paid

- Free Metrics
  - Automatically available for all customers
  - Contains around 28 usage metrics
  - Data is available for queries for 14 days

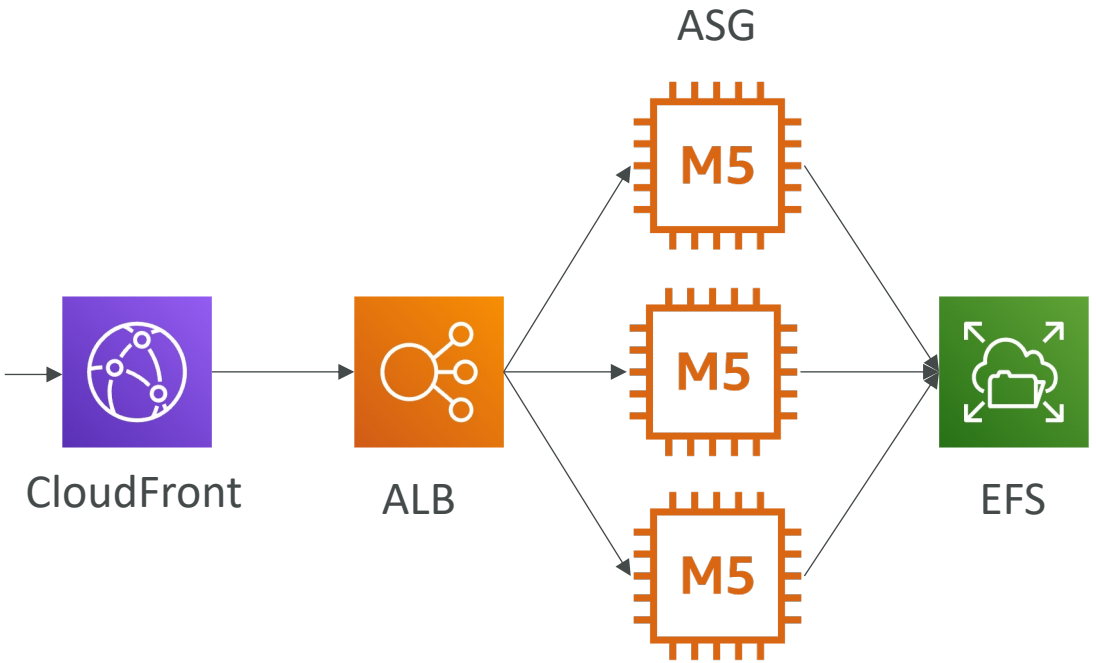
- Advanced Metrics and Recommendations
  - Additional paid metrics and features
  - **Advanced Metrics** – Activity, Advanced Cost Optimization, Advanced Data Protection, Status Code
  - **CloudWatch Publishing** – Access metrics in CloudWatch without additional charges
  - **Prefix Aggregation** – Collect metrics at the prefix level
  - Data is available for queries for 15 months

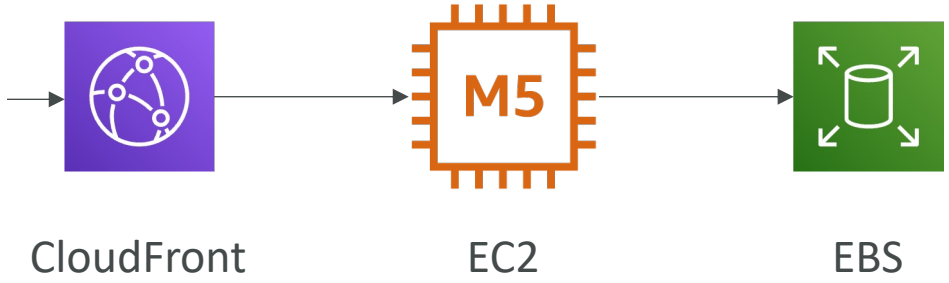


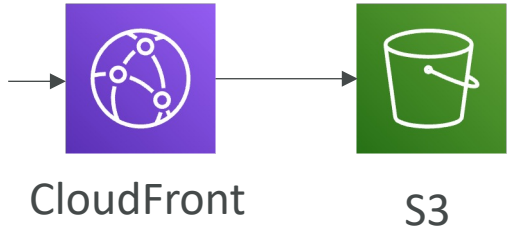
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Solution Architecture Exposing Static Objects



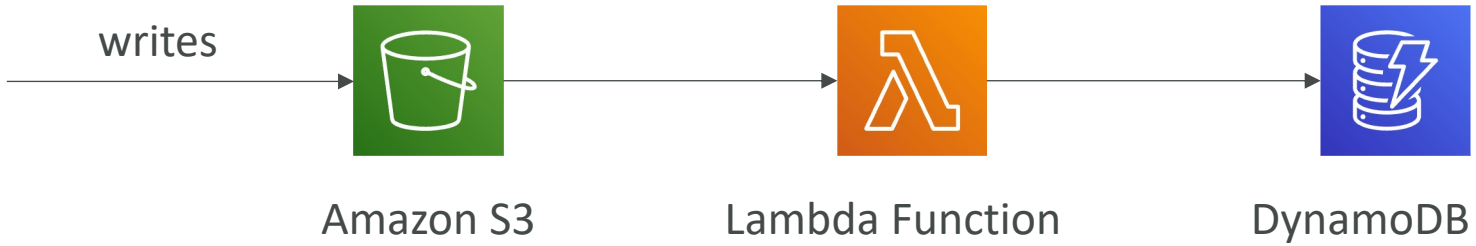




© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Solution Architecture
Indexing objects in DynamoDB



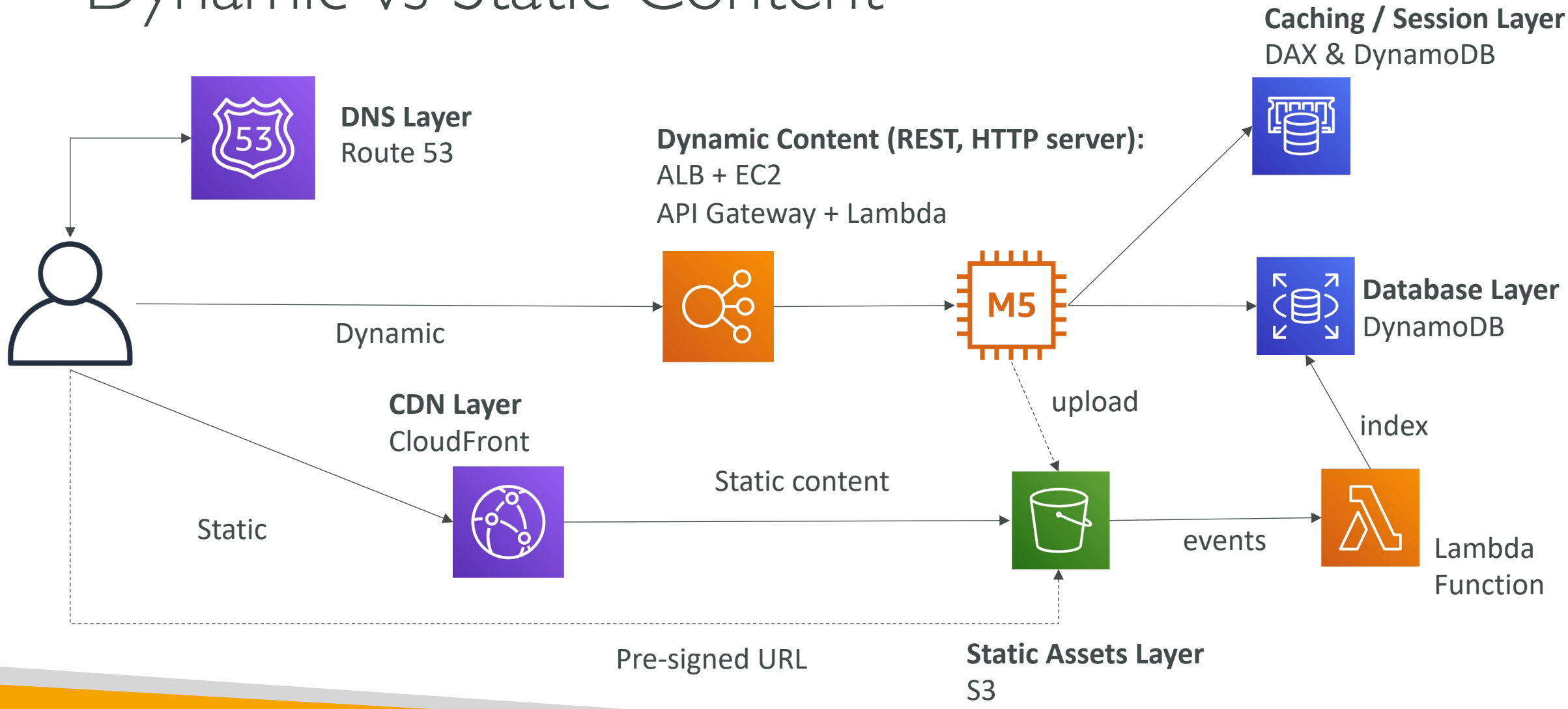
API for object metadata

- Search by date
- Total storage used by a customer
- List of all objects with certain attributes
- Find all objects uploaded within a date range

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Solution Architecture on AWS
Dynamic vs Static Content



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon FSx – Overview

- Launch 3rd party high-performance file systems on AWS
- Fully managed service

FSx for Lustre

FSx for Windows File Server

FSx for NetApp ONTAP

FSx for OpenZFS

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon FSx for Windows (File Server)

- **FSx for Windows** is a fully managed **Windows** file system share drive
- Supports SMB protocol & Windows NTFS
- Microsoft Active Directory integration, ACLs, user quotas
- **Can be mounted on Linux EC2 instances**
- Supports **Microsoft's Distributed File System (DFS) Namespaces** (group files across multiple FS)

- Scale up to 10s of GB/s, millions of IOPS, 100s PB of data
- Storage Options:
  - **SSD** – latency sensitive workloads (databases, media processing, data analytics, …)
  - **HDD** – broad spectrum of workloads (home directory, CMS, …)
- Can be accessed from your on-premises infrastructure (VPN or Direct Connect)
- Can be configured to be Multi-AZ (high availability)
- Data is backed-up daily to S3

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon FSx for Lustre

- Lustre is a type of parallel distributed file system, for large-scale computing
- The name Lustre is derived from "Linux" and "cluster

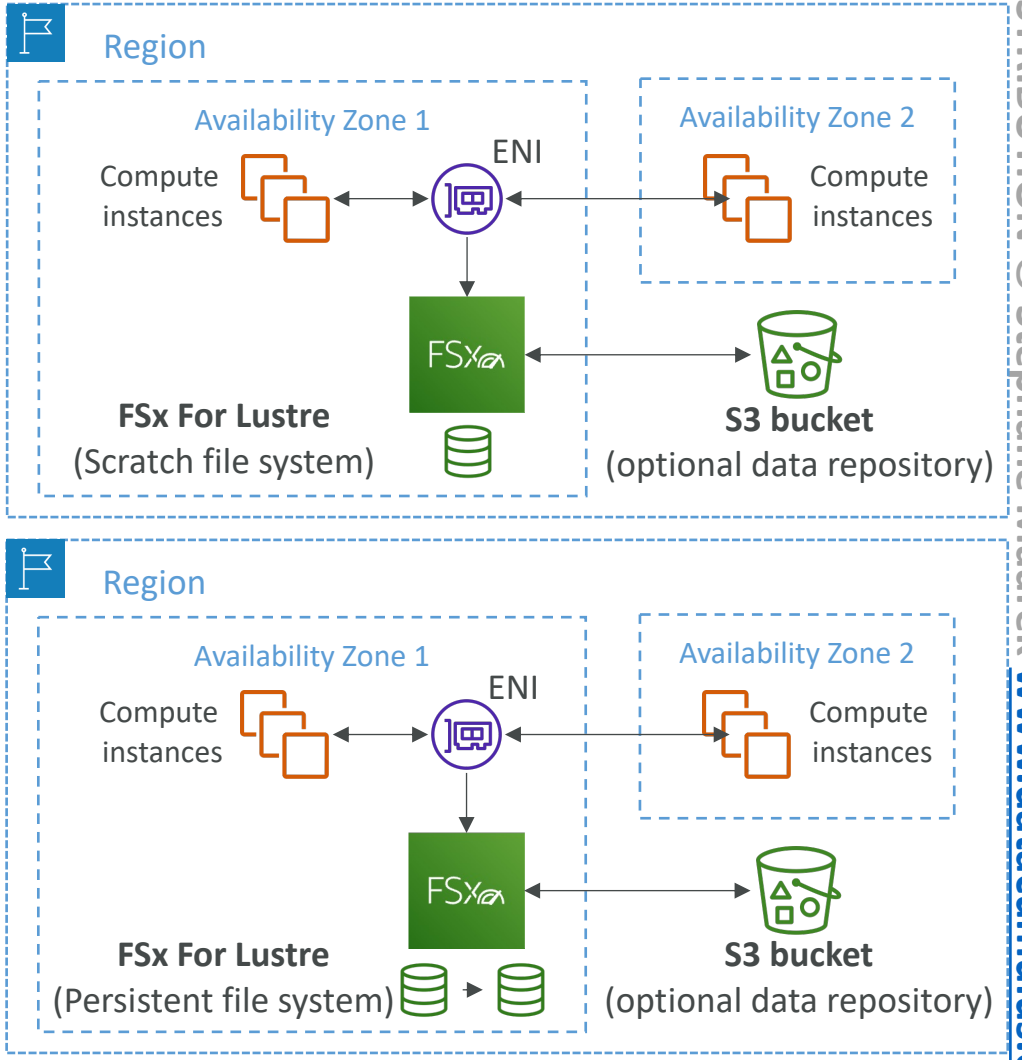
- Machine Learning, **High Performance Computing (HPC)**
- Video Processing, Financial Modeling, Electronic Design Automation
- Scales up to 100s GB/s, millions of IOPS, sub-ms latencies
- Storage Options:
  - **SSD** – low-latency, IOPS intensive workloads, small & random file operations
  - **HDD** – throughput-intensive workloads, large & sequential file operations
- **Seamless integration with S3**
  - Can "read S3" as a file system (through FSx)
  - Can write the output of the computations back to S3 (through FSx)
- **Can be used from on-premises servers (VPN or Direct Connect)**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

FSx Lustre - File System Deployment Options

- Scratch File System
  - Temporary storage
  - Data is not replicated (doesn't persist if file server fails)
  - High burst (6x faster, 200MBps per TiB)
  - Usage: short-term processing, optimize costs
- Persistent File System
  - Long-term storage
  - Data is replicated within same AZ
  - Replace failed files within minutes
  - Usage: long-term processing, sensitive data

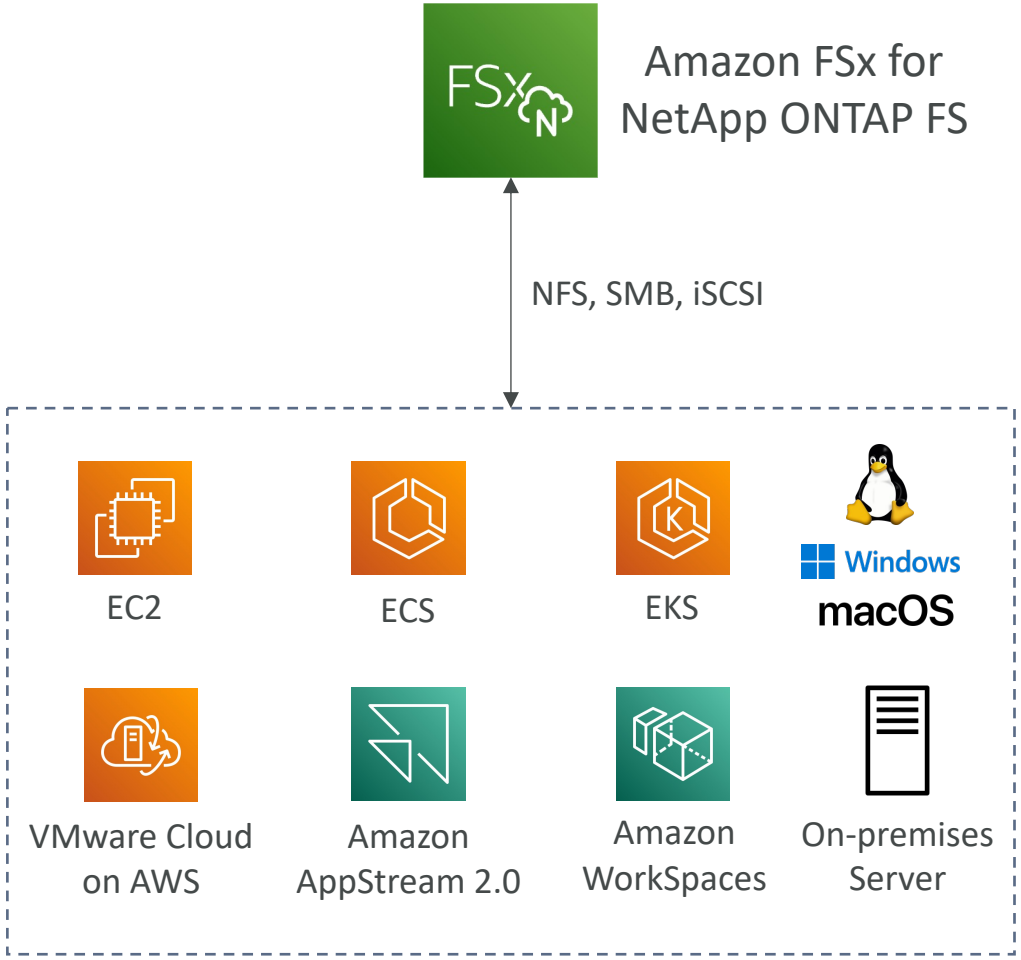


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Amazon FSx for NetApp ONTAP

- Managed NetApp ONTAP on AWS
- **File System compatible with NFS, SMB, iSCSI protocol**
- Move workloads running on ONTAP or NAS to AWS
- Works with:
  - Linux
  - Windows
  - MacOS
  - VMware Cloud on AWS
  - Amazon Workspaces & AppStream 2.0
  - Amazon EC2, ECS and EKS
- Storage shrinks or grows automatically
- Snapshots, replication, low-cost, compression and data de-duplication
- **Point-in-time instantaneous cloning (helpful for testing new workloads)**



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

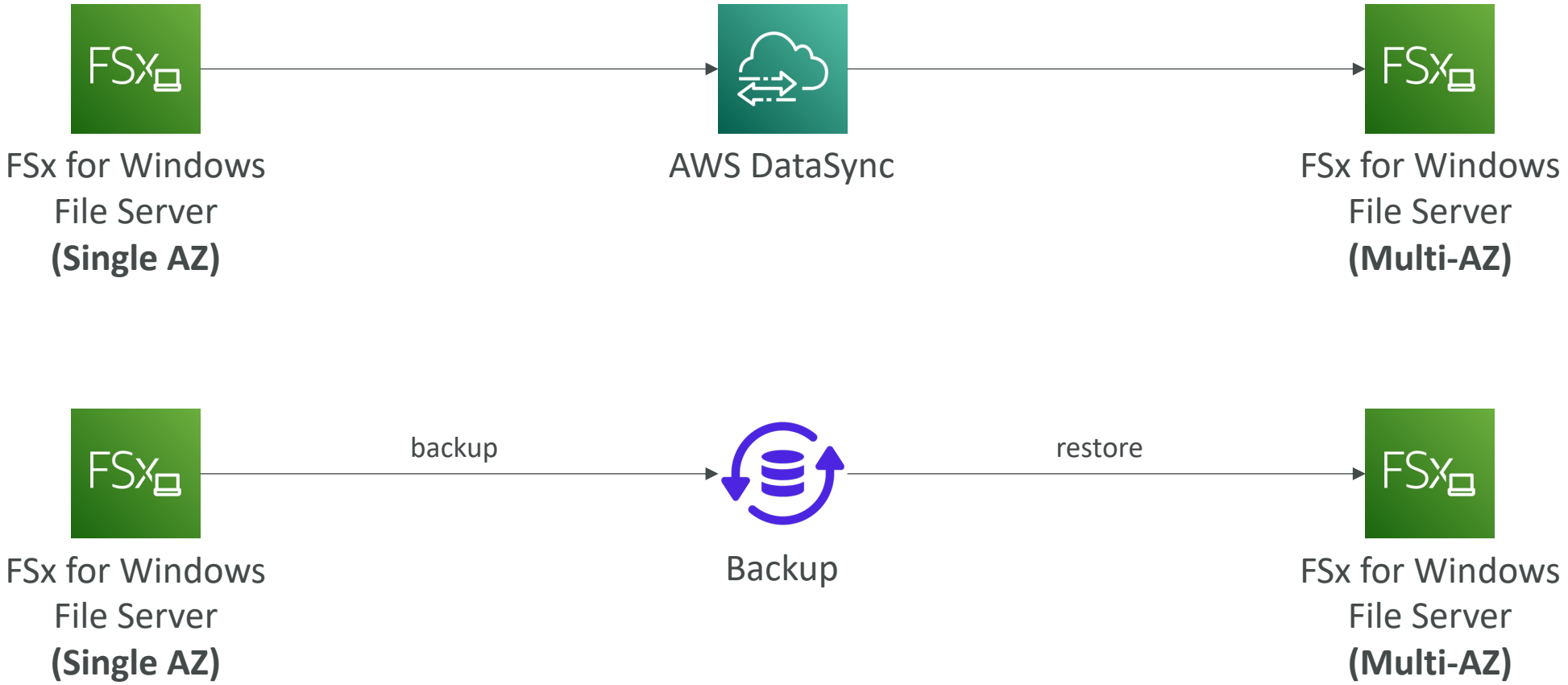
Amazon FSx for OpenZFS

- Managed OpenZFS file system on AWS
- File System compatible with NFS (v3, v4, v4.1, v4.2)
- Move workloads running on ZFS to AWS
- Works with:
  - Linux
  - Windows
  - MacOS
  - VMware Cloud on AWS
  - Amazon Workspaces & AppStream 2.0
  - Amazon EC2, ECS and EKS
- Up to 1,000,000 IOPS with < 0.5ms latency
- Snapshots, compression and low-cost
- Point-in-time instantaneous cloning (helpful for testing new workloads)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

FSx – Solution Architecture
Migration from Single AZ to Multi AZ

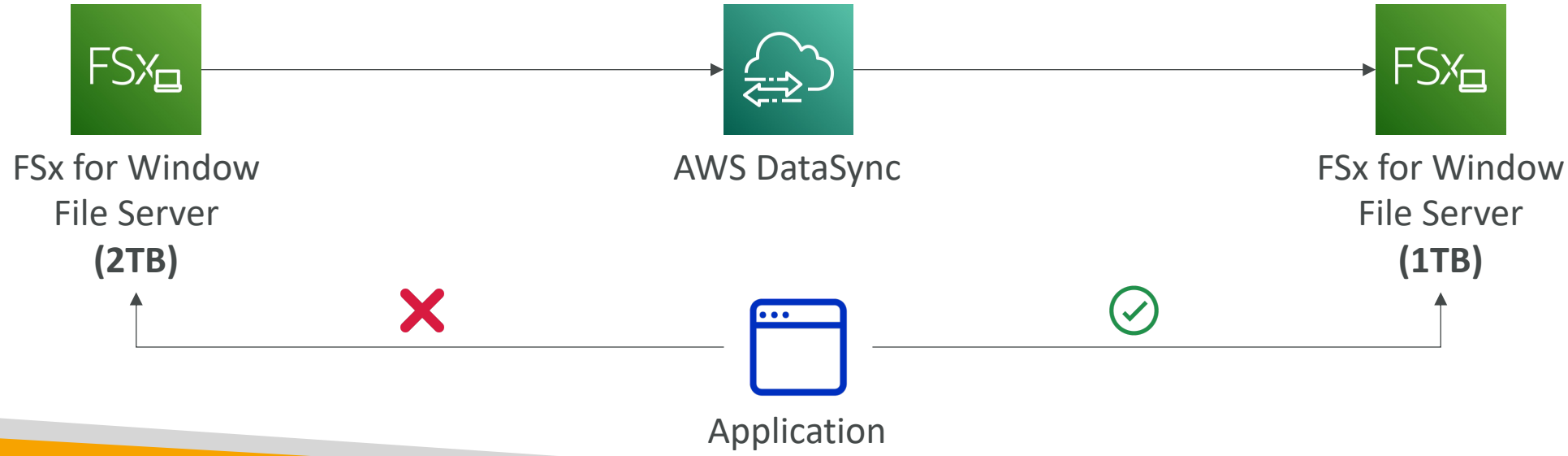


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

FSx – Solution Architecture
Decrease FSx Volume Size

- If you take a backup, you can only restore to a same size
- You can only *increase* the amount of storage capacity for a file system; you cannot decrease storage capacity.
- Instead, create a new FSx (smaller), use DataSync to sync data and then migrate your app over

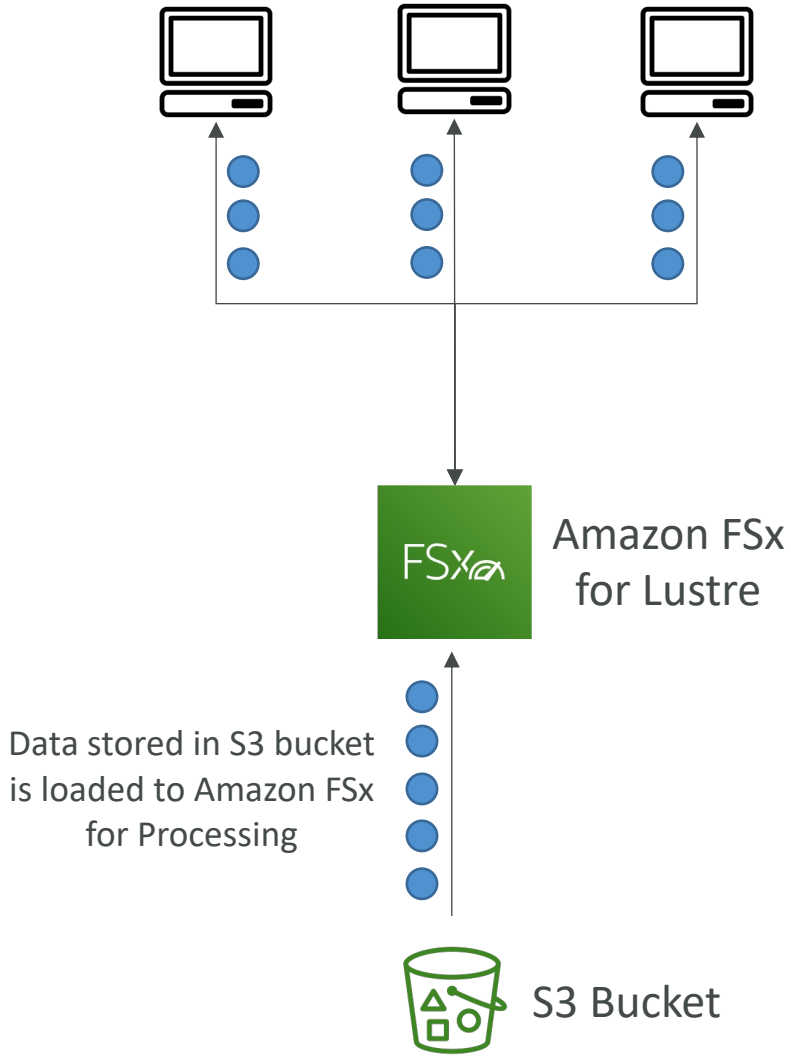


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

FSx for Lustre – Data Lazy Loading

- Any data processing job on Lustre with S3 as an input data source can be started without Lustre doing a full download of the dataset first
- Data is lazy loaded: only the data that is actually processed is loaded, meaning you can decrease your costs and latency
- Data is also loaded only once, therefore you reduce your requests on Amazon S3



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

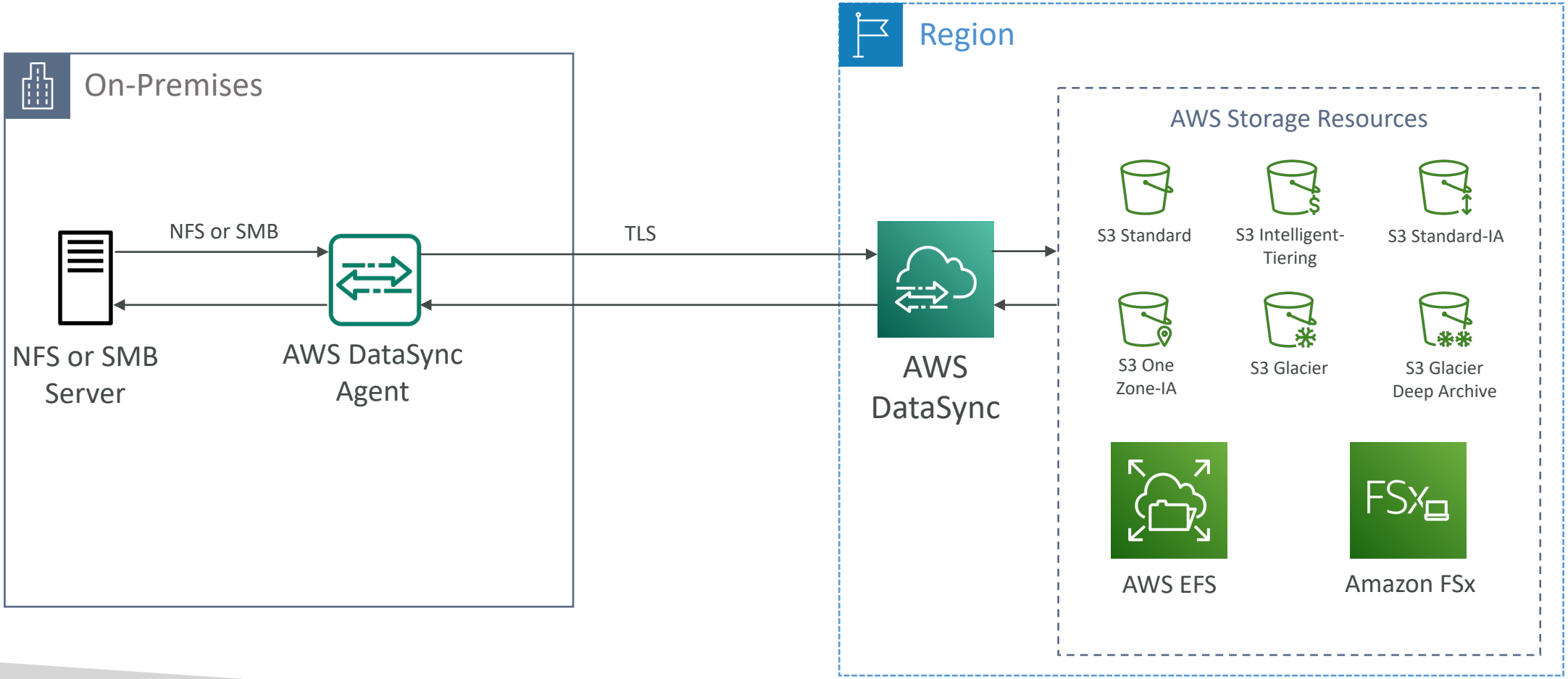
AWS DataSync

- Move large amount of data to and from
  - On-premises / other cloud to AWS (NFS, SMB, HDFS, S3 API…) – **needs agent**
  - AWS to AWS (different storage services) – no agent needed
- Can synchronize to:
  - Amazon S3 (any storage classes – including Glacier)
  - Amazon EFS
  - Amazon FSx (Windows, Lustre, NetApp, OpenZFS...)
- Replication tasks can be scheduled hourly, daily, weekly
- **File permissions and metadata are preserved** (NFS POSIX, SMB…)
- One agent task can use 10 Gbps, can setup a bandwidth limit

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

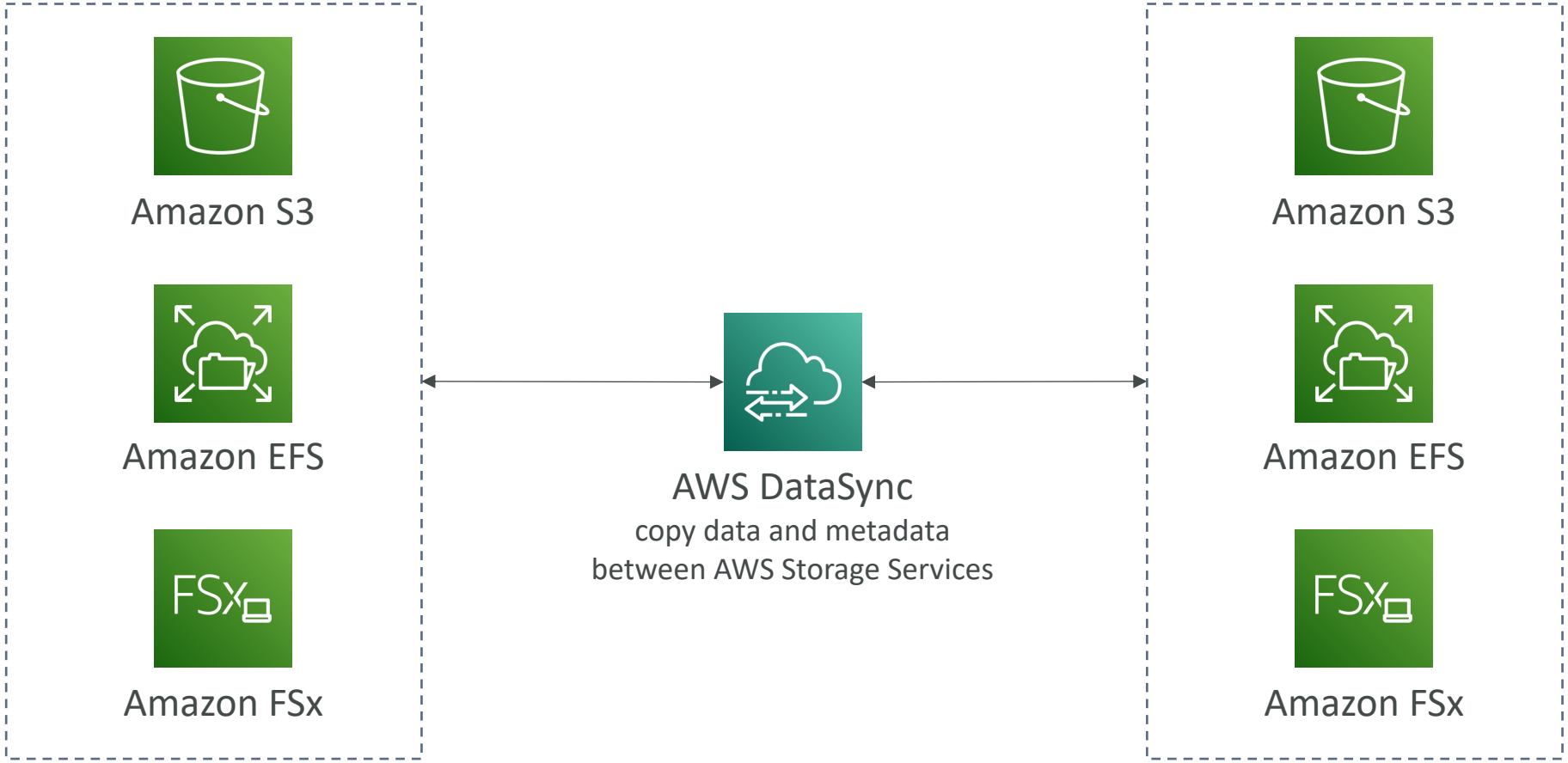
AWS DataSync
NFS / SMB to AWS (S3, EFS, FSx…)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

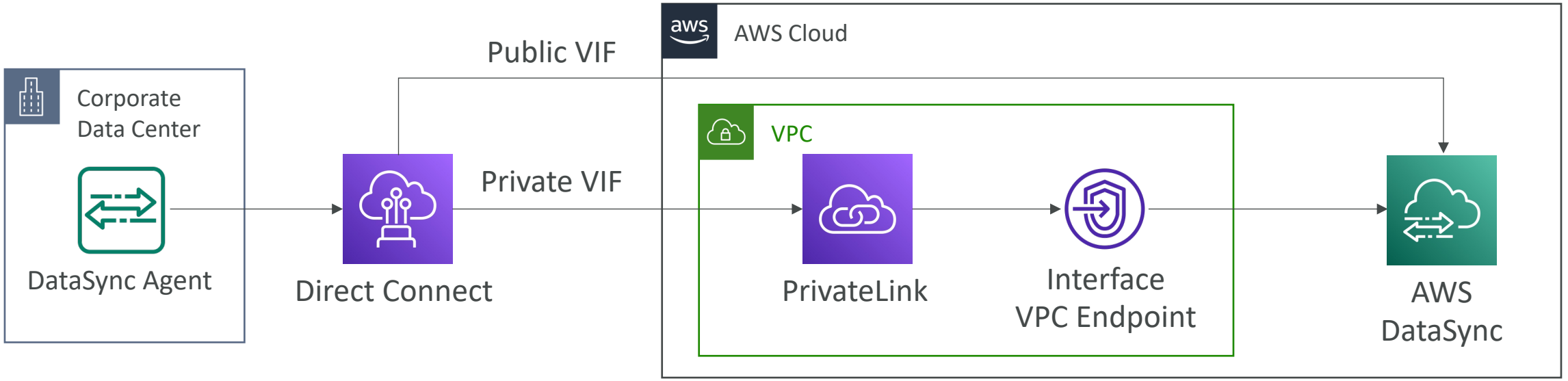
AWS DataSync
Transfer between AWS storage services



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS DataSync
Private VIF through Direct Connect



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

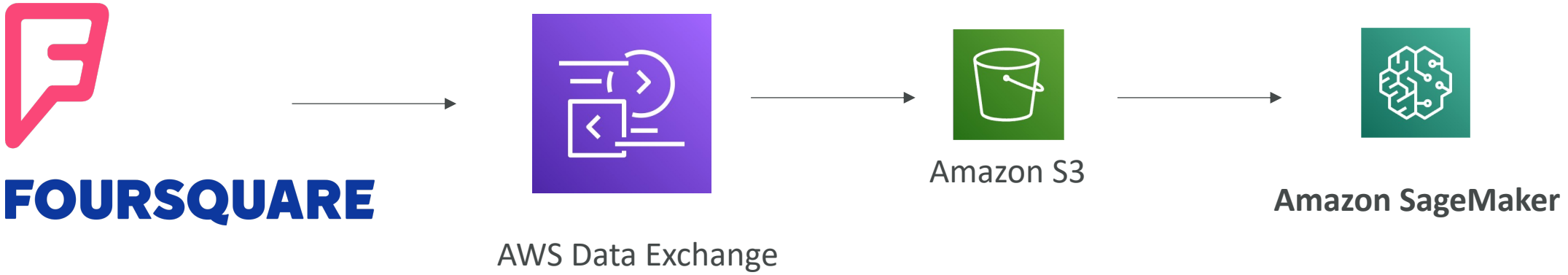
AWS Data Exchange

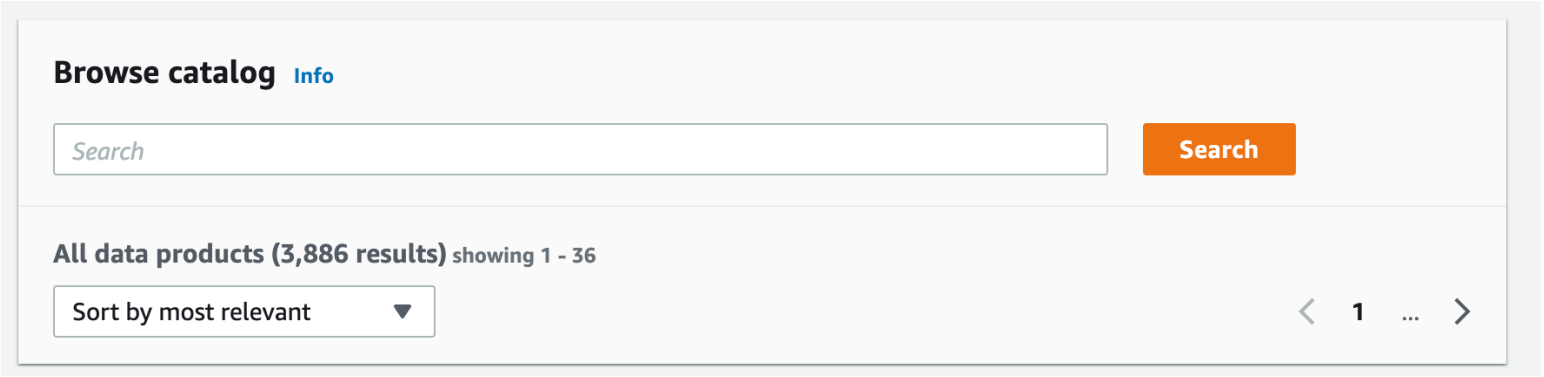
- Find, subscribe to, and use third-party data in the cloud
  - Reuters, who curate data from over 2.2 million unique news stories per year in multiple languages
  - Change Healthcare, who process and anonymize more than 14 billion healthcare transactions and $1 trillion in claims annually
  - Dun & Bradstreet, who maintain a database of more than 330 million global business records;
  - Foursquare, whose location data is derived from 220 million unique consumers and includes more than 60 million global commercial venues.
- Once subscribed to a data product, you can use the AWS Data Exchange API to load data directly into Amazon S3 and then analyze it with a wide variety of AWS analytics and machine learning services

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Data Exchange





NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

AWS Data Exchange – Other Products

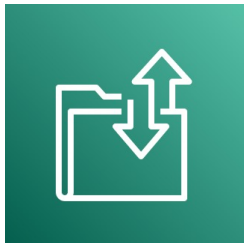
- AWS Data Exchange for Redshift
  - Find and subscribe to third-party data in AWS Data Exchange that you can query in an Amazon Redshift data warehouse in minutes
  - Easily license your data in Amazon Redshift through AWS Data Exchange

- AWS Data Exchange for APIs
  - Find and subscribe to third-party APIs with a consistent access using AWS SDKs
  - Consistent AWS-native authentication and governance

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Transfer Family

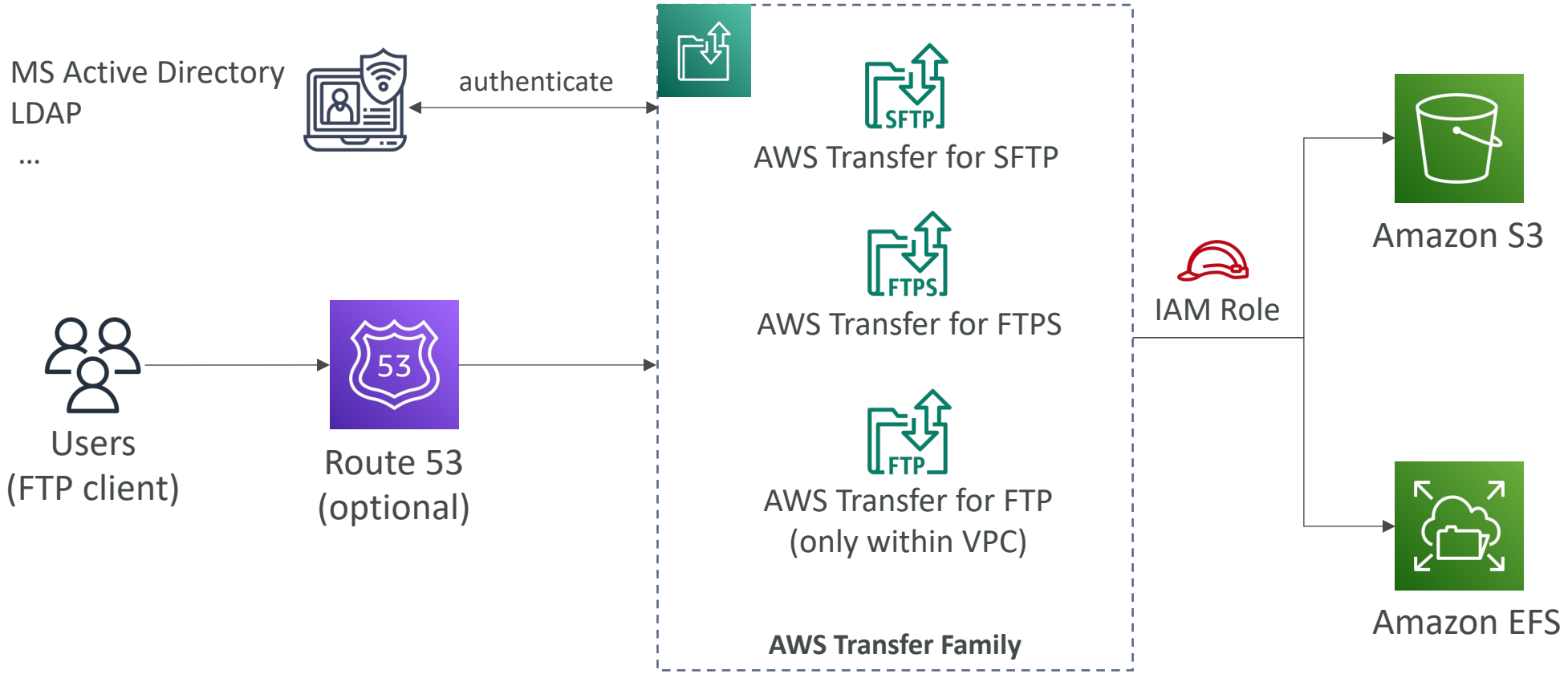


- A fully-managed service for file transfers into and out of Amazon S3 or Amazon EFS using the FTP protocol
- Supported Protocols
  - **AWS Transfer for FTP** (File Transfer Protocol (FTP))
  - **AWS Transfer for FTPS** (File Transfer Protocol over SSL (FTPS))
  - **AWS Transfer for SFTP** (Secure File Transfer Protocol (SFTP))
- Managed infrastructure, Scalable, Reliable, Highly Available (multi-AZ)
- Pay per provisioned endpoint per hour + data transfers in GB
- Store and manage users' credentials within the service
- Integrate with existing authentication systems (Microsoft Active Directory, LDAP, Okta, Amazon Cognito, custom)
- Usage: sharing files, public datasets, CRM, ERP, …

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Transfer Family

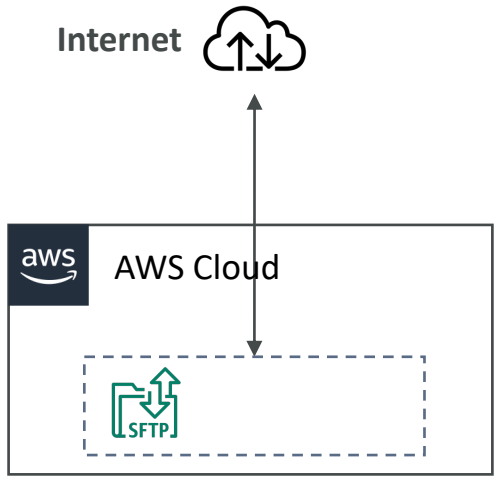


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

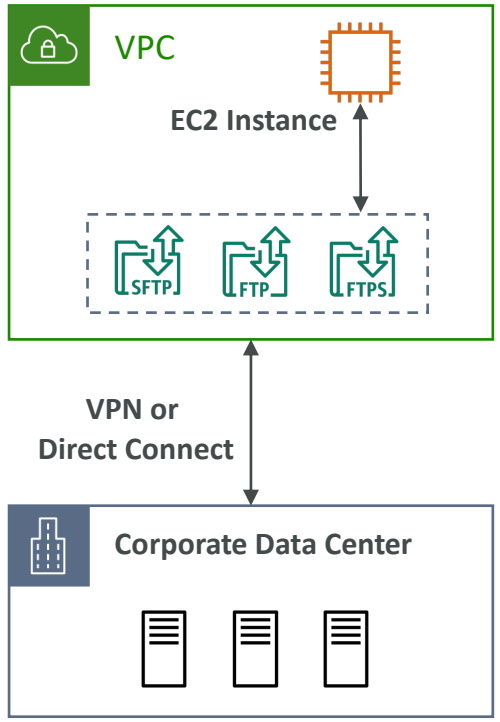
AWS Transfer Family – Endpoint Types

Public Endpoint



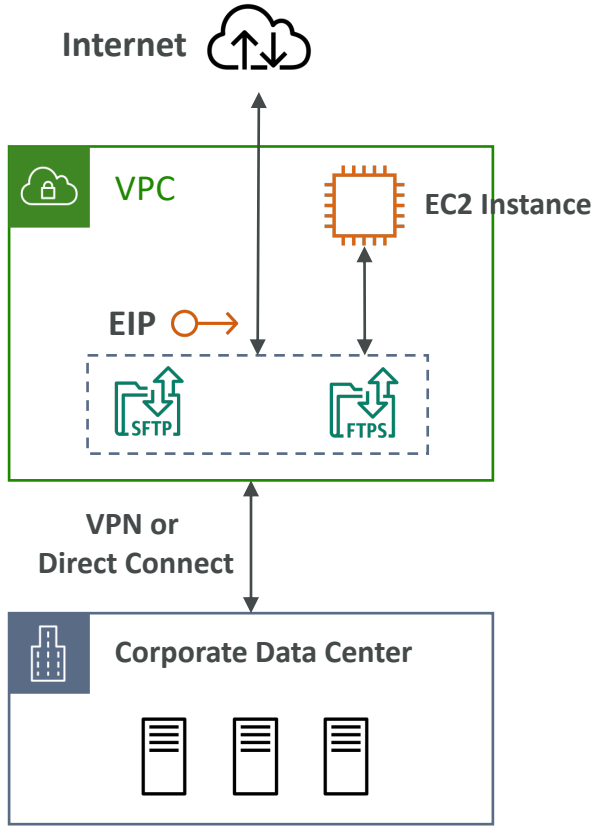
- IPs managed by AWS (subject to change, use DNS names)
- Can't setup allow lists by source IP addresses

VPC Endpoint with Internal Access



- Static private IPs
- Setup allow lists (SGs & NACLs)

VPC Endpoint with Internet-facing Access



- Static private IPs
- Static public IPs (EIPs)
- Setup Security Groups

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Price of Storage (GB Month)

S3 (GB/Month)

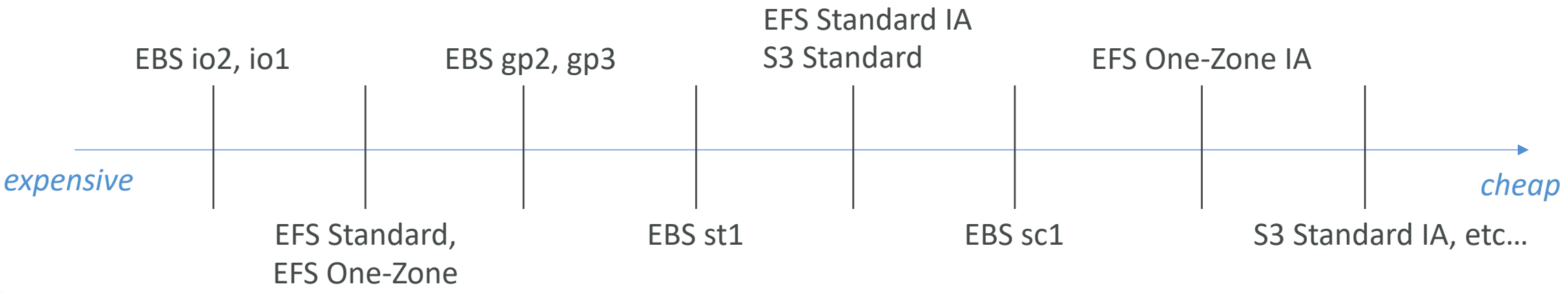
Class	Price
Standard	$0.023
Standard IA	$0.0125
One-Zone IA	$0.01
Intelligent Tiering	$0.023
Glacier Instant Retrieval	$0.004
Glacier Flexible Retrieval	$0.0036
Glacier Deep Arcvhive	$0.00099

EBS (GB/Month)

Class	Price
io1	$0.125 + $0.065/iops
io2	$0.125 + $0.065/iops
gp2	$0.10
gp3	$0.08
st1	$0.045
sc1	$0.015

EFS (GB/Month)

Class	Price
Standard	$0.30
One-Zone	$0.16
Standard IA	$0.025
One-Zone IA	$0.0133



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

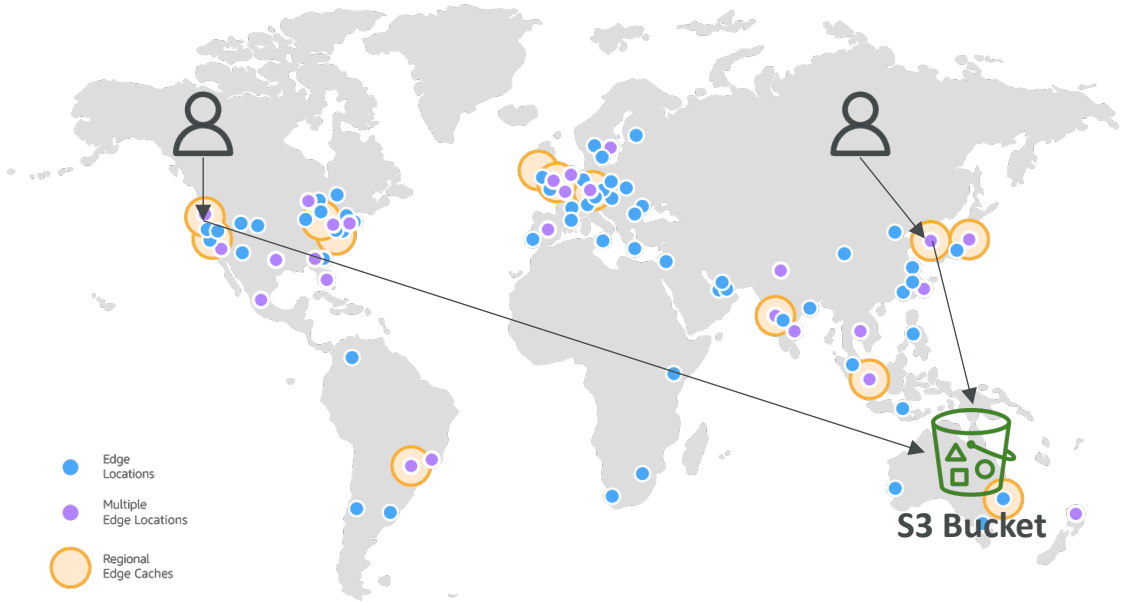
Caching

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon CloudFront

- Content Delivery Network (CDN)
- Improves read performance, content is cached at the edge
- 225+ Point of Presence globally (215+ Edge Locations & 13 Regional Edge Caches)
- Protect against Network and Application layer attacks (e.g., DDoS attacks)
- Integration with AWS Shield, AWS WAF, and Route 53
- Can expose external HTTPS and can talk to internal HTTPS backends
- Supports WebSocket protocol



Source: https://aws.amazon.com/cloudfront/features/

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

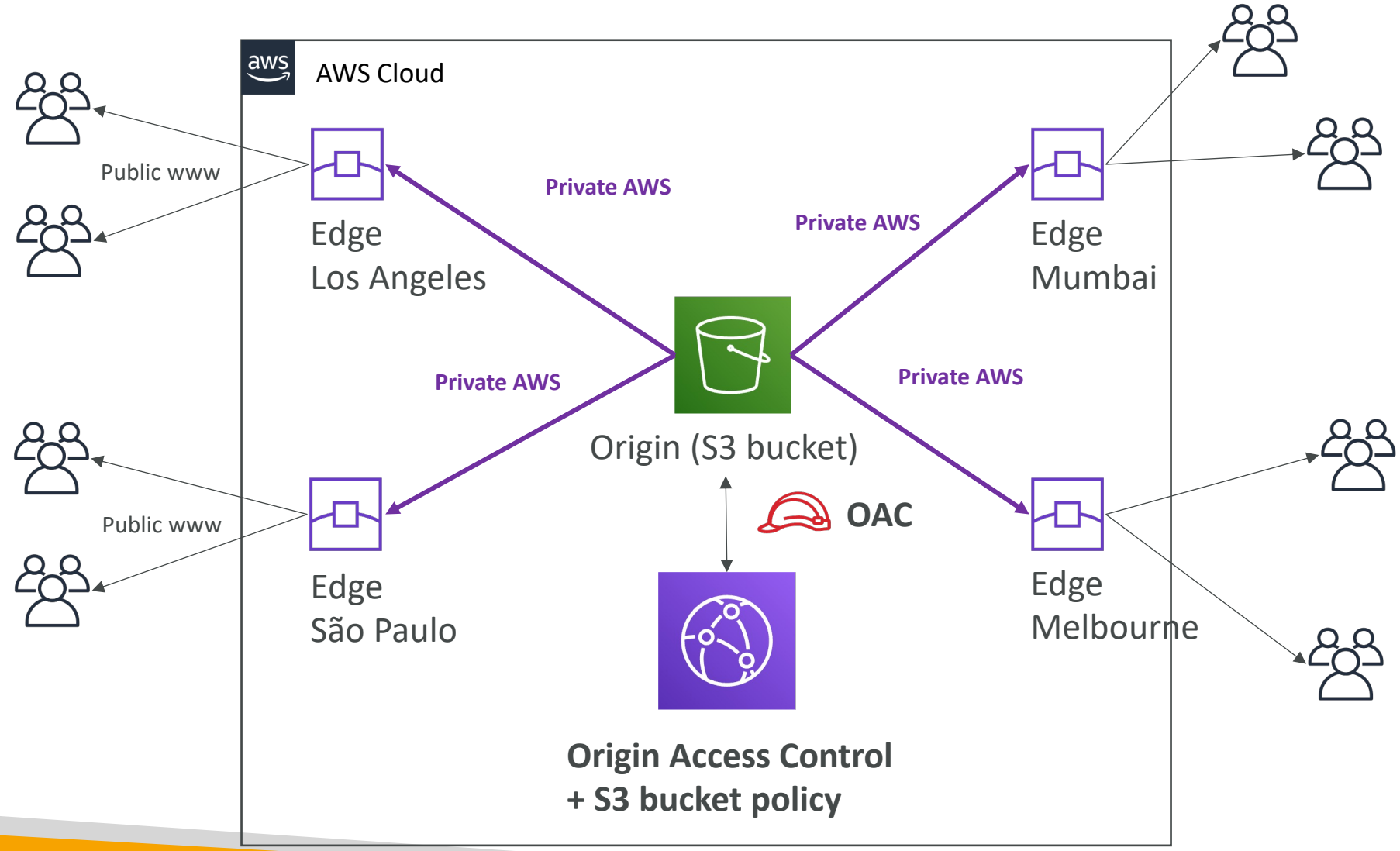
CloudFront – Origins

- S3 Bucket
  - For distributing files
  - For uploading files to S3 (using CloudFront as an ingres)
  - Enhanced security with CloudFront **Origin Access Control (OAC)**
- MediaStore Container & MediaPackage Endpoint
  - To deliver Video On Demand (VOD) or live streaming video using AWS Media Services
- VPC Origin
  - For applications hosted in VPC private subnets
  - Application Load Balancer / Network Load Balancer / EC2 Instances
- Custom Origin (HTTP)
  - API Gateway (for more control… otherwise use API Gateway Edge)
  - S3 Bucket configured as a website (enable Static Website hosting)
  - Any HTTP backend you want

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront – S3 as an Origin



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront vs S3 Cross Region Replication

- CloudFront:
  - Global Edge network
  - Files are cached for a TTL (maybe a day)
  - **Great for static content that must be available everywhere**

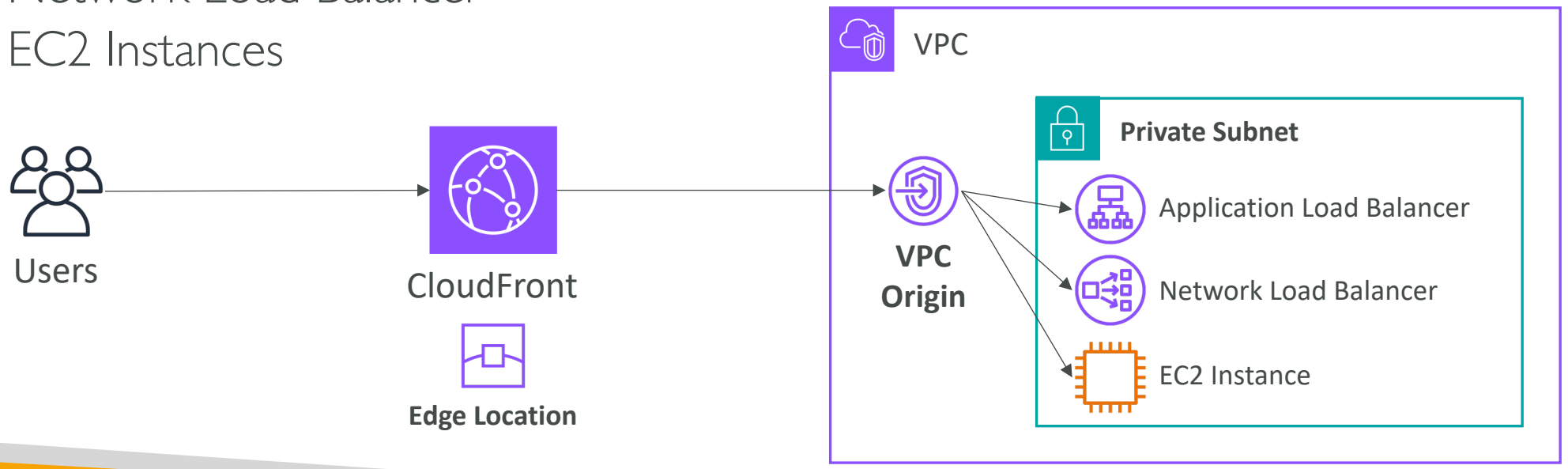
- S3 Cross Region Replication:
  - Must be setup for each region you want replication to happen
  - Files are updated in near real-time
  - Read only
  - **Great for dynamic content that needs to be available at low-latency in few regions**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront – ALB or EC2 as an origin Using VPC Origins

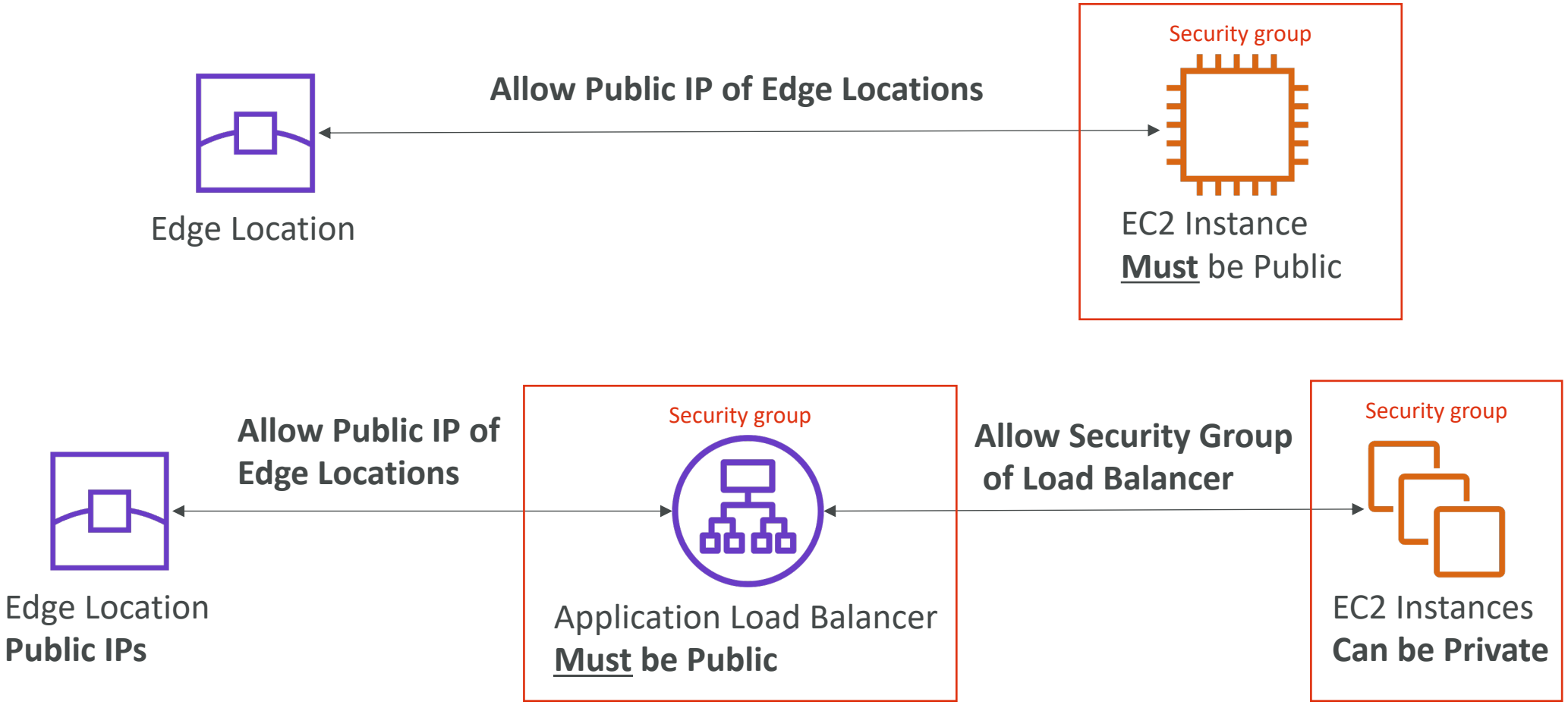
- Allows you to deliver content from your applications hosted in your VPC private subnets (no need to expose them on the Internet)
- Deliver traffic to **private**:
  - Application Load Balancer
  - Network Load Balancer
  - EC2 Instances



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront – EC2 or ALB as an origin



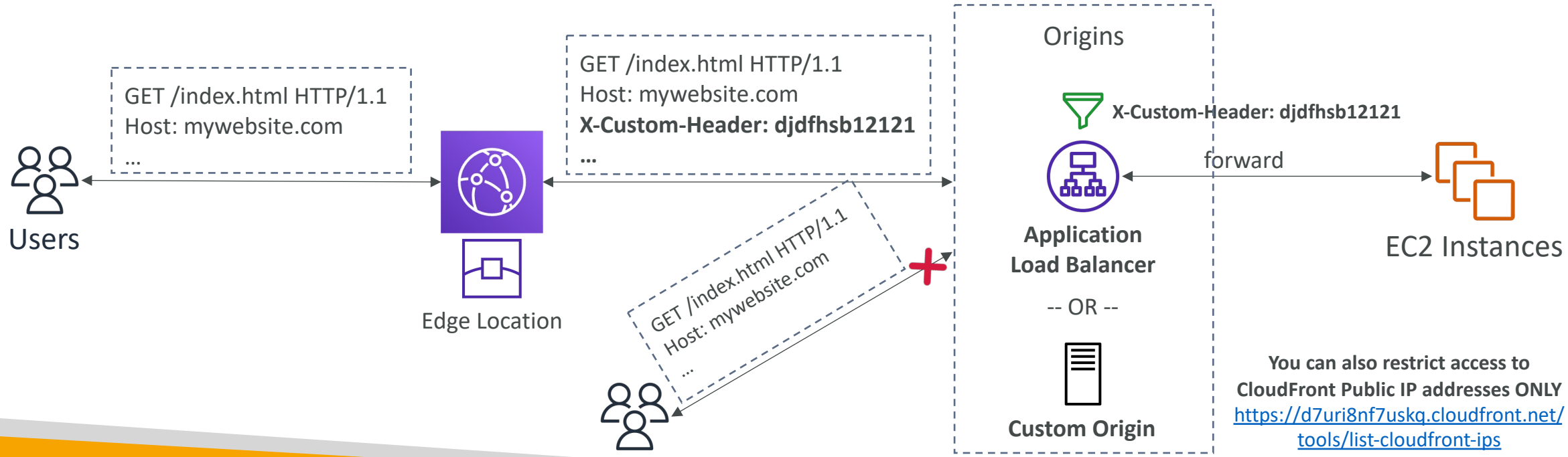
http://d7uri8nf7uskq.cloudfront.net/tools/list-cloudfront-ips

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront – Restrict Access to Application Load Balancers and Custom Origins

- Prevent direct access to your ALB or Custom Origins (only access through CloudFront)
- First, configure CloudFront to add a **Custom HTTP Header** to requests it sends to the ALB
- Second, configure the ALB to only forward requests that contain that Custom HTTP Header
- Keep the custom header name and value secret!

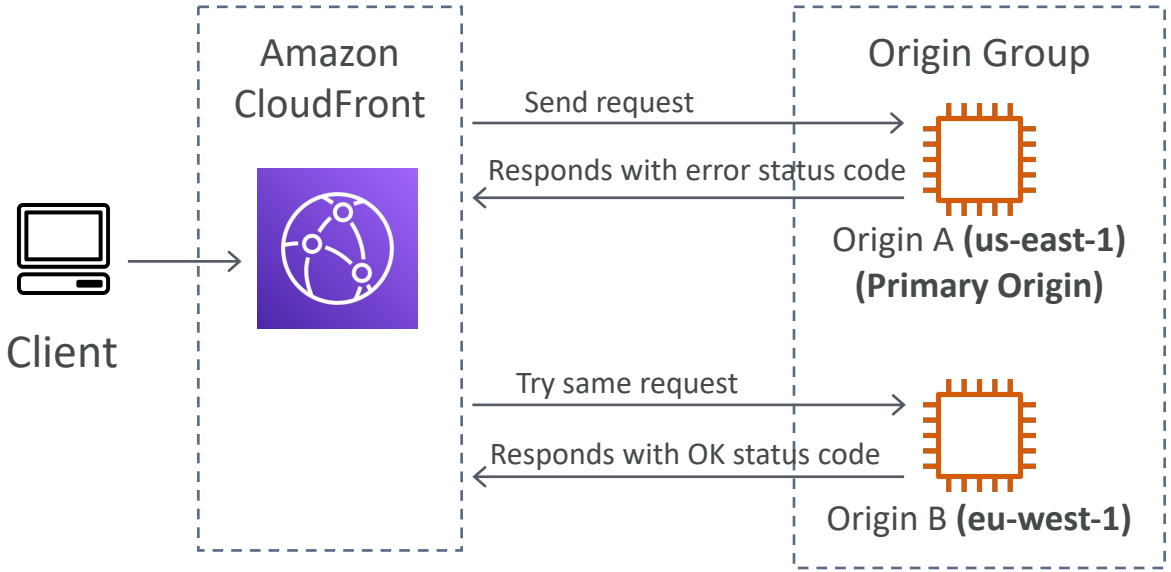


© Stephane Maarek

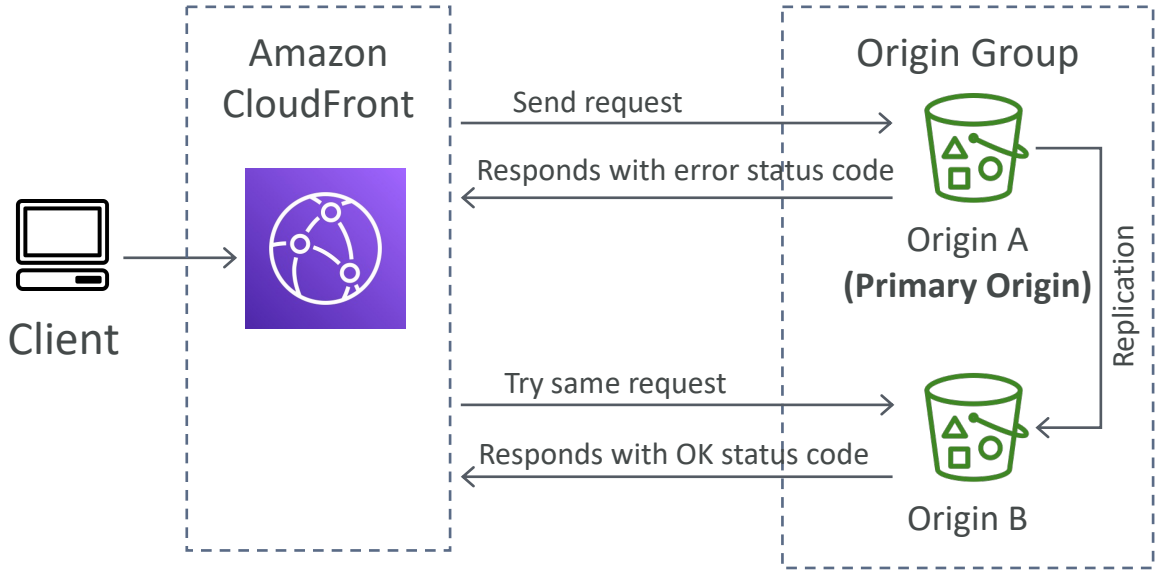
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront – Origin Groups

- To increase high-availability and do failover
- Origin Group: one primary and one secondary origin
- If the primary origin fails, the second one is used
- Origins can be cross AWS Regions



Cross-Region High Availability



S3 + CloudFront – Region-level High Availability

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront Geo Restriction

- You can restrict who can access your distribution
  - **Allow list:** Allow your users to access your content only if they're in one of the countries on a list of approved countries.
  - **Block list:** Prevent your users from accessing your content if they're in one of the countries on a blacklist of banned countries.
- The "country" is determined using a 3rd party Geo-IP database
- Use case: Copyright Laws to control access to content
- Note: the geo header *CloudFront-Viewer-Country* is in Lambda@Edge

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront – Pricing

- CloudFront Edge locations are all around the world
- The cost of data out per edge location varies

Per Month	United States, Mexico, & Canada	Europe & Israel	South Africa, Kenya, & Middle East	South America	Japan	Australia & New Zealand	Hong Kong, Philippines, Singapore, South Korea, Taiwan, & Thailand	India
First 10TB	$0.085	$0.085	$0.110	$0.110	$0.114	$0.114	$0.140	$0.170
Next 40TB	$0.080	$0.080	$0.105	$0.105	$0.089	$0.098	$0.135	$0.130
Next 100TB	$0.060	$0.060	$0.090	$0.090	$0.086	$0.094	$0.120	$0.110
Next 350TB	$0.040	$0.040	$0.080	$0.080	$0.084	$0.092	$0.100	$0.100
Next 524TB	$0.030	$0.030	$0.060	$0.060	$0.080	$0.090	$0.080	$0.100
Next 4PB	$0.025	$0.025	$0.050	$0.050	$0.070	$0.085	$0.070	$0.100
Over 5PB	$0.020	$0.020	$0.040	$0.040	$0.060	$0.080	$0.060	$0.100

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront – Price Classes

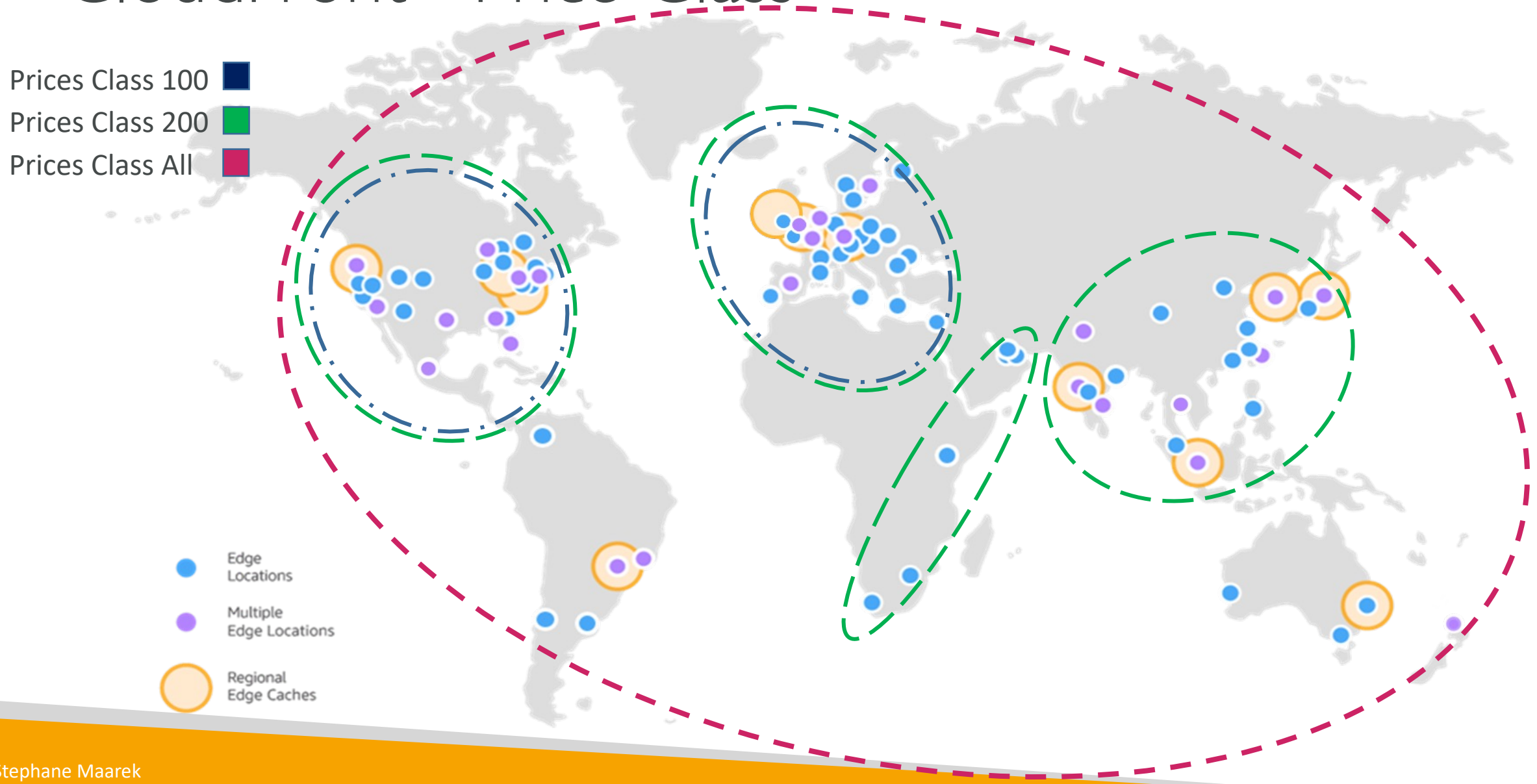
- You can reduce the number of edge locations for **cost reduction**
- Three price classes:
  1. Price Class All: all regions – best performance
  2. Price Class 200: most regions, but excludes the most expensive regions
  3. Price Class 100: only the least expensive regions

Edge Locations Included Within	**United States, Mexico, & Canada**	**Europe & Israel**	**South Africa, Kenya, & Middle East**	**South America**	**Japan**	**Australia & New Zealand**	**Hong Kong, Philippines, Singapore, South Korea, Taiwan, & Thailand**	**India**
Price Class All	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Price Class 200	Yes	Yes	Yes	x	Yes	x	Yes	Yes
Price Class 100	Yes	Yes	x	x	x	x	x	x

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront - Price Class

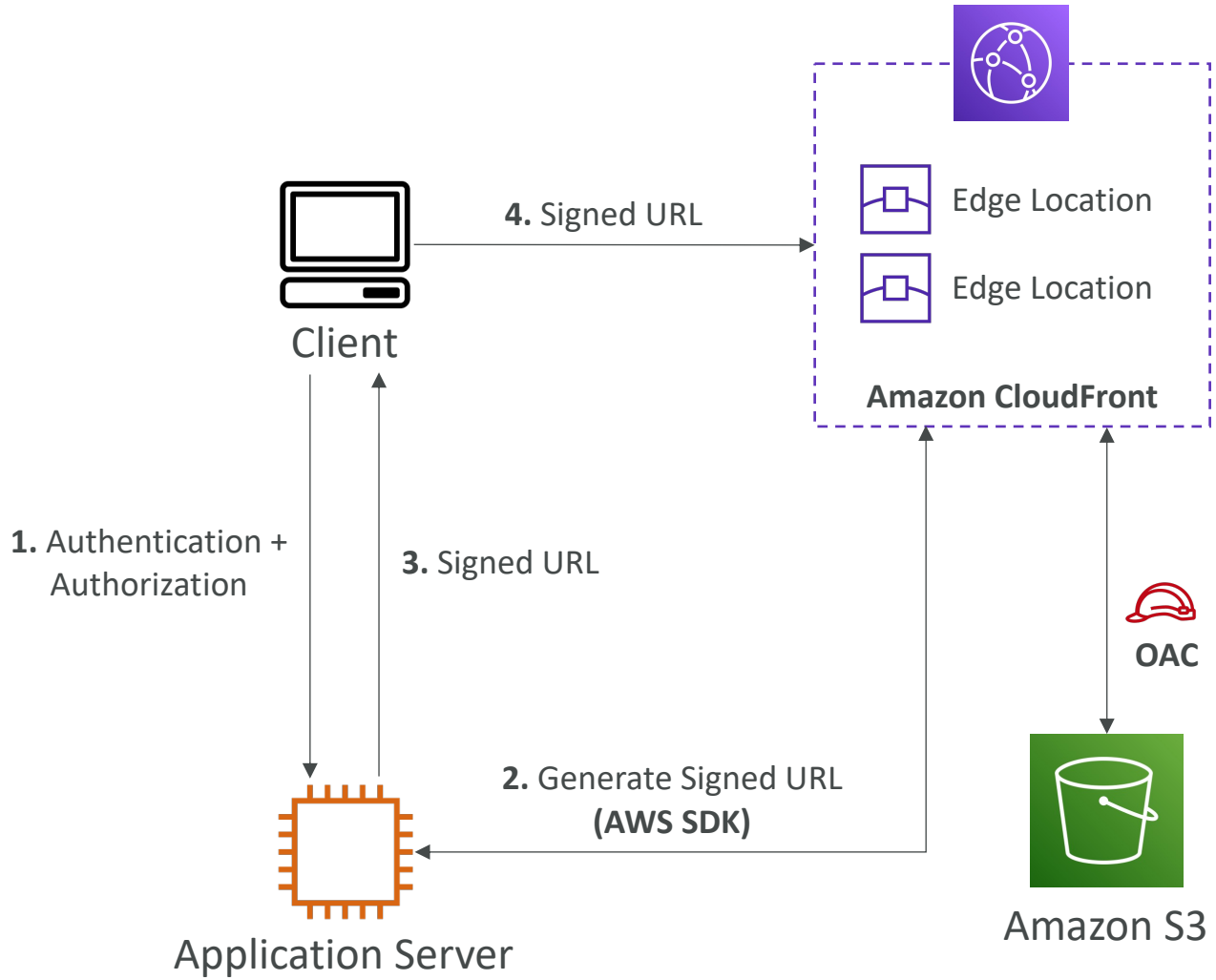


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront Signed URL Diagram

- Signed URL with expiration to control access to content in CloudFront
- The Signed URL are generated by an API call into CloudFront as a trusted signer

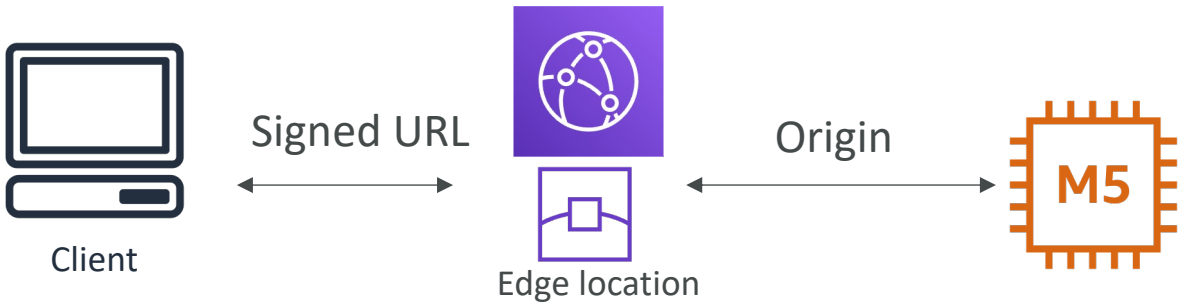


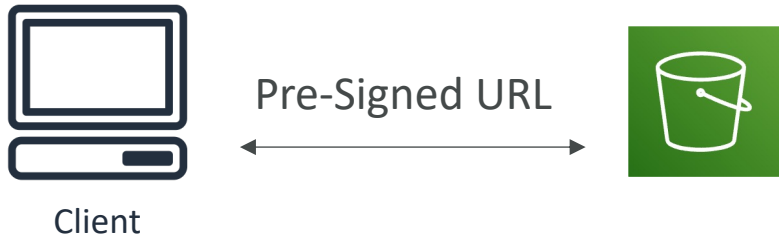
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront Signed URL vs S3 Pre-Signed URL

- CloudFront Signed URL:
  - Allow access to a path, no matter the origin
  - Account wide key-pair, only the root can manage it
  - Can filter by IP, path, date, expiration
  - Can leverage caching features
- S3 Pre-Signed URL:
  - Issue a request as the person who pre-signed the URL
  - Uses the IAM key of the signing IAM principal
  - Limited lifetime



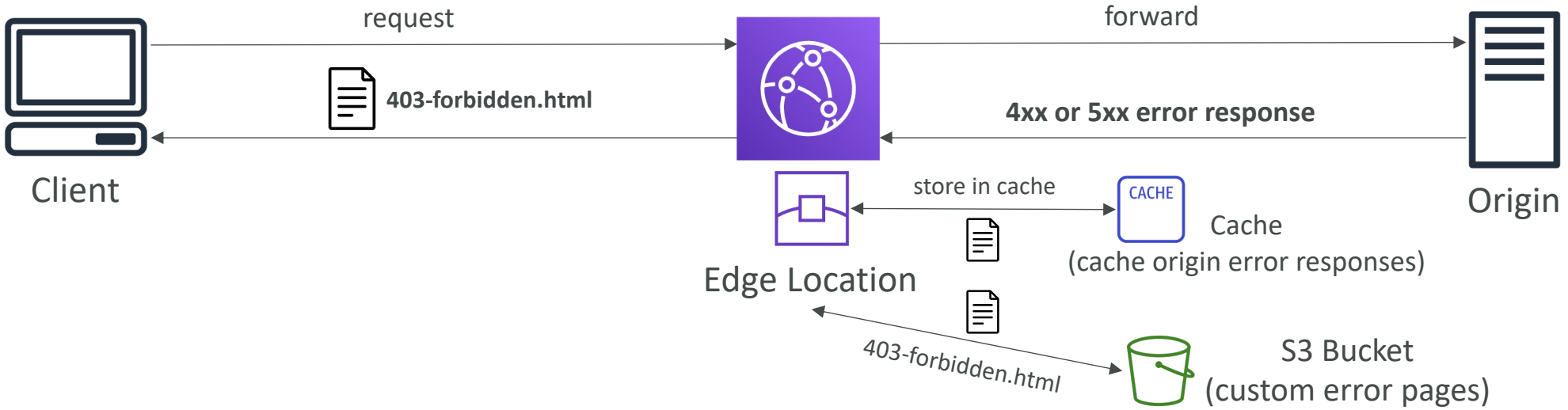


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront – Custom Error Pages

- Return an object to the viewer (e.g., .html) when your origin returns an HTTP 4xx or 5xx status code to CloudFront
- Use **Error Caching Minimum TTL** to specify how long CloudFront caches the custom error pages



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

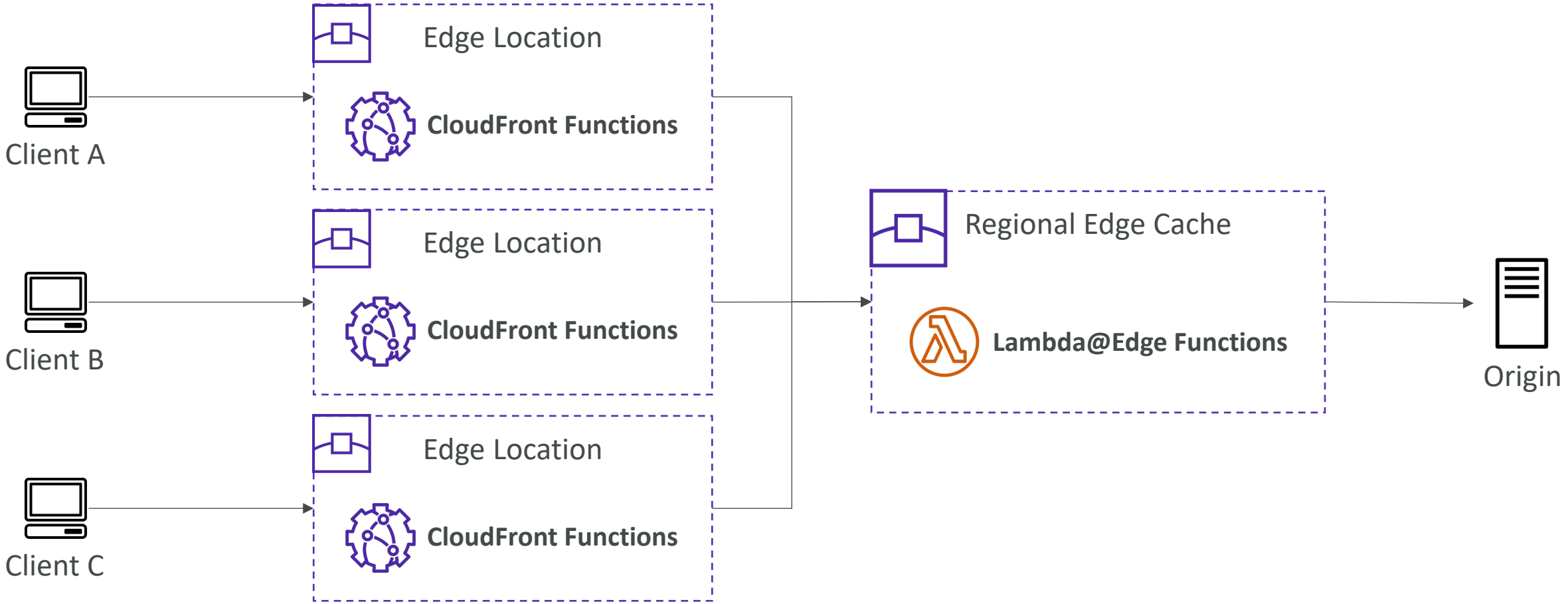
CloudFront – Customization At The Edge

- Many modern applications execute some form of the logic at the edge
- **Edge Function:**
  - A code that you write and attach to CloudFront distributions
  - Runs close to your users to minimize latency
  - Doesn't have any cache, only to change requests/responses
  - CloudFront provides two types: **CloudFront Functions** & **Lambda@Edge**
- Use cases:
  - Manipulate HTTP requests and responses
  - Implement request filtering before reaching your application
  - User authentication and authorization
  - Generate HTTP responses at the edge
  - A/B Testing
  - Bot mitigation at the edge
- You don't have to manage any servers, deployed globally

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront Functions & Lambda@Edge

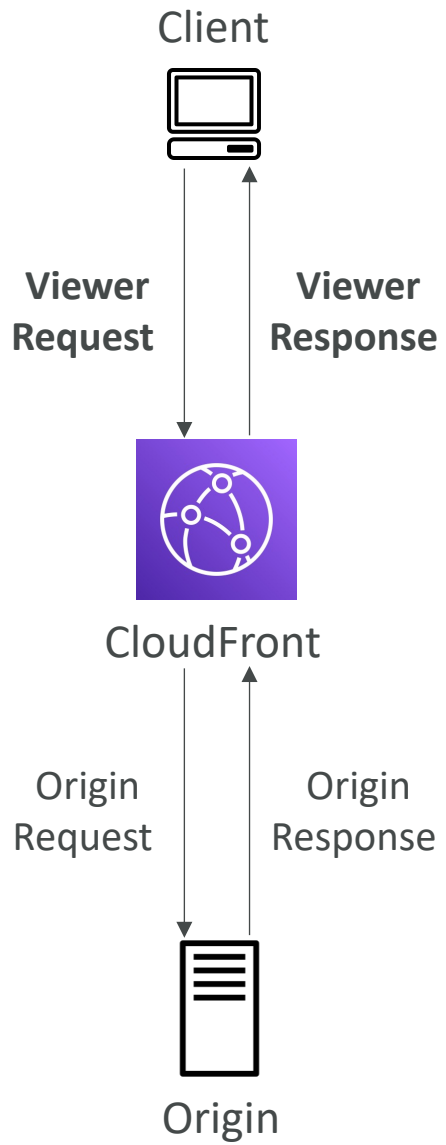


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront – CloudFront Functions

- Lightweight functions written in JavaScript
- For high-scale, latency-sensitive CDN customizations
- Sub-ms startup times, millions of requests/second
- RuN at Edge Locations
- Process-based isolation
- Used to change Viewer requests and responses:
  - **Viewer Request:** after CloudFront receives a request from a viewer
  - **Viewer Response:** before CloudFront forwards the response to the viewer
- Native feature of CloudFront (manage code entirely within CloudFront)

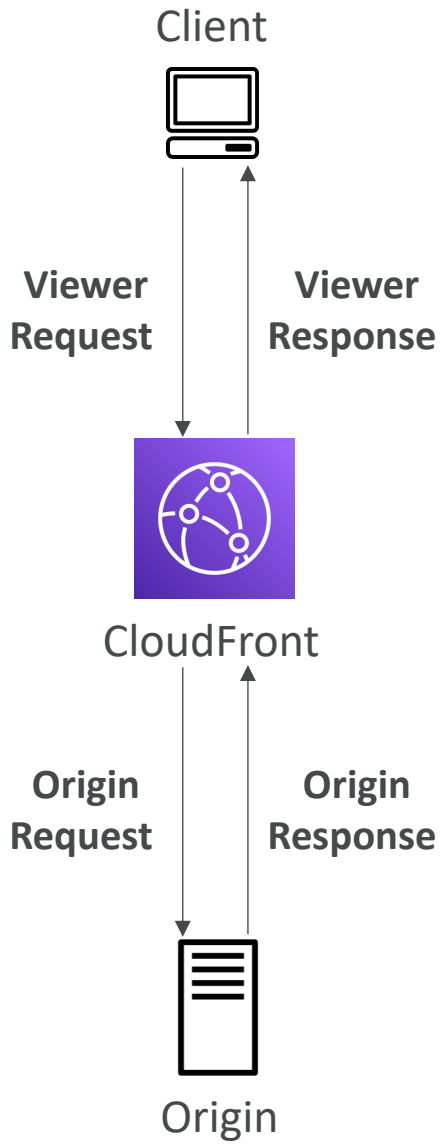


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront – Lambda@Edge

- Lambda functions written in NodeJS or Python
- Scales to 1000s of requests/second
- Runs at the nearest Regional Edge Cache
- VM-based isolation
- Used to change CloudFront requests and responses:
  - **Viewer Request** – after CloudFront receives a request from a viewer
  - **Origin Request** – before CloudFront forwards the request to the origin
  - **Origin Response** – after CloudFront receives the response from the origin
  - **Viewer Response** – before CloudFront forwards the response to the viewer
- Author your functions in one AWS Region (us-east-1), then CloudFront replicates to its locations

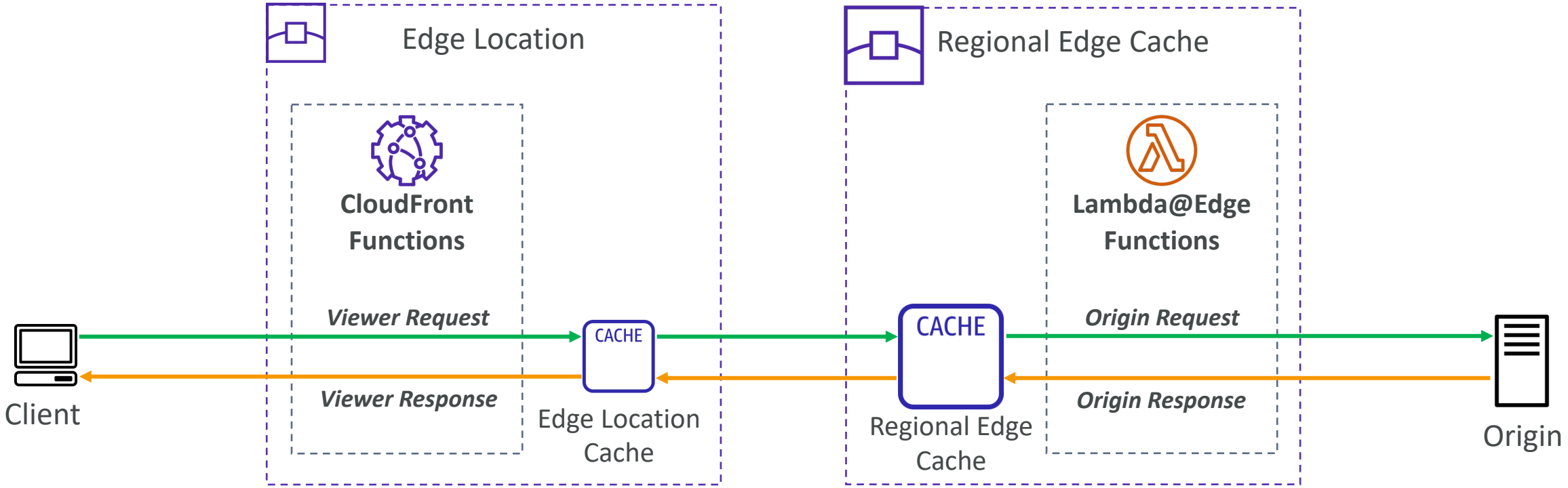


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront Functions with Lambda@Edge

CloudFront Functions and **Lambda@Edge** can be used together



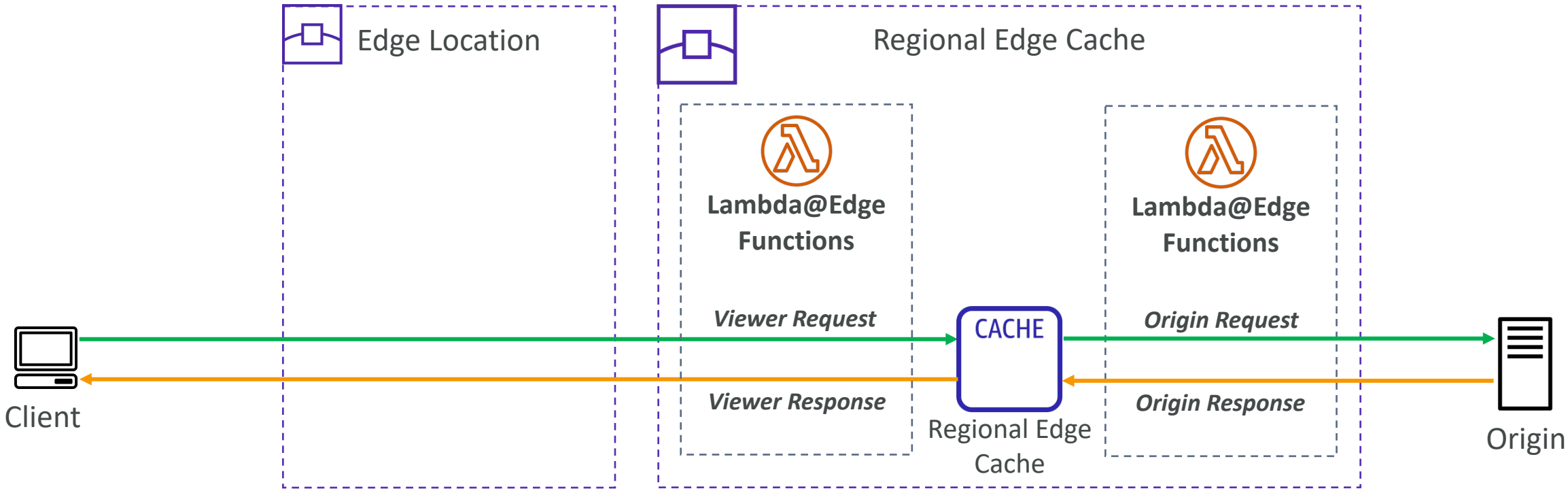
NOTE: You can't combine CloudFront Functions and Lambda@Edge in viewer events (viewer request & viewer response)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Using Lambda@Edge Only

Use when you need some of the capabilities of Lambda@Edge that aren't available with CloudFront Functions (e.g., longer execution time, network access, …)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront Functions vs. Lambda@Edge

	CloudFront Functions	Lambda@Edge
Runtime Support	JavaScript	Node.js, Python
Execution Location	Edge Locations	Regional Edge Caches
CloudFront Triggers	- Viewer Request/Response	- Viewer Request/Response - Origin Request/Response
Isolation	Process-based	VM-based
Max. Execution Time	< 1 ms	- 5 seconds (viewers triggers) - 30 seconds (origin triggers)
Max. Memory	2 MB	- 128 MB (viewer triggers) - 10 GB (origin triggers)
Total Package Size	10 KB	- 1 MB (viewer triggers) - 50 MB (origin t
Network Access, File System Access	No	Yes
Access to the Request Body	No	Yes
Pricing	Free tier available, 1/6th price of @Edge	No free tier, charged per request & duration

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFront Functions vs. Lambda@Edge – Use Cases

CloudFront Functions

- Cache key normalization
  - Transform request attributes (headers, cookies, query strings, URL) to create an optimal Cache Key
- Header manipulation
  - Insert/modify/delete HTTP headers in the request or response
- URL rewrites or redirects
- Request authentication & authorization
  - Create and validate user-generated tokens (e.g., JWT) to allow/deny requests

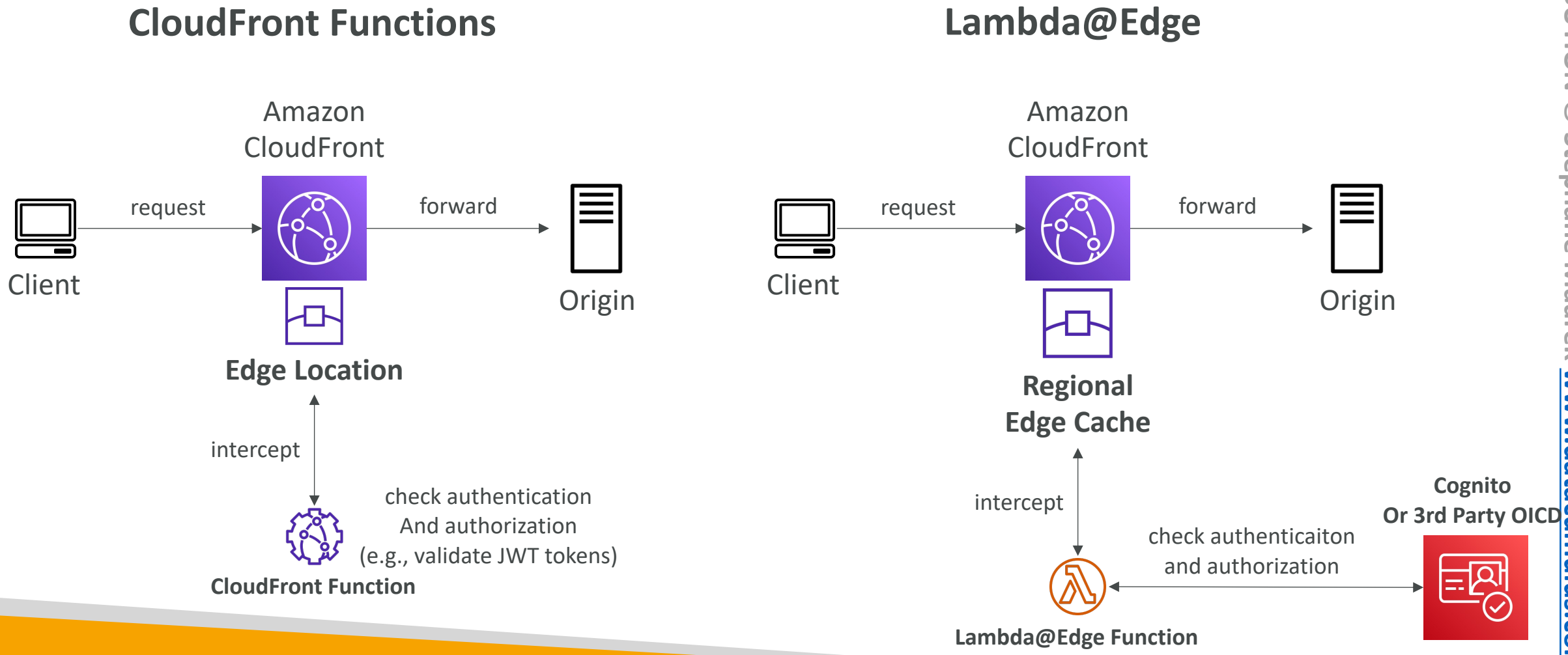
Lambda@Edge

- Longer execution time (several ms)
- Adjustable CPU or memory
- Your code depends on a 3rd libraries (e.g., AWS SDK to access other AWS services)
- Network access to use external services for processing
- File system access or access to the body of HTTP requests

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

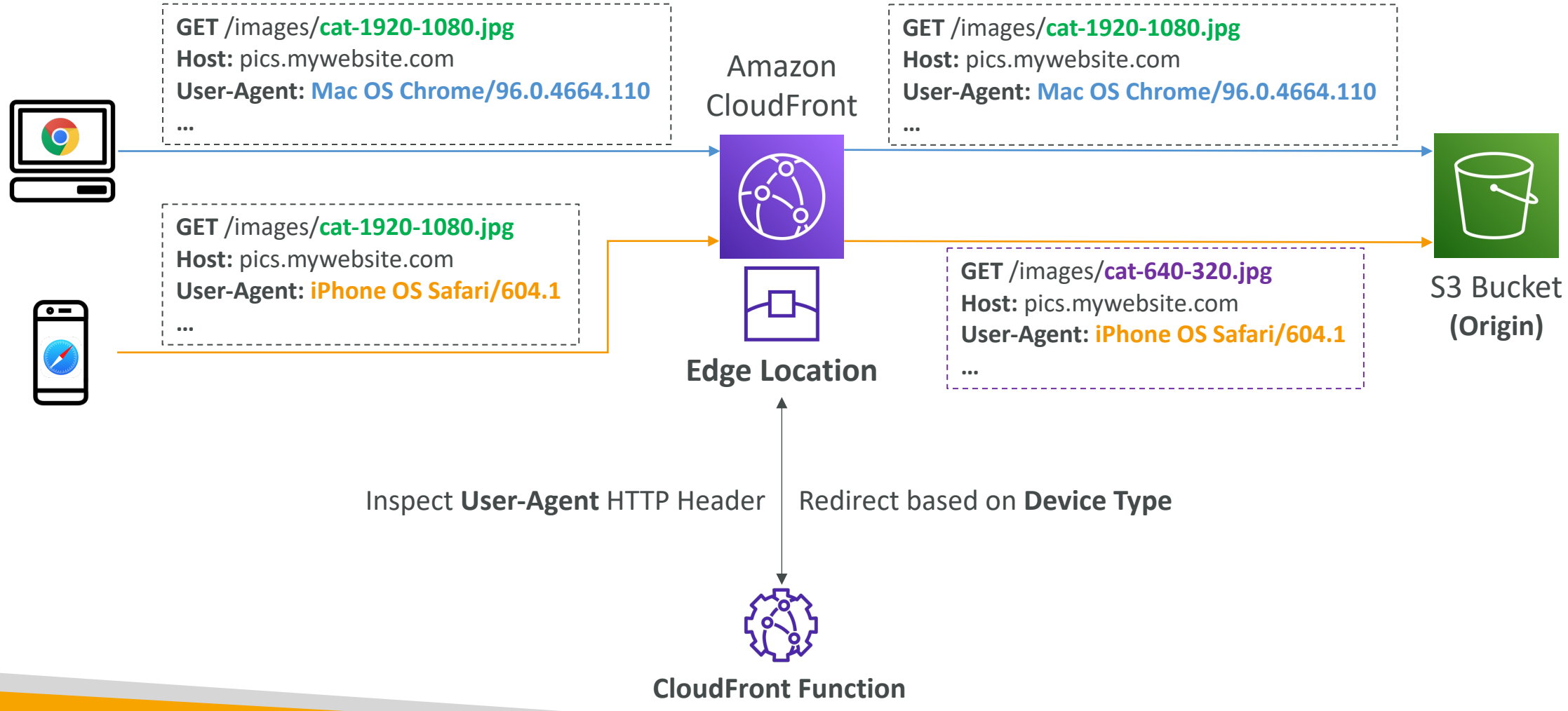
CloudFront Functions vs. Lambda@Edge – Authentication and Authorization



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

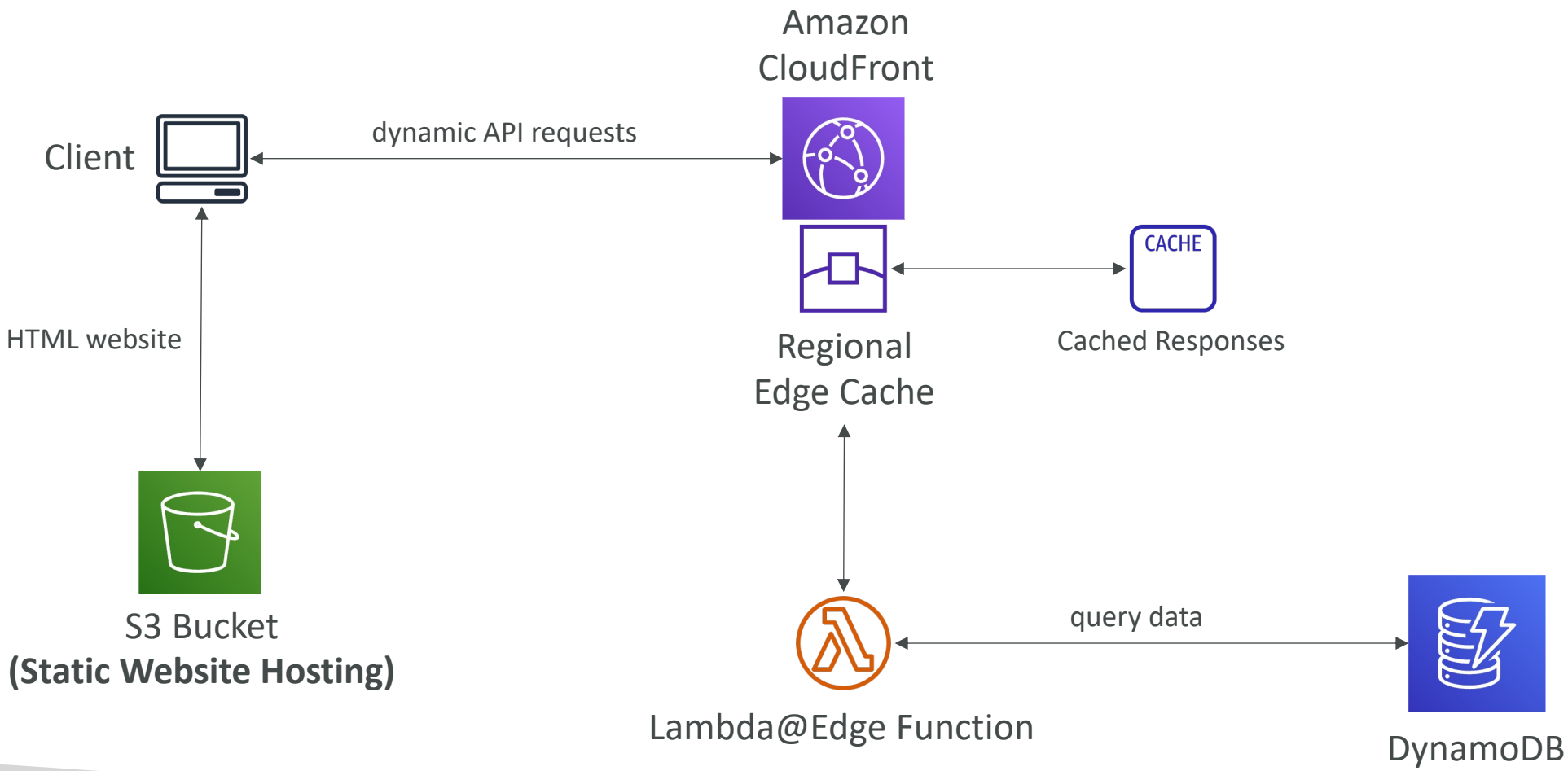
CloudFront Functions: Loading Content based on User-Agent



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

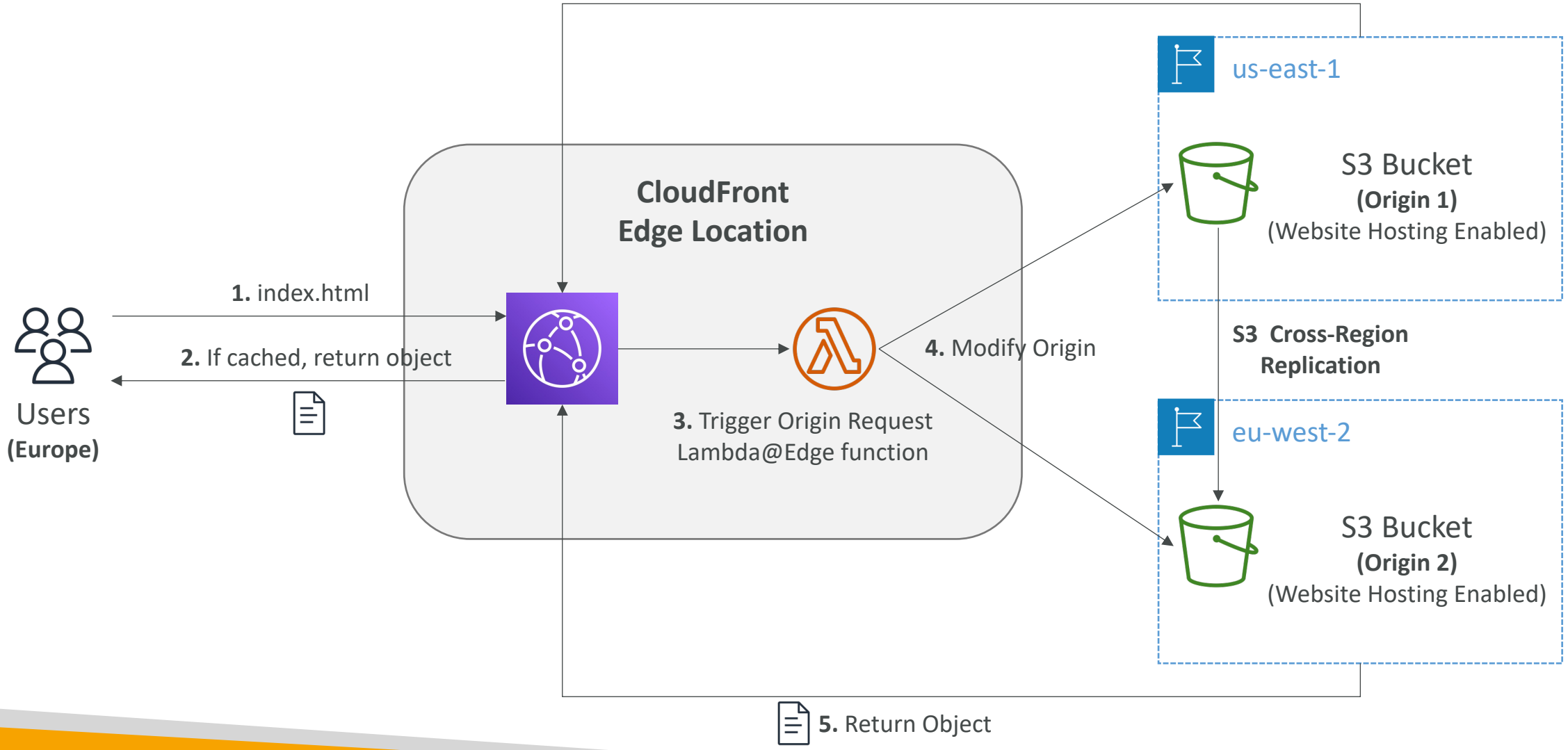
Lambda@Edge – Global Application



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Lambda@Edge – Route to Different Origin



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ElastiCache Overview

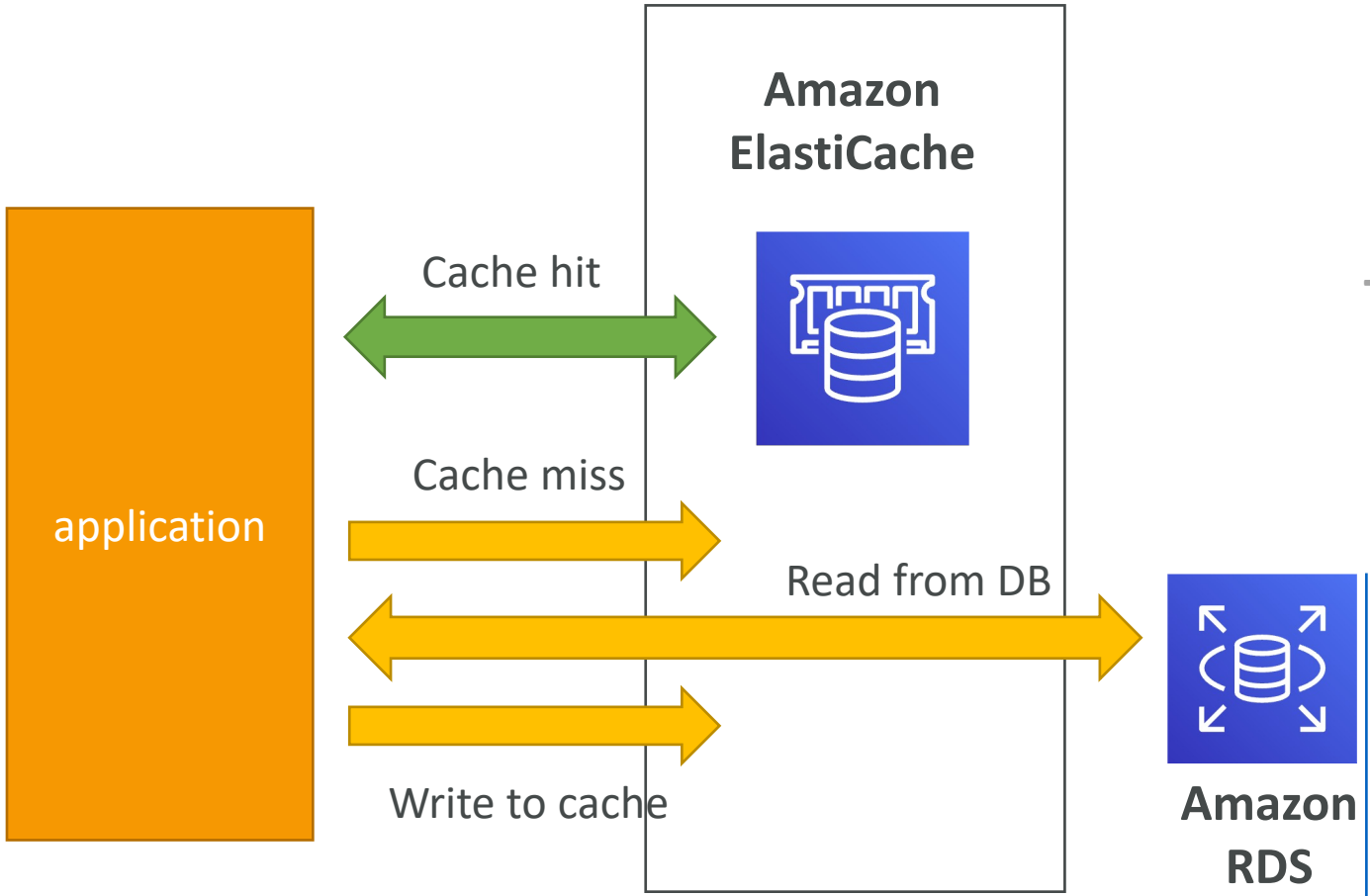
- The same way RDS is to get managed Relational Databases…
- ElastiCache is to get managed Redis or Memcached
- Caches are in-memory databases with really high performance, low latency
- Helps reduce load off of databases for read intensive workloads
- Helps make your application stateless
- AWS takes care of OS maintenance / patching, optimizations, setup, configuration, monitoring, failure recovery and backups
- Using ElastiCache involves heavy application code changes

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

ElastiCache Solution Architecture - DB Cache

- Applications queries ElastiCache, if not available, get from RDS and store in ElastiCache.
- Helps relieve load in RDS
- Cache must have an invalidation strategy to make sure only the most current data is used in there.

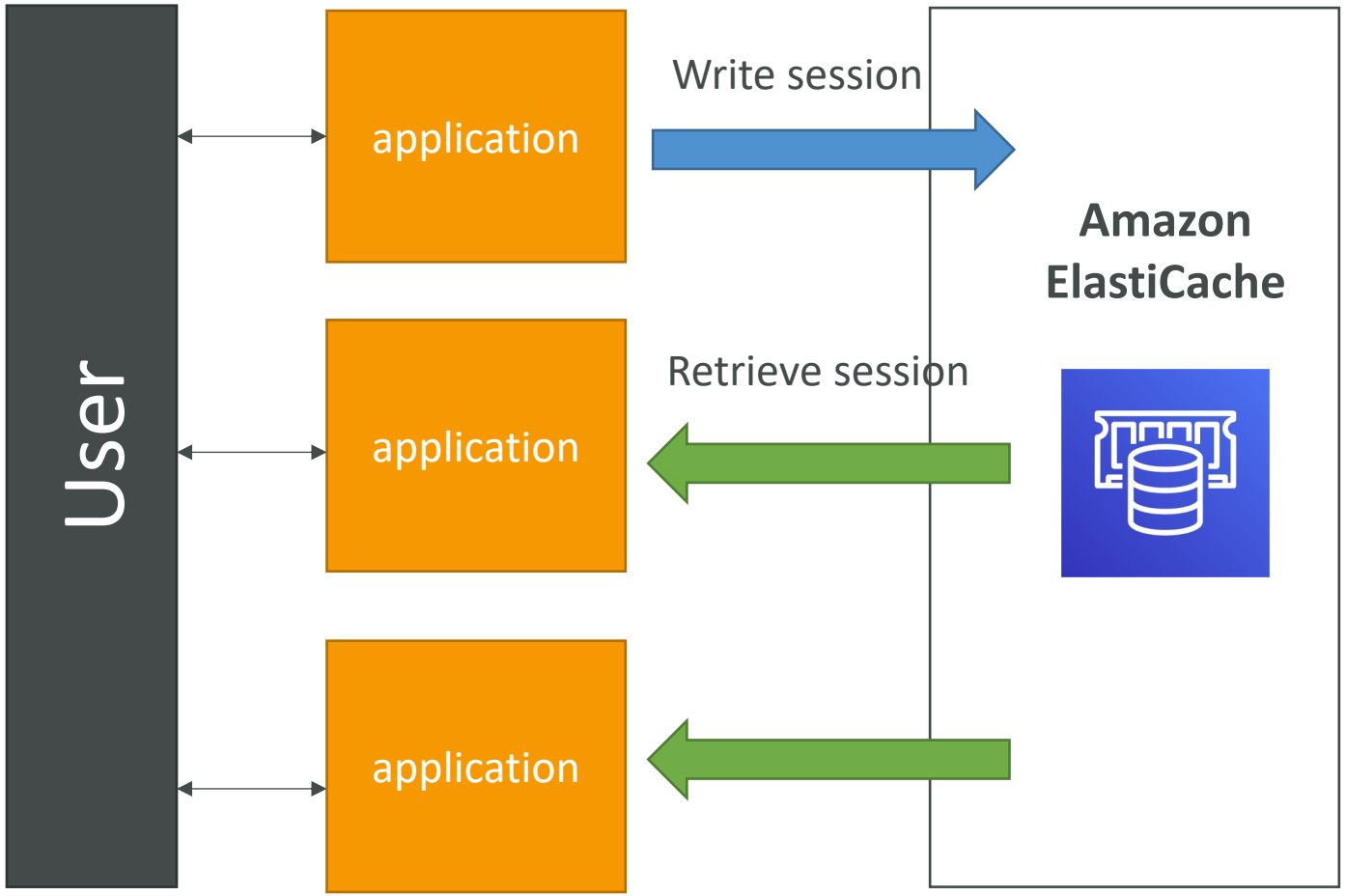


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

ElastiCache
Solution Architecture – User Session Store

- User logs into any of the application
- The application writes the session data into ElastiCache
- The user hits another instance of our application
- The instance retrieves the data and the user is already logged in



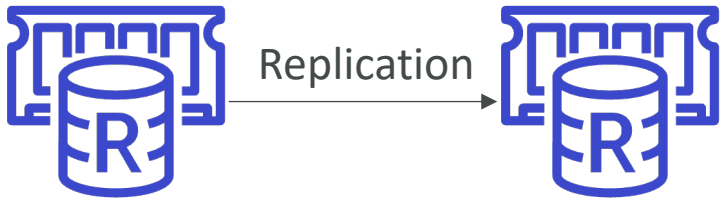
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

ElastiCache – Redis vs Memcached

REDIS

- **Multi AZ** with Auto-Failover
- **Read Replicas** to scale reads and have **high availability**
- Persistent, Data Durability: Append Only File (AOF), **backup and restore features**



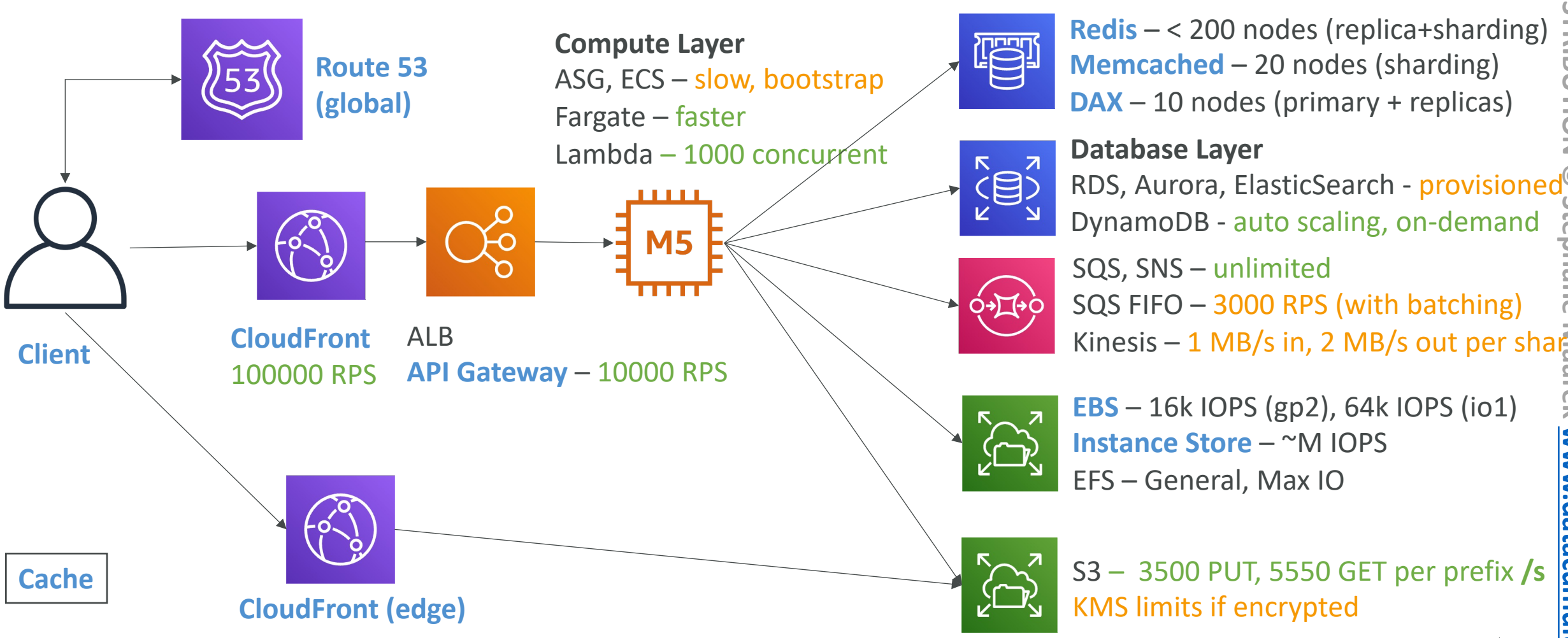
MEMCACHED

- Multi-node for partitioning of data (sharding)
- **Non persistent**
- **Backup and restore (Serverless)**
- Multi-threaded architecture

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Handling Extreme Rates



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Databases

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DynamoDB – in short

- NoSQL database, fully managed, massive scale (1,000,000 rps)
- Similar to Apache Cassandra (can migrate to DynamoDB)
- No disk space to provision, max object size is 400 KB
- Capacity: provisioned (WCU, RCU, & Auto Scaling) or on-demand
- Supports CRUD (Create Read Update Delete)
- Read: eventually or strong consistency
- Supports transactions across multiple tables (ACID support)
- Backups available, point in time recovery
- Table classes: Standard and Infrequent Access (IA)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DynamoDB - Basics

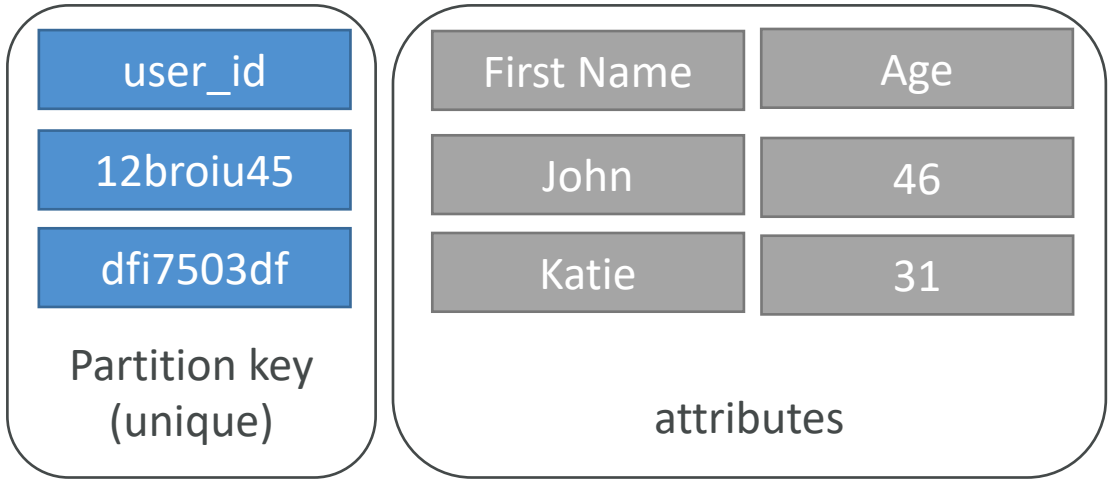
- DynamoDB is made of **tables**
- Each table has a **Primary Key** (must be decided at creation time)
- Each table can have an infinite number of items (= rows)
- Each item has **attributes** (can be added over time – can be null)
- Maximum size of a item is 400KB
- Data types supported are:
  - Scalar Types: String, Number, Binary, Boolean, Null
  - Document Types: List, Map
  - Set Types: String Set, Number Set, Binary Set

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DynamoDB – Primary Keys

- **Option 1: Partition Key only (HASH)**
- Partition key must be unique for each item
- Partition key must be "diverse" so that the data is distributed
- Example: user_id for a users table

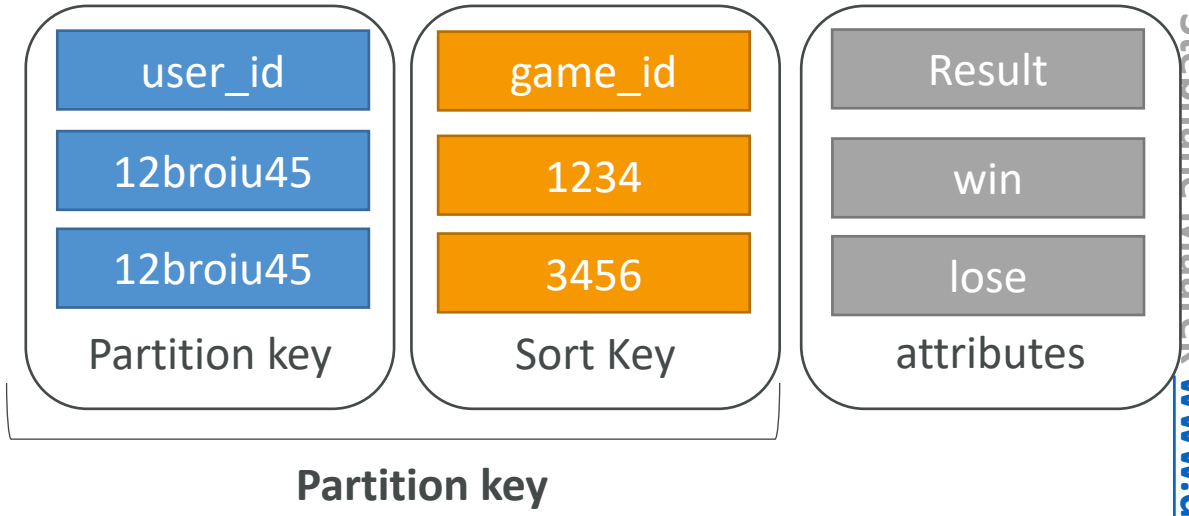


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DynamoDB – Primary Keys

- Option 2: Partition Key + Sort Key
- The combination must be unique
- Data is grouped by partition key
- Sort key == range key
- Example: users-games table
  - user_id for the partition key
  - game_id for the sort key



- Example good sort key: timestamp

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DynamoDB – Indexes

- Object = partition key + optional sort key + attributes
- LSI – Local Secondary Index
  - Keep the same partition key
  - Select an alternative sort key
  - Must be defined at table creation time
- GSI – Global Secondary Index
  - Change the partition key and optionally the sort key
  - Can be defined after the table is created

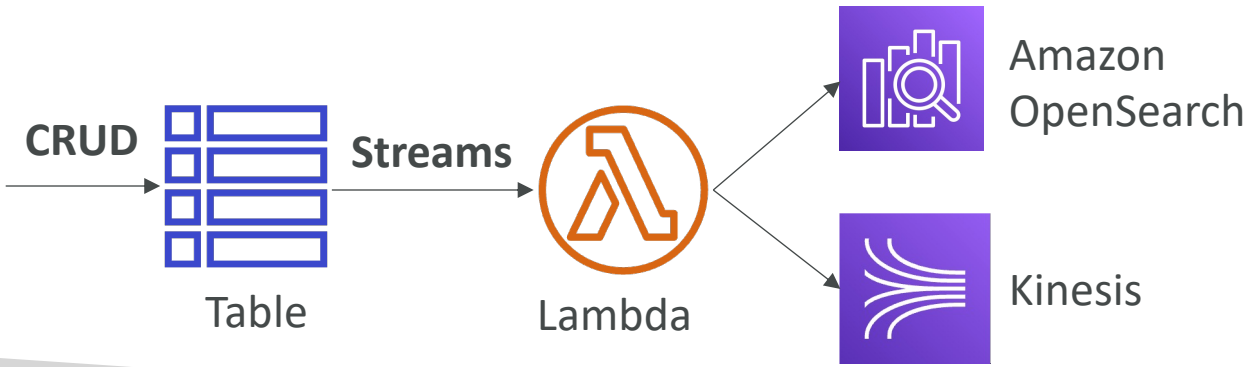
- You can only query by PK + sort key on the main table & indexes **(≠ RDS)**

© Stephane Maarek

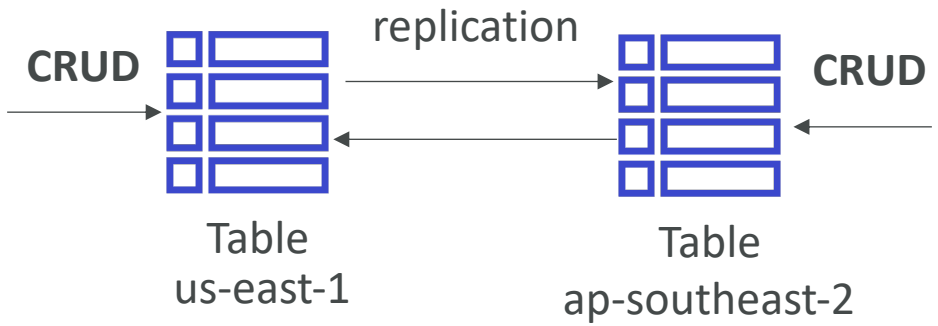
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DynamoDB – Important Features

- **TTL:** automatically expire row after a specified epoch date
- **DynamoDB Streams:**
  - react to changes to DynamoDB tables in real time
  - Can be read by AWS Lambda, EC2…
  - 24 hours retention of data



- **Global Tables:** (cross region replication)
  - Active Active replication, many regions
  - Must enable DynamoDB Streams
  - Useful for low latency, DR purposes

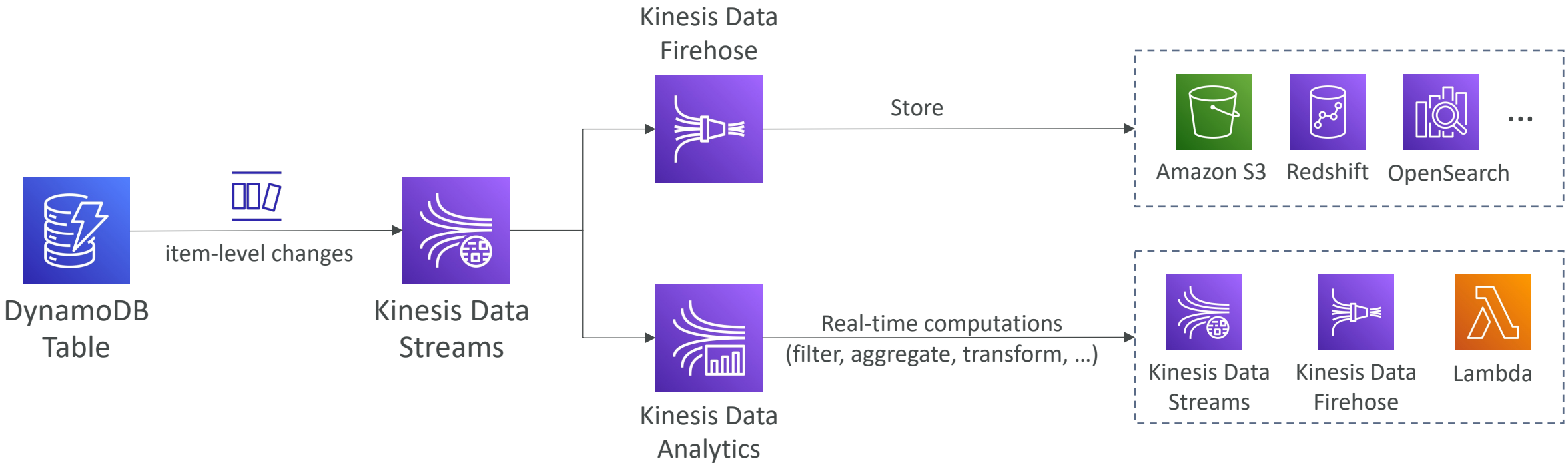


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Kinesis Data Streams for DynamoDB

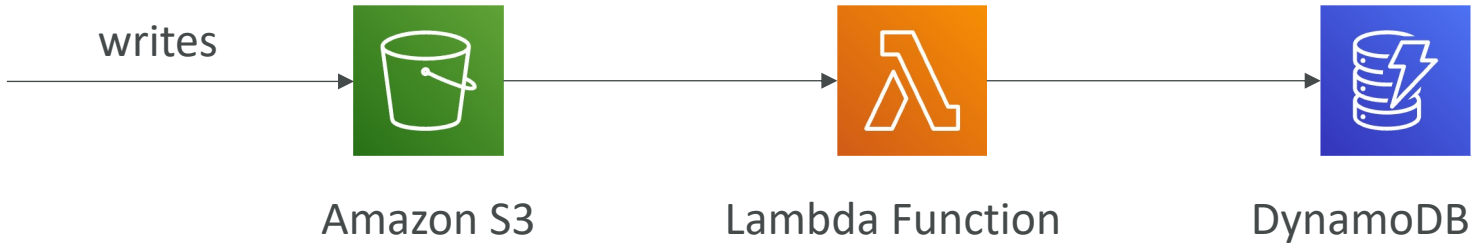
- You can use Kinesis Data Streams to capture item-level changes in DynamoDB
- Custom and longer data retention period (> 24 hours in DynamoDB Streams)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DynamoDB Solution Architecture
Indexing objects in DynamoDB



API for object metadata

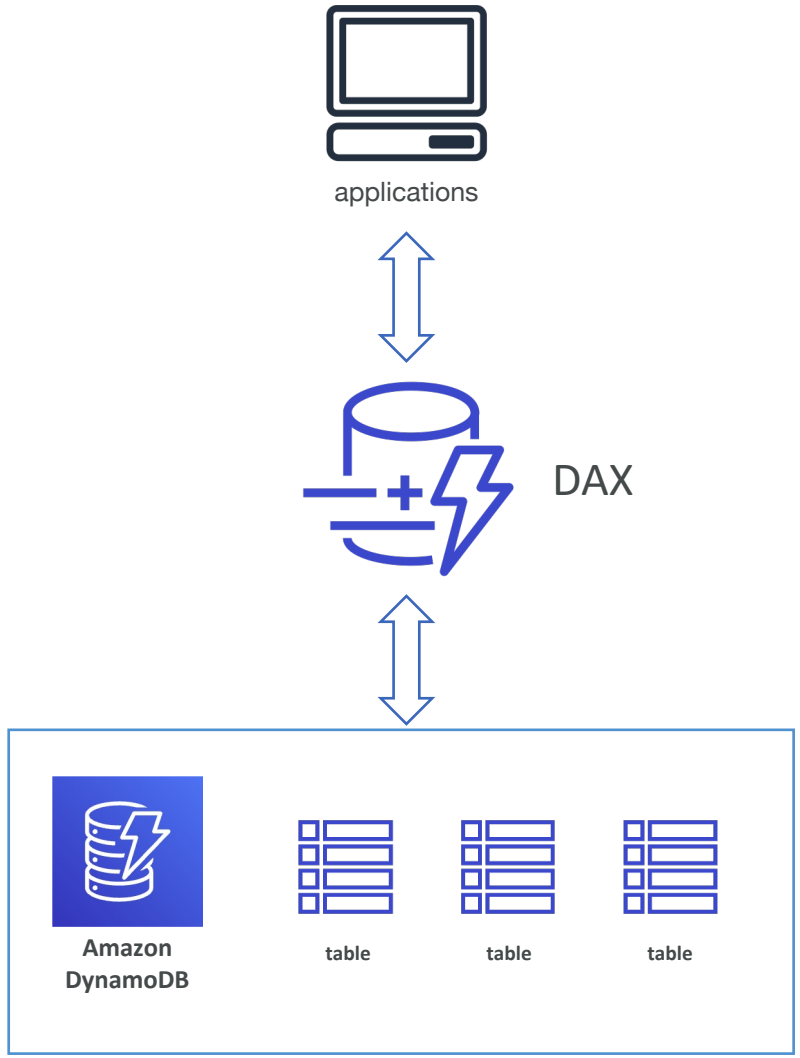
- Search by date
- Total storage used by a customer
- List of all objects with certain attributes
- Find all objects uploaded within a date range

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DynamoDB - DAX

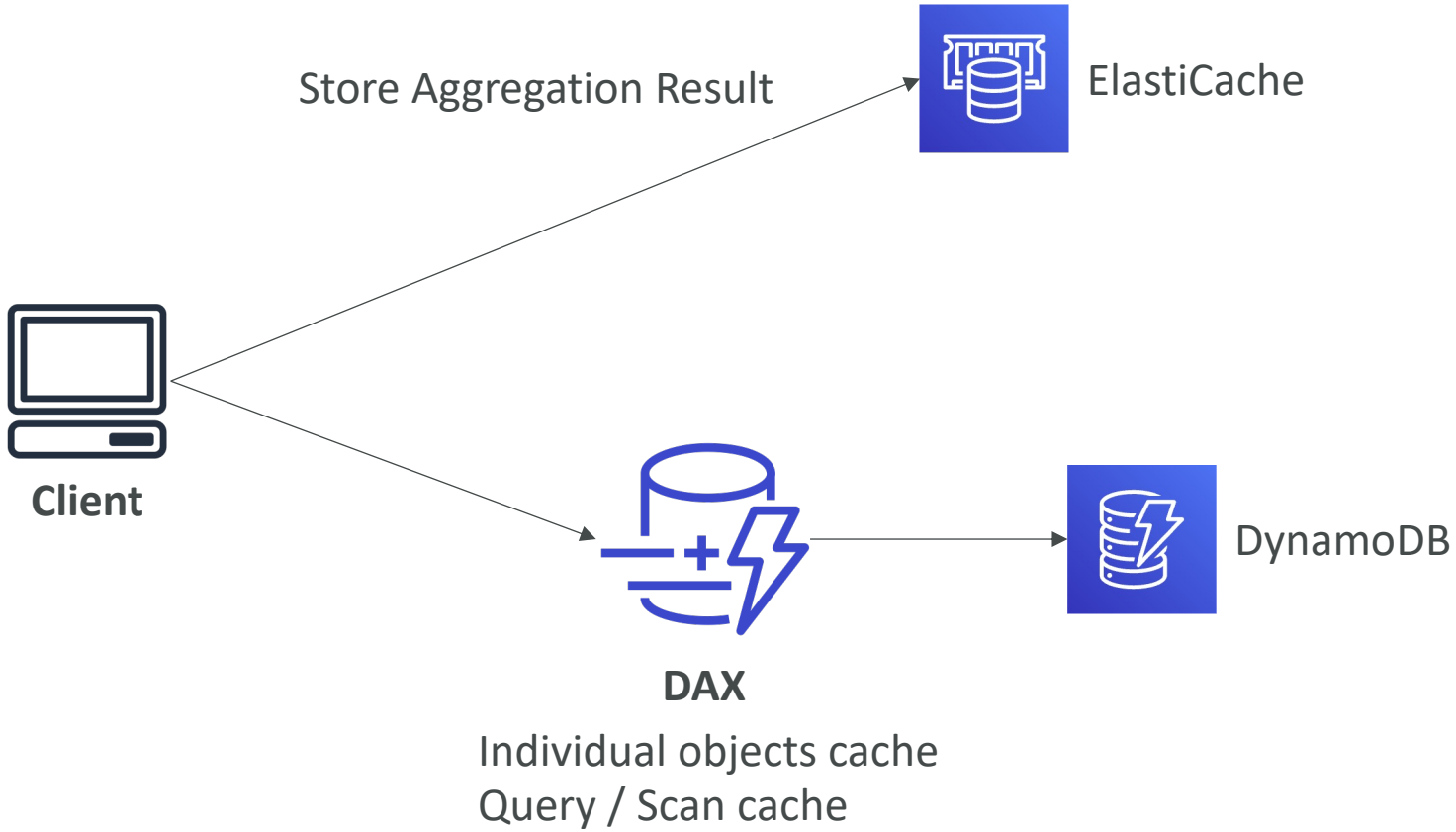
- DAX = DynamoDB Accelerator
- Seamless cache for DynamoDB, no application re-write
- Writes go through DAX to DynamoDB
- Micro second latency for cached reads & queries
- Solves the Hot Key problem (too many reads)
- 5 minutes TTL for cache by default
- Up to 10 nodes in the cluster
- Multi AZ (3 nodes minimum recommended for production)
- Secure (Encryption at rest with KMS, VPC, IAM, CloudTrail…)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DynamoDB – DAX vs ElastiCache



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon OpenSearch (ex ElasticSearch)

- New name is Amazon OpenSearch
- ElasticSearch => OpenSearch
- Kibana => OpenSearch Dashboards

- Managed version of OpenSearch (open-source project, fork of ElasticSearch)
- Two modes: **Managed** cluster or **Serverless** cluster
- Use cases:
  - Log Analytics
  - Real Time application monitoring
  - Security Analytics
  - Full Text Search
  - Clickstream Analytics
  - Indexing

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

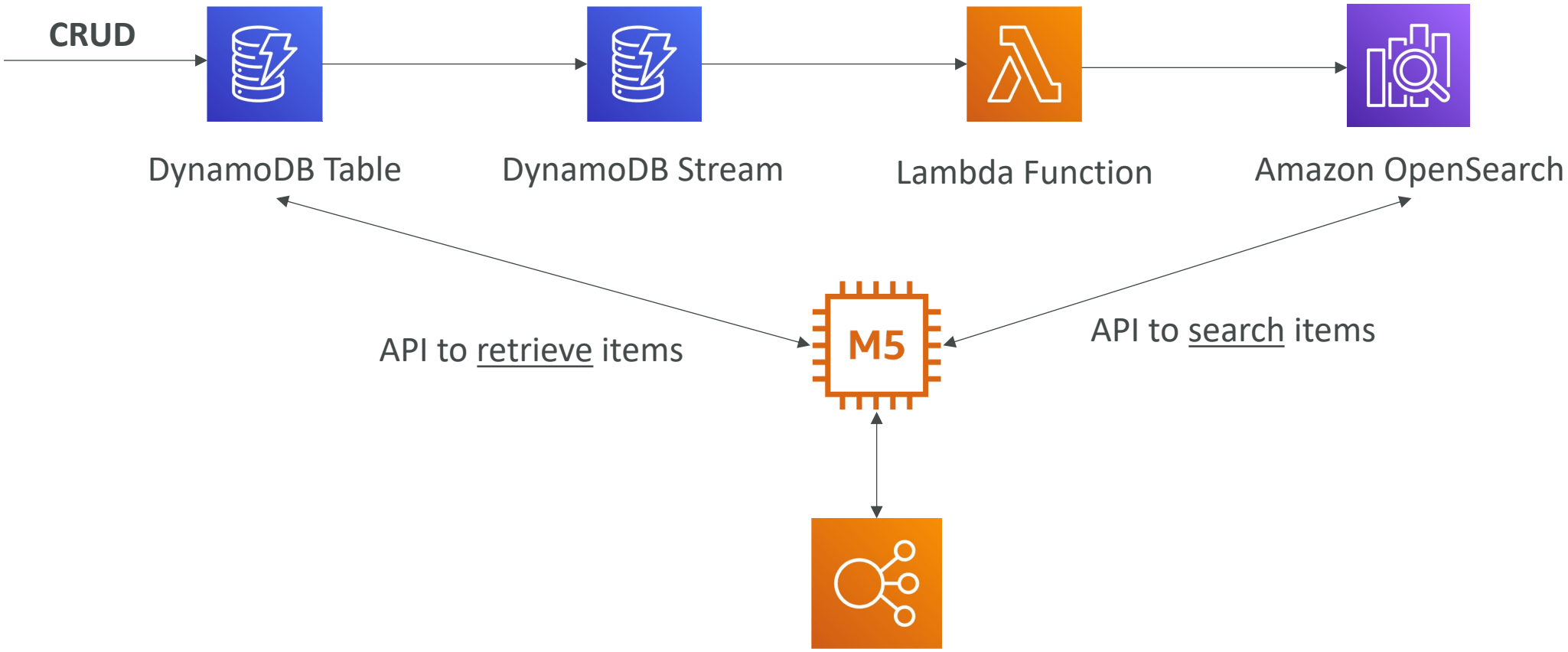
OpenSearch + OS Dashboards + Logstash

- OpenSearch (ex ElasticSearch): provide search and indexing capability
- OpenSearch Dashboards (ex Kibana):
  - Provide real-time dashboards on top of the data that sits in OpenSearch
  - Alternative to CloudWatch dashboards (more advanced capabilities)
- Logstash:
  - Log ingestion mechanism, use the "Logstash Agent"
  - Alternative to CloudWatch Logs (you decide on retention and granularity)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

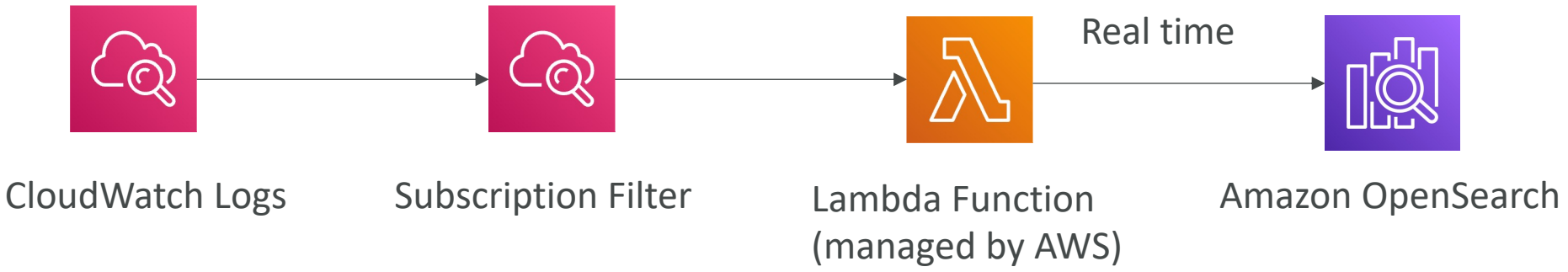
OpenSearch patterns DynamoDB



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

OpenSearch patterns CloudWatch Logs



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

RDS

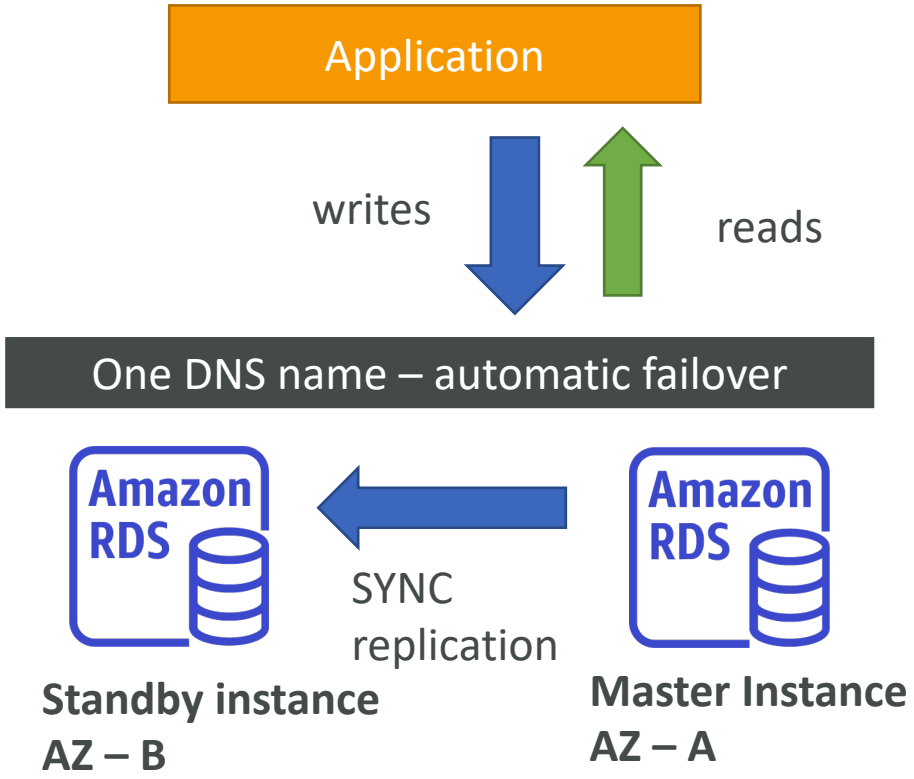
- **Engines:** PostgreSQL, MySQL, MariaDB, IBM DB2, Oracle, SQL Server
- **Managed DB:** provisioning, backups, patching, monitoring
- **Launched within a VPC**, usually in private subnet, control network access using security groups (important when using Lambda)
- **Storage by EBS**, can increase volume size with auto-scaling
- **Backups:** automated with point-in-time recovery. Backups expire
- **Snapshots:** manual, can make copies of snapshots cross region
- **RDS Events:** get notified via SNS for events (operations, outages…)

© Stephane Maarek

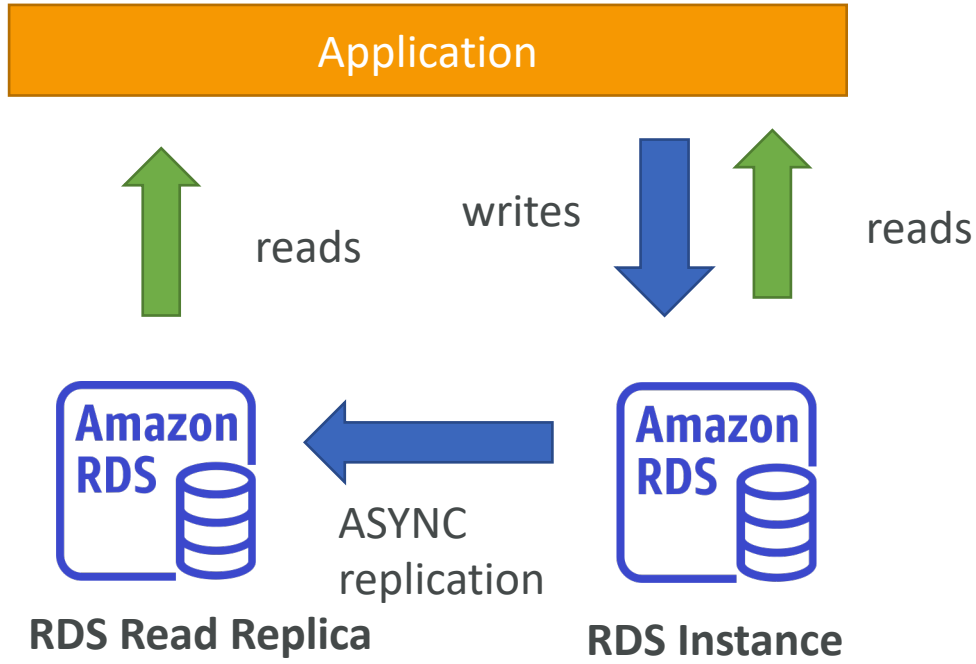
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

RDS – Multi AZ & Read Replicas

- **Multi-AZ:** Standby instance for failover in case of outage



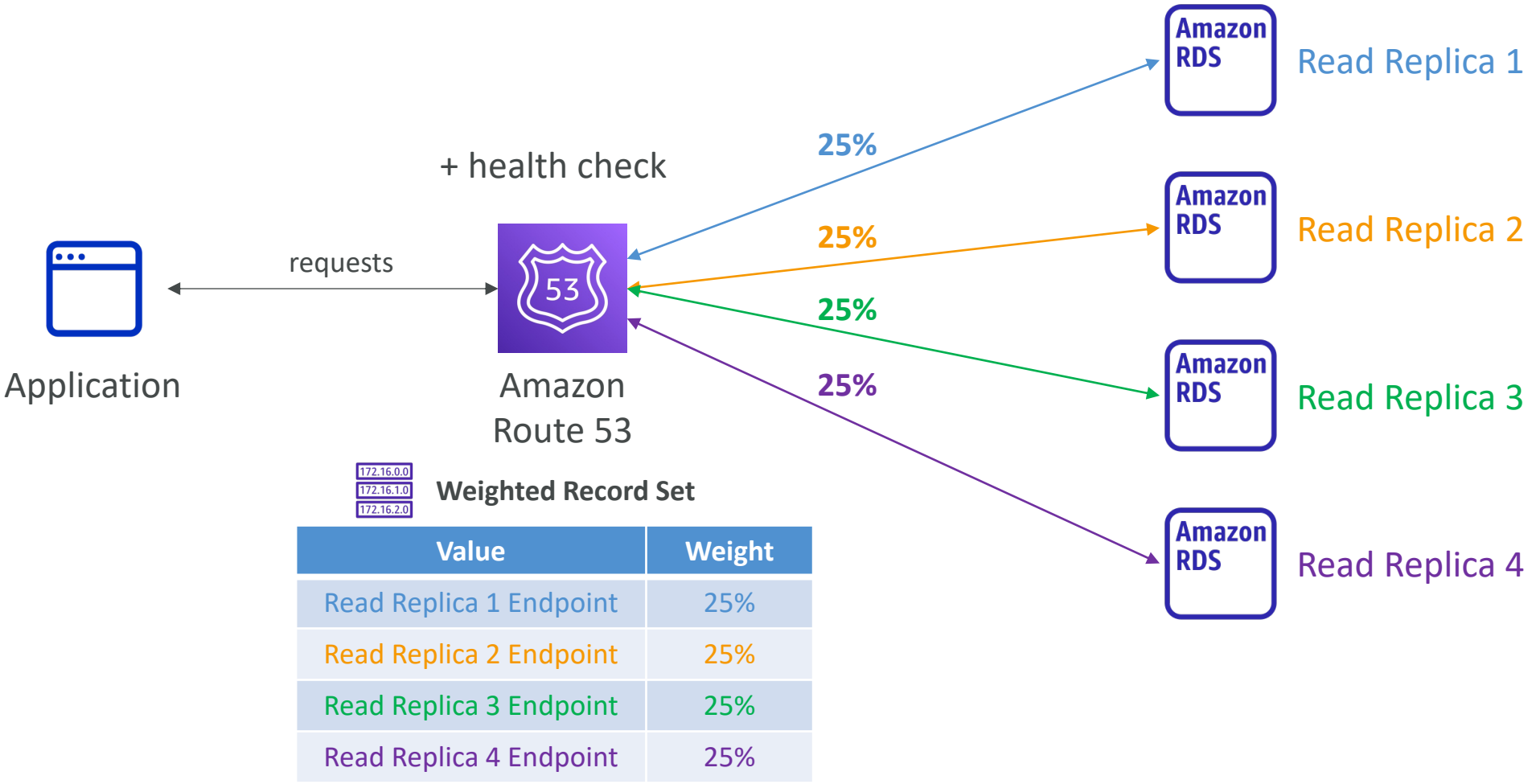
- **Read Replicas:** Increase read throughput. Eventual consistency. Can be cross-region



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

RDS – Distributing Reads across Replicas



172.16.0.0
172.16.1.0
172.16.2.0

Weighted Record Set

Value	Weight
Read Replica 1 Endpoint	25%
Read Replica 2 Endpoint	25%
Read Replica 3 Endpoint	25%
Read Replica 4 Endpoint	25%

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

RDS – Security (reminder)

- KMS encryption at rest for underlying EBS volumes / snapshots
- Transparent Data Encryption (TDE) for Oracle and SQL Server
- SSL encryption to RDS is possible for all DB (in-flight)
- IAM authentication for MySQL, PostgreSQL and MariaDB
- Authorization still happens within RDS (not in IAM)
- Can copy an un-encrypted RDS snapshot into an encrypted one
- CloudTrail cannot be used to track queries made within RDS

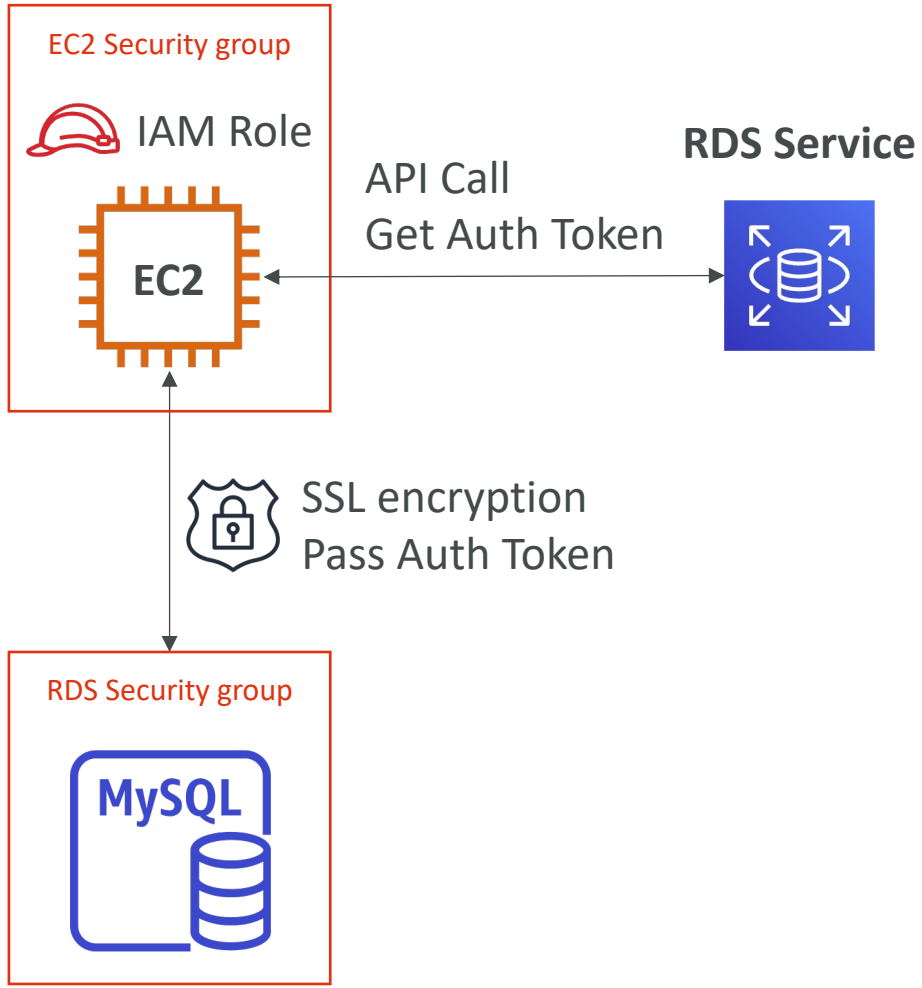
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

RDS - IAM Authentication

- IAM database authentication works with **MariaDB, MySQL and PostgreSQL**
- You don't need a password, just an authentication token obtained through IAM & RDS API calls
- Auth token has a lifetime of 15 minutes

- Benefits:
  - Network in/out must be encrypted using SSL
  - IAM to centrally manage users instead of DB
  - Can leverage IAM Roles and EC2 Instance profiles for easy integration



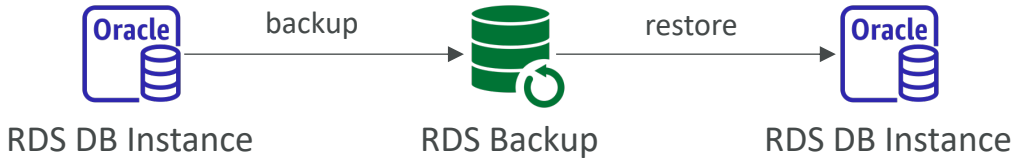
© Stephane Maarek

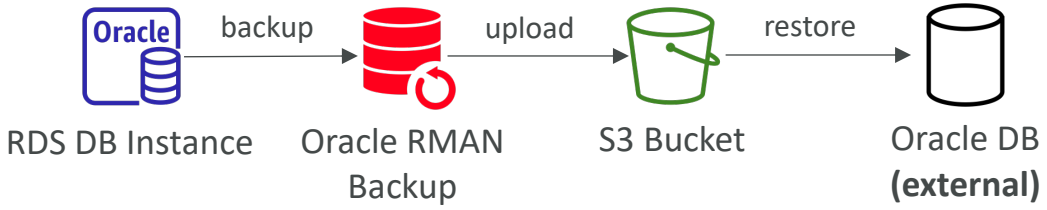
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

About RDS for Oracle – Exam Tips

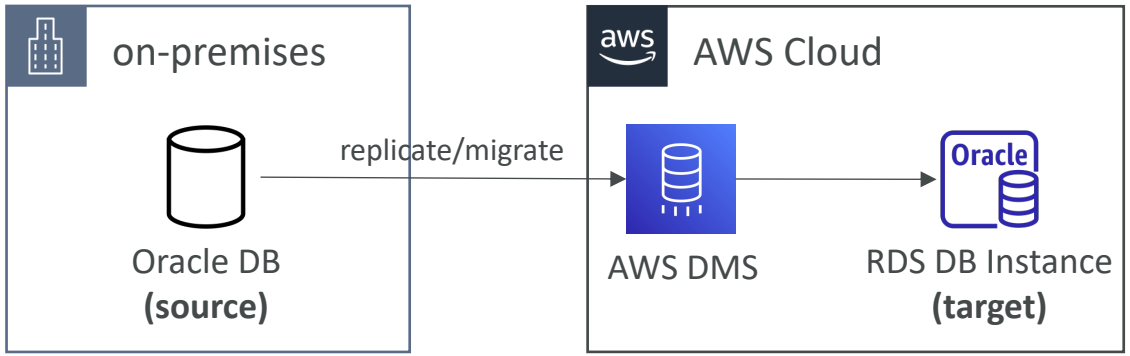
Backups

- Use RDS Backups for backups & restore to Amazon RDS for Oracle
- Use Oracle RMAN (Recovery Manager) for backups & restore to-non RDS (RDS not supported)





- Real Application Clusters (RAC)
  - RDS for Oracle does NOT support RAC
  - RAC is working on Oracle on EC2 Instances because you have full control
- RDS for Oracle supports Transparent Data Encryption (TDE) to encrypt data before it's written to storage
- DMS works on Oracle RDS

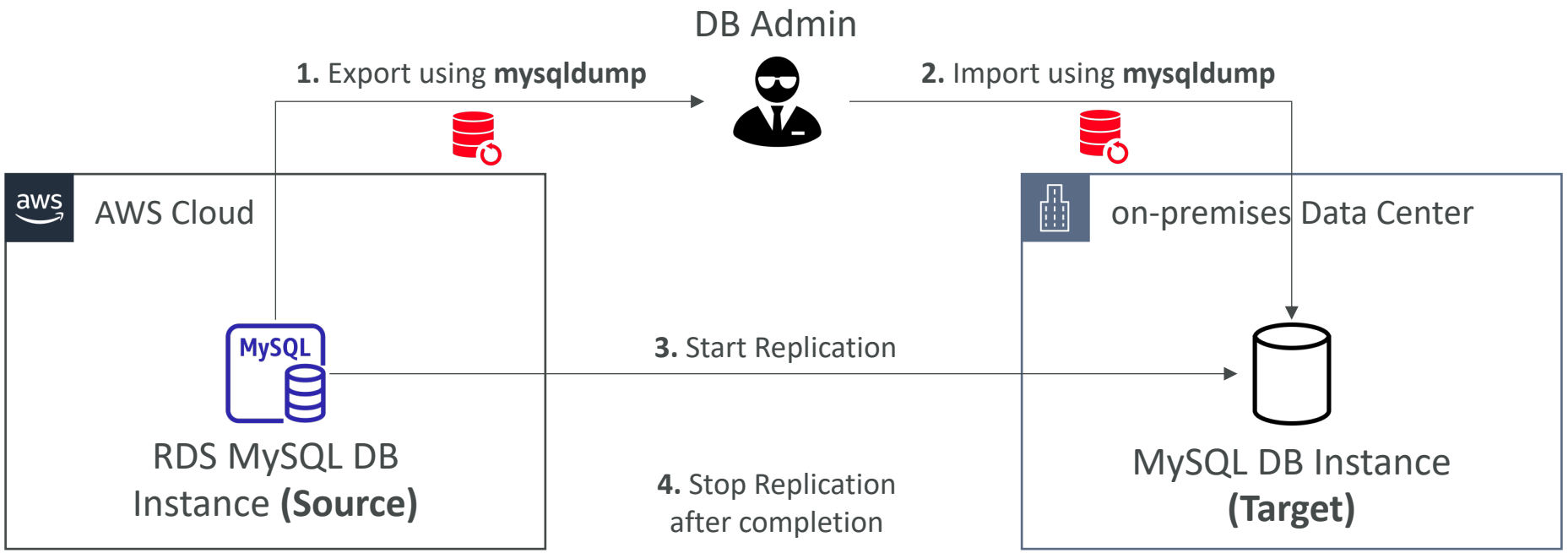


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

About RDS for MySQL

- You can use the native **mysqldump** to migrate a MySQL RDS DB to non-RDS
- The external MySQL database can run either on-premises in your data center, or on an Amazon EC2 instance

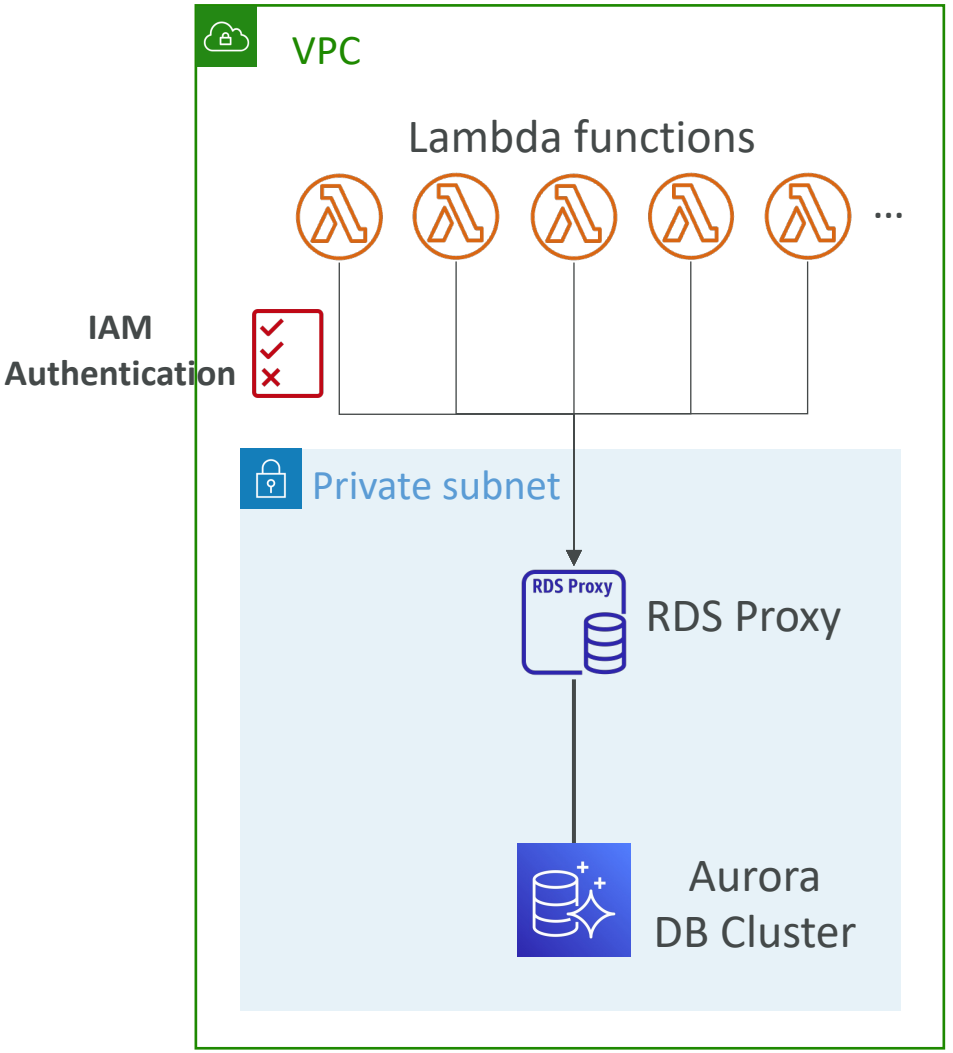


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

RDS Proxy for AWS Lambda

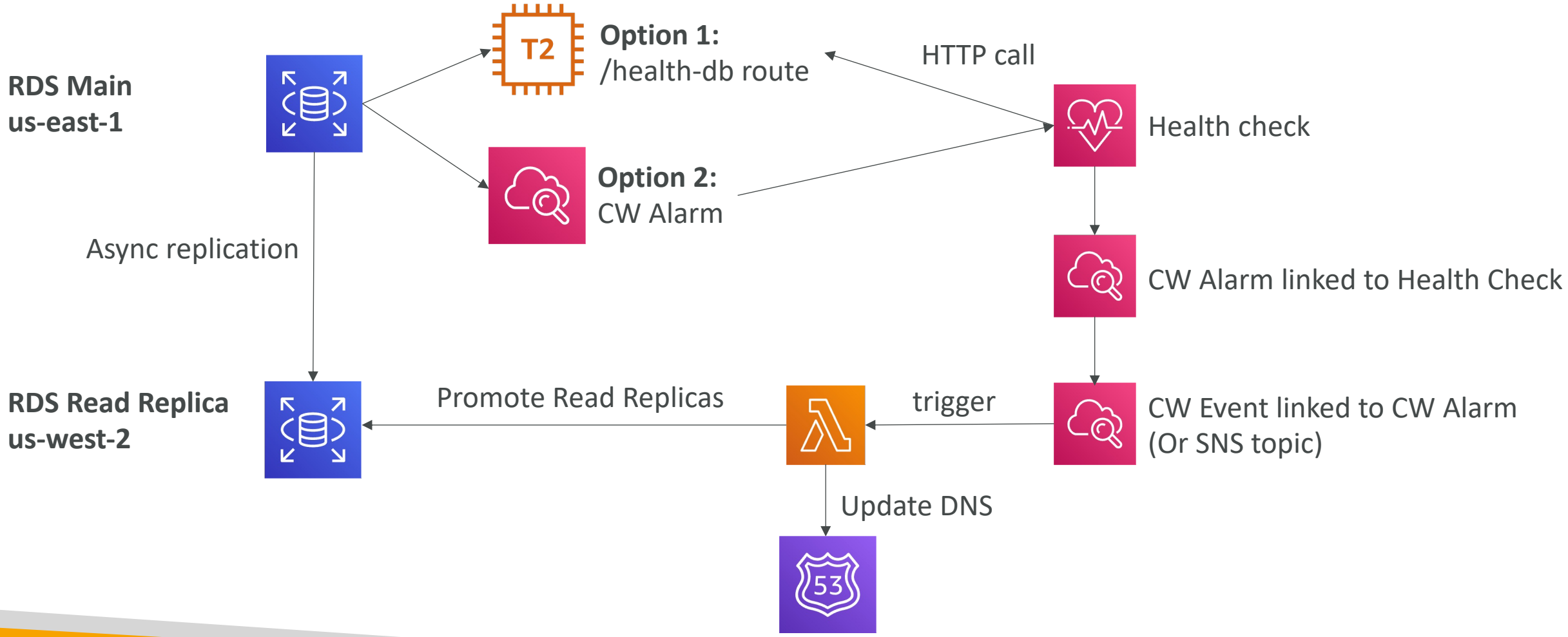
- When using Lambda functions with RDS, it opens and maintains a database connection
- This can result in a "TooManyConnections" exception
- With RDS Proxy, you no longer need code that handles cleaning up idle connections and managing connection pools
- Supports IAM authentication or DB authentication, auto-scaling
- The Lambda function must be deployed in your VPC, because RDS Proxy is never publicly accessible



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

RDS Solution Architecture
Cross Region Failover



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Aurora

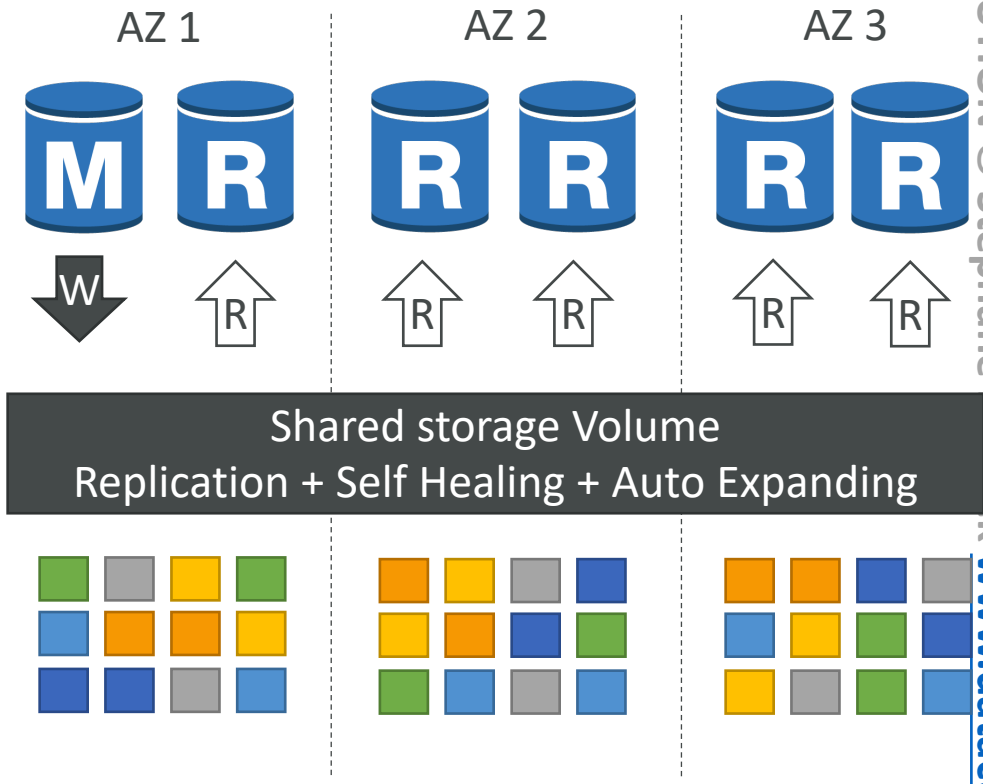
- **DB Engines:** PostgreSQL-compatible & MySQL-compatible
- **Storage:** automatically grows up to 256 TB, 6 copies of data, multi-AZ
- **Read Replicas:** up to 15 RR, reader endpoint to access them all
- **Cross Region RR:** entire database is copied (not select tables)
- **Load / Offload data directly from / to S3:** efficient use of resources
- **Backup, Snapshots & Restore:** same as RDS

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Aurora High Availability and Read Scaling

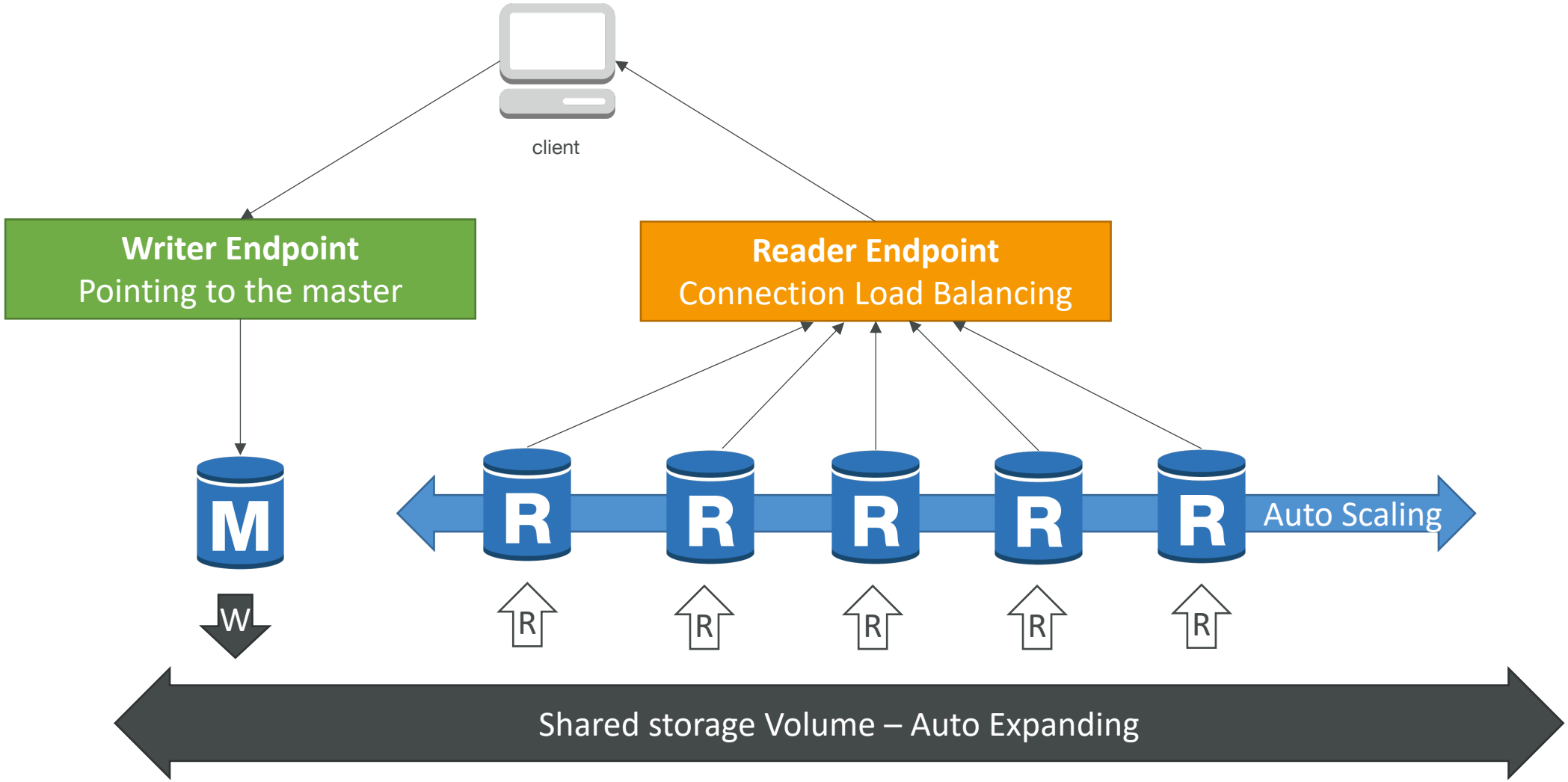
- 6 copies of your data across 3 AZ:
  - 4 copies out of 6 needed for writes
  - 3 copies out of 6 need for reads
  - Self healing with peer-to-peer replication
  - Storage is striped across 100s of volumes
- Automated failover for master in less than 30 seconds
- Master + up to 15 Aurora Read Replicas serve reads
- Support for Cross Region Replication



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Aurora DB Cluster



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Aurora Endpoints

- Endpoint = Host Address + Port

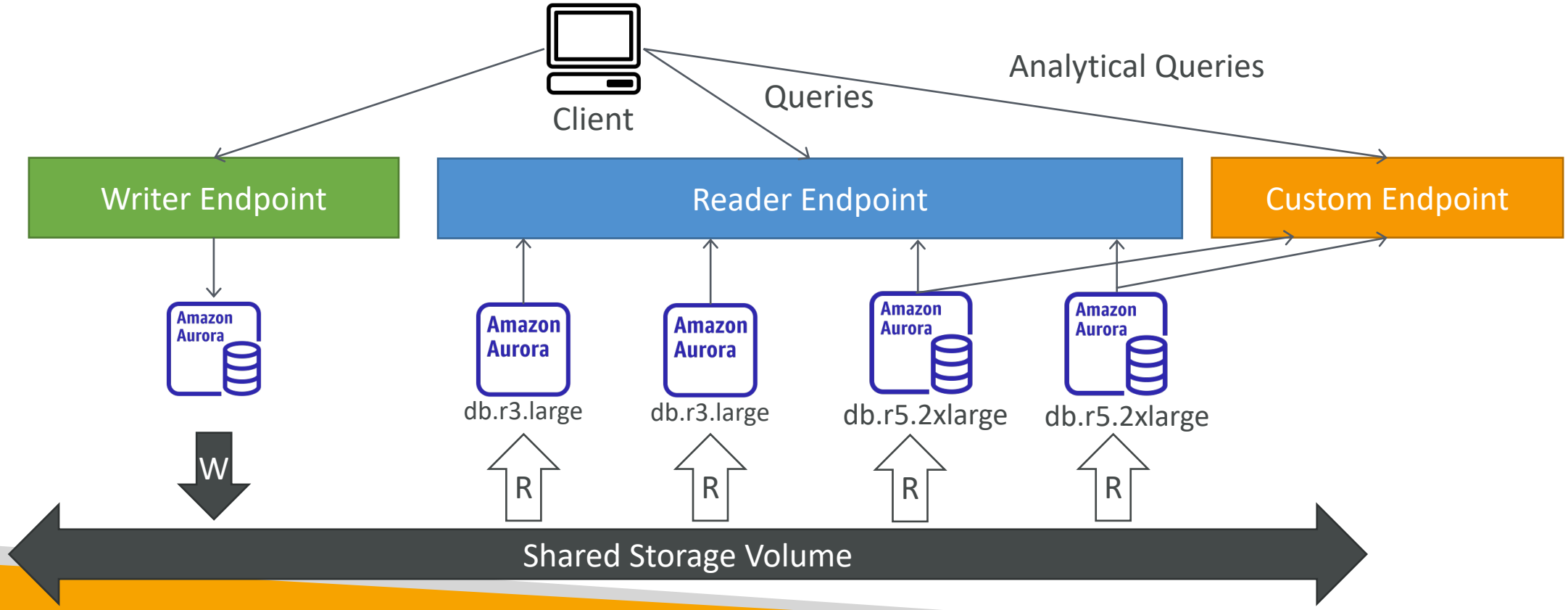
- Cluster Endpoint (Writer Endpoint)
  - Connects to the current primary DB instance in the Aurora cluster
  - Used for all write operations in the DB cluster (inserts, updates, deletes, and queries)
- Reader Endpoint
  - Provides load-balancing for read only connections to all Aurora Replicas in the Aurora cluster
  - Used only for read operations (queries)
- Custom Endpoint
  - Represents a set on DB instances that you choose in the Aurora cluster
  - Used when you want to connect to different subsets of DB instances with different capacities and configurations (e.g., different DB parameter group)
- Instance Endpoint
  - Connects to a specific DB instance in the Aurora cluster
  - Used when you want to diagnosis and fine tune a specific DB instance

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Aurora – Custom Endpoints

- Define a subset of Aurora Instances as a Custom Endpoint
- Example: Run analytical queries on specific replicas
- The Reader Endpoint is generally not used after defining Custom Endpoints



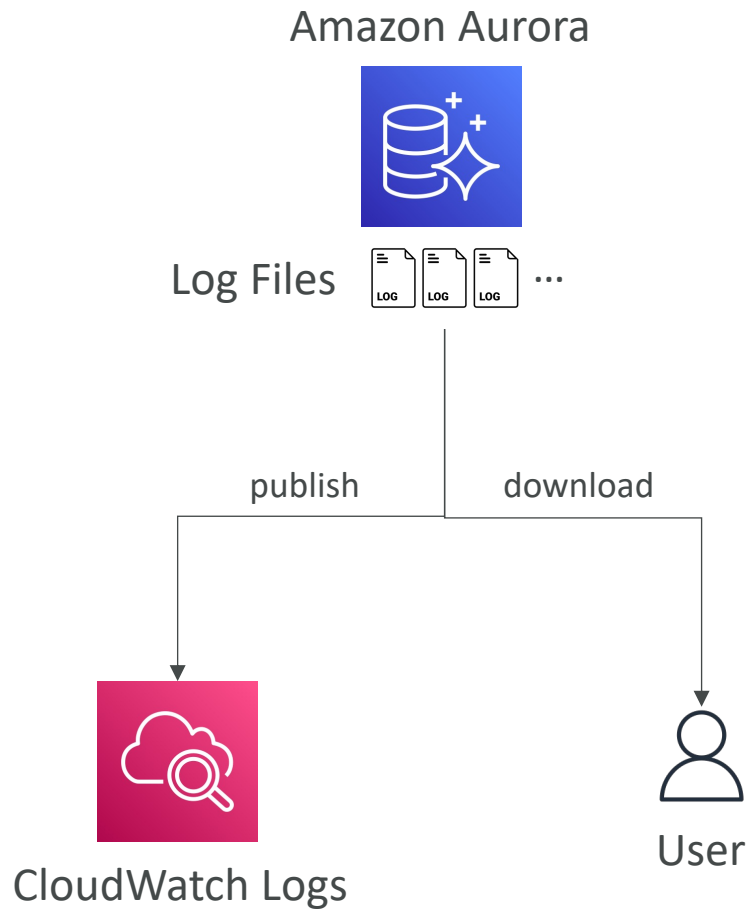
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Aurora Logs

- You can monitor the following types of Aurora MySQL log files:
  - Error log
  - Slow query log
  - General log
  - The audit log

- These log files are either downloaded or published to CloudWatch Logs

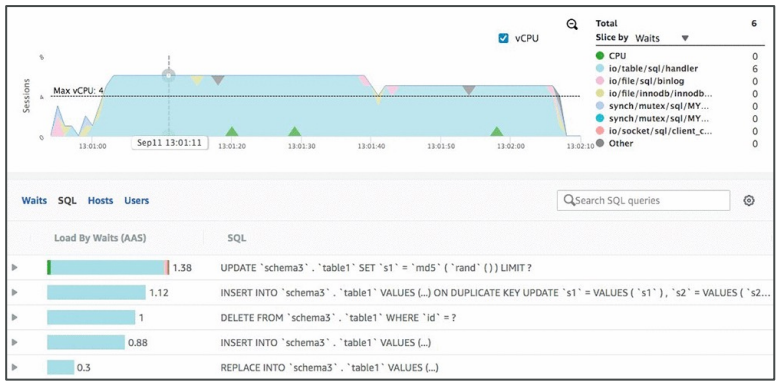


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Troubleshooting RDS & Aurora Performance

- **Performance Insights:** find issues by waits, SQL statements, hosts and users



SQL Statements

Users

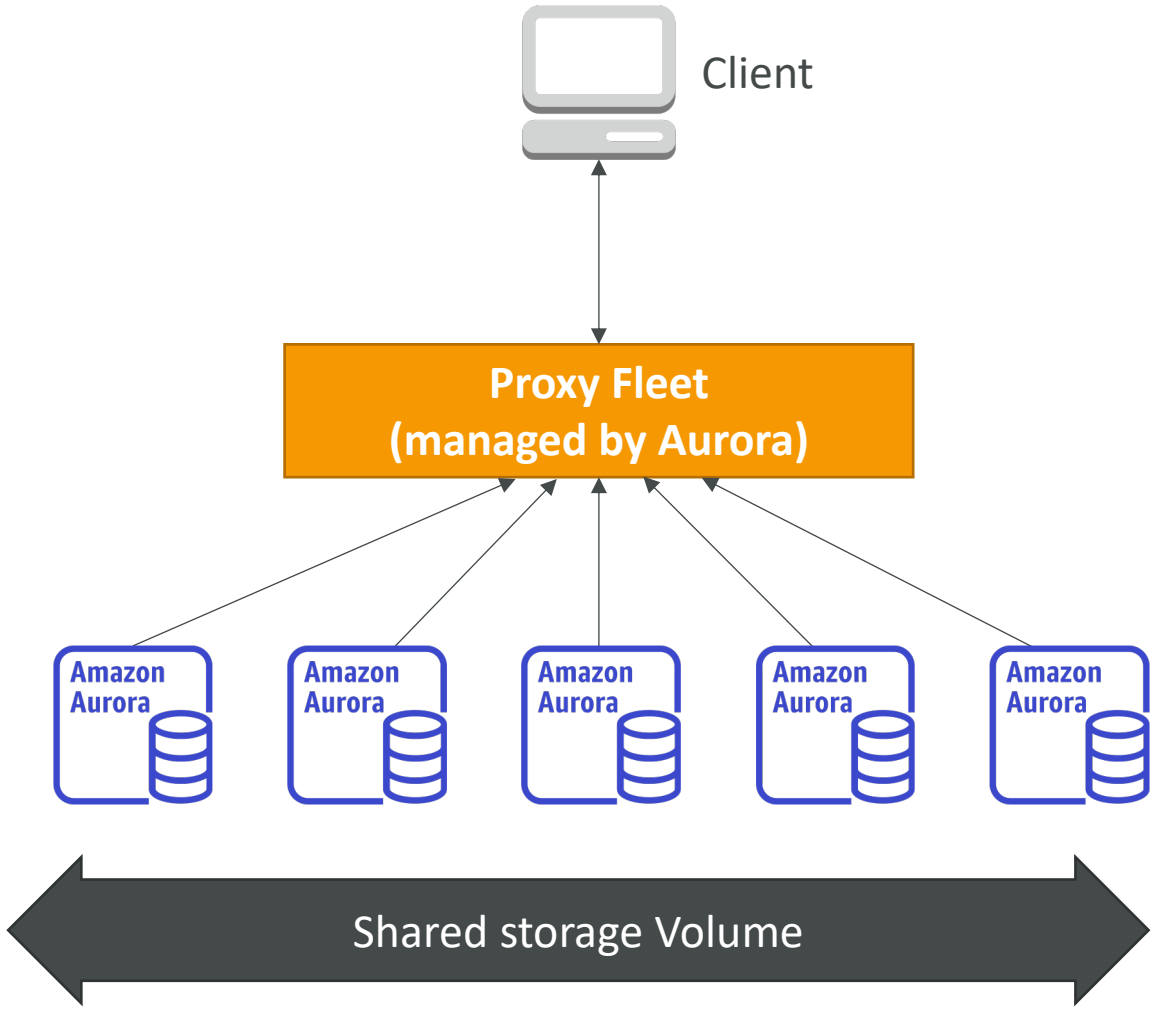
- **CloudWatch Metrics:** CPU, Memory, Swap Usage
- **Enhanced Monitoring Metrics:** at host level, process view, per-second metric
- Slow Query logs

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Aurora Serverless

- Automated database instantiation and auto-scaling based on actual usage
- Good for infrequent, intermittent or unpredictable workloads
- No capacity planning needed
- Pay per second, can be more cost-effective

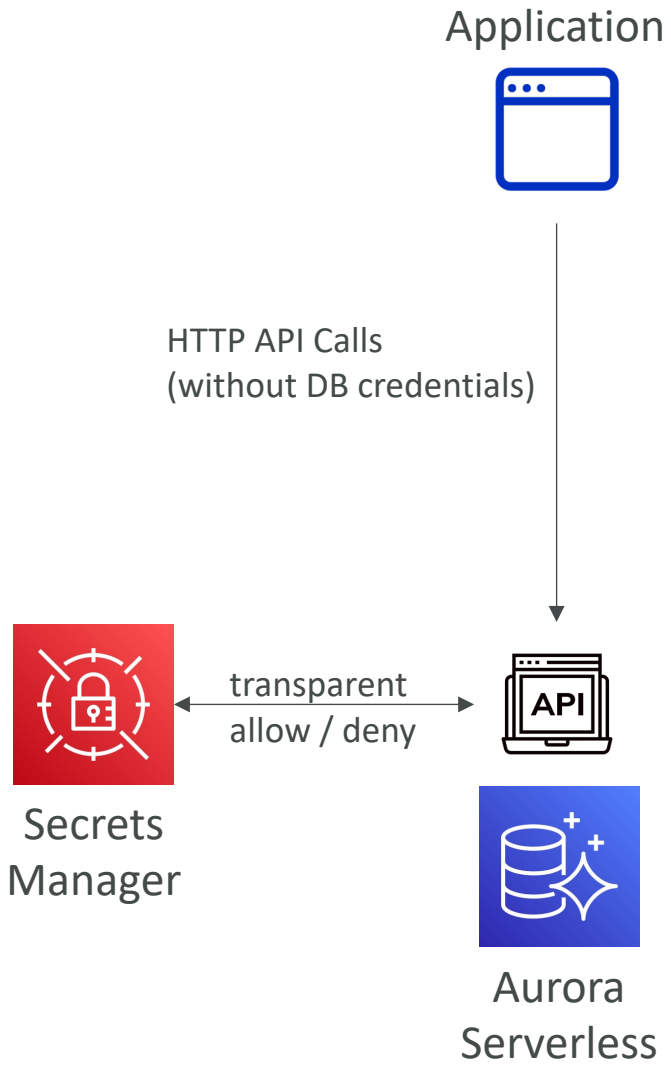


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Aurora Serverless – Data API

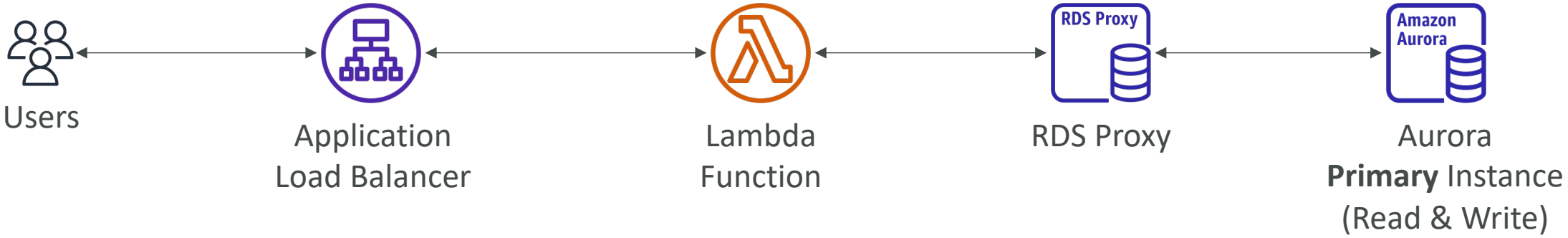
- Access Aurora Serverless DB with a simple API endpoint (no JDBC connection needed)
- **Secure HTTPS endpoint to run SQL statements**
- No persistent DB connections management
- Users must be granted permissions to Data API and Secrets Manager (where credentials are checked)



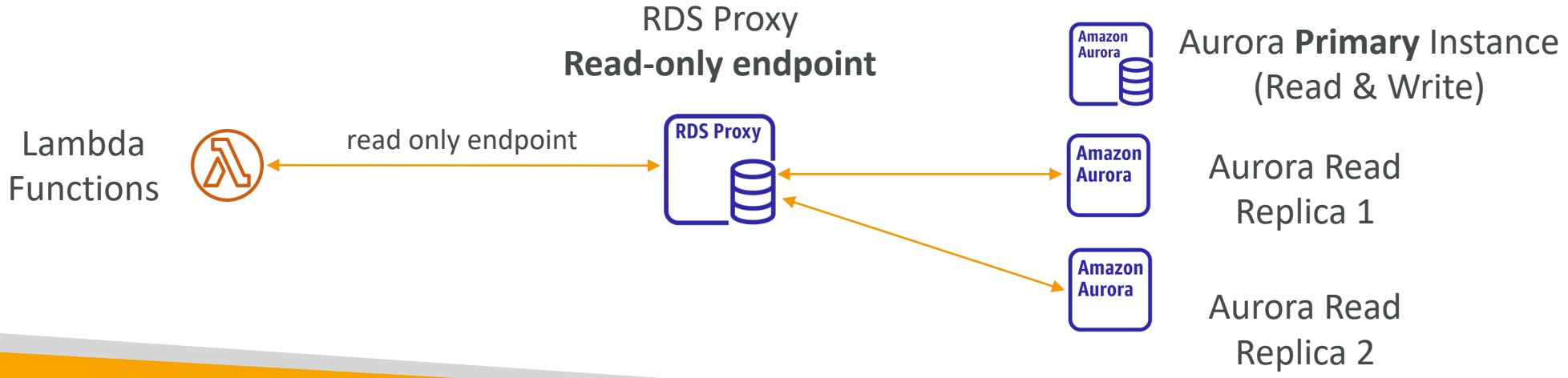
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

RDS Proxy for Aurora



- Ability to create an additional read-only endpoint that connects to Aurora Read Replicas only

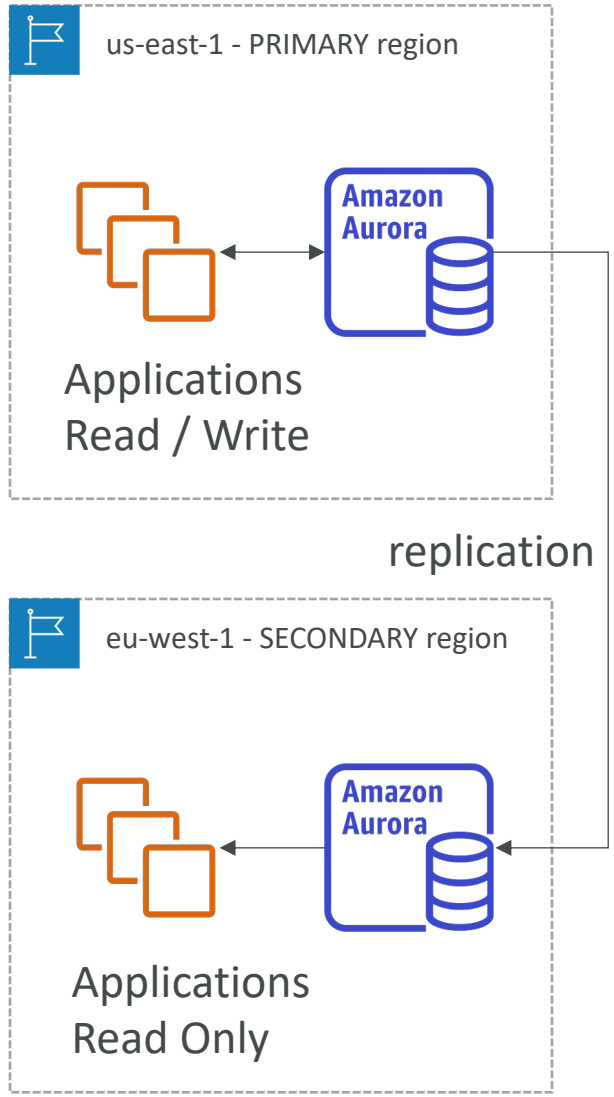


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Global Aurora

- Aurora Cross Region Read Replicas
  - Useful for disaster recovery
  - Simple to put in place
- Aurora Global Database (recommended)
  - 1 Primary Region (read / write)
  - Up to 10 secondary (read-only) regions, replication lag is less than 1 second
  - Up to 16 Read Replicas per secondary region
  - Helps for decreasing latency
  - Promoting another region (for disaster recovery) has an RTO of < 1 minute
  - Ability to manage the RPO in Aurora for PostgreSQL

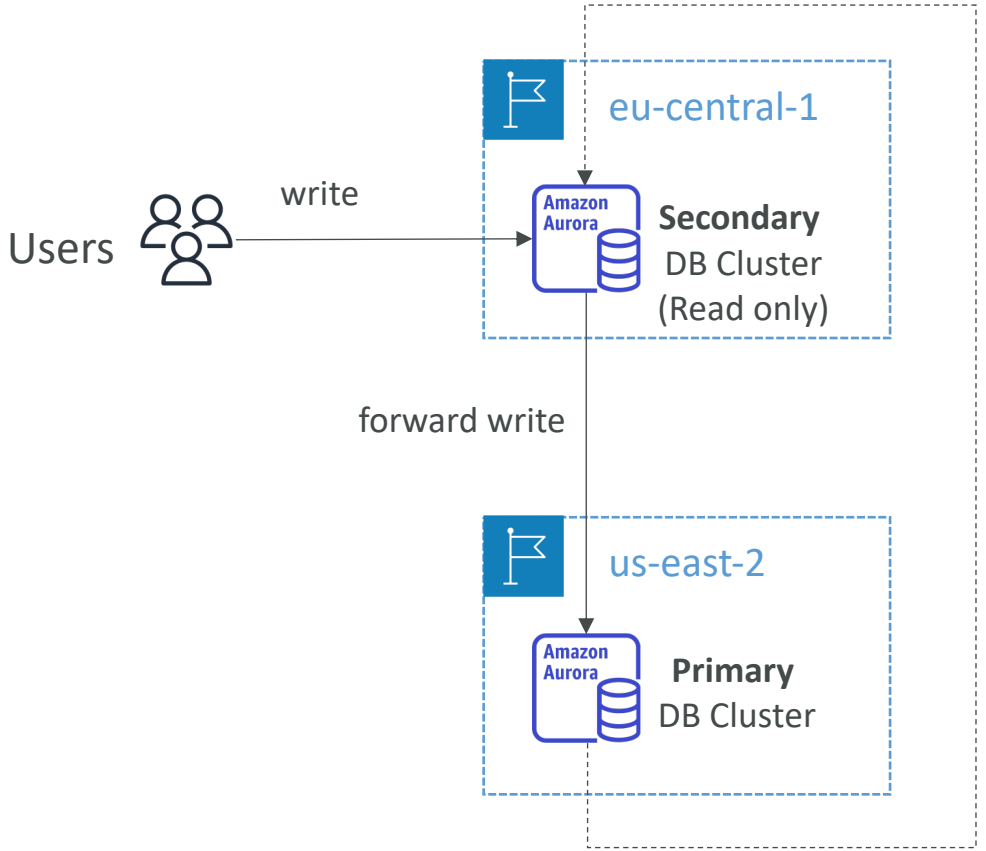


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Aurora Global – Write Forwarding

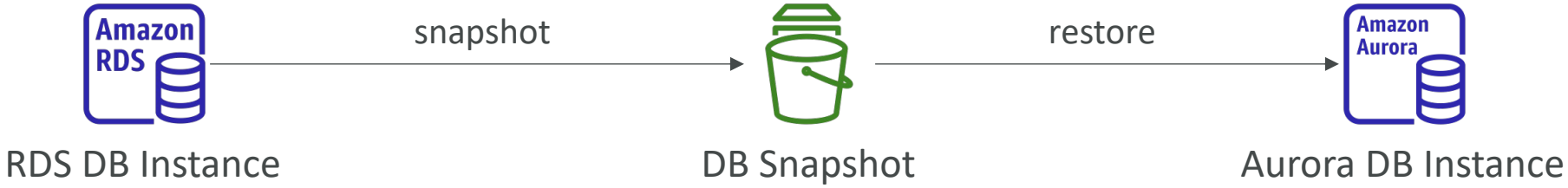
- Enables Secondary DB Clusters to forward SQL statements that perform write operations to the Primary DB Cluster
- Data is always changed first on the Primary DB Cluster, then replicated to the Secondary DB Clusters
- Primary DB Cluster always has an up-to-date copy of all data
- Reduces the number of endpoints to manage



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Convert RDS to Aurora



© Stephane Maarek

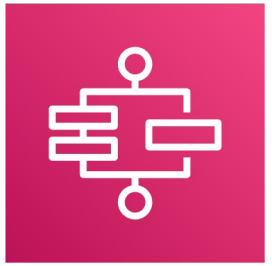
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Service Communication

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Step Functions



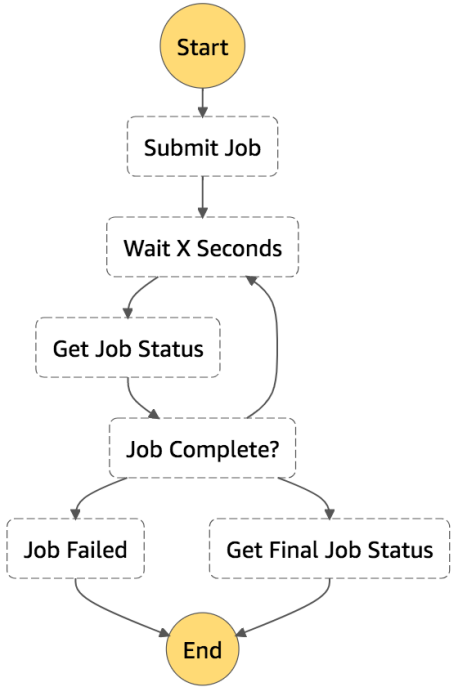
- Build serverless visual workflow to orchestrate your Lambda functions
- Represent flow as a JSON **state machine**
- Features: sequence, parallel, conditions, timeouts, error handling…
- Maximum execution time of 1 year
- Possibility to implement human approval feature

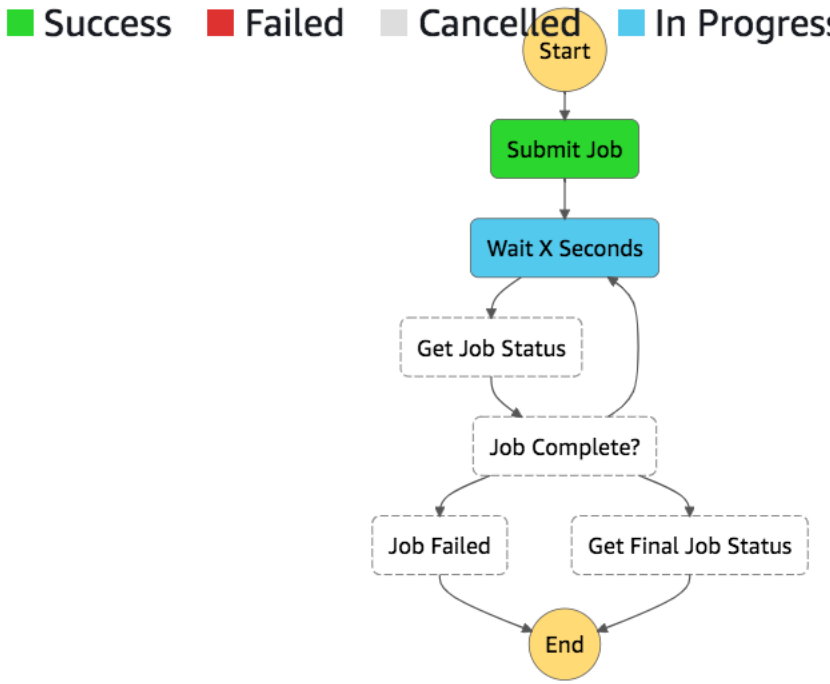
- If you chain Lambda functions using Step Functions, be mindful of the added latency to pass the calls.

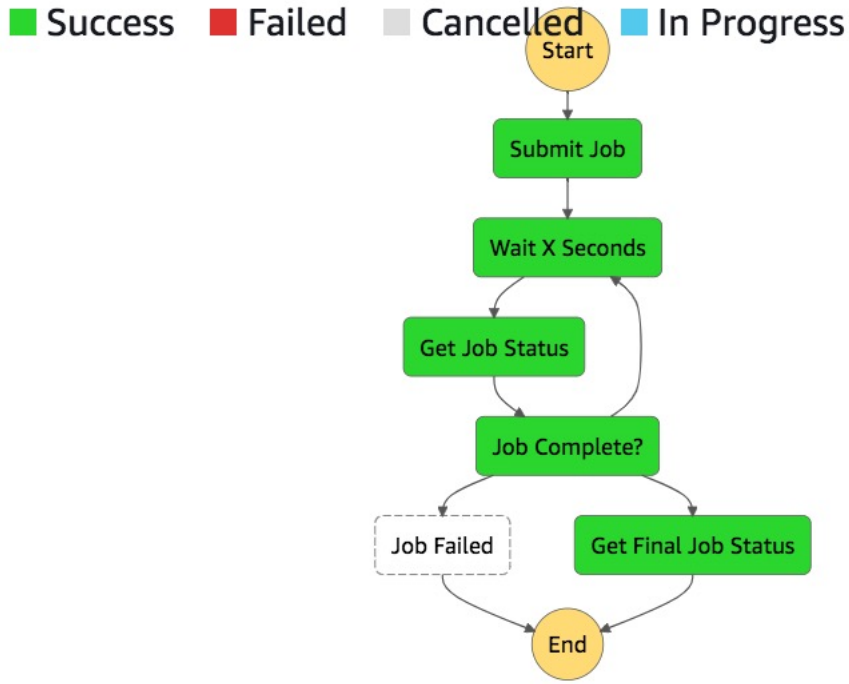
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Visual workflow in Step Functions







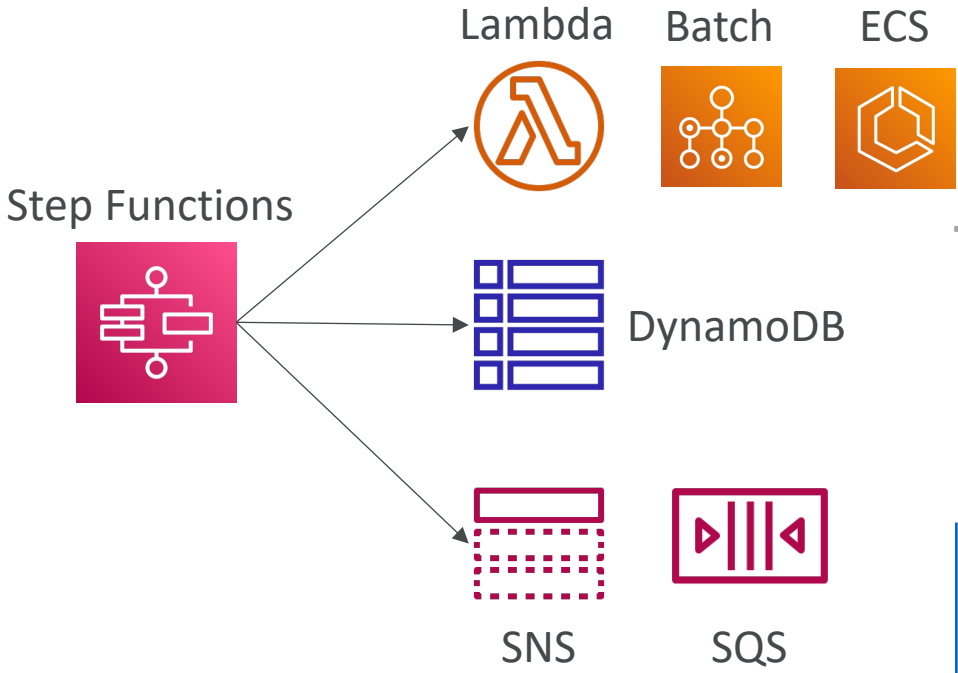
NOT FOR DISTRIBU

jmulus.com

© Stephane Maarek

Step Function Integrations

- Optimized Integrations
  - Can invoke a **Lambda function**
  - Run an AWS Batch job
  - Run an ECS task and wait for it to complete
  - Insert an item from DynamoDB
  - Publish message to SNS, SQS
  - Launch an EMR, Glue, or SageMaker jobs
  - Launch another Step Function workflow…
- AWS SDK Integrations
  - Access 200+ AWS services from your State Machine
  - Works like standard AWS SDK API call

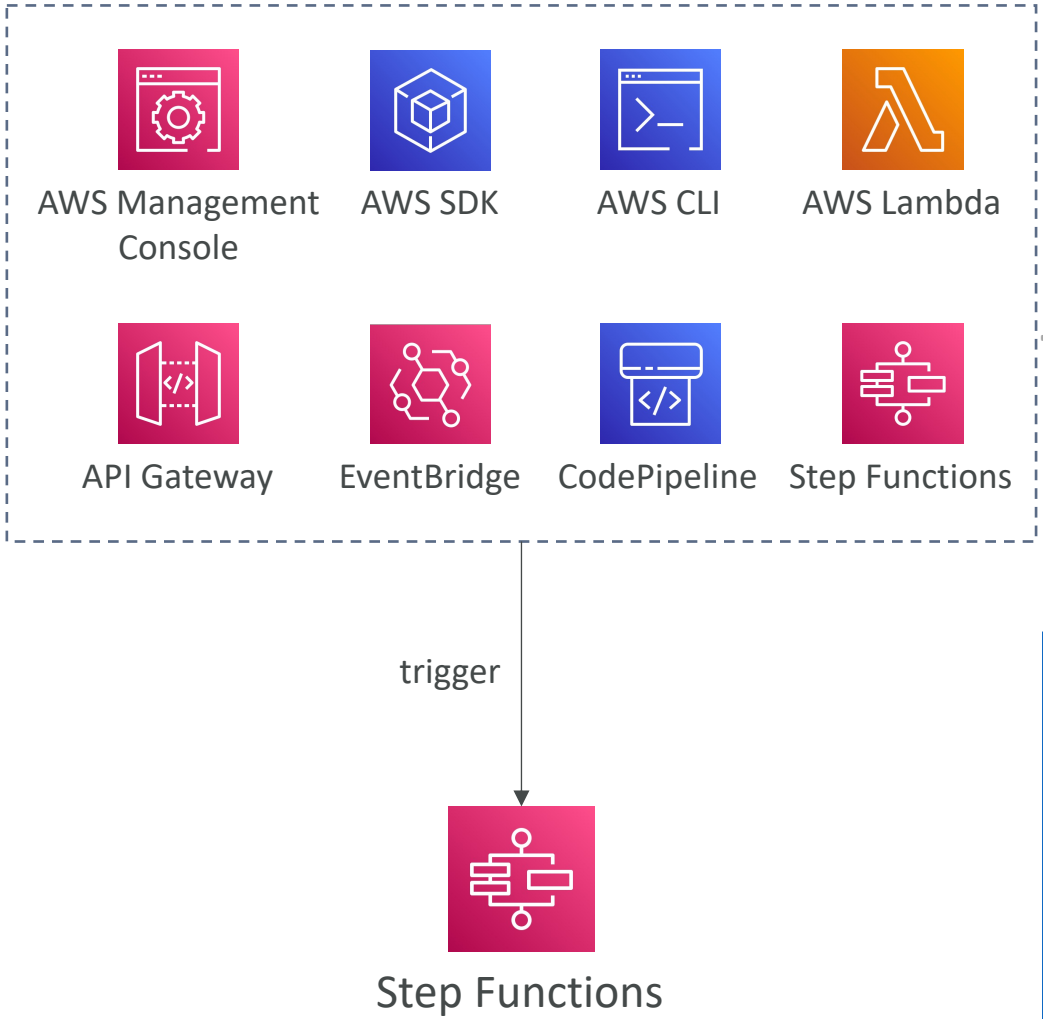


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Step Functions Workflow Triggers

- You can invoke a Step Function Workflow (State Machine) using:
  - AWS Management Console
  - AWS SDK (**StartExecution** API call)
  - AWS CLI (**start-execution**)
  - AWS Lambda (**StartExecution** API call)
  - API Gateway
  - EventBridge
  - CodePipeline
  - Step Functions

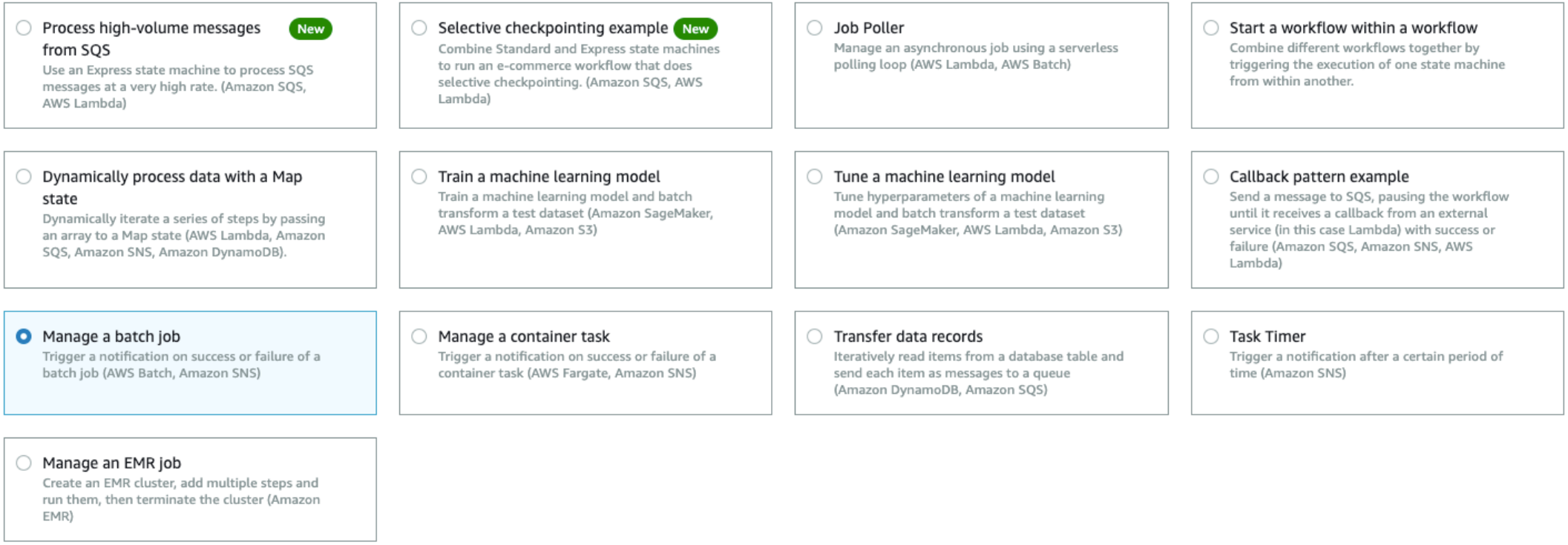


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Step Functions – Sample Projects

- https://console.aws.amazon.com/states/home?region=us-east-1#/sampleProjects



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Step Functions – Tasks

- Lambda Tasks:
  - Invoke a Lambda function
- Activity Tasks:
  - Activity worker (HTTP), EC2 Instances, mobile device, on premise DC
  - They poll the Step functions service
- Service Tasks:
  - Connect to a supported AWS service
  - Lambda function, ECS Task, Fargate, DynamoDB, Batch job, SNS topic, SQS queue
- Wait Task:
  - To wait for a duration or until a timestamp

- Note: Step Functions does not integrate natively with AWS Mechanical Turk

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Step Functions – Standard vs. Express

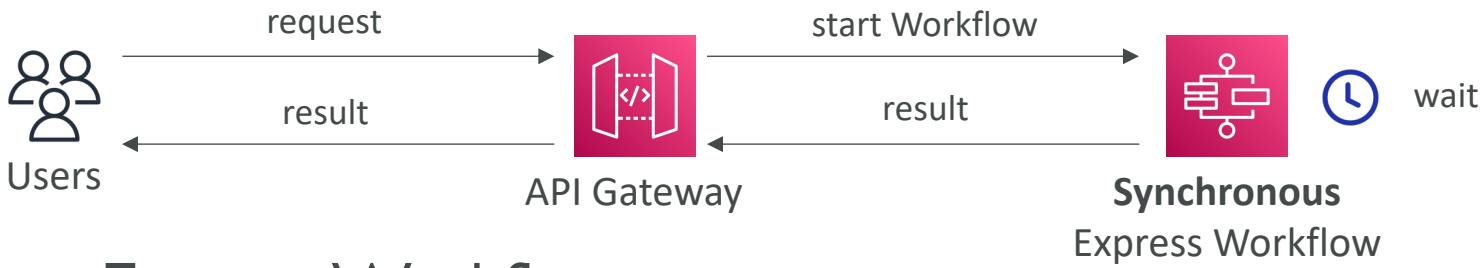
	Standard Workflows	Express Workflows
Maximum duration	1 year.	5 minutes.
Supported execution start rate	Over 2,000 per second	Over 100,000 per second
Supported state transition rate	Over 4,000 per second per account	Nearly unlimited
Pricing	Priced per state transition. A state transition is counted each time a step in your execution is completed (more expensive)	Priced by the number of executions you run, their duration, and memory consumption (cheaper)
Execution history	Executions can be listed and described with Step Functions APIs, and visually debugged through the console. They can also be inspected in CloudWatch Logs by enabling logging on your state machine.	Executions can be inspected in CloudWatch Logs by enabling logging on your state machine.
Execution semantics	Exactly-once workflow execution.	At-least-once workflow execution.

© Stephane Maarek

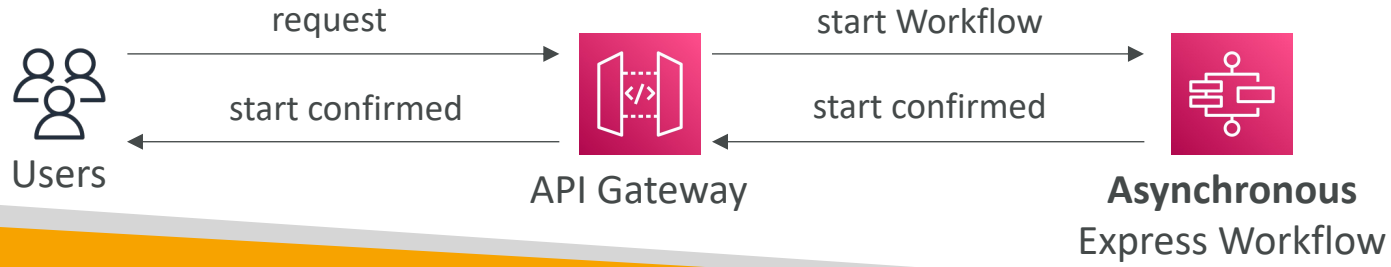
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Step Functions – Express Workflow – Synchronous vs. Asynchronous

- **Synchronous Express Workflows**
  - Wait until the Workflow completes, then return the result
  - Examples: orchestrate microservices, handle errors, retries, parallel tasks, …



- **Asynchronous Express Workflows**
  - Doesn't for the Workflow to complete
  - Examples: Workflows that don't require immediate response, messaging, …

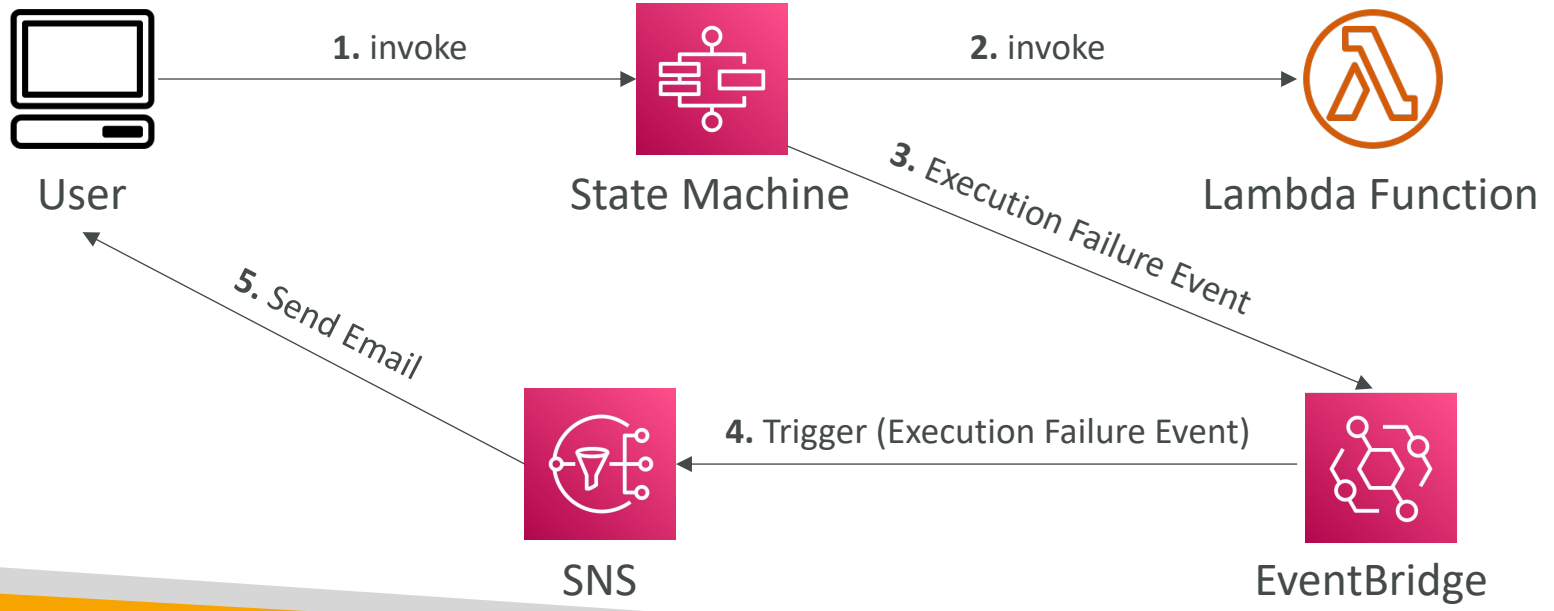


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Step Functions – Error Handling

- You can enable error handling, retries, and add alerting to Step Function State Machine
- Example: set up EventBridge to alert via email if a State Machine execution fails

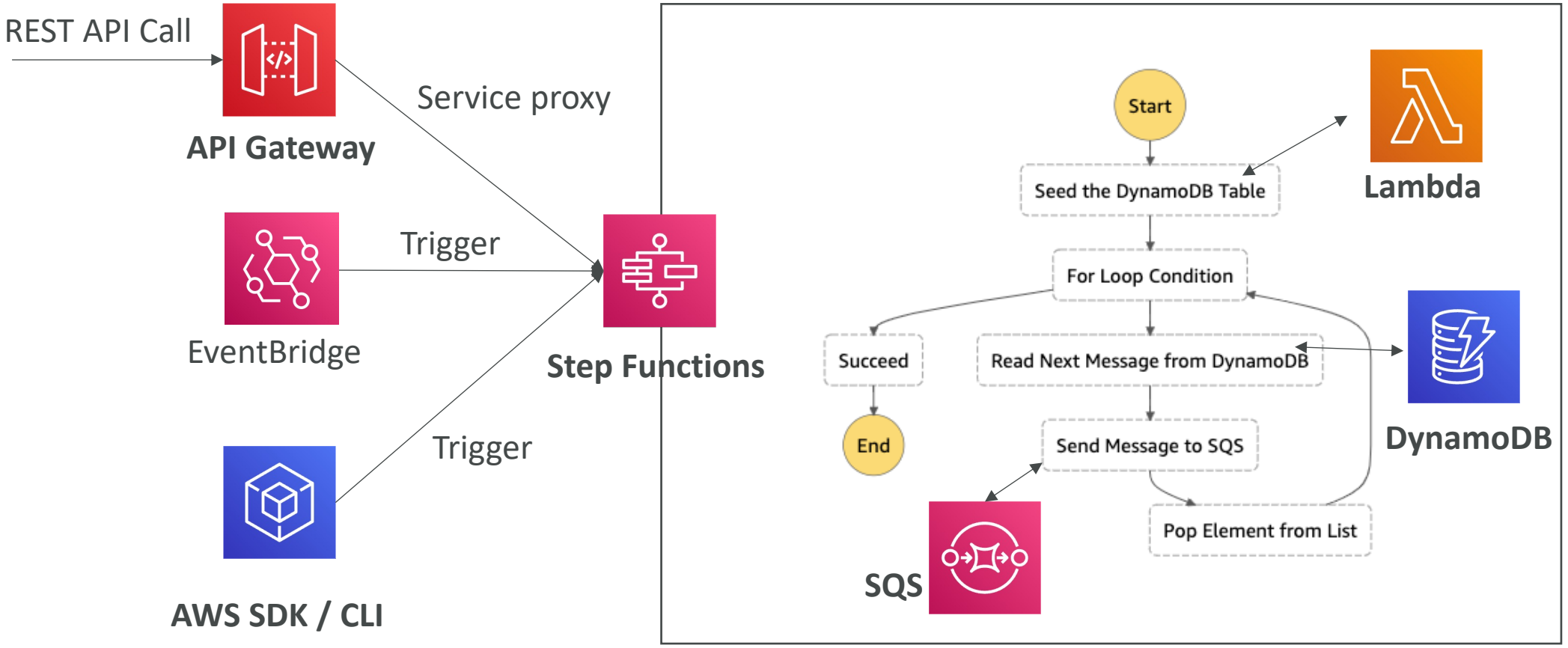



```
{
    "Retry": [
        {
            "ErrorEquals": ["States.ALL"],
            "IntervalSeconds": 3,
            "MaxAttempts": 2,
            "BackoffRate": 1.5
        }
    ]
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Step Functions – Solution Architecture



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon SQS

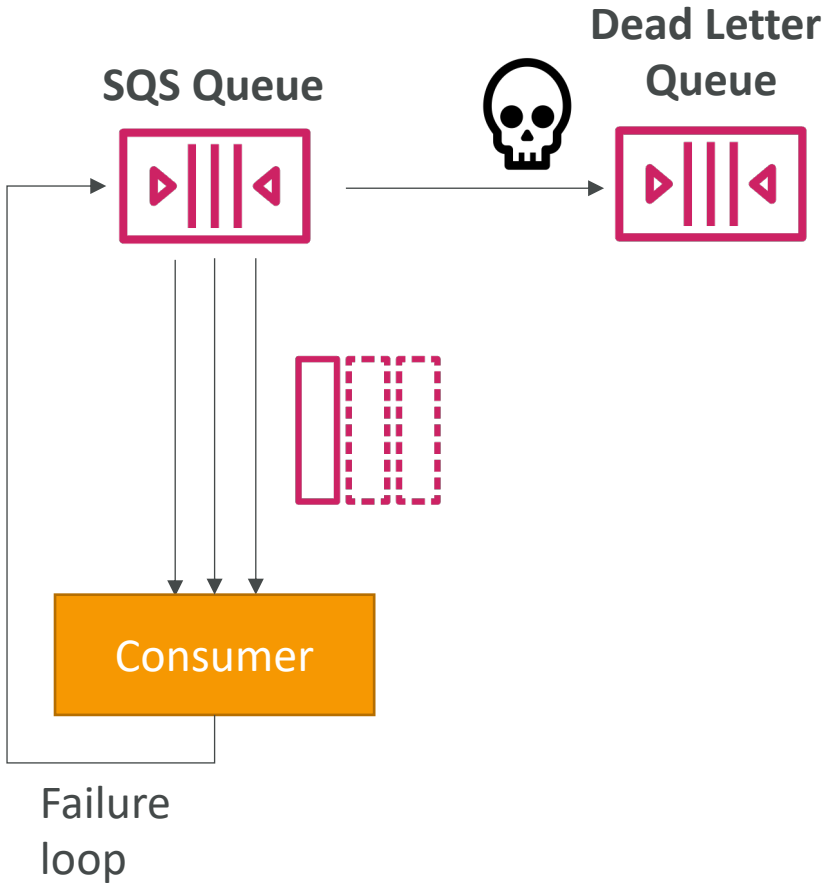
- Serverless, managed queue, integrated with IAM
- Can handle extreme scale, no provisioning required
- Used to **decouple** services
- Message size of max 1024 KB (use a pointer to S3 for large messages)
- Can be read from EC2 (optional ASG), Lambda
- SQS could be used as a write buffer for DynamoDB
- **SQS FIFO:**
  - receive messages in order they were sent
  - 300 messages/s without batching, 3000 /s with batching

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon SQS – Dead Letter Queue (DLQ)

- If a consumer fails to process a message within the Visibility Timeout…
  the message goes back to the queue!
- We can set a threshold of how many times a message can go back to the queue
- After the **MaximumReceives** threshold is exceeded, the message goes into a Dead Letter Queue (DLQ)
- Useful for debugging!
- **DLQ of a FIFO queue must also be a FIFO queue**
- **DLQ of a Standard queue must also be a Standard queue**
- Make sure to process the messages in the DLQ before they expire:
  - Good to set a retention of 14 days in the DLQ

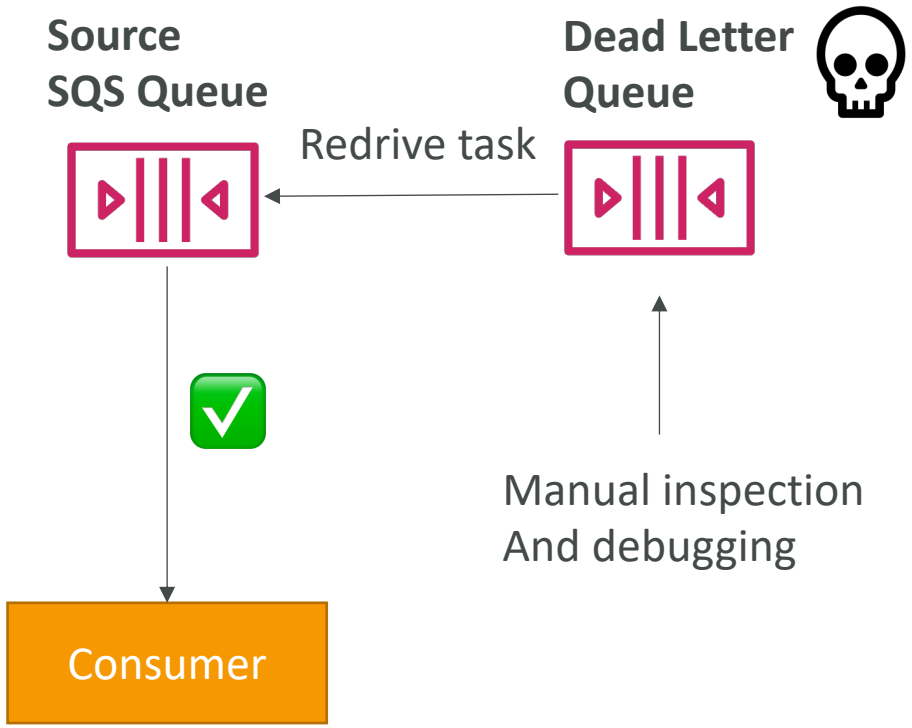


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SQS DLQ – Redrive to Source

- Feature to help consume messages in the DLQ to understand what is wrong with them
- When our code is fixed, we can redrive the messages from the DLQ back into the source queue (or any other queue) in batches without writing custom code

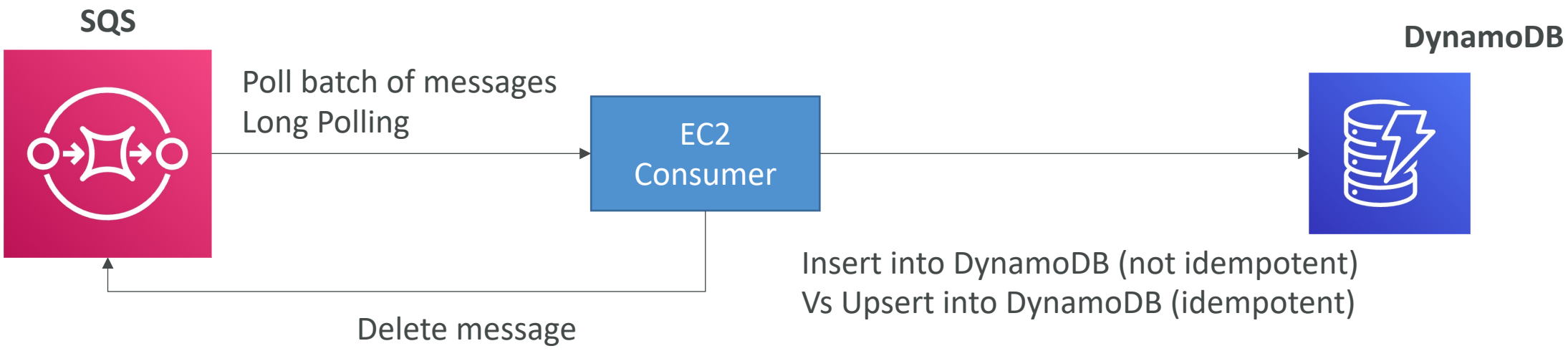


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SQS – Solution Architecture Idempotency

- Messages can be processed twice by consumer (in case of failures, timeouts, etc)
- To hedge against that problem, implement **idempotency** at the consumer level
- Means the same action done twice by the consumer won't duplicate the effect

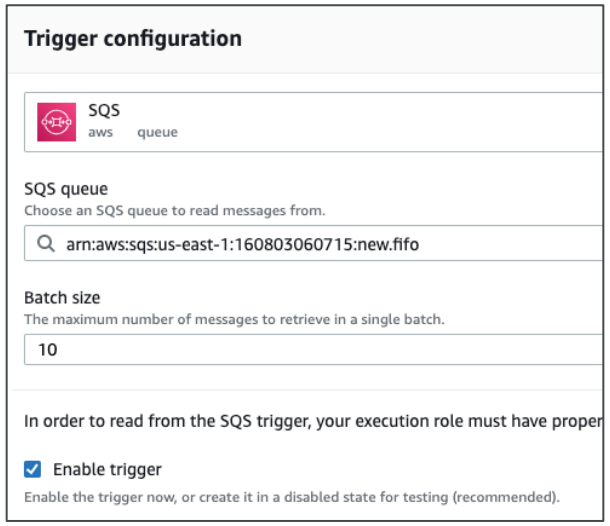


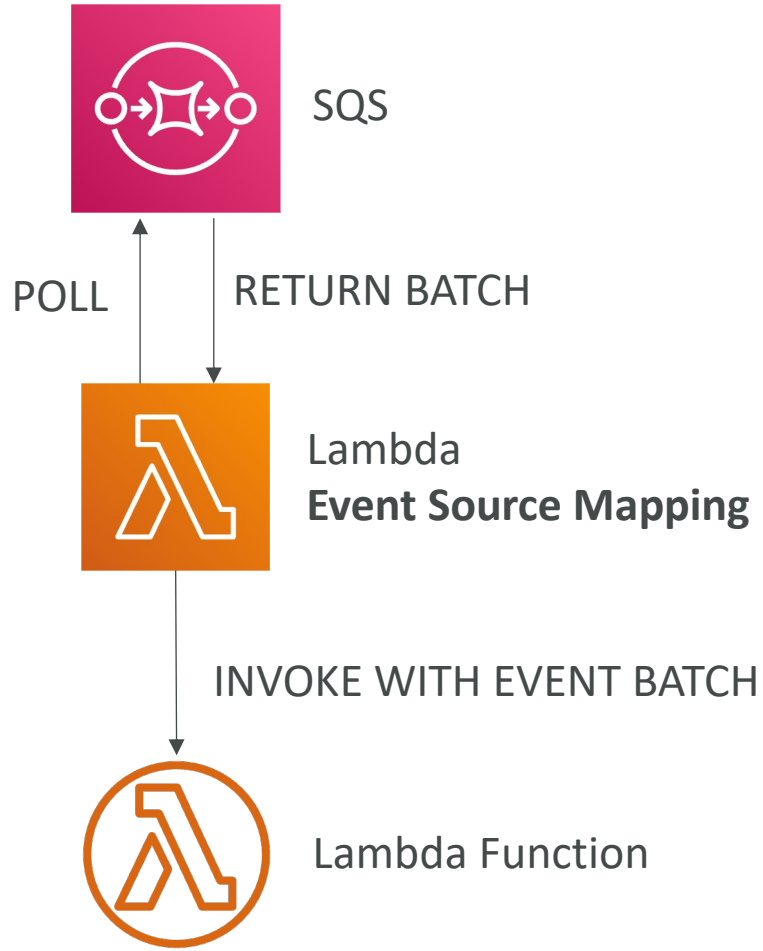
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Lambda – Event Source Mapping
SQS & SQS FIFO

- Event Source Mapping will poll SQS (Long Polling)
- Specify **batch size** (1-10 messages)
- Recommended: Set the queue visibility timeout to 6x the timeout of your Lambda function
- To use a DLQ
  - **set-up on the SQS queue, not Lambda** (DLQ for Lambda is only for async invocations)
  - Or use a Lambda destination for failures

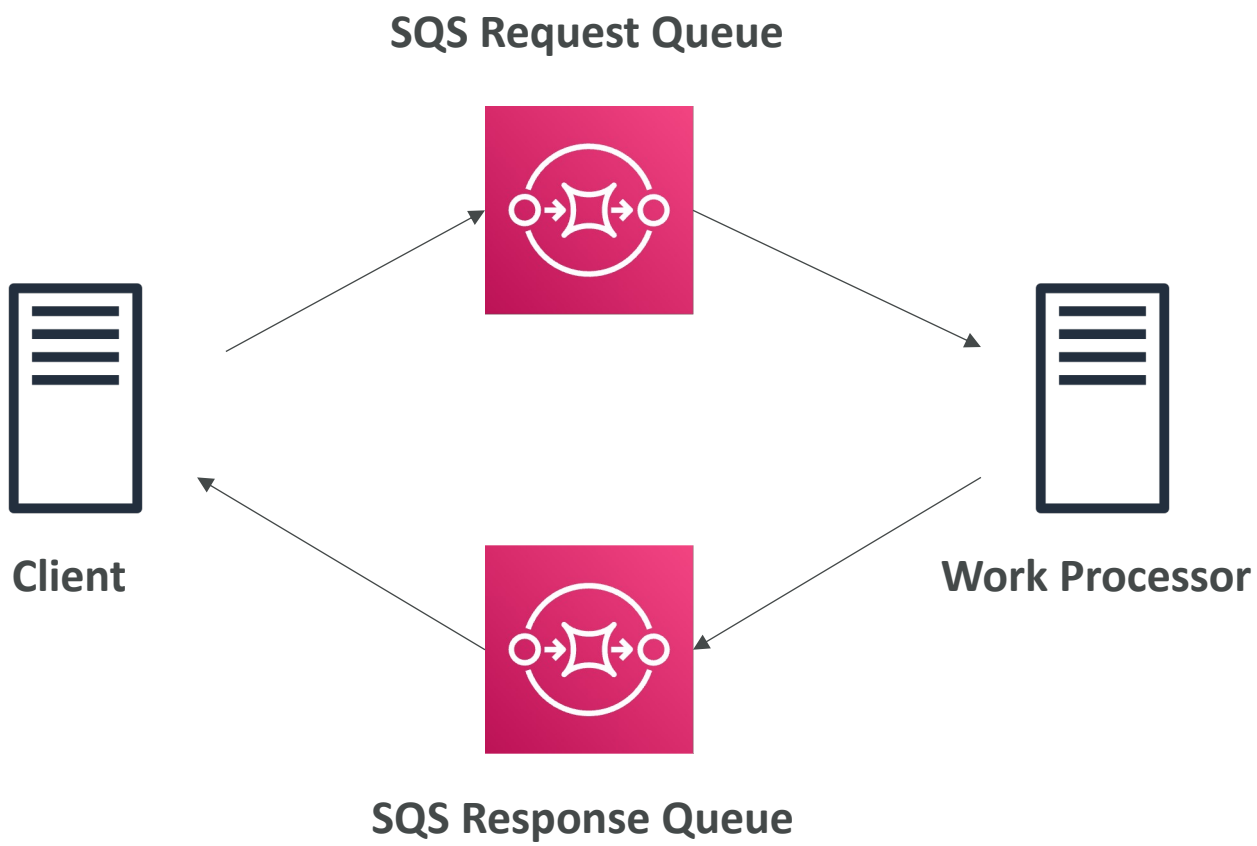




NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

SQS - Solution Architecture
Request / Response queue (async)



- Decoupling
- Fault-Tolerance
- Load Balancing

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon MQ

- SQS, SNS are "cloud-native" services: proprietary protocols from AWS
- Traditional applications running from on-premises may use open protocols such as: MQTT, AMQP, STOMP, Openwire, WSS
- **When migrating to the cloud**, instead of re-engineering the application to use SQS and SNS, we can use Amazon MQ
- **Amazon MQ is a managed message broker service for**

RabbitMQ™

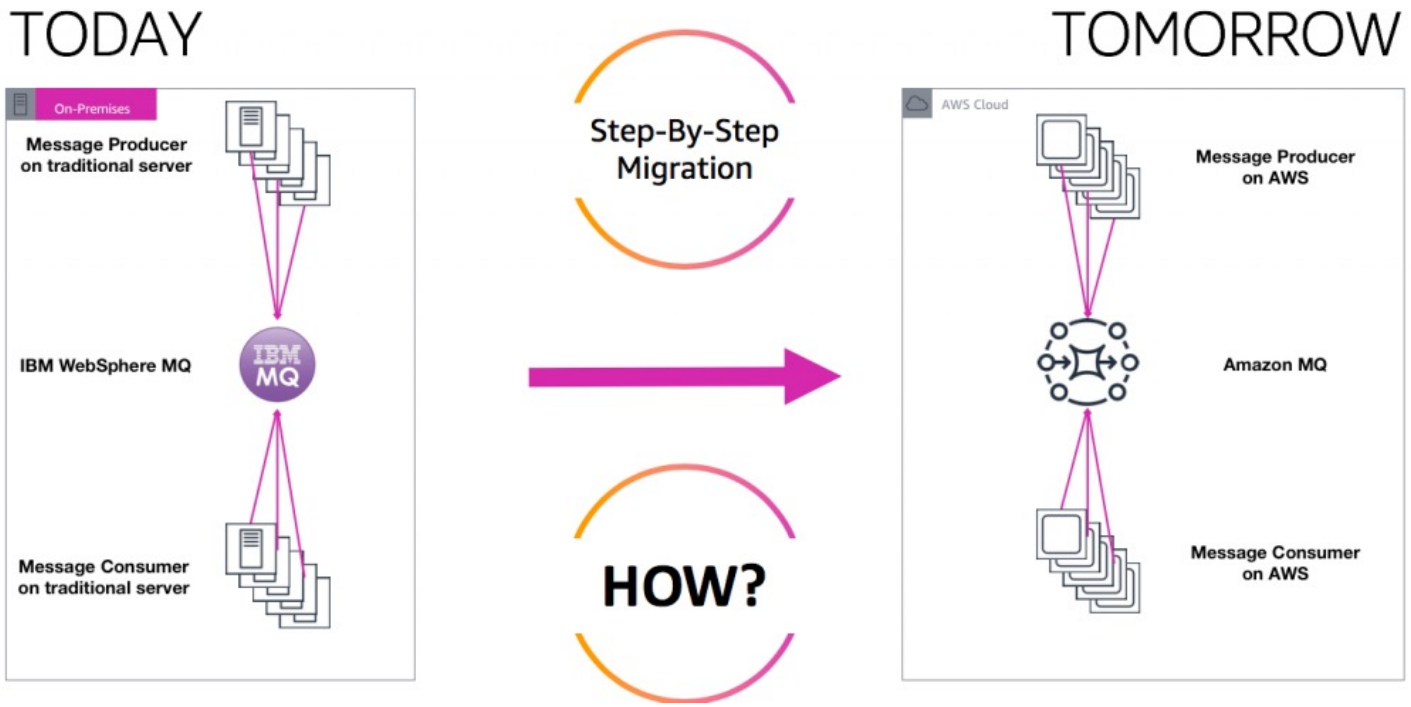
- Amazon MQ doesn't "scale" as much as SQS / SNS
- Amazon MQ runs on servers, can run in Multi-AZ with failover
- Amazon MQ has both queue feature (~SQS) and topic features (~SNS)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon MQ – Re-platform

- IBM MQ, TIBCO EMS, Rabbit MQ, and Apache ActiveMQ can be migrated to Amazon MQ



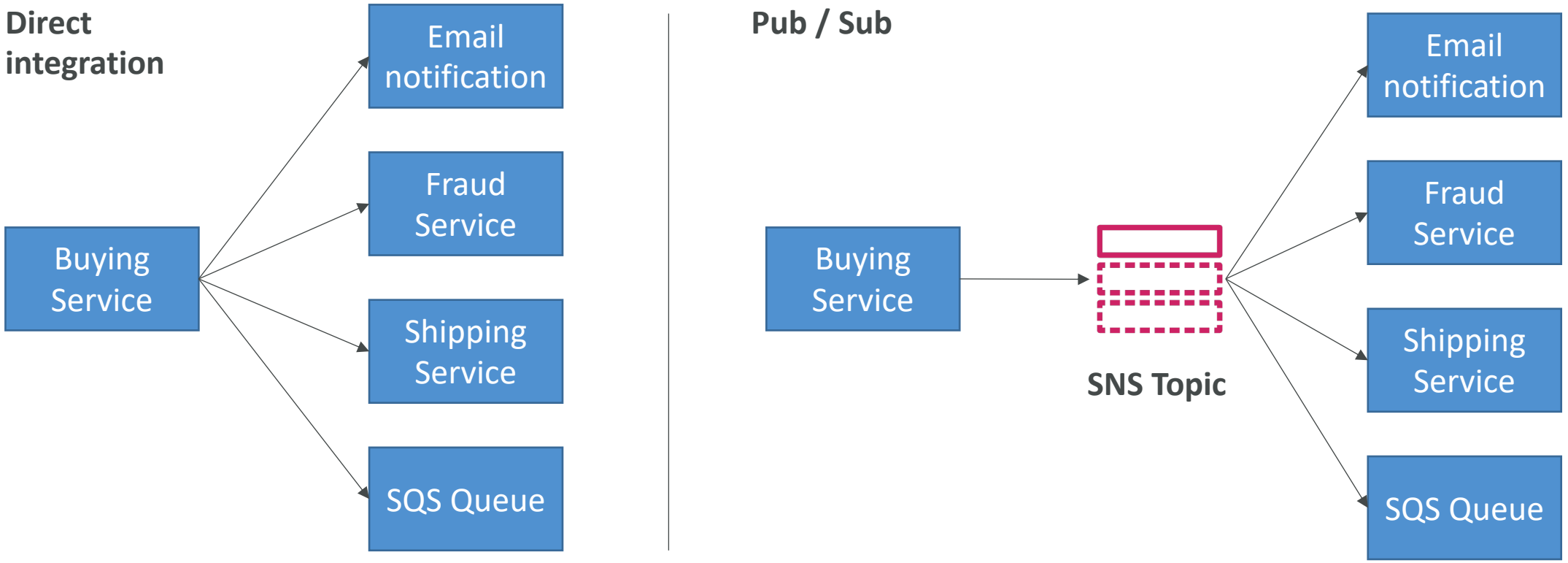
https://aws.amazon.com/blogs/compute/migrating-from-ibm-mq-to-amazon-mq-using-a-phased-approach/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon SNS

- What if you want to send one message to many receivers?

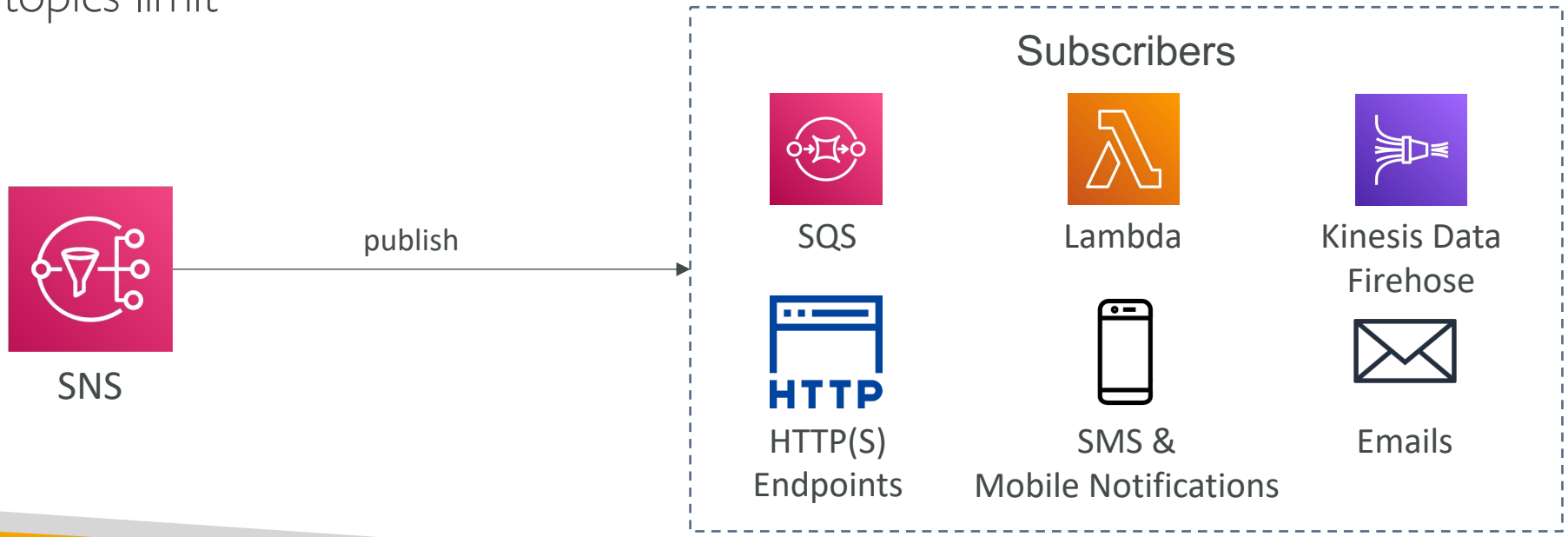


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon SNS

- The "event producer" only sends message to one SNS topic
- As many "event receivers" (subscriptions) as we want to listen to the SNS topic notifications
- Each subscriber to the topic will get all the messages (note: new feature to filter messages)
- Up to 12,500,000 subscriptions per topic
- 100,000 topics limit

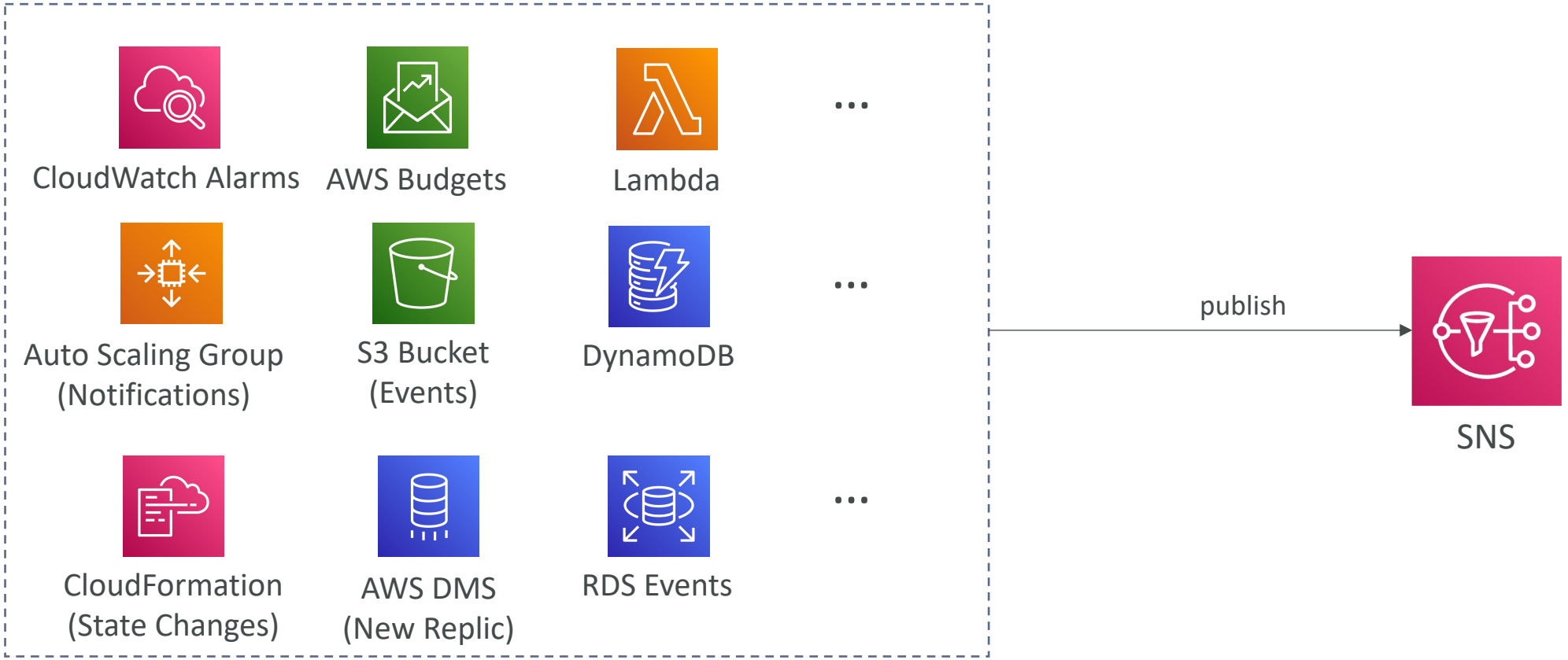


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SNS integrates with a lot of AWS services

- Many AWS services can send data directly to SNS for notifications



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon SNS – How to publish

- Topic Publish (using the SDK)
  - Create a topic
  - Create a subscription (or many)
  - Publish to the topic

- Direct Publish (for mobile apps SDK)
  - Create a platform application
  - Create a platform endpoint
  - Publish to the platform endpoint
  - Works with Google GCM, Apple APNS, Amazon ADM…

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

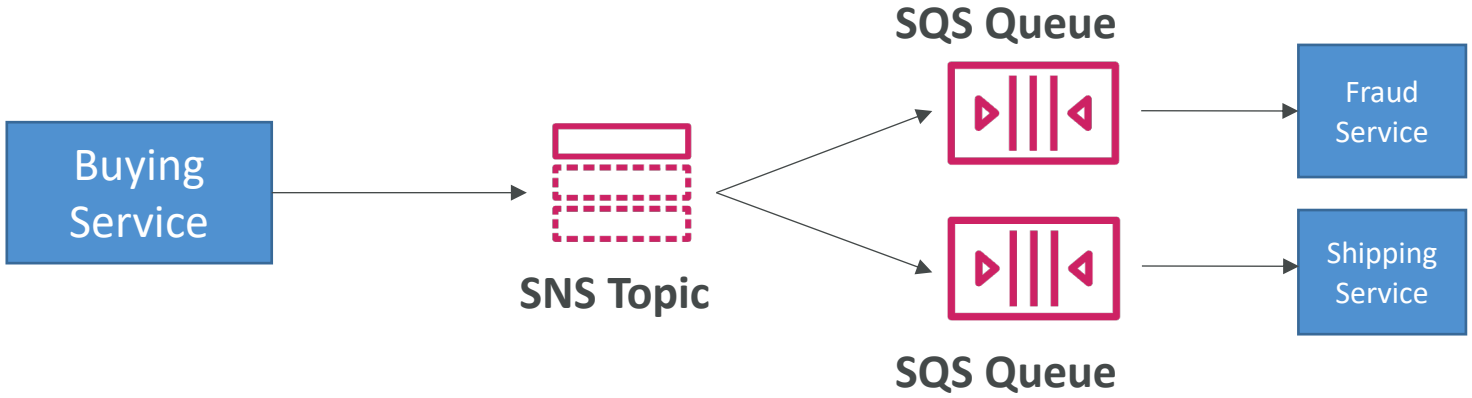
Amazon SNS – Security

- **Encryption:**
  - In-flight encryption using HTTPS API
  - At-rest encryption using KMS keys
  - Client-side encryption if the client wants to perform encryption/decryption itself
- **Access Controls:** IAM policies to regulate access to the SNS API
- **SNS Access Policies** (similar to S3 bucket policies)
  - Useful for cross-account access to SNS topics
  - Useful for allowing other services (S3…) to write to an SNS topic

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SNS + SQS: Fan Out



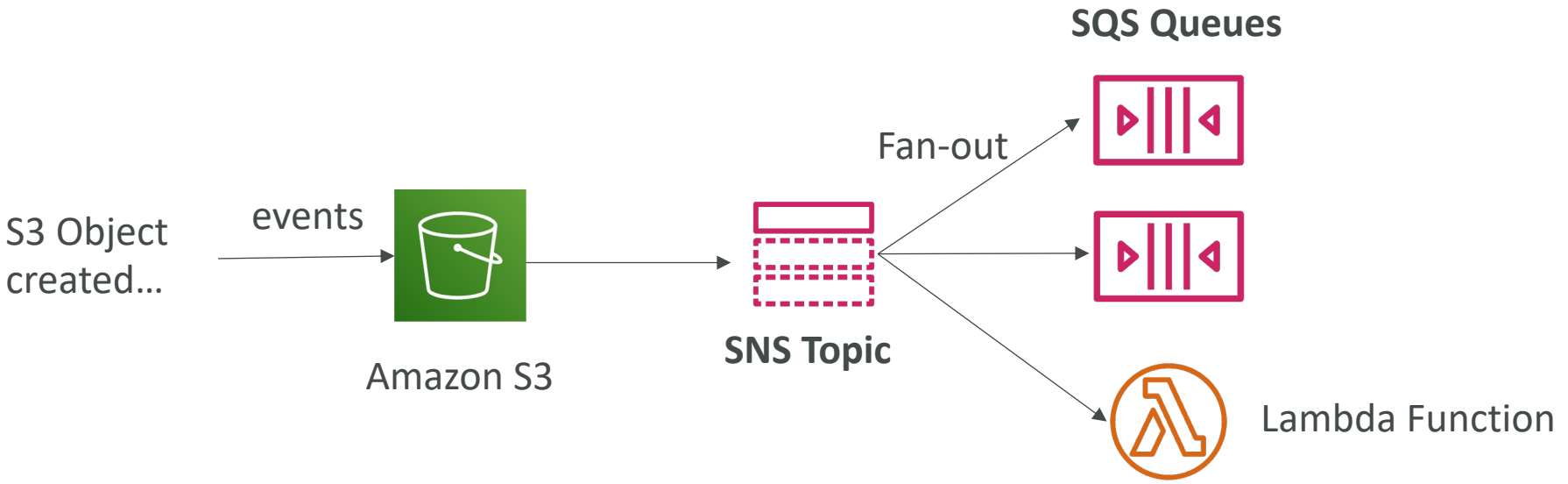
- Push once in SNS, receive in all SQS queues that are subscribers
- Fully decoupled, no data loss
- SQS allows for: data persistence, delayed processing and retries of work
- Ability to add more SQS subscribers over time
- Make sure your SQS queue **access policy** allows for SNS to write
- Cross-Region Delivery: works with SQS Queues in other regions

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Application: S3 Events to multiple queues

- For the same combination of: **event type** (e.g. object create) and **prefix** (e.g. images/) you can only have one S3 Event rule
- If you want to send the same S3 event to many SQS queues, use fan-out



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Application: SNS to Amazon S3 through Kinesis Data Firehose

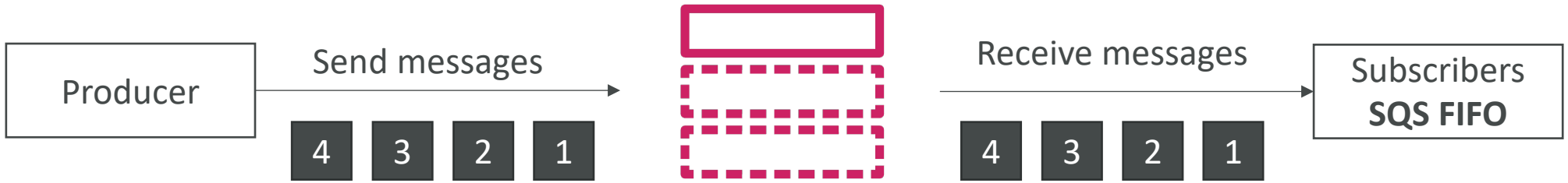
- SNS can send to Kinesis and therefore we can have the following solutions architecture:

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon SNS – FIFO Topic

- FIFO = First In First Out (ordering of messages in the topic)



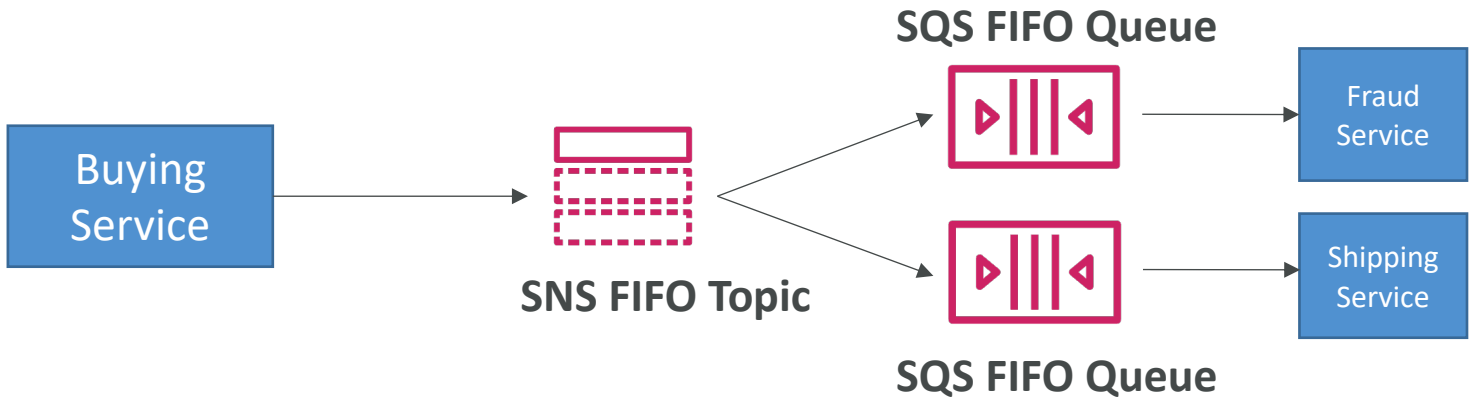
- Similar features as SQS FIFO:
  - **Ordering** by Message Group ID (all messages in the same group are ordered)
  - **Deduplication** using a Deduplication ID or Content Based Deduplication
- **Can have SQS Standard and FIFO queues as subscribers**
- Limited throughput (same throughput as SQS FIFO)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SNS FIFO + SQS FIFO: Fan Out

- In case you need fan out + ordering + deduplication

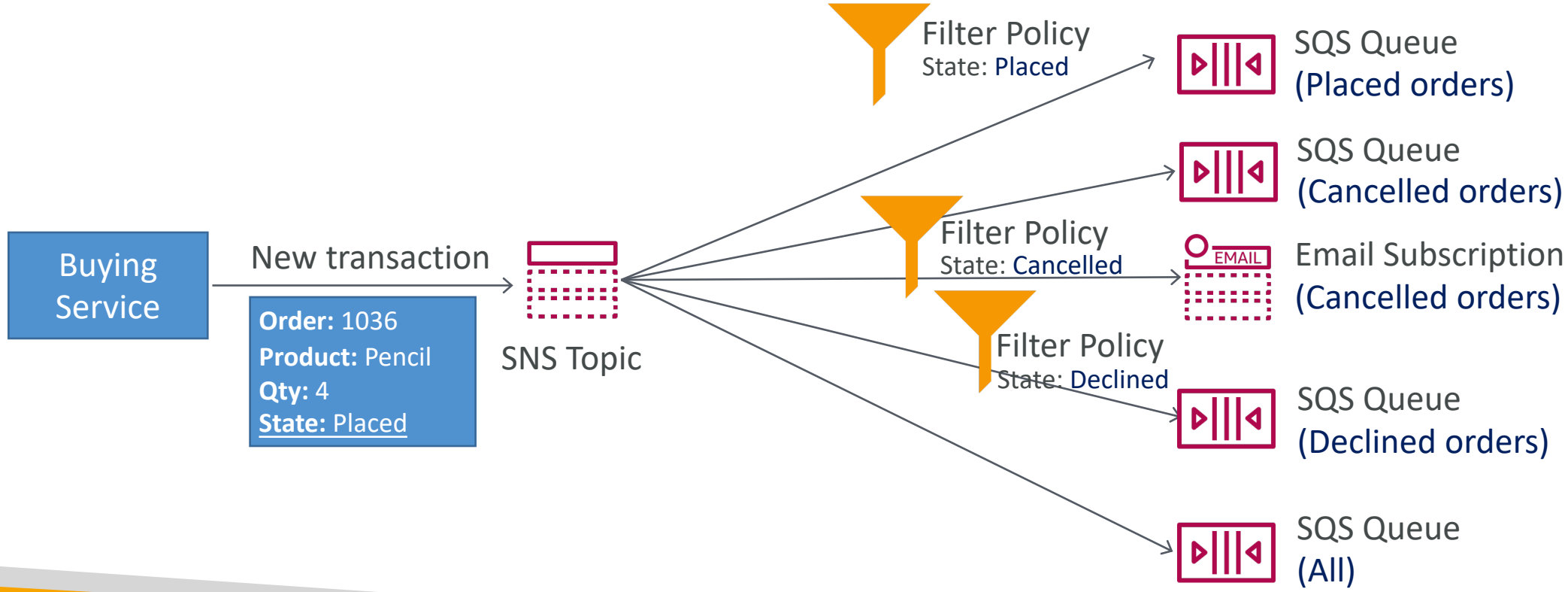


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SNS – Message Filtering

- JSON policy used to filter messages sent to SNS topic's subscriptions
- If a subscription doesn't have a filter policy, it receives every message

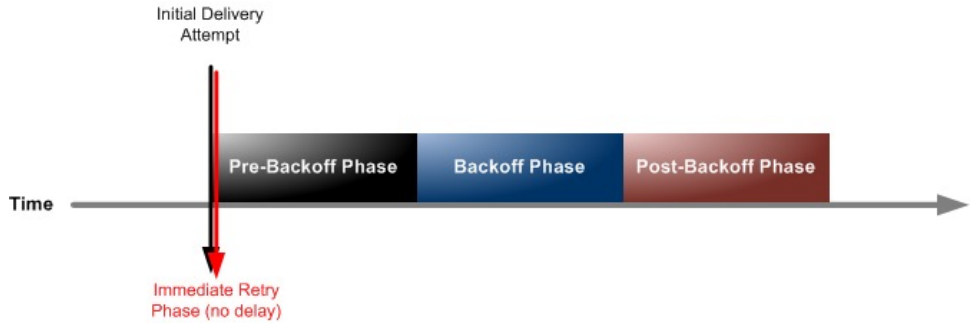


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SNS – Message Delivery Retries

- When a message is delivered to an SNS subscriber, in case of server-side errors, **a delivery policy** is applied



Endpoint type	Delivery protocols	Immediate retry (no delay) phase	Pre-backoff phase	Backoff phase	Post-backoff phase	Total attempts
AWS managed endpoints	Amazon Kinesis Data Firehose AWS Lambda Amazon SQS	3 times, without delay	2 times, 1 second apart	10 times, with exponential backoff, from 1 second to 20 seconds	100,000 times, 20 seconds apart	100,015 times, over 23 days
Customer managed endpoints	SMTP	0 times, without delay	2 times, 10 seconds apart	10 times, with exponential backoff, from 10 seconds to 600 seconds (10 minutes)	38 times, 600 seconds (10 minutes) apart	50 attempts, over 6 hours

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SNS – Custom Delivery Policies

- Only HTTP/S supports custom policies

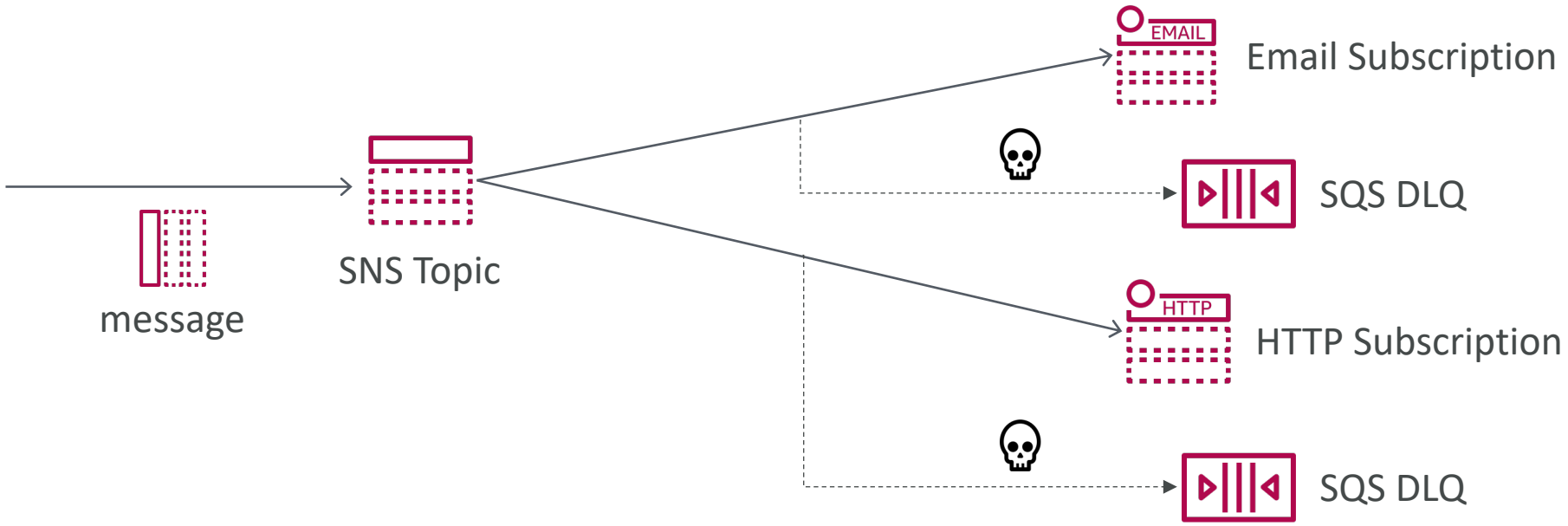
```
{
    "healthyRetryPolicy": {
        "minDelayTarget": 1,
        "maxDelayTarget": 60,
        "numRetries": 50,
        "numNoDelayRetries": 3,
        "numMinDelayRetries": 2,
        "numMaxDelayRetries": 35,
        "backoffFunction": "exponential"
    },
    "sicklyRetryPolicy": null,
    "throttlePolicy": {
        "maxReceivesPerSecond": 10
    }
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

SNS – Dead Letter Queues

- After exhausting the delivery policy, messages that haven't been delivered are discarded unless you set a DLQ (Dead Letter Queue)
- DLQ are Amazon SQS queues or Amazon SQS FIFO queues (for SNS FIFO)
- Dead Letter Queues are attached to a subscription (rather than a topic)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Data Engineering

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Kinesis Data Streams

- Collect and store streaming data in **real-time**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Kinesis Data Streams

- Retention between up to 365 days
- Ability to reprocess (replay) data by consumers
- Data can't be deleted from Kinesis (until it expires)
- Data up to 10MiB (typical use case is lot of "small" **real-time data**)
- Data ordering guarantee for data with the same "Partition ID"
- At-rest KMS encryption, in-flight HTTPS encryption
- Kinesis Producer Library (KPL) to write an optimized producer application
- Kinesis Client Library (KCL) to write an optimized consumer application

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Kinesis Data Streams – Capacity Modes

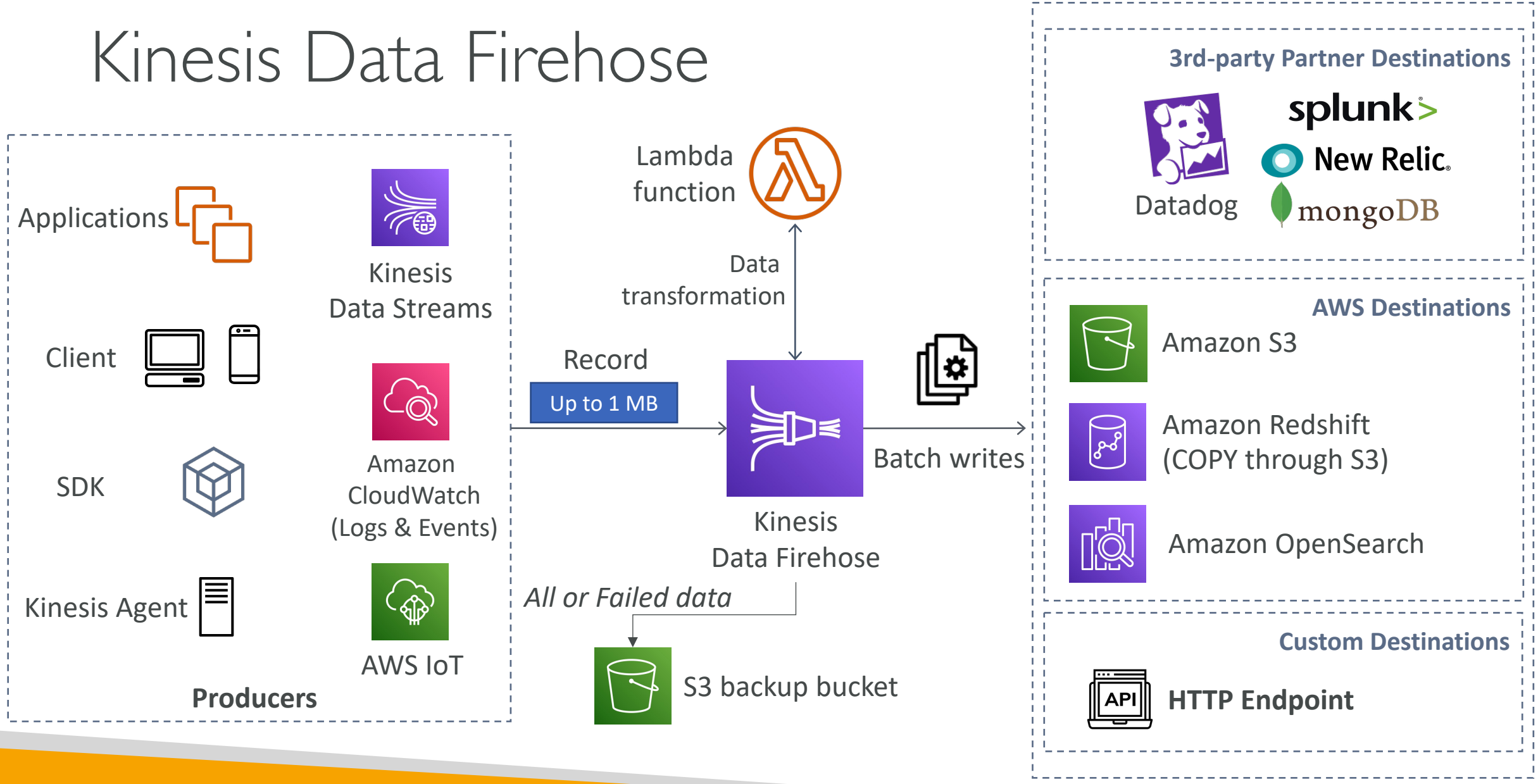
- Provisioned mode:
  - Choose number of shards
  - Each shard gets 1MB/s in (or 1000 records per second)
  - Each shard gets 2MB/s out
  - Scale manually to increase or decrease the number of shards
  - You pay per shard provisioned per hour

- On-demand mode:
  - No need to provision or manage the capacity
  - Default capacity provisioned (4 MB/s in or 4000 records per second)
  - Scales automatically based on observed throughput peak during the last 30 days
  - Pay per stream per hour & data in/out per GB

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

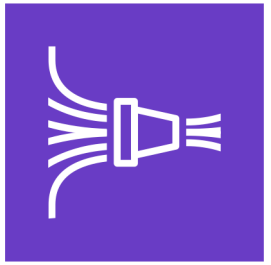
Kinesis Data Firehose



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Kinesis Data Firehose

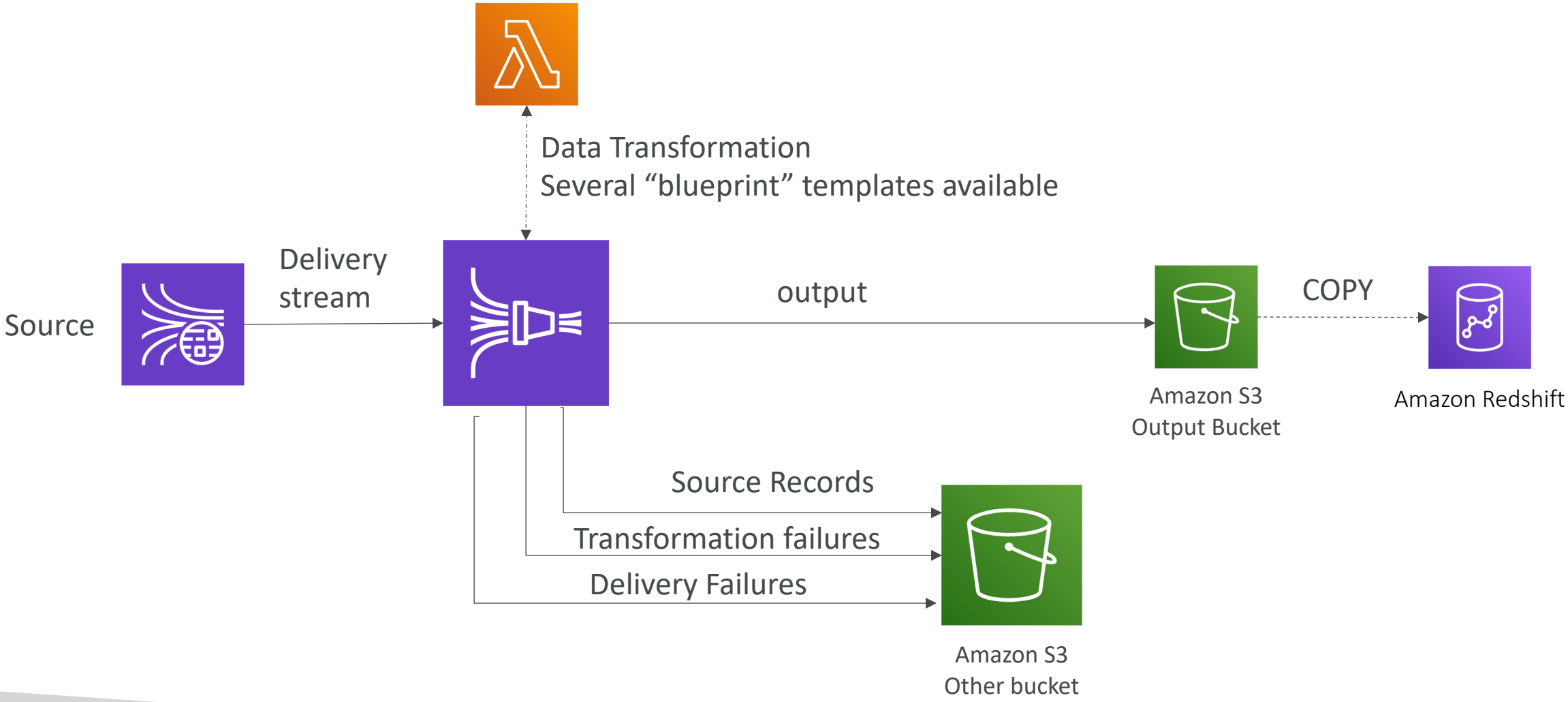


- Fully Managed Service, no administration
- **Near Real Time** (**Buffer based on time and size, optionally can be disabled**)
- Load data into Redshift / Amazon S3 / OpenSearch / Splunk
- Automatic scaling
- Supports many data formats
- Data Conversions from JSON to Parquet / ORC (only for S3)
- Data Transformation through AWS Lambda (ex: CSV => JSON)
- Supports **compression** when target is Amazon S3 (GZIP, ZIP, and SNAPPY)
- Only GZIP is the data is further loaded into Redshift
- Pay for the amount of data going through Firehose
- Spark / KCL do *not* read from KDF

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Kinesis Data Firehose Delivery Diagram



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Firehose Buffer Sizing

- Firehose accumulates records in a buffer
- The buffer is flushed based on time and size rules

- Buffer Size (ex: 32MB): if that buffer size is reached, it's flushed
- Buffer Time (ex: 2 minutes): if that time is reached, it's flushed
- Firehose can automatically increase the buffer size to increase throughput

- High throughput => Buffer Size will be hit
- Low throughput => Buffer Time will be hit

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Kinesis Data Streams vs Firehose

- Streams
  - Going to write custom code (producer / consumer)
  - Real time (~200 ms latency for classic, ~70 ms latency for enhanced fan-out)
  - On-demand or Provisioned mode
  - Data Storage for 1 to 365 days, replay capability, multi consumers
  - Use with Lambda to insert data in real-time to OpenSearch (for example)

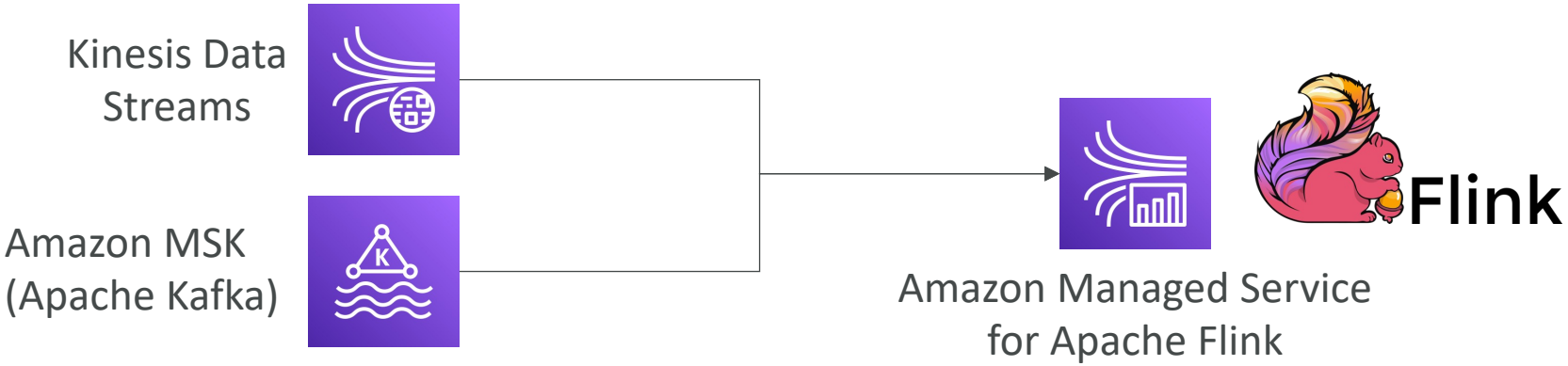
- Firehose
  - Fully managed, send to S3, Splunk, Redshift, OpenSearch
  - Serverless data transformations with Lambda
  - **Near** real time
  - Automated Scaling
  - No data storage

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Managed Service for Apache Flink

- *Previously named: Kinesis Data Analytics for Apache Flink*
- Flink (Java, Scala or SQL) is a framework for processing data streams

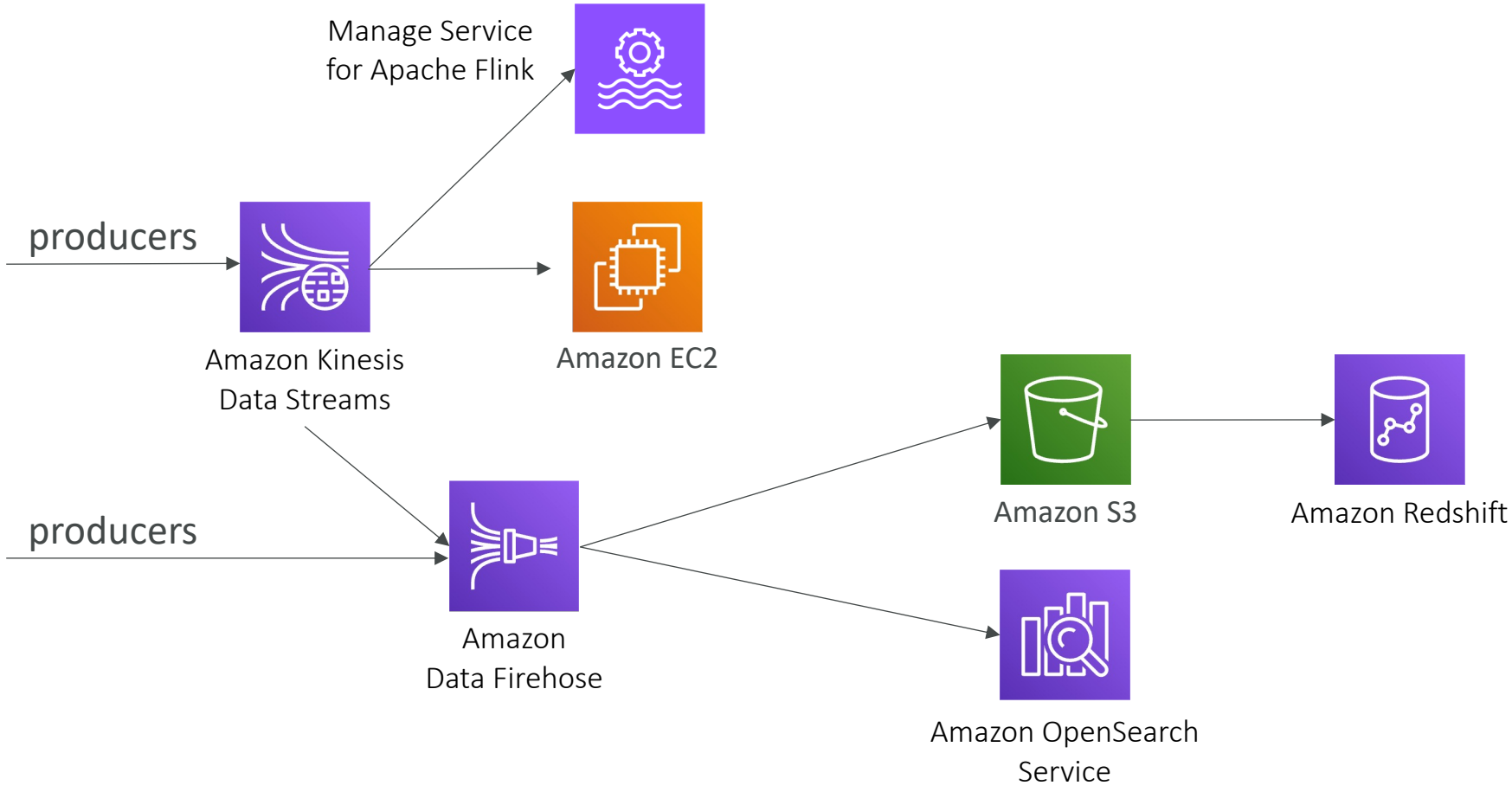


- Run any Apache Flink application on a managed cluster on AWS
  - Provisioned compute resources, parallel computation, automatic scaling
  - Application backups (implemented as checkpoints and snapshots)
  - Use any Apache Flink programming features to transform data
  - Important: Flink does not read from Amazon Data Firehose

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

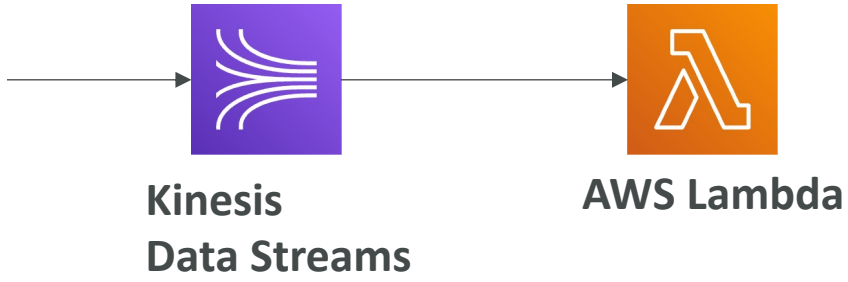
Full Data Engineering Pipeline
Real-Time Layer

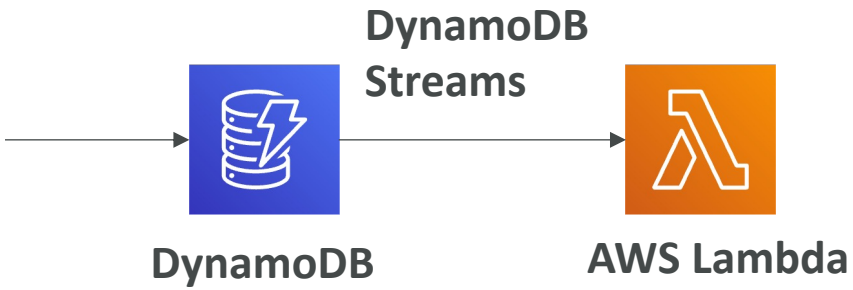


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Streaming Architectures
3000 messages of 1 KB per second





Kinesis

- 3 shards: 3MB/s in
- 3 * $0.015/hr = **$32.4/mth**
- Must use KDF for output to S3

DynamoDB + Streams

- 3000 WCU = 3 MB/s
- = **$1,450.90 / month**
- Storage in DynamoDB

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Comparison Charts

	Kinesis Data Streams	SQS	SQS FIFO	SNS	DynamoDB	S3
Data	Immutable	Immutable	Immutable	Immutable	Mutable	Mutable
Retention	1-365 days, export to S3 using KDF	1-14 days	1-14 days	No retention	Infinite or can implement TTL	Infinite, can setup lifecycle policies
Ordering	Per shard	No ordering	Per group-id	No ordering	No ordering	No ordering
Scalability	Provision shards	Soft limit	300 msg/s Or 3000 if batch	Soft limit	WCU & RCU On-demand	Infinite 3500 PUT 5500 GET / prefix
Readers	EC2, Lambda, KDF, KDA, KCL (checkpoint)	EC2, Lambda	EC2, Lambda	HTTP, Lambda, Email, SQS…	DynamoDB Streams	SDK, S3 Events
Latency	KDS (200 ms) KDF (1 min)	Low (10-100ms)	Low (10-100ms)	Low (10-100ms)	Low (10-100ms)	Low (10-100ms)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

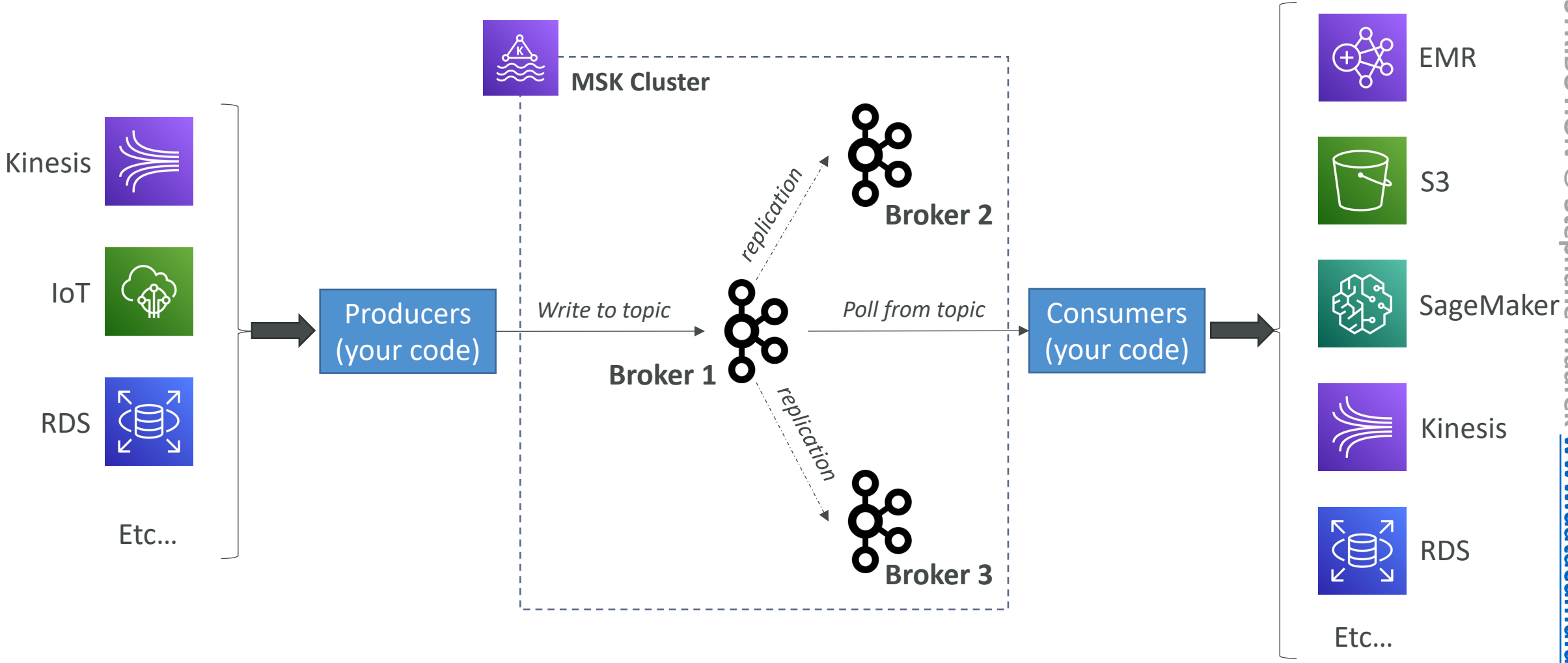
Amazon Managed Streaming for Apache Kafka (Amazon MSK)

- Alternative to Amazon Kinesis
- Fully managed Apache Kafka on AWS
  - Allow you to create, update, delete clusters
  - MSK creates & manages Kafka brokers nodes & Zookeeper nodes for you
  - Deploy the MSK cluster in your VPC, multi-AZ (up to 3 for HA)
  - Automatic recovery from common Apache Kafka failures
  - Data is stored on EBS volumes **for as long as you want**
- **MSK Serverless**
  - Run Apache Kafka on MSK without managing the capacity
  - MSK automatically provisions resources and scales compute & storage

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Apache Kafka at a high level



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Kinesis Data Streams vs. Amazon MSK

Kinesis Data Streams

- 1 MB message size limit
- Data Streams with Shards
- Shard Splitting & Merging
- TLS In-flight encryption
- KMS at-rest encryption

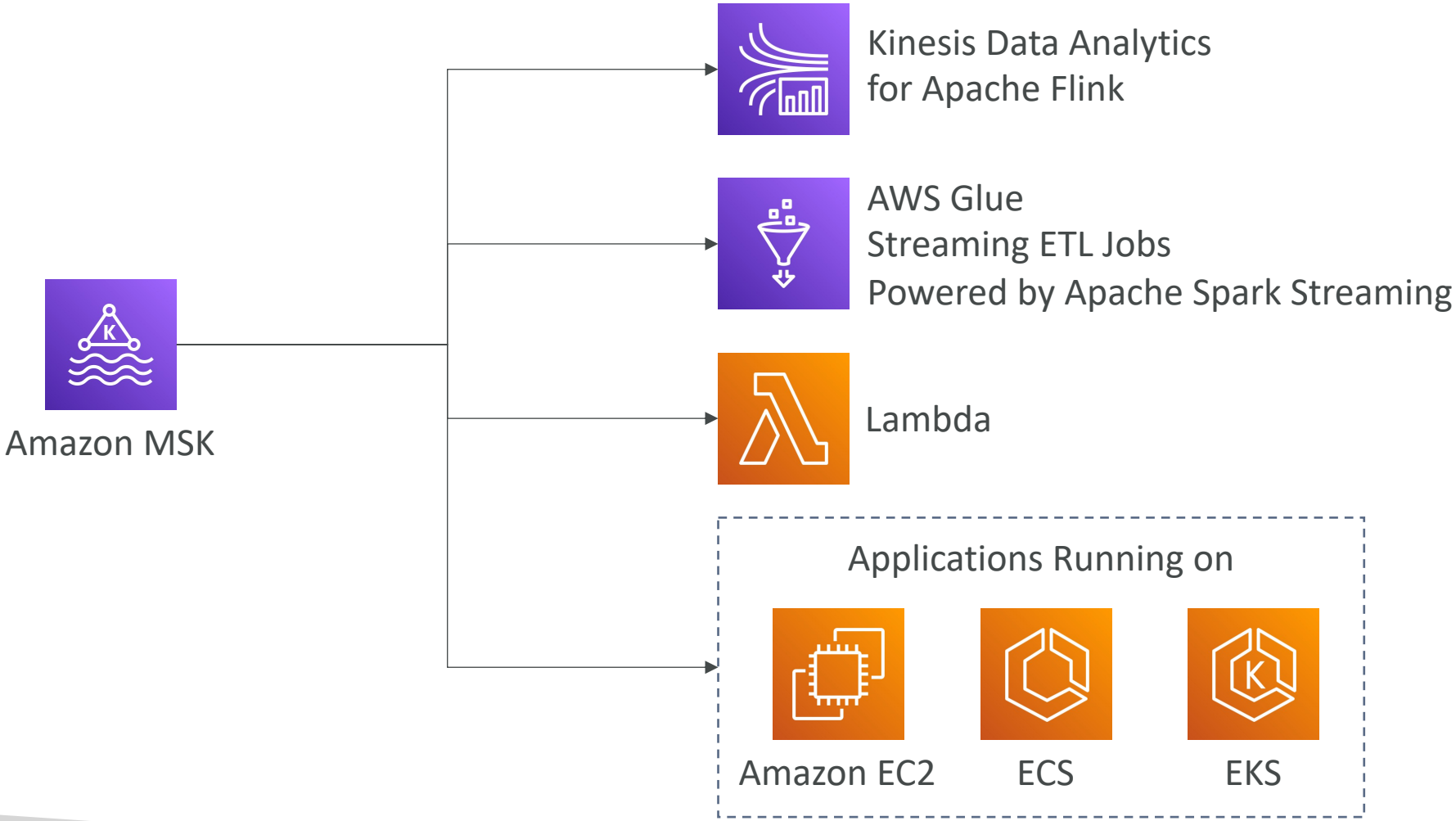
Amazon MSK

- 1MB default, configure for higher (ex: 10MB)
- Kafka Topics with Partitions
- Can only add partitions to a topic
- PLAINTEXT or TLS In-flight Encryption
- KMS at-rest encryption

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon MSK Consumers



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

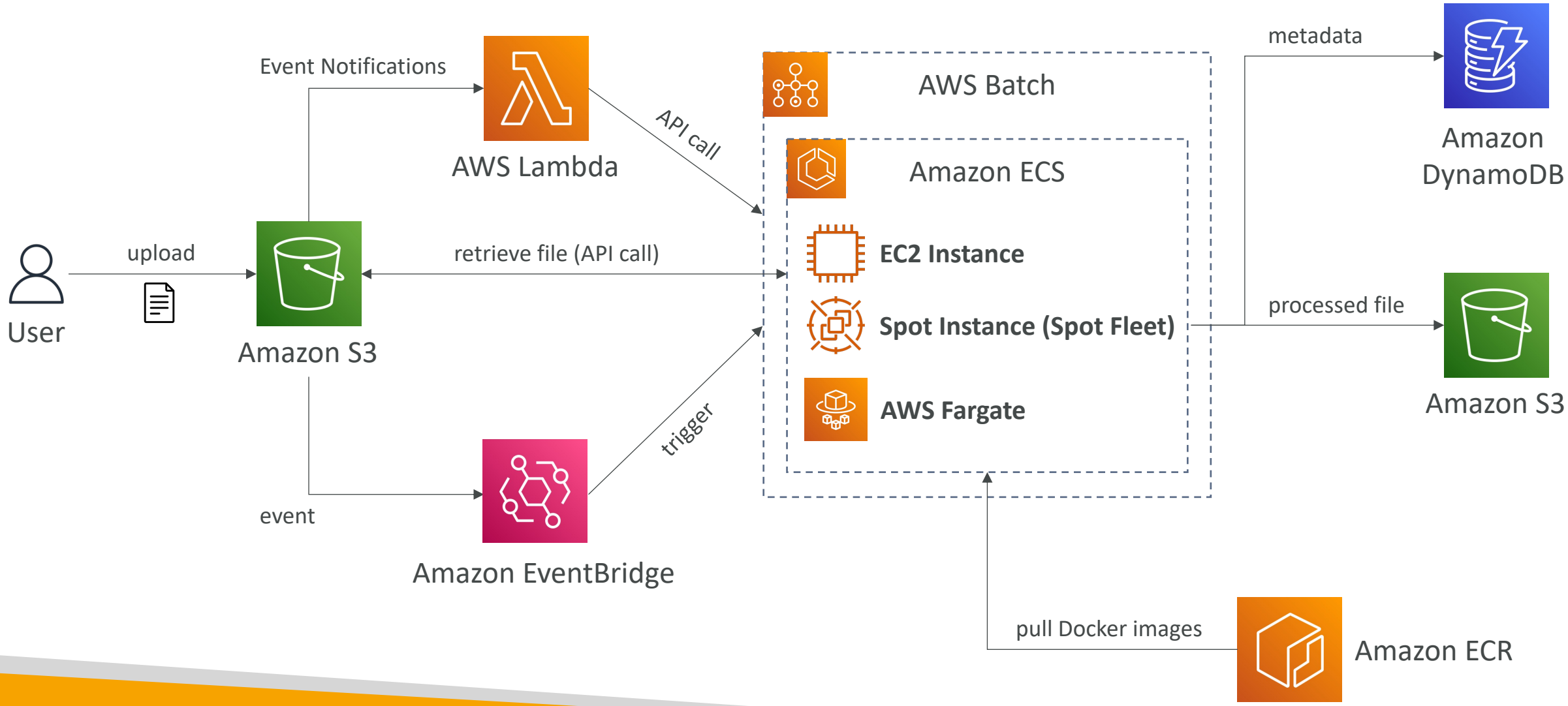
AWS Batch

- Run batch jobs as Docker images
- Two options:
  1. **Managed Compute Environment**
     - Dynamic provisioning of the instances (EC2 & **Spot Instances**) – in VPC
     - AWS Fargate (full serverless offering)
  2. **Unmanaged Compute Environment** (ECS & EKS)
- Optimal quantity and type based on volume and requirements
- You just pay for the underlying resources used
- Example: batch process of images, running thousands of concurrent jobs
- Schedule Batch Jobs using Amazon EventBridge
- Orchestrate Batch Jobs using AWS Step Functions

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Batch – Solution Architecture



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Batch vs. Lambda

- Lambda:
  - Time limit
  - Limited runtimes (built in runtimes, or Docker images built for Lambda)
  - Limited temporary disk space
  - Serverless

- Batch:
  - No time limit
  - Any runtime as long as it's packaged as a Docker image
  - Rely on EBS / instance store for disk space
  - Relies on EC2 (can be managed by AWS) or AWS Fargate

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Batch – Compute Environments

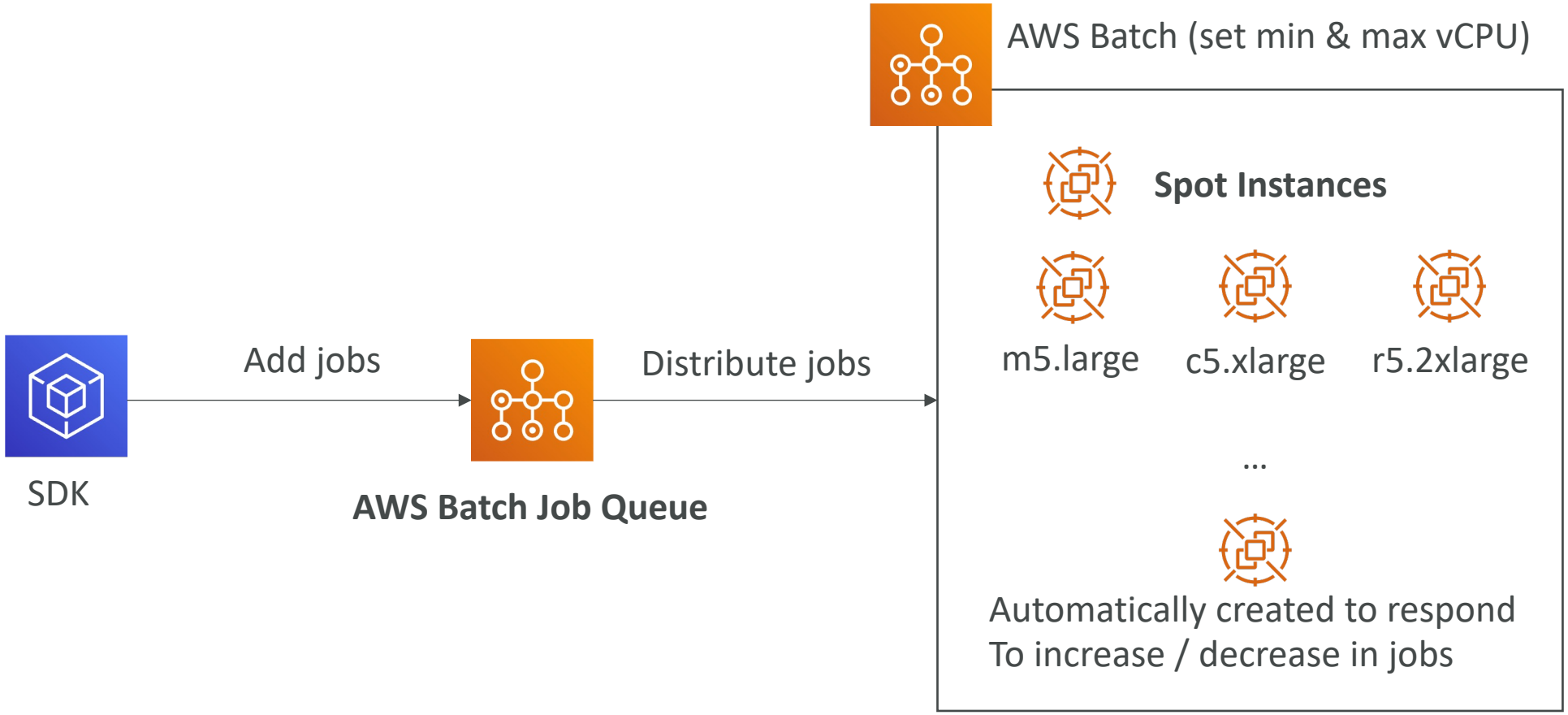
- Managed Compute Environment:
  - AWS Batch managed the capacity and instance types within the environment
  - **You can choose EC2 On-Demand or Spot Instances**
  - **You can choose Fargate On-Demand or Fargate Spot Instances**
  - You can set a maximum price for Spot Instances
  - Launched within your own VPC
    - If you launch within your own private subnet, make sure it has access to the ECS service
    - Either using a NAT gateway / instance or using VPC Endpoints for ECS

- Unmanaged Compute Environment (EC2, ECS, EKS)
  - You control and manage your compute resources, provisioning and scaling

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Batch – Managed Compute Environment

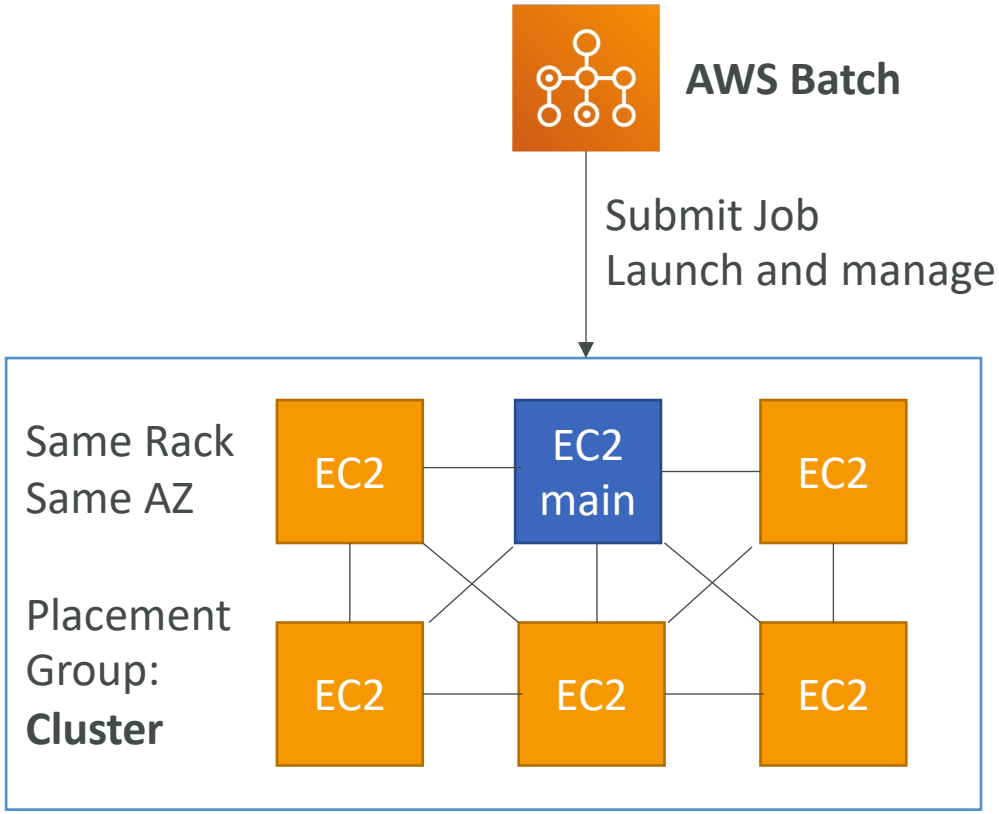


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Batch – Multi Node Mode

- **Multi Node:** large scale, good for HPC (high performance computing)
  - Leverages multiple EC2 / ECS instances at the same time
  - Good for tightly coupled workloads
  - Represents a single job, and specified how many nodes to create for the job
  - 1 main node, and many child node.
  - **Does not work with Spot Instances**
  - Works better if your EC2 launch mode is a placement group "cluster"



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

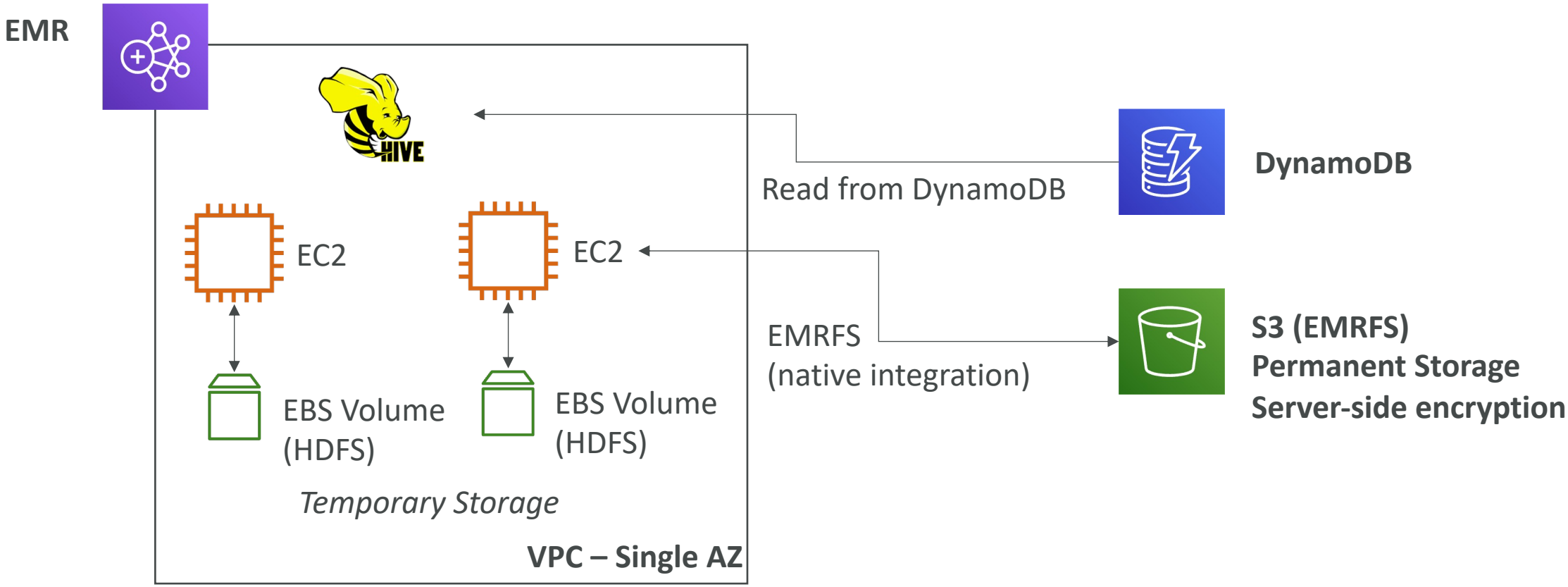
Amazon EMR

- EMR stands for "Elastic MapReduce"
- EMR helps creating **Hadoop clusters (Big Data)** to analyze and process vast amount of data
- The clusters can be made of hundreds of EC2 instances
- Also supports Apache Spark, HBase, Presto, Flink…
- EMR takes care of all the provisioning and configuration of EC2
- Auto-scaling with CloudWatch
- Use cases: data processing, machine learning, web indexing, big data…

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EMR – Integrations



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon EMR – Node types & purchasing

- **Master Node:** Manage the cluster, coordinate, manage health – long running
- **Core Node:** Run tasks and store data – long running
- **Task Node (optional):** Just to run tasks – usually Spot
- **Purchasing options:**
  - On-demand: reliable, predictable, won't be terminated
  - Reserved (min 1 year): cost savings (EMR will automatically use if available)
  - Spot Instances: cheaper, can be terminated, less reliable

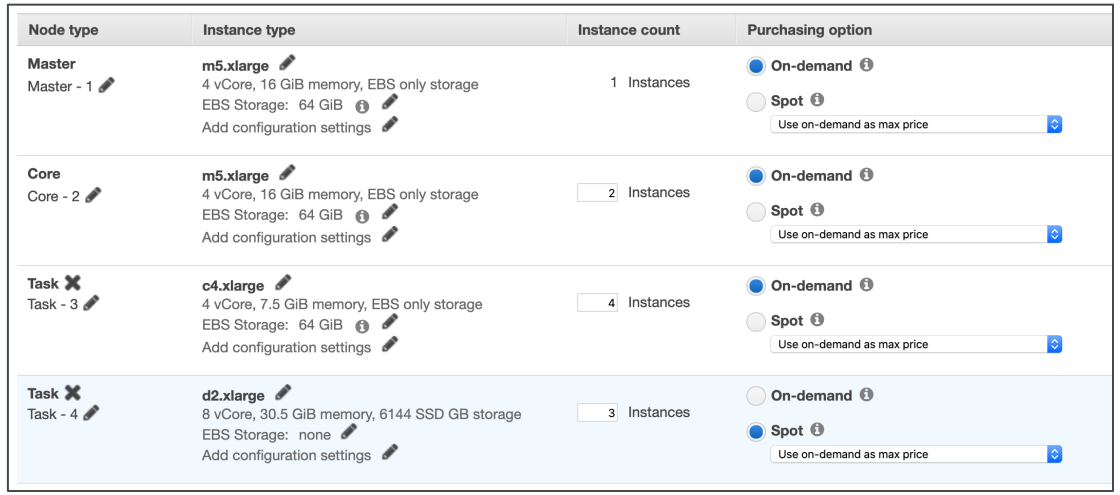
- Can have long-running cluster, or transient (temporary) cluster

© Stephane Maarek

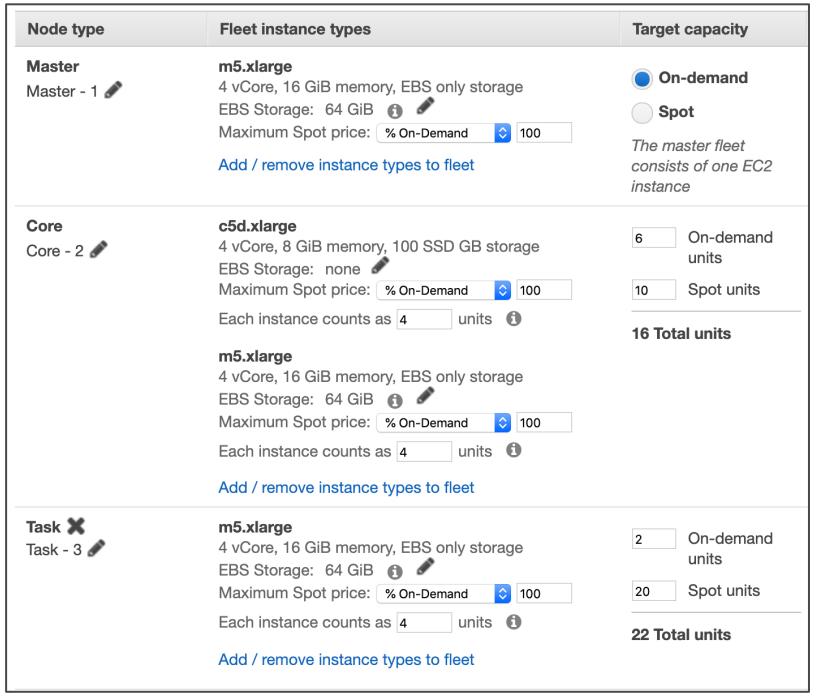
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon EMR – Instance Configuration

- **Uniform instance groups:** select a single instance type and purchasing option for each node



- **Instance fleet:** select target capacity, mix instance types and purchasing options



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Running Jobs on AWS

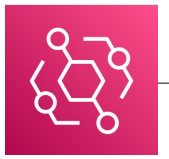
Strategy 1: Provision EC2 instance (long running - CRON jobs)

EC2

Strategy 3: Reactive Workflow

EventBridge
S3 Events
API Gateway
SQS, SNS
Etc…

Strategy 5: use Fargate



EventBridge

Fargate

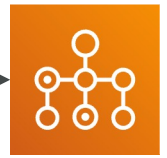
Strategy 2: Amazon EventBridge + Lambda (cron)

cron schedule

EventBridge

Strategy 4: use AWS Batch

EventBridge



Batch

Strategy 6: Use EMR (step execution or cluster)

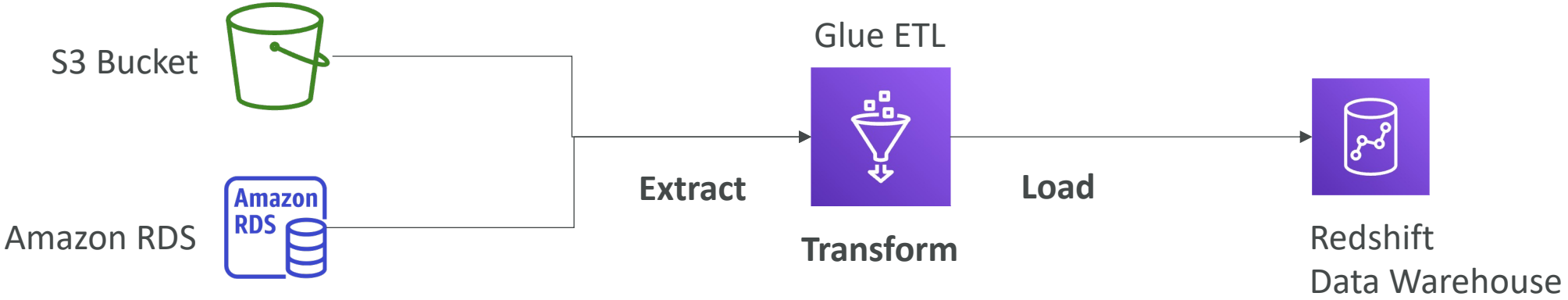
EMR

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Glue

- Managed **extract, transform, and load (ETL)** service
- Useful to prepare and transform data for analytics
- Fully **serverless** service

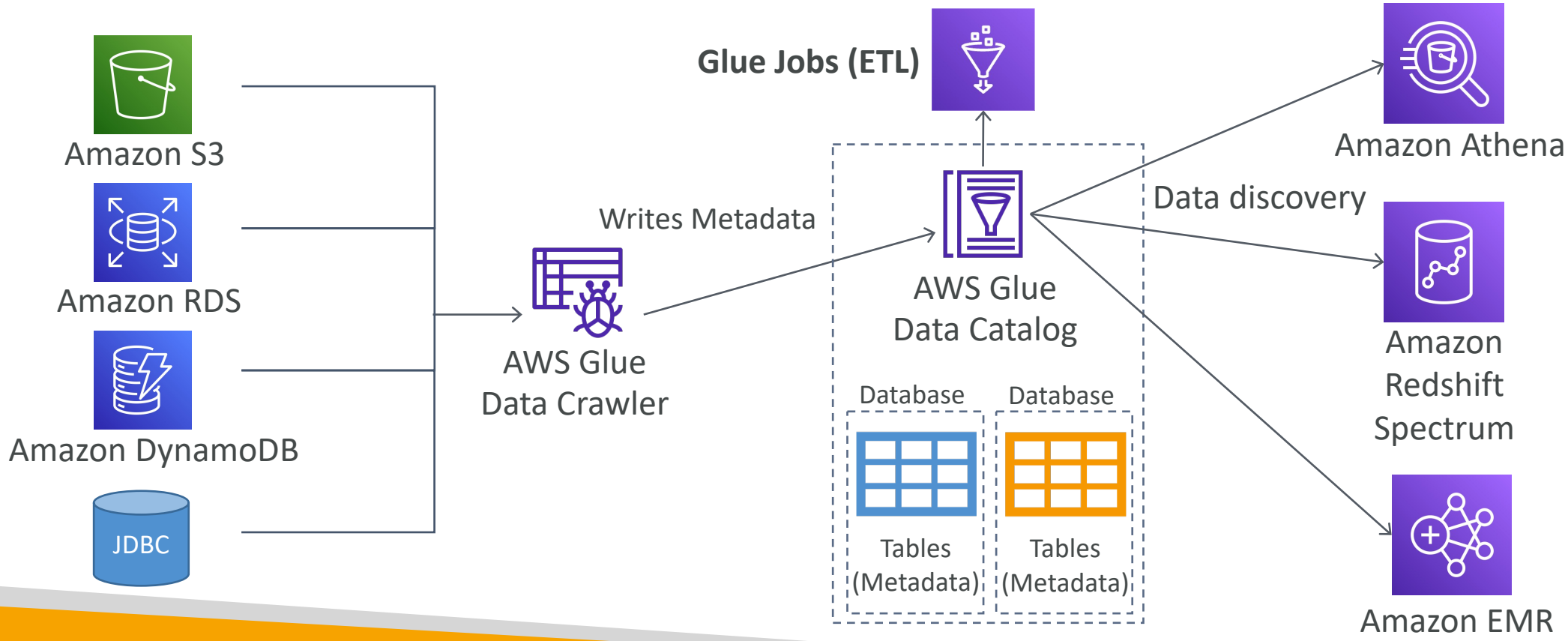


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Glue Data Catalog

- Glue Data Catalog: catalog of datasets



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Redshift Overview

- Redshift is based on PostgreSQL, but **it's not used for OLTP**
- **It's OLAP – online analytical processing (analytics and data warehousing)**
- 10x better performance than other data warehouses, scale to PBs of data
- **Columnar** storage of data (instead of row based)
- Massively Parallel Query Execution (MPP)
- Two modes: Provisioned cluster or Serverless cluster
- Has a SQL interface for performing the queries
- BI tools such as AWS Quicksight or Tableau integrate with it

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Redshift Continued…

- Data is loaded from **S3, Kinesis Firehose, DynamoDB, DMS…**
- Based on node type: up to 100+ nodes, up to 16 TB of space per node
- Can provision multiple nodes, with **Multi-AZ only for some clusters**
- Leader node: for query planning, results aggregation
- Compute node: for performing the queries, send results to leader
- Backup & Restore, Security VPC / IAM / KMS, Monitoring
- Redshift Enhanced VPC Routing: COPY / UNLOAD goes through VPC
- **Redshift is provisioned, so it's worth it when you have a sustained usage (use Athena if the queries are sporadic instead)**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Redshift – Snapshots & DR

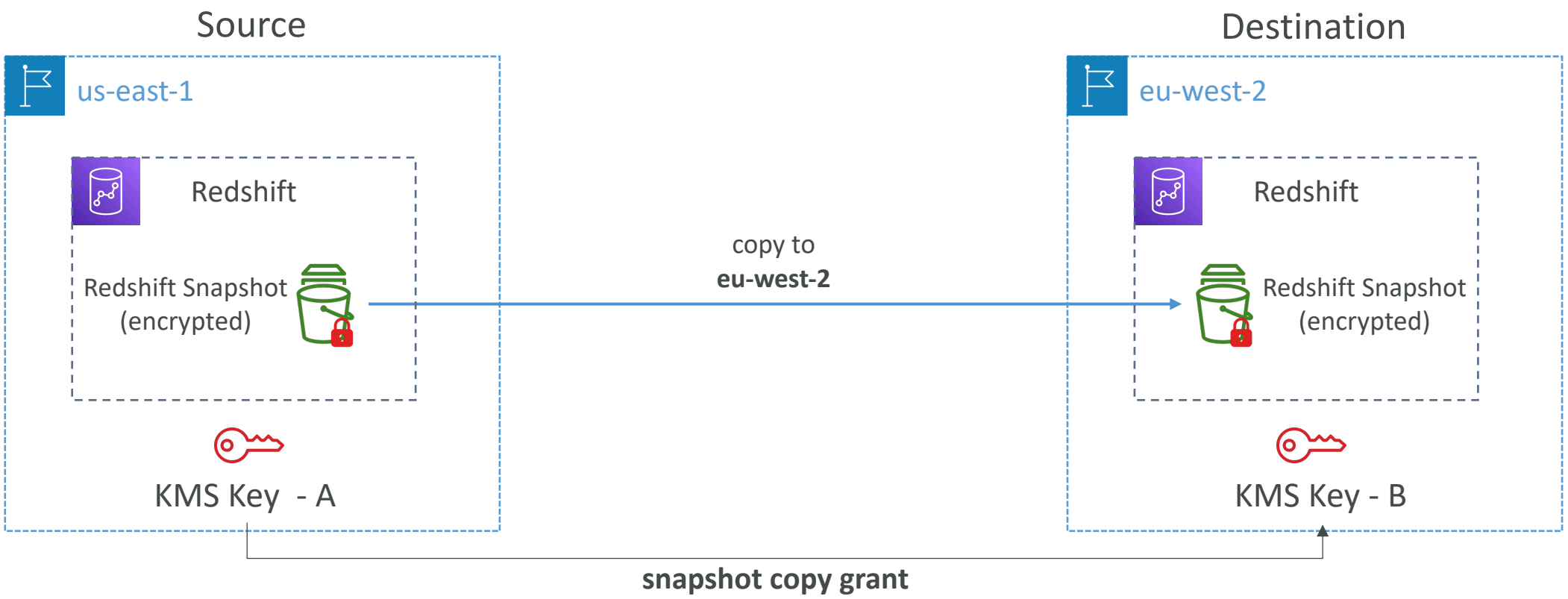
- Snapshots are point-in-time backups of a cluster, stored internally in S3
- Snapshots are incremental (only what has changed is saved)
- You can restore a snapshot into a **new cluster**
- **Automated: every 8 hours, every 5 GB, or on a schedule. Set retention**
- Manual: snapshot is retained until you delete it

- You can configure Amazon Redshift to automatically copy snapshots (automated or manual) of a cluster to another AWS Region

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cross-Region Snapshot Copy for an KMS-Encrypted Redshift



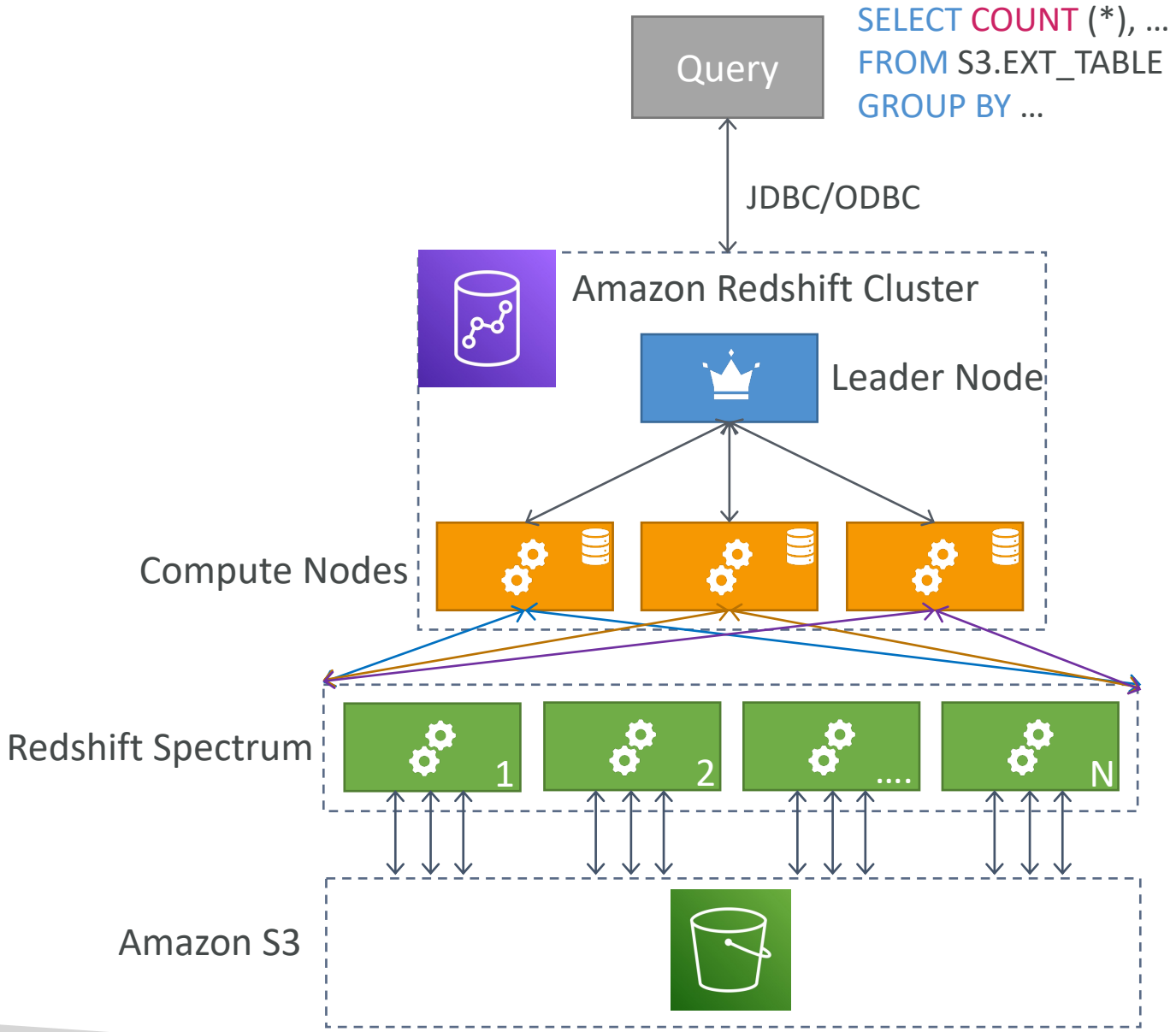
Enables Redshift to perform encryption operations in the destination Region

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Redshift Spectrum

- Query data that is already in S3 without loading it
- **Must have a Redshift cluster available to start the query**
- The query is then submitted to thousands of Redshift Spectrum nodes

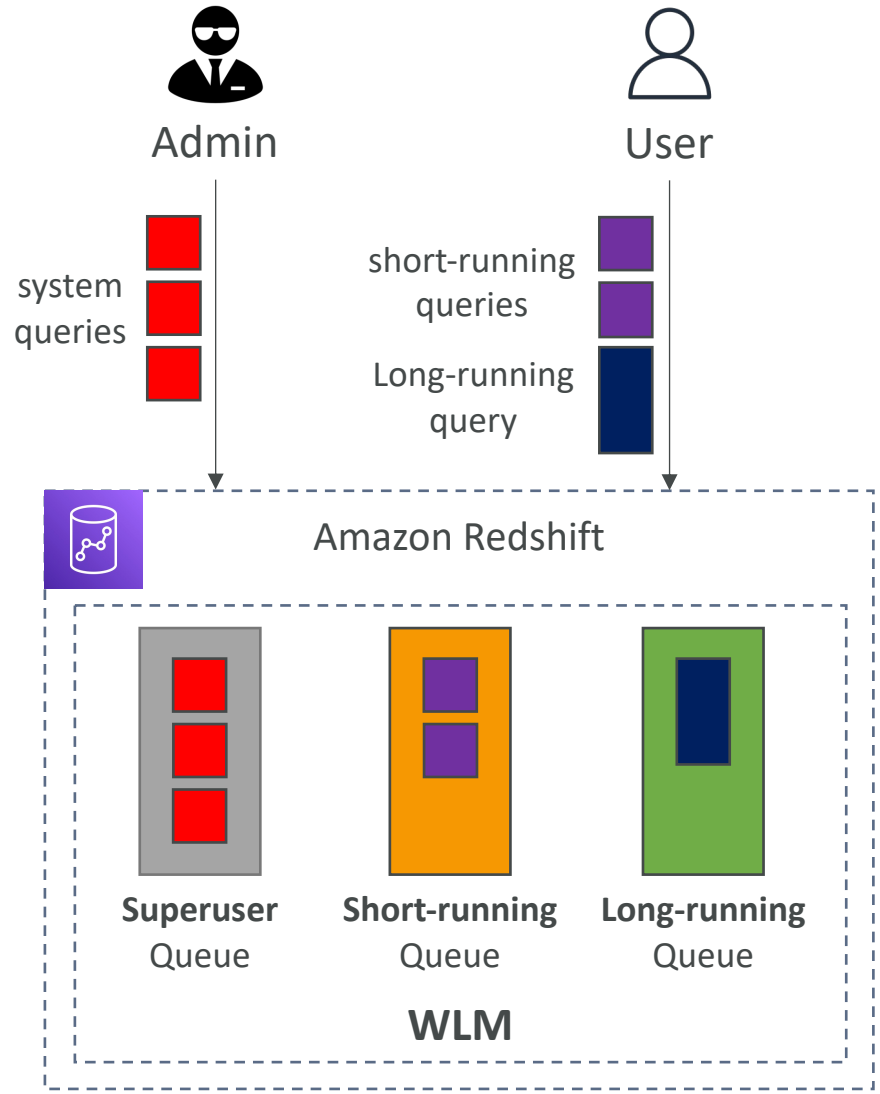


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Redshift Workload Management (WLM)

- Enables you to flexibly manage queries' priorities within workloads
- Example: prevent short, fast-running queries from getting stuck behind long-running queries
- Define multiple query queues (Superuser queue, User-defined queues)
- Route queries to the appropriate queues at runtime
- **Automatic WLM** – queues and resources managed by Redshift
- **Manual WLM** – queues and resources managed by you

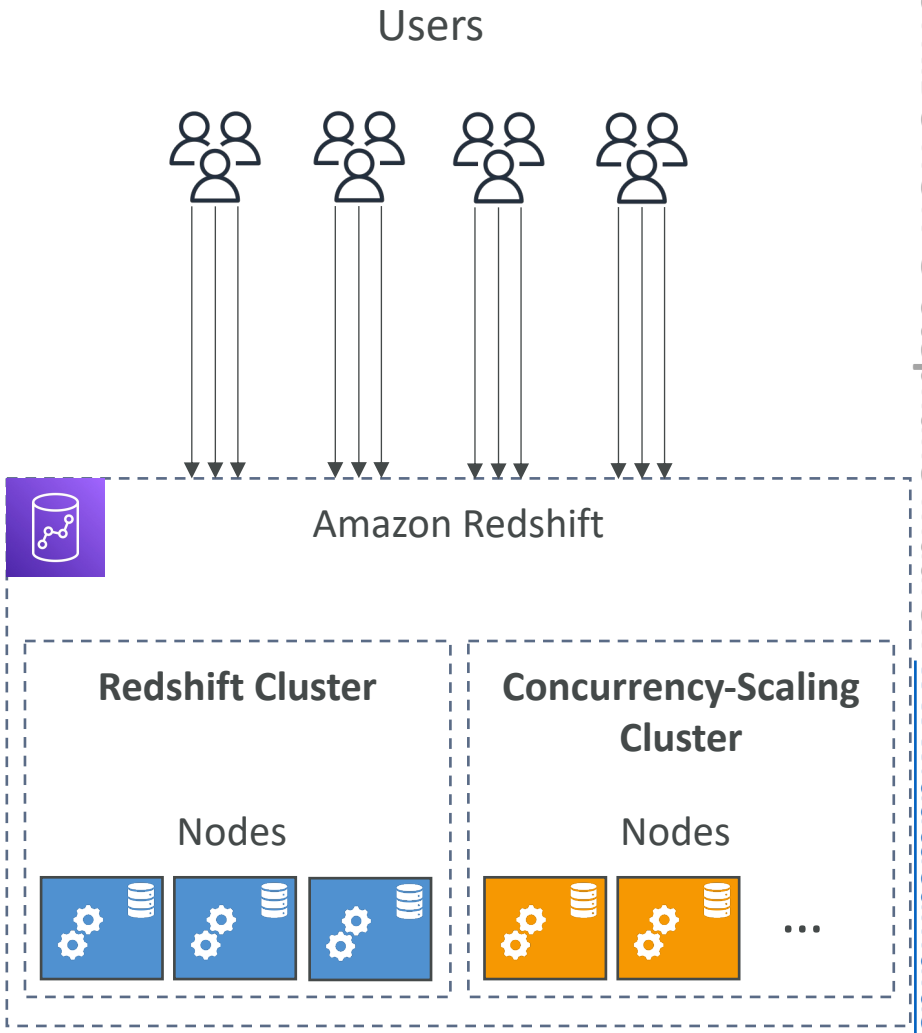


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Redshift Concurrency Scaling

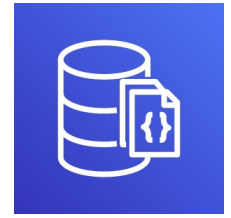
- Enables you to provide consistently fast performance with virtually unlimited concurrent users and queries
- Redshift automatically adds additional cluster capacity (**Concurrency-Scaling Cluster**) to process an increase in requests
- Ability to decide which queries sent to the concurrency-Scaling Cluster using WLM
- Charged per second



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

DocumentDB



- Aurora is an "AWS-implementation" of PostgreSQL / MySQL …
- **DocumentDB is the same for MongoDB (which is a NoSQL database)**
- MongoDB is used to store, query, and index JSON data

- Similar "deployment concepts" as Aurora
- Fully Managed, highly available with replication across 3 AZ
- DocumentDB storage automatically grows in increments of 10GB

- Automatically scales to workloads with millions of requests per seconds

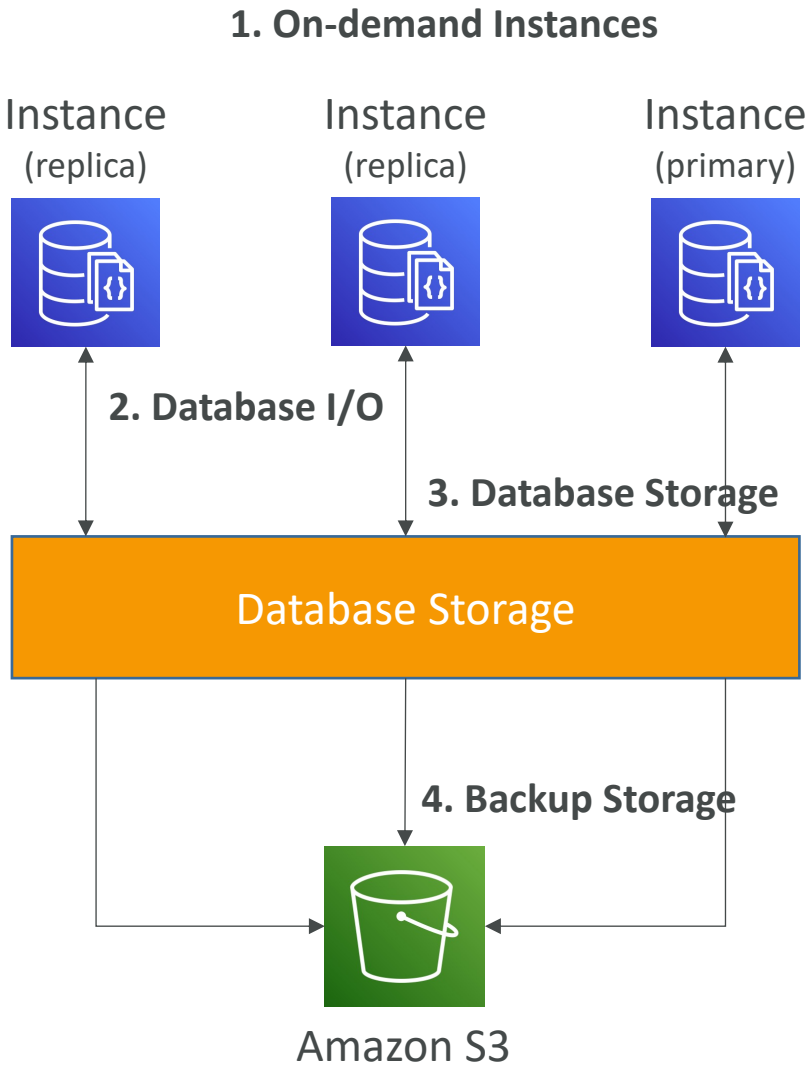
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DocumentDB – Pricing

- Pay for what you use, no upfront costs

- **On-demand Instances** (per second with minimum of 10 minutes)
- **Database I/O** – amount of I/O used when read and write (per million I/Os)
- **Database Storage** (per GB/month)
- **Backup Storage** (per GB/month)

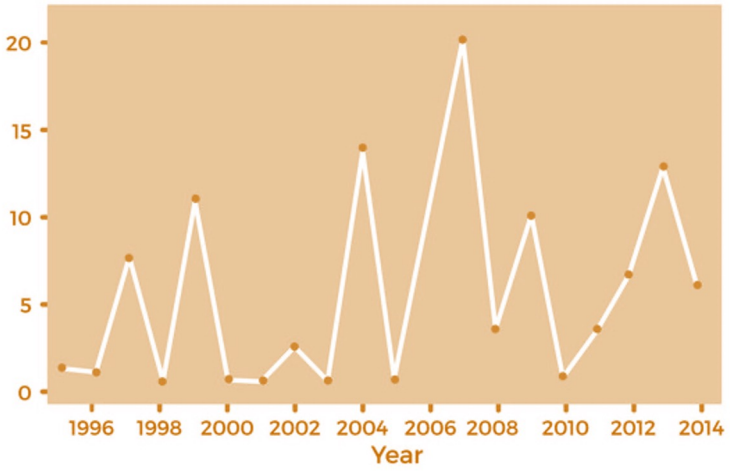


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Timestream

- Fully managed, fast, scalable, serverless **time series database**
- Automatically scales up/down to adjust capacity
- Store and analyze trillions of events per day
- 1000s times faster & 1/10th the cost of relational databases
- Scheduled queries, multi-measure records, SQL compatibility
- Data storage tiering: recent data kept in memory and historical data kept in a cost-optimized storage
- Built-in time series analytics functions (helps you identify patterns in your data in near real-time)
- Encryption in transit and at rest

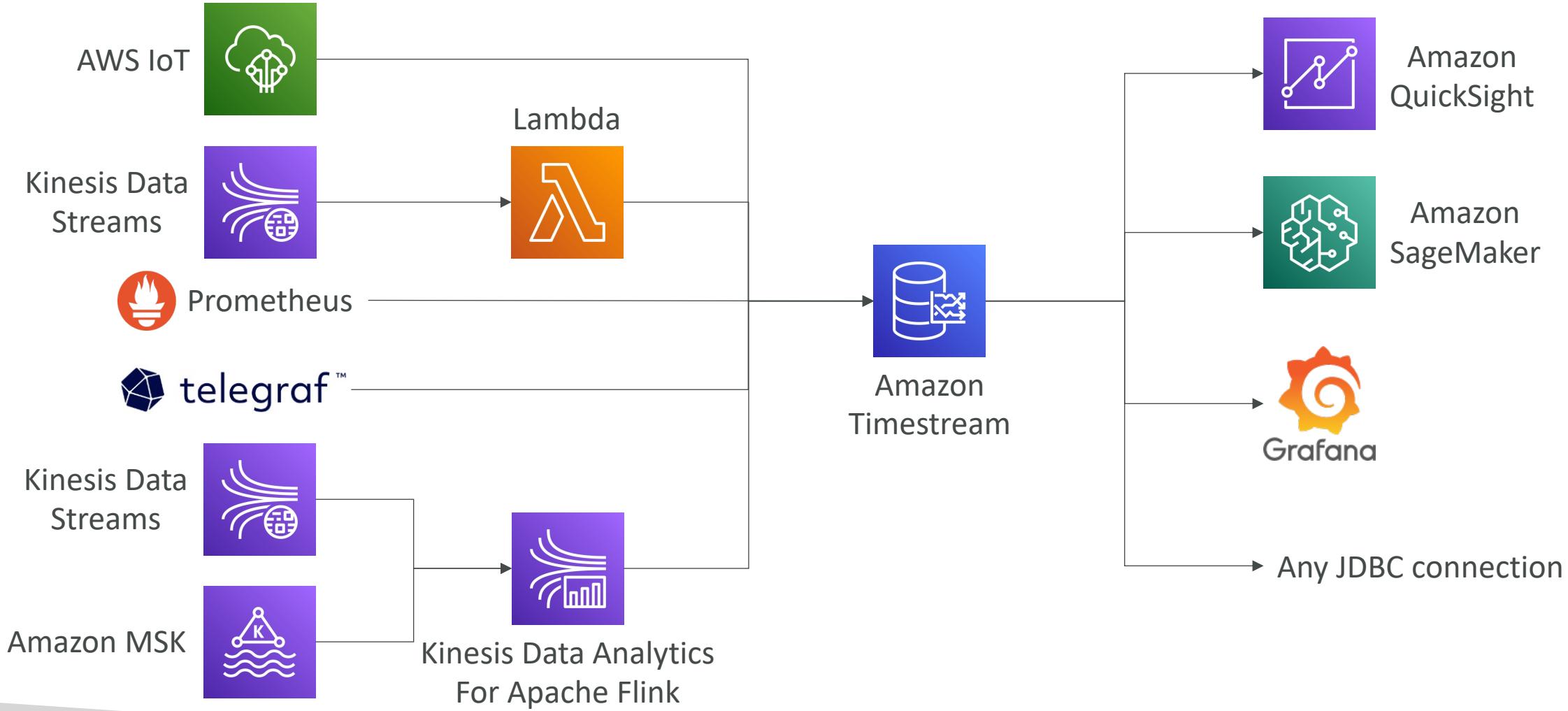


- Use cases: IoT apps, operational applications, real-time analytics, …

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Timestream – Architecture



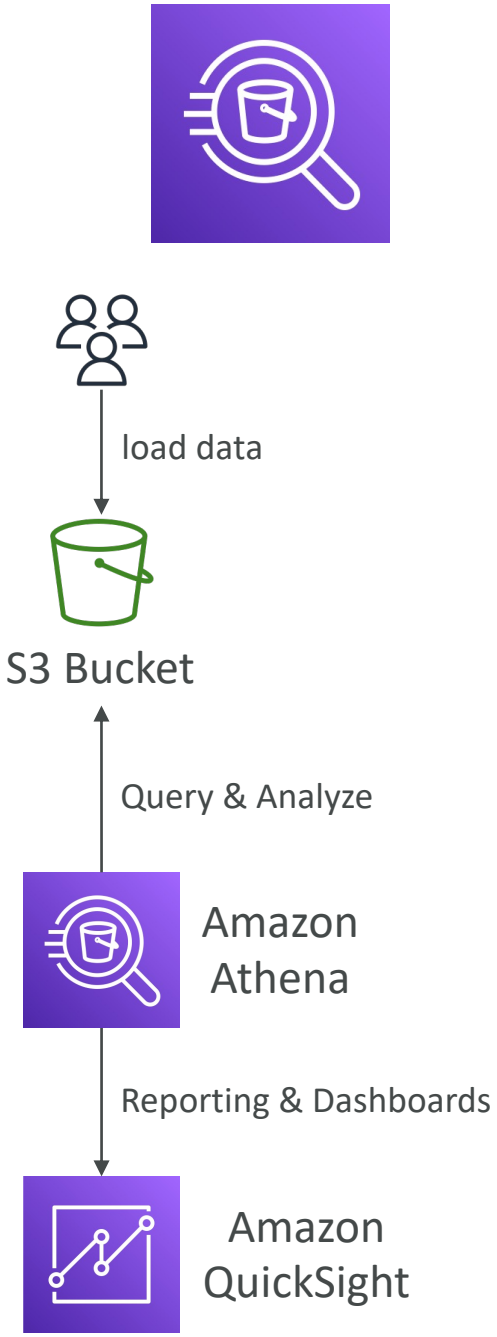
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Athena

- **Serverless** query service to analyze data stored in Amazon S3
- Uses standard SQL language to query the files (built on Presto)
- Supports CSV, JSON, ORC, Avro, and Parquet
- Pricing: $5.00 per TB of data scanned
- Commonly used with Amazon Quicksight for reporting/dashboards

- **Use cases:** Business intelligence / analytics / reporting, analyze & query VPC Flow Logs, ELB Logs, **CloudTrail trails**, etc...
- **Exam Tip:** analyze data in S3 using serverless SQL, use Athena



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Athena – Performance Improvement

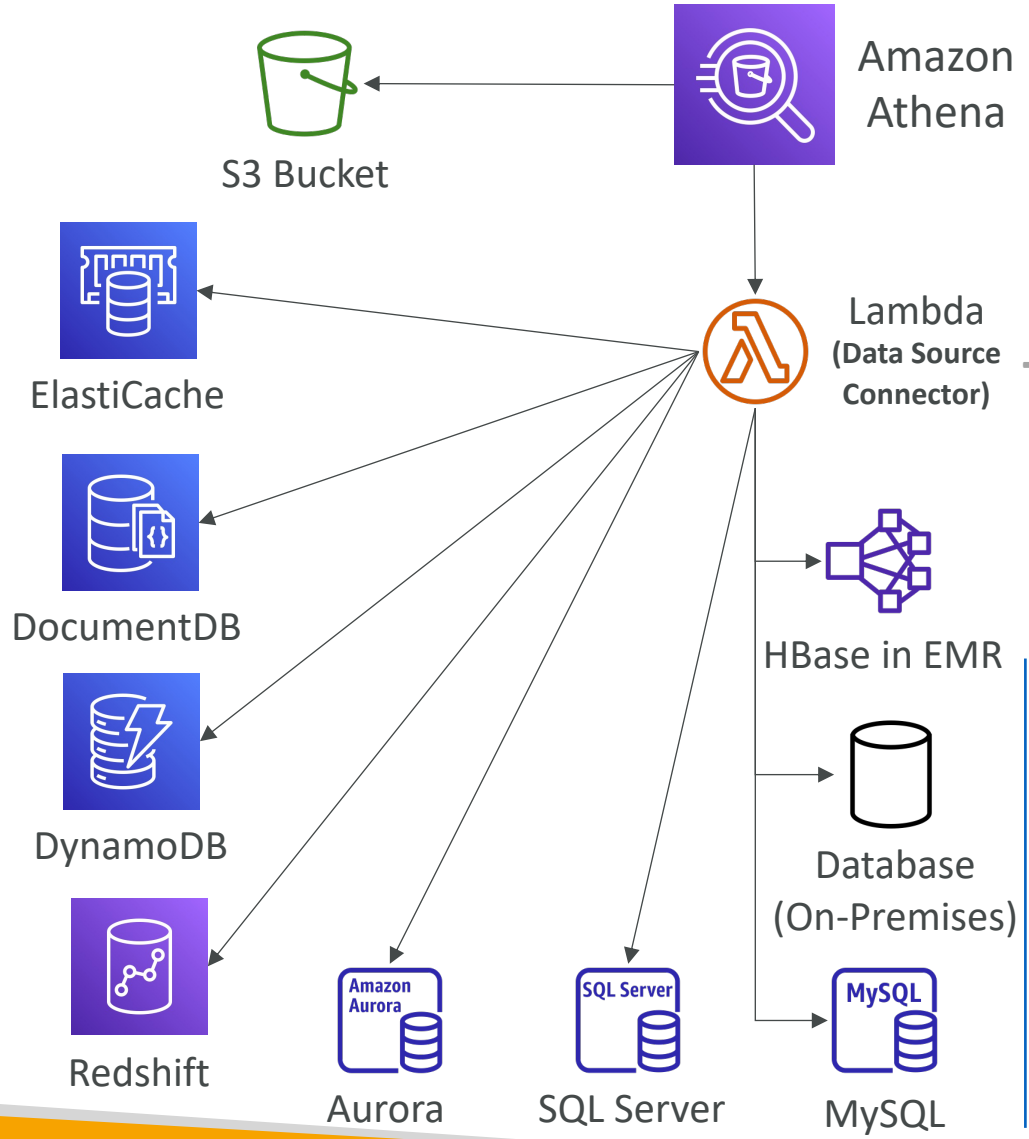
- Use **columnar data** for cost-savings (less scan)
  - Apache Parquet or ORC is recommended
  - Huge performance improvement
  - Use Glue to convert your data to Parquet or ORC
- **Compress data** for smaller retrievals (bzip2, gzip, lz4, snappy, zlip, zstd…)
- **Partition** datasets in S3 for easy querying on virtual columns
  - s3://yourBucket/pathToTable
/<PARTITION_COLUMN_NAME>=<VALUE>
/<PARTITION_COLUMN_NAME>=<VALUE>
/<PARTITION_COLUMN_NAME>=<VALUE>
/etc…
  - Example: s3://athena-examples/flight/parquet/year=1991/month=1/day=1/
- **Use larger files** (> 128 MB) to minimize overhead

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Athena – Federated Query

- Allows you to run SQL queries across data stored in relational, non-relational, object, and custom data sources (AWS or on-premises)
- Uses Data Source Connectors that run on AWS Lambda to run Federated Queries (e.g., CloudWatch Logs, DynamoDB, RDS, …)
- Store the results back in Amazon S3

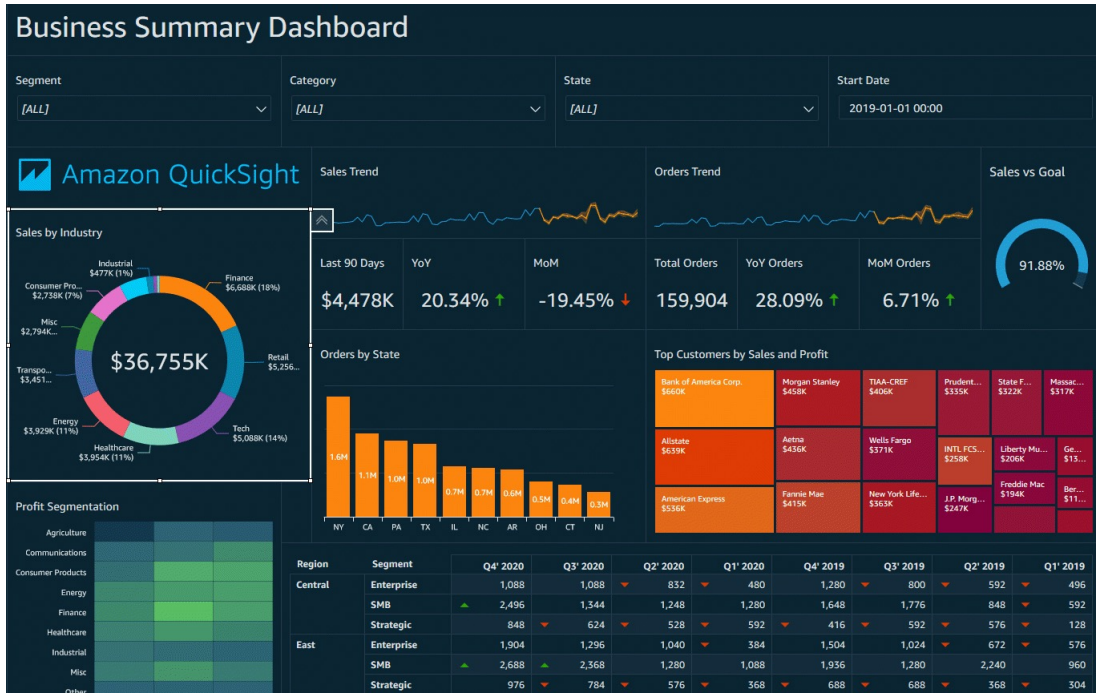


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Amazon QuickSight

- **Serverless machine learning-powered business intelligence service to create interactive dashboards**
- Fast, automatically scalable, embeddable, with per-session pricing
- Use cases:
  - Business analytics
  - Building visualizations
  - Perform ad-hoc analysis
  - Get business insights using data
- Integrated with RDS, Aurora, Athena, Redshift, S3…
- **In-memory computation using SPICE** engine if data is imported into QuickSight
- Enterprise edition: Possibility to setup **Column-Level security (CLS)**

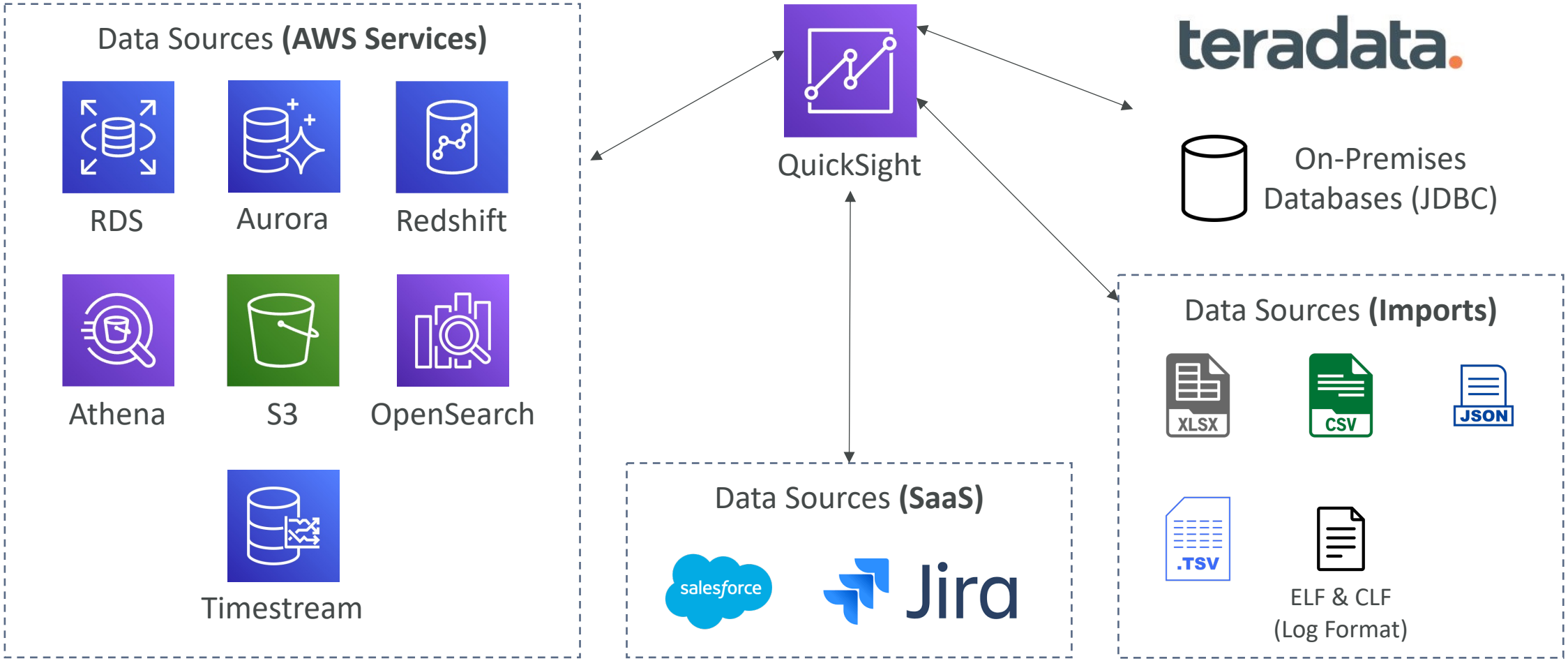


https://aws.amazon.com/quicksight/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

QuickSight Integrations



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

QuickSight – Dashboard & Analysis

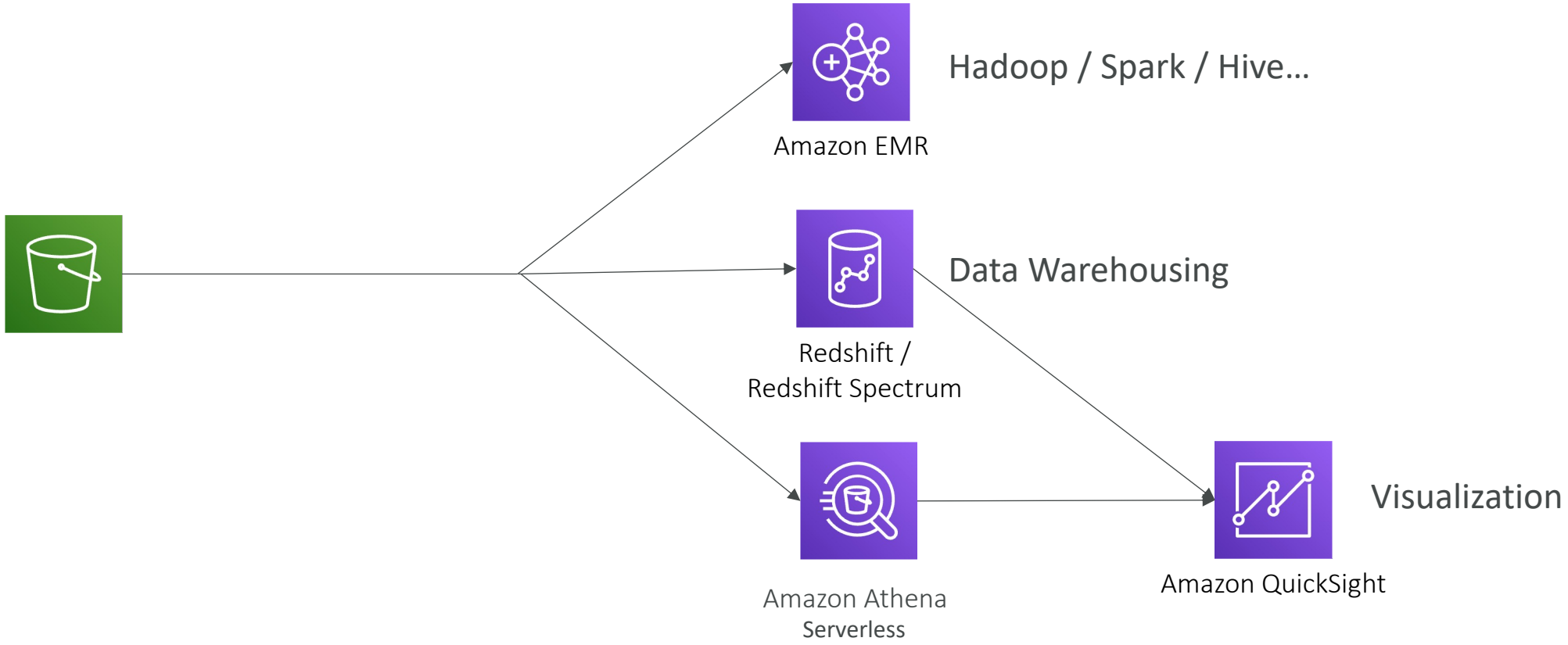
- Define Users (standard versions) and Groups (enterprise version)
  - These users & groups only exist within QuickSight, not IAM !!
- A *dashboard*…
  - is a read-only snapshot of an analysis that you can share
  - preserves the configuration of the analysis (filtering, parameters, controls, sort)

- **You can share the analysis or the dashboard with Users or Groups**
- To share a dashboard, you must first publish it
- Users who see the dashboard can also see the underlying data

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

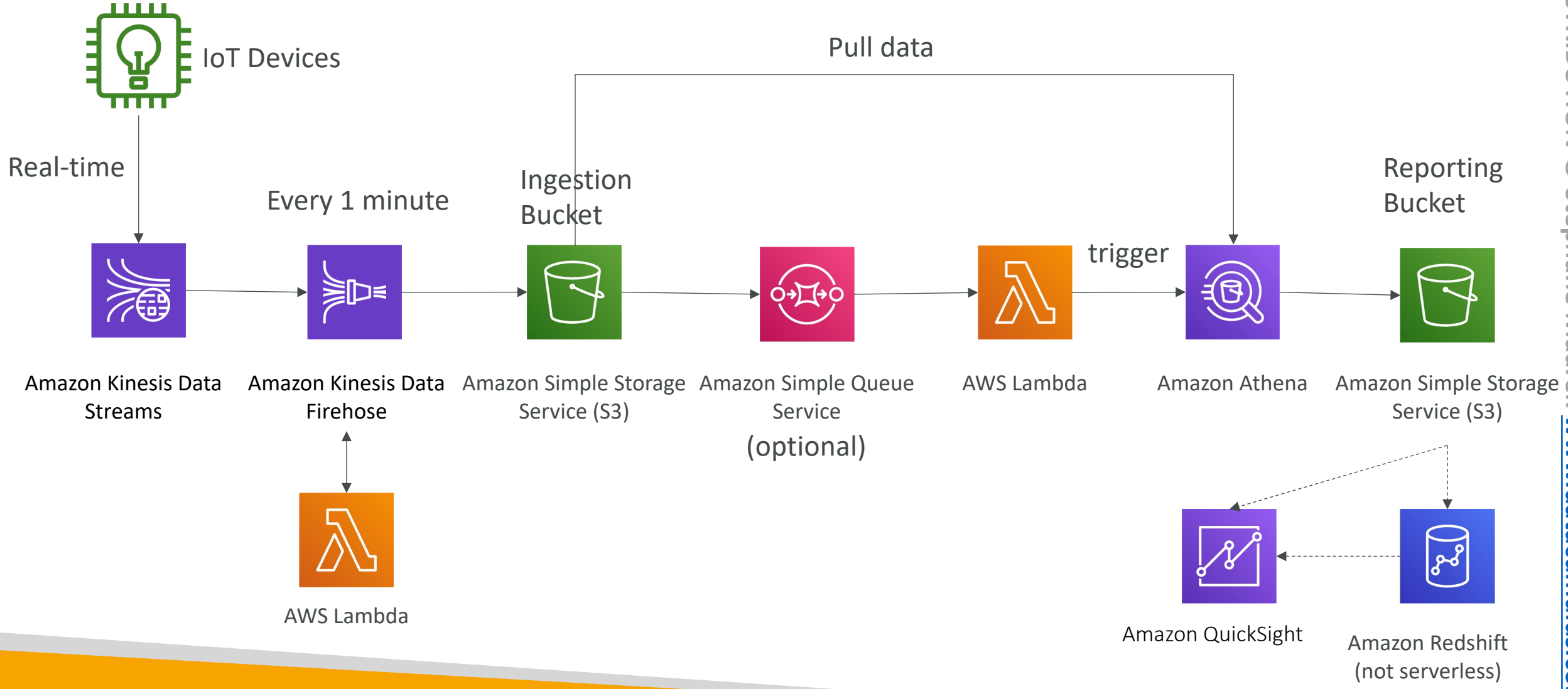
Full Data Engineering Pipeline
Analytics layer



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Big Data Ingestion Pipeline



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Comparison of warehousing technologies

- EMR
  - Need to use Big Data tools such as Apache Hive, Spark
  - One long-running cluster, many jobs, with auto-scaling, or one cluster per job?
  - Purchasing options – Spot, On Demand, Reserved Instances
  - Can access data in DynamoDB and / or S3
  - Scratch data on EBS disks (HDFS) and long term storage in S3 (EMRFS)
- Athena
  - Simple queries and aggregations, data must live in S3
  - Serverless, simple SQL queries, out-of-the-box queries for many services (cost & billing..)
  - Audit queries through CloudTrail
- Redshift
  - Advanced SQL queries, must provision servers
  - Can leverage Redshift Spectrum for serverless queries on S3

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Monitoring

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

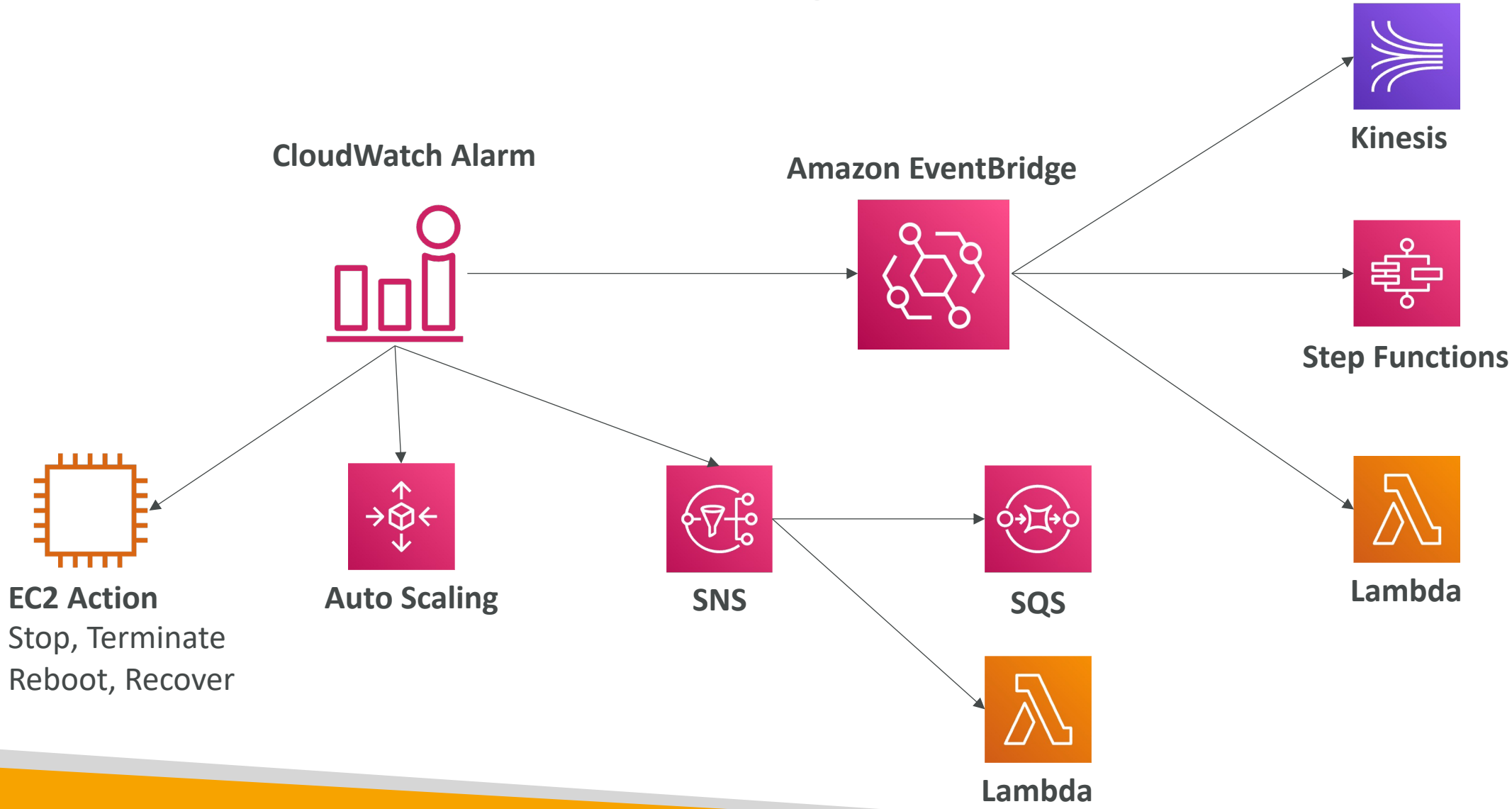
CloudWatch

- CloudWatch Metrics
  - Provided by many AWS services
  - EC2 standard: 5 minutes, detailed monitoring: 1 minute
  - EC2 RAM is not a built-in metric
  - Can create custom metrics: standard resolution 1 minute, high resolution 1 sec
- CloudWatch Alarms
  - Can trigger actions: EC2 action (reboot, stop, terminate, recover), Auto Scaling, SNS
  - Alarm events can be intercepted by Amazon EventBridge
- CloudWatch Dashboards
  - Display metrics and alarms
  - Can show metrics of multiple regions

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudWatch Alarms integrations

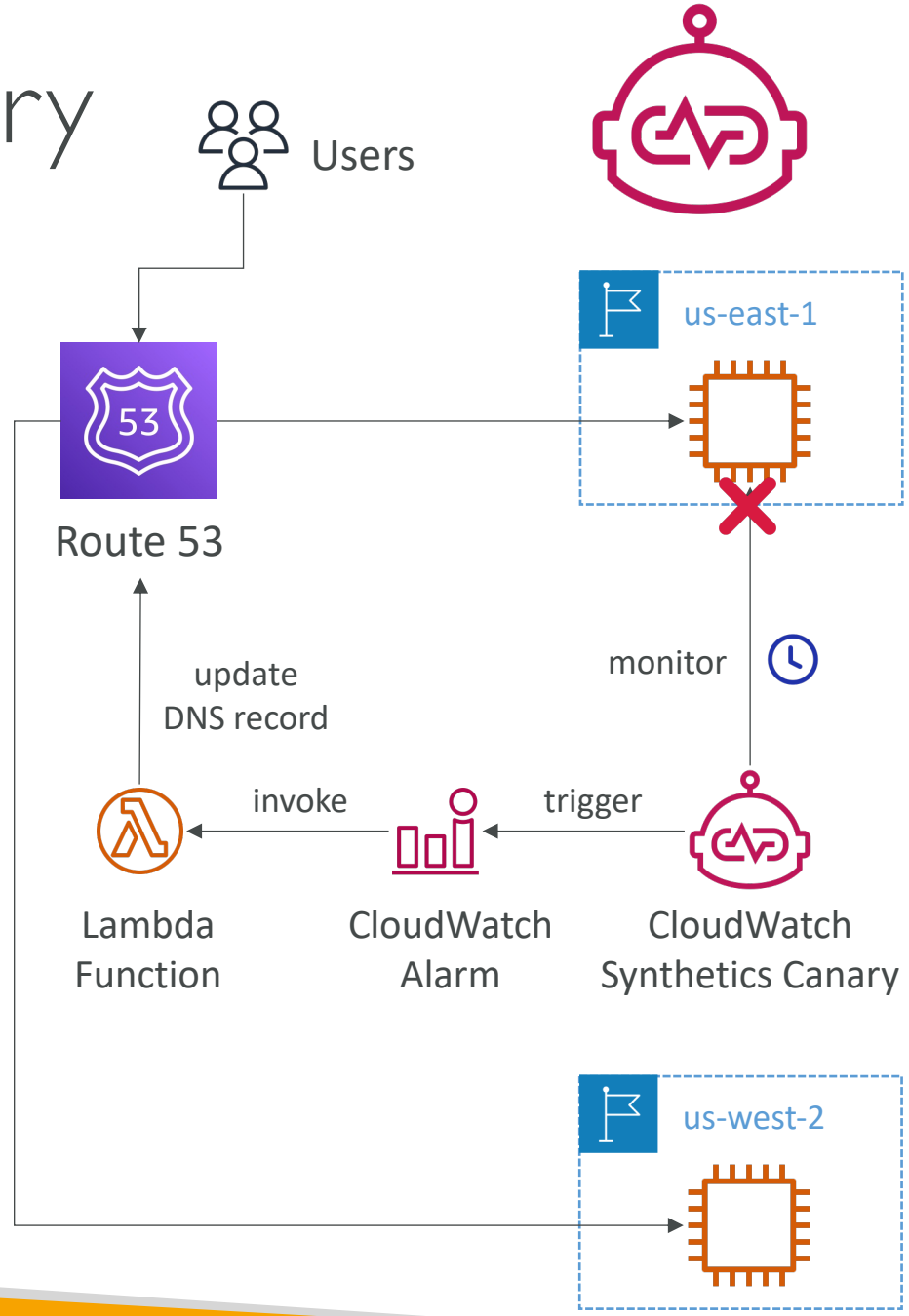


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudWatch Synthetics Canary

- Configurable script that monitor your APIs, URLs, Websites, …
- Reproduce what your customers do programmatically to find issues before customers are impacted
- Checks the availability and latency of your endpoints and can store load time data and screenshots of the UI
- Integration with CloudWatch Alarms
- Scripts written in Node.js or Python
- Programmatic access to a headless Google Chrome browser
- Can run once or on a regular schedule



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudWatch Synthetics Canary Blueprints

- **Heartbeat Monitor** – load URL, store screenshot and an HTTP archive file
- **API Canary** – test basic read and write functions of REST APIs
- **Broken Link Checker** – check all links inside the URL that you are testing
- **Visual Monitoring** – compare a screenshot taken during a canary run with a baseline screenshot
- **Canary Recorder** – used with CloudWatch Synthetics Recorder (record your actions on a website and automatically generates a script for that)
- **GUI Workflow Builder** – verifies that actions can be taken on your webpage (e.g., test a webpage with a login form)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS CloudWatch Logs - Sources

- SDK, CloudWatch Logs Agent, CloudWatch Unified Agent
- Elastic Beanstalk: collection of logs from application
- ECS: collection from containers
- AWS Lambda: collection from function logs
- VPC Flow Logs: VPC specific logs
- API Gateway
- CloudTrail based on filter
- CloudWatch log agents: for example on EC2 machines
- Route53: Log DNS queries

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudWatch Logs

- **Log groups:** arbitrary name, usually representing an application
- **Log stream:** instances within application / log files / containers
- Can define log expiration policies (never expire, 30 days, etc..)
- Optional KMS encryption
- **CloudWatch Logs can send logs to:**
  - Amazon S3 (exports)
  - Kinesis Data Streams
  - Kinesis Data Firehose
  - AWS Lambda
  - ElasticSearch

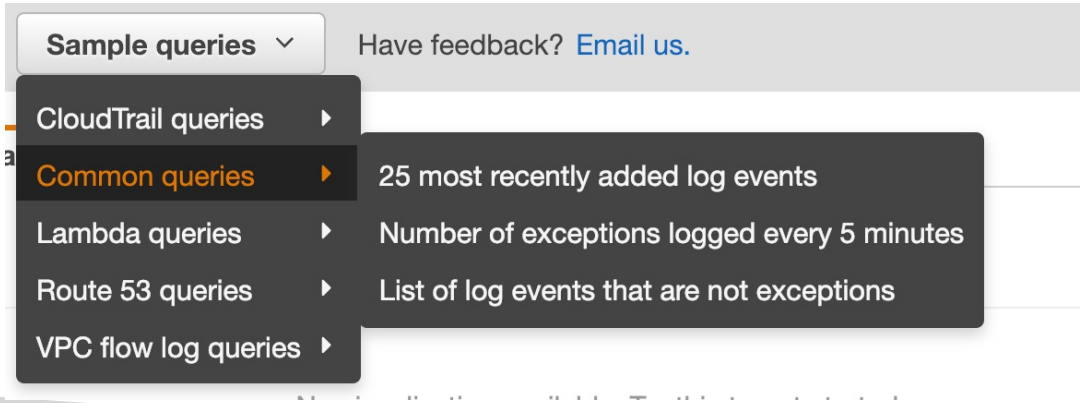
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudWatch Logs Metric Filter & Insights

- CloudWatch Logs can use filter expressions
  - For example, find a specific IP inside of a log
  - Or count occurrences of "ERROR" in your logs
  - Metric filters can be used to trigger alarms

- CloudWatch Logs Insights can be used to query logs and add queries to CloudWatch Dashboards



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudWatch Logs – S3 Export

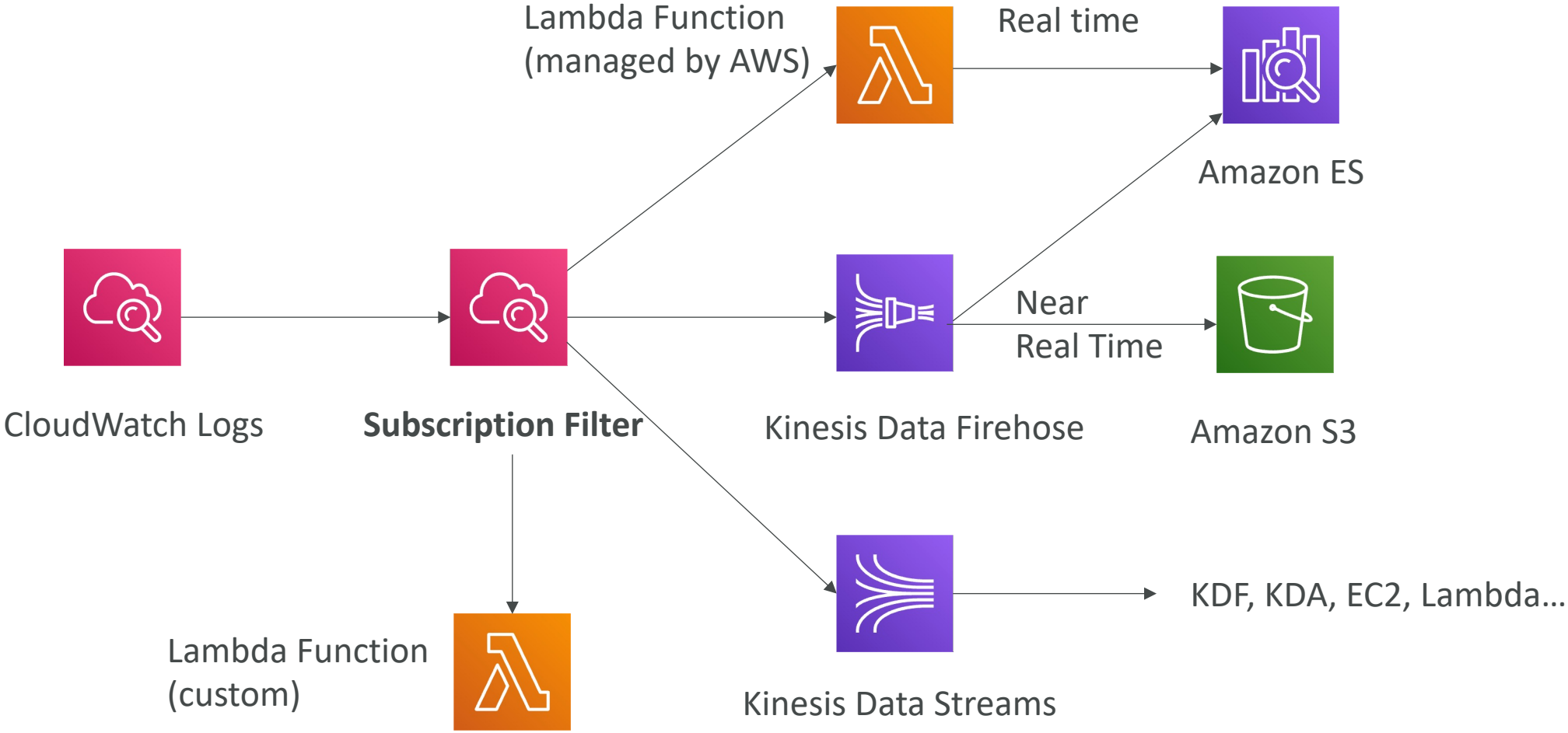
- S3 buckets must be encrypted with AES-256 (SSE-S3) or SSE-KMS
- Log data can take **up to 12 hours** to become available for export
- The API call is **CreateExportTask**

- Not near-real time or real-time… use Logs Subscriptions instead

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

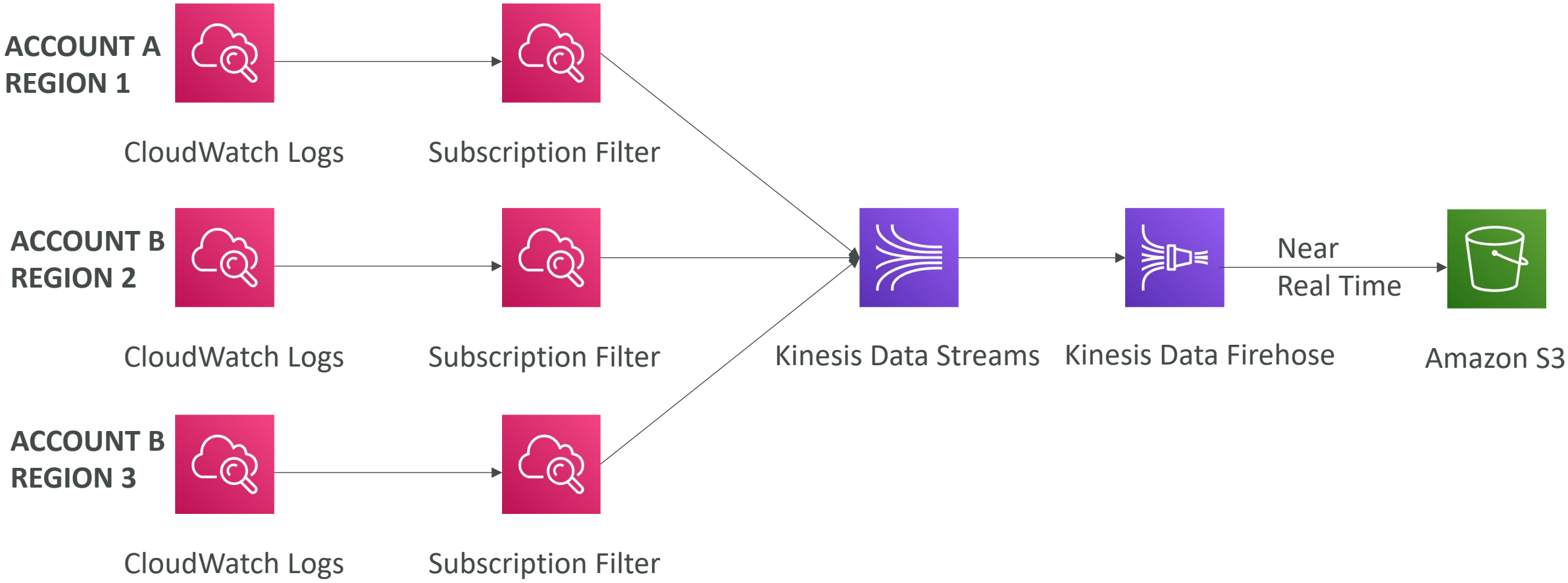
CloudWatch Logs Subscriptions



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudWatch Logs Aggregation Multi-Account & Multi Region



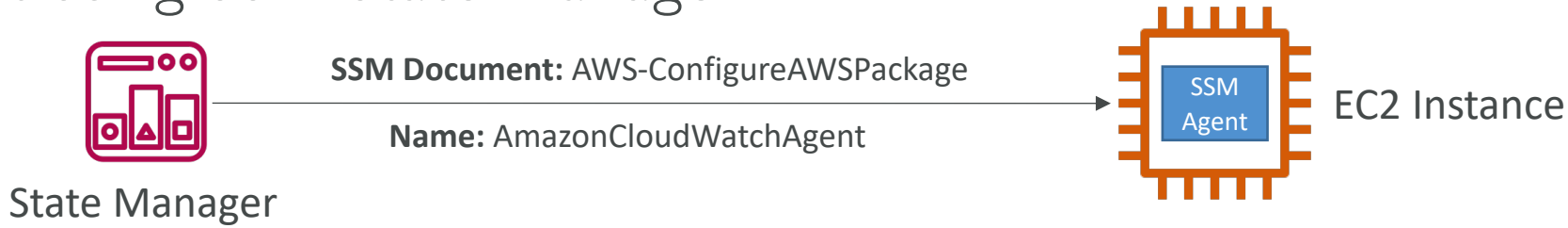
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudWatch Agent – Integration with SSM

- Install CW Agent using SSM Run Command

- Install CW Agent using SSM State Manager



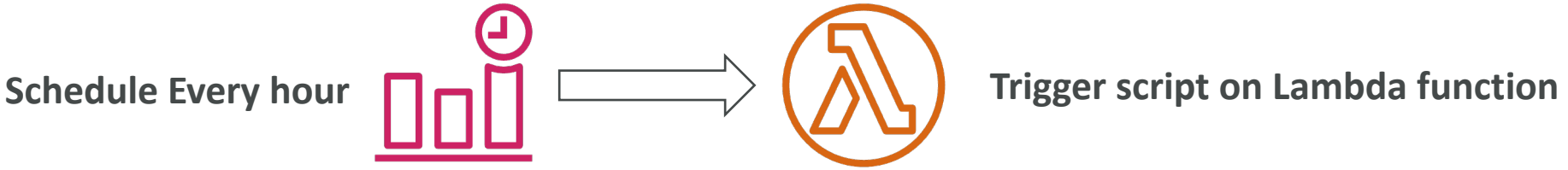
- Configure CW Agent by storing config. in SSM Parameter Store

© Stephane Maarek

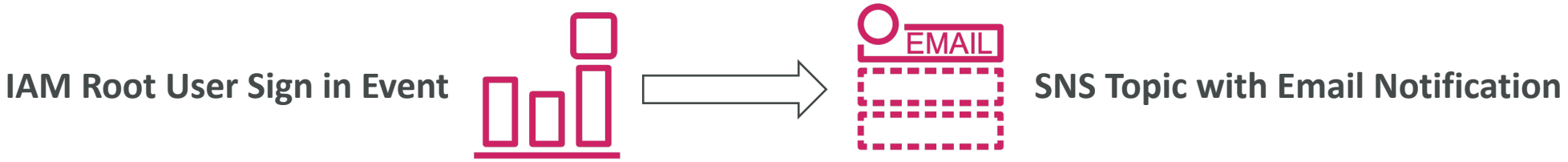
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon EventBridge (formerly CloudWatch Events)

- Schedule: Cron jobs (scheduled scripts)



- Event Pattern: Event rules to react to a service doing something

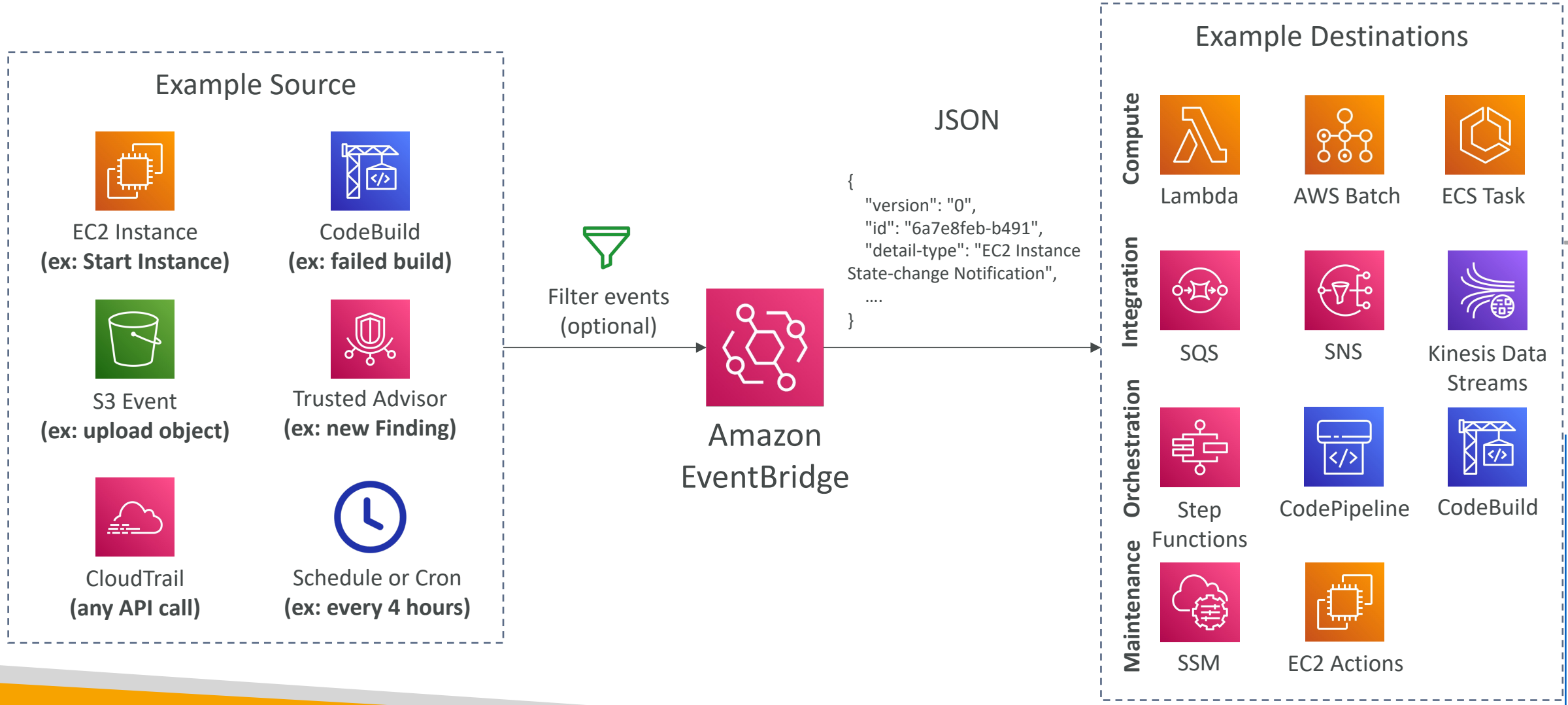


- Trigger Lambda functions, send SQS/SNS messages…

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

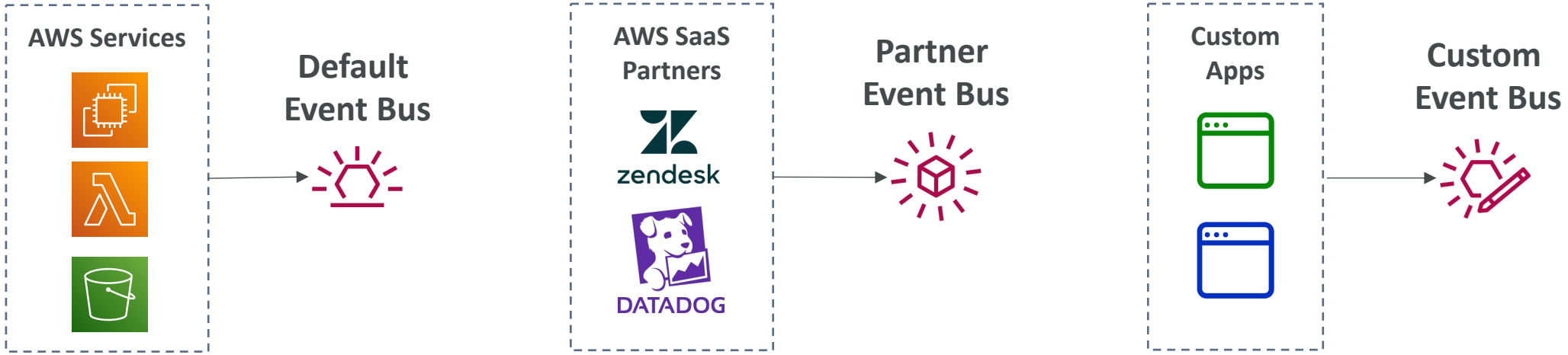
Amazon EventBridge Rules



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Amazon EventBridge



- Event buses can be accessed by other AWS accounts using Resource-based Policies

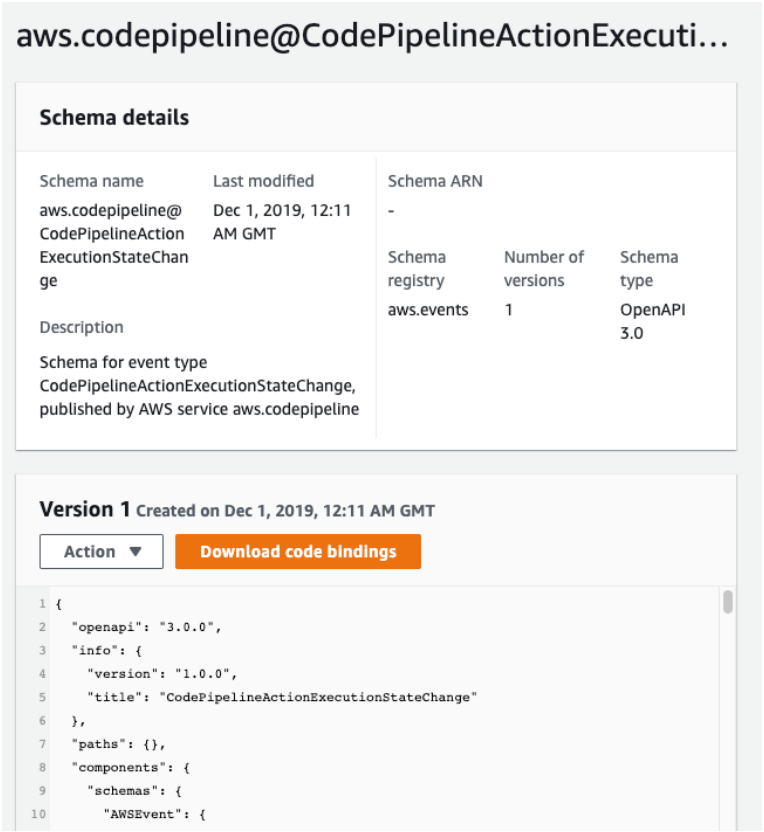
- You can **archive events** (all/filter) sent to an event bus (indefinitely or set period)
- Ability to **replay archived events**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon EventBridge – Schema Registry

- EventBridge can analyze the events in your bus and infer the **schema**
- The **Schema Registry** allows you to generate code for your application, that will know in advance how data is structured in the event bus
- Schema can be versioned



© Stephane Maarek

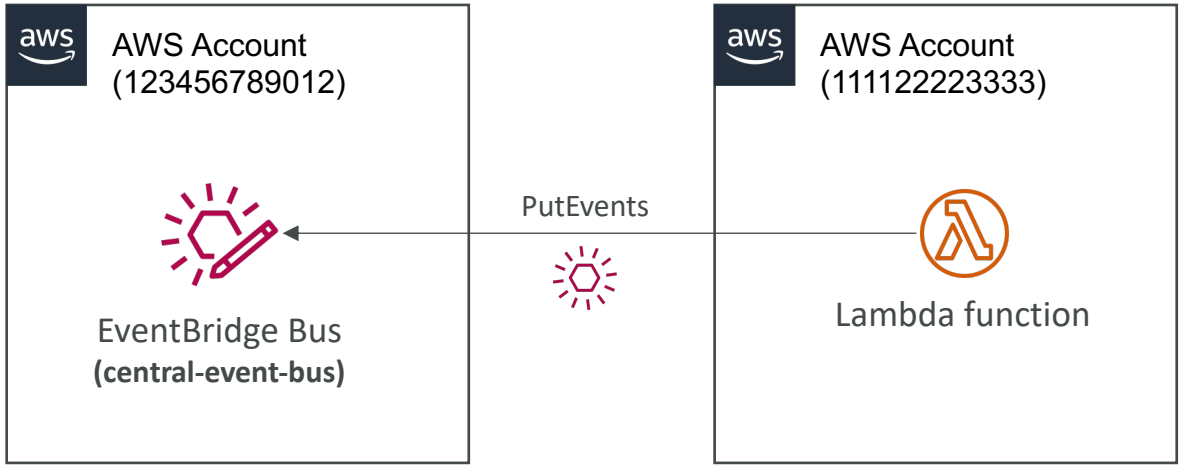
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon EventBridge – Resource-based Policy

- Manage permissions for a specific Event Bus
- Example: allow/deny events from another AWS account or AWS region
- Use case: aggregate all events from your AWS Organization in a single AWS account or AWS region

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "events:PutEvents",
            "Principal": { "AWS": "111122223333" },
            "Resource": "arn:aws:events:us-east-1:123456789012:
event-bus/central-event-bus"
        }
    ]
}
```

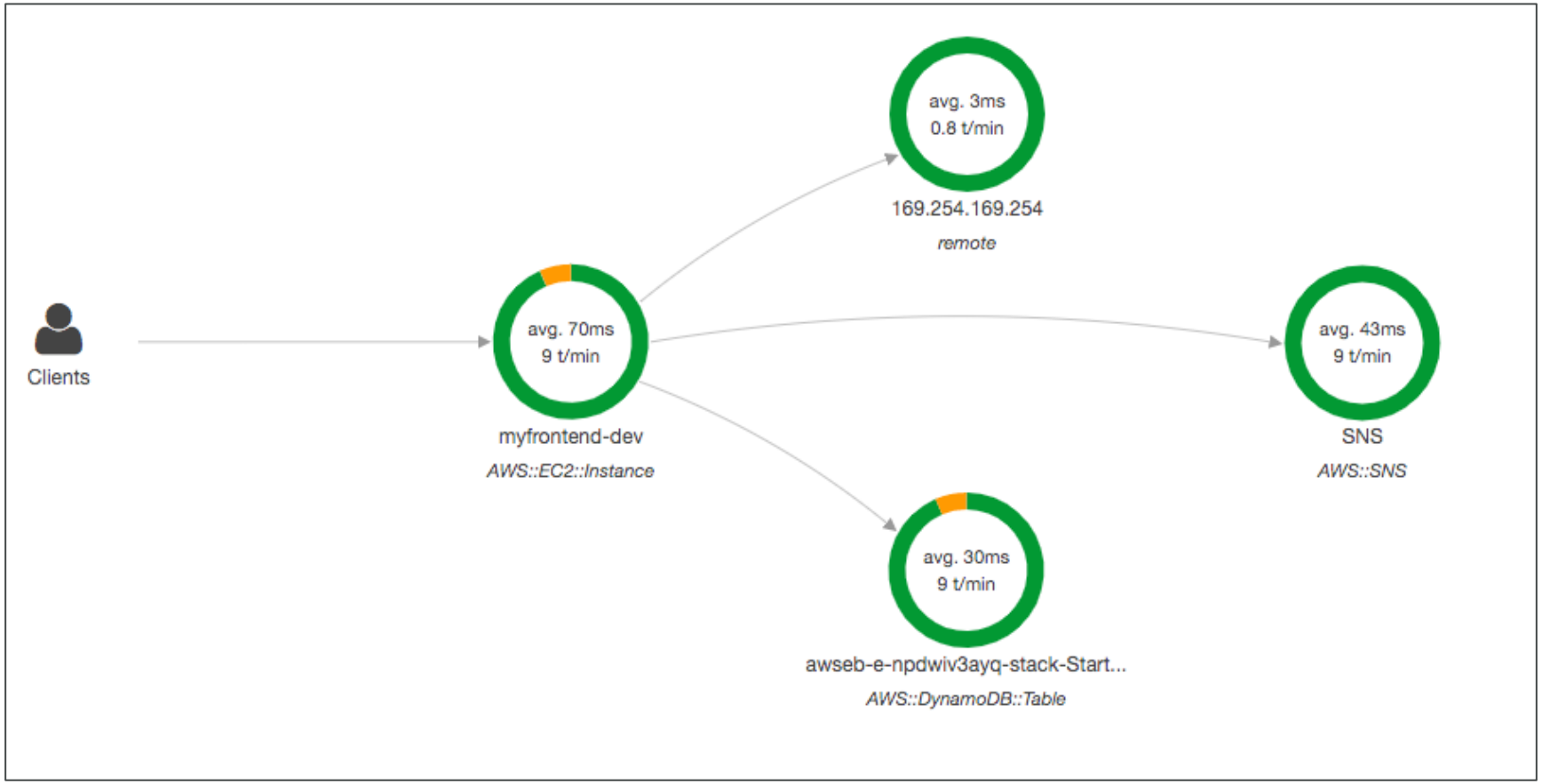
Allow **events** from another AWS account



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS X-Ray
Visual analysis of our applications



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

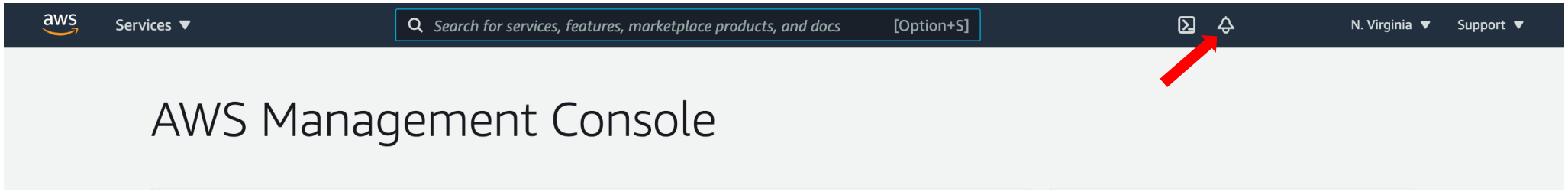
X-Ray

- Tracing requests across your microservices (distributed systems)
- Integrations with:
  - EC2 – install the X-Ray agent
  - ECS – install the X-Ray agent or Docker container
  - Lambda
  - Beanstalk - agent is automatically installed
  - API Gateway – helpful to debug errors (such as 504)
- The X-Ray agent or services need IAM permissions to X-Ray

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Personal Health Dashboard



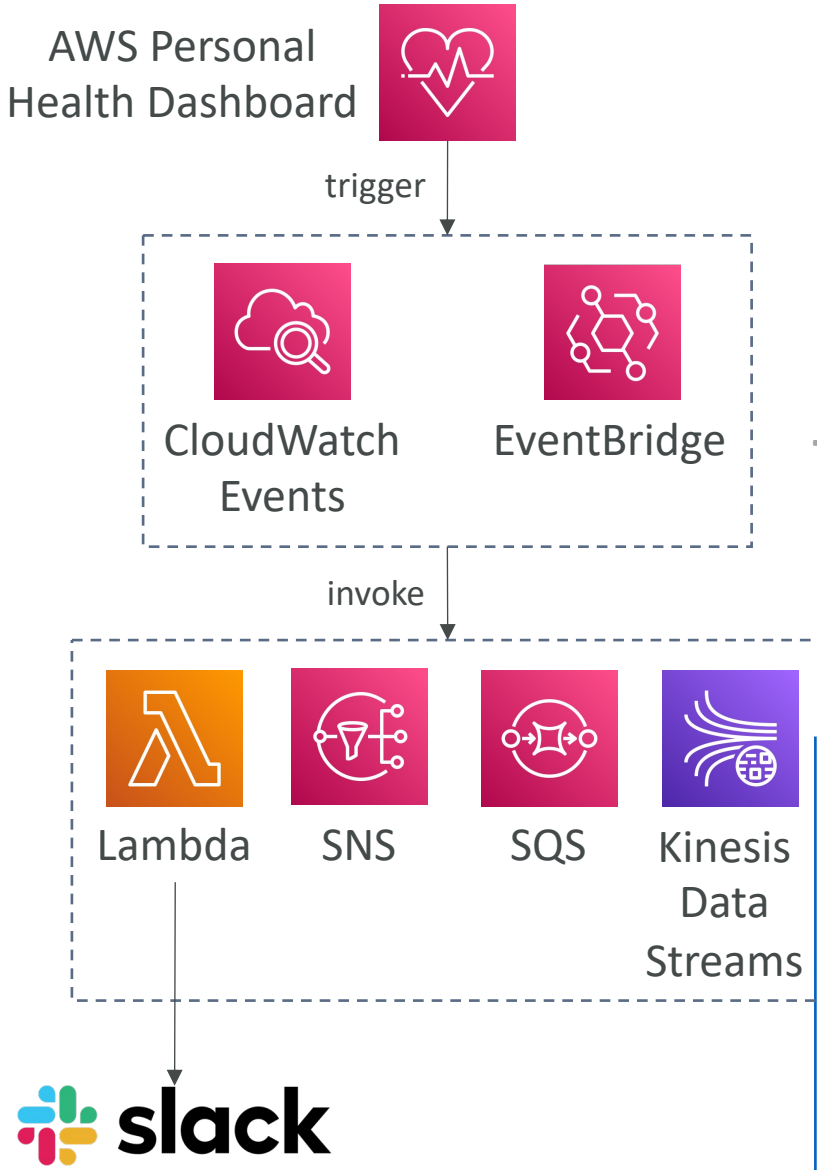
- Global service
- Show how AWS outages directly impact you
- Show the impact on your resources
- List issues and actions you can do to remediate them
- Will show **maintenance** events from AWS
- Programmatically accessible through the **AWS Health API**
- Aggregations across multiple accounts of an **AWS Organization**
- https://phd.aws.amazon.com/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Health Event Notifications

- Use EventBridge (CloudWatch Events) to react to changes for AWS Health events in your AWS account
- Example: receive email notifications when EC2 instances in your AWS account are scheduled for updates
- Can't be used to return public events from the Service Health Dashboard
- Use cases: send notifications, capture event information, take corrective action, …



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Deployment & Instance Management

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Elastic Beanstalk Overview

- Elastic Beanstalk is a developer centric view of deploying an application on AWS

- It uses all the component's we've seen before:
  EC2, Auto Scaling Group, Elastic Load Balancers, RDS, etc…
- But it's all in one view that's easy to make sense of!
- **We still have full control over the configuration of each component**

- Beanstalk is free but you pay for the underlying instances

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Elastic Beanstalk

- Support for many platforms:
  - Go
  - Java SE
  - **Java with Tomcat**
  - .NET on Windows Server with IIS
  - Node.js
  - PHP
  - Python
  - Ruby
  - Packer Builder
  - **Single Container Docker**
  - **Multicontainer Docker**
  - **Preconfigured Docker**
- Beanstalk is great to "**Replatform**" your application from on-premises to the cloud

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Elastic Beanstalk

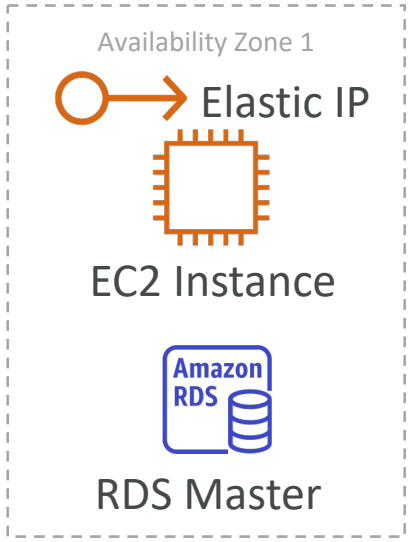
- Managed service
  - Instance configuration / OS is handled by Beanstalk
  - Deployment strategy is configurable but performed by Elastic Beanstalk
- Just the application code is the responsibility of the developer
- Three architecture models:
  - Single Instance deployment: good for dev
  - LB + ASG: great for production or pre-production web applications
  - ASG only: great for non-web apps in production (workers, etc..)

© Stephane Maarek

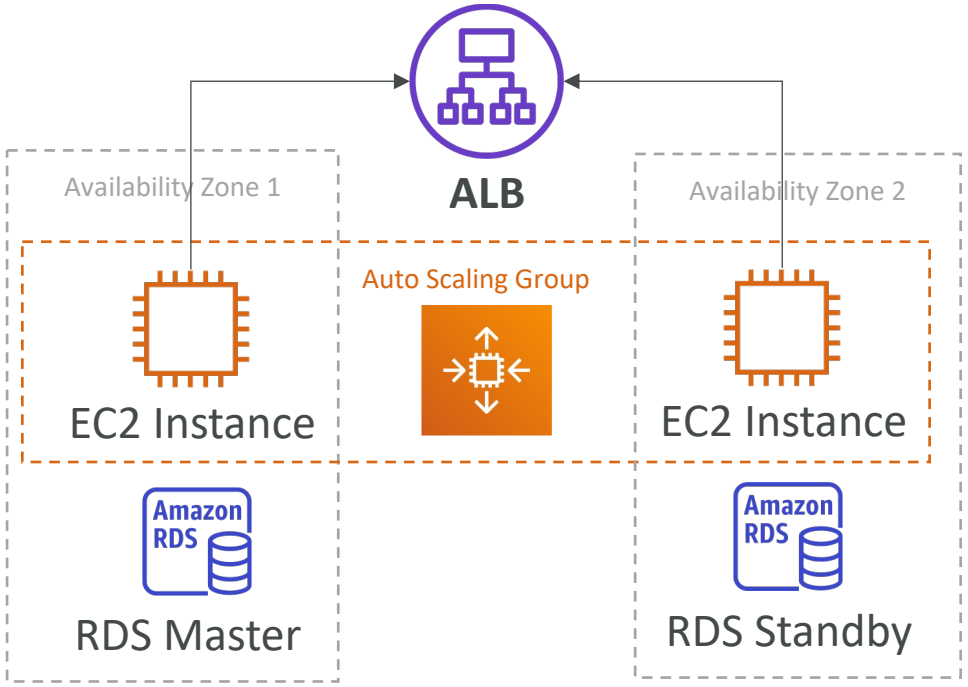
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Beanstalk Environments

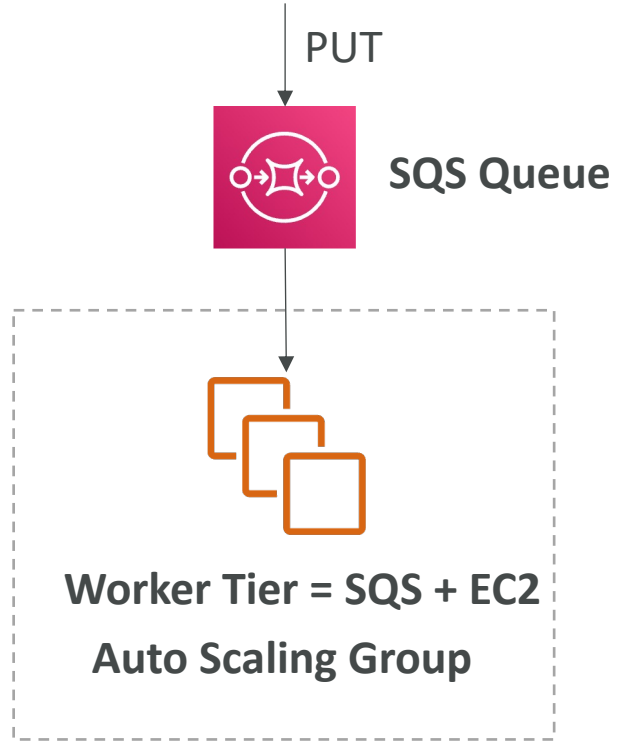
Single Instance
Great for dev



High Availability with Load Balancer
Great for prod



Worker Tier

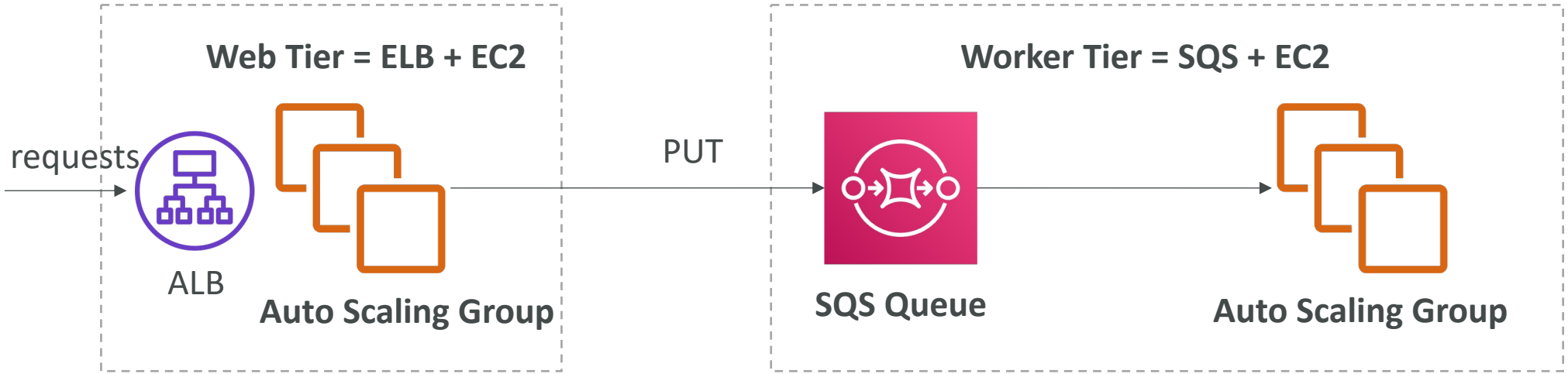


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Web Server vs Worker Environment

- If your application performs tasks that are long to complete, offload these tasks to a dedicated **worker environment**
- **Decoupling** your application into two tiers is common
- Example: processing a video, generating a zip file, etc
- You can define periodic tasks in a file **cron.yaml**

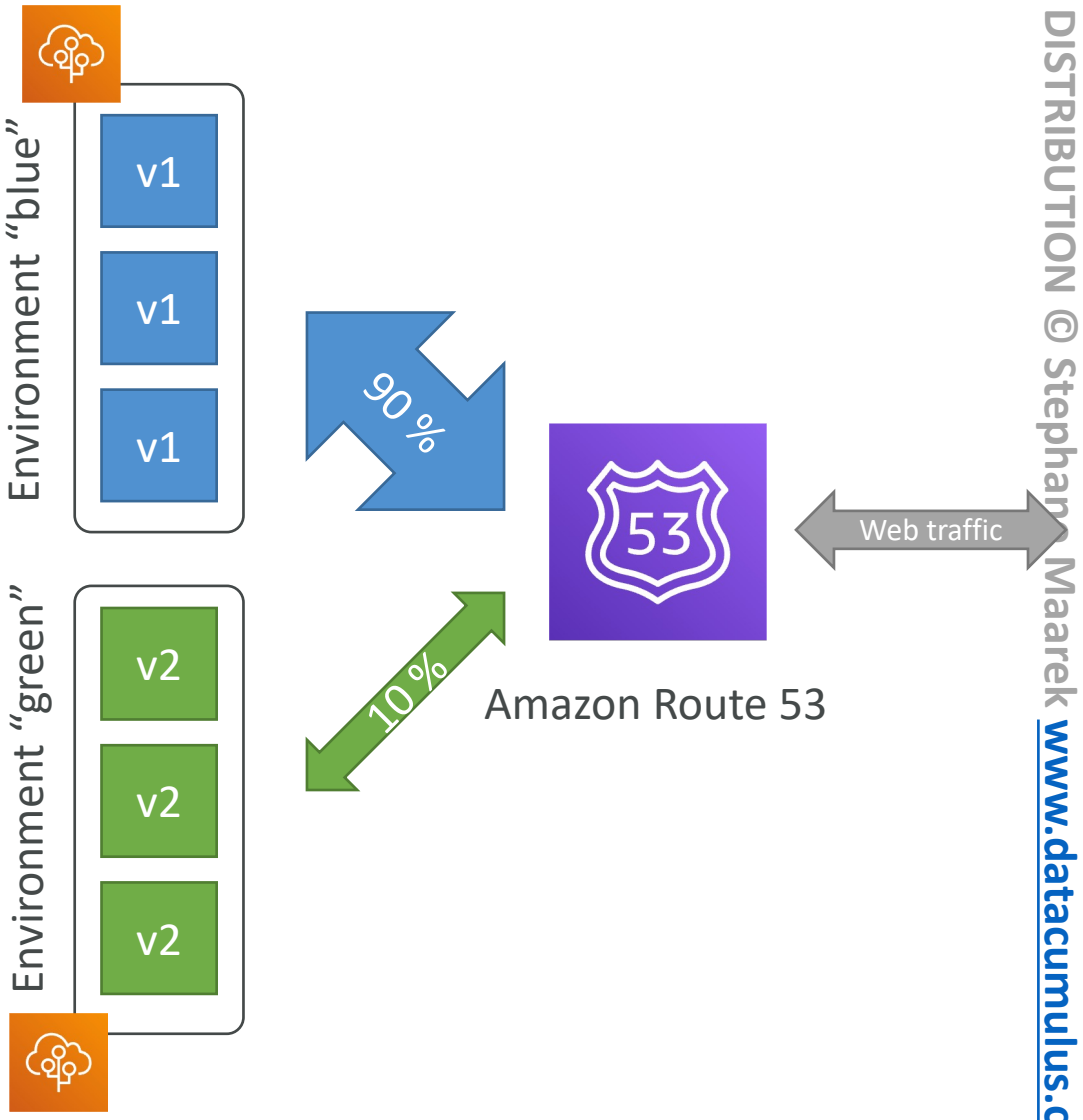


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Elastic Beanstalk Deployment Blue / Green

- Not a "direct feature" of Elastic Beanstalk
- Zero downtime and release facility
- Create a new "stage" environment and deploy v2 there
- The new environment (green) can be validated independently and roll back if issues
- Route 53 can be setup using weighted policies to redirect a little bit of traffic to the stage environment
- Using Beanstalk, "swap URLs" (DNS swap) when done with the environment test

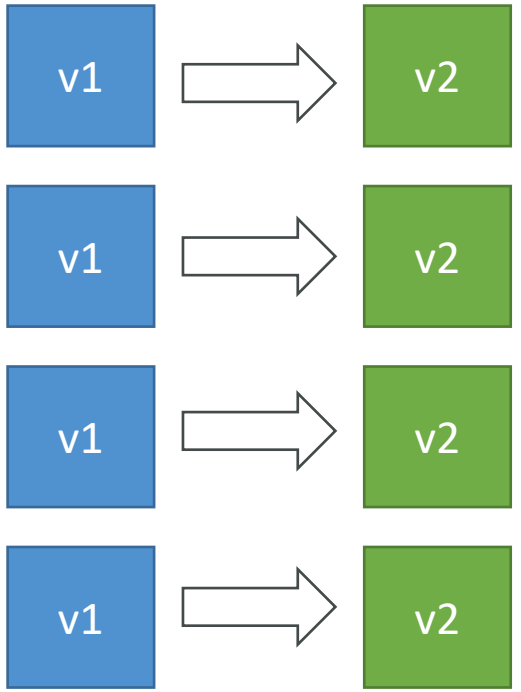


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS CodeDeploy

- We want to deploy our application automatically to many EC2 instances
- These instances are not managed by Elastic Beanstalk
- There are several ways to handle deployments using open source tools (Ansible, Terraform, Chef, Puppet, etc…)
- We can use the managed Service AWS CodeDeploy
- CodeDeploy can deploy to: **EC2, ASG, ECS & Lambda**

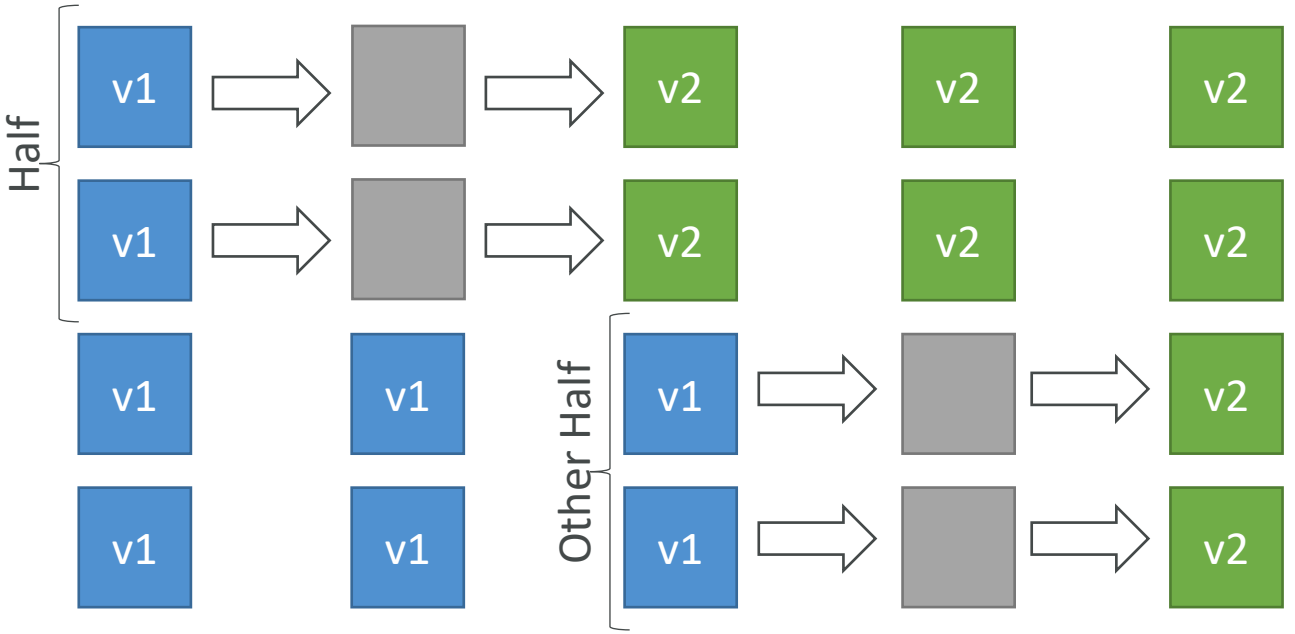


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CodeDeploy to EC2

- Define how to deploy the application using appspec.yml + deployment strategy
- Will do in-place update to your fleet of EC2 instances
- Can use hooks to verify the deployment after each deployment phase

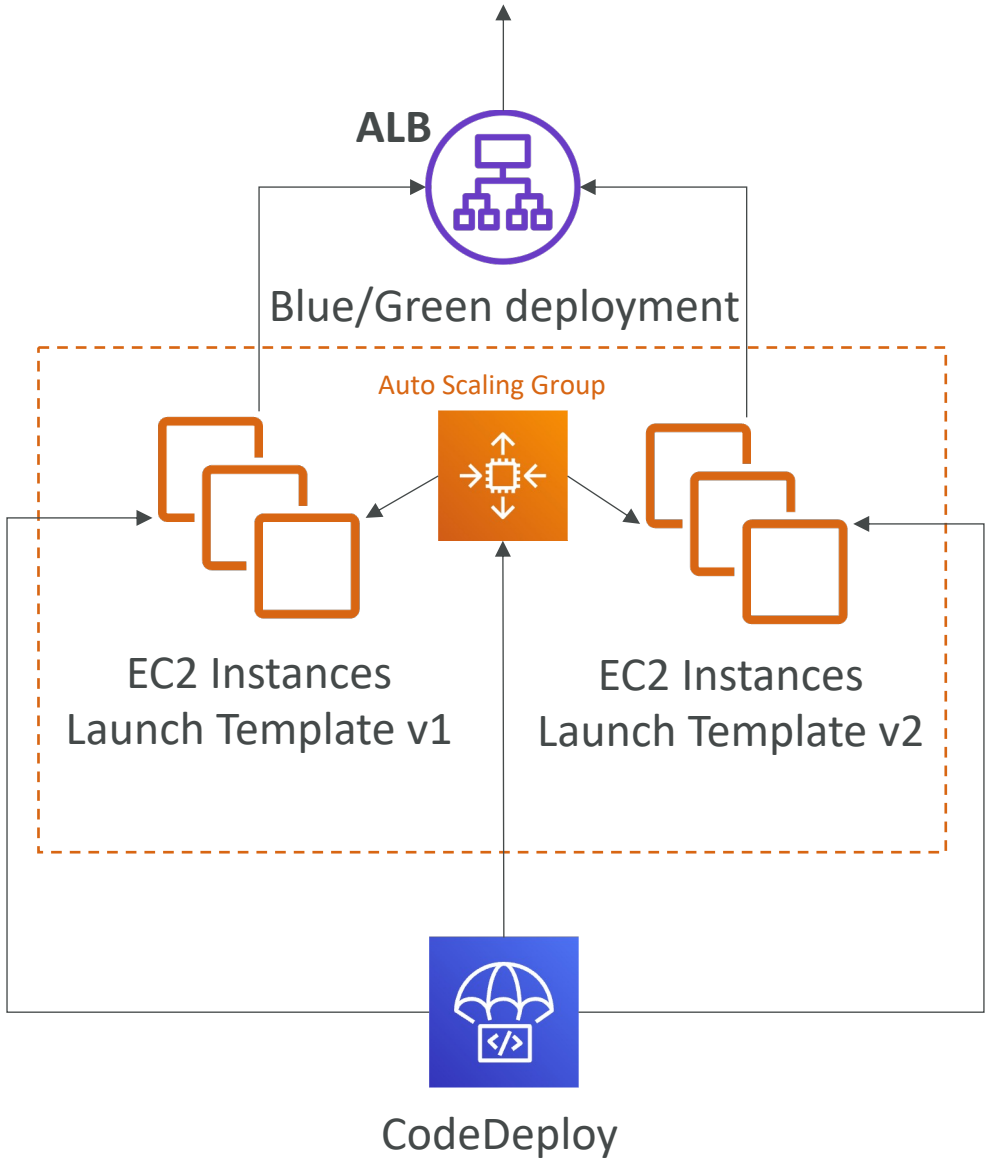


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CodeDeploy to ASG

- In place updates:
  - Updates current existing EC2 instances
  - Instances newly created by an ASG will also get automated deployments
- Blue / green deployment:
  - A new auto-scaling group is created (settings are copied)
  - Choose how long to keep the old instances
  - Must be using an ELB

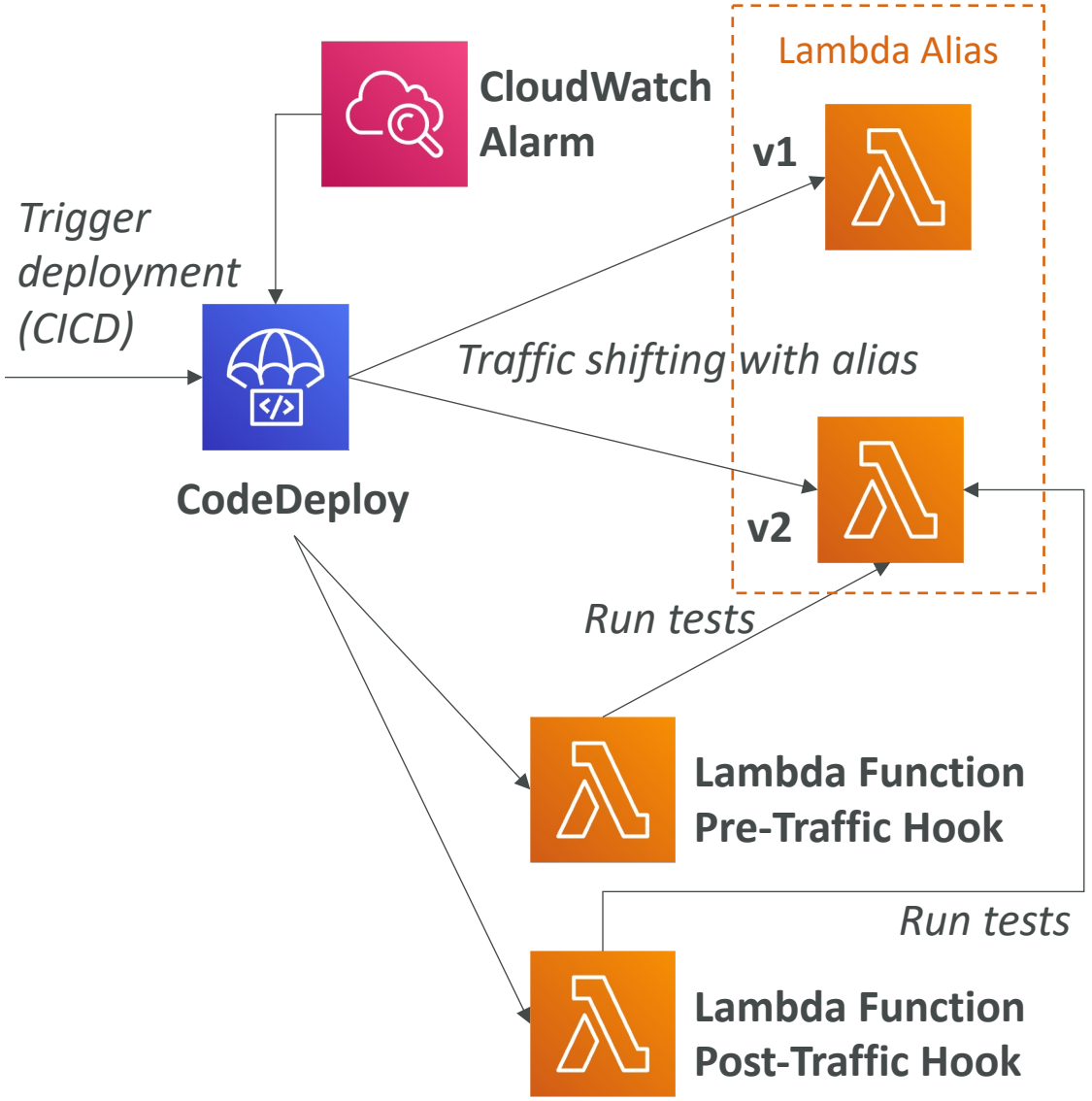


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CodeDeploy to AWS Lambda

- **Traffic Shifting** feature
- **Pre and Post traffic hooks** features to validate deployment (before the traffic shift starts and after it ends)
- Easy & automated rollback using CloudWatch Alarms
- SAM framework natively uses CodeDeploy

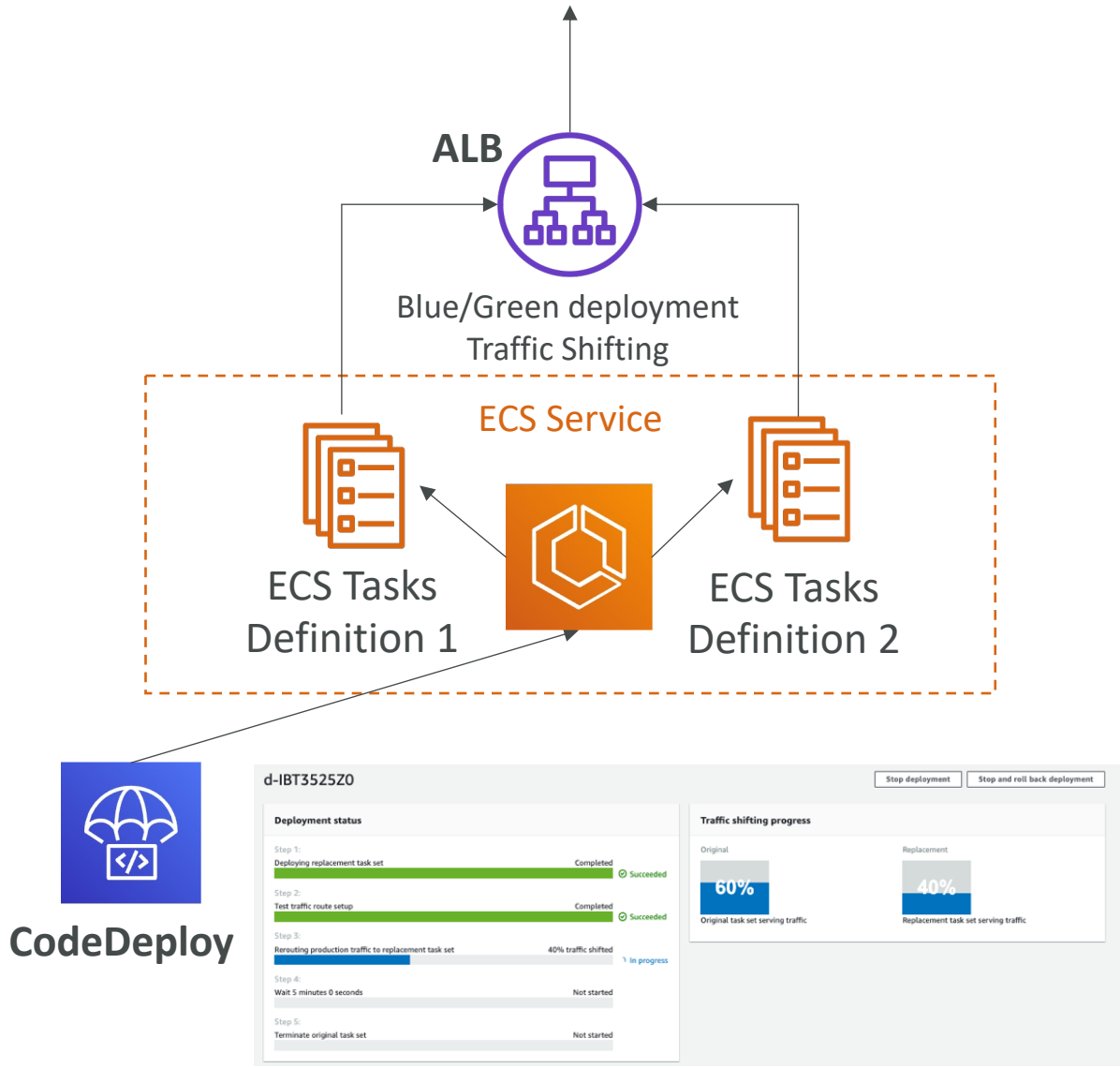


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CodeDeploy to ECS

- Support for Blue/Green deployments for Amazon ECS and AWS Fargate
- Setup is done within the ECS service definition
- A new task set is created, and traffic is re-routed to the new task test.
- Then if everything is stable for X minutes, the old task set is terminated (so you have time to notice issues)
- Supports Canary deployments (Canary10Percent5Minutes)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS CloudFormation

- Infrastructure as code (IaC) in AWS
- Portability of stacks across multiple accounts and regions

- Backbone of the Elastic Beanstalk service
- Backbone of the Service Catalog service
- Backbone of the SAM (Serverless Application Model) framework

- Must-know service as a developer / sysops / devops

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Retaining Data on Deletes

- You can put a DeletionPolicy on any resource to control what happens when the CloudFormation template is deleted
- **DeletionPolicy=Retain**:
  - Specify on resources to preserve / backup in case of CloudFormation deletes
  - To keep a resource, specify **Retain** (works for any resource / nested stack)
- **DeletionPolicy=Snapshot:**
  - EBS Volume, ElastiCache Cluster, ElastiCache ReplicationGroup
  - RDS DBInstance, RDS DBCluster, Redshift Cluster
- **DeletePolicy=Delete (default behavior)**:
  - Note: for **AWS::RDS::DBCluster** resources, the default policy is Snapshot
  - Note: to delete an S3 bucket, you need to first empty the bucket of its content

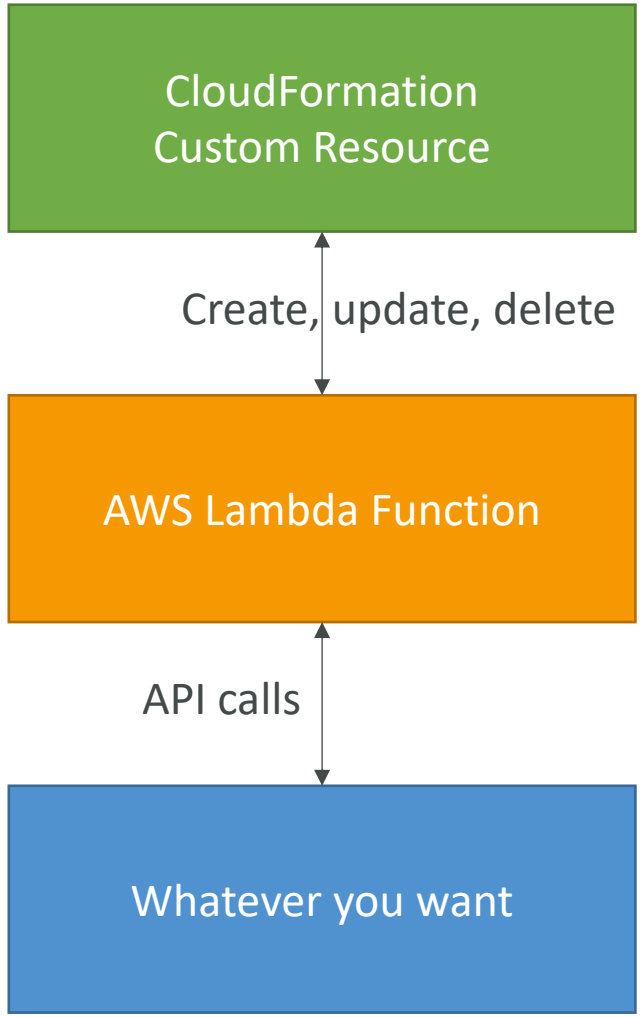
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFormation Custom Resources (Lambda)

- You can define a Custom Resource in CloudFormation to address any of these use cases:

- An AWS resource is **not yet supported** (new service for example)
- An **on-premises resource**
- Emptying an S3 bucket before being deleted
- Fetch an AMI id
- Anything you want…!

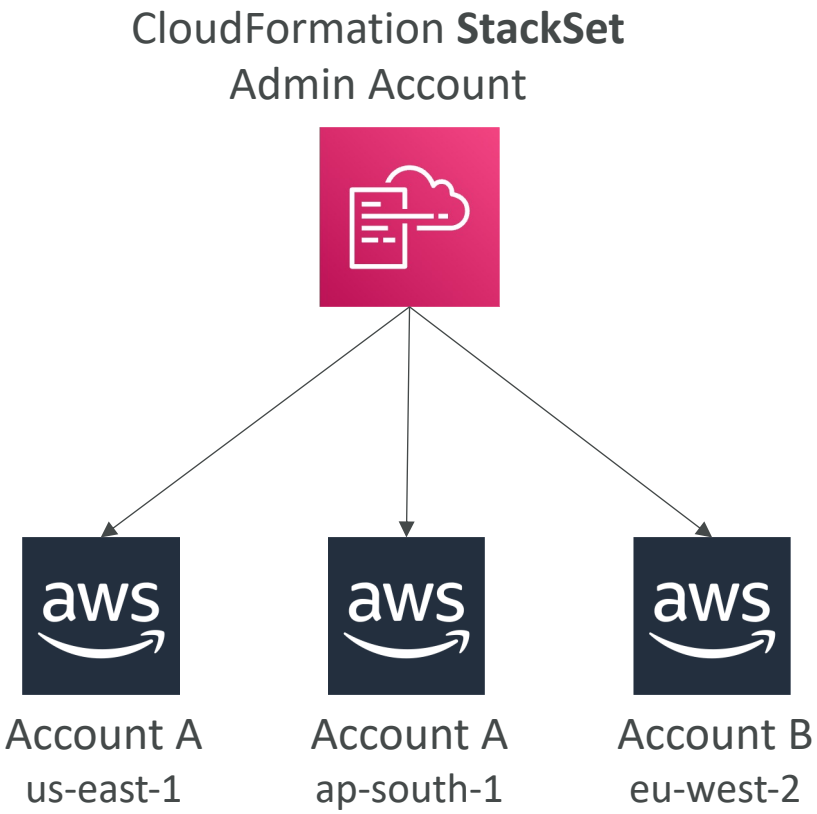


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFormation – StackSets

- Create, update, or delete stacks across **multiple accounts and regions** with a single operation
- Administrator account to create StackSets
- Trusted accounts to create, update, delete stack instances from StackSets
- When you update a stack set, *all* associated stack instances are updated throughout all accounts and regions
- Enable **Automatic Deployment** feature to automatically deploy to accounts in AWS Organization or OUs



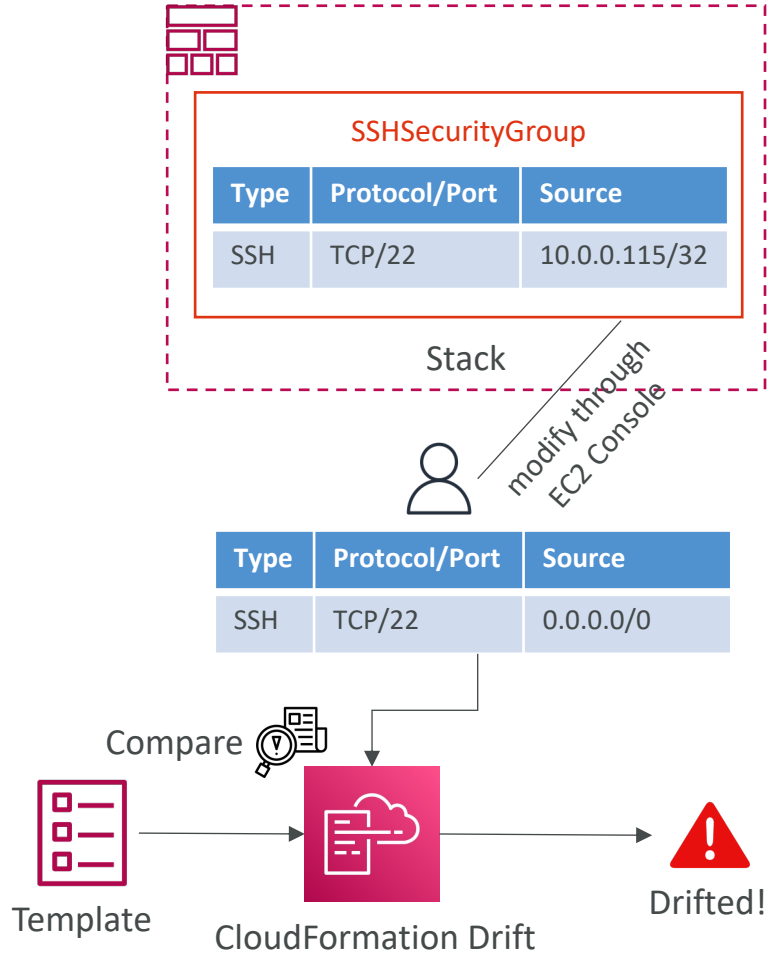
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CloudFormation Drift Overview

- CloudFormation allows you to create infrastructure
- But it doesn't protect you against manual configuration changes
- How do we know if our resources have **drifted**?

- We can use CloudFormation Drift!
- Detect drift on an entire stack or on individual resources within a stack



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

CloudFormation – Integration with Secrets Manager

```yaml
Resources:
  # Secret resource with a randomly generated password in its SecureString JSON
  MyRDSDBInstanceRotationSecret:
    Type: AWS::SecretsManager::Secret
    Properties:
      GenerateSecretString:
        SecretStringTemplate: '{"username": "admin"}'
        GenerateStringKey: password
        PasswordLength: 16
        ExcludeCharacters: "\"@/\\"

  # RDS Instance resource. Its master username and password use dynamic references
  # to resolve values from Secrets Manager
  MyRDSDBInstance:
    Type: AWS::RDS::DBInstance
    Properties:
      DBInstanceClass: db.t2.micro
      Engine: mysql
      MasterUsername: !Sub "{{resolve:secretsmanager:${MyRDSDBInstanceRotationSecret}::username}}"
      MasterUserPassword: !Sub "{{resolve:secretsmanager:${MyRDSDBInstanceRotationSecret}::password}}"

  # SecretTargetAttachment resource which updates the referenced Secret with properties
  # about the referenced RDS instance
  SecretRDSDBInstanceAttachment:
    Type: AWS::SecretsManager::SecretTargetAttachment
    Properties:
      TargetType: AWS::RDS::DBInstance
      SecretId: !Ref MyRDSDBInstanceRotationSecret
      TargetId: !Ref MyRDSDBInstance
```

secret is generated

reference secret in RDS DB instance

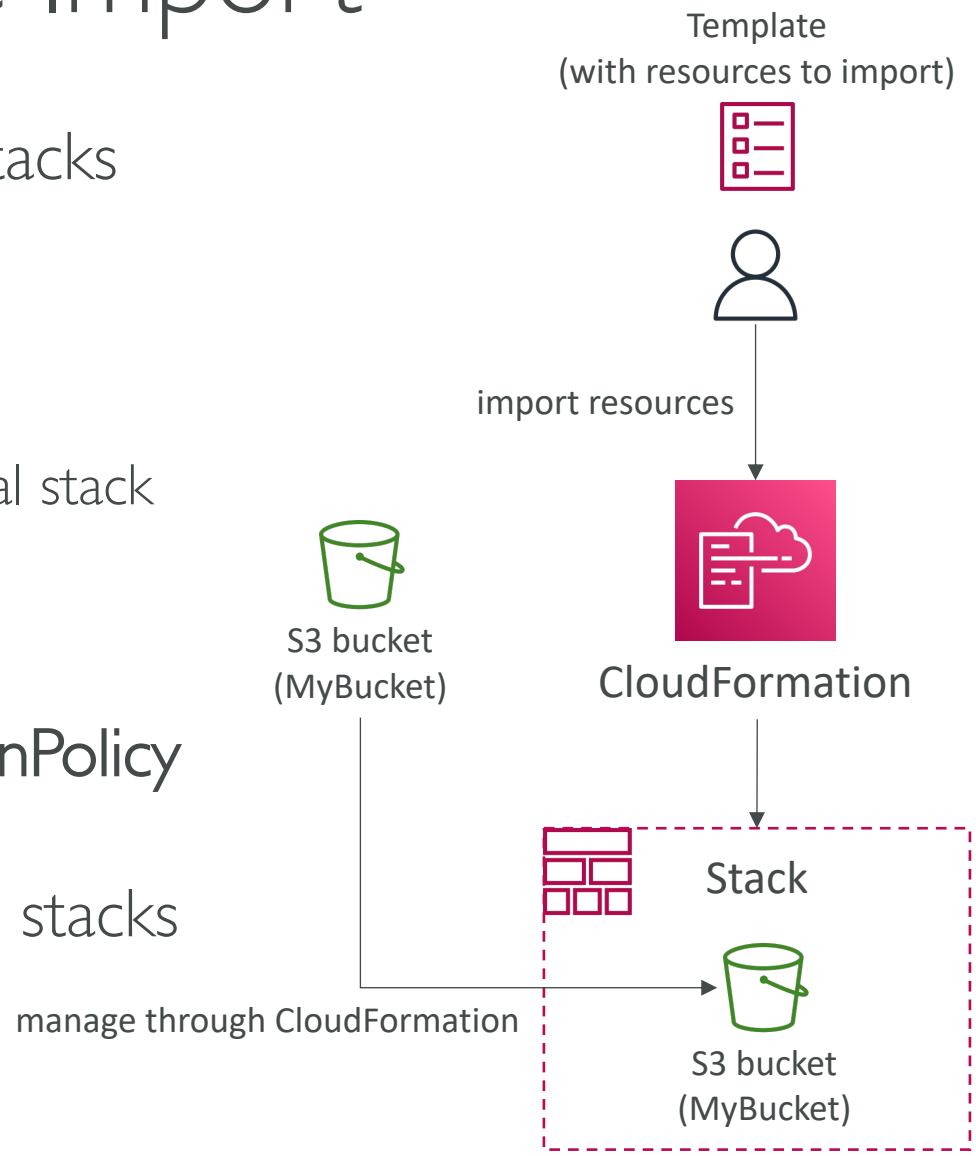
link the secret to RDS DB instance

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

CloudFormation - Resource Import

- Import existing resources into existing/new stacks
- You don't need to delete and re-create the resources as part of a CloudFormation stack
- During import operation, you'll need
  - A template that describes the entire stack (original stack resources & target resources to import)
  - A Unique identifier for each target resource (ex. BucketName for S3 buckets)
- Each resource to import must have a **DeletionPolicy** attribute (any value) & Identifier
- Can't import the same resource into multiple stacks



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Service Catalog

- Users that are new to AWS have too many options, and may create stacks that are not compliant / in line with the rest of the organization
- Some users just want a quick **self-service portal** to launch a set of **authorized products** pre-defined **by admins**
- Includes: virtual machines, databases, storage options, etc…
- Enter AWS Service Catalog!

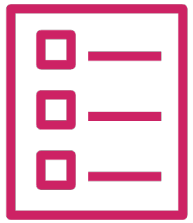
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Service Catalog diagram

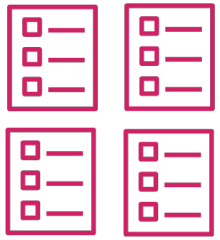
ADMIN TASKS

Product



CloudFormation Templates

Portfolio



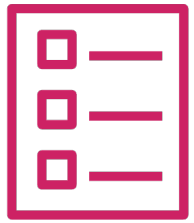
Collection of Products

Control

IAM Permissions to Access Portfolios

USER TASKS

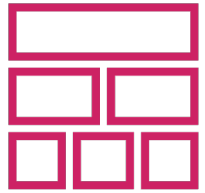
Product List



Authorized by IAM

launch

Provisioned Products



Ready to use
Properly Configured
Properly Tagged

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Service Catalog

- Create and manage catalogs of IT services that are approved on AWS
- The "products" are CloudFormation templates
- Ex: Virtual machine images, Servers, Software, Databases, Regions, IP address ranges
- **CloudFormation helps ensure consistency, and standardization by Admins**
- They are assigned to Portfolios (teams)
- Teams are presented a self-service portal where they can launch the products
- All the deployed products are centrally managed deployed services
- **Helps with governance, compliance, and consistency**
- Can give user access to launching products without requiring deep AWS knowledge
- Integrations with "self-service portals" such as ServiceNow

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

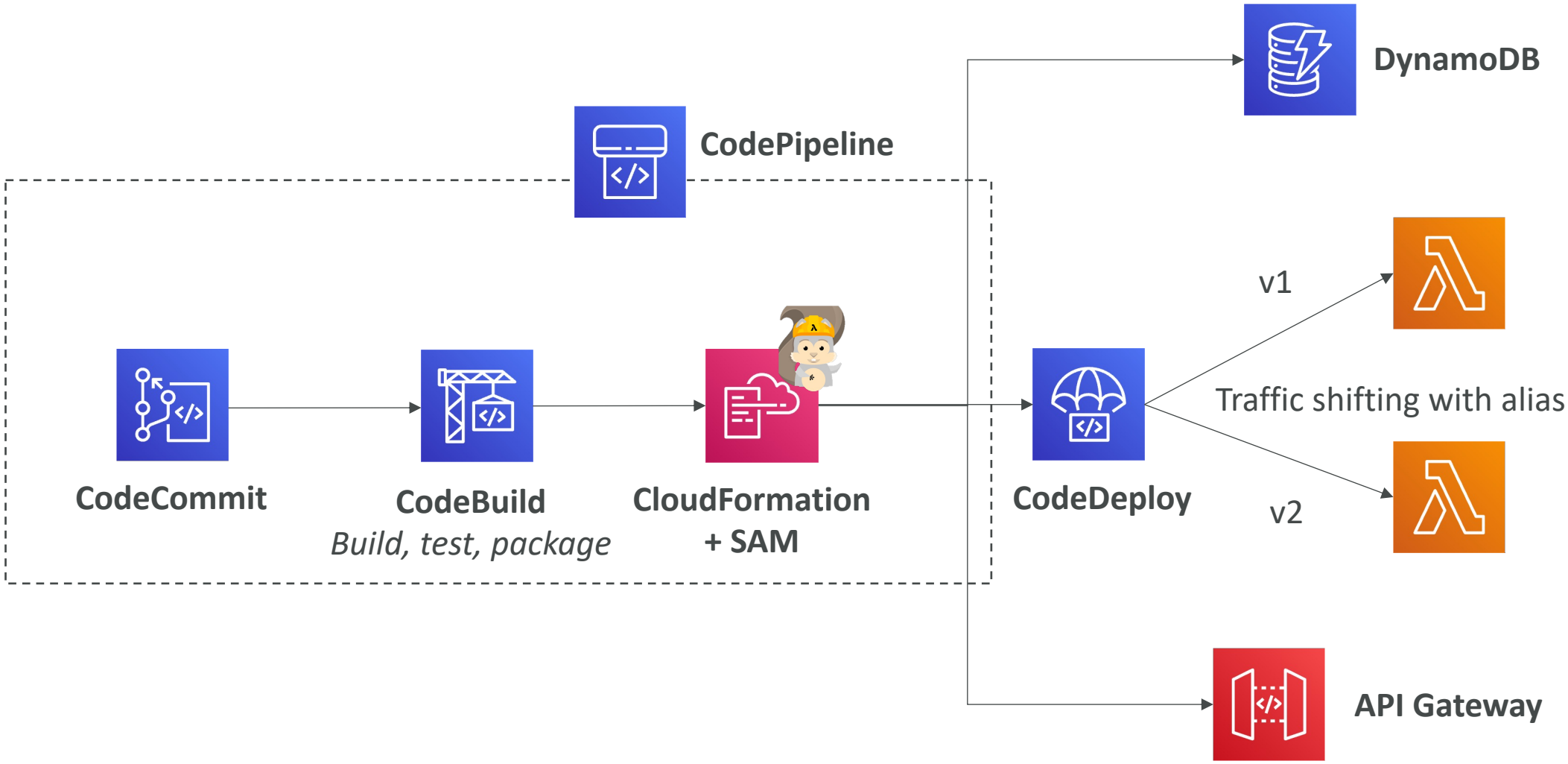
AWS SAM - Serverless Application Model

- SAM = Serverless Application Model
- Framework for developing and deploying serverless applications
- All the configuration is YAML code. Examples:
  - Lambda Functions (AWS::Serverless::Function)
  - DynamoDB tables (AWS::Serverless::SimpleTable)
  - API Gateway (AWS::Serverless::API)
  - StepFunction - State Machine (AWS::Serverless::StateMachine)
- SAM can help you to run Lambda, API Gateway, DynamoDB locally
- SAM can use CodeDeploy to deploy Lambda functions (traffic shifting)
- Leverages CloudFormation in the backend

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CICD Architecture for SAM

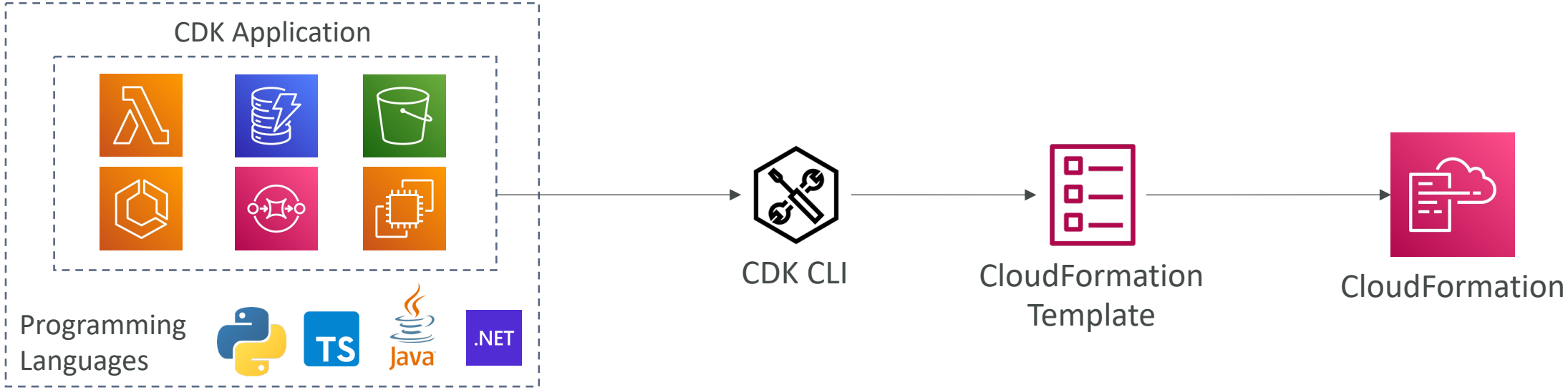


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Cloud Development Kit (CDK)

- Define your cloud infrastructure using a familiar language:
  - JavaScript/TypeScript, Python, Java, and .NET
- The code is "compiled" into a CloudFormation template (JSON/YAML)
- You can therefore deploy infrastructure and application runtime code together
  - Great for Lambda functions
  - Great for Docker containers in ECS / EKS



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CDK Example

```
export class MyEcsConstructStack extends core.Stack {
  constructor(scope: core.App, id: string, props?: core.StackProps)
    super(scope, id, props);

    const vpc = new ec2.Vpc(this, "MyVpc", {
      maxAzs: 3 // Default is all AZs in region
    });

    const cluster = new ecs.Cluster(this, "MyCluster", {
      vpc: vpc
    });

    // Create a load-balanced Fargate service and make it public
    new ecs_patterns.ApplicationLoadBalancedFargateService(this, "My
      cluster: cluster, // Required
      cpu: 512, // Default is 256
      desiredCount: 6, // Default is 1
      taskImageOptions: { image: ecs.ContainerImage.fromRegistry("am
      memoryLimitMiB: 2048, // Default is 512
      publicLoadBalancer: true // Default is false
    });
  }
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Systems Manager Overview

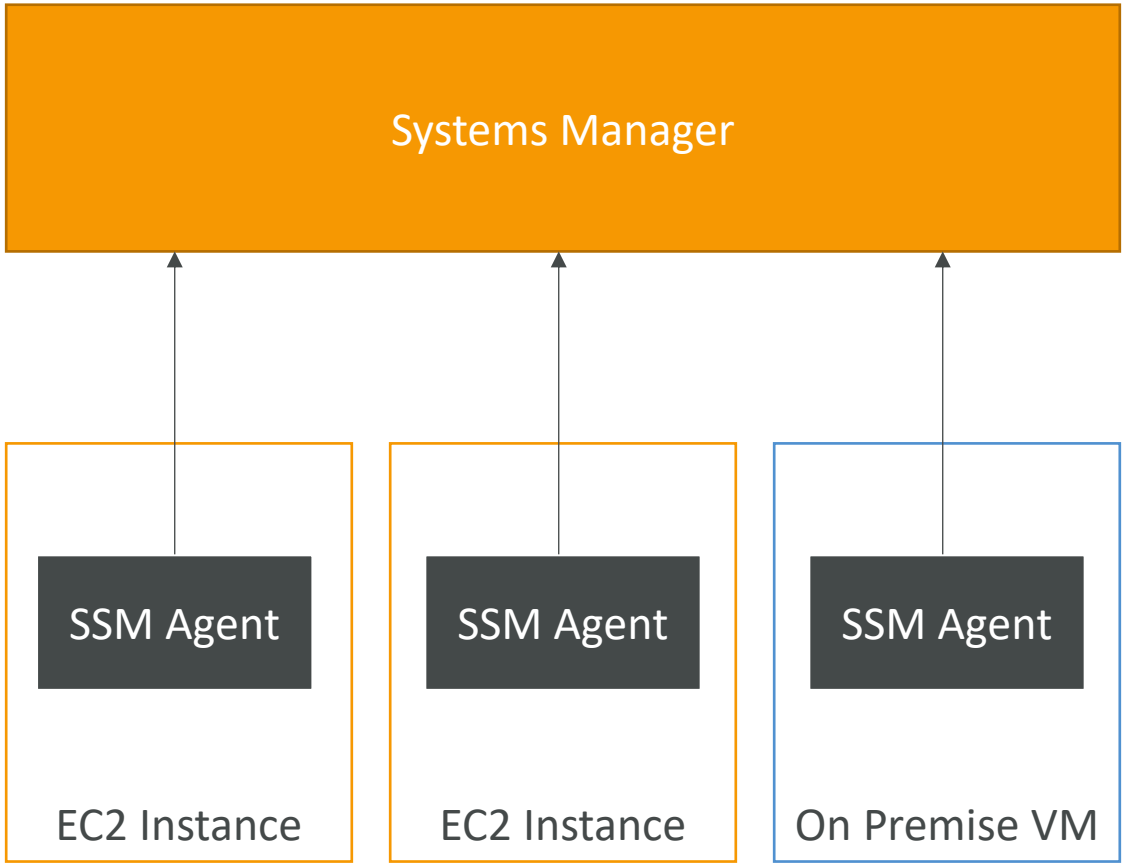
- Helps you manage your **EC2 and on-premises** systems at scale
- Get operational insights about the state of your infrastructure
- Easily detect problems
- **Patching automation for enhanced compliance**
- Works for both Windows and Linux OS
- Integrated with CloudWatch metrics / dashboards
- Integrated with AWS Config
- Free service

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

How Systems Manager works

- We need to install the SSM agent onto the systems we control
- Installed by default on Amazon Linux AMI & some Ubuntu AMI
- If an instance can't be controlled with Systems Manager, it's probably an issue with the SSM agent!
- Make sure the EC2 instances have a proper IAM role to allow Systems Manager actions

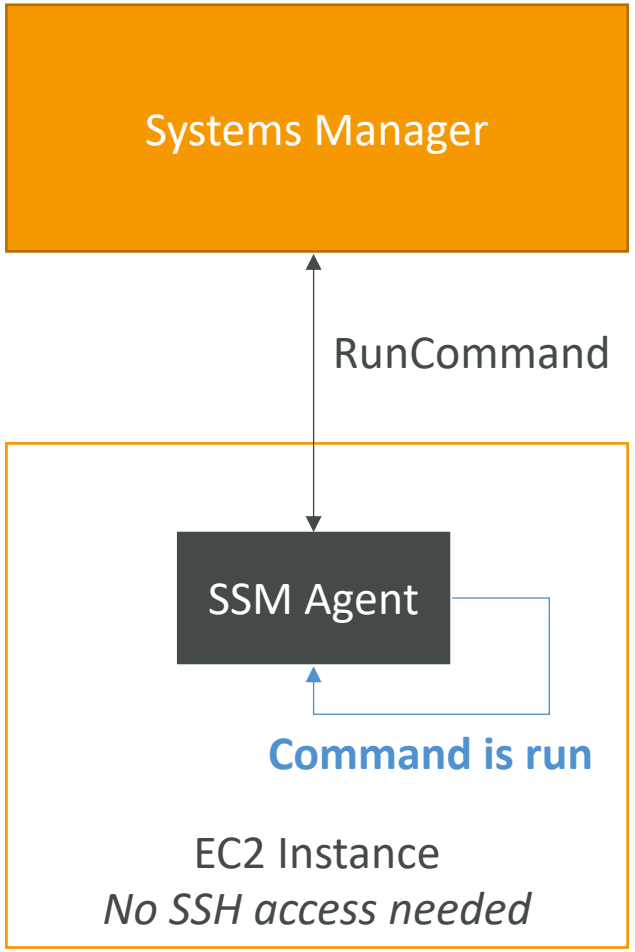


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Systems Manager Run Command

- Execute a document (= script) or just run a command
- Run command across multiple instances (using resource groups)
- Rate Control / Error Control
- Integrated with IAM & CloudTrail
- No need for SSH
- Results in the console

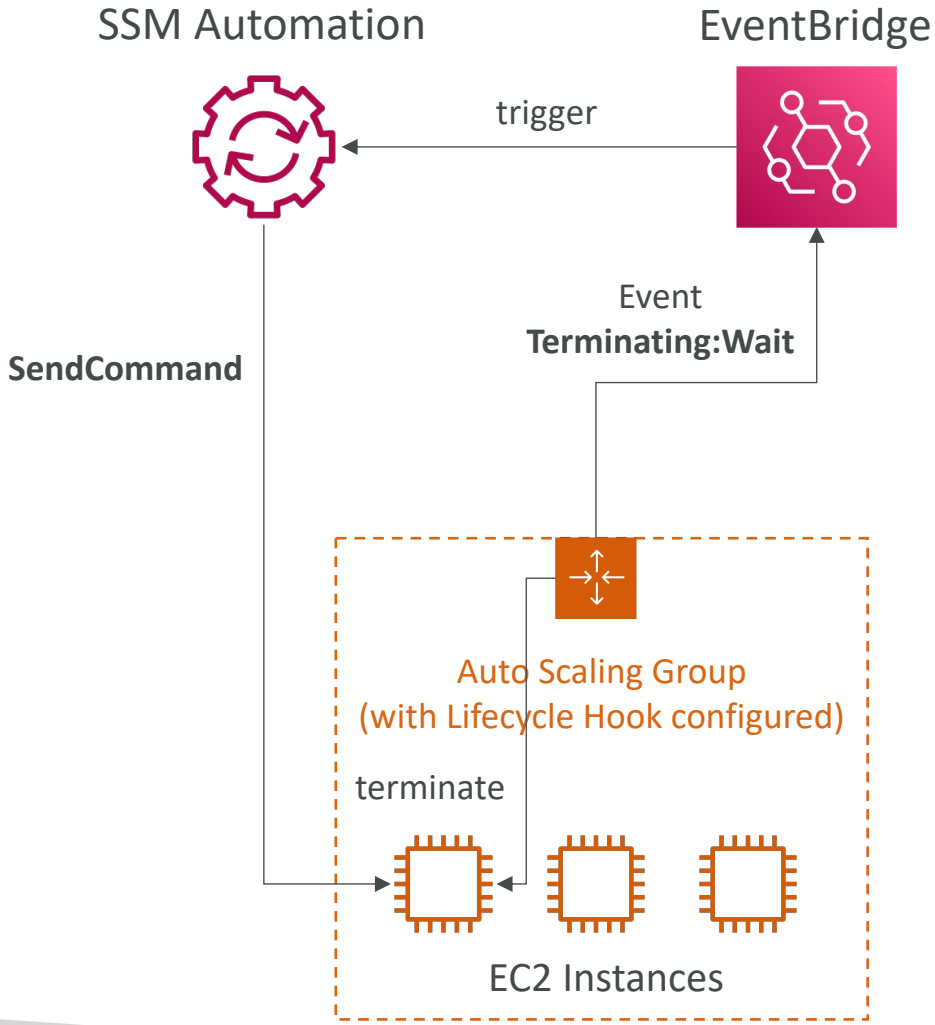


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Systems Manager – Send Command before an ASG Instance is Terminated

- Perform any action before the ASG terminates an EC2 instance
- Create a ASG Lifecycle Hook that puts the instance in **Terminating:Wait** state
- Monitor the **Terminating:Wait** state using EventBridge
- Trigger a SSM Automation Document to perform the actions on the instance before termination

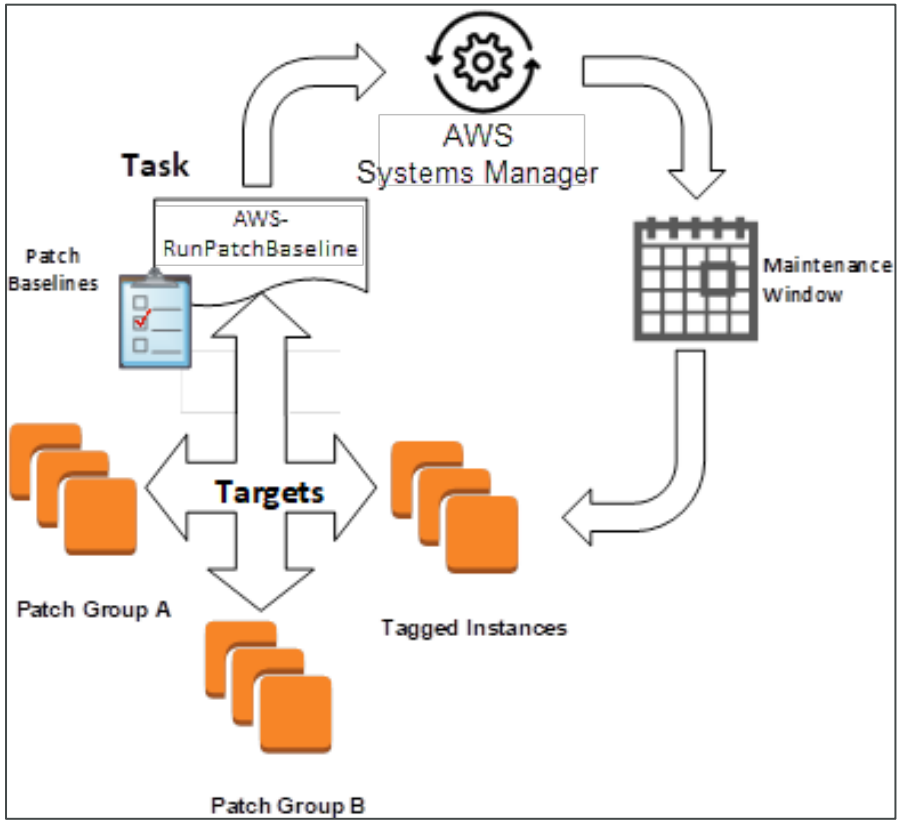


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Systems Manager Patch Managers – Steps

1. Define a **patch baseline** to use (or multiple if you have multiple environments)
2. Define patch groups: define based on tags, example different environments (dev, test, prod) – use tag **Patch Group**
3. Define **Maintenance Windows** (schedule, duration, registered targets/patch groups and registered tasks)
4. Add the **AWS-RunPatchBaseline Run Command** as part of the registered tasks of the Maintenance Window (works cross platform Linux & Windows)
5. Define **Rate Control** (concurrency & error threshold) for the task
6. Monitor **Patch Compliance** using **SSM Inventory**



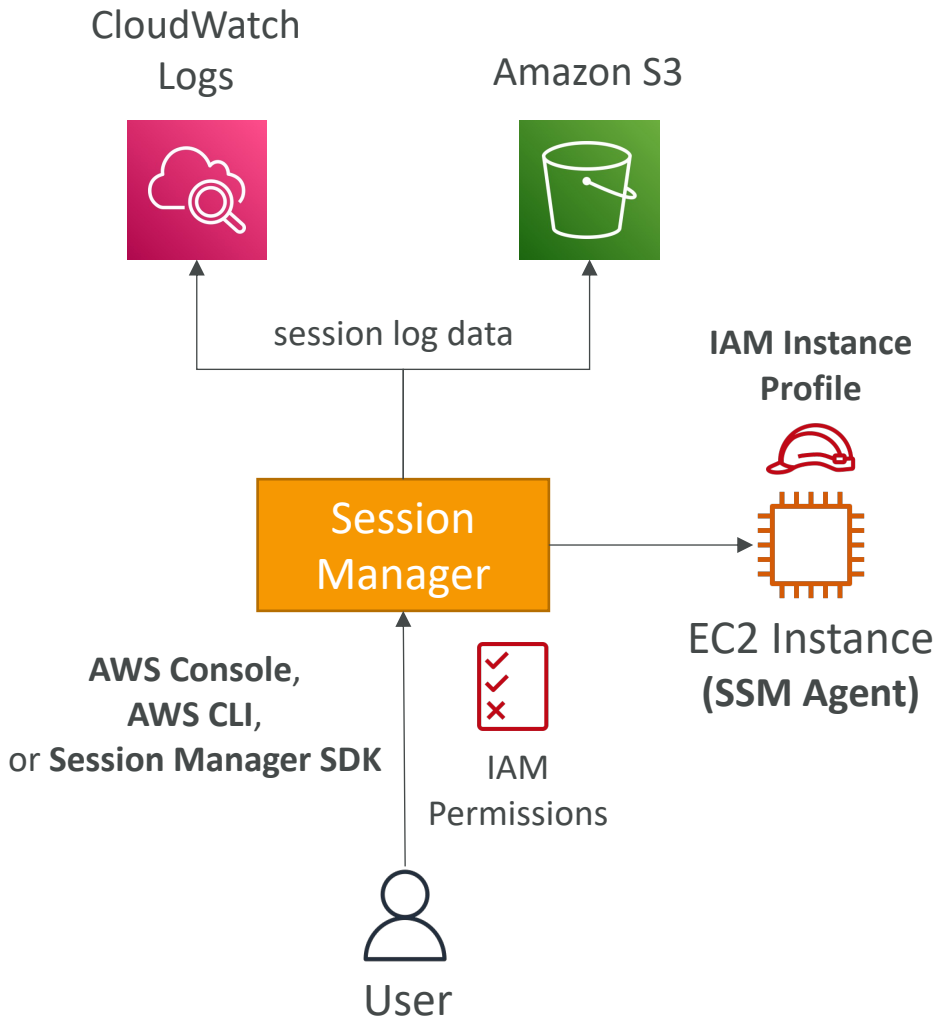
https://aws.amazon.com/blogs/mt/patching-your-windows-ec2-instances-using-aws-systems-manager-patch-manager/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Systems Manager Session Manager

- Allows you to start a secure shell on your EC2 and on-premises servers
- Access through AWS Console, AWS CLI, or Session Manager SDK
- **Does not need SSH access, bastion hosts, or SSH keys**
- Supports Linux, macOS, and Windows
- Log connections to your instances and executed commands
- Session log data can be sent to S3 or CloudWatch Logs
- CloudTrail can intercept **StartSession** events



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Systems Manager OpsCenter

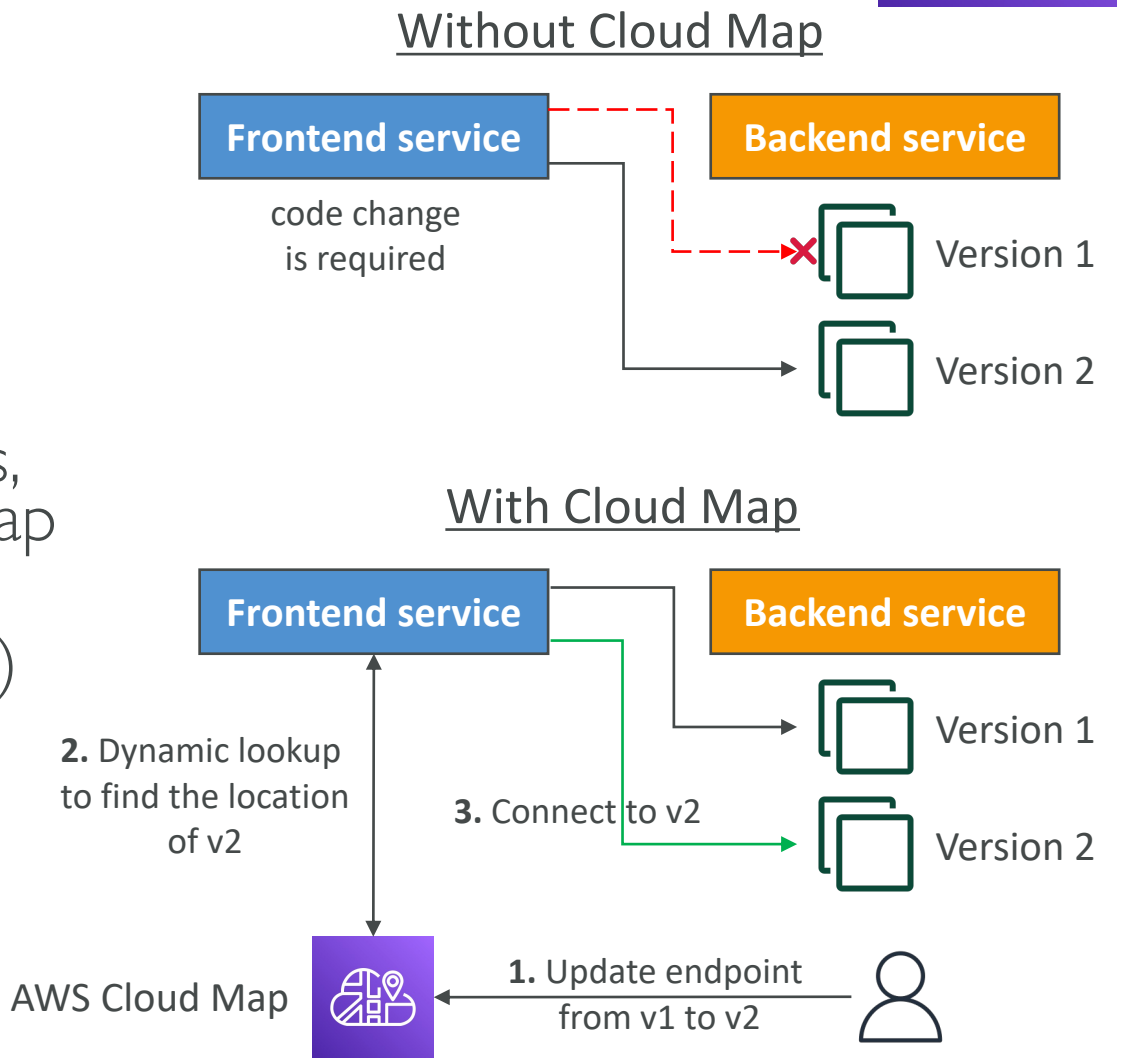
- Resolve Operational Issues (OpsItems) related to AWS resources
- **OpsItems** – issues, events and alerts
- Aggregates information to resolve issues on each OpsItems such as:
  - AWS Config changes and relationships
  - CloudTrail Logs
  - CloudWatch Alarms
  - CloudFormation Stack information
- Provides Automation Runbooks that you can use to resolve issues
- EventBridge or CloudWatch Alarms can create OpsItems

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Cloud Map

- A fully managed resource discovery service
- Creates a map of the backend services/resources that your applications depend on
- You register your application components, their locations, attributes, and health status with AWS Cloud Map
- Integrated health checking (stop sending traffic to unhealthy endpoints)
- Your applications can query AWS Cloud Map using AWS SDK, API, or DNS



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cost Control

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Cost Allocation Tags

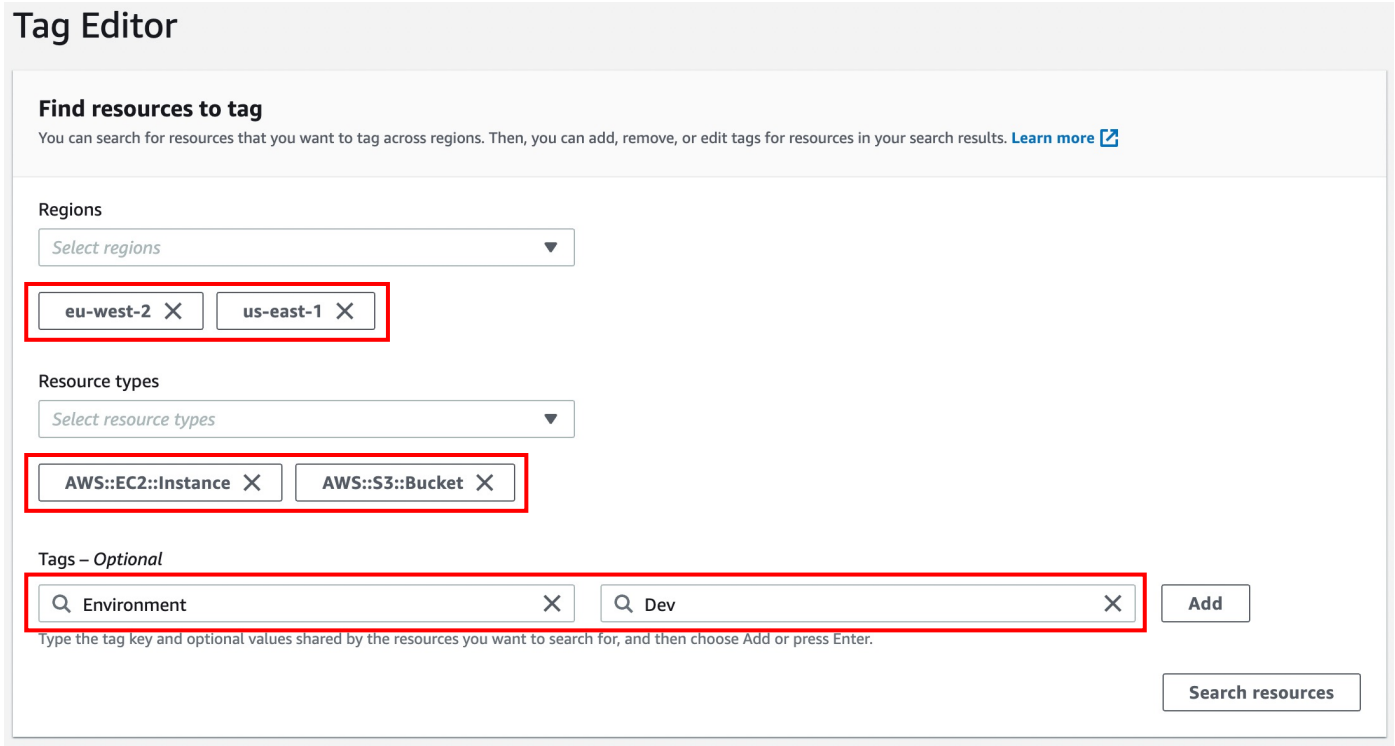
- With Tags we can track resources that relate to each other
- With Cost Allocation Tags we can enable detailed costing reports
- **Just like Tags, but they show up as columns in Reports**
- AWS Generated Cost Allocation Tags
  - Automatically applied to the resource you create
  - Starts with Prefix **aws:** **(e.g. aws: createdBy)**
  - They're not applied to resources created before the activation
- User tags
  - Defined by the user
  - Starts with Prefix **user:**
- **Cost Allocation Tags just appear in the Billing Console**
- Takes up to 24 hours for the tags to show up in the report

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Tag Editor

- Allows you to manage tags of multiple resources at once
- You can add/update/delete tags
- Search tagged/untagged resources in all AWS Regions



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Trusted Advisor

- No need to install anything – high level AWS account assessment
- Analyze your AWS accounts and provides recommendation:
  - Cost Optimization
  - Performance
  - Security
  - Fault Tolerance
  - Service Limits
  - Operational Excellence
- Core Checks and recommendations – all customers
- Can enable weekly email notification from the console
- Full Trusted Advisor – Available for Business & Enterprise support plans
  - Ability to set CloudWatch alarms when reaching limits
  - Programmatic Access using AWS Support API

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Support Plans

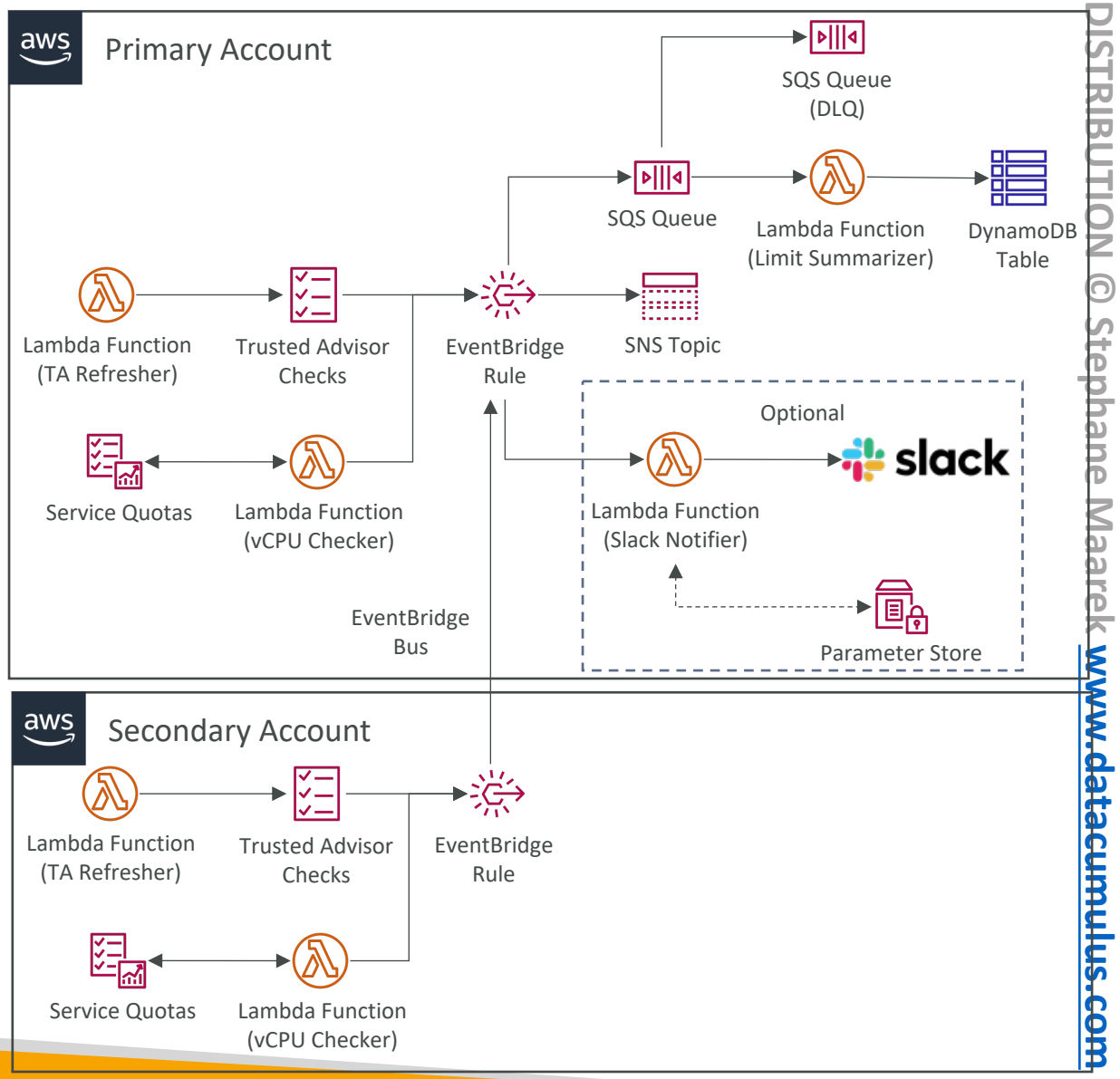
	Basic Support	Developer	Business	Enterprise
	*included for all AWS customers and **free***	*Recommended if you are experimenting or testing in AWS.*	*Recommended if you have production workloads in AWS.*	*Recommended if you have business and/or mission critical workloads in AWS.*
AWS Trusted Advisor Best Practice Checks	7 Core checks	7 Core checks	**Full set of checks + Programmatic Access using AWS Support API**	**Full set of checks + Programmatic Access using AWS Support API**
Enhanced Technical Support	24x7 access to customer service, documentation, whitepapers, and support forums.	Business hours email access to Cloud Support Associates Unlimited cases / 1 primary contact	24x7 phone, email, and chat access to Cloud Support Engineers Unlimited cases / unlimited contacts (IAM supported)	24x7 phone, email, and chat access to Cloud Support Engineers Unlimited cases / unlimited contacts (IAM supported)
Case Severity / Response Times		General guidance: < 24 business hours** System impaired: < 12 business hours**	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour Business-critical system down: < 15 minutes

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Trusted Advisor – Good to know

- Can check if an S3 bucket is made public
  - But cannot check for S3 objects that are public inside of your bucket!
  - Use Amazon EventBridge / S3 Events instead / AWS Config Rules
- Service Limits
  - **Limits can only be monitored** in Trusted Advisor (cannot be changed)
  - Cases must be created manually in **AWS Support Centre** to increase limits
  - OR use the **AWS Service Quotas** service

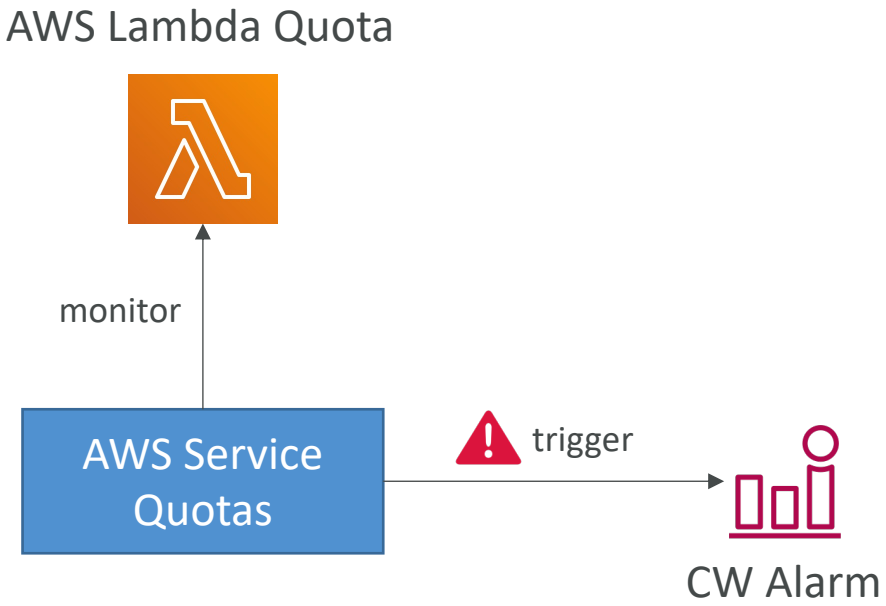


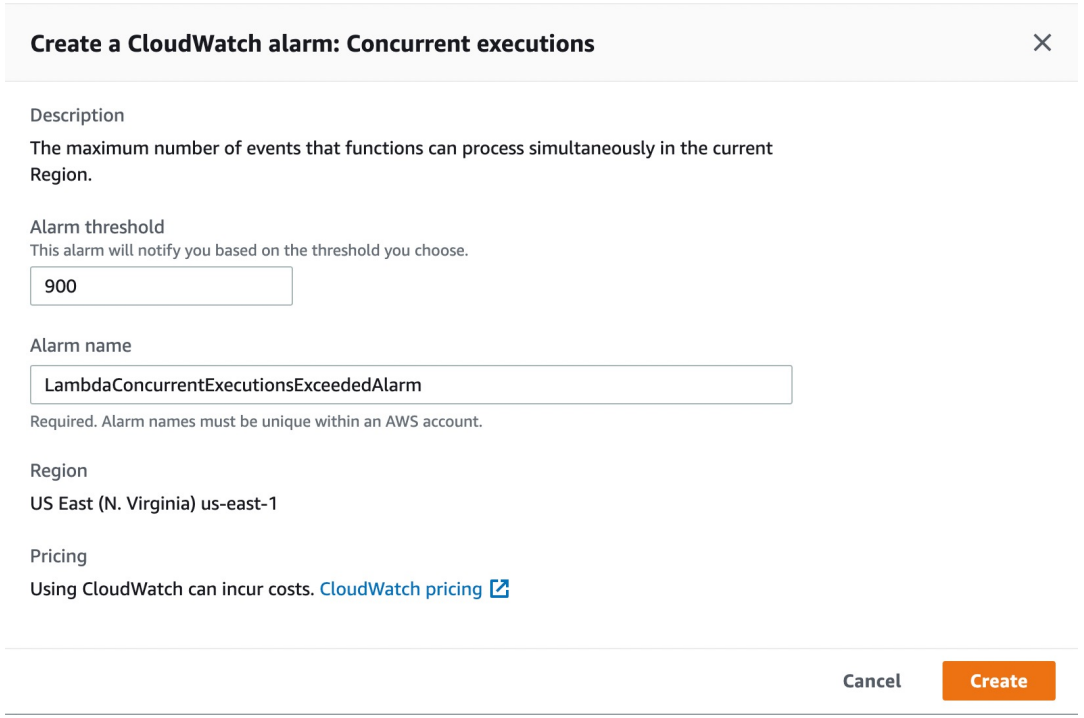
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

AWS Service Quotas

- Notify you when you're close to a service quota value threshold
- Create CloudWatch Alarms on the Service Quotas console
- Example: Lambda concurrent executions
- Helps you know if you need to request a quota increase or shutdown resources before limit is reached





© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EC2 Instance Launch Types

- **On Demand Instances:** short workload, predictable pricing, reliable
- **Spot Instances:** short workloads, for cheap, can lose instances (not reliable)
- **Reserved:** (MINIMUM 1 year)
  - **Reserved Instances:** long workloads
  - **Convertible Reserved Instances:** long workloads with flexible instances
- **Dedicated Instances:** no other customers will share your hardware
- **Dedicated Hosts:** book an entire physical server, control instance placement
  - Great for software licenses that operate at the core, or socket level
  - Can define **host affinity** so that instance reboots are kept on the same host

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Savings Plan

- New pricing model to get a discount based on long-term usage
- Commit to a certain type of usage: ex $10 per hour for 1 to 3 years
- Any usage beyond the savings plan is billed at the on-demand price

- **EC2 Instance Savings plan** (up to 72% - same discount as Standard RIs)
  - Select instance family (e.g. M5, C5…), and locked to a specific region
  - Flexible across size (m5.large to m5.4xlarge), OS (Windows to Linux), tenancy (dedicated or default)
- **Compute Savings plan** (up to 66% - same discount as Convertible RIs)
  - Ability to move between instance family (move from C5 to M5), region (Ireland to US), compute type (EC2, Fargate, Lambda), OS & tenancy
- **SageMaker Savings plan** (up to 64% off)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Storage Classes

- Amazon S3 Standard - General Purpose
- Amazon S3 Standard-Infrequent Access (IA)
- Amazon S3 One Zone-Infrequent Access
- Amazon S3 Glacier Instant Retrieval
- Amazon S3 Glacier Flexible Retrieval
- Amazon S3 Glacier Deep Archive
- Amazon S3 Intelligent Tiering

- Can move between classes manually or using S3 Lifecycle configurations

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 – Other Cost Savings

- **S3 Lifecycle Rules:** transition objects between tiers
- **Compress objects** to save space
- **S3 Requester Pays:**
  - In general, bucket owners pay for all Amazon S3 storage and data transfer costs associated with their bucket
  - With Requester Pays buckets, the requester instead of the bucket owner pays the cost of the request and the data download from the bucket
  - The bucket owner always pays the cost of storing data
  - Helpful when you want to share large datasets with other accounts
  - **If an IAM role is assumed, the owner account of that role pays for the request**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Storage Classes

- Amazon S3 Standard - General Purpose
- Amazon S3 Standard-Infrequent Access (IA)
- Amazon S3 One Zone-Infrequent Access
- Amazon S3 Glacier Instant Retrieval
- Amazon S3 Glacier Flexible Retrieval
- Amazon S3 Glacier Deep Archive
- Amazon S3 Intelligent Tiering

- Can move between classes manually or using S3 Lifecycle configurations

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Durability and Availability

- Durability:
  - High durability (99.999999999%, 11 9's) of objects across multiple AZ
  - If you store 10,000,000 objects with Amazon S3, you can on average expect to incur a loss of a single object once every 10,000 years
  - Same for all storage classes

- Availability:
  - Measures how readily available a service is
  - Varies depending on storage class
  - Example: S3 standard has 99.99% availability = not available 53 minutes a year

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Standard – General Purpose

- 99.99% Availability
- Used for frequently accessed data
- Low latency and high throughput
- Sustain 2 concurrent facility failures

- Use Cases: Big Data analytics, mobile & gaming applications, content distribution…

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Storage Classes – Infrequent Access

- For data that is less frequently accessed, but requires rapid access when needed
- Lower cost than S3 Standard

- Amazon S3 Standard-Infrequent Access (S3 Standard-IA)
  - 99.9% Availability
  - Use cases: Disaster Recovery, backups

- Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)
  - High durability (99.999999999%) in a single AZ; data lost when AZ is destroyed
  - 99.5% Availability
  - Use Cases: Storing secondary backup copies of on-premise data, or data you can recreate

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon S3 Glacier Storage Classes

- Low-cost object storage meant for archiving / backup
- Pricing: price for storage + object retrieval cost

- **Amazon S3 Glacier Instant Retrieval**
  - Millisecond retrieval, great for data accessed once a quarter
  - Minimum storage duration of 90 days
- **Amazon S3 Glacier Flexible Retrieval** (formerly Amazon S3 Glacier):
  - Expedited (1 to 5 minutes), Standard (3 to 5 hours), Bulk (5 to 12 hours) – free
  - Minimum storage duration of 90 days
- **Amazon S3 Glacier Deep Archive – for long term storage:**
  - Standard (12 hours), Bulk (48 hours)
  - Minimum storage duration of 180 days

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Intelligent-Tiering

- Small monthly monitoring and auto-tiering fee
- Moves objects automatically between Access Tiers based on usage
- There are no retrieval charges in S3 Intelligent-Tiering

- *Frequent Access tier (automatic):* default tier
- *Infrequent Access tier (automatic)*: objects not accessed for 30 days
- *Archive Instant Access tier (automatic):* objects not accessed for 90 days
- *Archive Access tier (optional):* configurable from 90 days to 700+ days
- *Deep Archive Access tier (optional):* config. from 180 days to 700+ days

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Storage Classes Comparison

	Standard	Intelligent-Tiering	Standard-IA	One Zone-IA	Glacier Instant Retrieval	Glacier Flexible Retrieval	Glacier Deep Archive
Durability	99.999999999% == (11 9's)						
Availability	99.99%	99.9%	99.9%	99.5%	99.9%	99.99%	99.99%
Availability SLA	99.9%	99%	99%	99%	99%	99.9%	99.9%
Availability Zones	>= 3	>= 3	>= 3	1	>= 3	>= 3	>= 3
Min. Storage Duration Charge	None	None	30 Days	30 Days	90 Days	90 Days	180 Days
Min. Billable Object Size	None	None	128 KB	128 KB	128 KB	40 KB	40 KB
Retrieval Fee	None	None	Per GB retrieved	Per GB retrieved	Per GB retrieved	Per GB retrieved	Per GB retrieved

https://aws.amazon.com/s3/storage-classes/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

S3 Storage Classes – Price Comparison Example: us-east- l

	Standard	Intelligent-Tiering	Standard-IA	One Zone-IA	Glacier Instant Retrieval	Glacier Flexible Retrieval	Glacier Deep Archive
Storage Cost (per GB per month)	$0.023	$0.0025 - $0.023	$0.0125	$0.01	$0.004	$0.0036	$0.00099
Retrieval Cost (per 1000 request)	**GET:** $0.0004 **POST:** $0.005	**GET:** $0.0004 **POST:** $0.005	**GET:** $0.001 **POST:** $0.01	**GET:** $0.001 **POST:** $0.01	**GET:** $0.01 **POST:** $0.02	**GET:** $0.0004 **POST:** $0.03 **Expedited:** $10 **Standard:** $0.05 **Bulk:** free	**GET:** $0.0004 **POST:** $0.05 **Standard:** $0.10 **Bulk:** $0.025
Retrieval Time	Instantaneous					**Expedited** (1 – 5 mins) **Standard** (3 – 5 hours) Bulk (5 – 12 hours)	**Standard** (12 hours) **Bulk** (48 hours)
Monitoring Cost (per 1000 objects)		$0.0025					

https://aws.amazon.com/s3/pricing/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Budgets

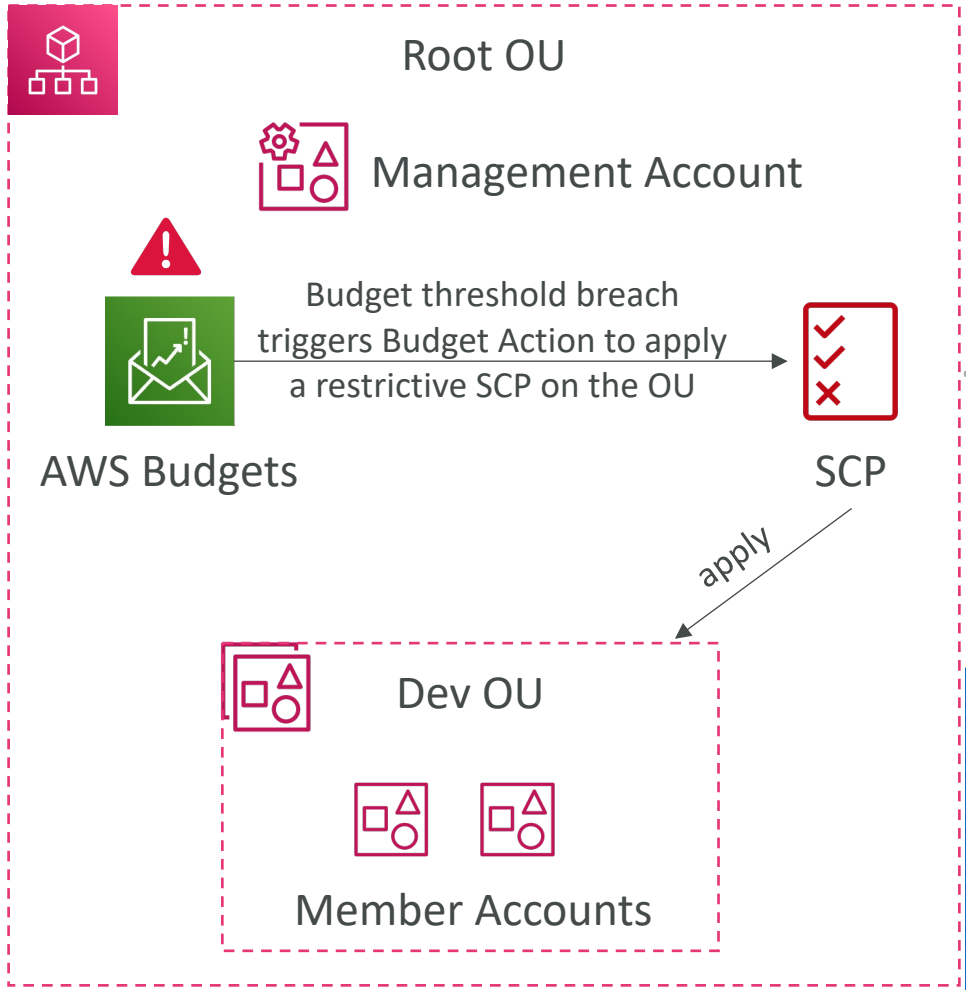
- Create budget and **send alarms when costs exceeds the budget**
- 4 types of budgets: Usage, Cost, Reservation, Savings Plans
- For Reserved Instances (RI)
  - Track utilization
  - Supports EC2, ElastiCache, RDS, Redshift
- Up to 5 SNS notifications per budget
- Can filter by: Service, Linked Account, Tag, Purchase Option, Instance Type, Region, Availability Zone, API Operation, etc…
- Same options as AWS Cost Explorer!
- 2 budgets are free, then $0.02/day/budget

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Budget Actions

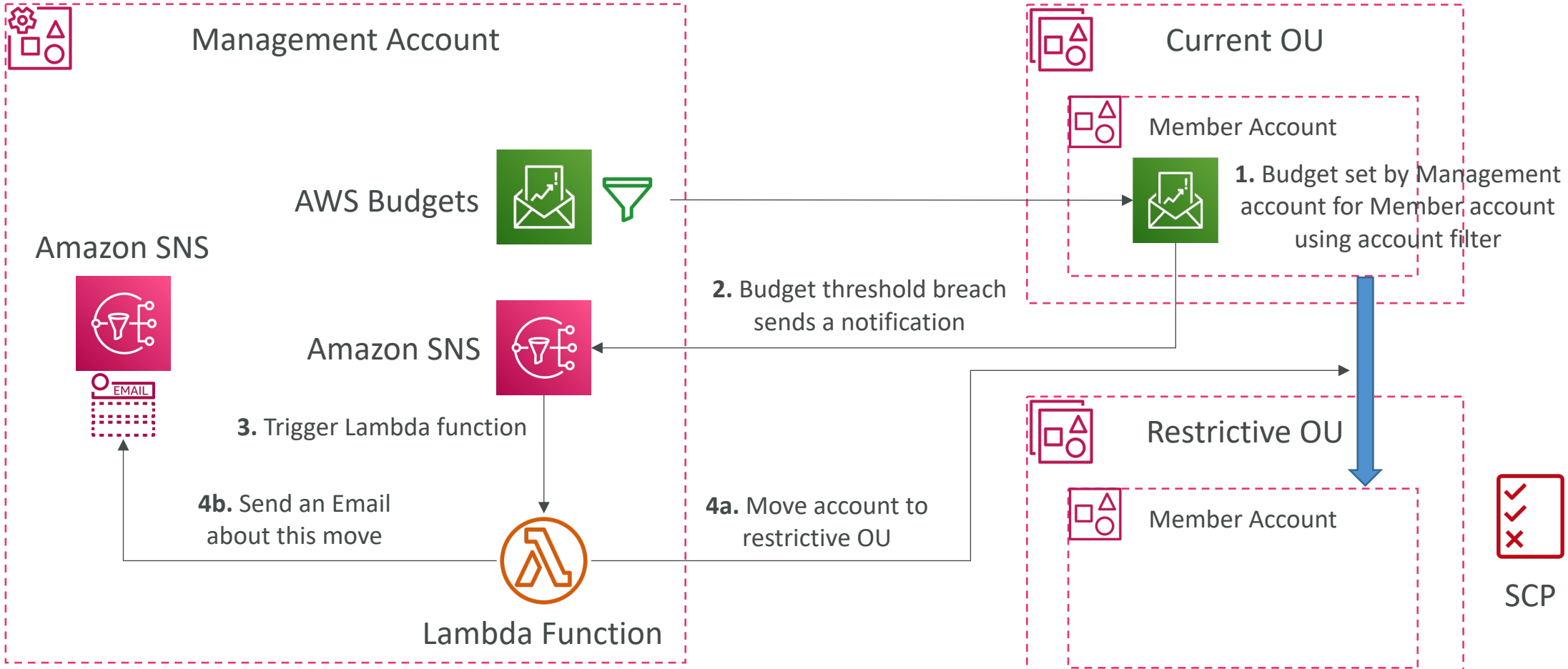
- Run actions on your behalf when a budget exceeds a certain cost or usage threshold
- Supports 3 action types:
  - Applying an IAM Policy to a user, group, or IAM role
  - Applying Service Control Policy (SCP) to an OU
  - Stop EC2 or RDS Instances
- Actions can be executed automatically or require a workflow approval process
- Reduces unintentional overspending in your account



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

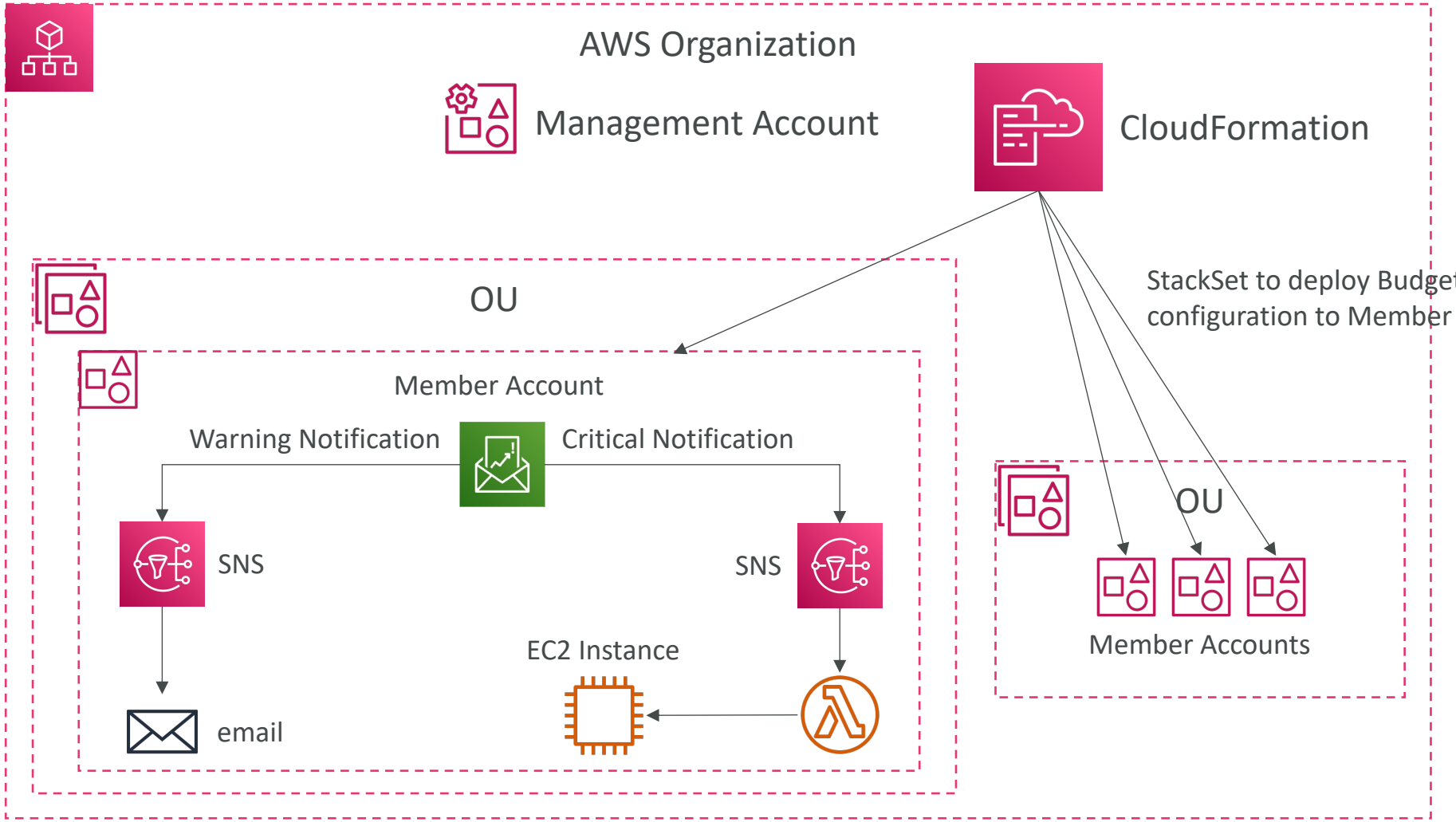
Centralized Budget Management



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Decentralized Budget Management



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

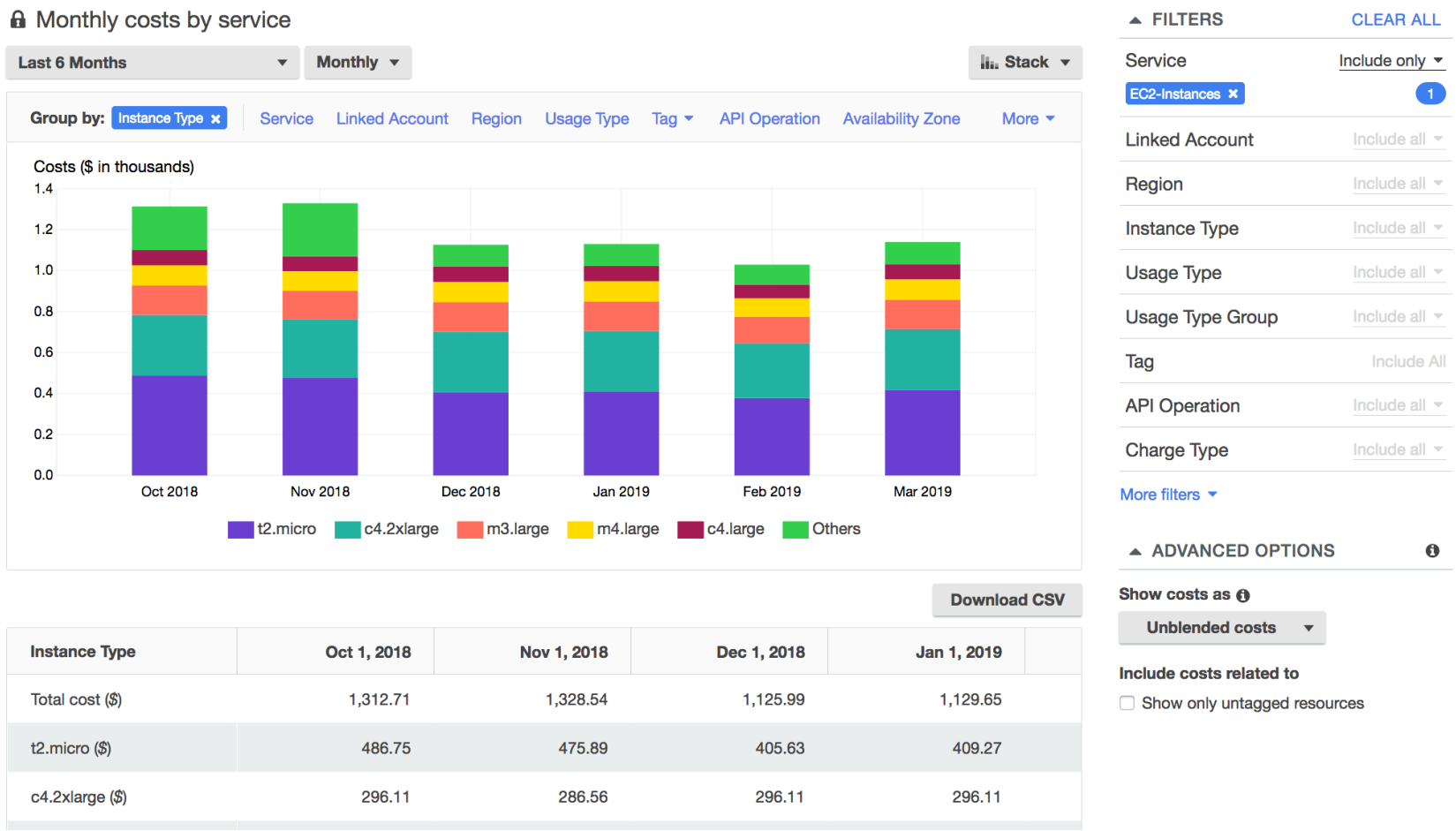
Cost Explorer

- Visualize, understand, and manage your AWS costs and usage over time
- Create custom reports that analyze cost and usage data.
- Analyze your data at a high level: total costs and usage across all accounts
- Or Monthly, hourly, resource level granularity
- Choose an optimal **Savings Plan** (to lower prices on your bill)
- **Forecast usage up to 12 months based on previous usage**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cost Explorer – Example



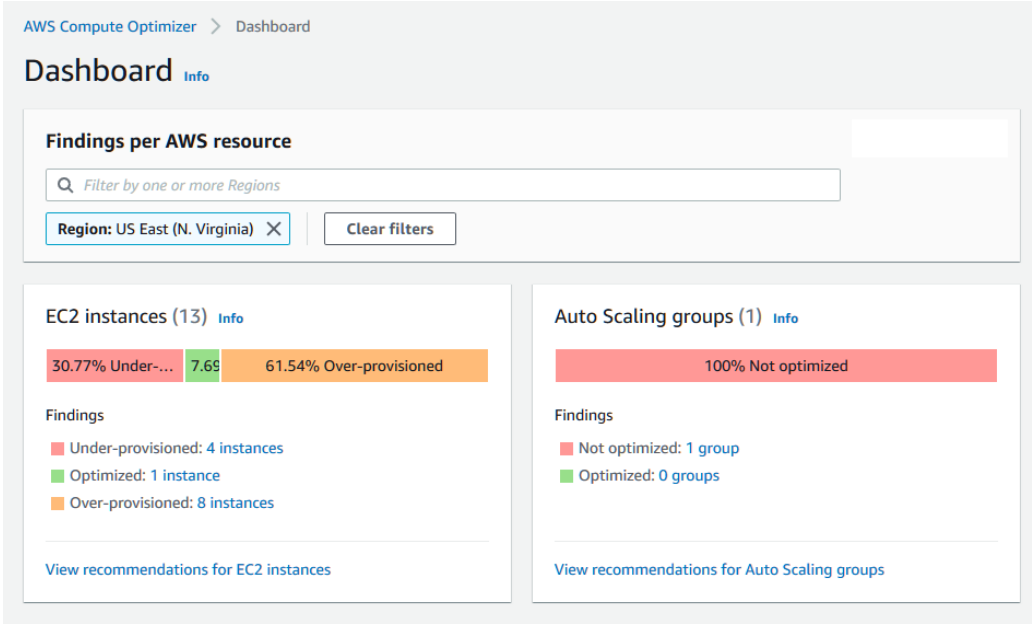
Instance Type	Oct 1, 2018	Nov 1, 2018	Dec 1, 2018	Jan 1, 2019
Total cost ($)	1,312.71	1,328.54	1,125.99	1,129.65
t2.micro ($)	486.75	475.89	405.63	409.27
c4.2xlarge ($)	296.11	286.56	296.11	296.11

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Compute Optimizer

- **Reduce costs** and **improve performance by** recommending optimal AWS resources for your workloads
- Helps you choose optimal configurations and right-size your workloads (over/under provisioned)
- Uses Machine Learning to analyze your **resources' configurations** and their **utilization CloudWatch metrics**
- Supported resources
  - EC2 instances
  - EC2 Auto Scaling Groups
  - EBS volumes
  - Lambda functions
- Lower your costs by up to 25%
- Recommendations can be exported to S3

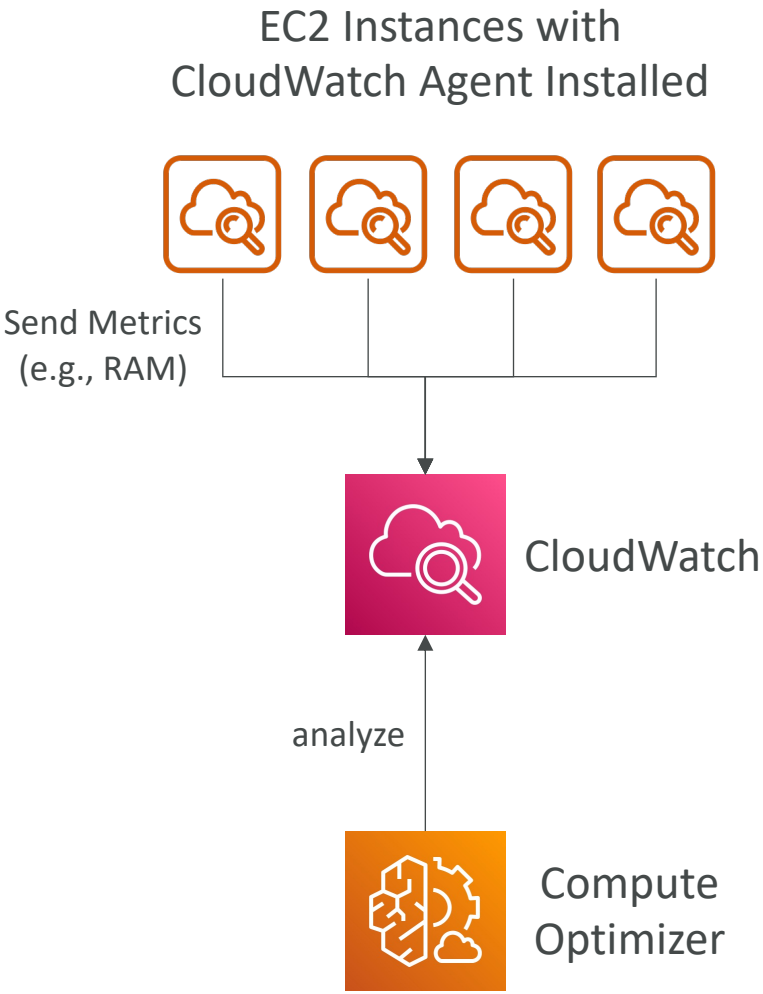


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Compute Optimizer – CloudWatch Agent

- Needed to analyze Memory Utilization
- Not needed for CPU, NetworkIn/Out, DiskReadOps, DiskWriteOps, …



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Reserved Instances

- Reserved Instances in an AWS Organization
  - all accounts share the Reserved Instances and Savings Plan
  - The payer account (Management account) of an organization can turn off Reserved Instance (RI) discount and Savings Plans discount sharing for any accounts in that organization, including the payer account

- Renewal of Reserved Instances
  - You can queue (schedule and reserve ahead of time) your reserved instances
  - To renew a RI, just queue an RI purchase whenever the previous one expires

© Stephane Maarek

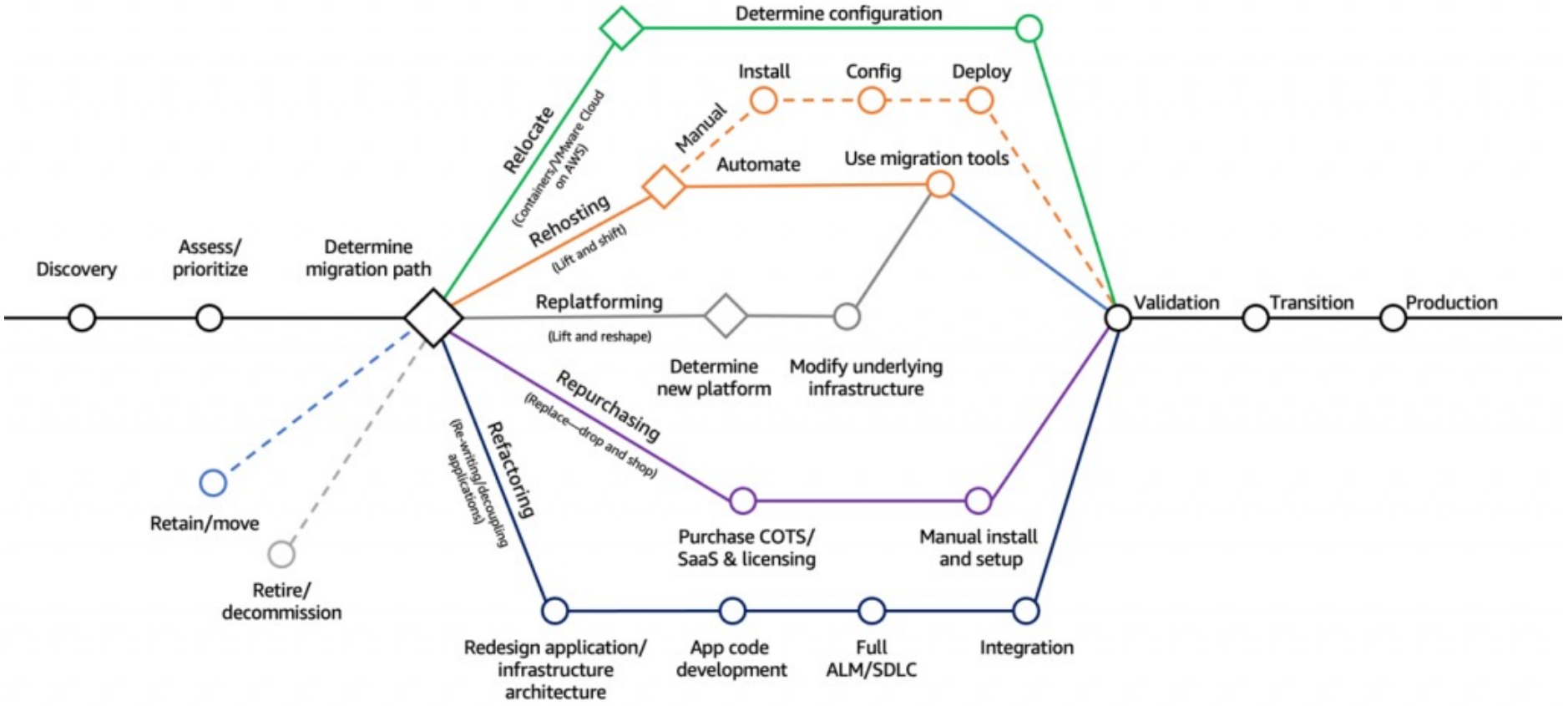
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Migrations

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cloud Migration Strategies: The 7Rs



https://aws.amazon.com/blogs/enterprise-strategy/new-possibilities-seven-strategies-to-accelerate-your-application-migration-to-aws/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cloud Migration Strategies: The 7Rs

- Retire
  - Turn off things you don't need (maybe as a result of Re-architecting)
  - Helps with reducing the surface areas for attacks (more security)
  - Save cost, maybe up to 10% to 20%
  - Focus your attention on resources that must be maintained

- Retain
  - Do nothing for now (it's still a decision to make in a Cloud Migration)
  - Security, data compliance, performance, unresolved dependencies
  - No business value to migrate, mainframe or mid-range and non-x86 Unix apps

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cloud Migration Strategies: The 7Rs

- Relocate
  - Move apps from on-premises to its Cloud version
  - Move EC2 instances to a different VPC, AWS account or AWS Region
  - Example: transfer servers from VMware Software-defined Data Center (SSDC) to VMware Cloud on AWS

- Rehost *"lift and shift"*
  - Simple migrations by re-hosting on AWS (applications, databases, data…)
  - Migrate machines (physical, virtual, another Cloud) to AWS Cloud
  - No cloud optimizations being done, applications is migrated as is
  - Could save as much as 30% on cost
  - Example: Migrate using AWS Application Migration Service

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cloud Migration Strategies: The 7Rs

- Replatform *"lift and reshape"*
  - Example: migrate your database to RDS
  - Example: migrate your application to Elastic Beanstalk
  - Not changing the core architecture, but leverage some Cloud optimizations
  - Save time and money by moving to a fully managed service or Serverless

- Repurchase *"drop and shop"*
  - Moving to a different product while moving to the Cloud
  - Often you move to a SaaS platform
  - Expensive in the short term, but quick to deploy
  - Example: CRM to Salesforce.com, HR to Workday, CMS to Drupal

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cloud Migration Strategies: The 7Rs

- **Refactor / Re-architect**
  - Reimagining how the application is architected using Cloud Native features
  - Driven by the need of the business to add features and improve scalability, performance, security, and agility
  - Move from a monolithic application to micro-services
  - Example: move an application to Serverless architectures, use AWS S3

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Storage Gateway

- Bridge between on-premises data and cloud data
- Use cases:
  - disaster recovery
  - backup & restore
  - tiered storage
  - on-premises cache & low-latency files access
- Types of Storage Gateway:
  - S3 File Gateway
  - Volume Gateway
  - Tape Gateway

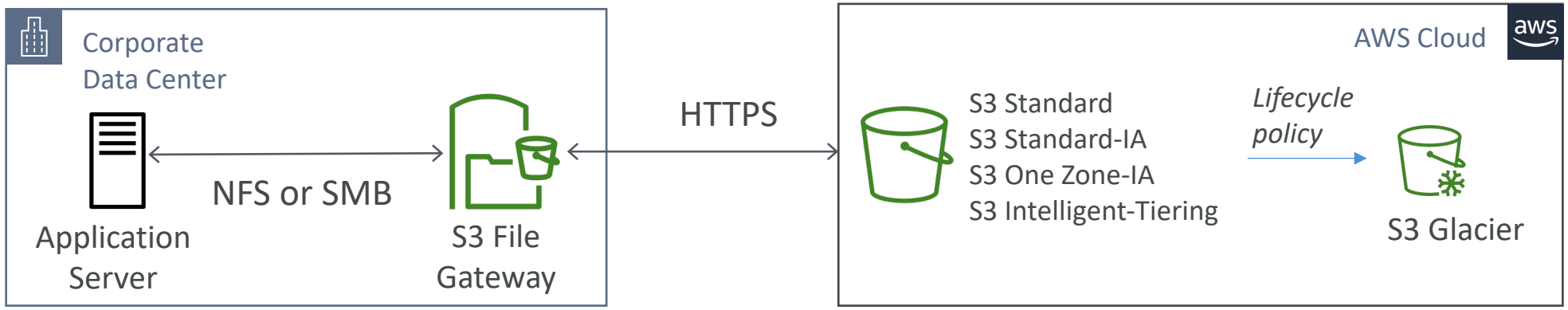
Storage Gateway

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon S3 File Gateway

- Configured S3 buckets are accessible using the NFS and SMB protocol
- **Most recently used data is cached in the file gateway**
- Supports S3 Standard, S3 Standard IA, S3 One Zone A, S3 Intelligent Tiering
- **Transition to S3 Glacier using a Lifecycle Policy**
- Bucket access using IAM roles for each File Gateway
- SMB Protocol has integration with Active Directory (AD) for user authentication

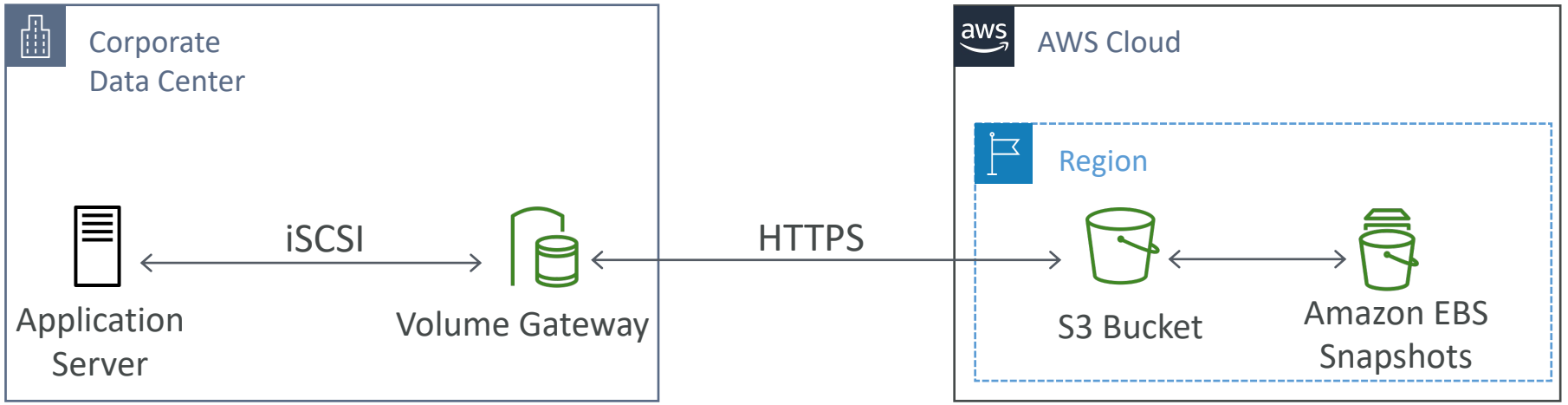


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Volume Gateway

- Block storage using iSCSI protocol backed by S3
- Backed by EBS snapshots which can help restore on-premises volumes!
- **Cached volumes:** low latency access to most recent data
- **Stored volumes:** entire dataset is on premise, scheduled backups to S3

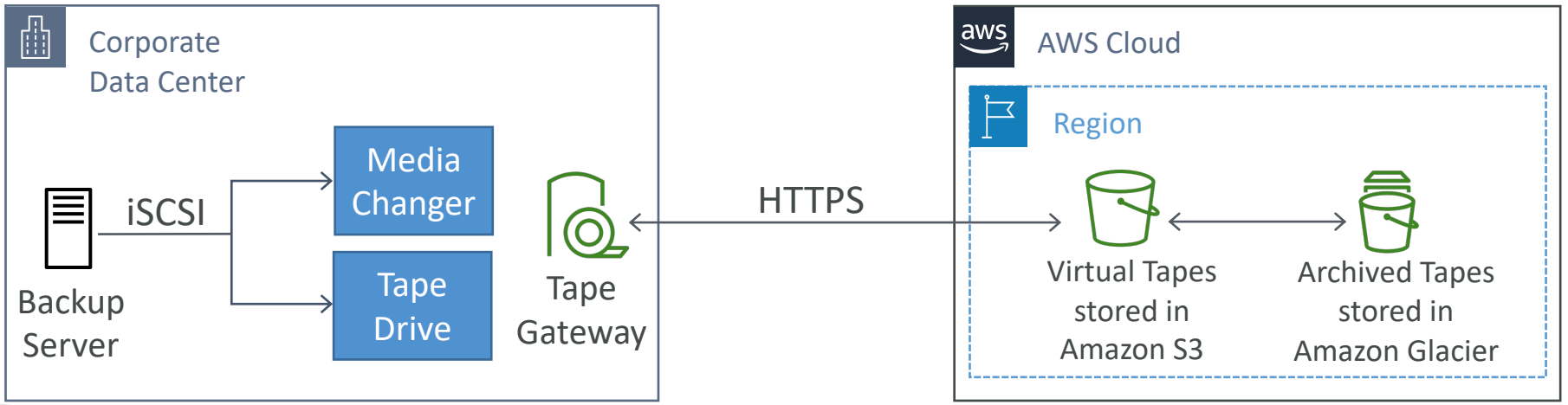


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Tape Gateway

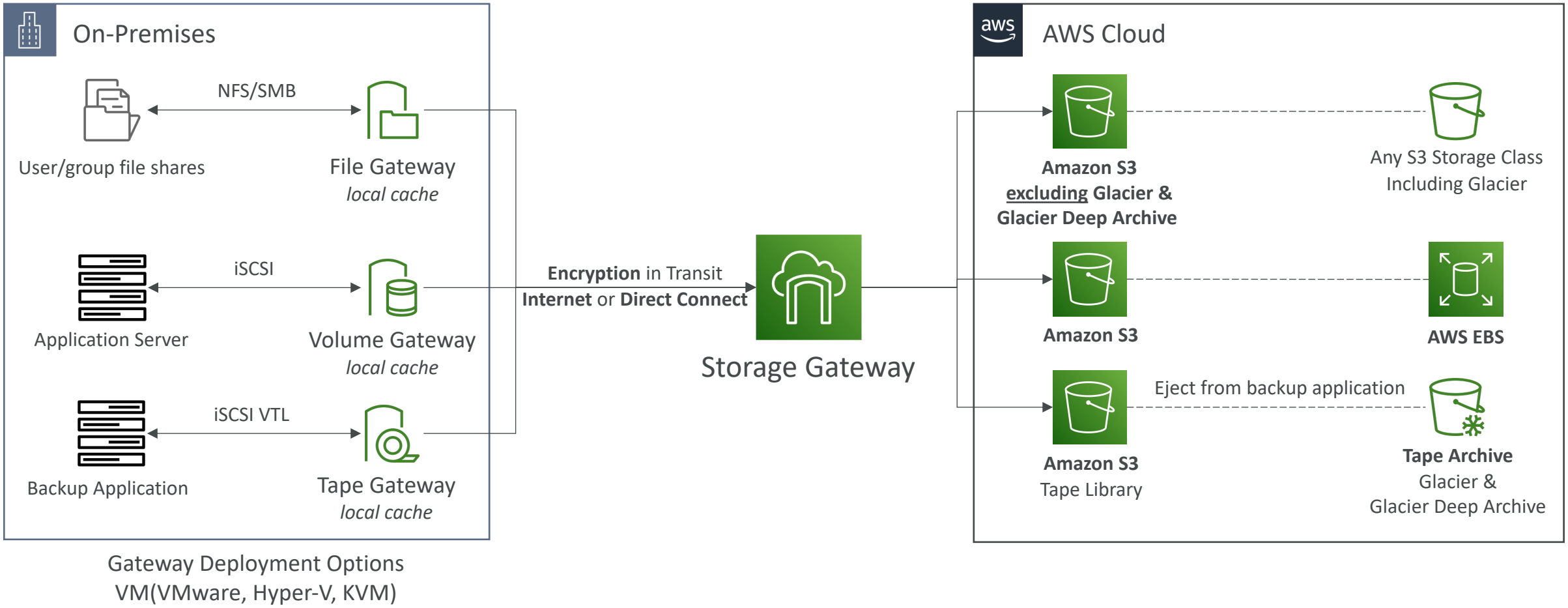
- Some companies have backup processes using physical tapes (!)
- With Tape Gateway, companies use the same processes but, in the cloud
- Virtual Tape Library (VTL) backed by Amazon S3 and Glacier
- Back up data using existing tape-based processes (and iSCSI interface)
- Works with leading backup software vendors



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Storage Gateway

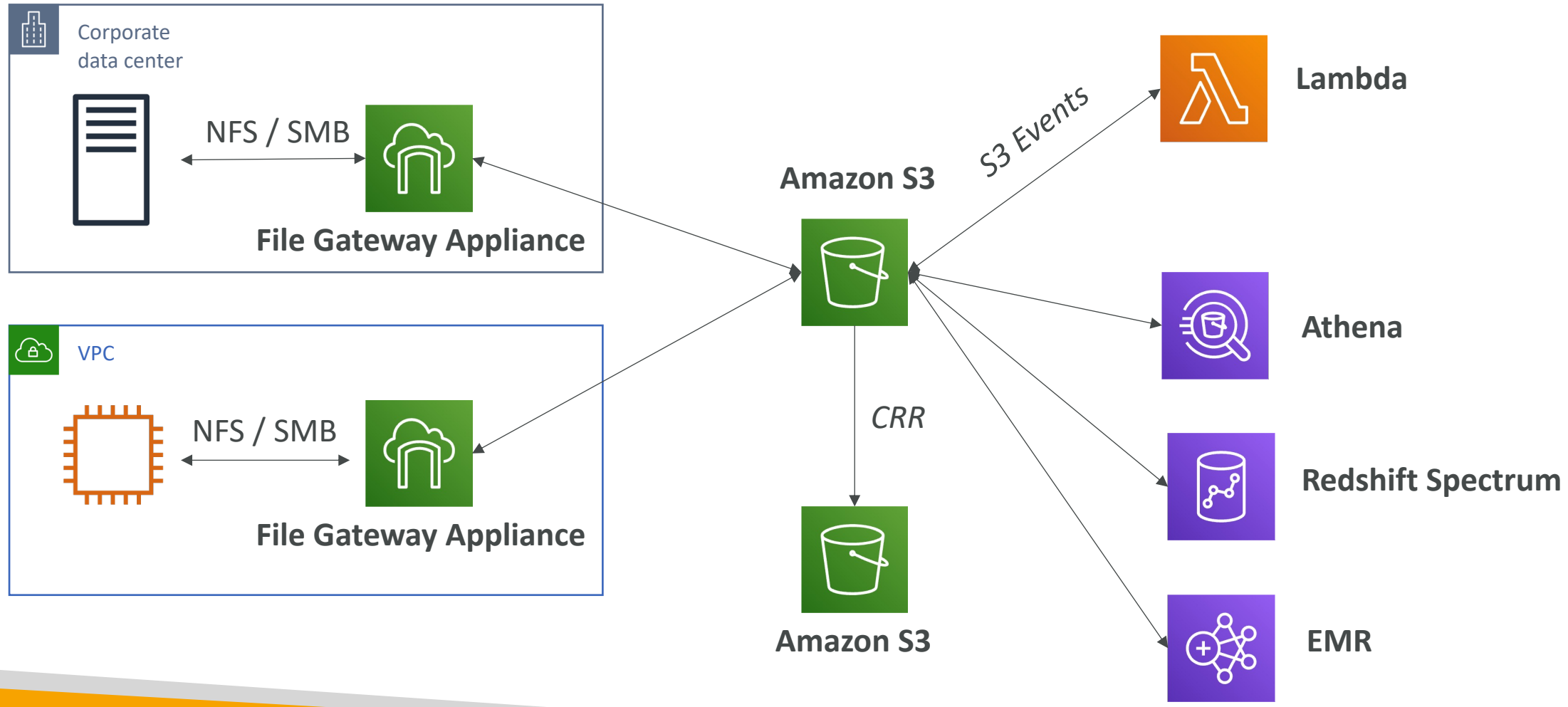


Gateway Deployment Options
VM(VMware, Hyper-V, KVM)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

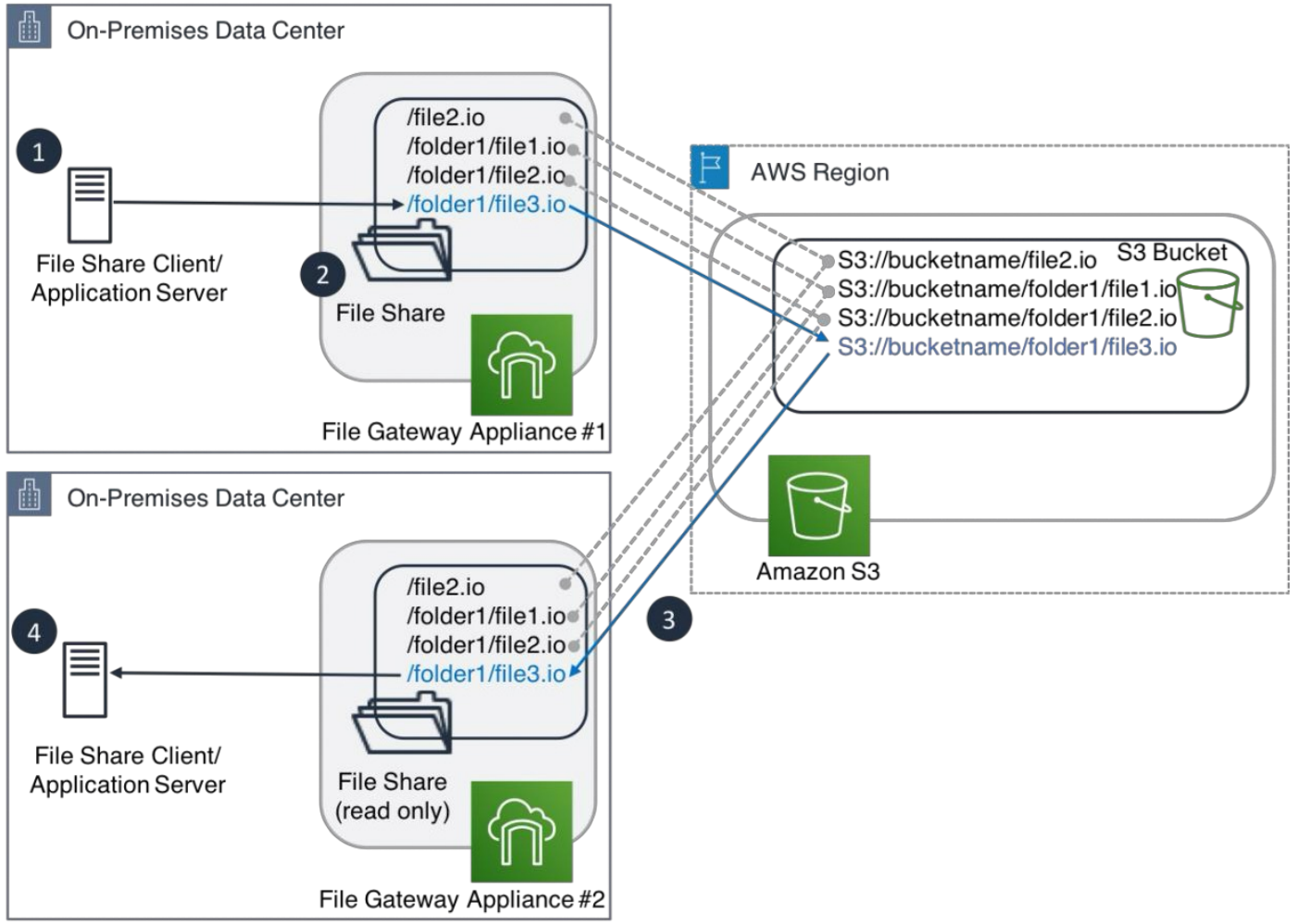
File Gateway: Extensions



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

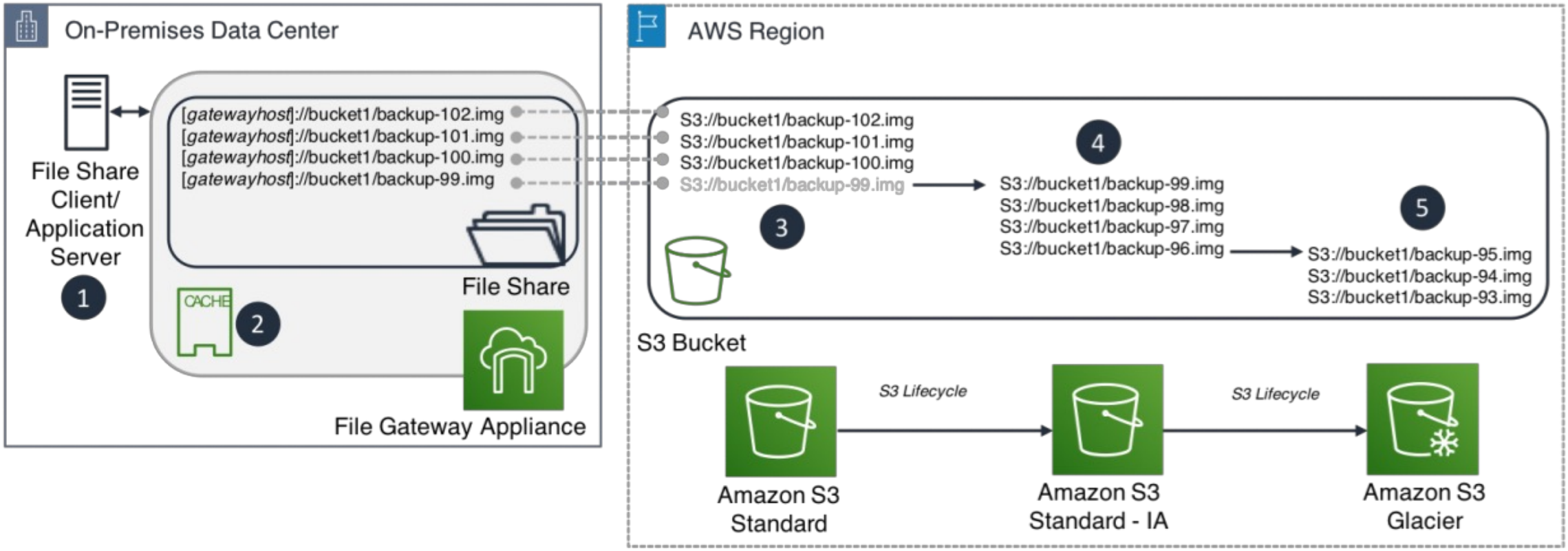
File Gateway: Read Only Replicas



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

File Gateway: Backup and Lifecycle Policies



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

File Architectures: Other possibilities

- Amazon S3 Object Versioning
  - Ability to store multiple object versions as they are modified
  - Helpful to restore a file to a previous version
  - Could restore an entire file system to a previous version
  - Must use the "RefreshCache" API on the Gateway to be notified of restore

- Amazon S3 Object Lock
  - Enables to have the File Gateway for Write Once Read Many (WORM) data
  - If there are file modifications or renames in the file share clients, the file gateway creates a new version of the object without affecting priori versions, and the original locked version will remain unchanged

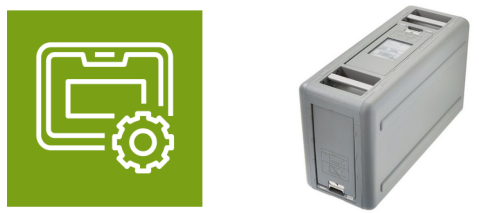
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Snowball

- Highly-secure, portable devices to **collect and process data at the edge**, and **migrate data into and out of AWS**
- Helps migrate up to Petabytes of data

Device	Compute	Memory	Storage (SSD)
Snowball Edge **Storage** Optimized	104 vCPUs	416 GB	**210 TB**
Snowball Edge **Compute** Optimized	104 vCPUs	416 GB	**28 TB**



Snowball Edge

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Data Migrations with Snowball

	Time to Transfer		
	100 Mbps	**1Gbps**	**10Gbps**
10 TB	12 days	30 hours	3 hours
100 TB	124 days	12 days	30 hours
1 PB	3 years	124 days	12 days

Challenges:

- Limited connectivity
- Limited bandwidth
- High network cost
- Shared bandwidth (can't maximize the line)
- Connection stability

AWS Snowball: offline devices to perform data migrations

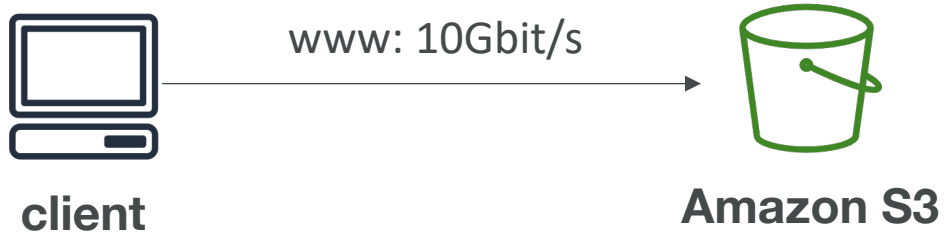
If it takes more than a week to transfer over the network, use Snowball devices!

© Stephane Maarek

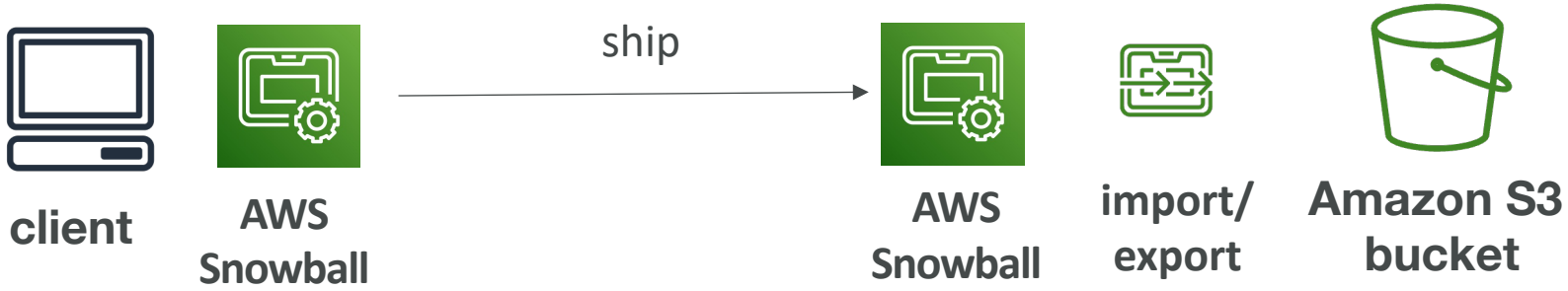
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Diagrams

- Direct upload to S3:



- With Snowball:

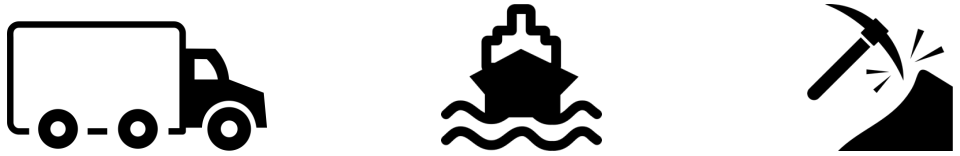


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

What is Edge Computing?

- Process data while it's being created on **an edge location**
  - A truck on the road, a ship on the sea, a mining station underground...



- These locations may have limited internet and no access to computing power
- We setup a **Snowball Edge** device to do edge computing
  - Snowball Edge Compute Optimized (dedicated for that use case) & Storage Optimized
  - Run EC2 Instances or Lambda functions at the edge
- Use cases: preprocess data, machine learning, transcoding media

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Snow Family – Improving Transfer Performance

- Most impactful to least:
  - **Perform multiple write operations at one time** - from multiple terminals
  - **Transfer small files in batches** – zip up small files until at least 1MB
  - **Don't perform other operations on files during transfer**
  - **Reduce local network use**
  - **Eliminate unnecessary hops** – directly connect to the computer

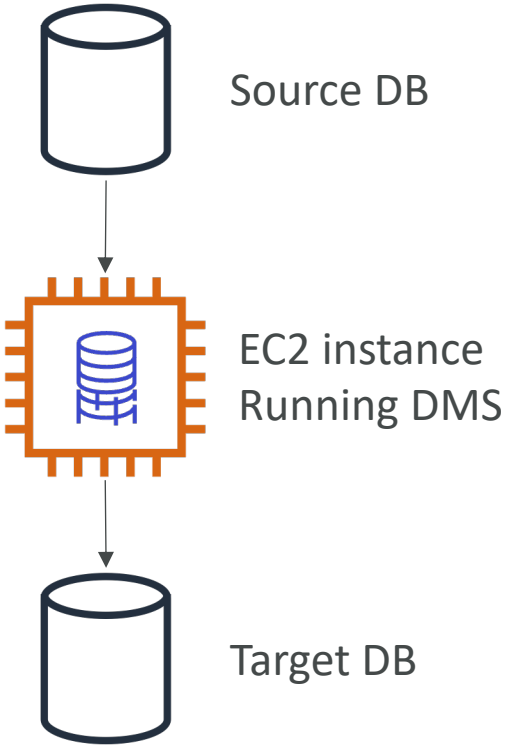
- The data transfer rate using the file interface is typically between 25 MB/s and 40 MB/s. If you need to transfer data faster than this, use the **Amazon S3 Adapter for Snowball**, which has a data transfer rate typically between 250 MB/s and 400 MB/s

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DMS – Database Migration Service

- Quickly and securely migrate databases to AWS, resilient, self healing
- The source database remains available during the migration
- Supports:
  - Homogeneous migrations: ex Oracle to Oracle
  - Heterogeneous migrations: ex Microsoft SQL Server to Aurora
- Continuous Data Replication using CDC
- You must create an EC2 instance to perform the replication tasks



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DMS Sources and Targets

SOURCES:

- On-premises and EC2 instances databases: *Oracle, MS SQL Server, MySQL, MariaDB, PostgreSQL, MongoDB, SAP, DB2*
- Azure: *Azure SQL Database*
- Amazon RDS: all including Aurora
- Amazon S3
- DocumentDB

TARGETS:

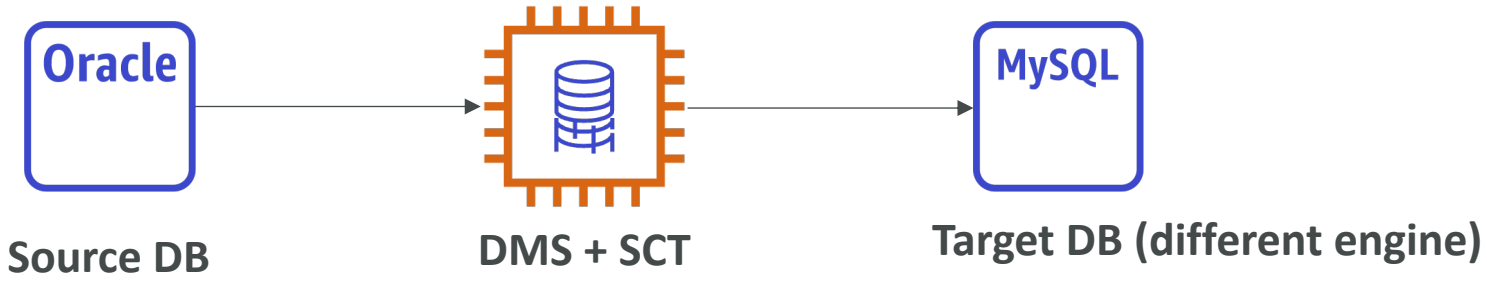
- On-premises and EC2 instances databases: Oracle, MS SQL Server, MySQL, MariaDB, PostgreSQL, SAP
- Amazon RDS including Aurora
- Amazon Redshift
- Amazon DynamoDB
- Amazon S3
- OpenSearch Service
- Kinesis Data Streams
- DocumentDB

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Schema Conversion Tool (SCT)

- Convert your Database's Schema from one engine to another
- Example OLTP: (SQL Server or Oracle) to MySQL, PostgreSQL, Aurora
- Example OLAP: (Teradata or Oracle) to Amazon Redshift



- You do not need to use SCT if you are migrating the same DB engine
  - Ex: on-premises PostgreSQL => RDS PostgreSQL
  - The DB engine is still PostgreSQL (RDS is the platform)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DMS – Good things to know

- Works over VPC Peering, VPN (site to site, software), Direct Connect
- Supports Full Load, Full Load + CDC, or CDC only
- **Oracle:**
  - Source: Supports TDE for the source using "BinaryReader"
  - Target: Supports BLOBs in tables that have a primary key, and TDE
- **OpenSearch:**
  - Source: does not exist
  - Target: possible to migrate from a relational database using DMS
  - Therefore, DMS cannot be used to replicate OpenSearch data

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Snowball + Database Migration Service (DMS)

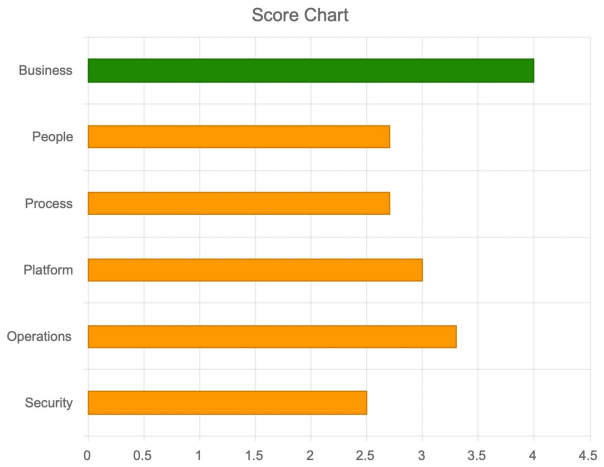
- Larger data migrations can include many terabytes of information.
- Can be limited due to network bandwidth or size of data
- AWS DMS can use Snowball Edge & Amazon S3 to speed up migration
- Following stages:
  1. You use the AWS Schema Conversion Tool (AWS SCT) to extract the data locally and move it to an Edge device.
  2. You ship the Edge device or devices back to AWS.
  3. After AWS receives your shipment, the Edge device automatically loads its data into an Amazon S3 bucket.
  4. AWS DMS takes the files and migrates the data to the target data store. If you are using change data capture (CDC), those updates are written to the Amazon S3 bucket and then applied to the target data store.

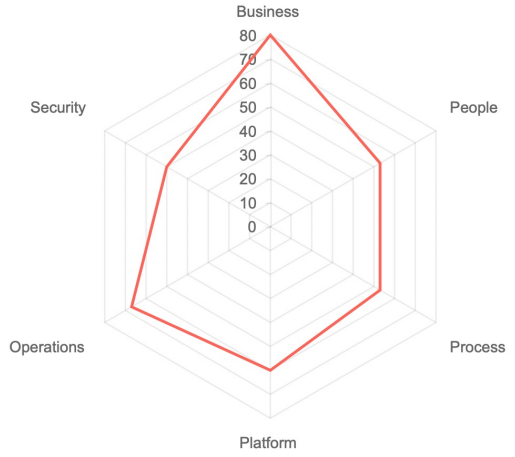
© Stephane Maarek

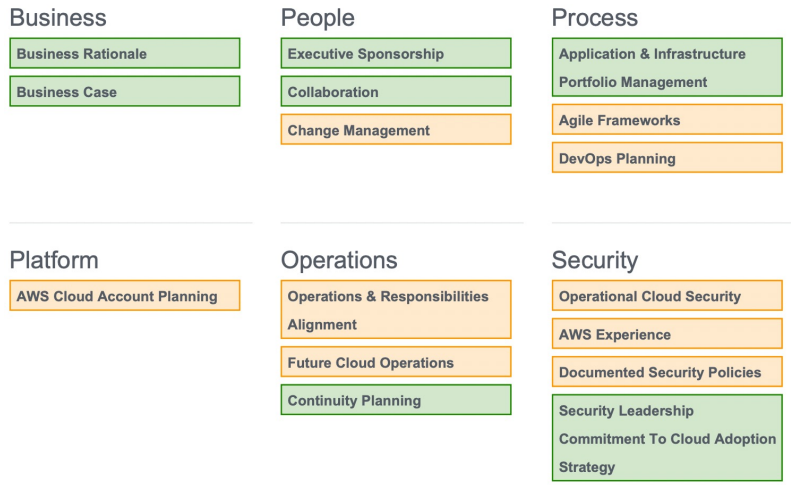
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Cloud Adoption Readiness Tool (CART)

- Helps organizations develop efficient and effective plans for cloud adoption and migrations
- Transforms your idea of moving to the cloud into a detailed plan that follows AWS best practices
- Answer a set of questions across six perspectives (business, people, process, platform, operations, security)
- Generates a custom report on your level of migration readiness







© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

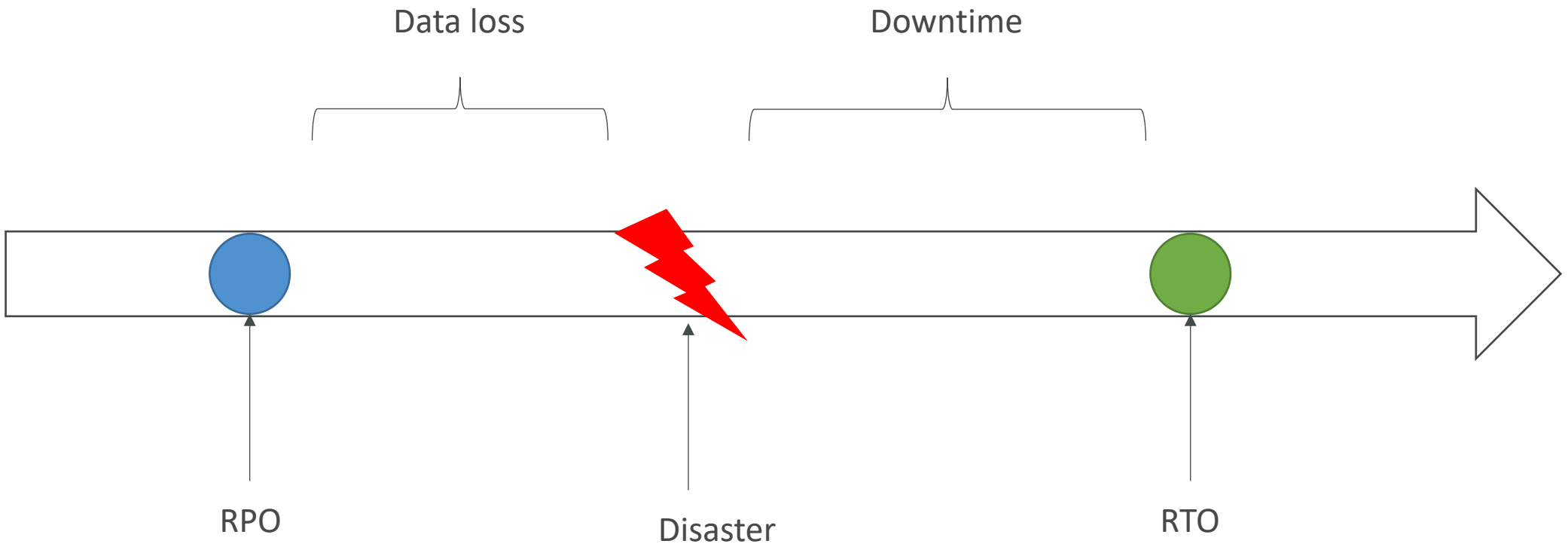
Disaster Recovery Overview

- Any event that has a negative impact on a company's business continuity or finances is a disaster
- Disaster recovery (DR) is about preparing for and recovering from a disaster
- What kind of disaster recovery?
  - on-premises => on-premises: traditional DR, and very expensive
  - on-premises => AWS Cloud: hybrid recovery
  - AWS Cloud Region A => AWS Cloud Region B
- Need to define two terms:
  - RPO: Recovery Point Objective
  - RTO: Recovery Time Objective

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

RPO and RTO

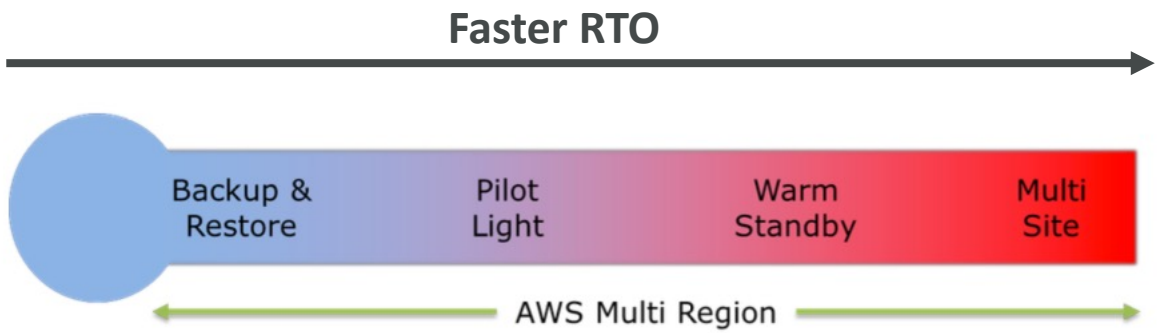


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Disaster Recovery Strategies

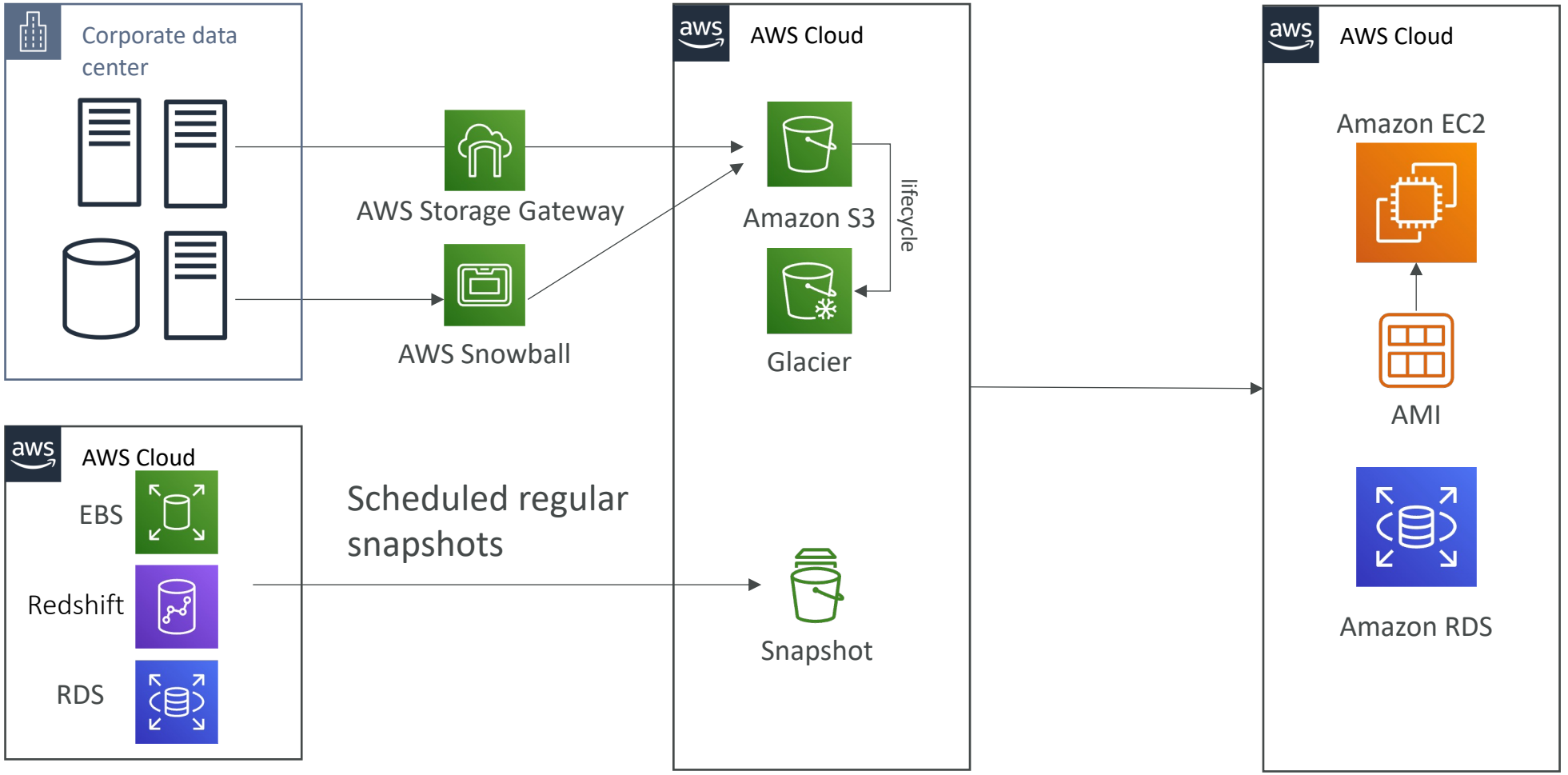
- Backup and Restore
- Pilot Light
- Warm Standby
- Hot Site / Multi Site Approach



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Backup and Restore (High RPO)

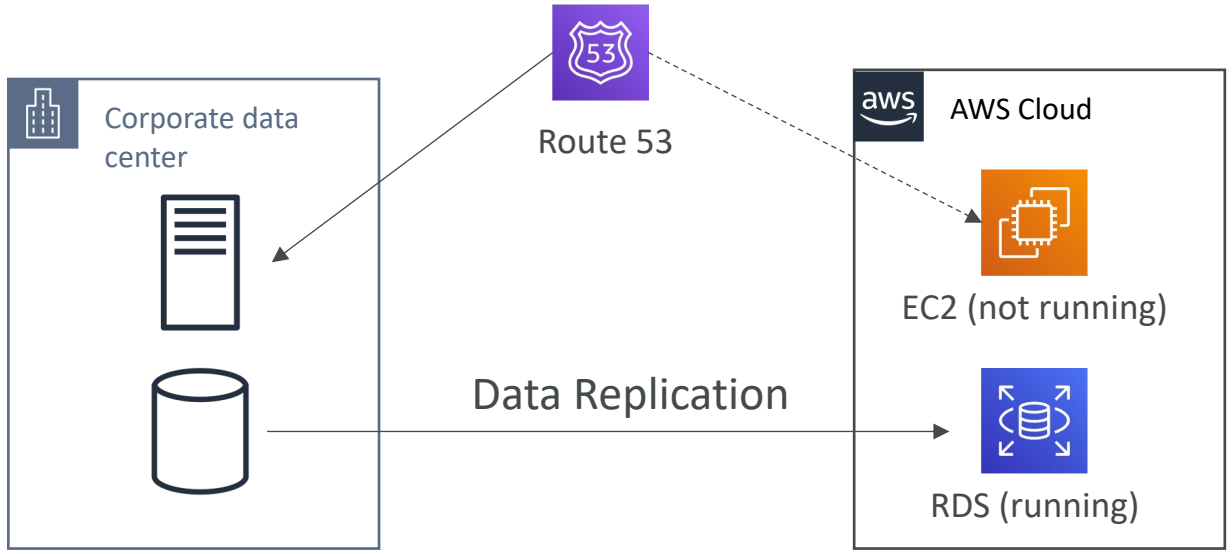


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Disaster Recovery – Pilot Light

- A small version of the app is always running in the cloud
- Useful for the critical core (pilot light)
- Very similar to Backup and Restore
- Faster than Backup and Restore as critical systems are already up

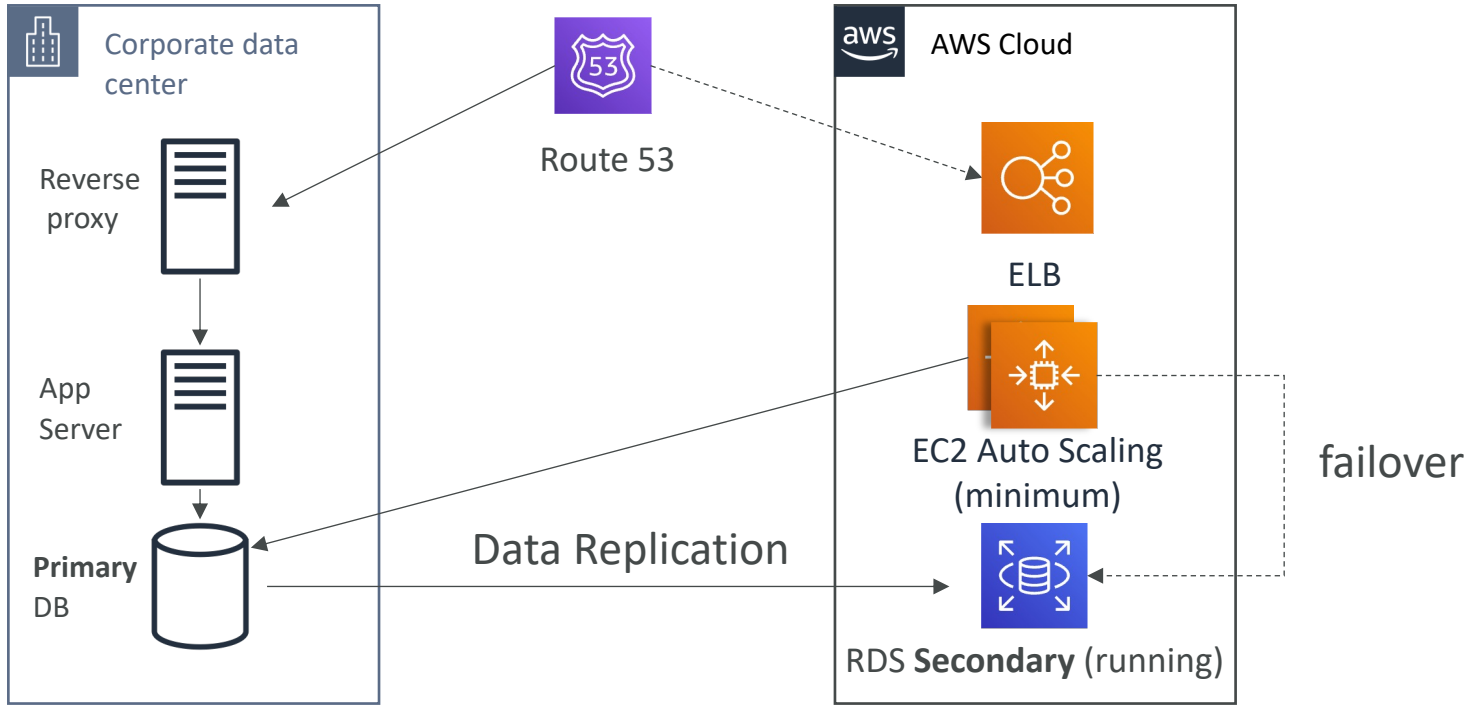


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Warm Standby

- Full system is up and running, but at minimum size
- Upon disaster, we can scale to production load

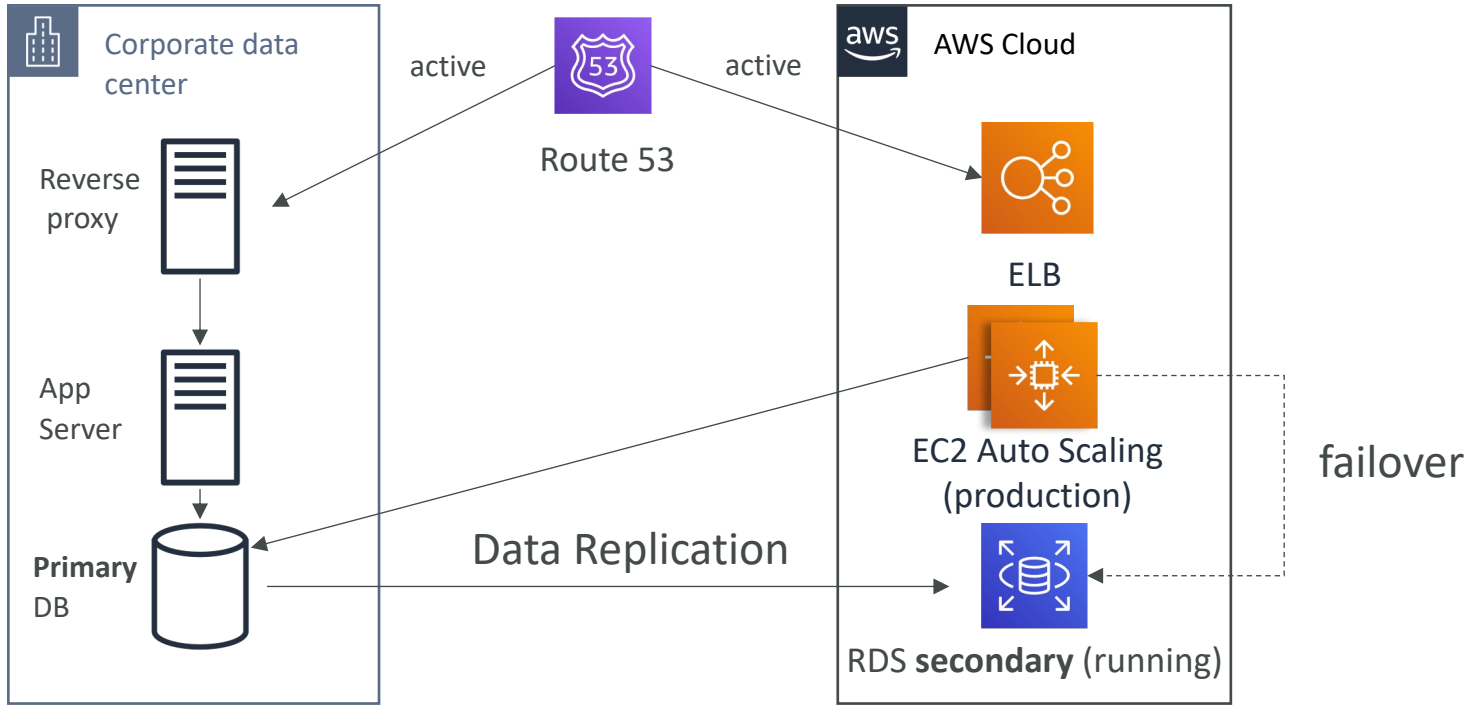


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Multi Site / Hot Site Approach

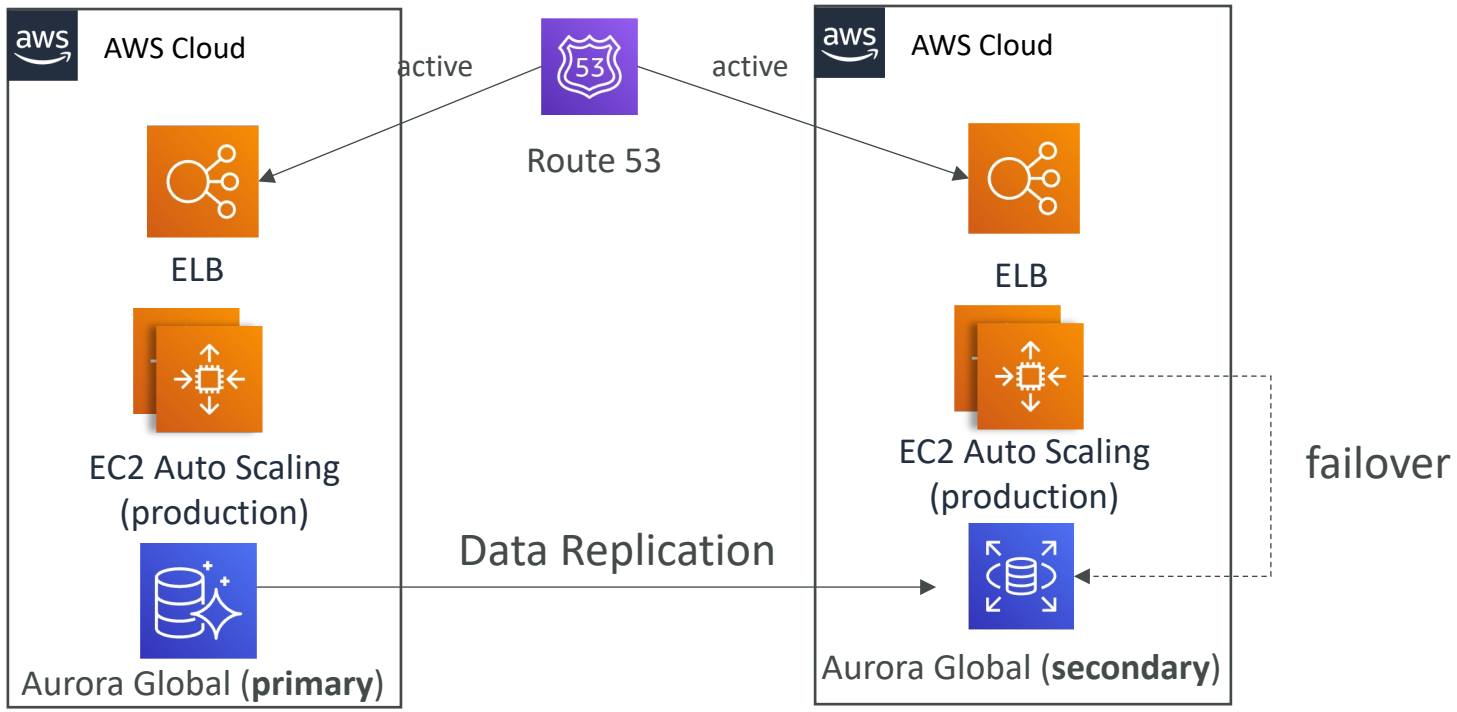
- Very low RTO (minutes or seconds) – very expensive
- Full Production Scale is running AWS and On Premise



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

All AWS Multi Region



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Disaster Recovery Tips

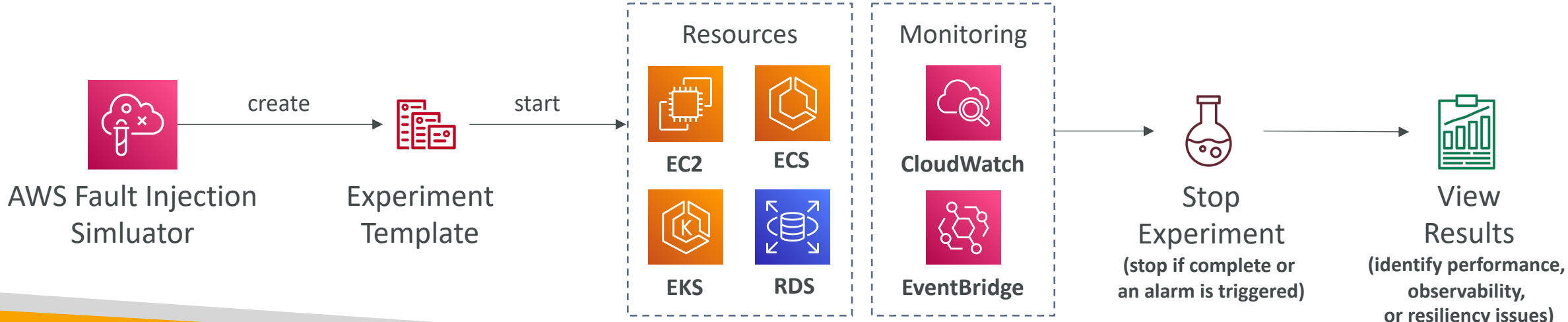
- Backup
  - EBS Snapshots, RDS automated backups / Snapshots, etc…
  - Regular pushes to S3 / S3 IA / Glacier, Lifecycle Policy, Cross Region Replication
  - From on-premises: Snowball or Storage Gateway
- High Availability
  - Use Route53 to migrate DNS over from Region to Region
  - RDS Multi-AZ, ElastiCache Multi-AZ, EFS, S3
  - Site to Site VPN as a recovery from Direct Connect
- Replication
  - RDS Replication (Cross Region), AWS Aurora + Global Database
  - Database replication from on-premises to RDS
  - Storage Gateway
- Automation
  - CloudFormation / Elastic Beanstalk to re-create a whole new environment
  - Recover / Reboot EC2 instances with CloudWatch if alarms fail
  - AWS Lambda functions for customized automations
- Chaos
  - Netflis has a "simian-army" randomly terminating EC2

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Fault Injection Simulator (FIS)

- A fully managed service for running fault injection experiments on AWS workloads
- Based on **Chaos Engineering** – stressing an application by creating disruptive events (e.g., sudden increase in CPU or memory), observing how the system responds, and implementing improvements
- Helps you uncover hidden bugs and performance bottlenecks
- Supports the following AWS services: EC2, ECS, EKS, RDS…
- Use pre-built templates that generate the desired disruptions



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Application Discovery Service

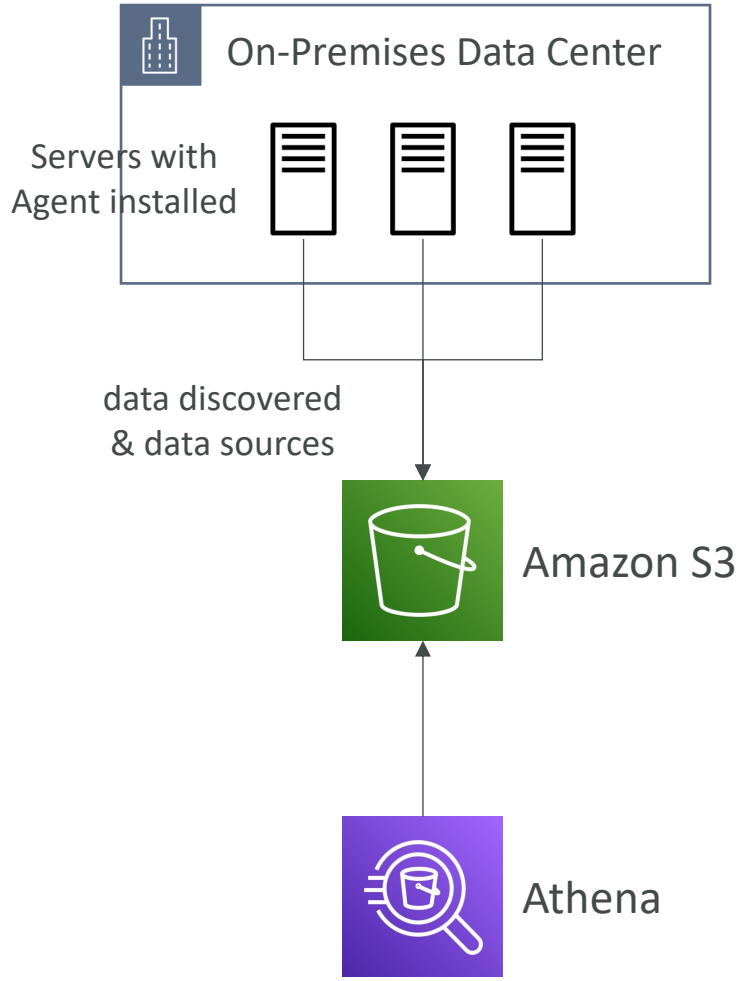
- Plan migration projects by gathering information about on-premises data centers
- Server utilization data and dependency mapping are important for migrations
- Agentless Discovery (AWS Agentless Discovery Connector)
  - Open Virtual Appliance (OVA) package that can be deployed to a VMware host
  - VM inventory, configuration, and performance history such as CPU, memory, and disk usage
  - OS agnostic
- Agent-based Discovery (AWS Application Discovery Agent)
  - System configuration, system performance, running processes, and details of the network connections between systems
  - Supports Microsoft Server, Amazon Linux, Ubuntu, RedHat, CentOS, SUSE…
- Resulting data can be exported as CSV or viewed within AWS Migration Hub
- Data can be explorer using pre-defined queries in Amazon Athena

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Application Discovery Service – Migration Hub Data Exploration

- Allows you to use Amazon Athena to analyze data collected from on-premises servers during discovery
- Data is automatically stored in S3 bucket at regular intervals
- Use Pre-defined or custom queries in Amazon Athena to analyze data
- Example: type of processes running on each server
- Ability to upload additional data sources such as Configuration Management Database (CMDB) exports
- Integrate Athena with QuickSight to visualize data



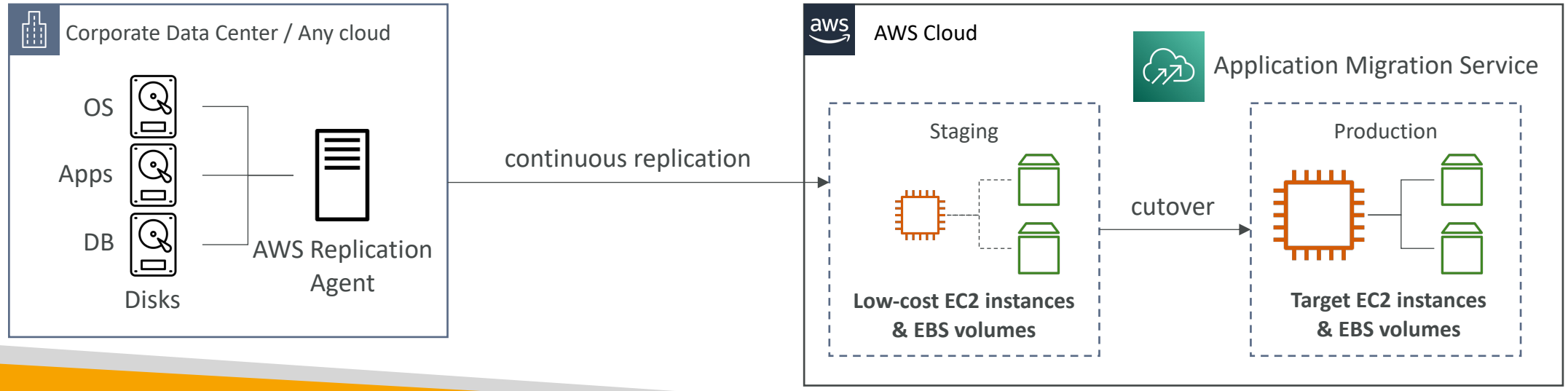
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Application Migration Service (MGN)

- *The "AWS evolution" of CloudEndure Migration, replacing AWS Server Migration Service (SMS)*

- Lift-and-shift (rehost) solution which simplify **migrating** applications to AWS
- Converts your physical, virtual, and cloud-based servers to run natively on AWS
- Supports wide range of platforms, Operating Systems, and databases
- Minimal downtime, reduced costs



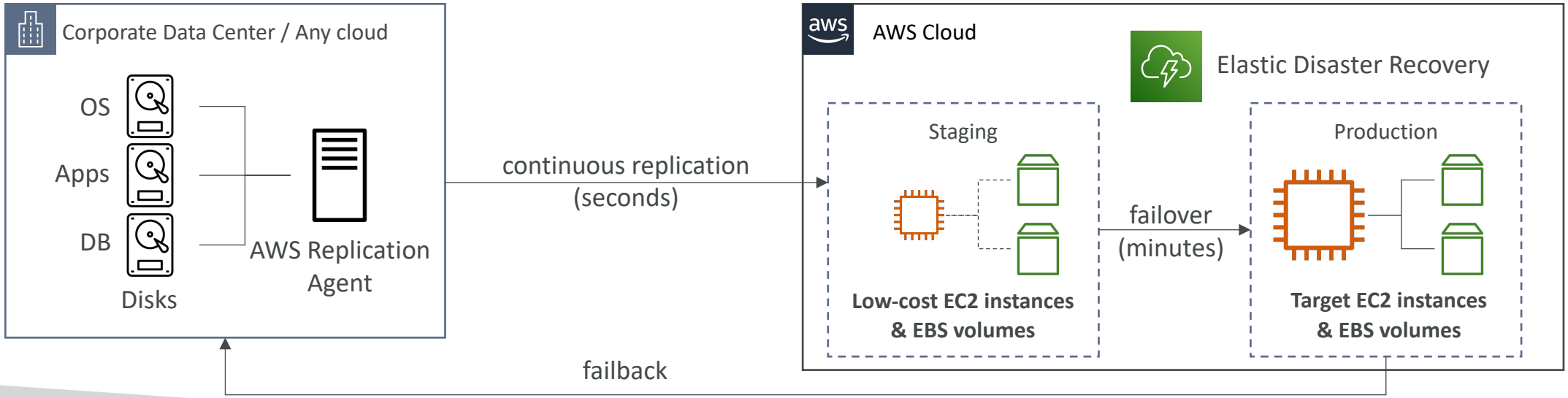
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Elastic Disaster Recovery (DRS)

- *Used to be named "CloudEndure Disaster Recovery"*

- Quickly and easily **recover** your physical, virtual, and cloud-based servers into AWS
- Example: protect your most critical databases (including Oracle, MySQL, and SQL Server), enterprise apps (SAP), protect your data from ransomware attacks, …
- Continuous block-level replication for your servers



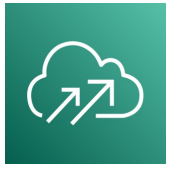
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

On-premises strategy with AWS

- Ability to download Amazon Linux 2 AMI as a VM (.iso format)
  - VMWare, KVM, VirtualBox (Oracle VM), Microsoft Hyper-V
- AWS Application Discovery Service
  - Gather information about your on-premises servers to plan a migration
  - Server utilization and dependency mappings
  - Track with AWS Migration Hub
- AWS Application Migration Service (MGN)
  - Replacing AWS Server Migration Services & CloudEndure Migration
  - Incremental replication of on-premises live servers to AWS
  - Migrates the entire VM into AWS
- AWS Elastic Disaster Recovery (DRS)
  - Replacing CloudEndure Disaster Recovery
  - Recover on-premises workloads onto AWS
- AWS Database Migration Service (DMS)
  - replicate on-premises => AWS , AWS => AWS, AWS => on-premises
  - Works with various database technologies (Oracle, MySQL, DynamoDB, etc..)

AWS Application Discovery Service



AWS Application Migration Service

AWS Elastic Disaster Recovery

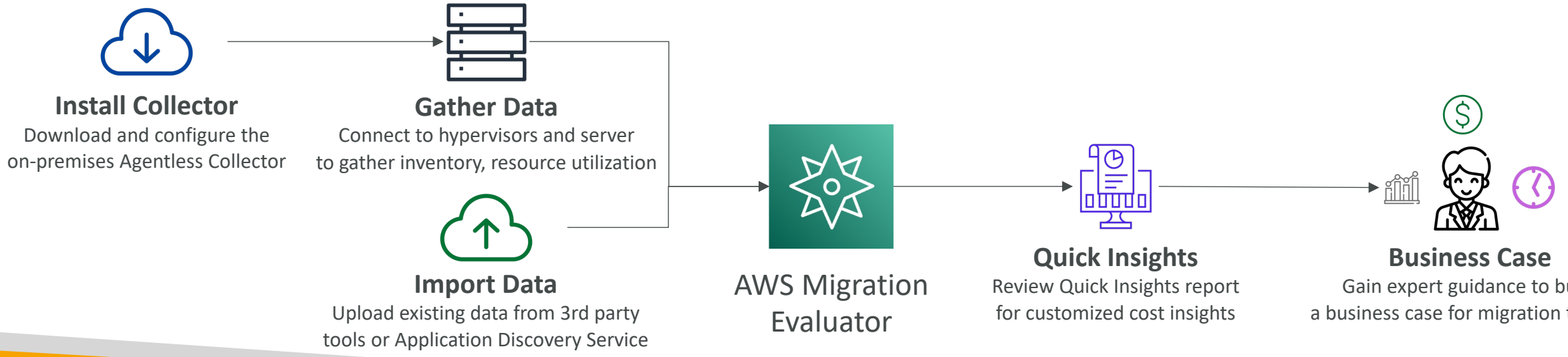
AWS Database Migration Service

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Migration Evaluator

- Helps you build a data-driven business case for migration to AWS
- Provides a clear baseline of what your organization is running today
- Install Agentless Collector to conduct broad-based discovery
- Take a snapshot of on-premises foot-print, server dependepncies, …
- Analyze current state, define target state, then develop migration plan



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Backup

- Fully managed service
- Centrally manage and automate backups across AWS services
- No need to create custom scripts and manual processes
- Supported services:
  - Amazon EC2 / Amazon EBS
  - Amazon S3
  - Amazon RDS (all DBs engines) / Amazon Aurora / Amazon DynamoDB
  - Amazon DocumentDB / Amazon Neptune
  - Amazon EFS / Amazon FSx (Lustre & Windows File Server)
  - AWS Storage Gateway (Volume Gateway)
- Supports cross-region backups
- Supports cross-account backups

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

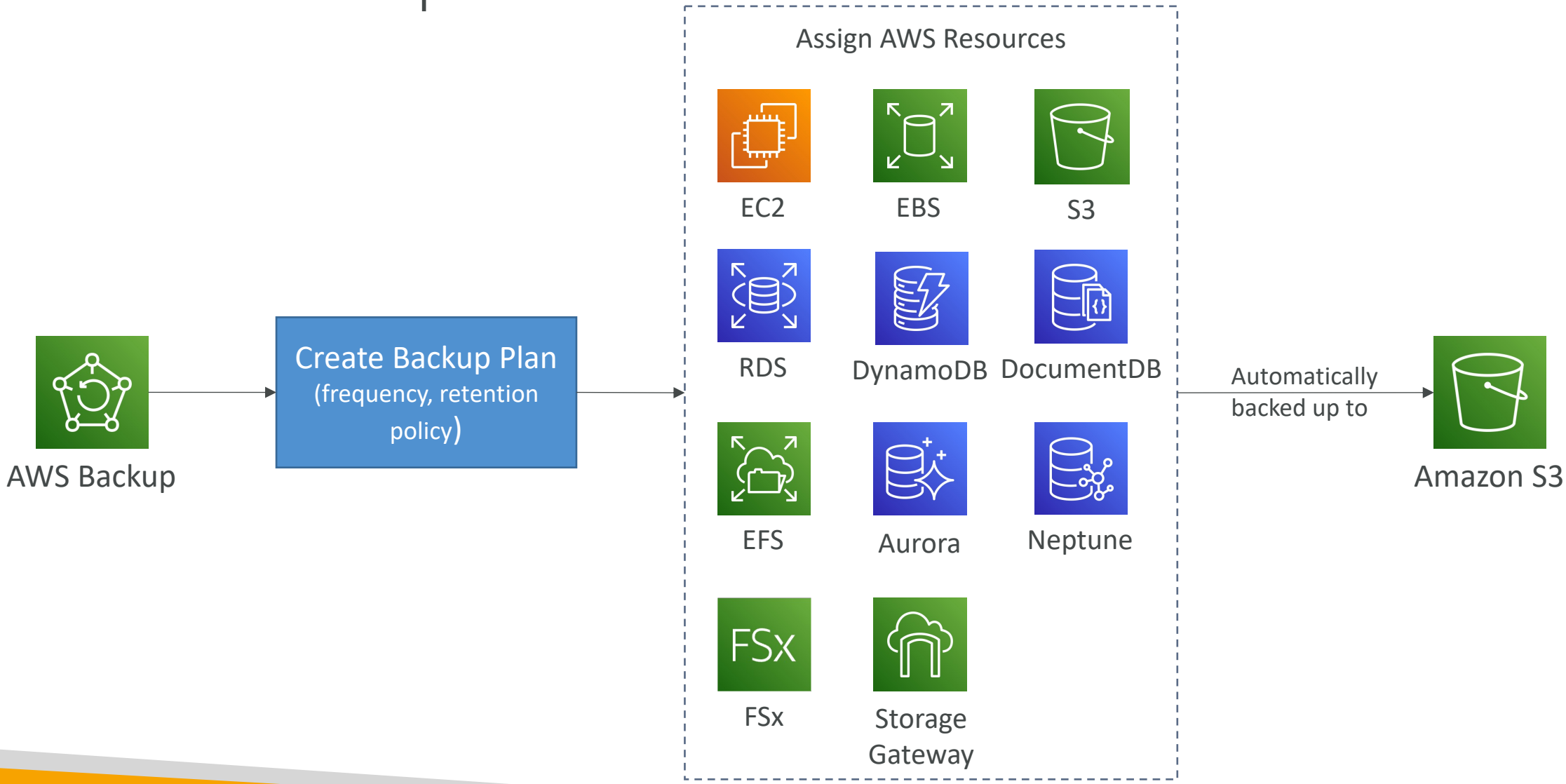
AWS Backup

- Supports PITR for supported services
- On-Demand and Scheduled backups
- Tag-based backup policies
- You create backup policies known as **Backup Plans**
  - Backup frequency (every 12 hours, daily, weekly, monthly, cron expression)
  - Backup window
  - Transition to Cold Storage (Never, Days, Weeks, Months, Years)
  - Retention Period (Always, Days, Weeks, Months, Years)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Backup

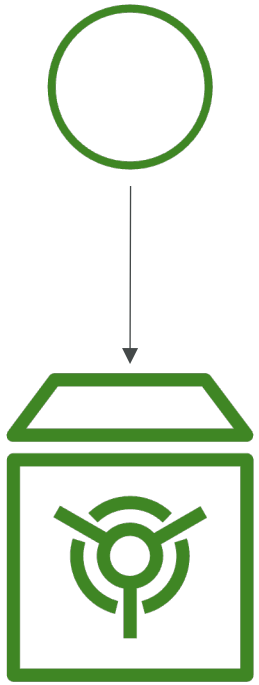


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Backup Vault Lock

- Enforce a WORM (Write Once Read Many) state for all the backups that you store in your AWS Backup Vault
- Additional layer of defense to protect your backups against:
  - Inadvertent or malicious delete operations
  - Updates that shorten or alter retention periods
- Even the root user cannot delete backups when enabled



backup

Backup Vault Lock Policy
Backups can't be deleted

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon VPC

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Basics

- CIDR: Block of IP address
  - Example: 192.168.0.0/26: 192.168.0.0 – 192.168.0.63 (64 IP)
  - Used for security groups, route tables, VPC, subnets, etc…
- Private IP
  - 10.0.0.0 – 10.255.255.255 (10.0.0.0/8) <= in big networks
  - 172.16.0.0 – 172.31.255.255 (172.31.0.0/12)
  - 192.168.0.0 – 192.168.255.255 (192.168.0.0/16) <= example: home networks
- Public IP
  - All the rest

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Basics

- VPC
  - A VPC must have a defined list of CIDR blocks, that cannot be changed
  - Each CIDR within VPC: min size is /28, max size is /16 (65536 IP addresses)
  - VPC is private, so only Private IP CIDR ranges are allowed
- Subnets
  - Within a VPC, defined as a CIDR that is a subset of the VPC CIDR
  - All instances within subnets get a private IP
  - First 4 IP and last one in every subnet is reserved by AWS
- Route Tables
  - Used to control where the network traffic is directed to
  - Can be associated with specific subnets
  - The "most specific" routing rule is always followed (192.168.0.1/24 beats 0.0.0.0/0)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Basics

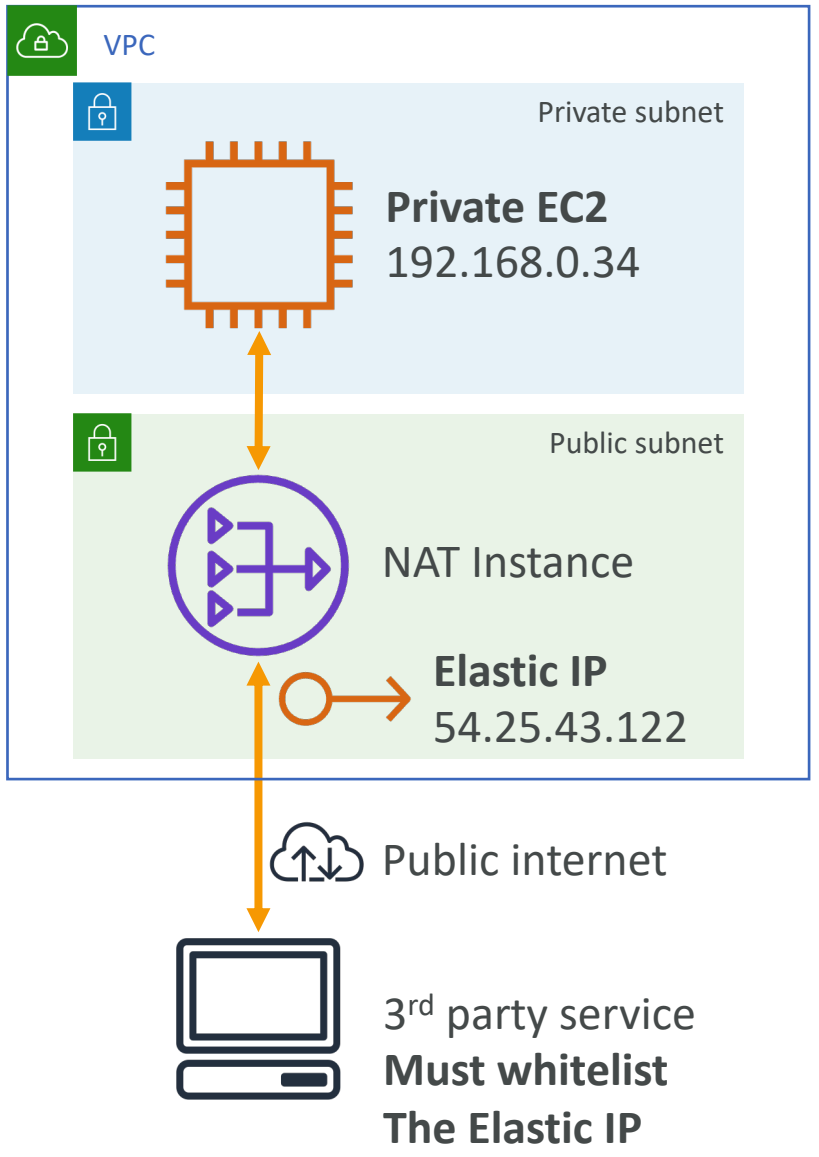
- Internet Gateway (IGW)
  - Helps our VPC connect to the internet, HA, scales horizontally
  - Acts as a NAT for instances that have a public IPv4 or public IPv6
- Public Subnets
  - Has a route table that sends 0.0.0.0/0 to an IGW
  - Instances must have a public IPv4 to talk to the internet
- Private Subnets
  - Access internet with a NAT Instance or NAT Gateway setup in a public subnet
  - Must edit routes so that 0.0.0.0/0 routes traffic to the NAT

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Basics - NAT Instance

- EC2 instance you deploy in a public subnet
- Edit the route in your private subnet to route 0.0.0.0/0 to your NAT instance
- Not resilient to failure, limited bandwidth based on instance type, cheap
- Must manage failover yourself
- Must disable Source/Destination Check (EC2 setting)

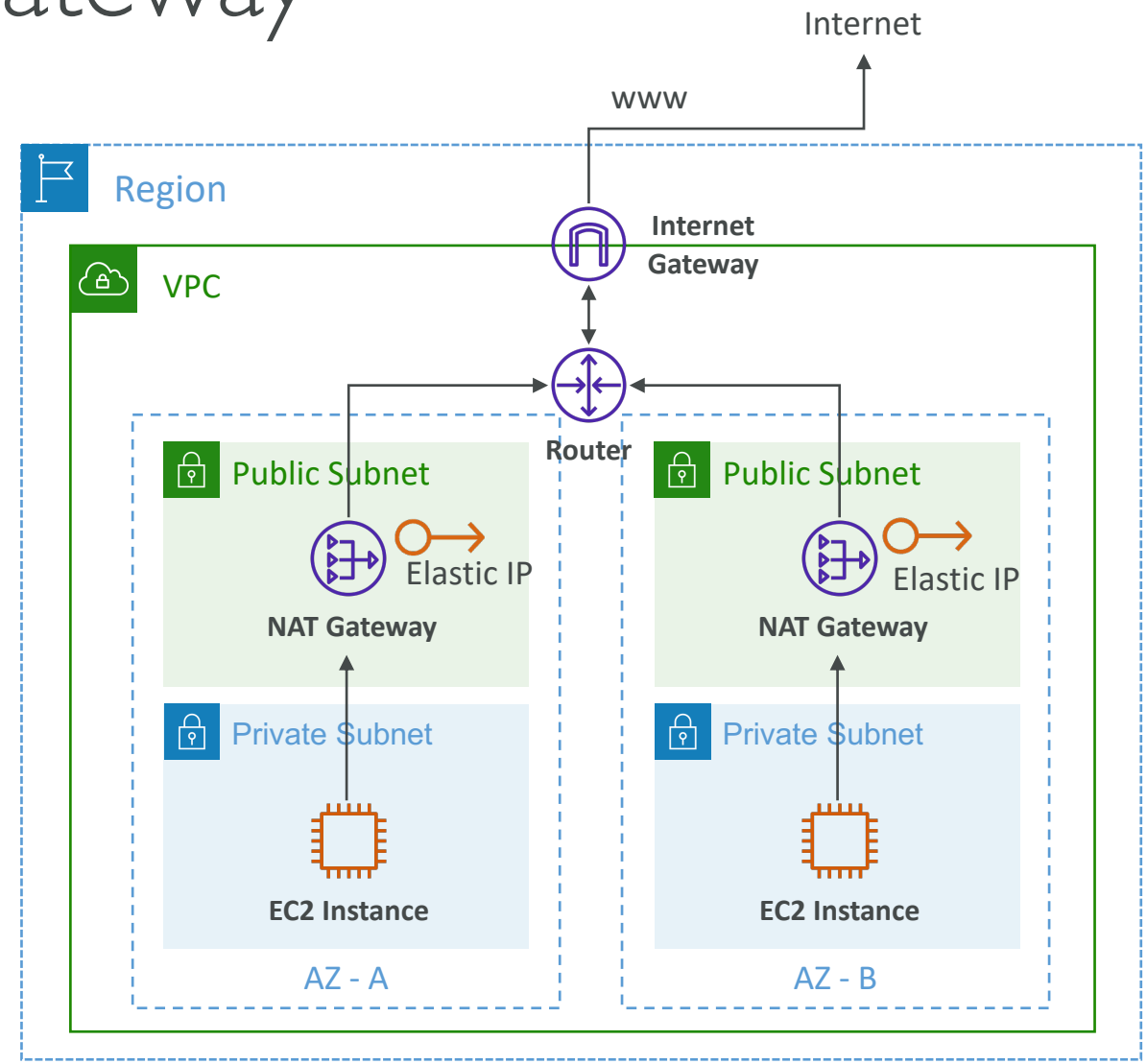


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Basics – NAT Gateway

- Managed NAT solution, bandwidth scales automatically
- Resilient to failure within a single AZ
- Must deploy multiple NAT Gateways in multiple AZ for HA
- Has an Elastic IP, external services see the IP of the NAT Gateway as the source



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Basics

- Network ACL (NACL)
  - Stateless firewall defined at the subnet level, applies to all instances within
  - Support for allow and deny rules
  - Stateless = return traffic must be explicitly allowed by rules
  - Helpful to quickly and cheaply block specific IP addresses

- Security Groups
  - Applied at the instance level, only support for allow rules, no deny rules
  - Stateful = return traffic is automatically allowed, regardless of rules
  - Can reference other security groups in the same region (peered VPC, cross-account)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Basics

- VPC Flows Logs
  - Log internet traffic going through your VPC
  - Can be defined at the VPC level, Subnet level, or ENI-level
  - Helpful to capture "denied internet traffic"
  - Can be sent to CloudWatch Logs and Amazon S3

- Bastion Hosts
  - SSH into private EC2 instances through a public EC2 instance (bastion host)
  - You must manage these instances yourself (failover, recovery)
  - SSM Session Manager is a more secure way to remote control without SSH

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Basics

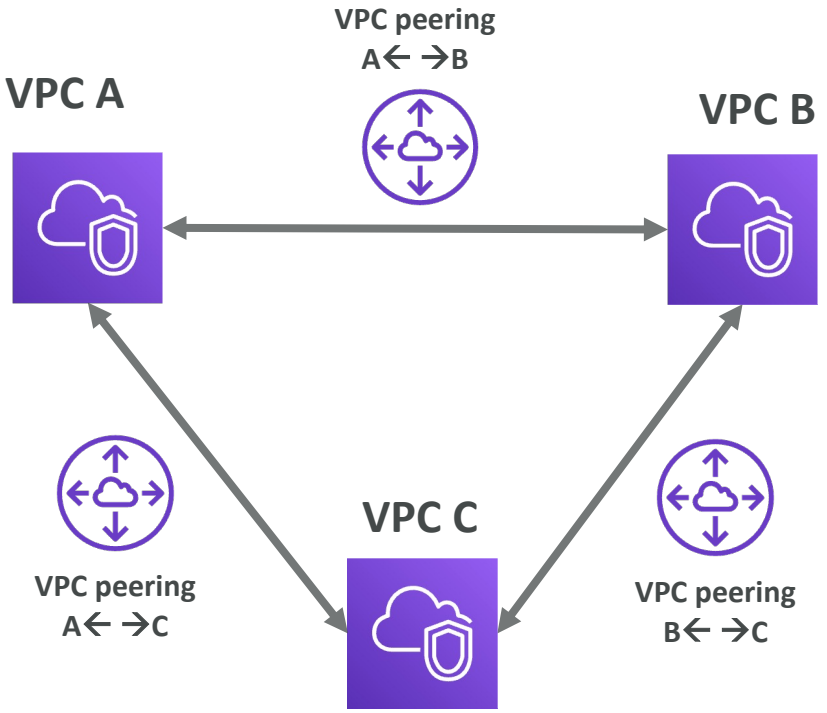
- IPv6 in short
  - All IPv6 addresses are public, total 3.4×10^{38} addresses (vs 4.3 billion IPv4)
  - Example CIDR: 2600:1f18:80c:a900::/56
  - Addresses are "random" and can't be scanned online (because too many)
- VPC support for IPv6
  - Create an IPv6 CIDR for VPC & use an IGW (supports IPv6)
  - Public subnet:
    - Create an instance with IPv6 support
    - Create a route table entry to ::/0 (IPv6 "all") to the IGW
  - Private subnet (instances cannot be reached by IPv6 but can reach IPv6):
    - Create an **Egress-Only Internet Gateway** in the VPC
    - Add a route table entry for the private subnet from ::/0 to the Egress-Only IGW

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Peering

- Connect two VPC, privately using AWS' network
- Make them behave as if they were in the same network
- Must not have overlapping CIDR
- VPC Peering connection is **not transitive** (must be established for each VPC that need to communicate with one another)
- You can do VPC peering with another AWS account
- **You must update route tables in each VPC's subnets to ensure instances can communicate**



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Peering – Good to know

- VPC peering can work **inter-region, cross-account**
- You can reference a security group of a peered VPC (works cross account)

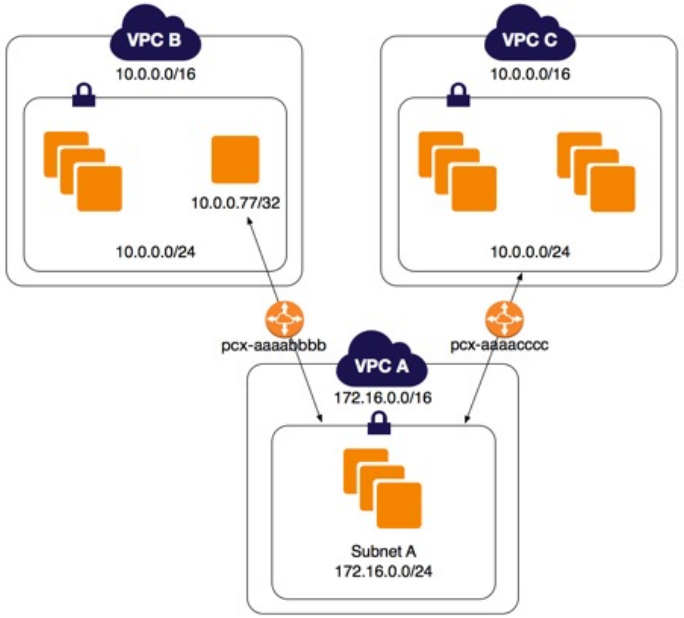
Type	Protocol	Port Range	Source
HTTP	TCP	80	sg-00d2b0f5fd6de757e
HTTP	TCP	80	sg-013347765f7a63aae/12356788

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Peering – Longest Prefix Match

- VPC uses the longest prefix match to select the most specific route



Route Table	Destination	Target
VPC A	172.16.0.0/16	Local
	10.0.0.77/32	pcx-aaaabbbb
	10.0.0.0/16	pcx-aaaacccc
VPC B	10.0.0.0/16	Local
	172.16.0.0/16	pcx-aaaabbbb
VPC C	10.0.0.0/16	Local
	172.16.0.0/16	pcx-aaaacccc

- Here the longest prefix for 10.0.0.77 is **10.0.0.77/32** (route table VPC A)
- (other way of saying it is "most specific route")

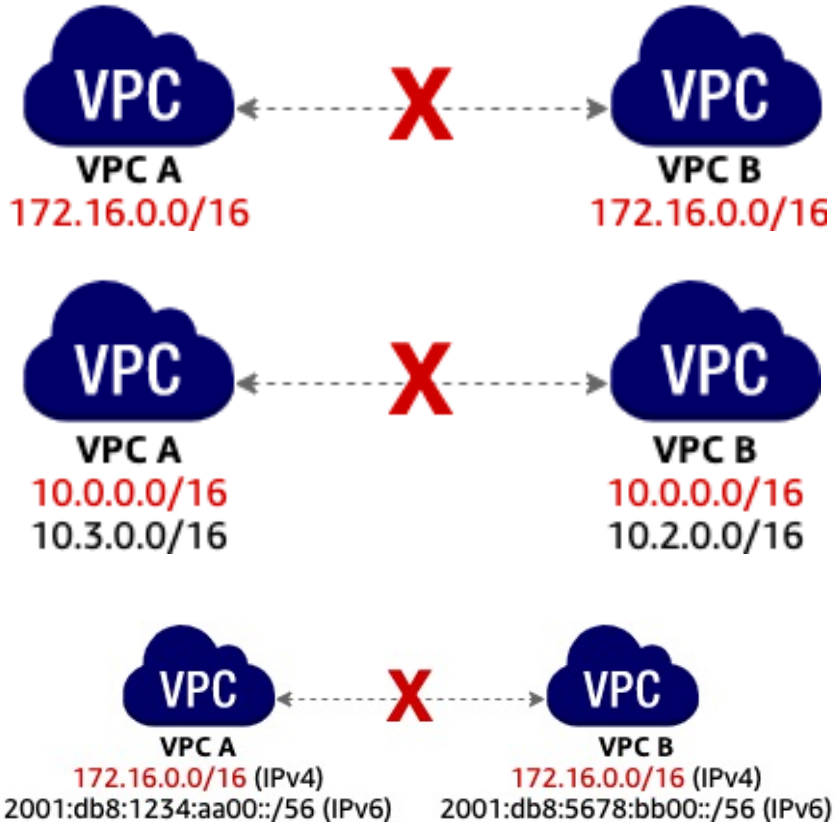
https://docs.aws.amazon.com/vpc/latest/peering/peering-configurations-partial-access.html#one-to-two-vpcs-lpm

© Stephane Maarek

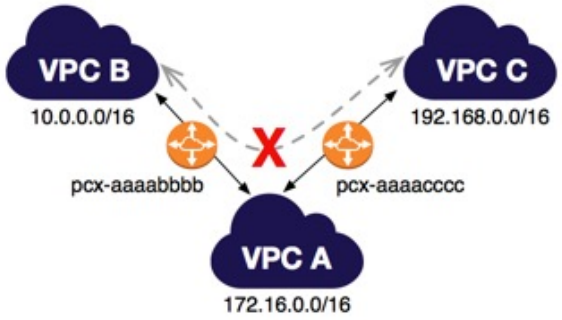
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Peering – Invalid Configurations

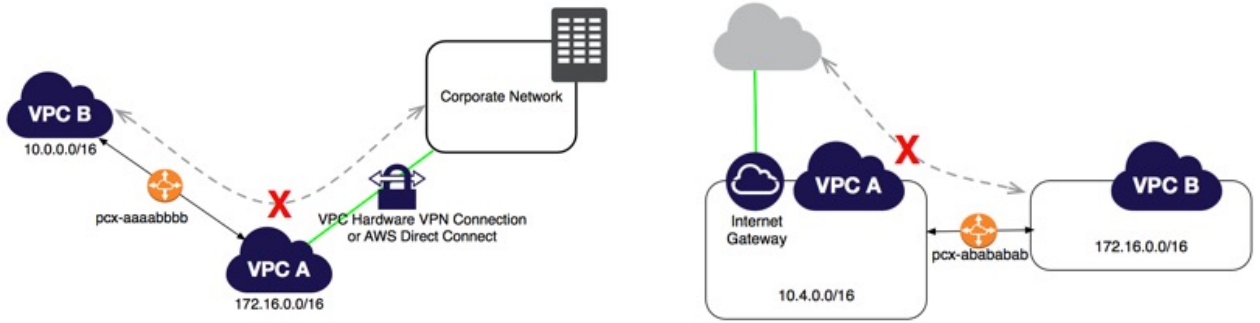
Overlapping CIDR for IPv4



No Transitive VPC Peering



No Edge to Edge Routing



VPN, Direct Connect, IGW, NAT, Gateway VPC Endpoint (S3 & DynamoDB)

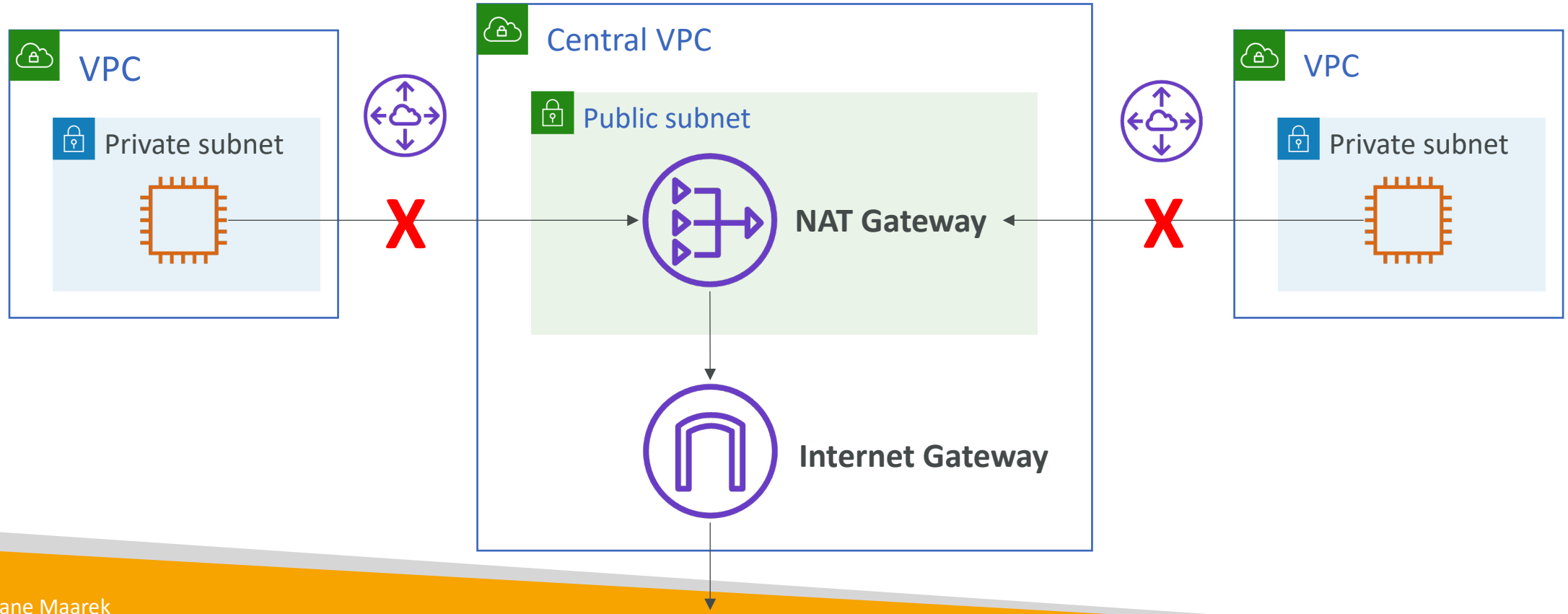
https://docs.aws.amazon.com/vpc/latest/peering/invalid-peering-configurations.html

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Peering – Invalid Configuration
No edge to edge routing

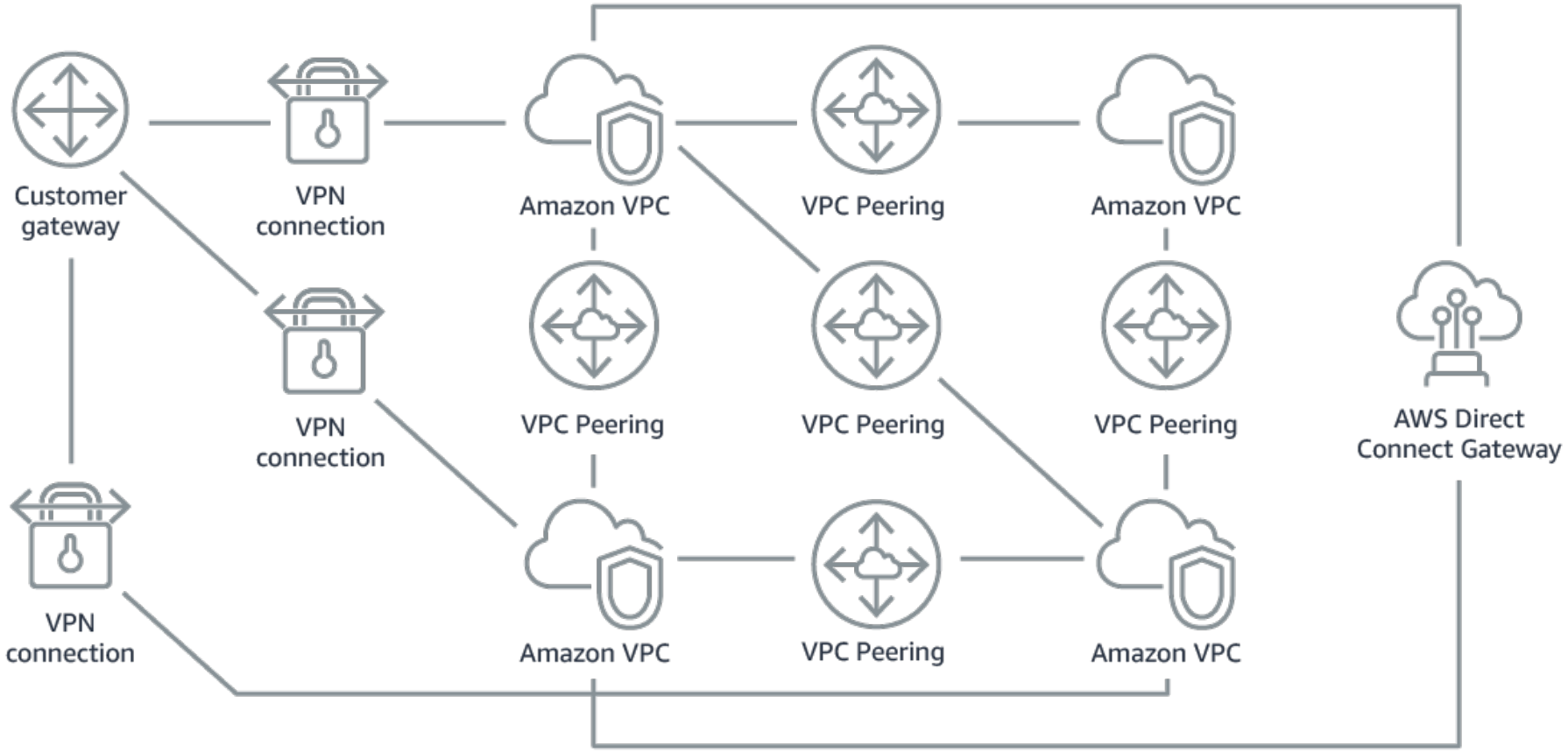
- This is an invalid configuration
- VPC Peering does not support edge to edge routing for NAT devices



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Network topologies can become complicated

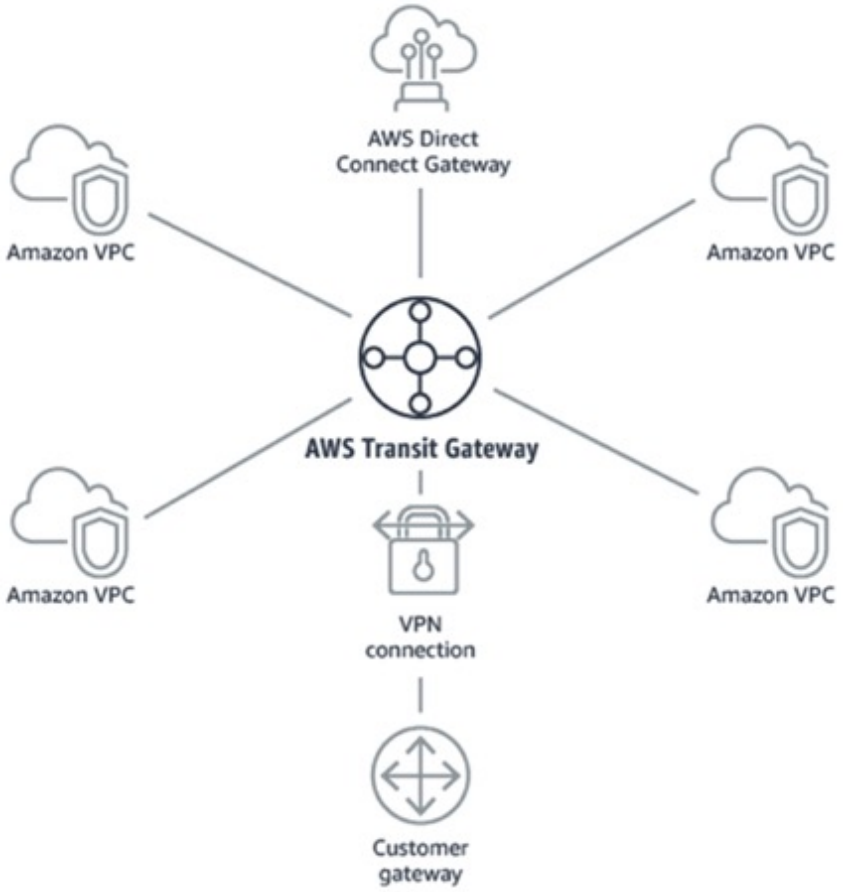


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Transit Gateway

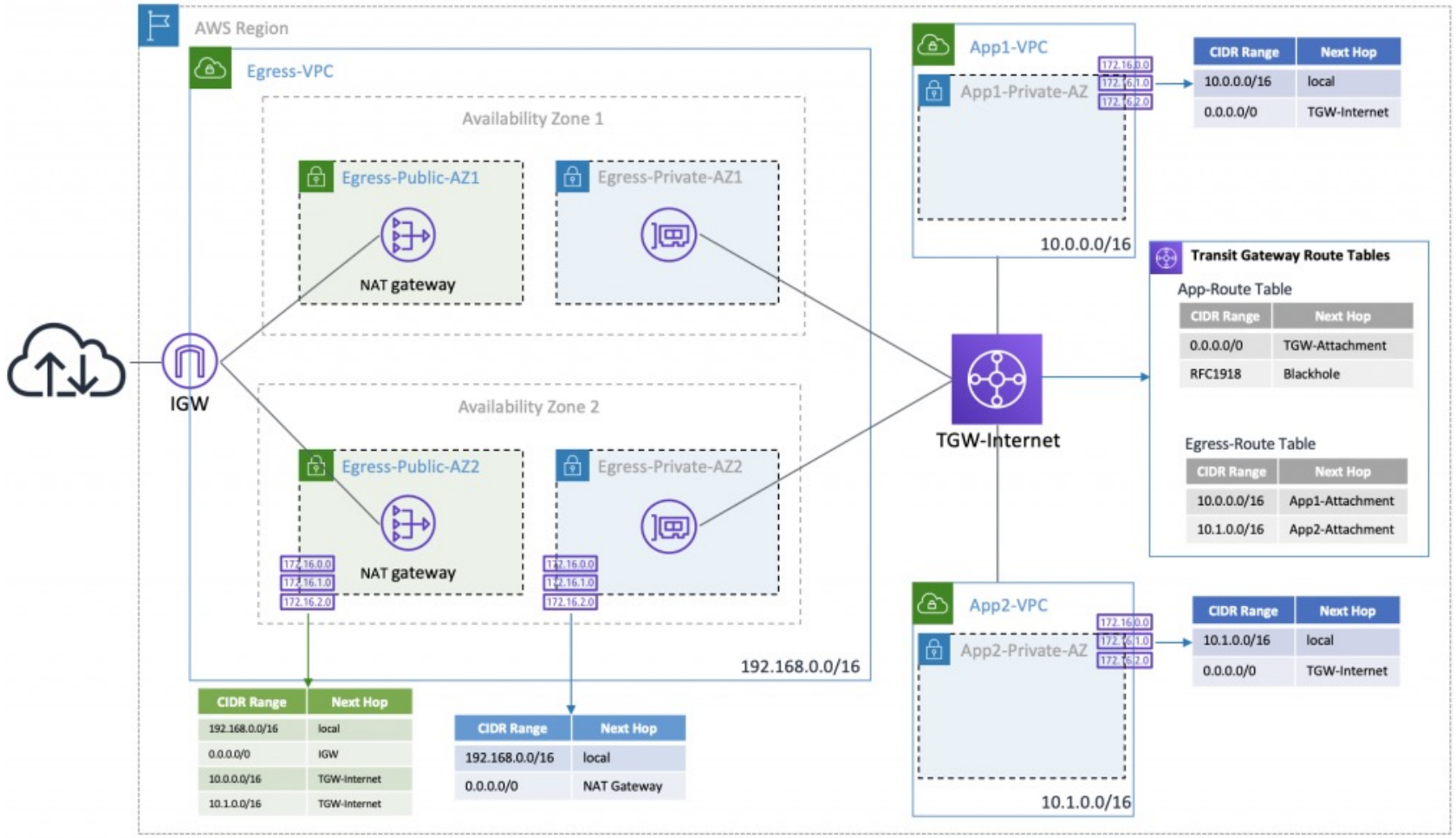
- For having transitive peering between thousands of VPC and on-premises, hub-and-spoke (star) connection
- Regional resource, can work cross-region
- Share cross-account using Resource Access Manager (RAM)
- You can peer Transit Gateways across regions
- Route Tables: limit which VPC can talk with other VPC
- Works with Direct Connect Gateway, VPN connections
- Supports **IP Multicast** (not supported by any other AWS service)
- Instances in a VPC can access a NAT Gateway, NLB, PrivateLink, and EFS in others VPCs attached to the AWS Transit Gateway.



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Transit Gateway – Central NAT Gateway



- The NAT Gateway is shared in the Egress-VPC
- The private App VPC can access internet through the TGW
- In this example: the App VPCs cannot communicate with each other based on the TGW route table

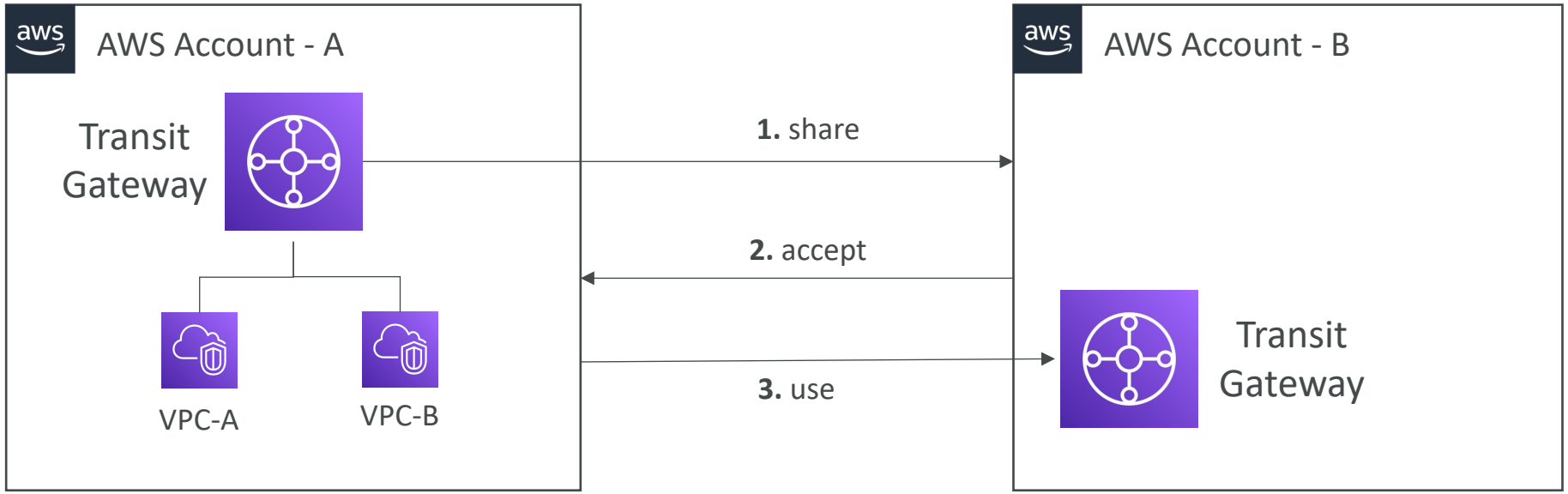
https://aws.amazon.com/blogs/networking-and-content-delivery/creating-a-single-internet-exit-point-from-multiple-vpcs-using-aws-transit-gateway/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Transit Gateway – Sharing through RAM

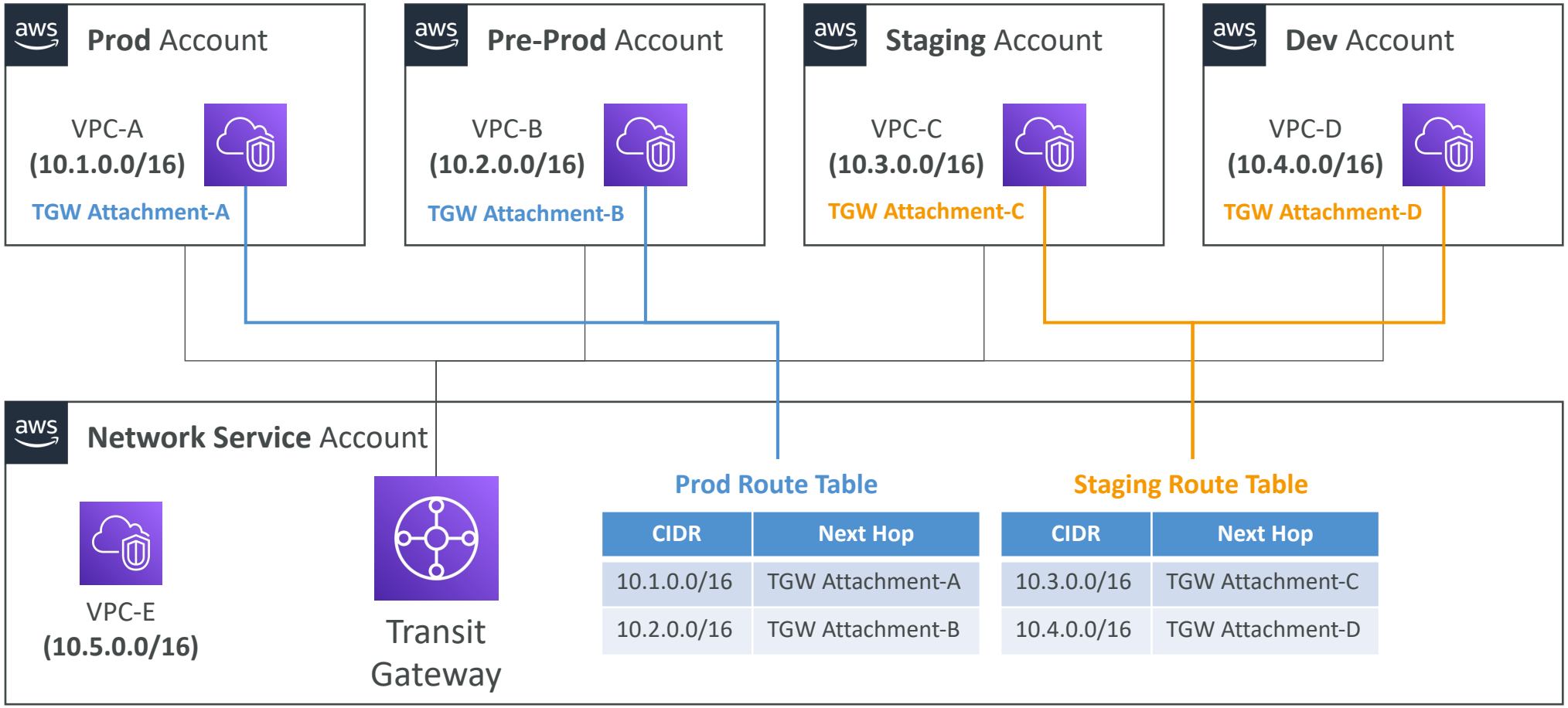
- You can use AWS RAM to share a Transit Gateway for VPC attachments across accounts or across AWS Organization



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

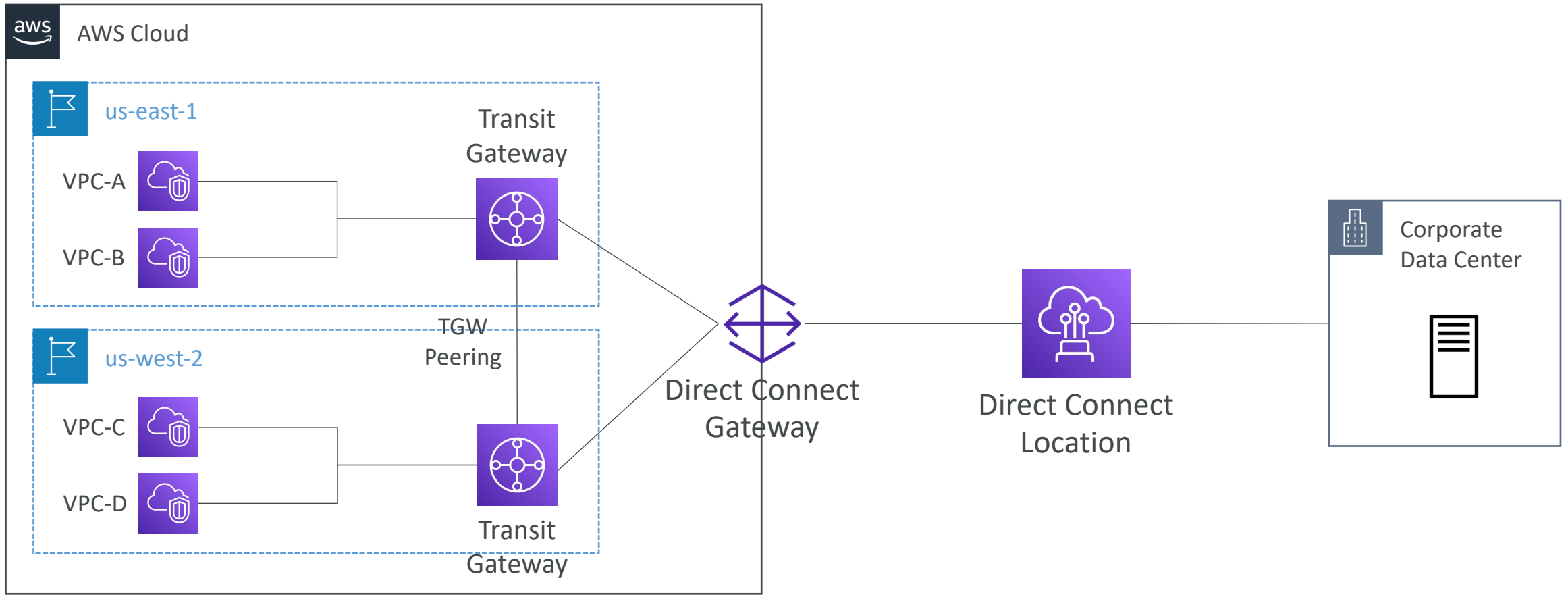
Transit Gateway – Use Different Route Tables to Prevent VPC from Communicating



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

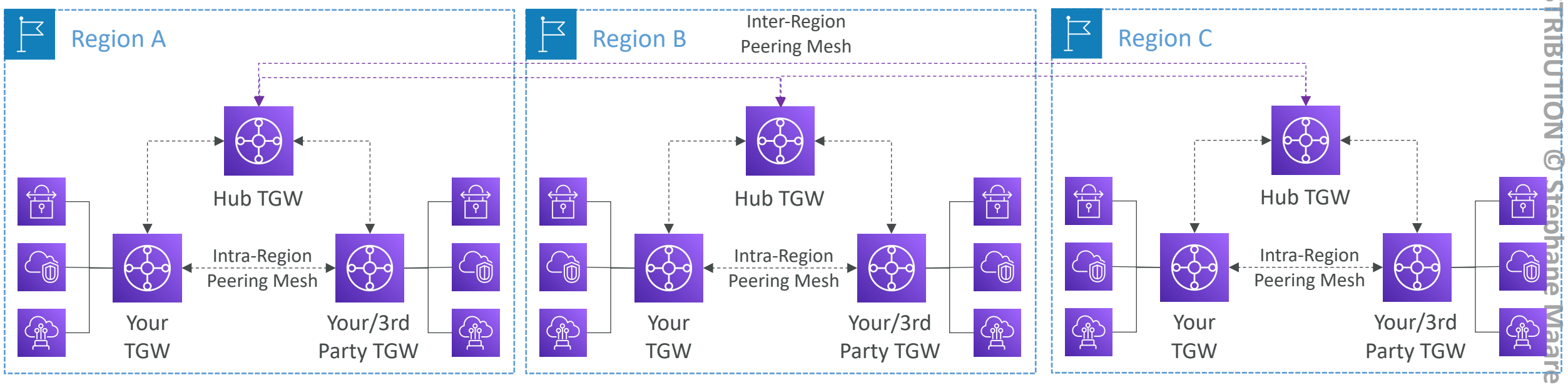
Transit Gateway to Direct Connect Gateway



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Transit Gateway – Inter & Intra Region Peering



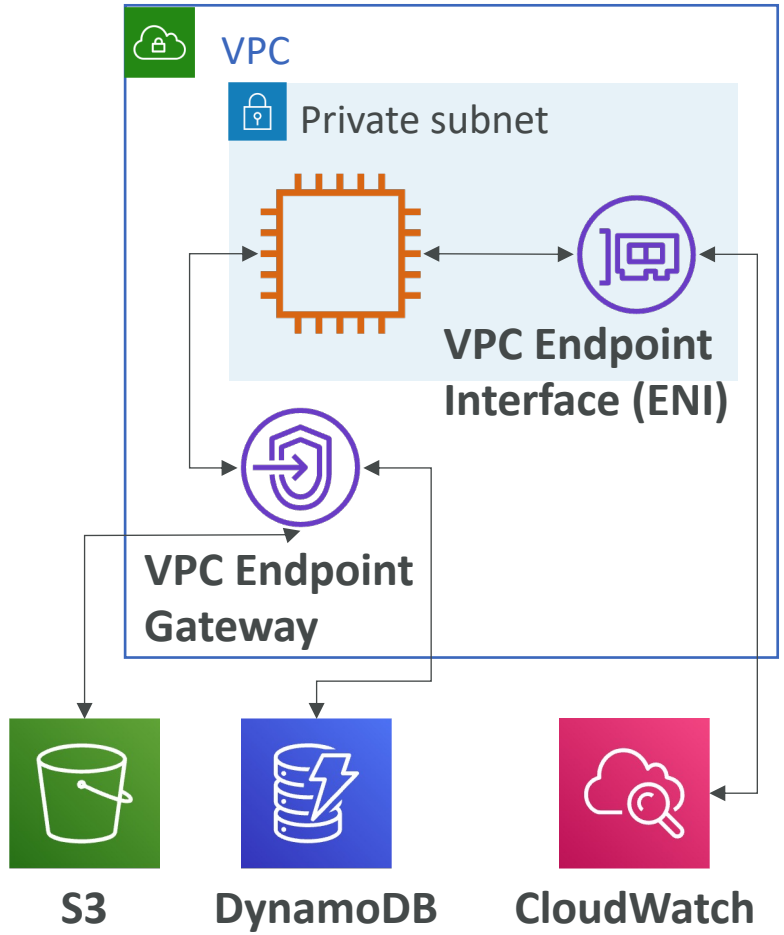
- Billed hourly for each peering attachment, no data processing charges
- If data goes cross-region through the attachment, standard charges
- Specify the Transit Gateway ID

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Endpoints

- Endpoints allow you to connect to AWS Services using a private network instead of the public www network
- They scale horizontally and are redundant
- No more IGW, NAT, etc… to access AWS Services
- VPC Endpoint Gateway (S3 & DynamoDB)
- VPC Endpoint Interface (all services)
- In case of issues:
  - Check DNS Setting Resolution in your VPC
  - Check Route Tables
- Supports cross-region endpoints

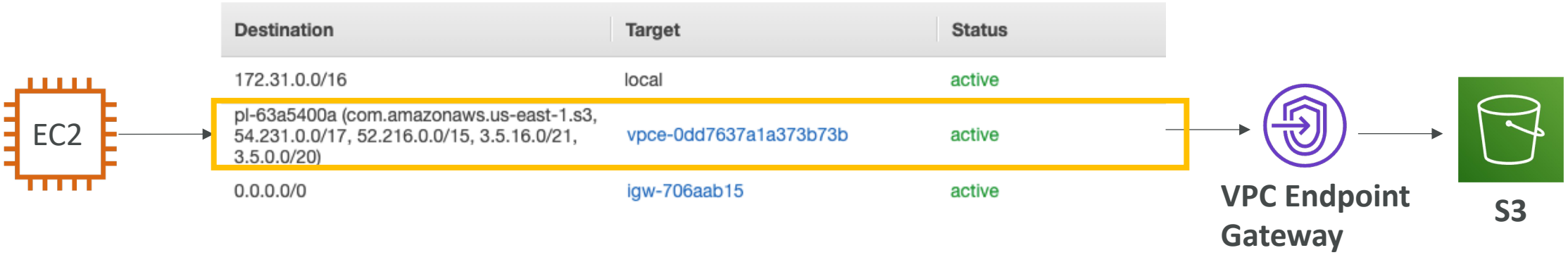


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Endpoint Gateway

- Only works for S3 and DynamoDB, must create one gateway per VPC
- Must update route tables entries
- Gateway is defined at the VPC level



- DNS resolution must be enabled in the VPC
- The same public hostname for S3 can be used
- Gateway endpoint cannot be extended out of a VPC (VPN, DX, TGW, peering)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Endpoints Interface

- Provision an ENI that will have a private endpoint interface hostname
- Leverage Security Groups for security
- Private DNS (setting when you create the endpoint)
  - The public hostname of a service will resolve to the private Endpoint Interface hostname
  - VPC Setting: "Enable DNS hostnames" and "Enable DNS Support" must be 'true'
  - Example for Athena:
    - vpce-0b7d2995e9dfe5418-mwrths3x.athena.us-east-1.vpce.amazonaws.com
    - vpce-0b7d2995e9dfe5418-mwrths3x-us-east-1a.athena.us-east-1.vpce.amazonaws.com
    - vpce-0b7d2995e9dfe5418-mwrths3x-us-east-1b.athena.us-east-1.vpce.amazonaws.com
    - **athena.us-east-1.amazonaws.com** (private DNS name)
- Interface can be accessed from Direct Connect and Site-to-Site VPN

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Endpoint Policies

- Endpoint Policies are JSON documents to control access to services
- Does not override or replace IAM user policies or service-specific policies (such as S3 bucket policies)

```
{
   "Statement": [{
      "Action": ["sqs:SendMessage"],
      "Effect": "Allow",
      "Resource": "arn:aws:sqs:us-east-2:123456789012:MyQueue",
      "Principal": {
        "AWS": "arn:aws:iam:123456789012:user/MyUser"
      }
   }]
}
```

- **Note:** the IAM user can still use other SQS API from outside the VPC Endpoint
- You could add an SQS queue policy to deny any action not done through the VPC endpoint

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Endpoint Policy & S3 bucket policy

- VPC Endpoint Policy to restrict access to bucket "my_secure_bucket"

```
{
  "Statement": [
    {
      "Sid": "Access-to-specific-bucket-only",
      "Principal": "*",
      "Action": [
        "s3:GetObject",
        "s3:PutObject"
      ],
      "Effect": "Allow",
      "Resource": ["arn:aws:s3:::my_secure_bucket",
                   "arn:aws:s3:::my_secure_bucket/*"]
    }
  ]
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Endpoint Policy & S3 bucket policy

- VPC Endpoint Policy to allow access to Amazon Linux 2 repositories

```
{
  "Statement": [
    {
      "Sid": "AmazonLinux2AMIRepositoryAccess",
      "Principal": "*",
      "Action": [
        "s3:GetObject"
      ],
      "Effect": "Allow",
      "Resource": [
        "arn:aws:s3:::amazonlinux.*.amazonaws.com/*"
      ]
    }
  ]
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Endpoint Policy & S3 bucket policy

- S3 bucket policy may have
  - **Condition: "aws:sourceVpce": "vpce-1a2b3c4d"** to Deny any traffic that doesn't come from a specific VPC endpoint (more secure)
  - **Condition: "aws:sourceVpc": "vpc-111bbb22"** for a specific VPC
- The **aws:sourceVpc** condition only works for VPC Endpoints, in case you have multiple endpoints and want to manage access to your S3 buckets for all your endpoints
- The S3 bucket policies can restrict access only from a specific **public IP** address or an elastic IP address. You can't restrict based on private IP
- Therefore aws:SourceIp condition doesn't apply for VPC endpoints

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Example S3 bucket policies

- S3 bucket policy to restrict to one specific VPC Endpoint

```
{
  "Version": "2012-10-17",
  "Id": "Policy1415115909152",
  "Statement": [
    {
      "Sid": "Access-to-specific-VPCE-only",
      "Principal": "*",
      "Action": "s3:*",
      "Effect": "Deny",
      "Resource": ["arn:aws:s3:::my_secure_bucket",
                   "arn:aws:s3:::my_secure_bucket/*"],
      "Condition": {
        "StringNotEquals": {
          "aws:sourceVpce": "vpce-1a2b3c4d"
        }
      }
    }
  ]
}
```

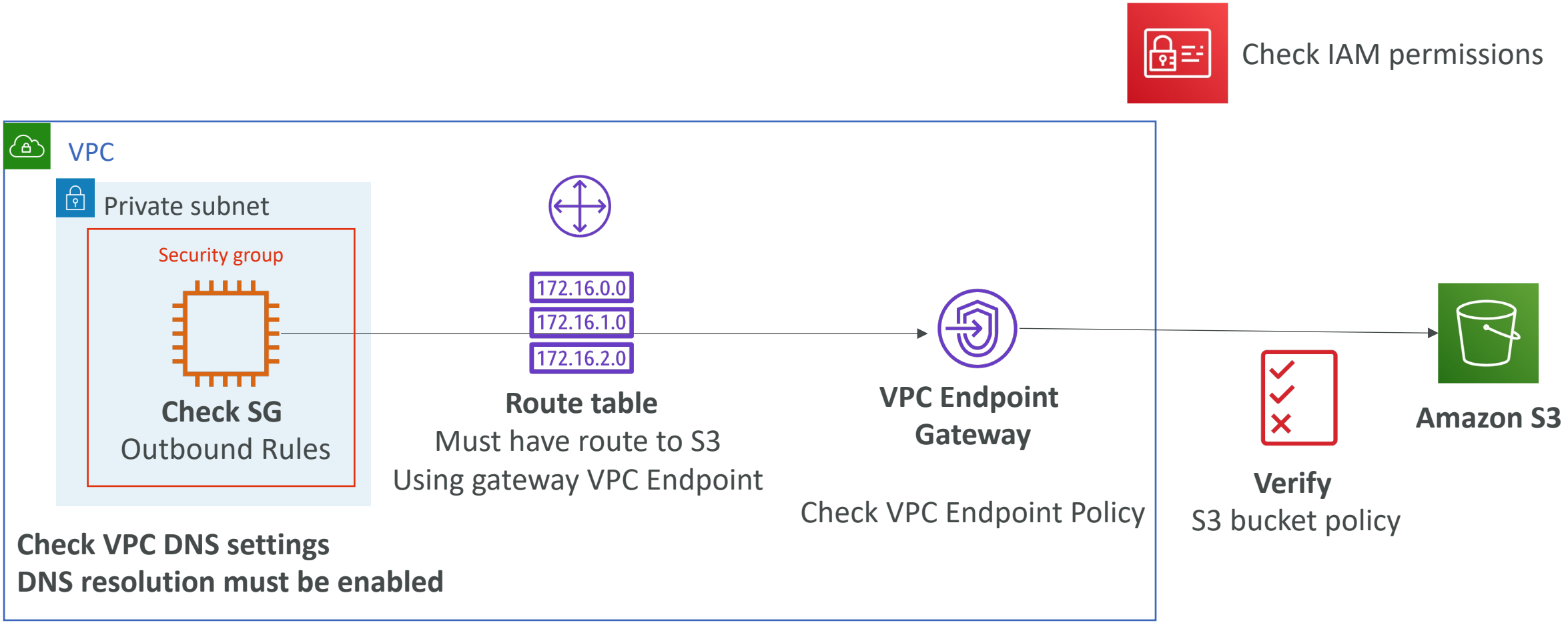
- S3 bucket policy to restrict to an entire VPC (multiple VPC Endpoints)

```
{
  "Version": "2012-10-17",
  "Id": "Policy1415115909152",
  "Statement": [
    {
      "Sid": "Access-to-specific-VPC-only",
      "Principal": "*",
      "Action": "s3:*",
      "Effect": "Deny",
      "Resource": ["arn:aws:s3:::my_secure_bucket",
                   "arn:aws:s3:::my_secure_bucket/*"],
      "Condition": {
        "StringNotEquals": {
          "aws:sourceVpc": "vpc-111bbb22"
        }
      }
    }
  ]
}
```

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Endpoint Policies for S3 Troubleshooting

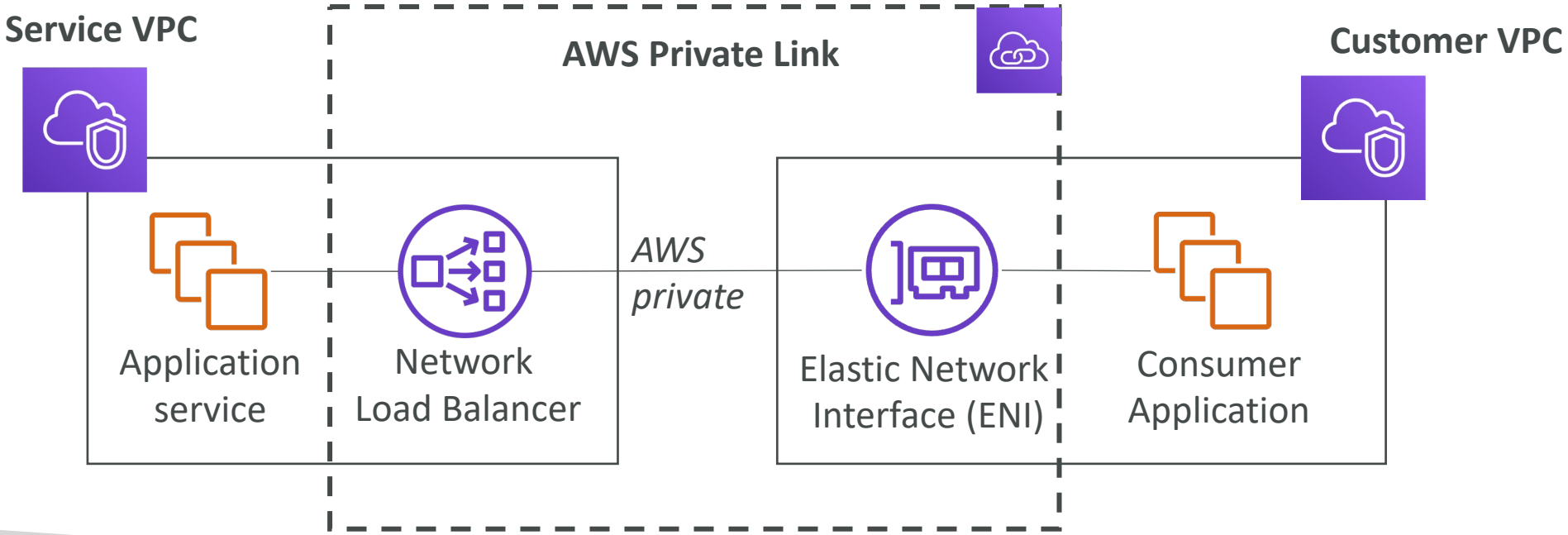


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS PrivateLink (VPC Endpoint Services)

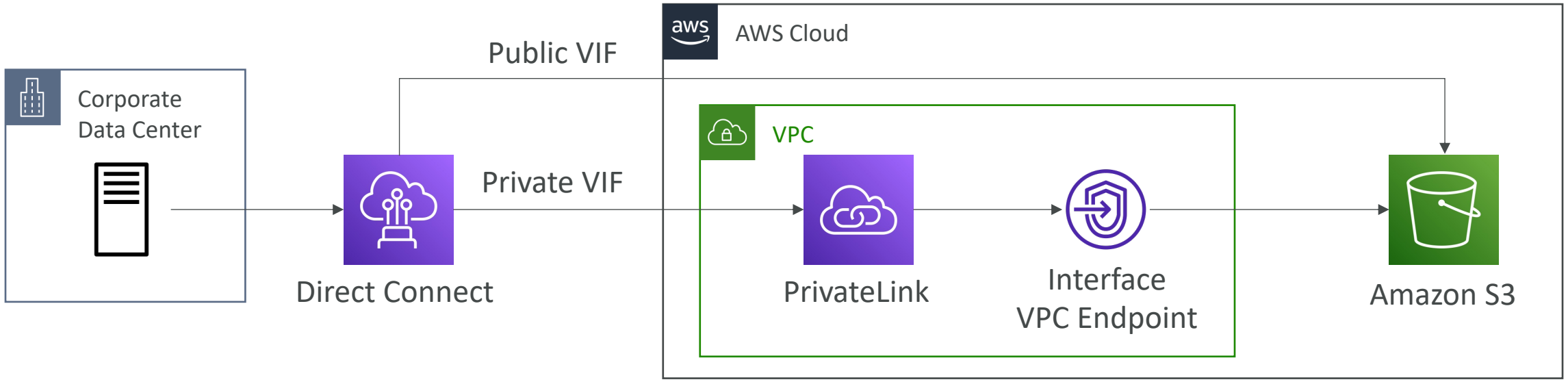
- Most secure & scalable way to expose a service to 1000s of VPC (own or other accounts)
- Does not require VPC peering, internet gateway, NAT, route tables…
- Requires a network load balancer (Service VPC) and ENI (Customer VPC)
- If the NLB is in multiple AZ, and the ENI in multiple AZ, the solution is fault tolerant!



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

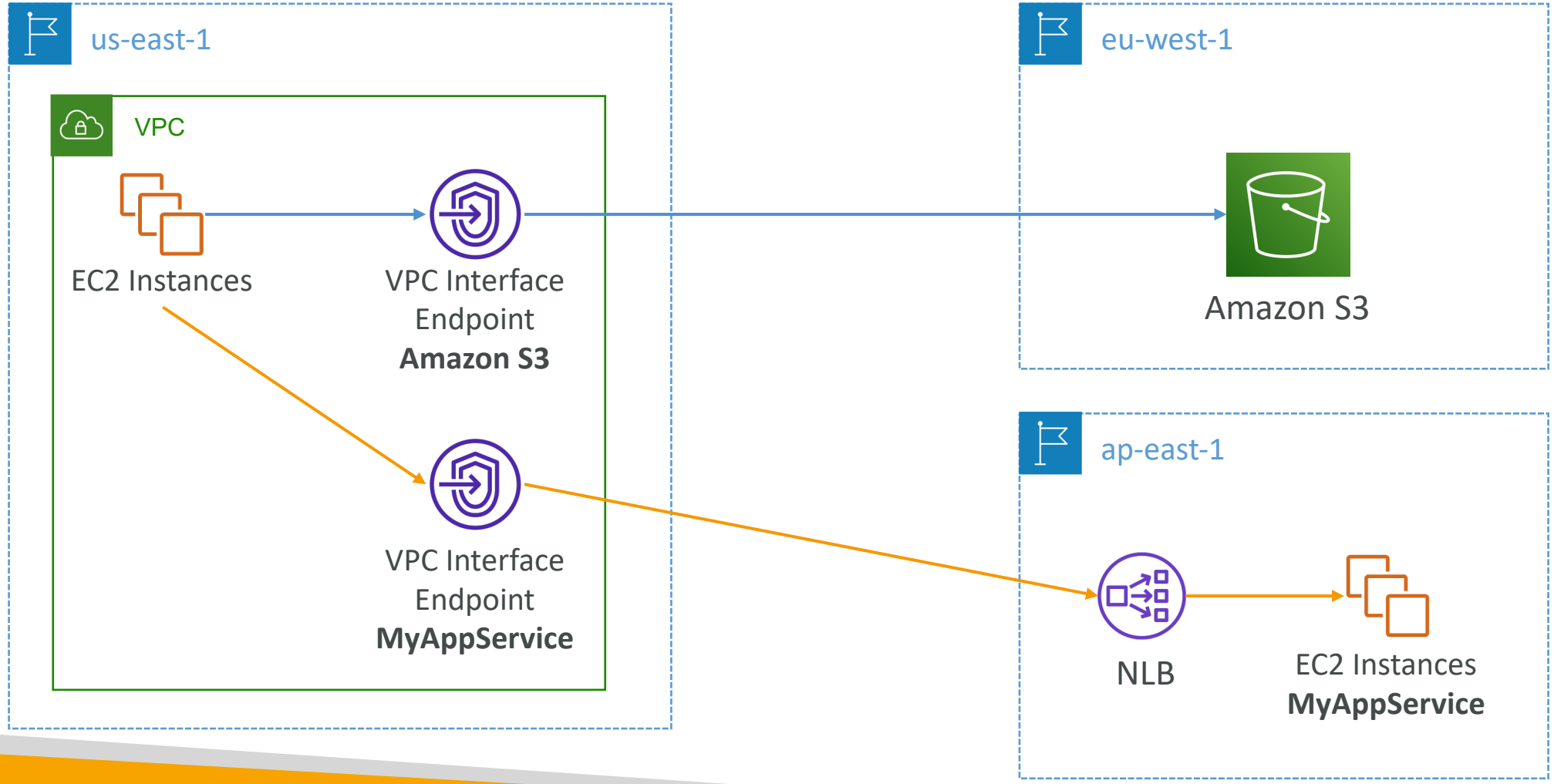
PrivateLink for Amazon S3 with Direct Connect



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Endpoints / PrivateLink Cross-Region Connectivity

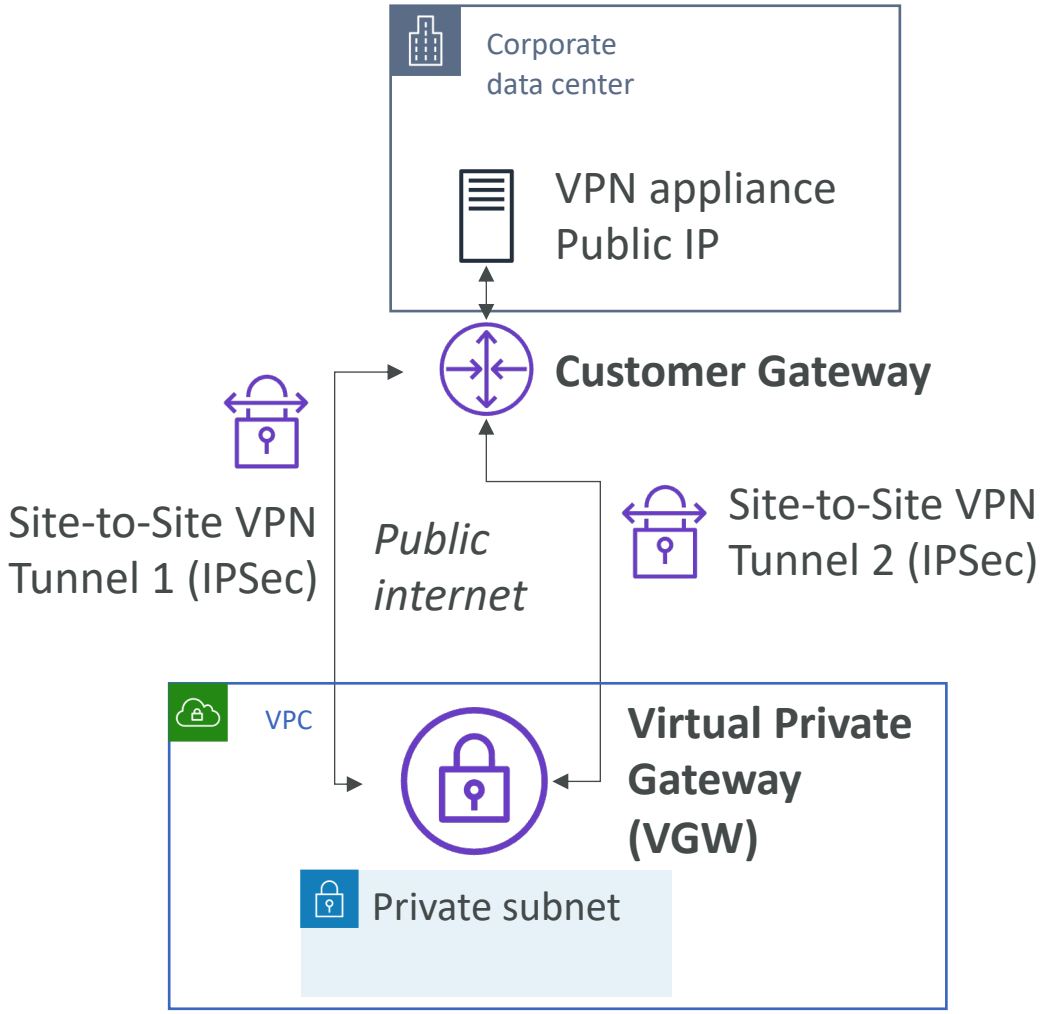


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Site to Site VPN (AWS Managed VPN)

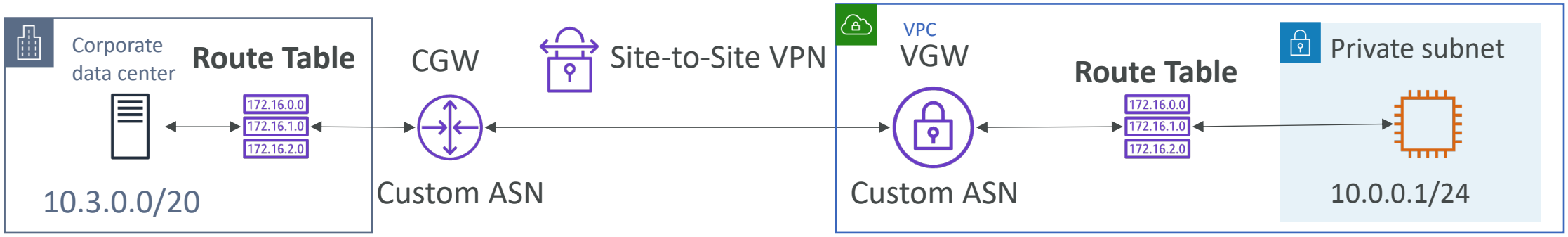
- on-premises:
  - Setup a software or hardware VPN appliance to your on-premises network.
  - The on-premises VPN should be accessible using a public IP
- AWS-side:
  - Setup a Virtual Private Gateway (VGW) and attach to your VPC
  - Setup a Customer Gateway to point the on-premises VPN appliance
- Two VPN connections (tunnels) are created for redundancy, encrypted using IPSec
- Can optionally accelerate it using Global Accelerator (for worldwide networks)



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Route Propagation in Site-to-Site VPN



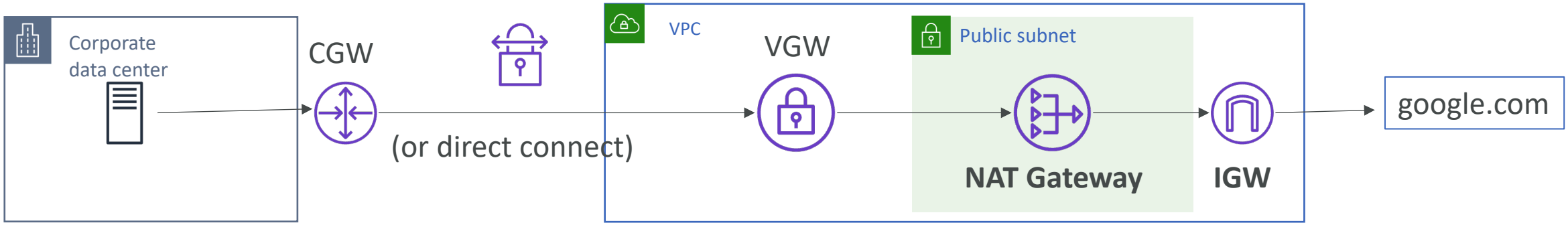
- Static Routing:
  - Create static route in corporate data center for 10.0.0.1/24 through the CGW
  - Create static route in AWS for 10.3.0.0/20 through the VGW
- Dynamic Routing (BGP):
  - Uses BGP (Border Gateway Protocol) to share routes automatically (eBGP for internet)
  - We don't need to update the routing tables, it will be done for us dynamically
  - Just need to specify the ASN (Autonomous System Number) of the CGW and VGW

© Stephane Maarek

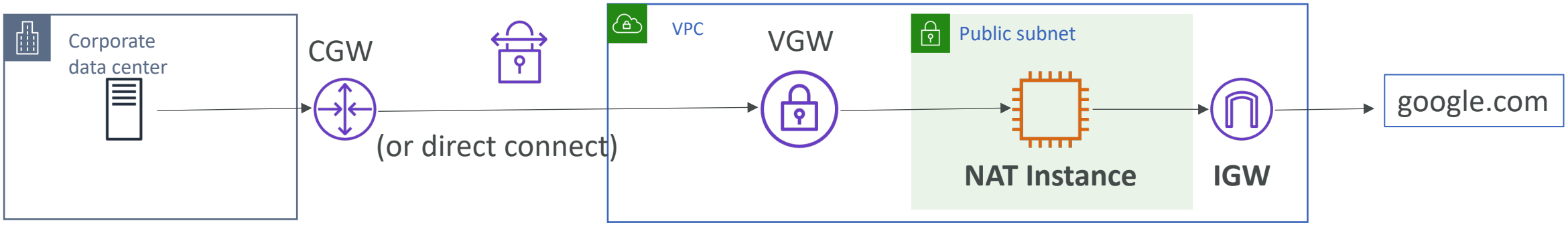
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Site to Site VPN and Internet Access

- NOT OKAY (blocked by NAT Gateway restrictions)



- OKAY (self managed NAT Instance – more control)

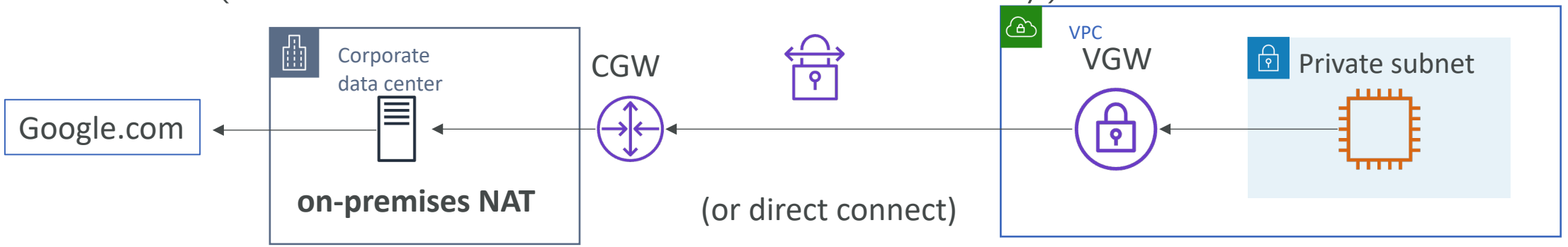


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Site to Site VPN and Internet Access

- OKAY (alternative to NAT Instances / Gateway)

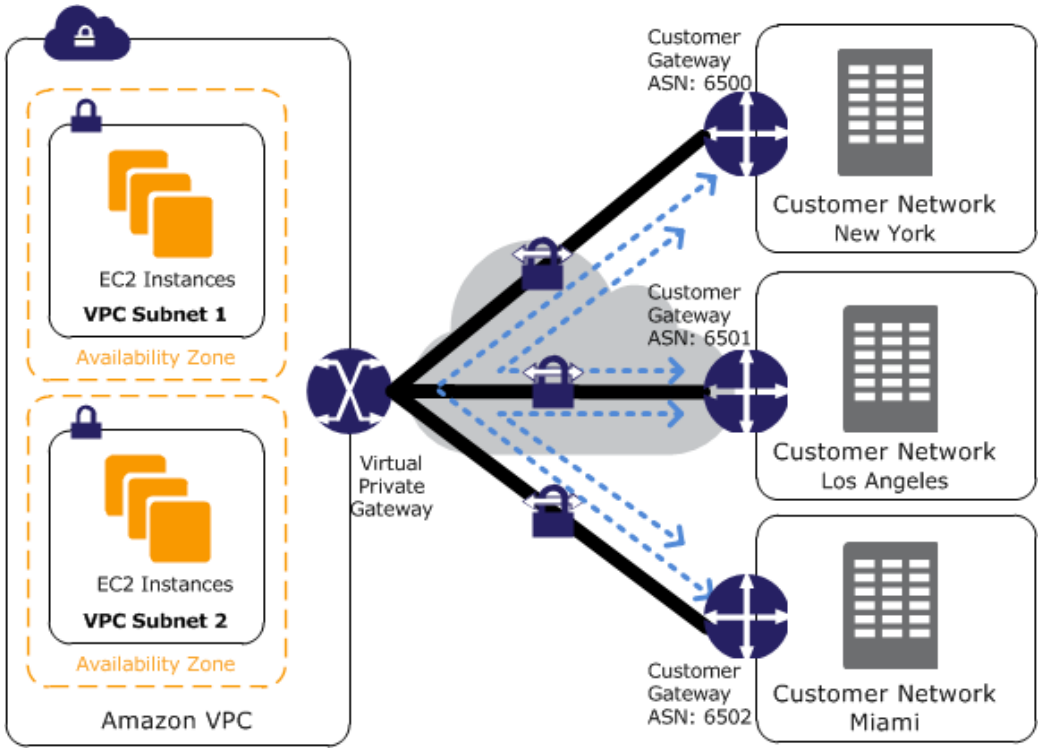


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS VPN CloudHub

- Can connect up to 10 Customer Gateway for each Virtual Private Gateway (VGW)
- Low cost hub-and-spoke model for primary or secondary network connectivity between locations
- Provide secure communication between sites, if you have multiple VPN connections
- It's a VPN connection so it goes over the public internet
- Can be a **failover connection** between your on-premises locations

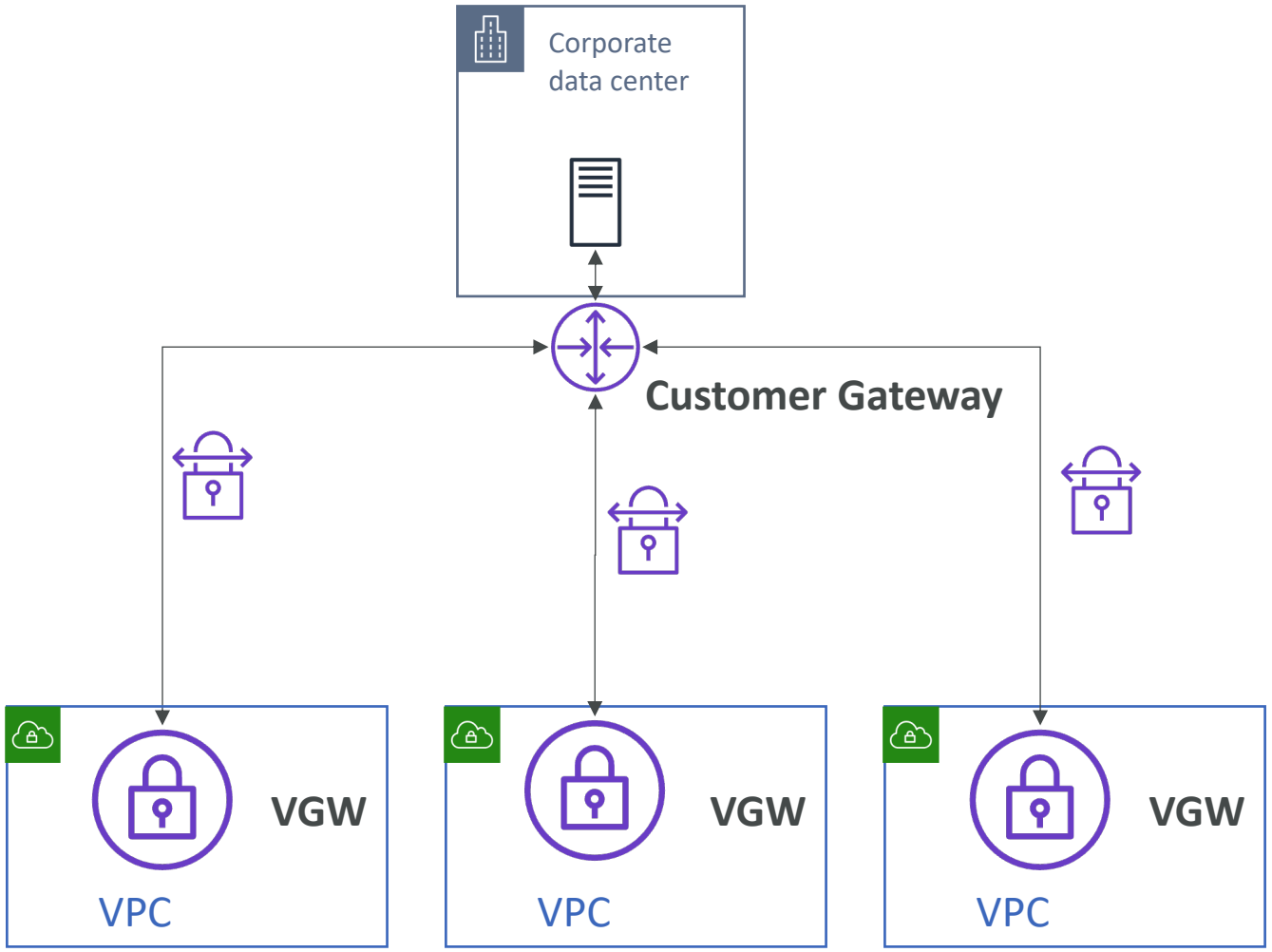


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPN to multiple VPC

- For VPN-based customers, AWS recommends creating a separate VPN connection for each customer VPC.
- Direct Connect is recommended because it has a Direct Connect Gateway

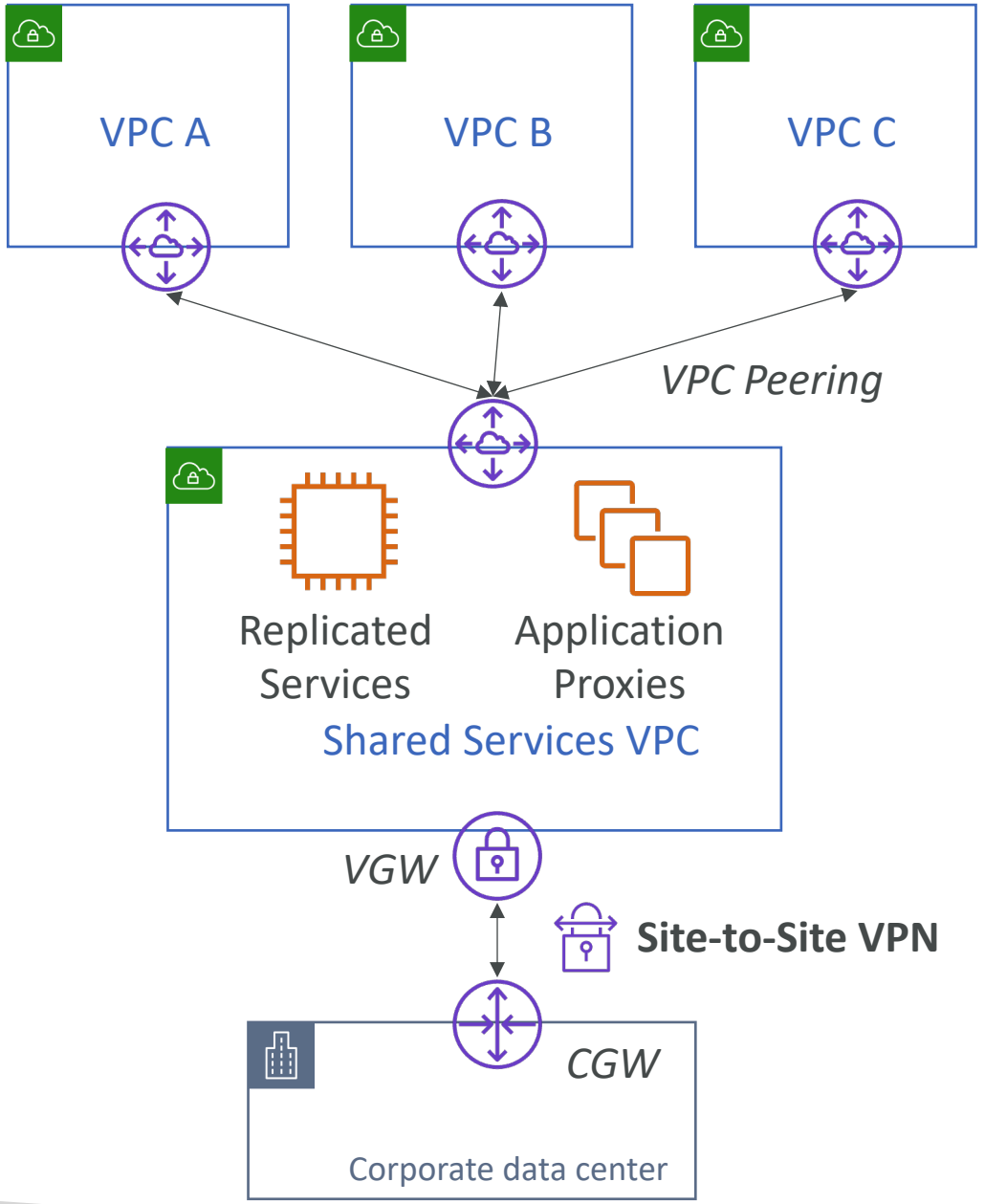


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Shared Services VPC

- Create a VPN connection between on-premises and shared service VPC
- Replicate services, applications, databases between on-premises and the Shared Services VPC or deploy proxies in the shared service VPC
- Do VPC peering between the VPC and the shared service VPC
- VPCs can directly access the Shared Service VPC services and do not need VPN connections to on-premises

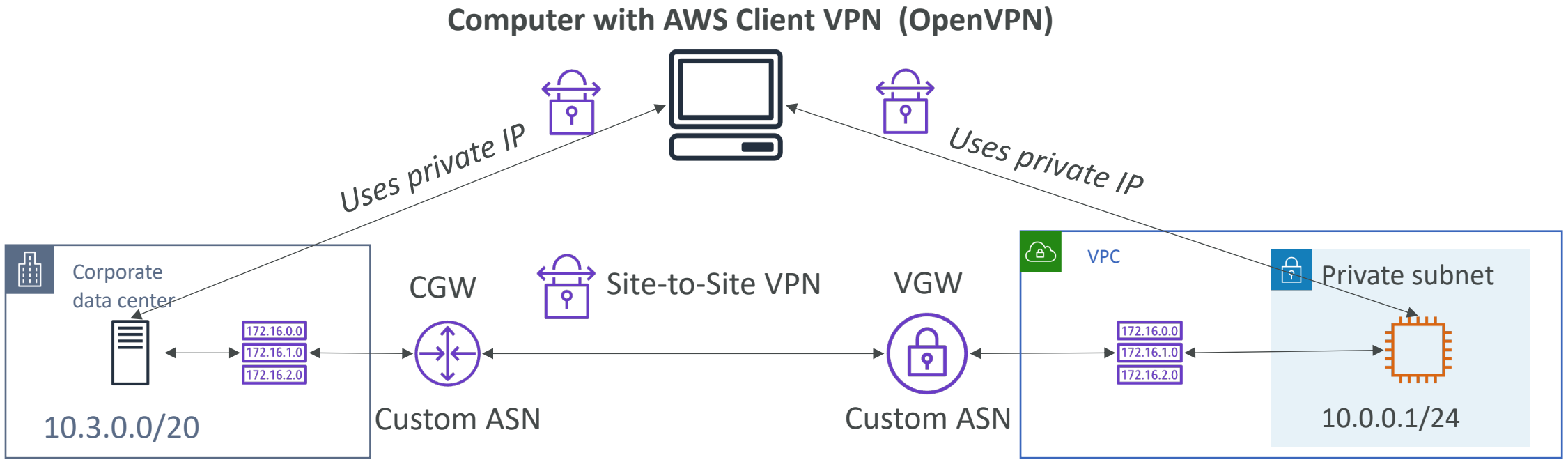


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Client VPN

- Connect from your computer using OpenVPN to your private network in AWS and on-premises

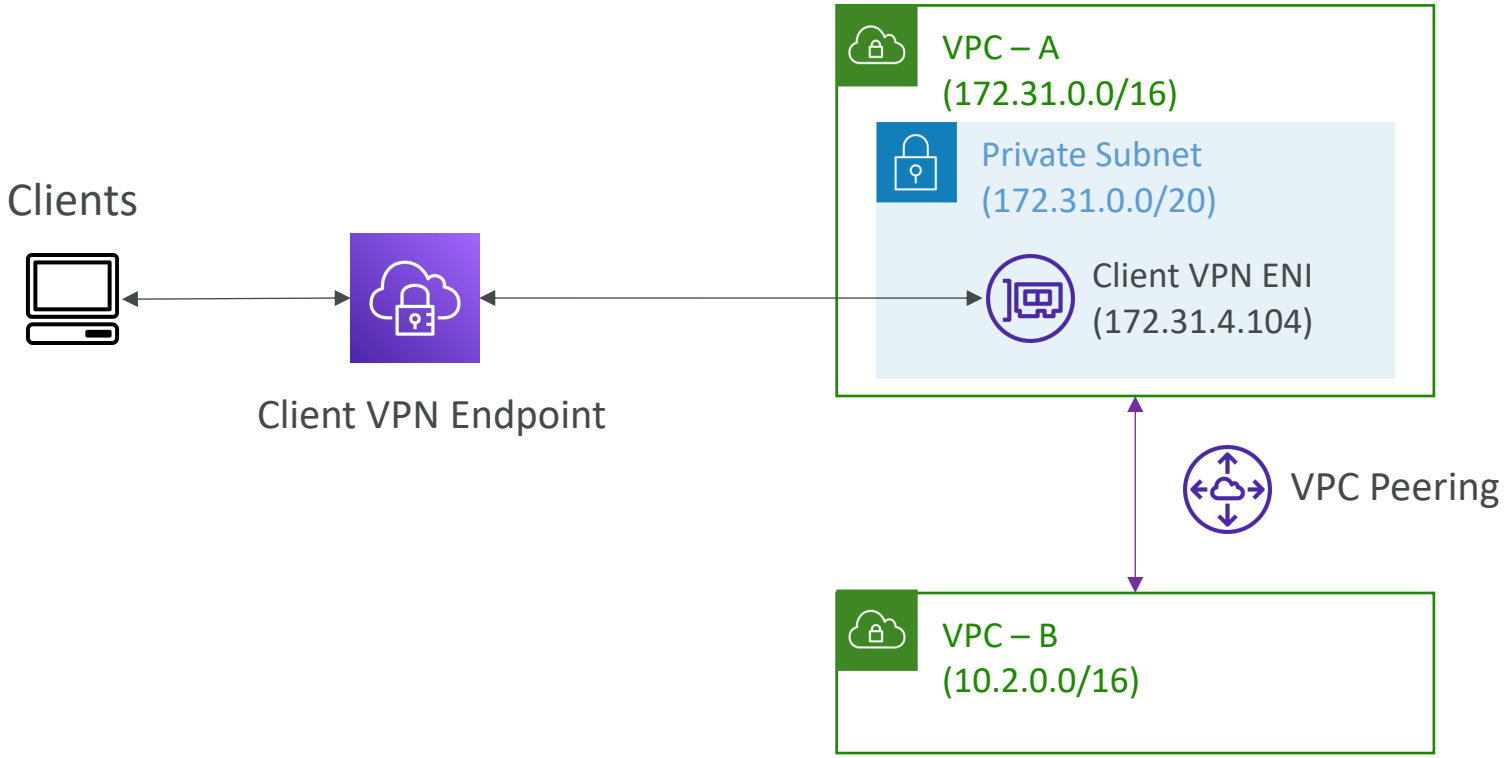


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Client VPN – Peered VPC

- Client VPN is compatible with VPC peering

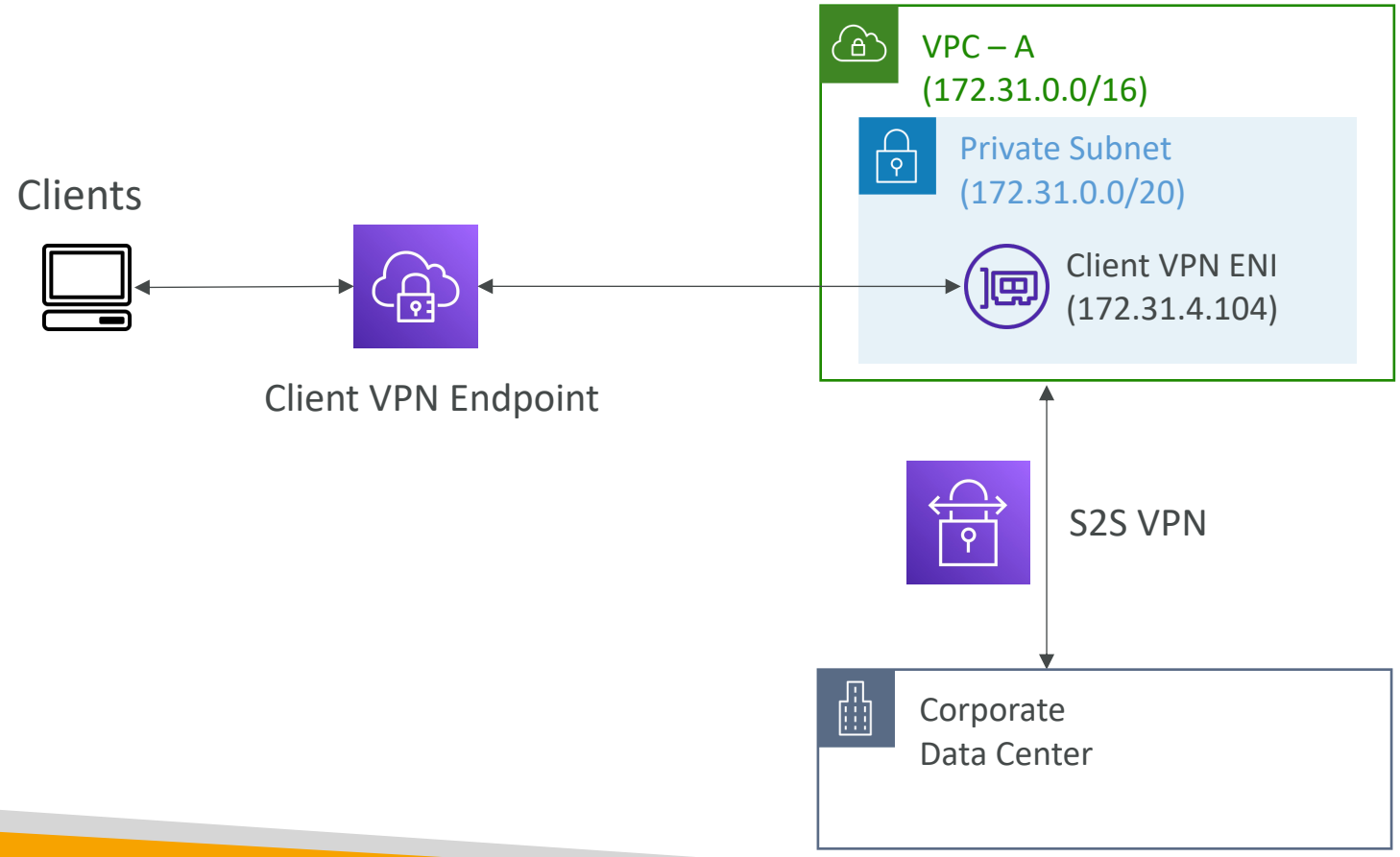


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Client VPN – Access to On-Premises

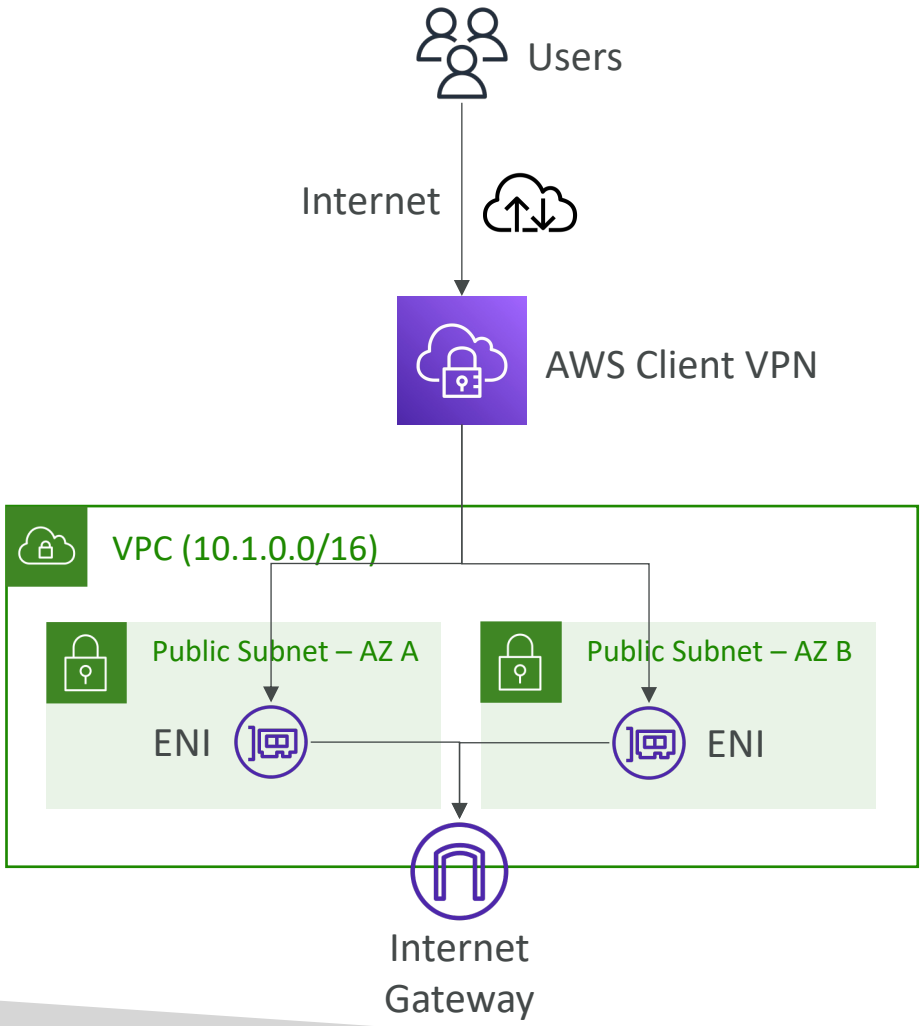
- Access On-Premises resources through AWS with Client VPN

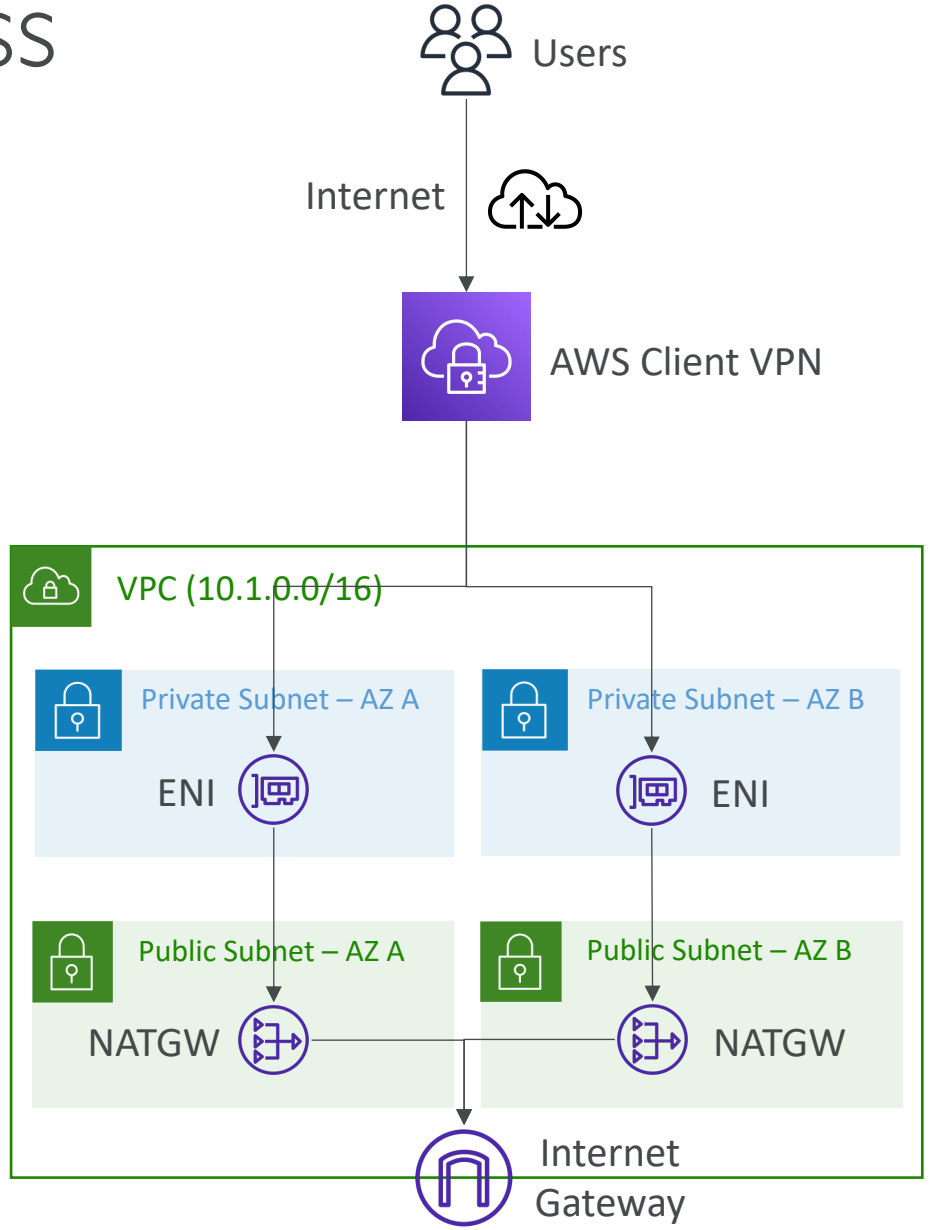


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Client VPN – Internet Access

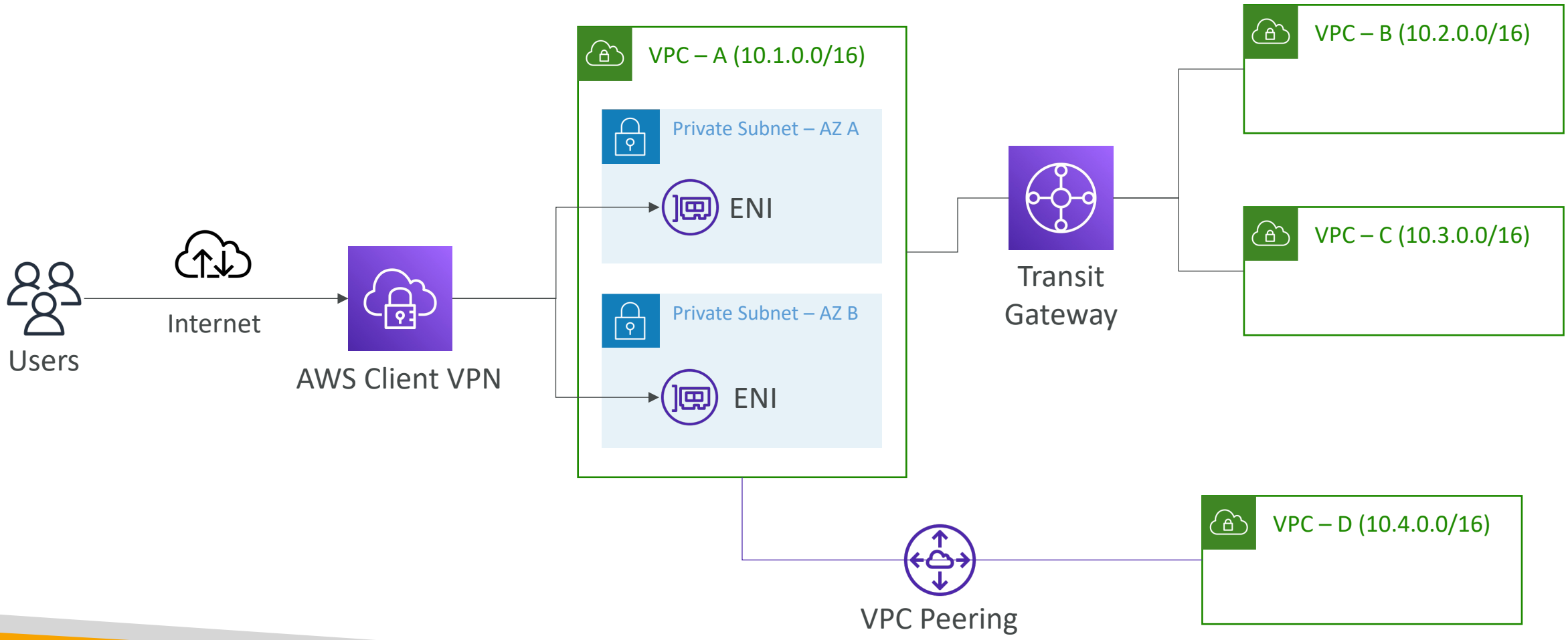




© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

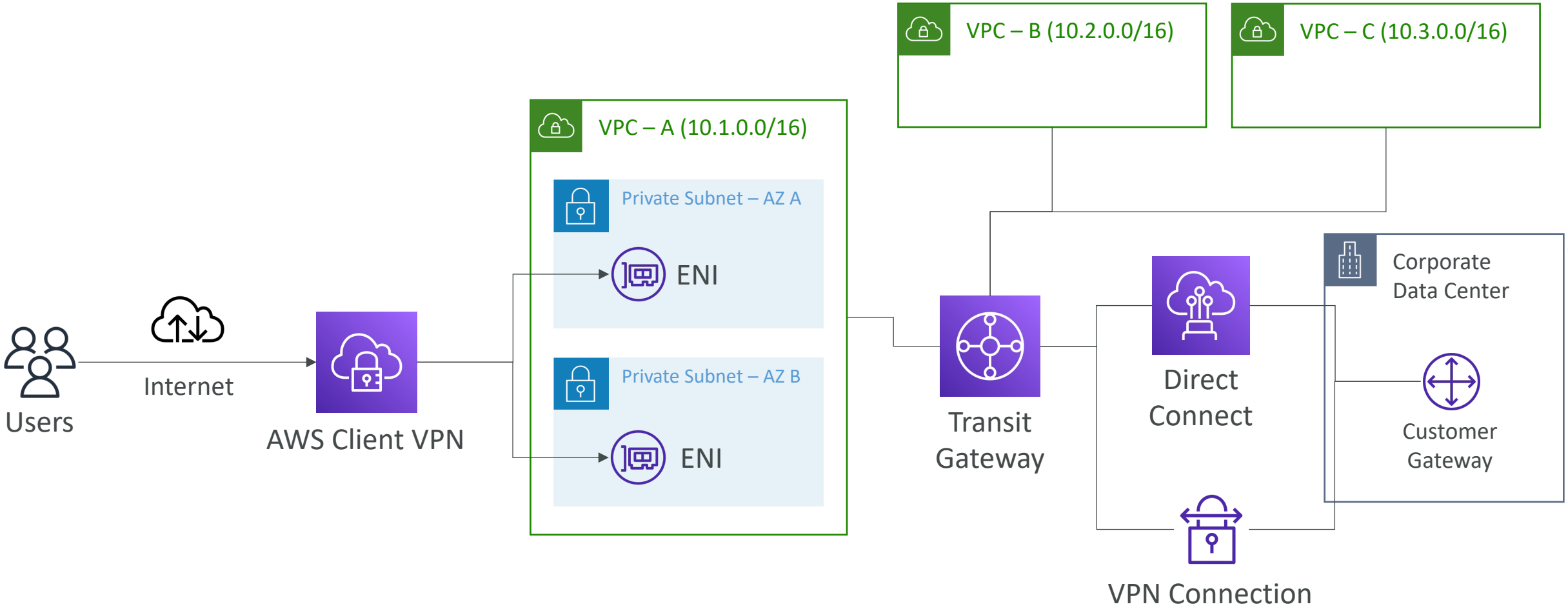
Client VPN – Transit Gateway



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Client VPN – Transit Gateway



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Direct Connect

- Provides a dedicated private connection from a remote network to your VPC
- Dedicated connection must be setup between your DC and AWS Direct Connect locations
- More expensive than running a VPN solution
- Private access to AWS services through VIF
- Bypass ISP, reduce network cost, increase bandwidth and stability
- Not redundant by default (must setup a failover DX or VPN)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Direct Connect – Virtual Interfaces (VIF)

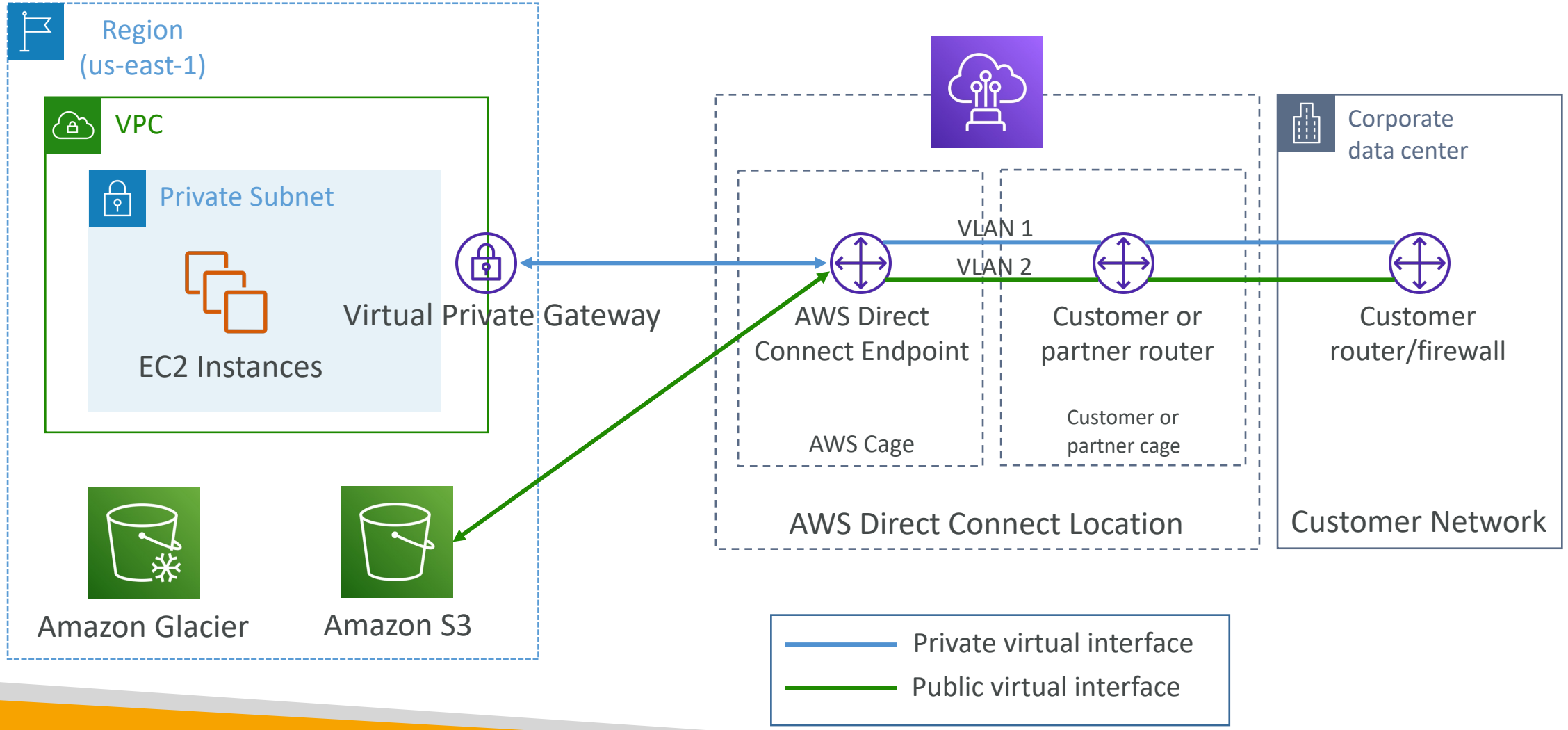
- **Public VIF** – connect to Public AWS Endpoints (S3 buckets, EC2 service, anything AWS …)
- **Private VIF** – connect to resources in your VPC (EC2 instances, ALB, …)
- **Transit Virtual Interface** – connect to resources in a VPC using a Transit Gateway

- VPC Interface Endpoints can be accessed through Private VIF

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Direct Connect Diagram



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Direct Connect – Connection Types

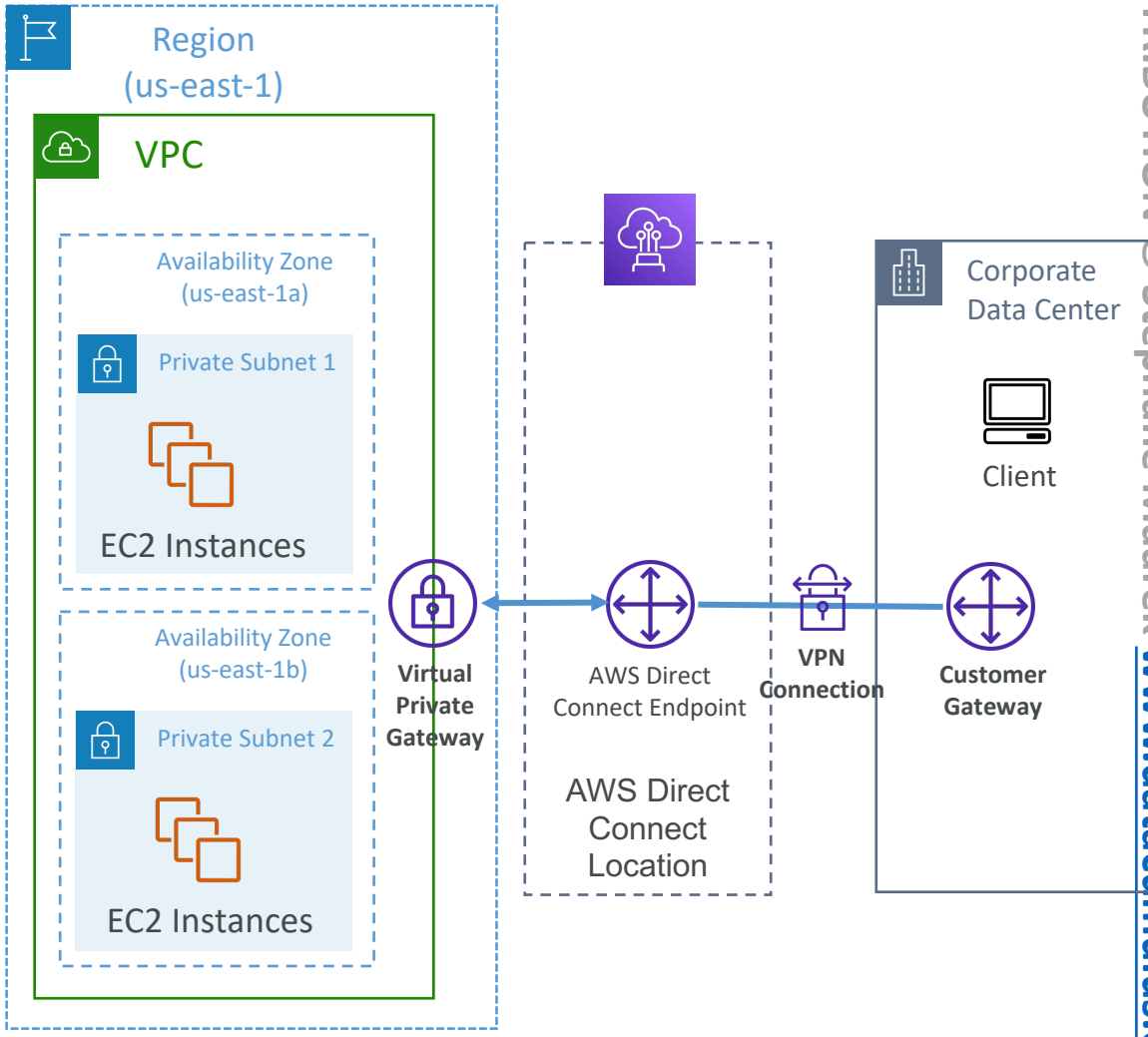
- **Dedicated Connections:** from 1 Gbps to 400 Gbps
  - Physical ethernet port dedicated to a customer
  - Request made to AWS first, then completed by AWS Direct Connect Partners
- **Hosted Connections:** from 50Mbps to 25 Gbps
  - Connection requests are made via AWS Direct Connect Partners
  - Capacity can be **added or removed on demand**
- Lead times are often longer than 1 month to establish a new connection

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Direct Connect – Encryption

- Data in transit is not encrypted but is private
- AWS Direct Connect + VPN provides an IPsec-encrypted private connection
- VPN over Direct Connect connection Uses Public VIF
- Good for an extra level of security, but slightly more complex to put in place

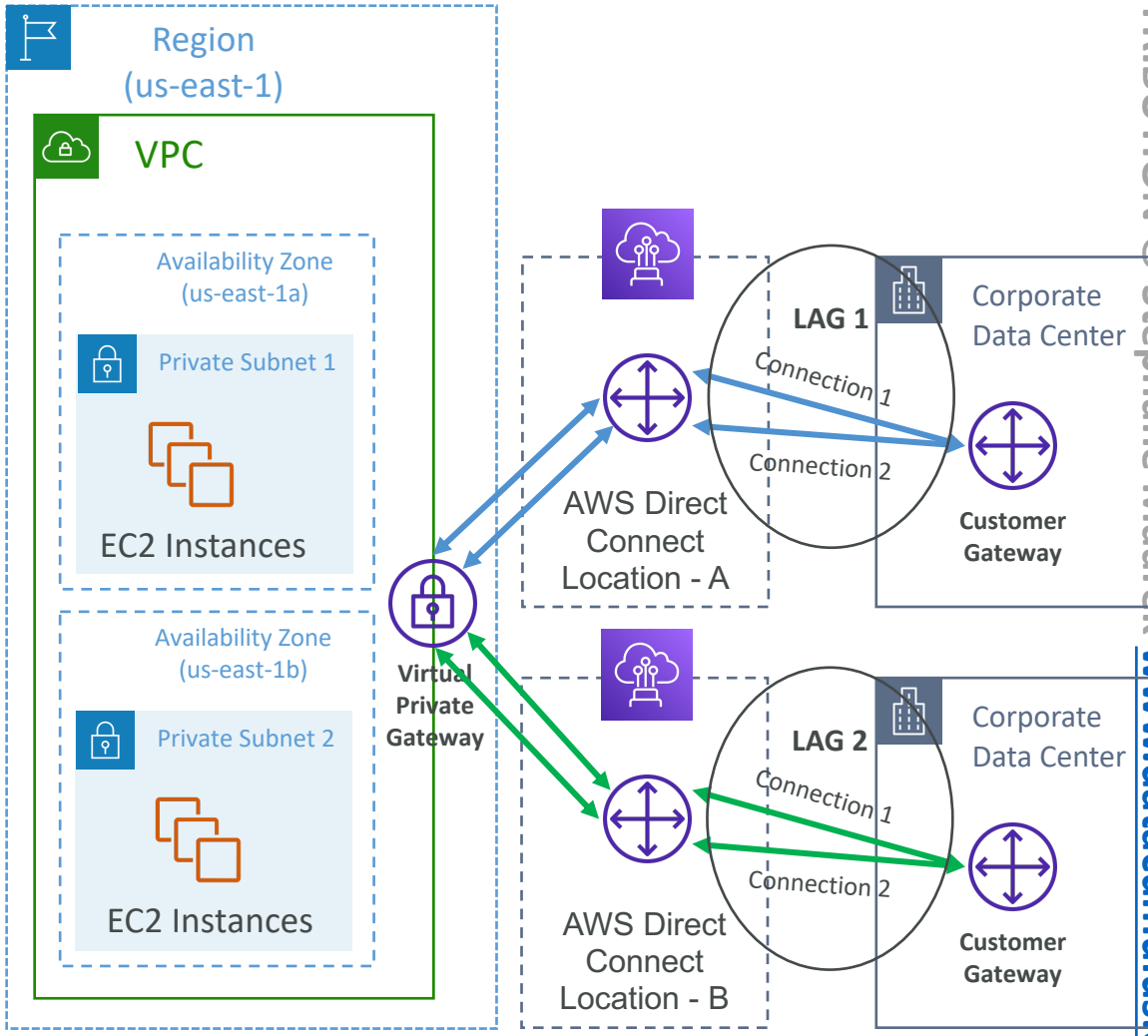


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Direct Connect – Link Aggregation Groups (LAG)

- Get **increased speed** and **failover** by summing up existing DX connections into a **logical one**
- Can aggregate up to 4 connections (active-active mode)
- Can add connections over time to the LAG
- All connections in the LAG:
  - Must be dedicated connections
  - Must have the same bandwidth
  - Must terminate at the same AWS Direct Connect Endpoint
- Can set a minimum number of connections for the LAG to function

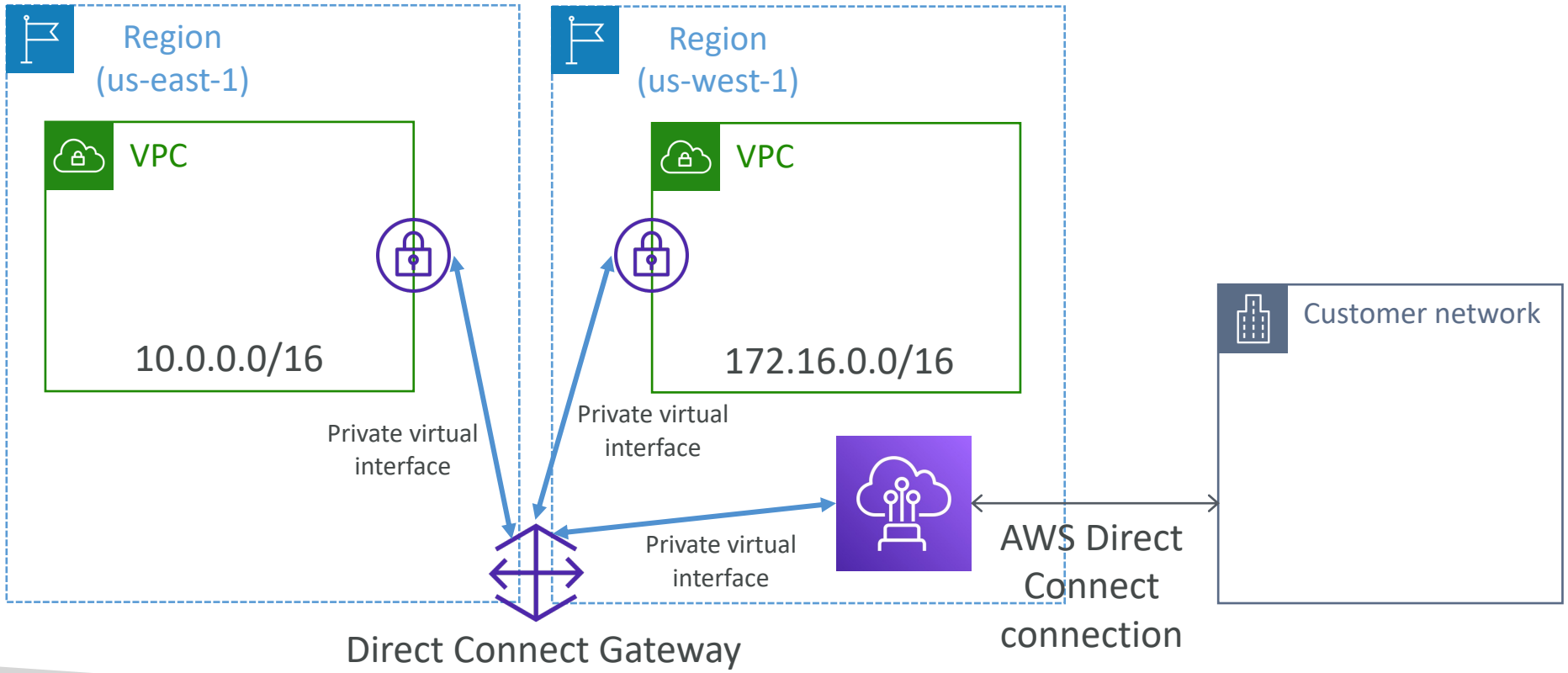


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Direct Connect Gateway

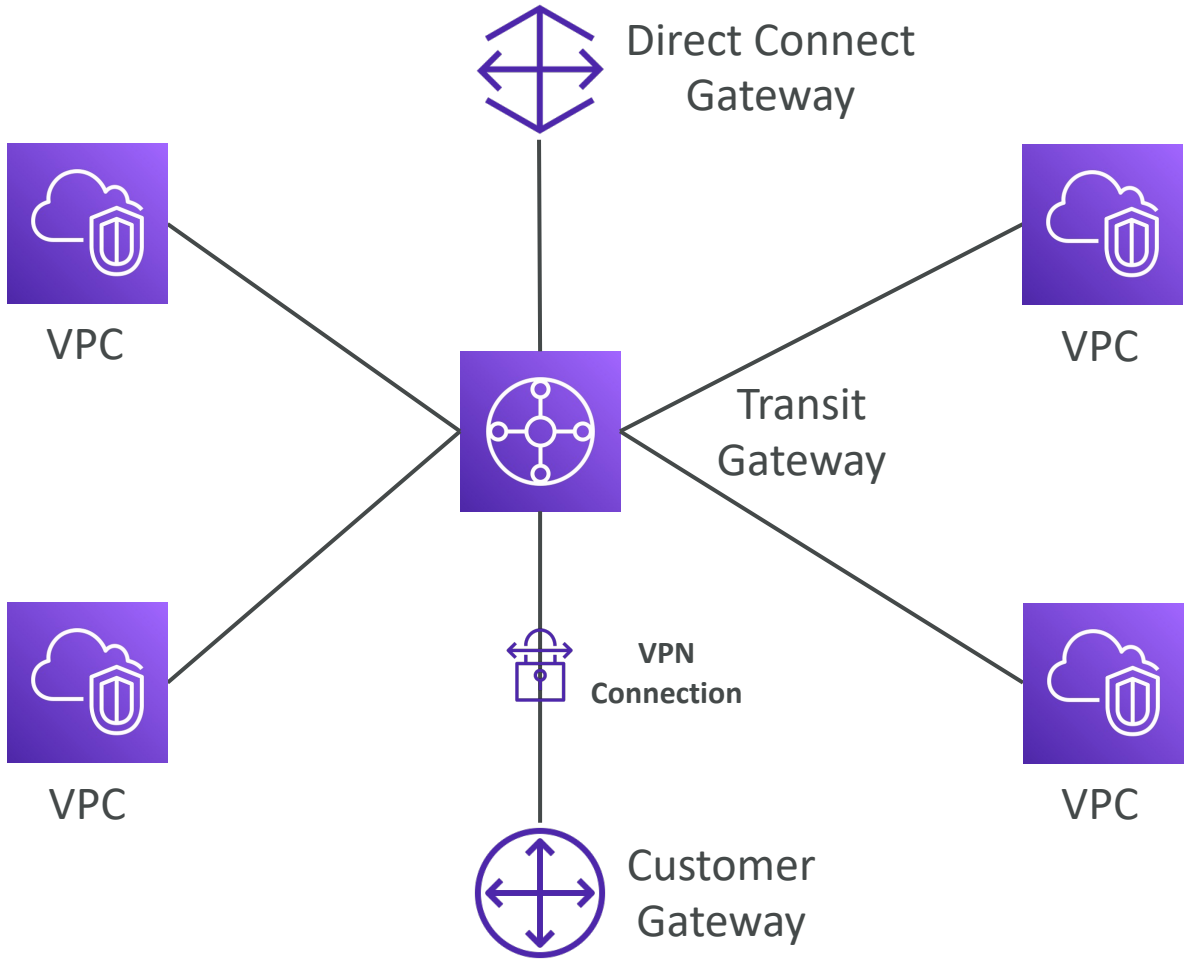
- If you want to setup a Direct Connect to one or more VPC in many different regions (same/cross account), you must use a Direct Connect Gateway



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

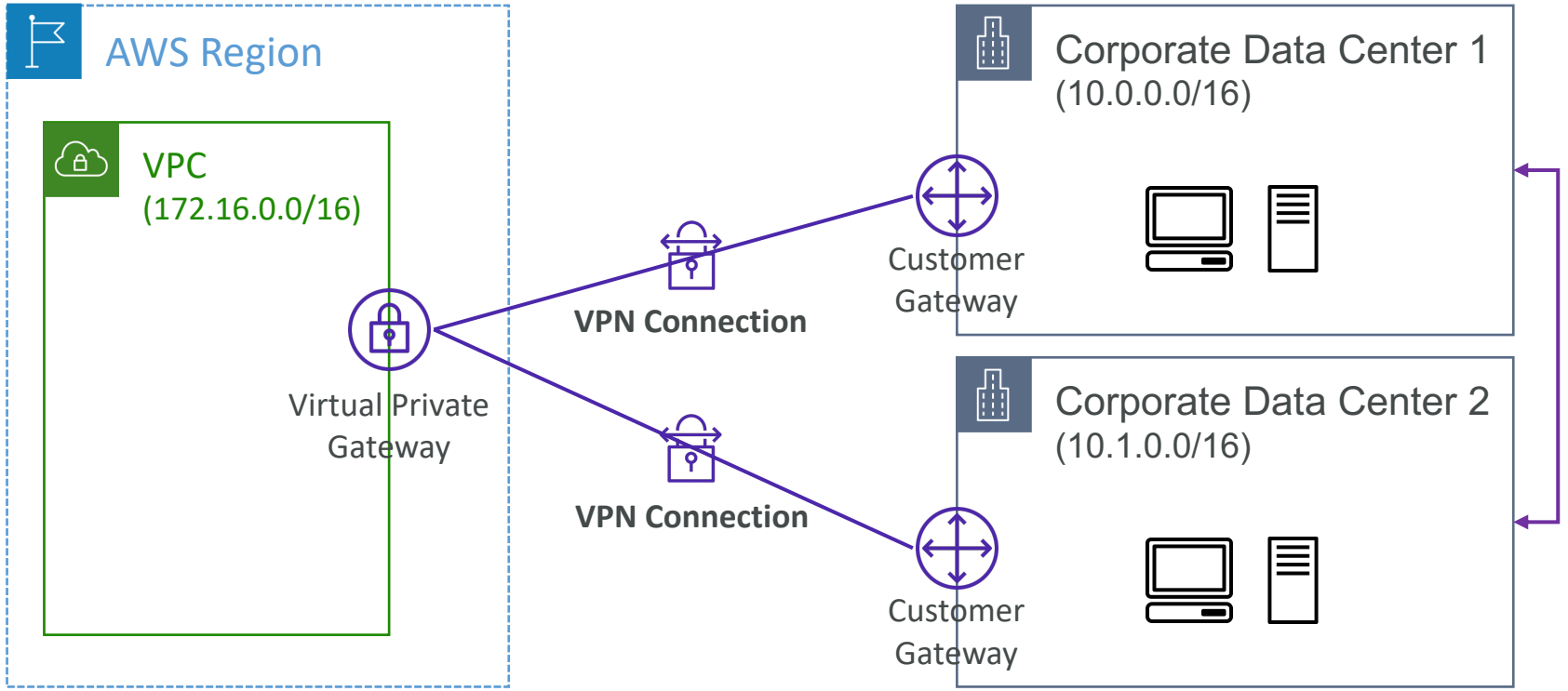
Direct Connect Gateway + Transit Gateway



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Site-to-Site Active-Active Connection

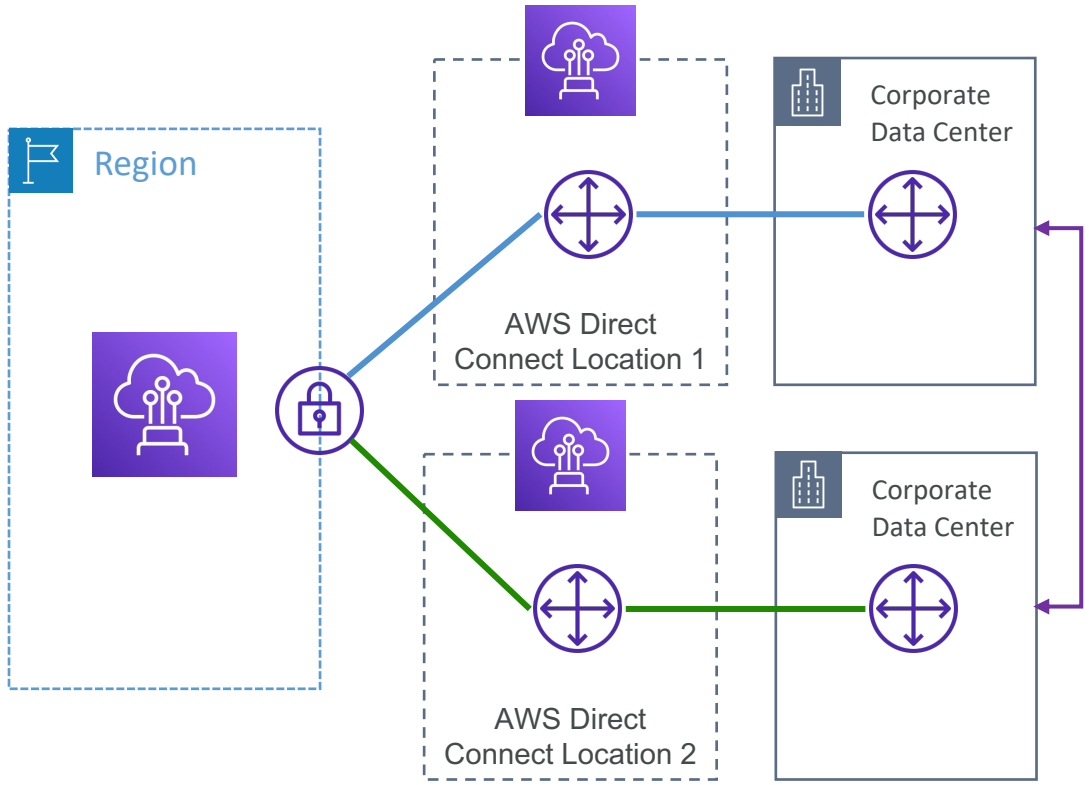


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Direct Connect – High Availability

Multiple connections at multiple AWS Direct Connect locations

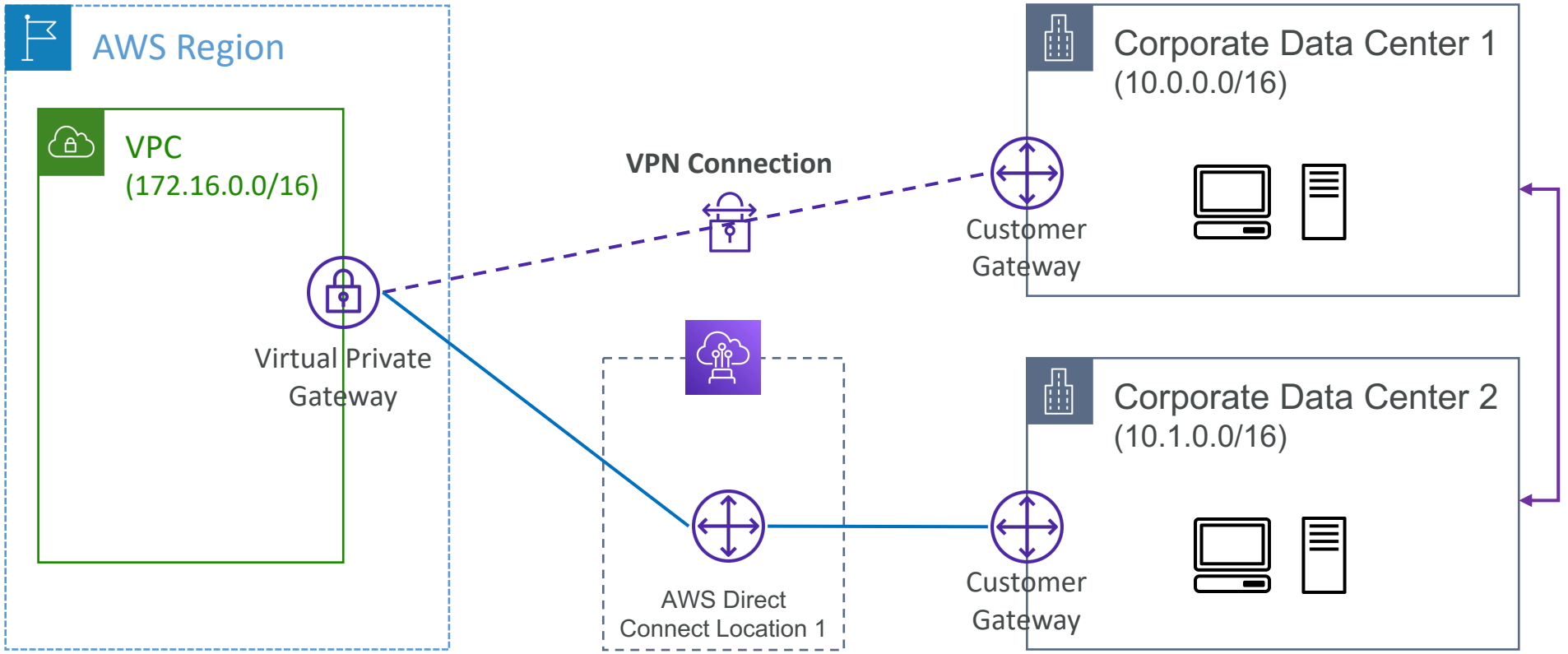


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Direct Connect – High Availability

Backup VPN Connection

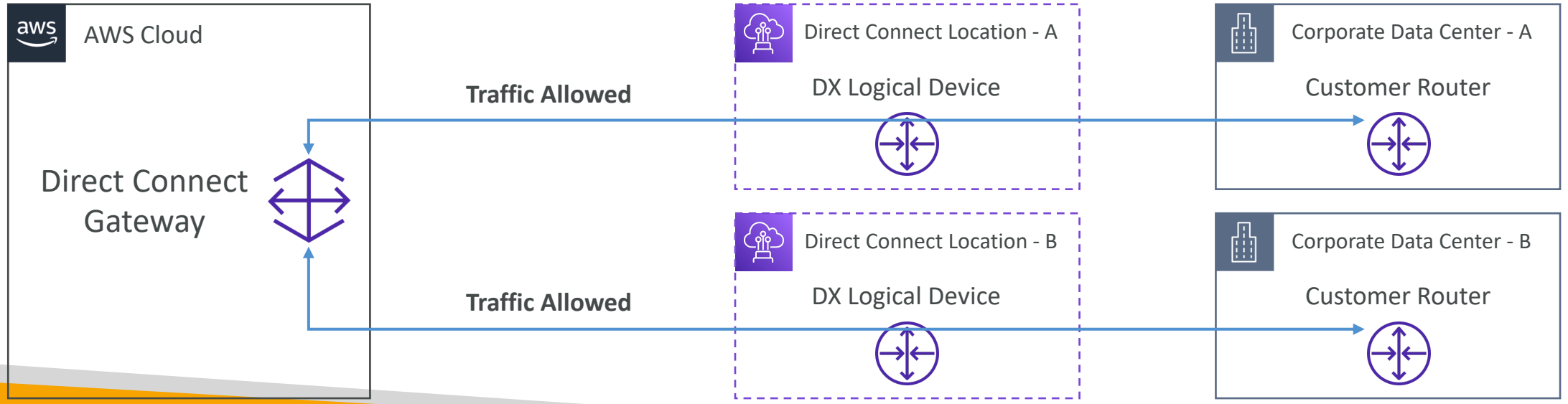


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Direct Connect Gateway – SiteLink

- Allows you to send data from on Direct Connect location to another
- Bypassing AWS Regions
- Use cases: **create private network connections between on-premises data centers by connecting them to Direct Connect locations**
- Data is sent over the fastest path between Direct Connect locations



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

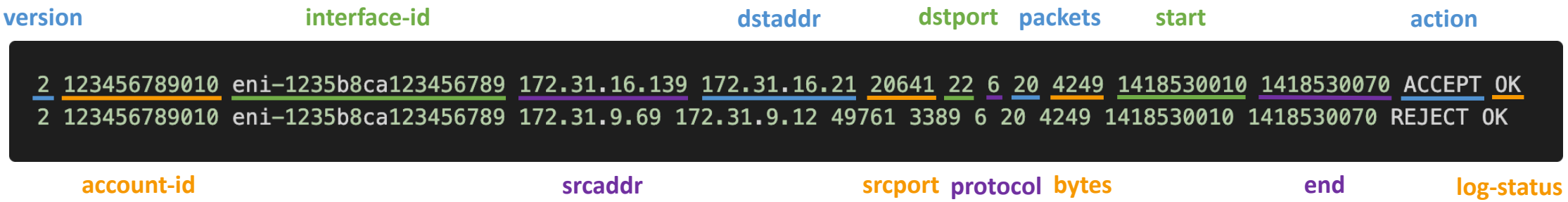
VPC Flow Logs

- Capture information about IP traffic going into your interfaces:
  - VPC Flow Logs
  - Subnet Flow Logs
  - Elastic Network Interface (ENI) Flow Logs
- Helps to monitor & troubleshoot connectivity issues
- Flow logs data can go to S3, CloudWatch Logs, and Kinesis Data Firehose
- Captures network information from AWS managed interfaces too: ELB, RDS, ElastiCache, Redshift, WorkSpaces, NATGW, Transit Gateway…

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Flow Logs Syntax



- **srcaddr & dstaddr** – help identify problematic IP
- **srcport & dstport** – help identity problematic ports
- **Action** – success or failure of the request due to Security Group / NACL
- Can be used for analytics on usage patterns, or malicious behavior
- **Query VPC flow logs using Athena on S3 or CloudWatch Logs Insights**
- Flow Logs examples: https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs-records-examples.html

© Stephane Maarek

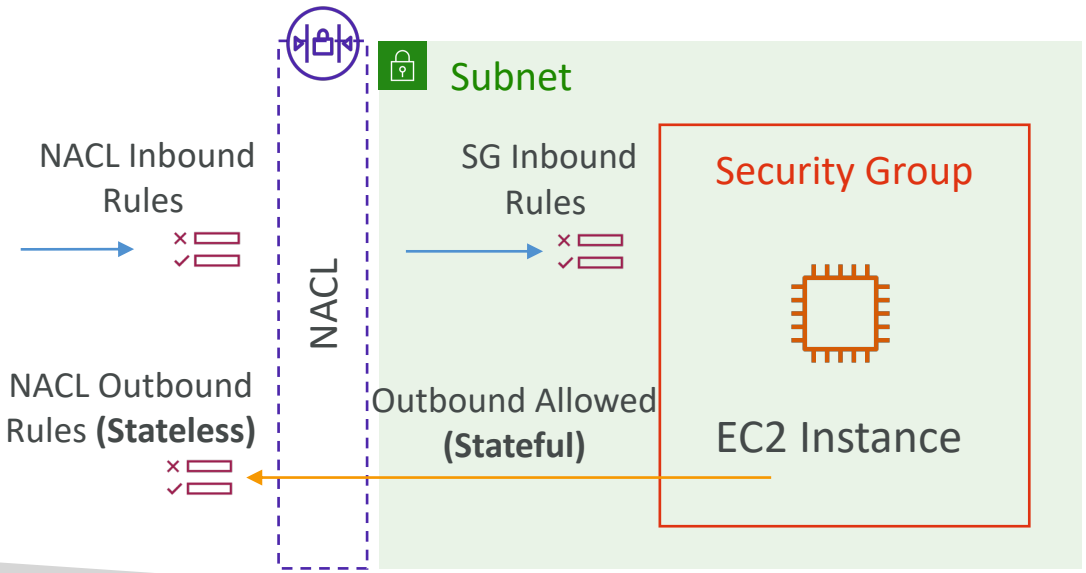
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Flow Logs – Troubleshoot SG & NACL issues

Look at the **"ACTION"** field

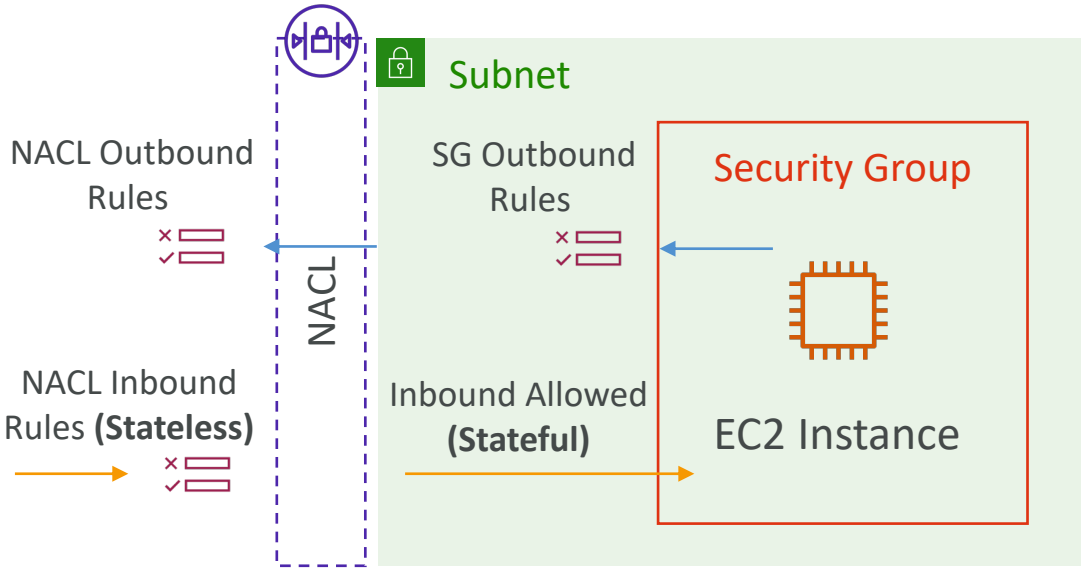
Incoming Requests

- Inbound REJECT => NACL or SG
- Inbound ACCEPT, Outbound REJECT => NACL



Outgoing Requests

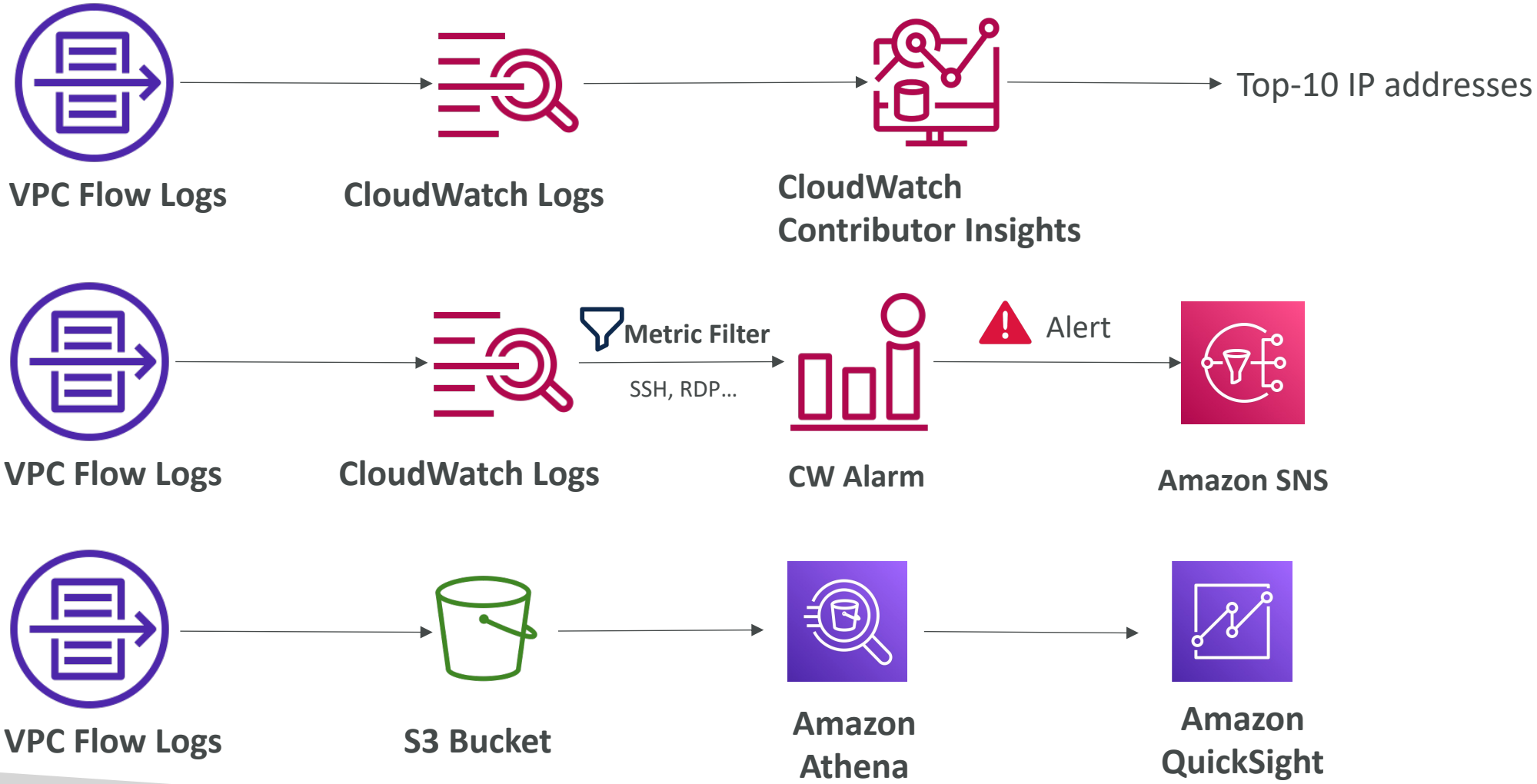
- Outbound REJECT => NACL or SG
- Outbound ACCEPT, Inbound REJECT => NACL



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Flow Logs – Architectures



© Stephane Maarek

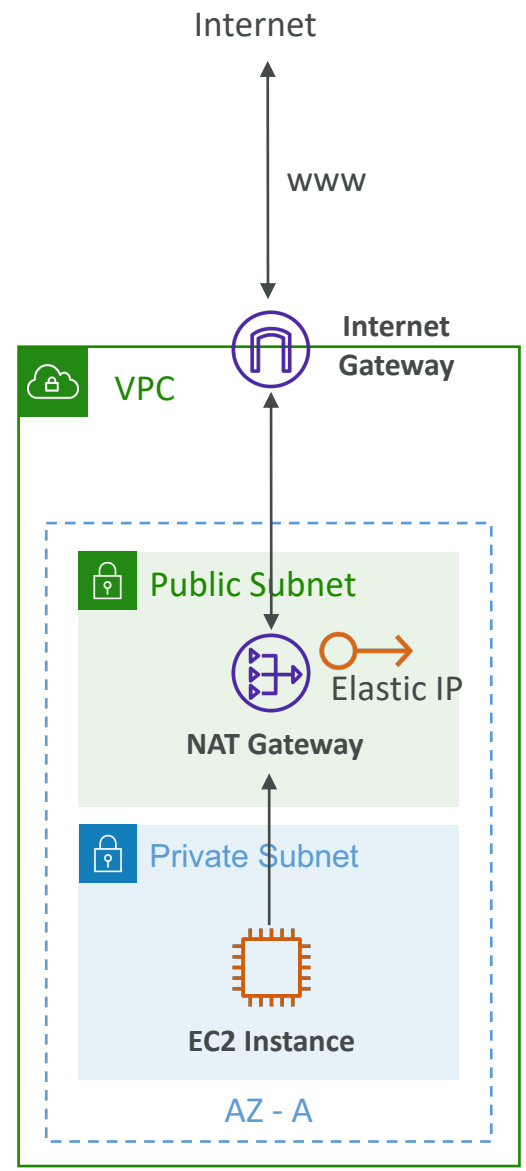
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

VPC Flow Logs – with NAT Gateway

- *My Virtual Private Cloud (VPC) flow logs show Action = ACCEPT for inbound traffic coming from public IP addresses. However, my understanding of network address translation (NAT) gateways was that they don't accept traffic from the internet. Is my NAT gateway accepting inbound traffic from the internet?*
- Inbound traffic is permitted by Security Group or NACLs
  - Traffis isn't permitted by the NAT Gateway, it's dropped
  - To confirm run the following query in CloudWatch Log Group

```
filter (dstAddr like 'xxx.xxx' and srcAddr like 'public IP')
| stats sum(bytes) as bytesTransferred by srcAddr, dstAddr
| limit 10
```

- Make 'xxx.xxx' the first two octets of your VPC CIDR
- Replace Public IP with the IP you see in logs
- You will see traffic on the Private IP of the NAT Gateway but nowhere else: traffic was unsolicited and then dropped



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Network Protection on AWS

- To protect network on AWS, we've seen
  - Network Access Control Lists (NACLs)
  - Amazon VPC security groups
  - AWS WAF (protect against malicious requests)
  - AWS Shield & AWS Shield Advanced
  - AWS Firewall Manager (to manage them across accounts)

- But what if we want to protect in a sophisticated way our entire VPC?

© Stephane Maarek

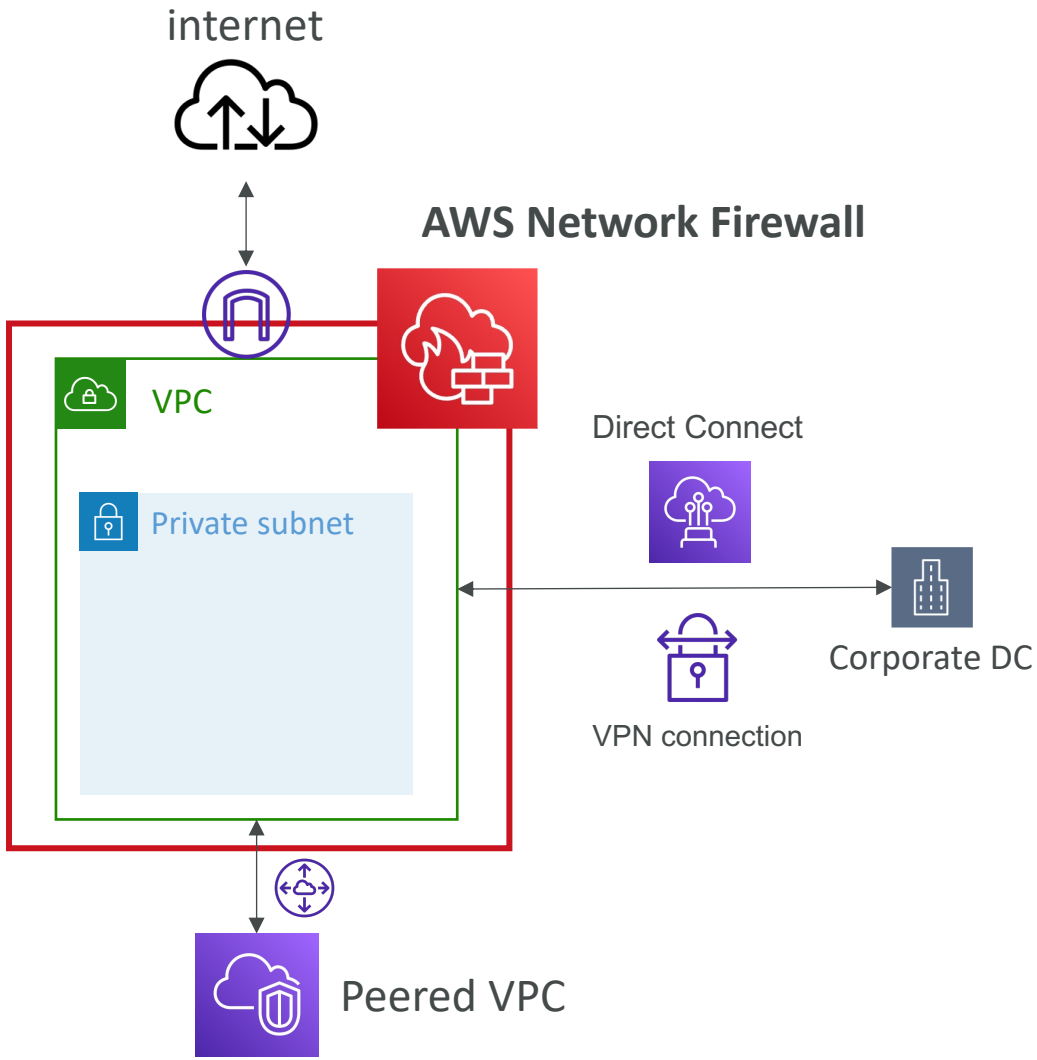
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Network Firewall

- Protect your entire Amazon VPC

- From Layer 3 to Layer 7 protection
- Any direction, you can inspect
  - VPC to VPC traffic
  - Outbound to internet
  - Inbound from internet
  - To / from Direct Connect & Site-to-Site VPN

- Internally, the AWS Network Firewall uses the AWS Gateway Load Balancer
- Rules can be centrally managed cross-account by AWS Firewall Manager to apply to many VPCs



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Network Firewall – Fine Grained Controls

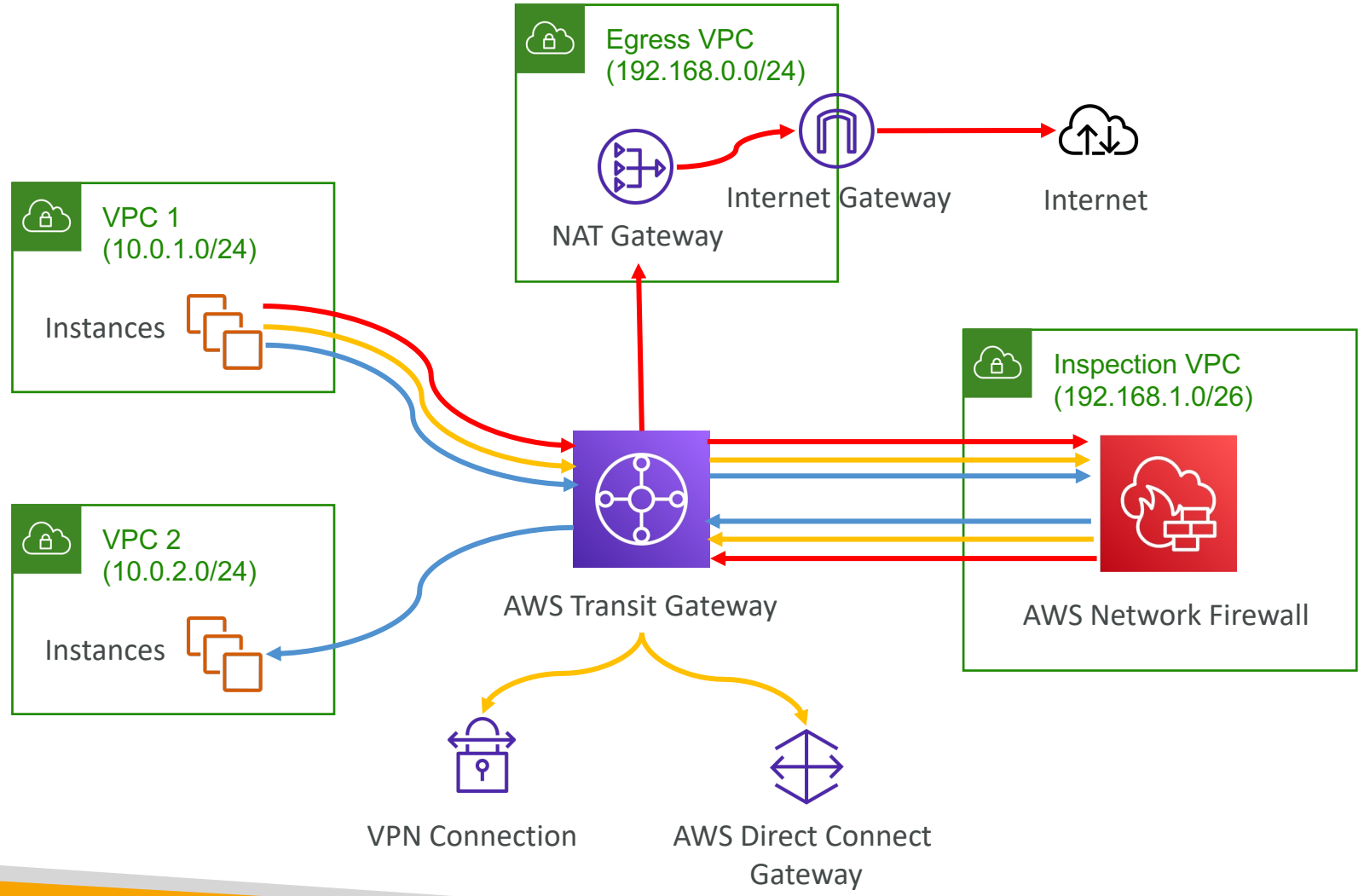
- Supports 1000s of rules
  - IP & port - example: 10,000s of IPs filtering
  - Protocol – example: block the SMB protocol for outbound communications
  - Stateful domain list rule groups: only allow outbound traffic to *.mycorp.com or third-party software repo
  - General pattern matching using regex
- **Traffic filtering: Allow, drop, or alert for the traffic that matches the rules**
- **Active flow inspection** to protect against network threats with intrusion-prevention capabilities (like Gateway Load Balancer, but all managed by AWS)
- Send logs of rule matches to Amazon S3, CloudWatch Logs, Kinesis Data Firehose

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Network Firewall - Architecture

- North-South VPC egress to Internet
- North-South VPC to VPN or Direct Connect
- East-West VPC to VPC



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

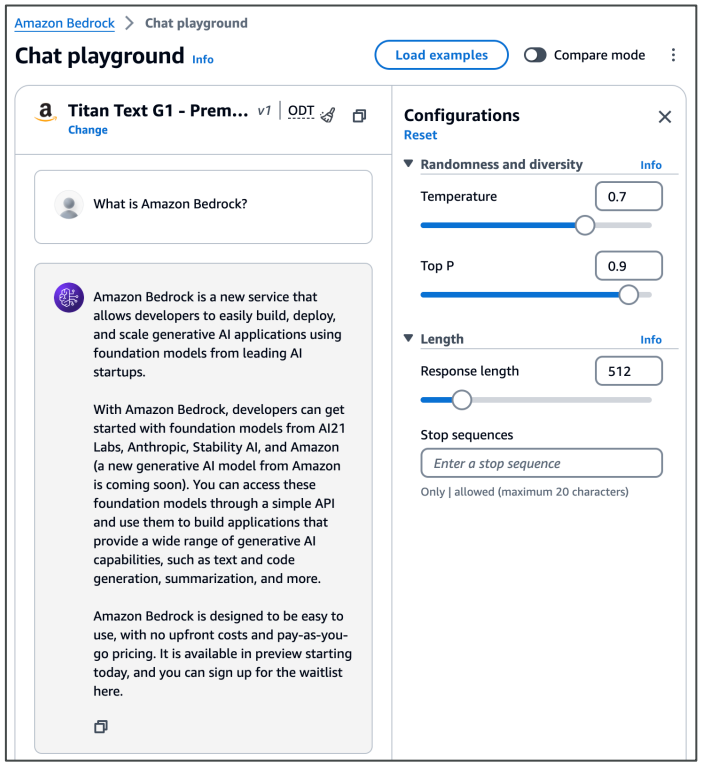
Machine Learning

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Bedrock

- Build Generative AI (Gen-AI) applications on AWS
- Fully-managed service, no servers for you to manage
- Keep control of your data used to train the model
- Pay-per-use pricing model
- Unified APIs
- Leverage a wide array of foundation models
- Out-of-the box features: RAG, LLM Agents…
- Security, Privacy, Governance and Responsible AI features



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Bedrock – Foundation Models

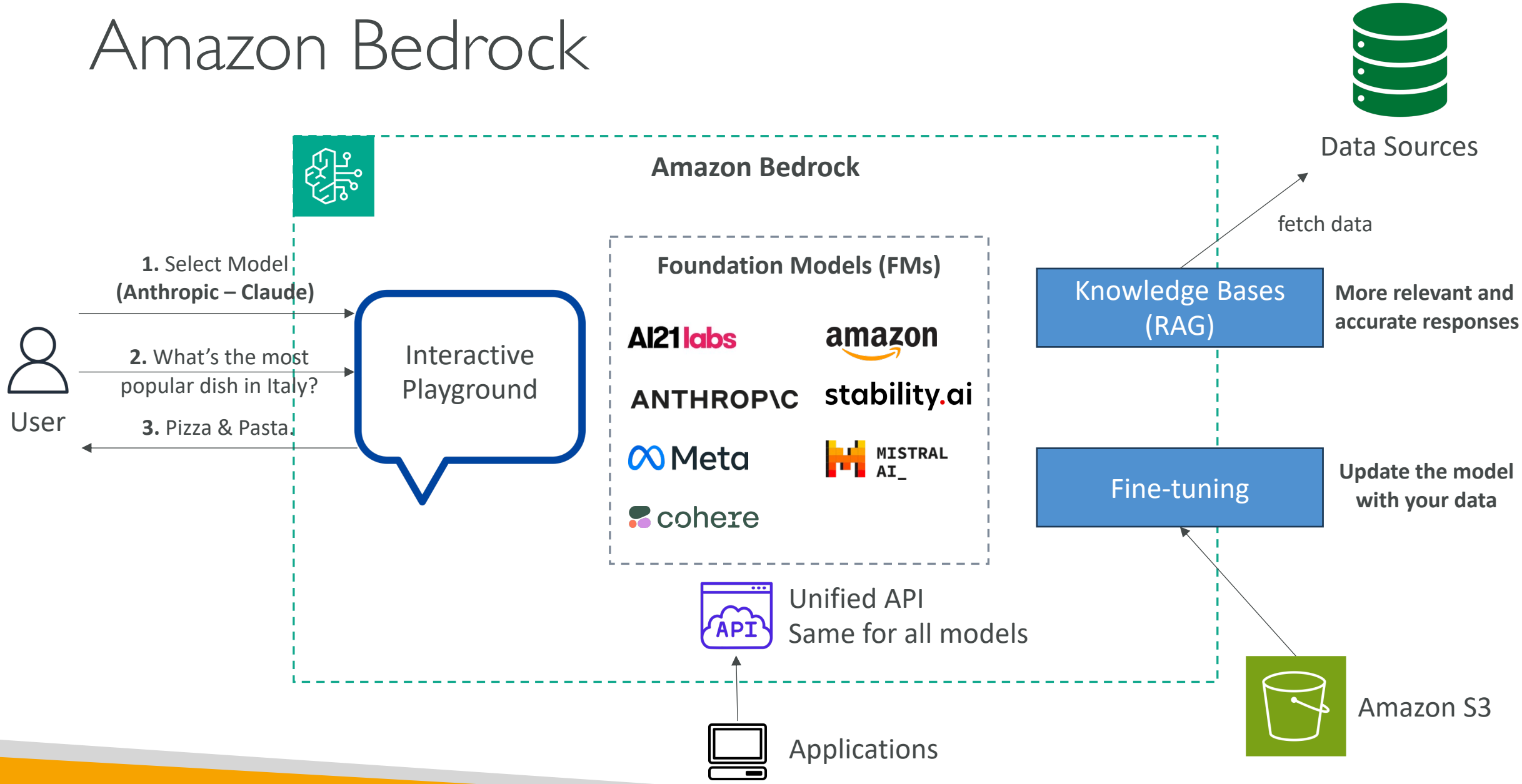
- Access to a wide range of Foundation Models (FM)

- Amazon Bedrock makes a copy of the FM, available only to you, which you can further fine-tune with your own data
- None of your data is used to train the FM

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Bedrock



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Amazon Rekognition

- Find **objects, people, text, scenes** in **images and videos** using ML
- **Facial analysis** and **facial search** to do user verification, people counting
- Create a database of "familiar faces" or compare against celebrities
- Use cases:
  - Labeling
  - Content Moderation
  - Text Detection
  - Face Detection and Analysis (gender, age range, emotions…)
  - Face Search and Verification
  - Celebrity Recognition
  - Pathing (ex: for sports game analysis)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Rekognition – Content Moderation

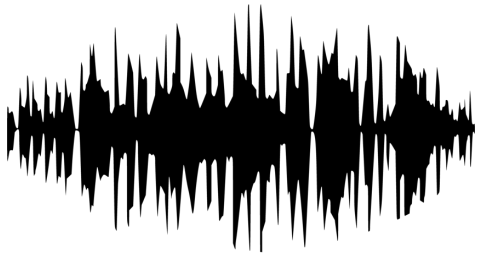
- Detect content that is inappropriate, unwanted, or offensive (image and videos)
- Used in social media, broadcast media, advertising, and e-commerce situations to create a safer user experience
- **Set a Minimum Confidence Threshold for items that will be flagged**
- Flag sensitive content for manual review in Amazon Augmented AI (A2I)
- Help comply with regulations

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Transcribe

- Automatically **convert speech to text**
- Uses a **deep learning process** called **automatic speech recognition** (ASR) to convert speech to text quickly and accurately
- **Automatically remove Personally Identifiable Information (PII) using Redaction**
- **Supports Automatic Language Identification for multi-lingual audio**
- Use cases:
  - transcribe customer service calls
  - automate closed captioning and subtitling
  - generate metadata for media assets to create a fully searchable archive



"Hello my name is Stéphane.
I hope you're enjoying the course!

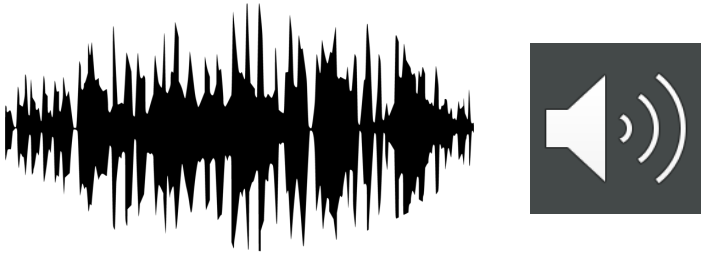
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Polly

- Turn text into lifelike speech using deep learning
- Allowing you to create applications that talk

Hi! My name is Stéphane
and this is a demo of Amazon Polly



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Polly – Lexicon & SSML

- Customize the pronunciation of words with **Pronunciation lexicons**
  - Stylized words: St3ph4ne => "Stephane"
  - Acronyms: AWS => "Amazon Web Services"
- Upload the lexicons and use them in the **SynthesizeSpeech** operation

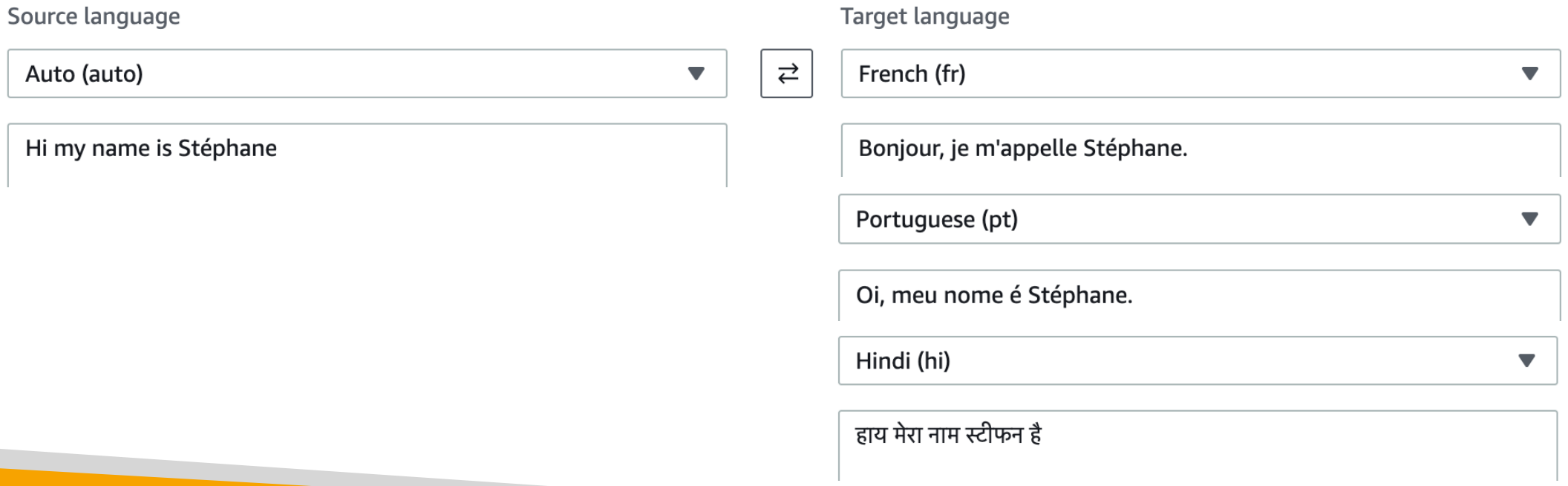
- Generate speech from plain text or from documents marked up with **Speech Synthesis Markup Language (SSML)** – enables more customization
  - emphasizing specific words or phrases
  - using phonetic pronunciation
  - including breathing sounds, whispering
  - using the Newscaster speaking style

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Translate

- Natural and accurate **language translation**
- Amazon Translate allows you to **localize content** - such as websites and applications - for **international users**, and to easily translate large volumes of text efficiently.

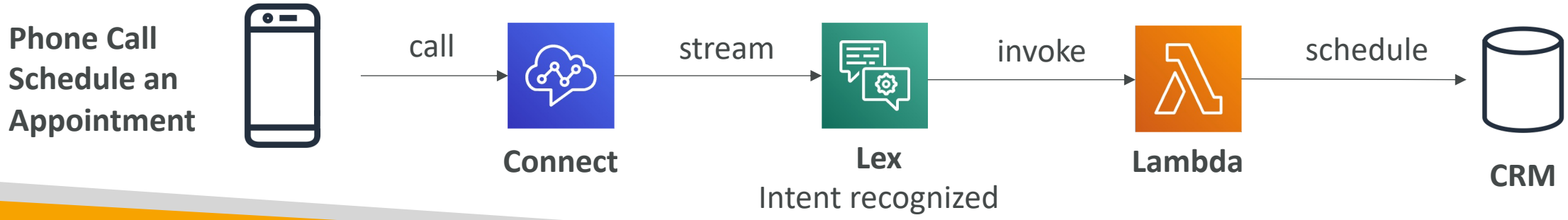


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Lex & Connect

- **Amazon Lex:** (same technology that powers Alexa)
  - Automatic Speech Recognition (ASR) to convert speech to text
  - Natural Language Understanding to recognize the intent of text, callers
  - Helps build chatbots, call center bots
- **Amazon Connect:**
  - Receive calls, create contact flows, cloud-based **virtual contact center**
  - Can integrate with other CRM systems or AWS
  - No upfront payments, 80% cheaper than traditional contact center solutions



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Comprehend

- For **Natural Language Processing – NLP**
- Fully managed and serverless service
- Uses machine learning to find insights and relationships in text
  - Language of the text
  - Extracts key phrases, places, people, brands, or events
  - Understands how positive or negative the text is
  - Analyzes text using tokenization and parts of speech
  - Automatically organizes a collection of text files by topic
- Sample use cases:
  - analyze customer interactions (emails) to find what leads to a positive or negative experience
  - Create and groups articles by topics that Comprehend will uncover

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Comprehend Medical

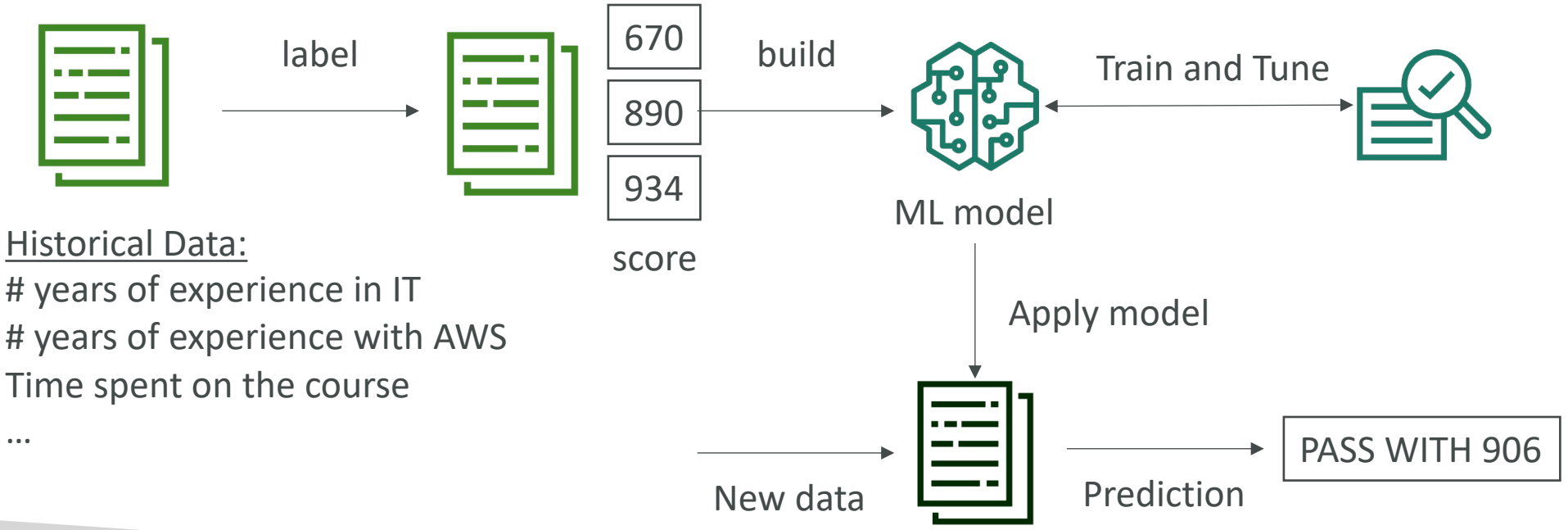
- Amazon Comprehend Medical detects and returns useful information in unstructured clinical text:
  - Physician's notes
  - Discharge summaries
  - Test results
  - Case notes
- **Uses NLP to detect Protected Health Information (PHI)** – *DetectPHI* API
- Store your documents in Amazon S3, analyze real-time data with Kinesis Data Firehose, or use Amazon Transcribe to transcribe patient narratives into text that can be analyzed by Amazon Comprehend Medical.

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon SageMaker AI

- Fully managed service for developers / data scientists to build ML models
- Typically, difficult to do all the processes in one place + provision servers
- Machine learning process (simplified): predicting your exam score

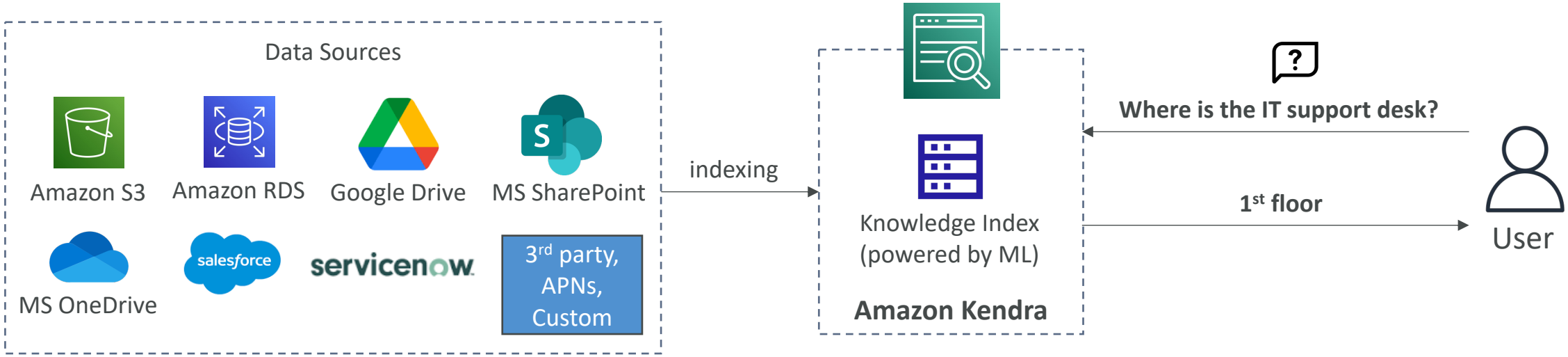


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Kendra

- Fully managed **document search service** powered by Machine Learning
- Extract answers from within a document (text, pdf, HTML, PowerPoint, MS Word, FAQs…)
- Natural language search capabilities
- Learn from user interactions/feedback to promote preferred results **(Incremental Learning)**
- Ability to manually fine-tune search results (importance of data, freshness, custom, …)

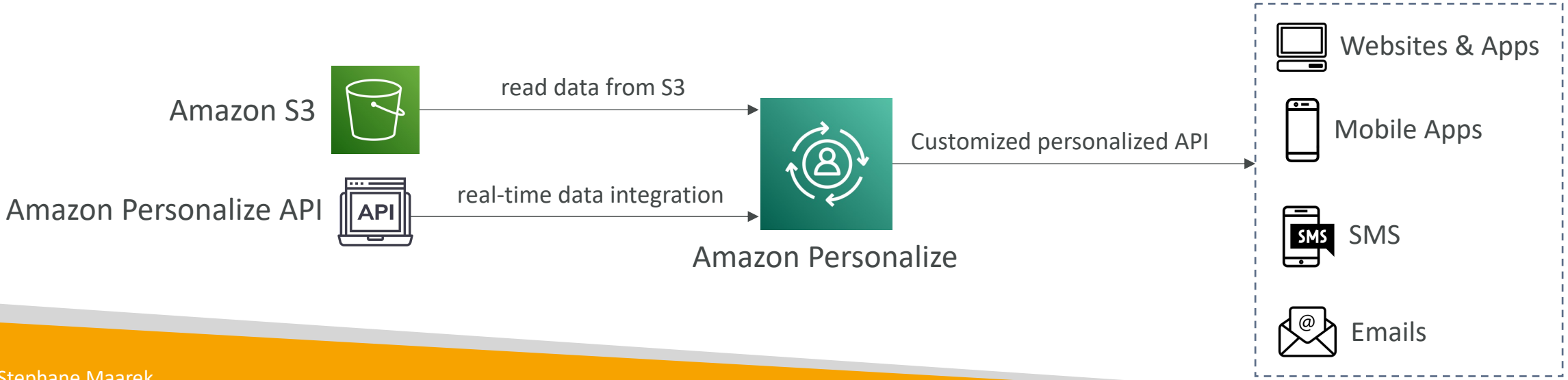


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Personalize

- Fully managed ML-service to build apps with real-time personalized recommendations
- Example: personalized product recommendations/re-ranking, customized direct marketing
  - Example: User bought gardening tools, provide recommendations on the next one to buy
- Same technology used by Amazon.com
- Integrates into existing websites, applications, SMS, email marketing systems, …
- Implement in days, not months (you don't need to build, train, and deploy ML solutions)
- Use cases: retail stores, media and entertainment…



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Textract

- Automatically extracts text, handwriting, and data from any scanned documents using AI and ML

analyze

Amazon Textract

result

```
{
   "Document ID": "123456789-005",
   "Name": "",
   "SEX": "F",
   "DOB": "23.05.1997",
   ...
}
```

- Extract data from forms and tables
- Read and process any type of document (PDFs, images, …)
- Use cases:
  - Financial Services (e.g., invoices, financial reports)
  - Healthcare (e.g., medical records, insurance claims)
  - Public Sector (e.g., tax forms, ID documents, passports)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Machine Learning - Summary

- **Rekognition:** face detection, labeling, celebrity recognition
- **Transcribe:** audio to text (ex: subtitles)
- **Polly:** text to audio
- **Translate:** translations
- **Lex:** build conversational bots – chatbots
- **Connect:** cloud contact center
- **Comprehend:** natural language processing
- **SageMaker:** machine learning for every developer and data scientist
- **Kendra:** ML-powered search engine
- **Personalize:** real-time personalized recommendations
- **Textract:** detect text and data in documents

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Other Services

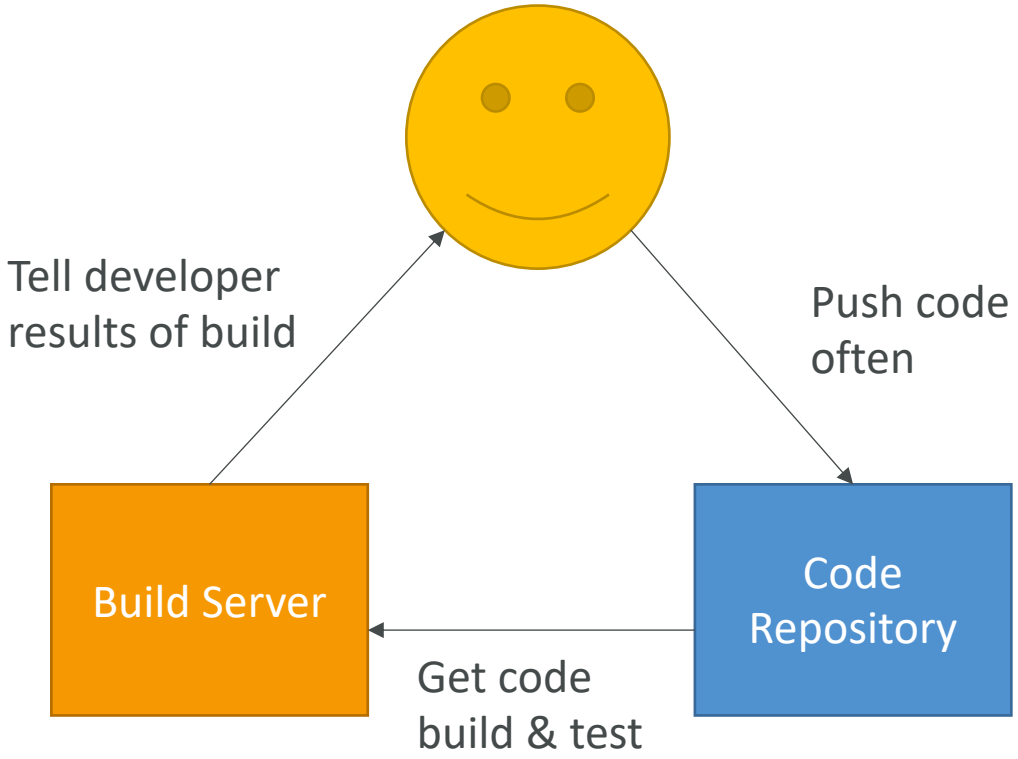
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Continuous Integration

- Developers push the code to a code repository often (GitHub / CodeCommit / Bitbucket / etc…)
- A testing / build server checks the code as soon as it's pushed (CodeBuild / Jenkins CI / etc…)
- The developer gets feedback about the tests and checks that have passed / failed

- Find bugs early, fix bugs
- Deliver faster as the code is tested
- Deploy often
- Happier developers, as they're unblocked

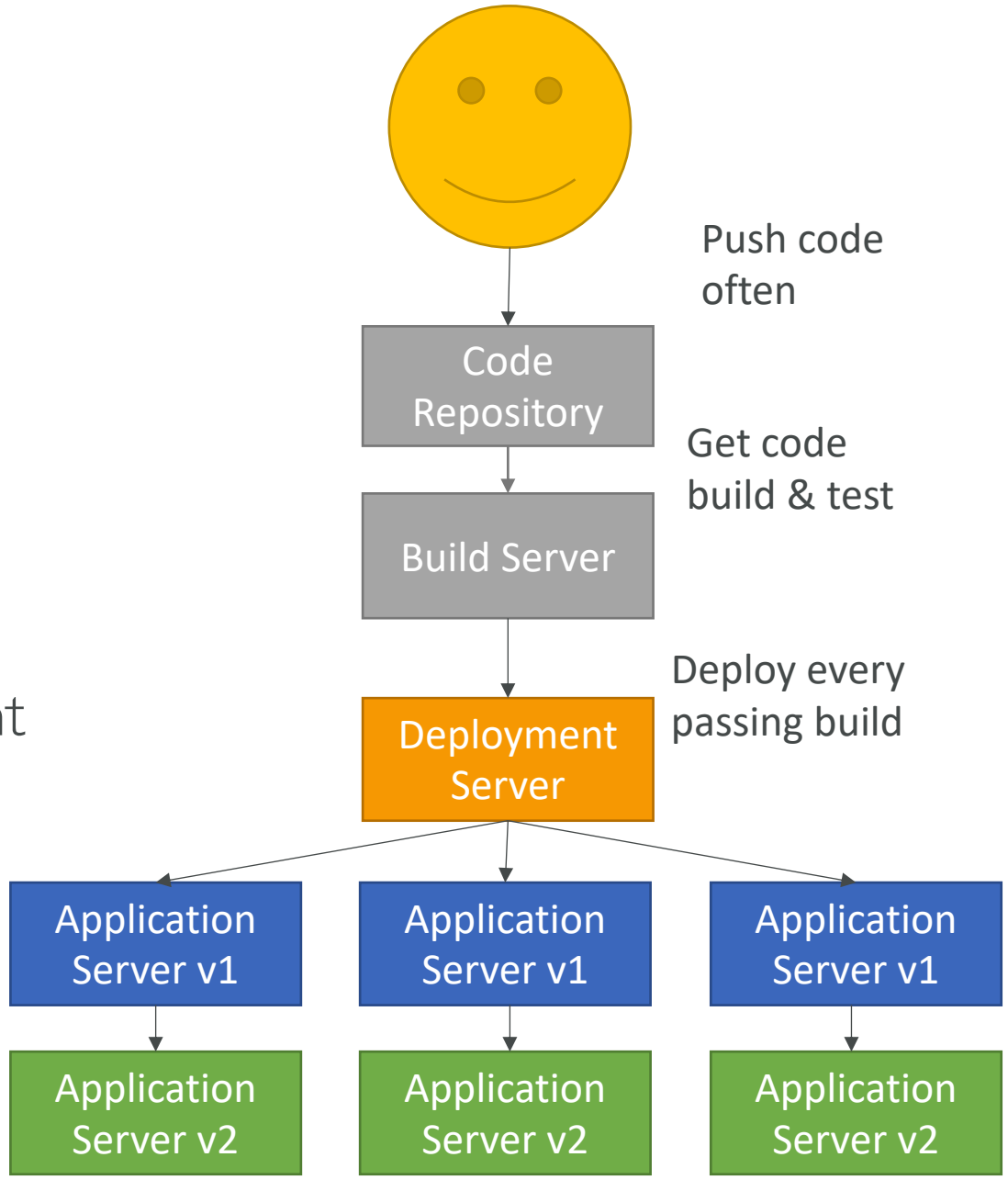


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Continuous Delivery

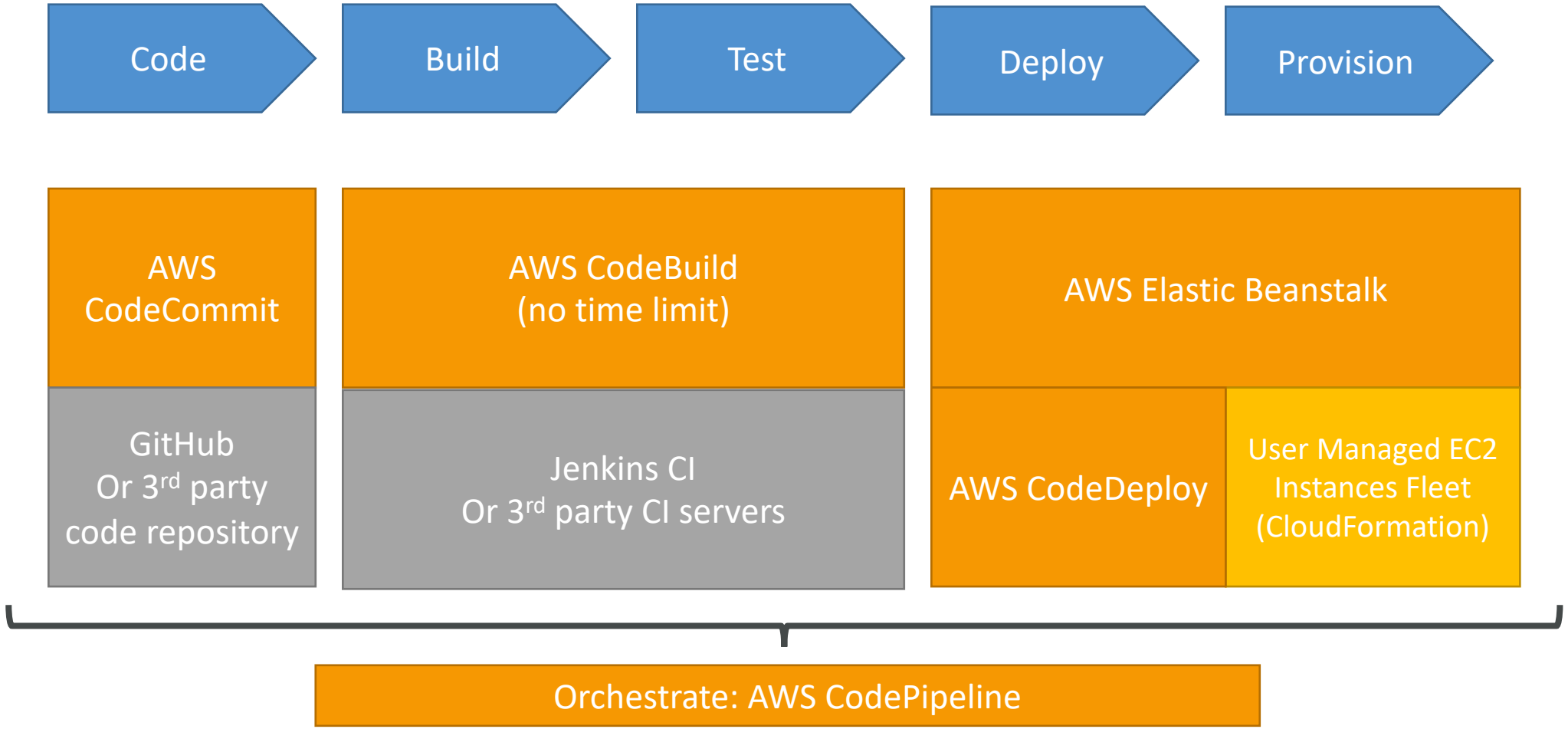
- Ensure that the software can be released reliably whenever needed.
- Ensures deployments happen often and are quick
- Shift away from "one release every 3 months" to "5 releases a day"
- That usually means automated deployment
  - CodeDeploy
  - Jenkins CD
  - Spinnaker
  - Etc…



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

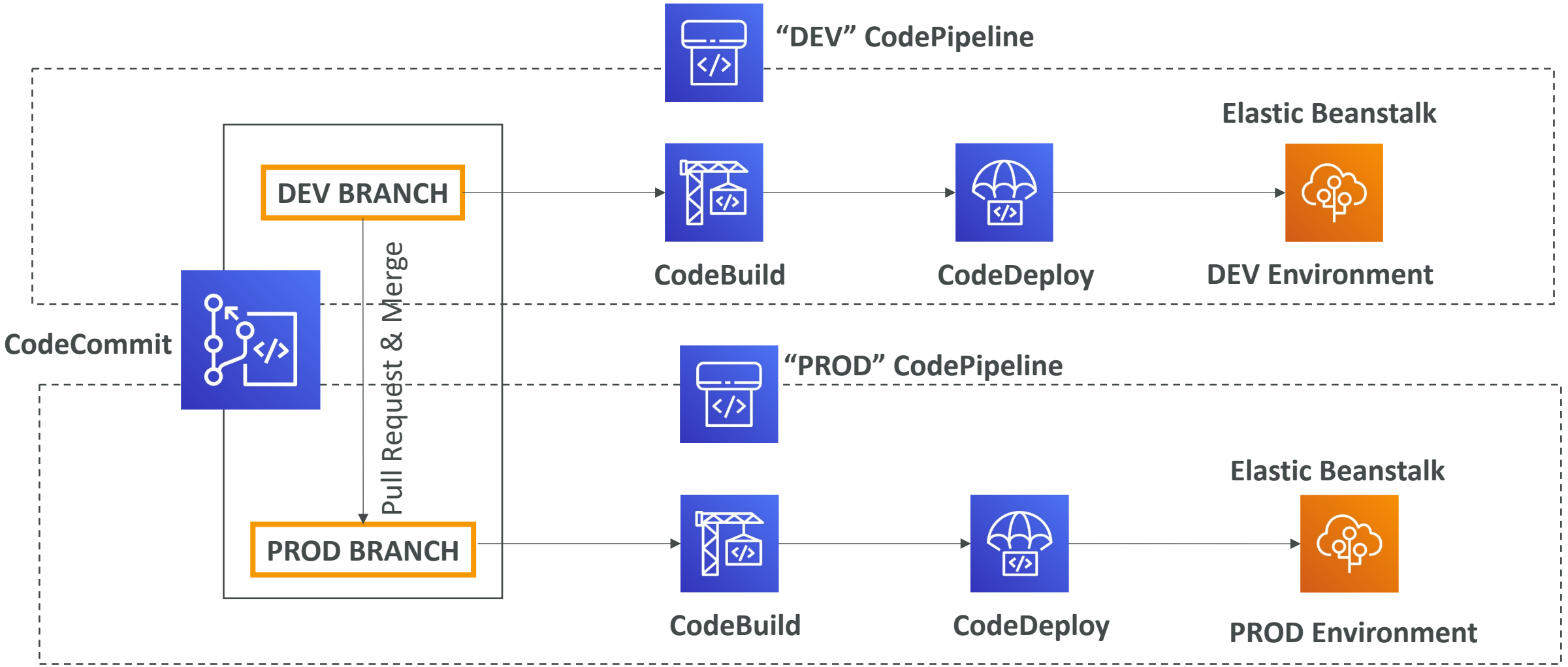
Technology Stack for CICD



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CICD Architecture

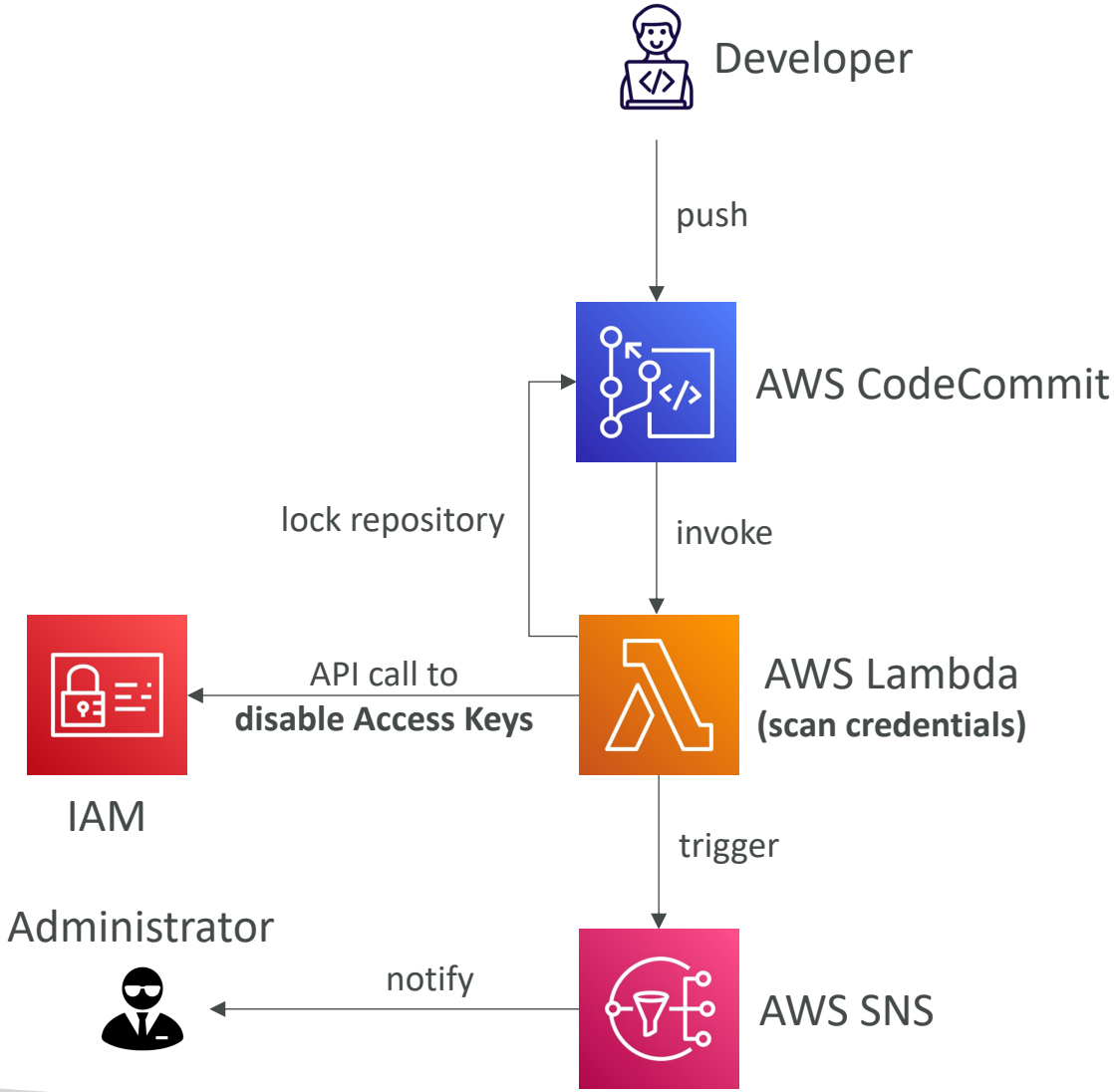


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CodeCommit Trigger for AWS Lambda

- Every push to CodeCommit can trigger a Lambda function
- The Lambda function can scan for leaked AWS credentials on every code push, and disable them automatically to remedy the issue



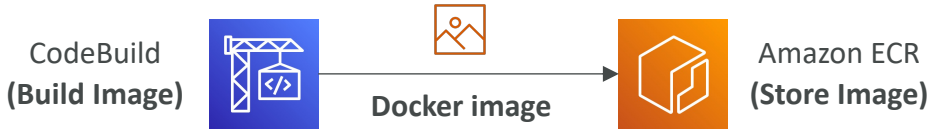
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Good to know – CICD

- You can use a **manual approval stage** in CodePipeline
- Running unit tests CodeCommit + CodeBuild + Code Pipeline

- Build and Store Docker Images: CodeBuild + ECR

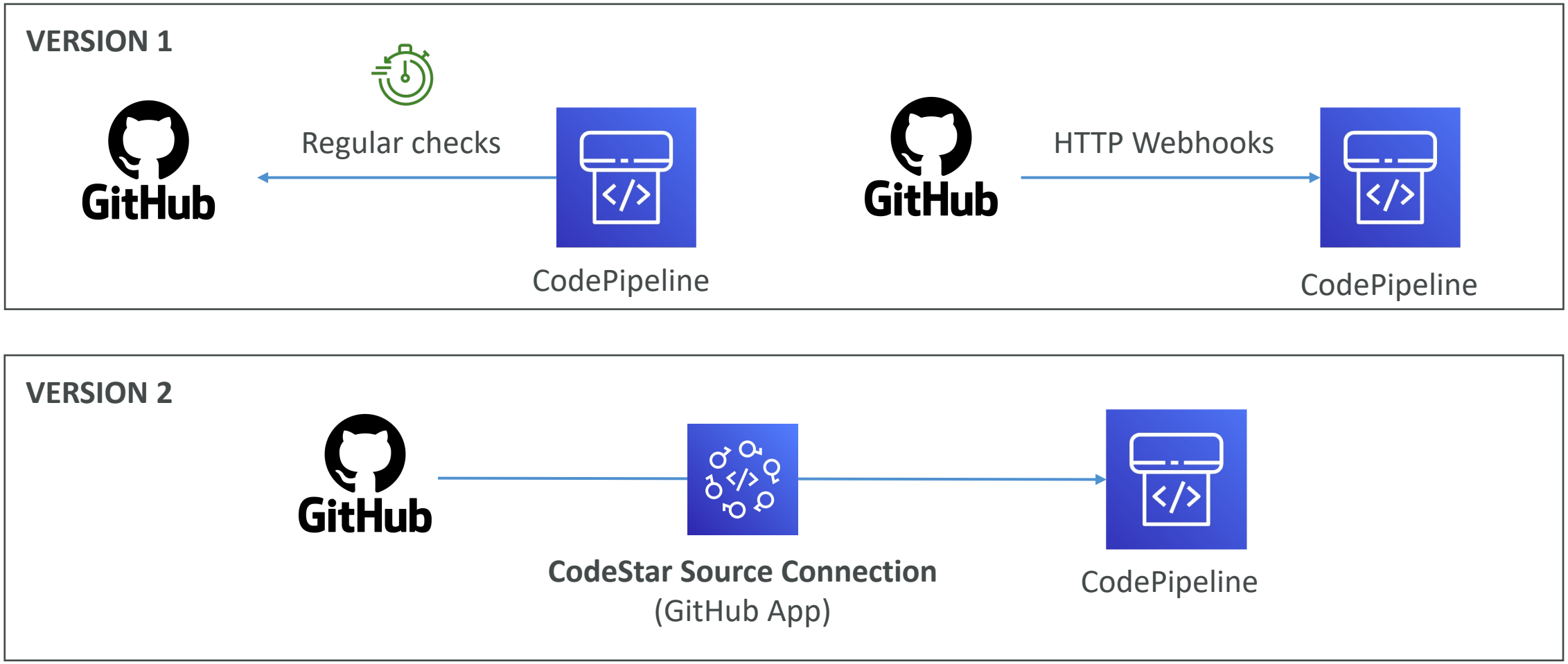


- Automated CloudFormation deployment: CodePipeline

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CodePipeline – GitHub integration

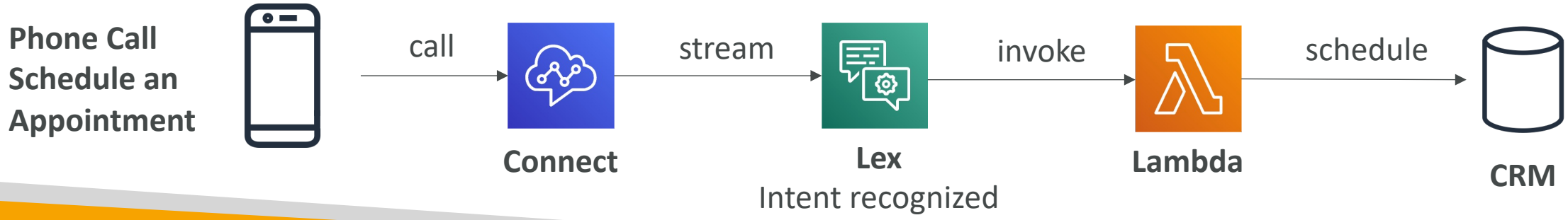


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Alexa for Business, Lex & Connect

- Alexa for Business:
  - Use Alexa to help employees be more productive in meeting rooms and their desk
  - Measure and increase the utilization of meeting rooms in their workplace
- Amazon Lex: (same technology that powers Alexa)
  - Automatic Speech Recognition (ASR) to convert speech to text
  - Natural Language Understanding to recognize the intent of text, callers
  - Helps build chatbots, call center bots
- Amazon Connect:
  - Receive calls, create contact flows, cloud-based virtual contact center
  - Can integrate with other CRM systems or AWS



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

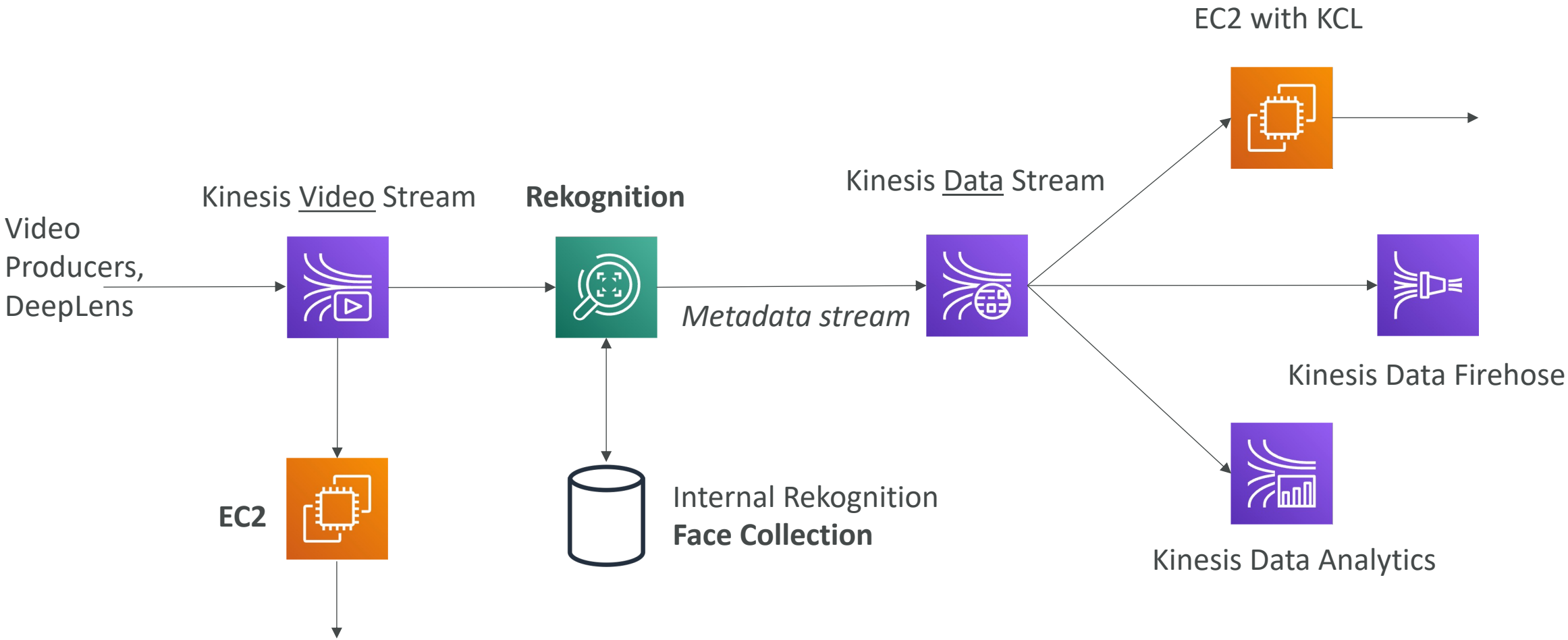
Kinesis Video Streams

- One video stream per streaming device (producers)
  - Security cameras, body worn camera, smartphone
  - Can use a Kinesis Video Streams Producer library
- Underlying data is stored in S3 (but we don't have access to it)
- Cannot output the stream data to S3 (must build custom solution)
- Consumers:
  - Consumed by EC2 instances for real time analysis, or in batch
  - Can leverage the Kinesis Video Stream Parser Library
  - Integration with AWS Rekognition for facial detection

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

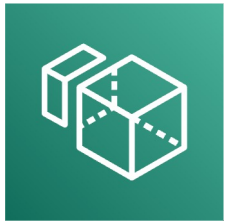
Video Streaming & Rekognition



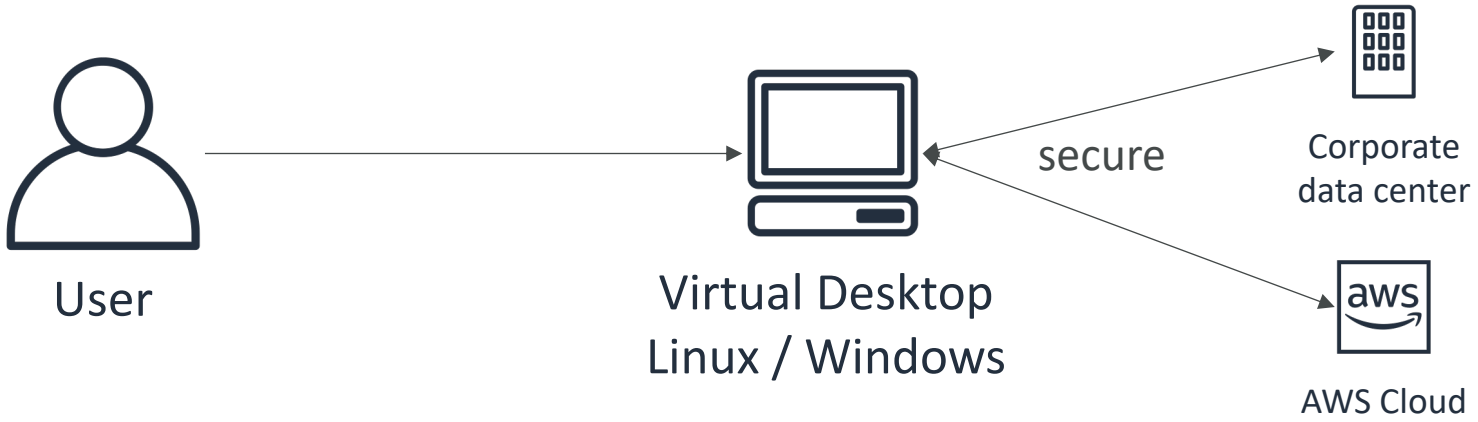
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon WorkSpaces



- Managed, Secure Cloud Desktop
- Great to eliminate management of on-premises VDI (Virtual Desktop Infrastructure)
- Pricing is either on-demand (pay per hour) or monthly subscription
- Secure, Encrypted, Network Isolation
- Integrated with Microsoft Active Directory



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

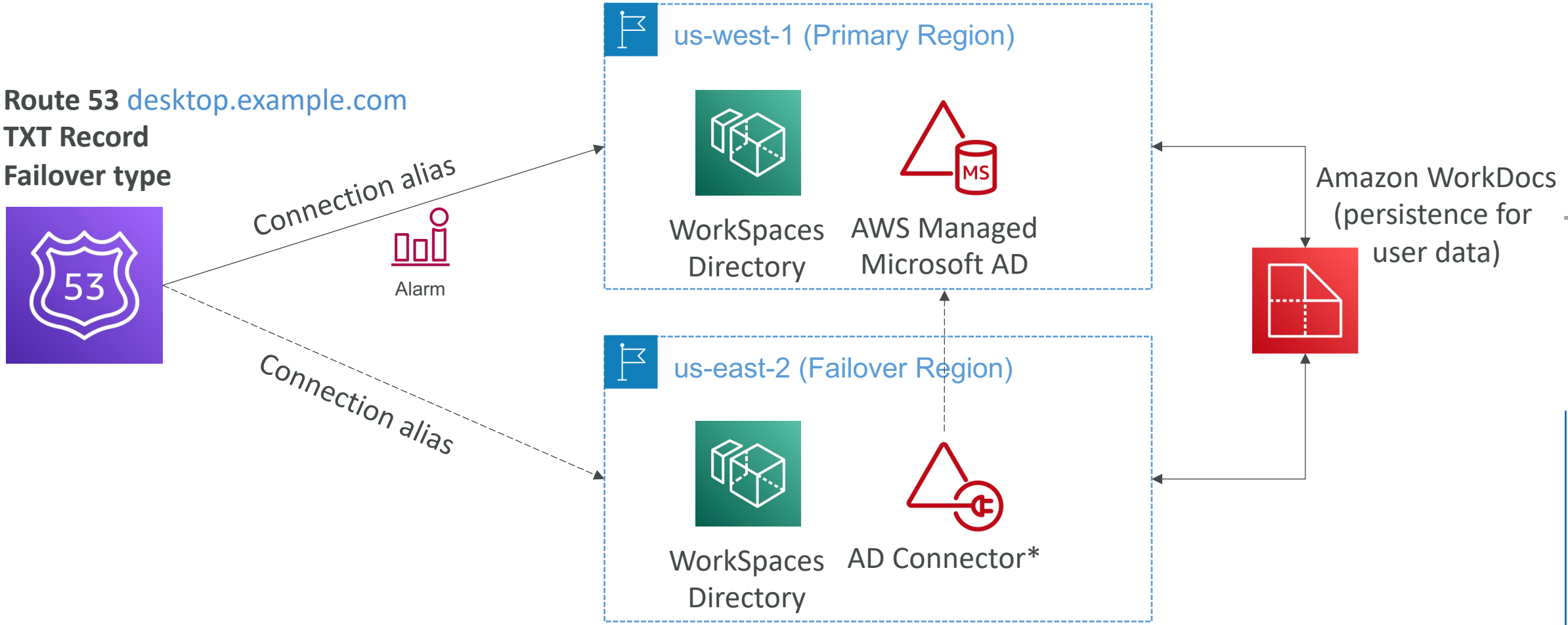
Amazon WorkSpaces

- WorkSpaces Application Manager (WAM)
  - Deploy and Manage applications as virtualized application containers
  - Provision at scale, and keep the applications updated using WAM
- Windows Updates
  - By default, Amazon Workspaces are configured to install software updates
  - Amazon WorkSpaces with Windows will have Windows Update turned on
  - You have full control over the Windows Update frequency
- Maintenance Windows
  - Updates are installed during maintenance windows (you define them)
  - Always On WorkSpaces: default is from 00h00 to 04h00 on Sunday morning
  - AutoStop WorkSpaces: automatically starts once a month to install updates
  - Manual maintenance: you define your windows and perform maintenance

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon WorkSpaces - Cross Region Redirection



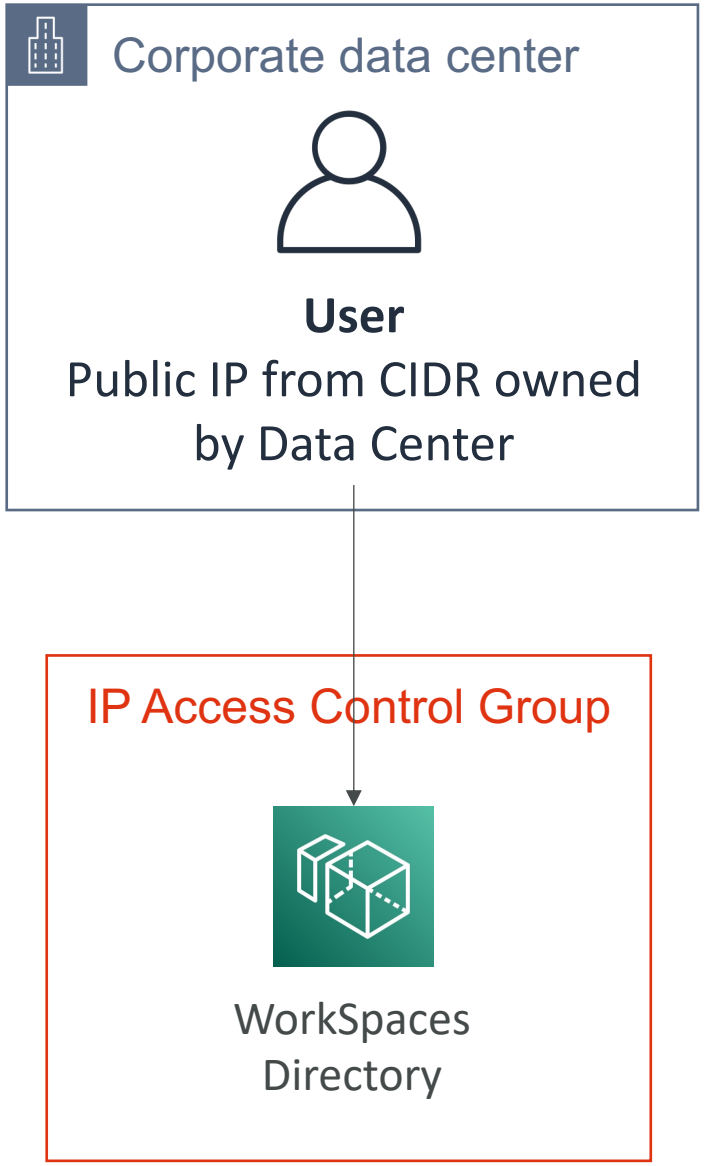
*must use AD Connector, can't use multi-region AWS Managed Microsoft AD

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon WorkSpaces IP Access Control Groups

- Similar to security groups for Amazon WorkSpaces
- List of IP addresses / CIDR address ranges that users are authorized to connect from
- If users access WorkSpaces through VPN or NAT, the IP Access Control Group must authorize the public IP of these

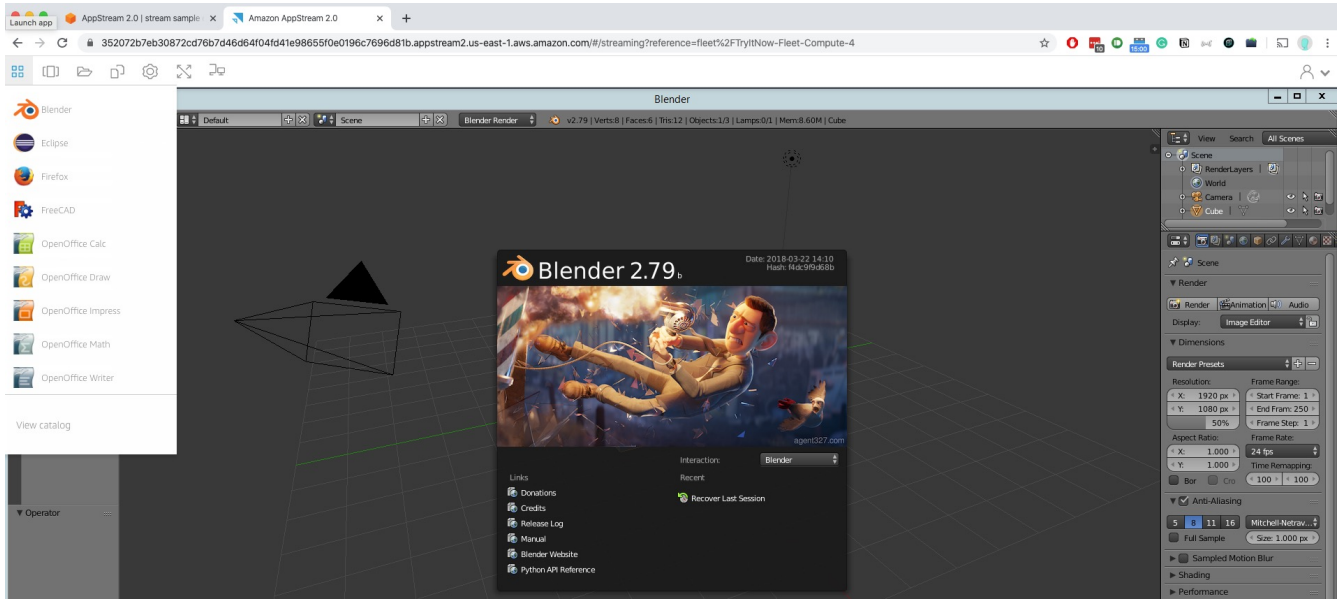


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon AppStream 2.0

- Desktop Application Streaming Service
- Deliver to any computer, without acquiring, provisioning infrastructure
- The application is delivered from within a web browser



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon AppStream 2.0 vs WorkSpaces

- Workspaces
  - Fully managed VDI and desktop available
  - The users connect to the VDI and open native or WAM applications
  - Workspaces are on-demand or always on

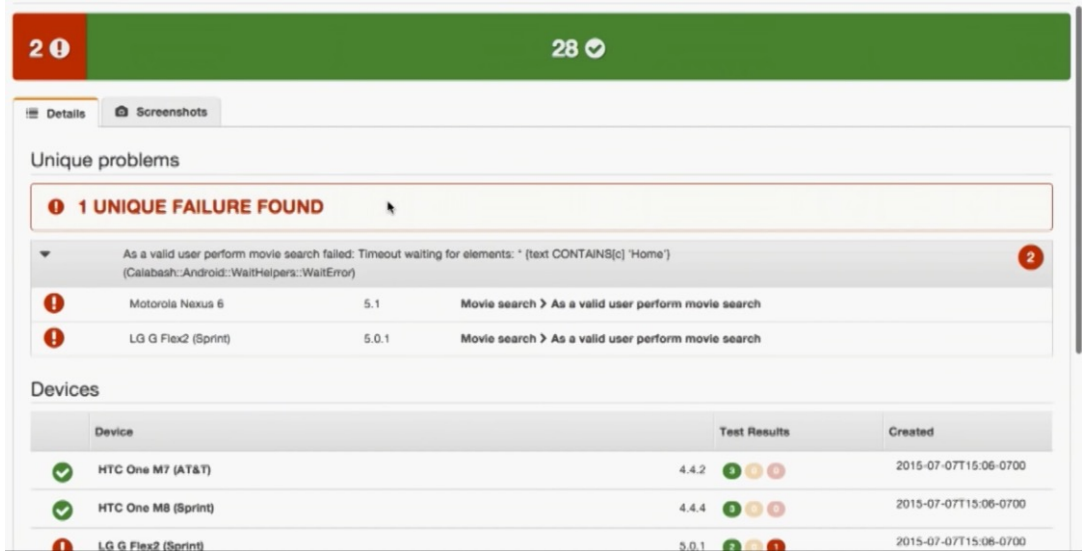
- AppStream 2.0
  - Stream a desktop application to web browsers (no need to connect to a VDI)
  - Works with any device (that has a web browser)
  - Allow to configure an instance type per application type (CPU, RAM, GPU)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Device Farm

- Application testing service for your mobile and web applications
- Test across **real browsers** and **real mobiles devices**
- **Fully automated** using framework
- Improve the quality of web and mobile apps
- Generates videos and logs to document the issues encountered
- Can **remotely log-in** to devices for debugging

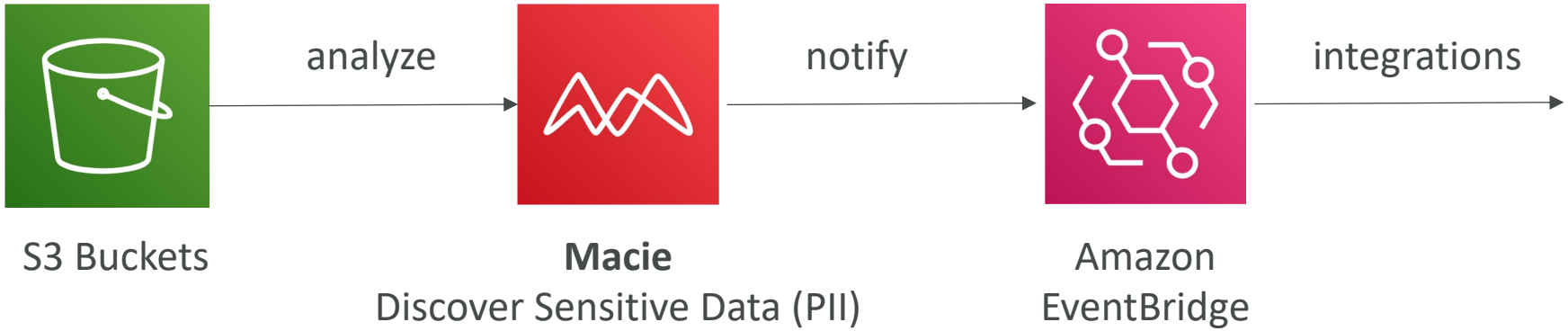


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Macie

- Amazon Macie is a fully managed data security and data privacy service that uses **machine learning and pattern matching to discover and protect your sensitive data in AWS.**
- Macie helps identify and alert you to **sensitive data, such as personally identifiable information (PII)**



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Simple Email Service (Amazon SES)

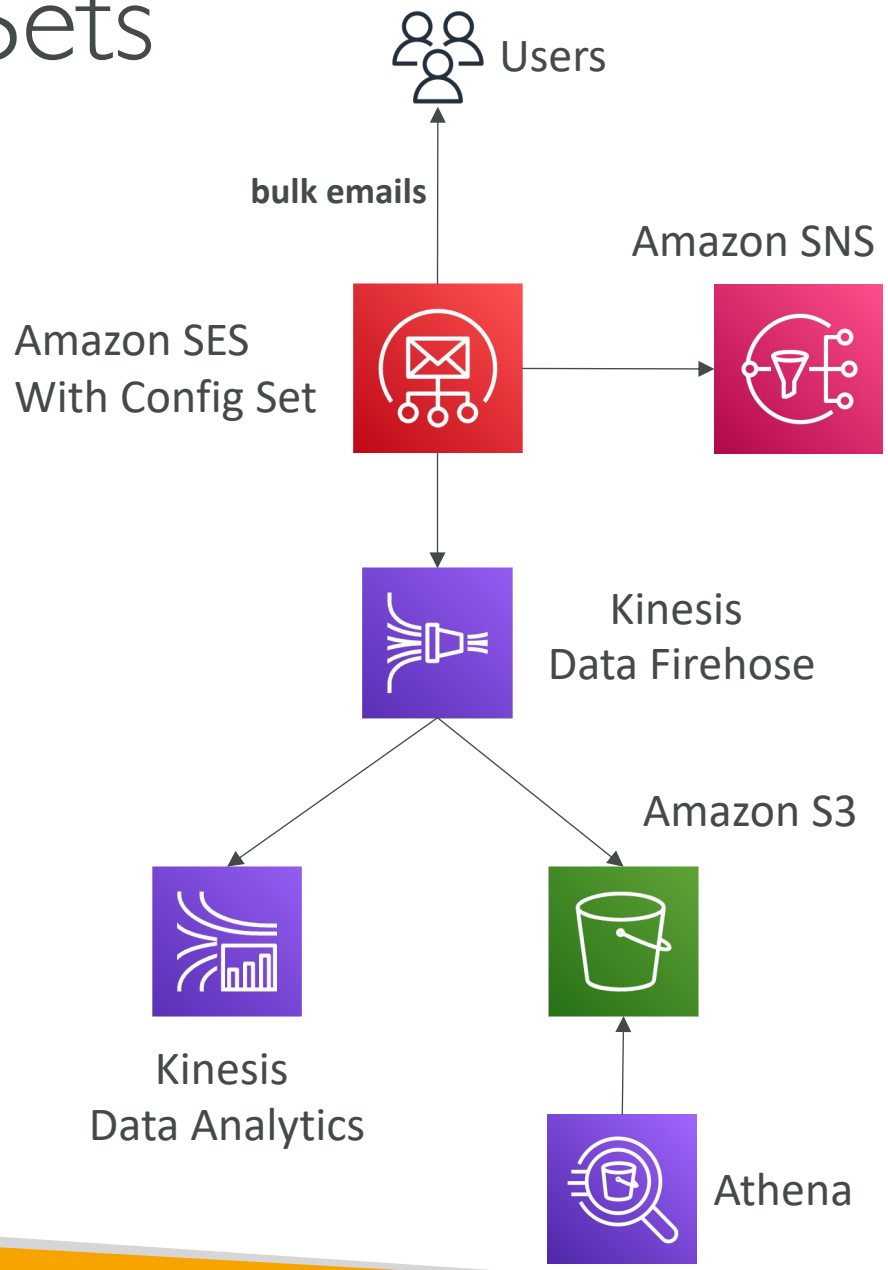
- **Fully managed service to send emails securely, globally and at scale**
- Allows inbound/outbound emails
- Reputation dashboard, performance insights, anti-spam feedback
- Provides statistics such as email deliveries, bounces, feedback loop results, email open
- Supports DomainKeys Identified Mail (DKIM) and Sender Policy Framework (SPF)
- Flexible IP deployment: shared, dedicated, and customer-owned IPs
- Send emails using your application using AWS Console, APIs, or SMTP
- Use cases: transactional, marketing and bulk email communications

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon SES – Configuration Sets

- Configuration sets help you customize and analyze your email send events
- **Event destinations:**
  - Kinesis Data Firehose: receives metrics (numbers of sends, deliveries, opens, clicks, bounces, and complaints) for each email
  - SNS: for immediate feedback on bounce and complaint information
- **IP pool management**: use IP pools to send particular types of emails

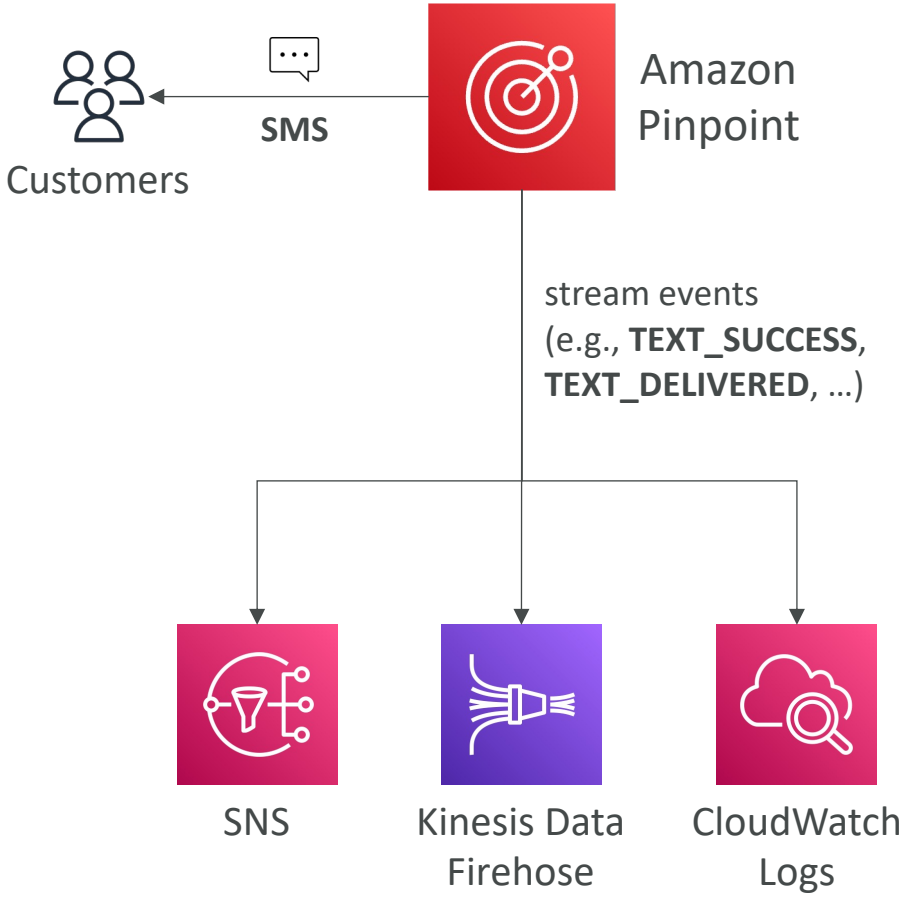


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Pinpoint

- Scalable **2-way (outbound/inbound)** marketing communications service
- Supports email, SMS, push, voice, and in-app messaging
- Ability to segment and personalize messages with the right content to customers
- Possibility to receive replies
- Scales to billions of messages per day
- Use cases: run campaigns by sending marketing, bulk, transactional SMS messages
- **Versus Amazon SNS or Amazon SES**
  - In SNS & SES you managed each message's audience, content, and delivery schedule
  - In Amazon Pinpoint, you create message templates, delivery schedules, highly-targeted segments, and full campaigns

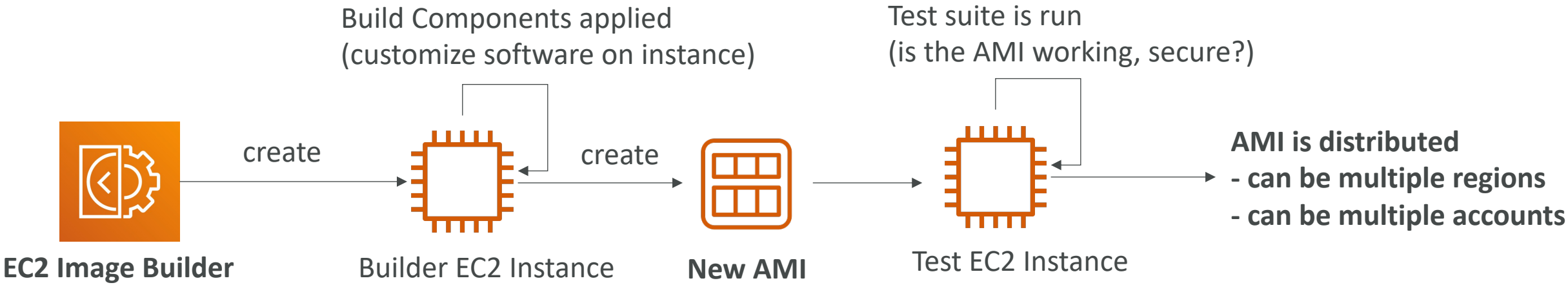


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EC2 Image Builder

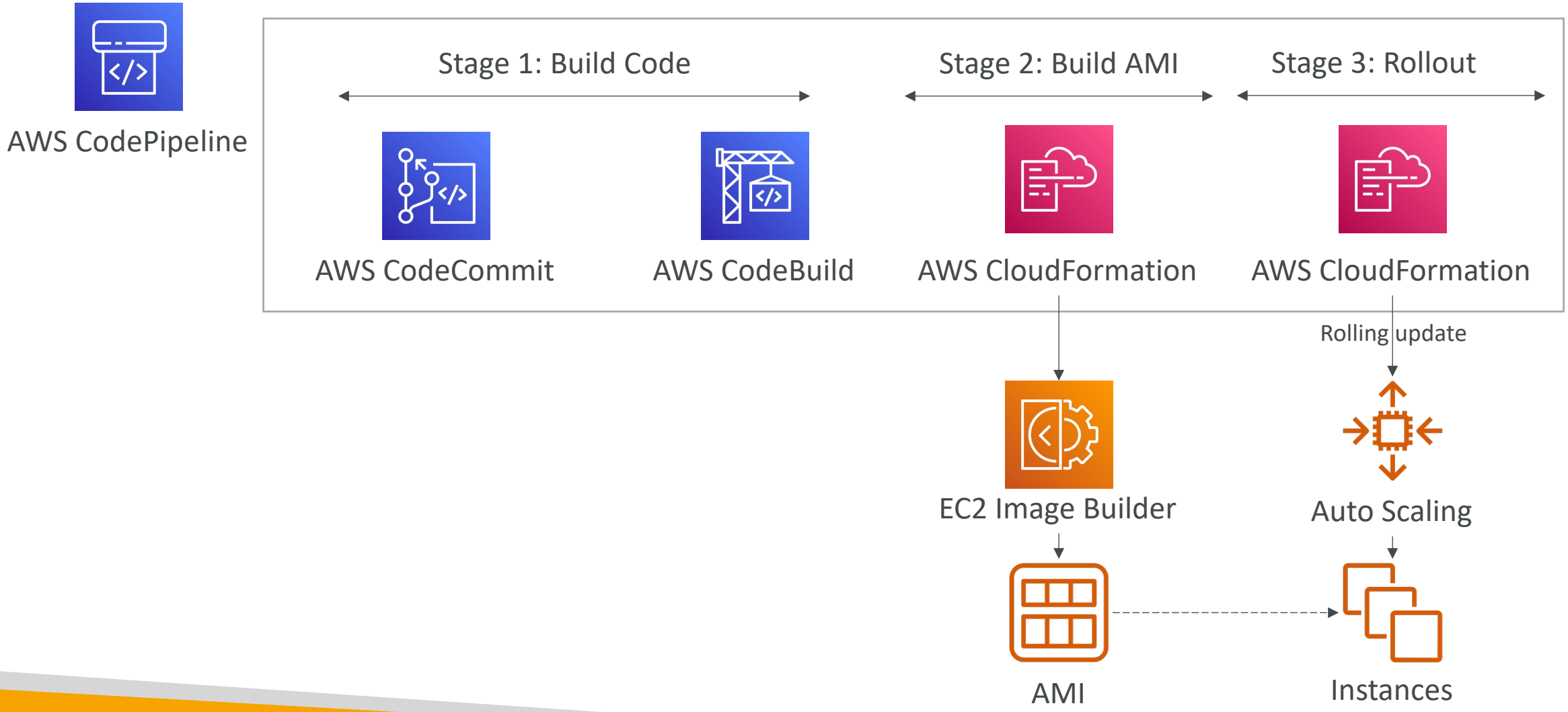
- Used to automate the creation of Virtual Machines or container images
- => Automate the creation, maintain, validate and test **EC2 AMIs**
- Can be run on a schedule (weekly, whenever packages are updated, etc…)
- Free service (only pay for the underlying resources)
- **Can publish AMI to multiple regions and multiple accounts**



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EC2 Image Builder – CICD Architecture

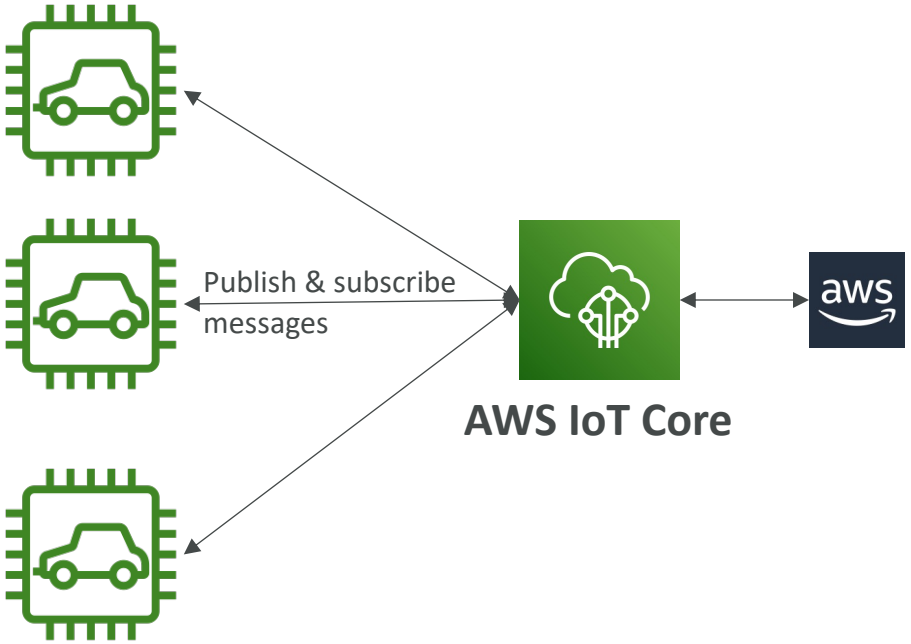


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS IoT Core

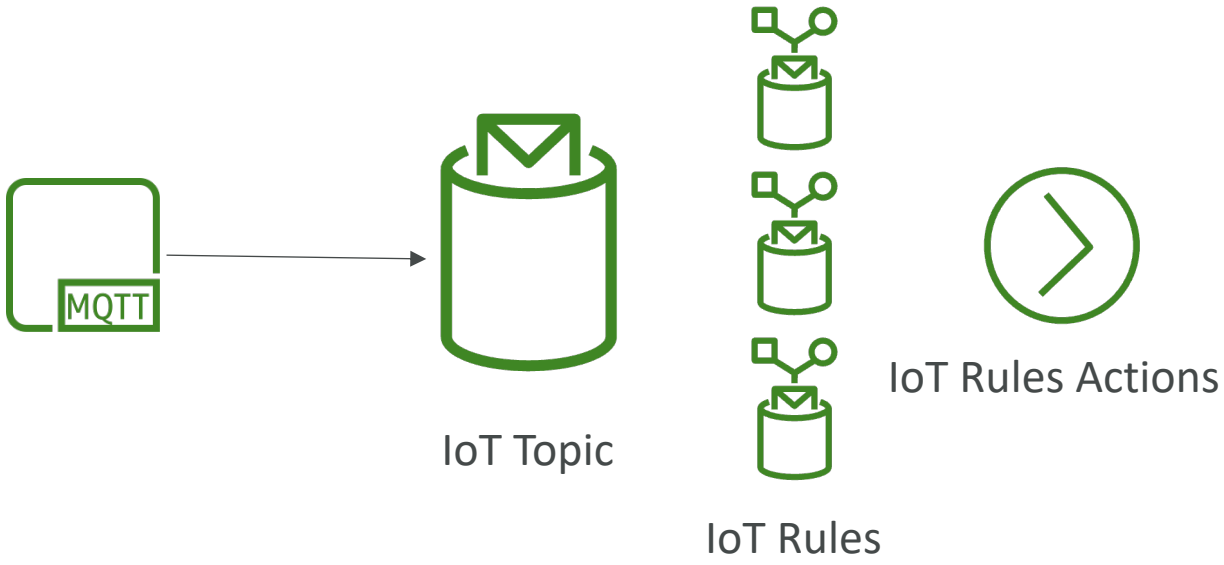
- IoT stands for "Internet of Things" – the network of internet-connected devices that are able to collect and transfer data
- AWS IoT Core allows you to **easily connect IoT devices to the AWS Cloud**
- **Serverless, secure & scalable** to billions of devices and trillions of messages
- Integrates with a lot of AWS services (Lambda, S3, SageMaker, etc.)
- Build IoT applications that gather, process, analyze, and act on data

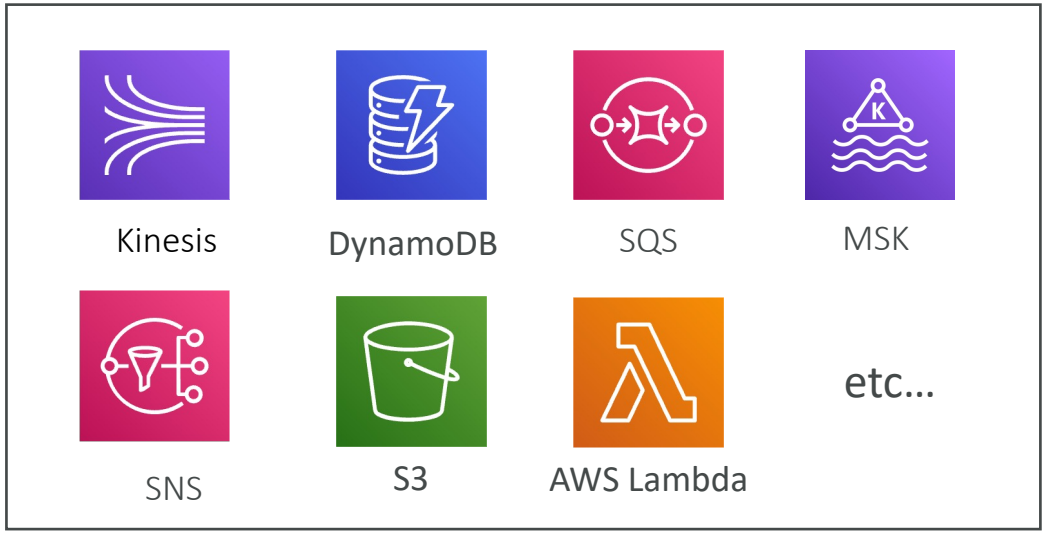


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IoT Core - Integrations

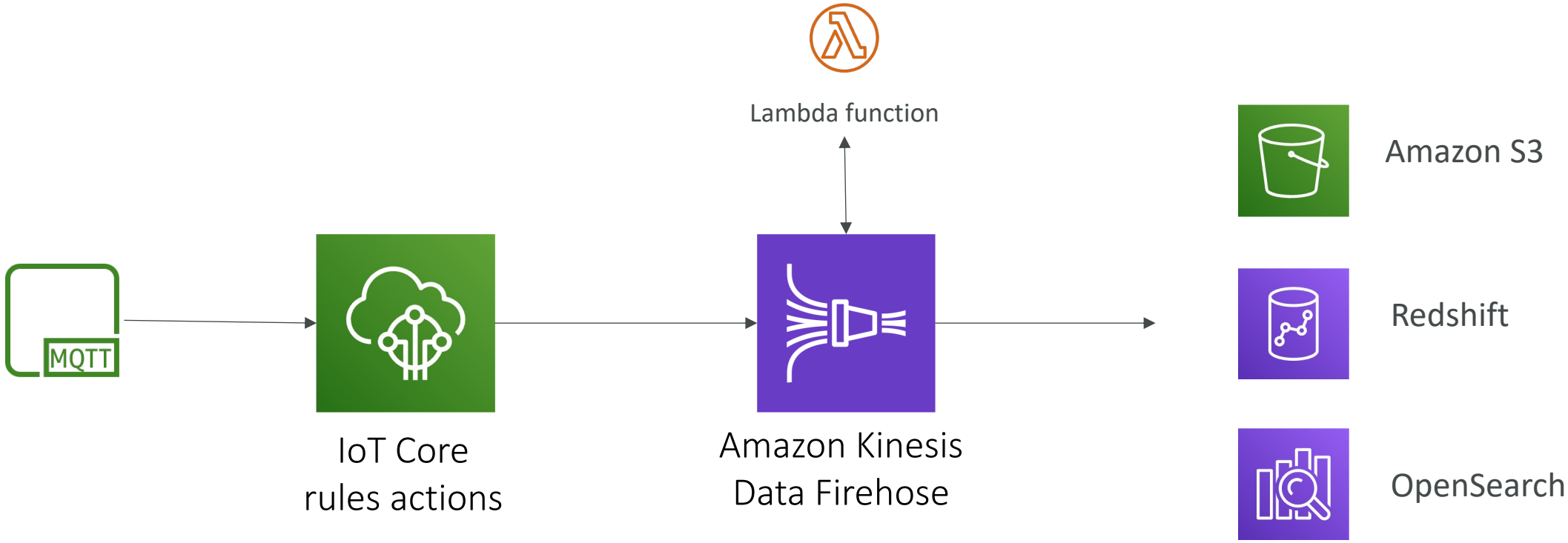




© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

IoT Core – Kinesis Data Firehose



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Exam Preparation

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Analysis from the Practice Sample questions

- Exam Page: https://aws.amazon.com/certification/certified-solutions-architect-professional/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

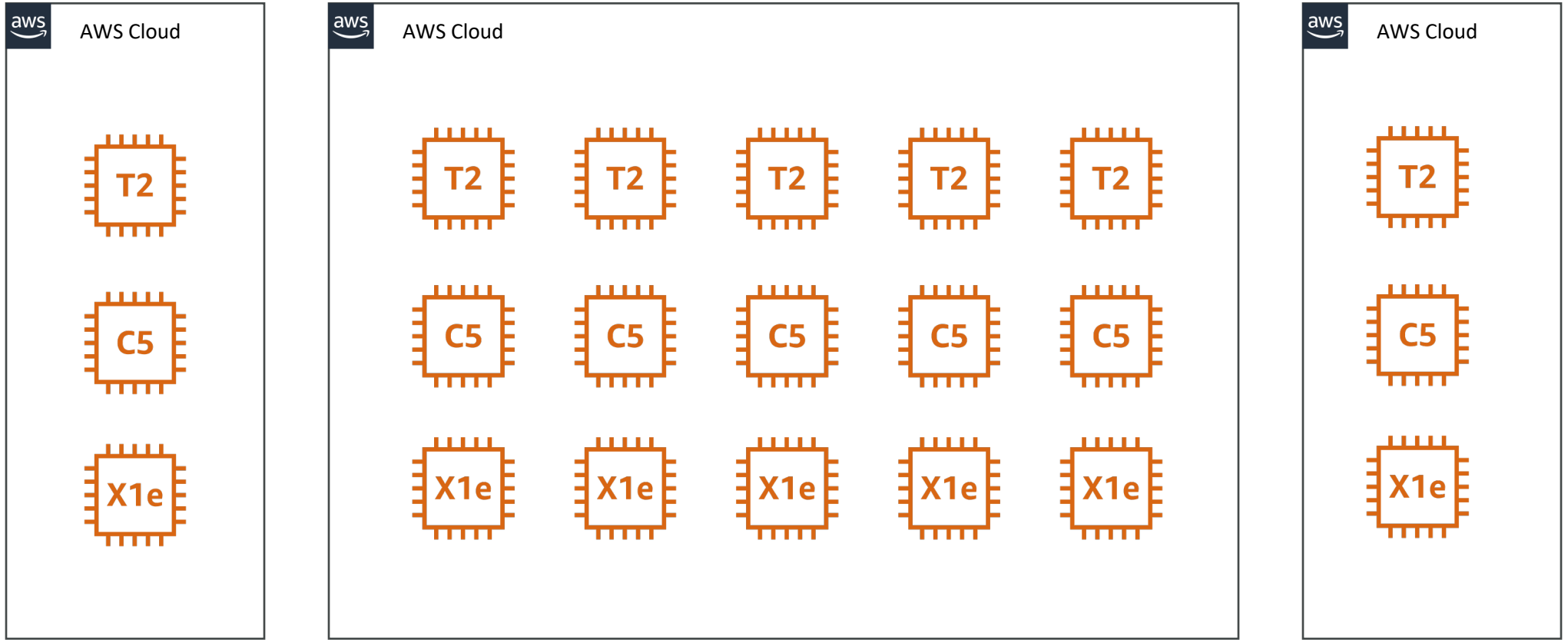
Question 1

- An enterprise has a large number of AWS accounts owned by separate business groups. One of the accounts was recently compromised. The attacker launched a large number of instances, resulting in a high bill for that account.
- The security breach was addressed, but management has asked a solutions architect to develop a solution to **help prevent excessive spending** in all accounts. **Each business group wants to retain full control over its AWS account.**
- Which solution should the solutions architect recommend to meet these requirements?

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Question 1 – Architecture Diagram

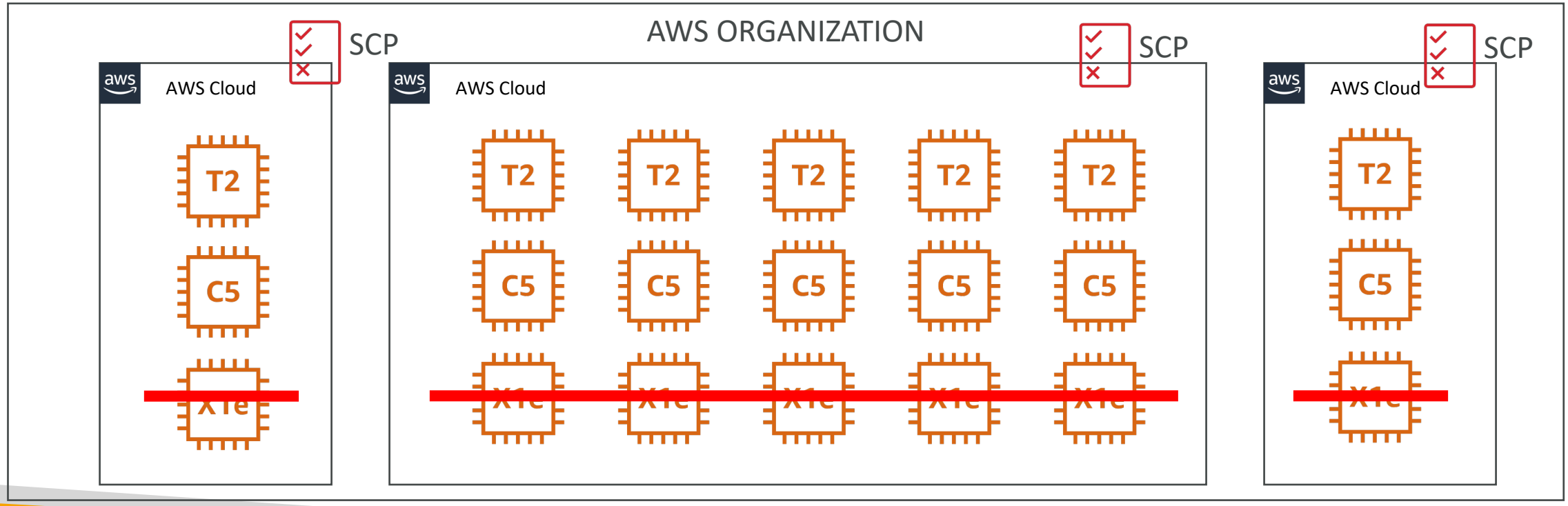


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option A

- Use AWS Organizations to add each AWS account to the master account. Create a service control policy (SCP) that uses the `ec2:instanceType` condition key to prevent the launch of high-cost instance types in each account.

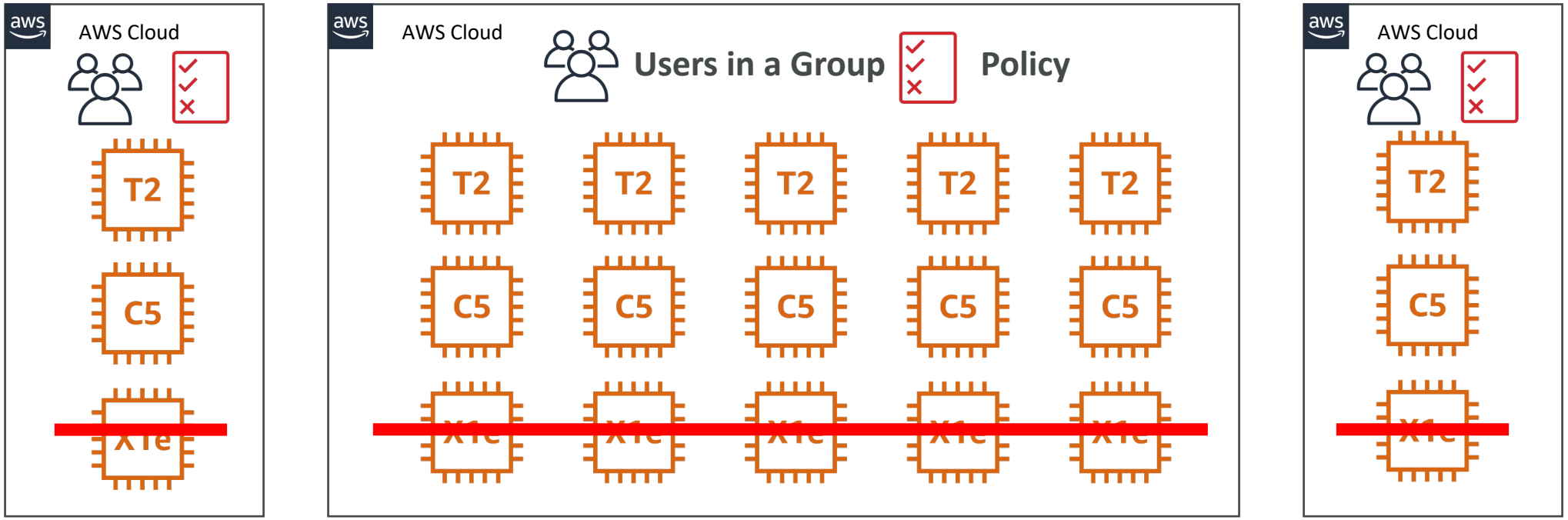


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option B

- Attach a new customer-managed IAM policy to an IAM group in each account that uses the ec2:instanceType condition key to prevent the launch of high-cost instance types. Place all of the existing IAM users in each group.



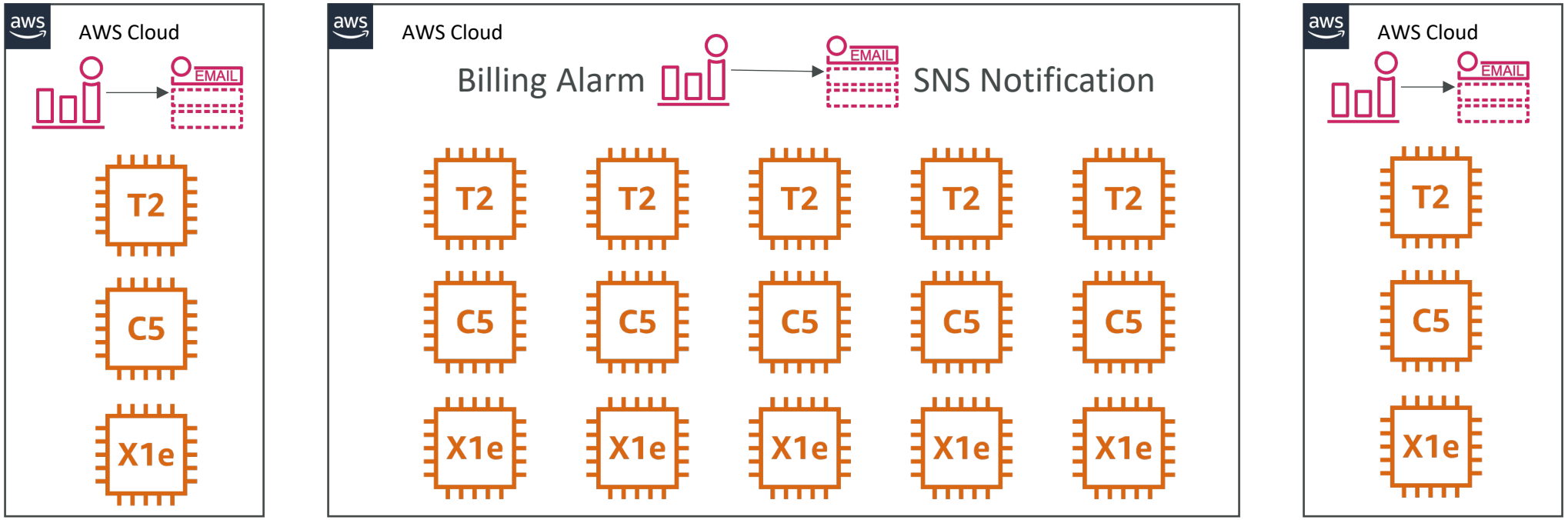
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option C

CORRECT ANSWER

- Enable billing alerts on each AWS account. Create Amazon CloudWatch alarms that send an Amazon SNS notification to the account administrator whenever their account exceeds the spending budget.

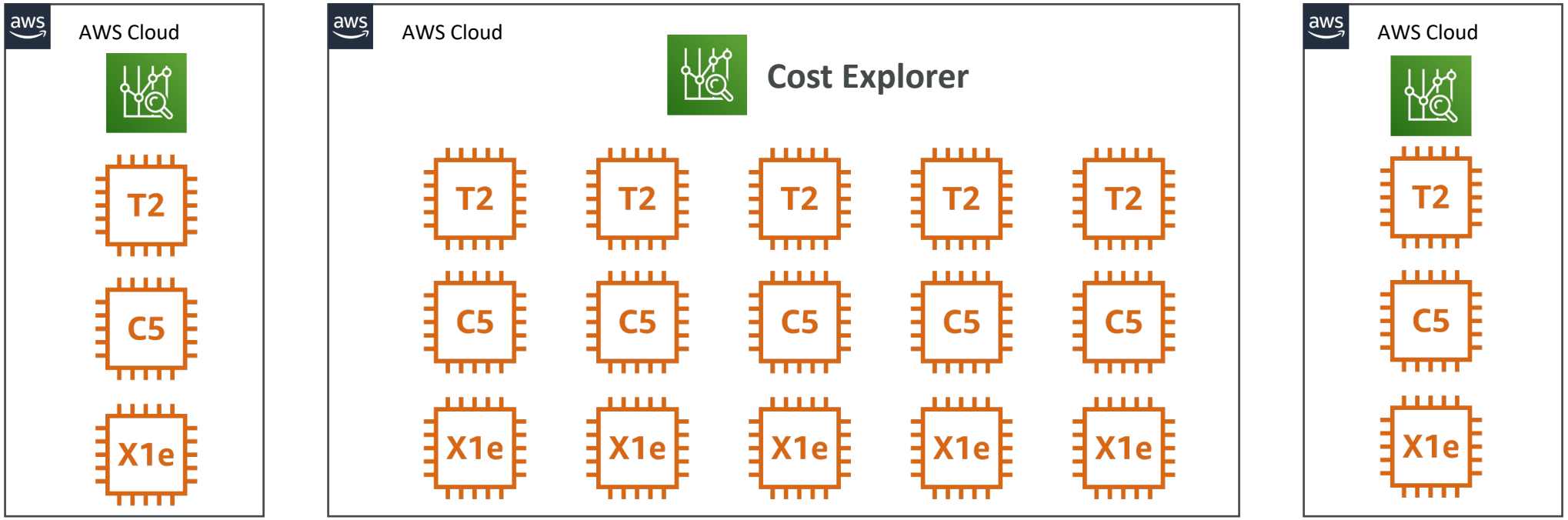


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option D

- Enable AWS Cost Explorer in each account. Regularly review the Cost Explorer reports for each account to ensure spending does not exceed the planned budget



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

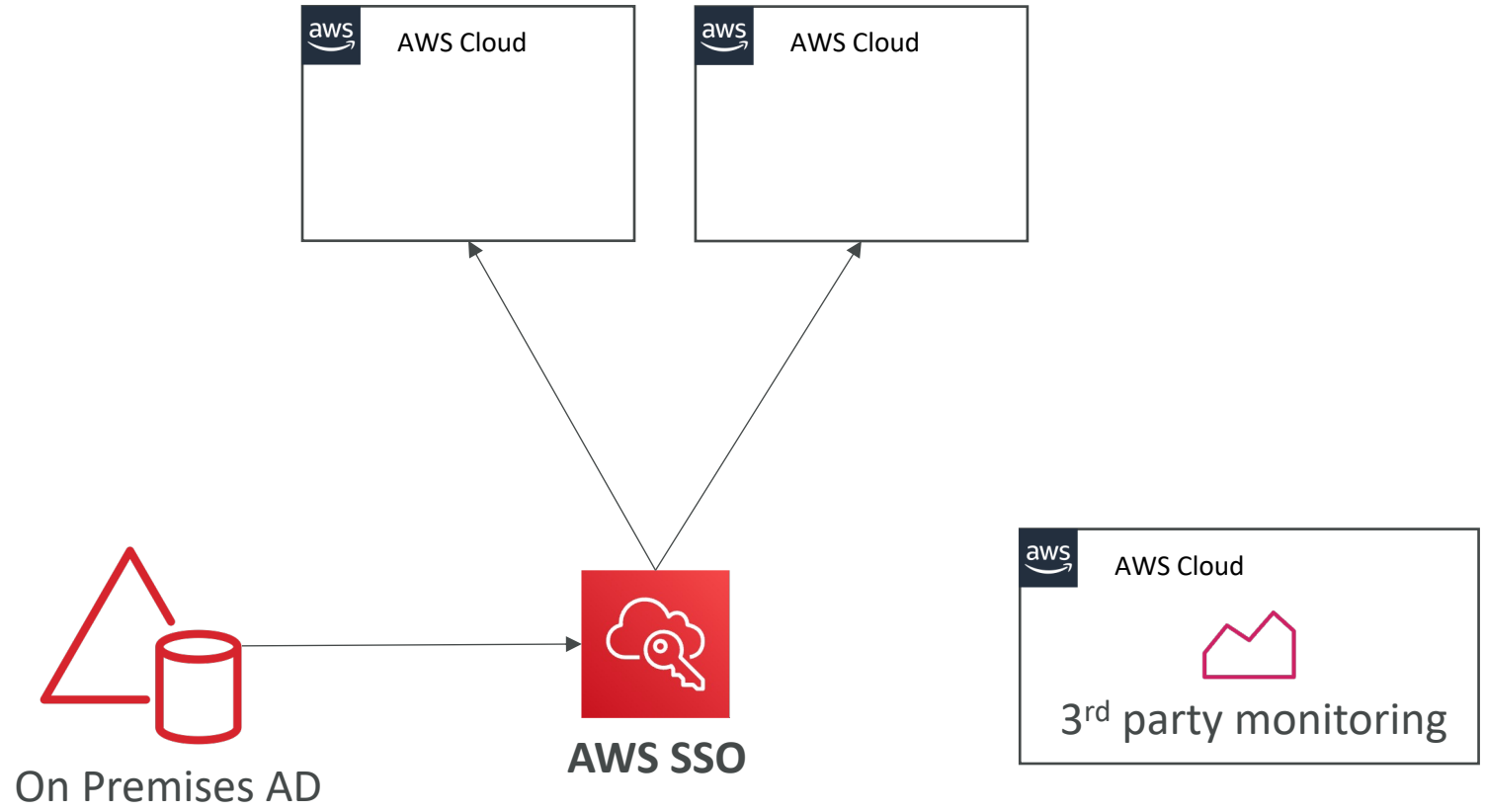
Question 2

- A company has multiple AWS accounts. The company has integrated its on-premises Active Directory (AD) with **AWS SSO to grant AD users least privilege abilities to manage infrastructure across all the accounts.**
- A solutions architect must integrate a third-party monitoring solution that requires **read-only access across all AWS accounts.** The monitoring solutions will run **in its own AWS account.**
- How can the monitoring solution be given the required permissions?

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Question 2 - Architecture

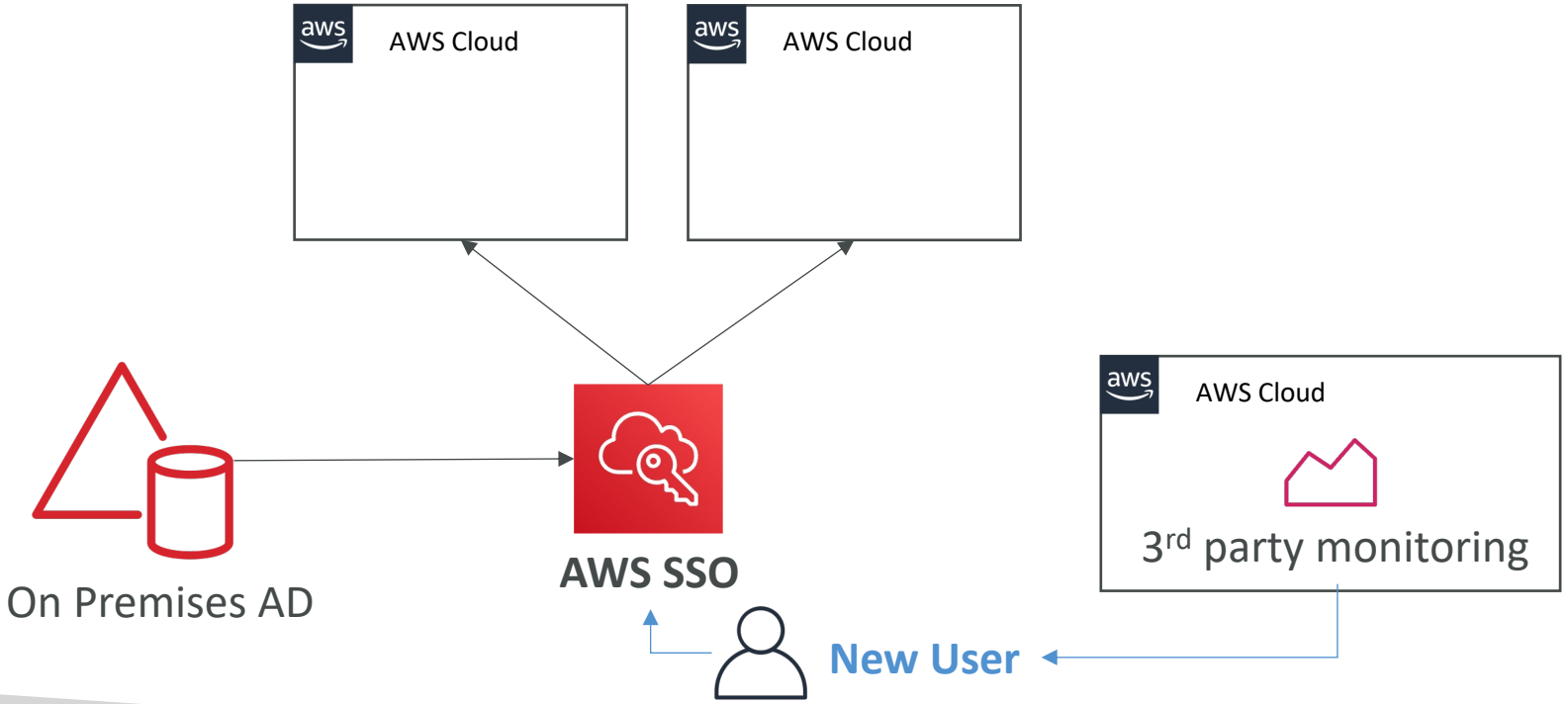


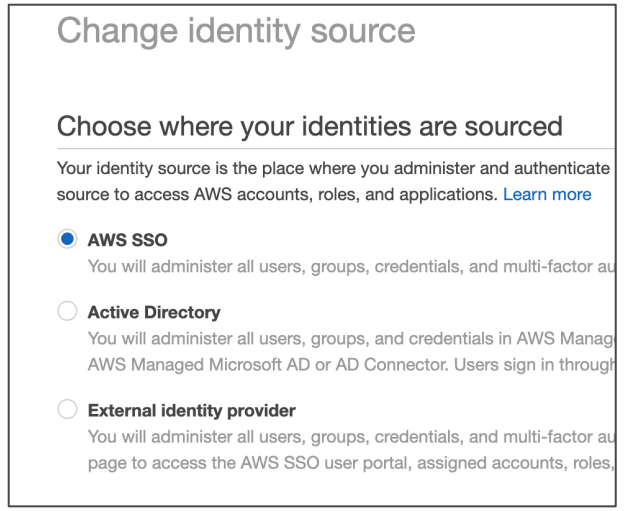
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option A

- Create a user in an AWS SSO directory and assign a read-only permissions set. Assign all AWS accounts to be monitored to the new user. Provide the third-party monitoring solution with the user name and password.





© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Note on option A

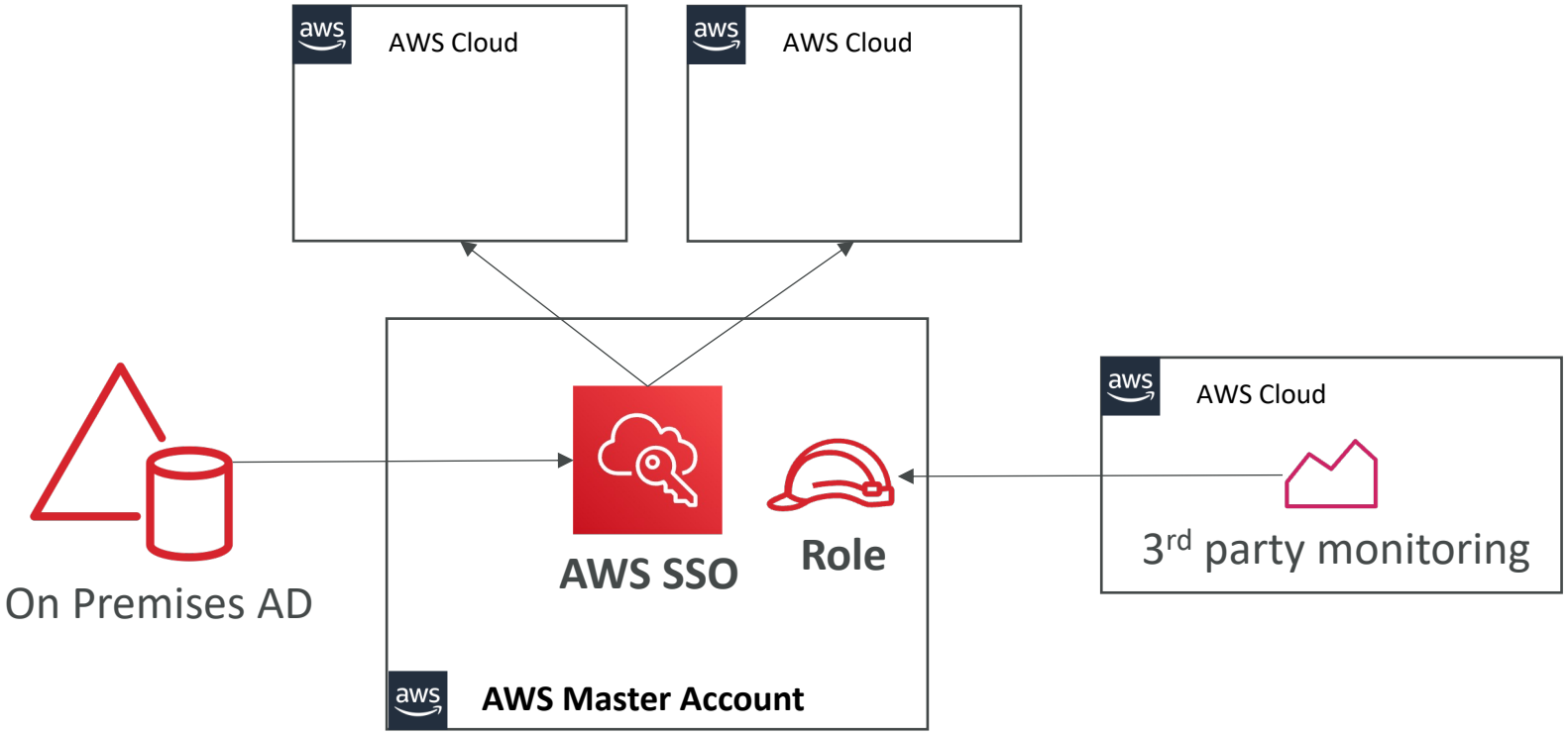
- Currently, the sample question PDF says:
- "A is incorrect **because credentials supplied by AWS SSO are temporary**, so the application would lose permissions and have to re-login"
- That is **wrong**.
- Users created in AWS SSO have a password that doesn't change and must respect the password policy defined.
- Here Option A is wrong because you can't have both users defined in AWS SSO and in Active Directory. SSO only allows for one Identity source (SSO, AD or IdP).

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option B

- Create an AWS IAM role in the organization's master account. Allow the AWS account of the third-party monitoring solution to assume the role.

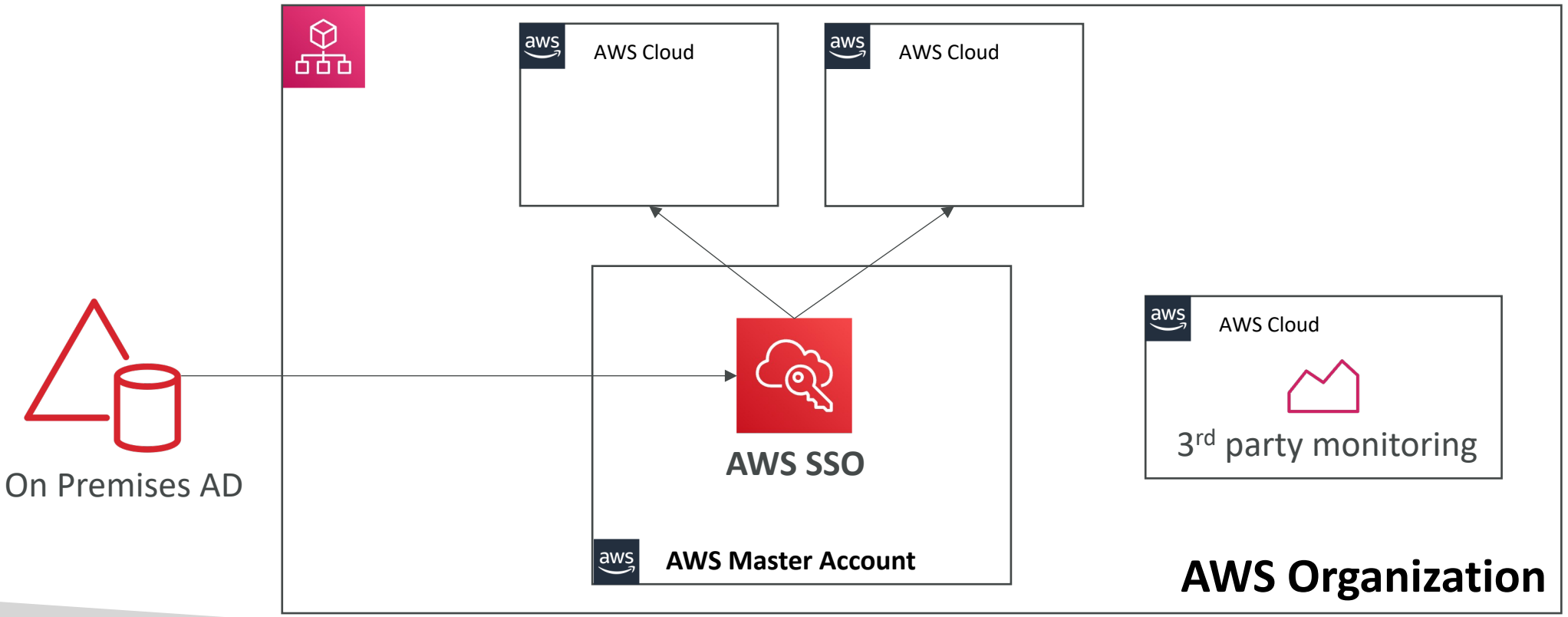


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option C

- Invite the AWS account of the third-party monitoring solution to join the organization. Enable all features



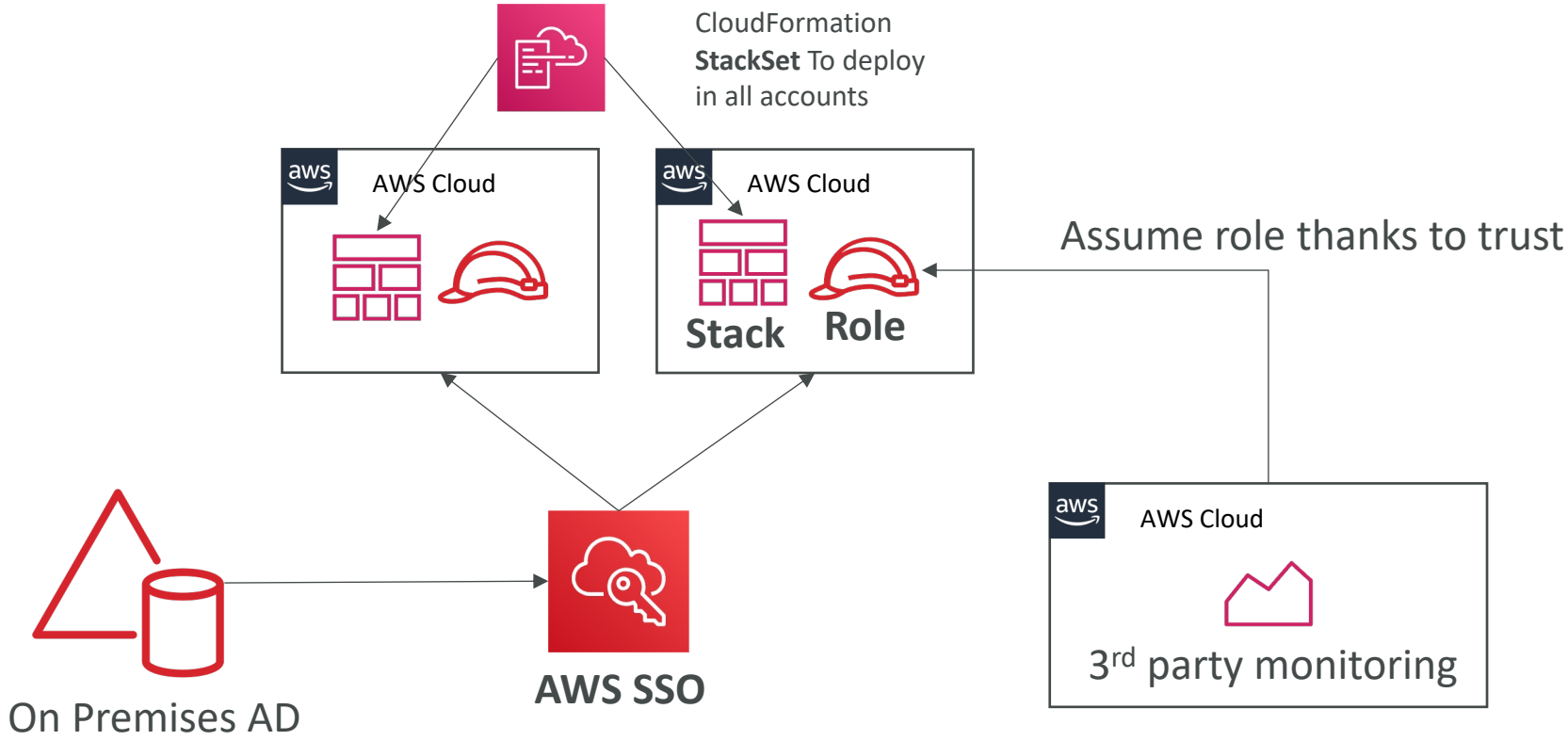
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

CORRECT ANSWER

Option D

- Create an AWS CloudFormation template that defines a new AWS IAM role for the third-party monitoring solution with the account of the third party listed in the trust policy. Create the IAM role across all linked AWS accounts by using a stack set.



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

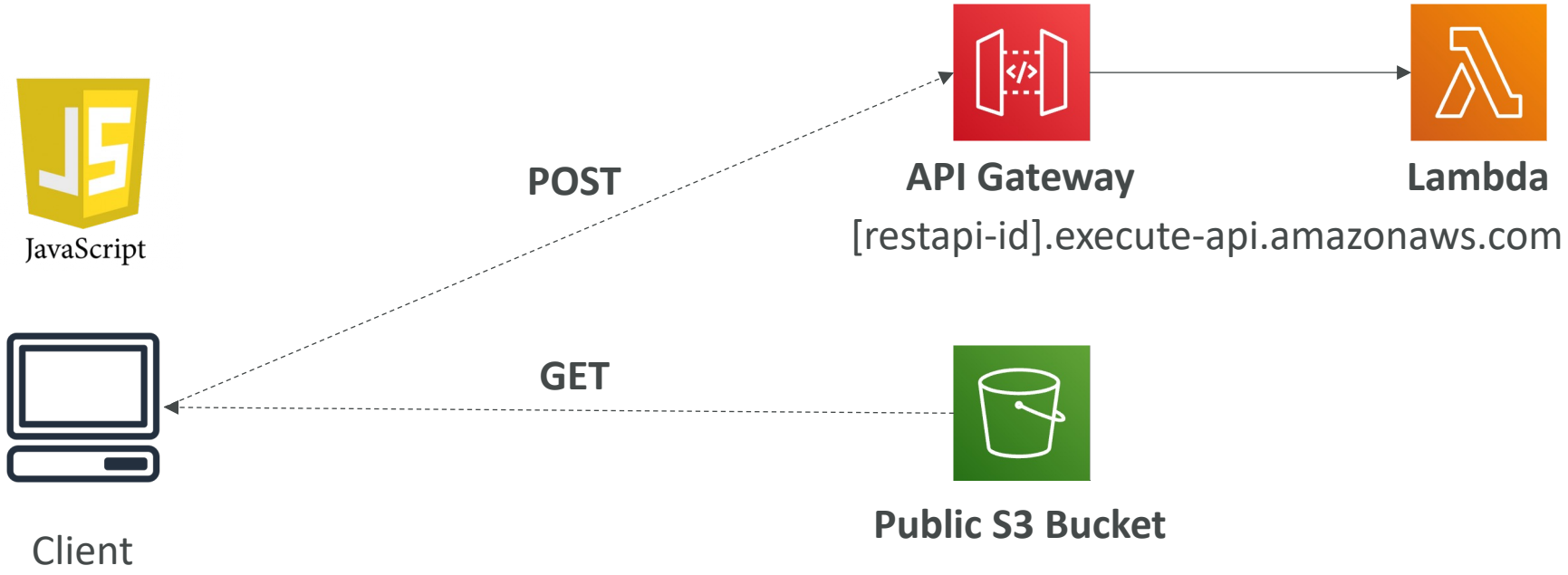
Question 3

- A team is building an **HTML form** hosted in a **public Amazon S3 bucket.** The form uses **JavaScript to post data to an Amazon API Gateway** endpoint. The endpoint is integrated with AWS Lambda functions. The team has tested each method in the API Gateway console and received valid responses.
- Which combination of steps must be completed for the form **to successfully post to the API Gateway** and receive a valid response? (Select TWO.)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Question 3 – Architecture



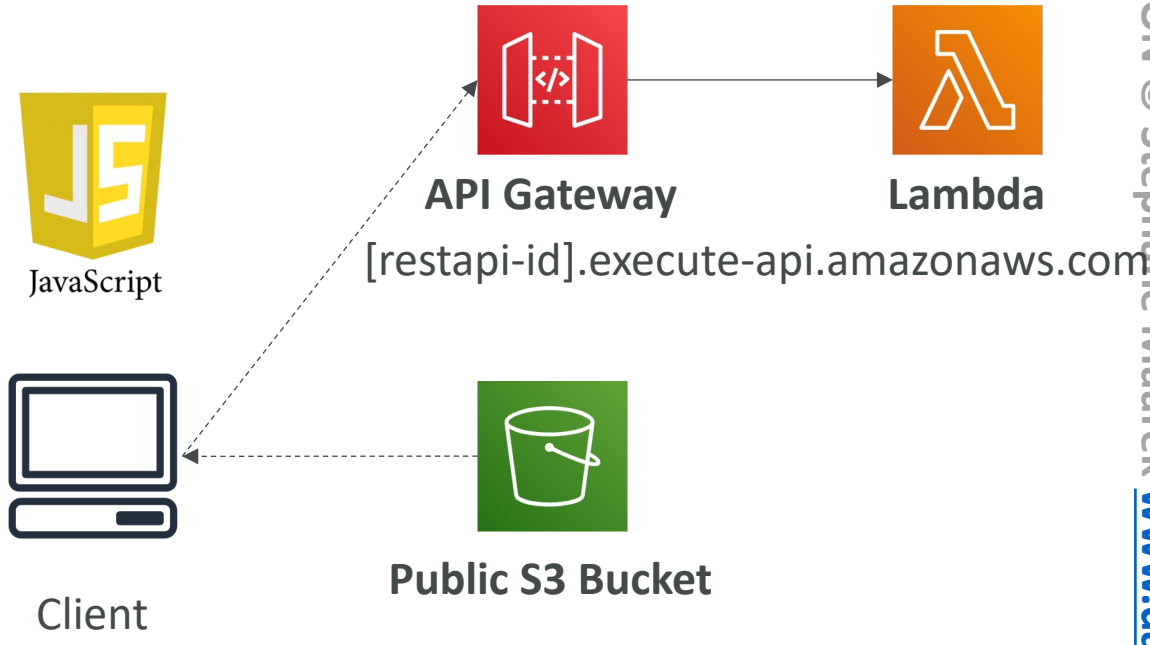
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Options

CORRECT ANSWERS D,E

- A) Configure the S3 bucket to allow cross-origin resource sharing (CORS).
- B) Host the form on Amazon EC2 rather than Amazon S3.
- C) Request a limit increase for API Gateway.
- D) Enable cross-origin resource sharing (CORS) in API Gateway.
- E) Configure the S3 bucket for web hosting.

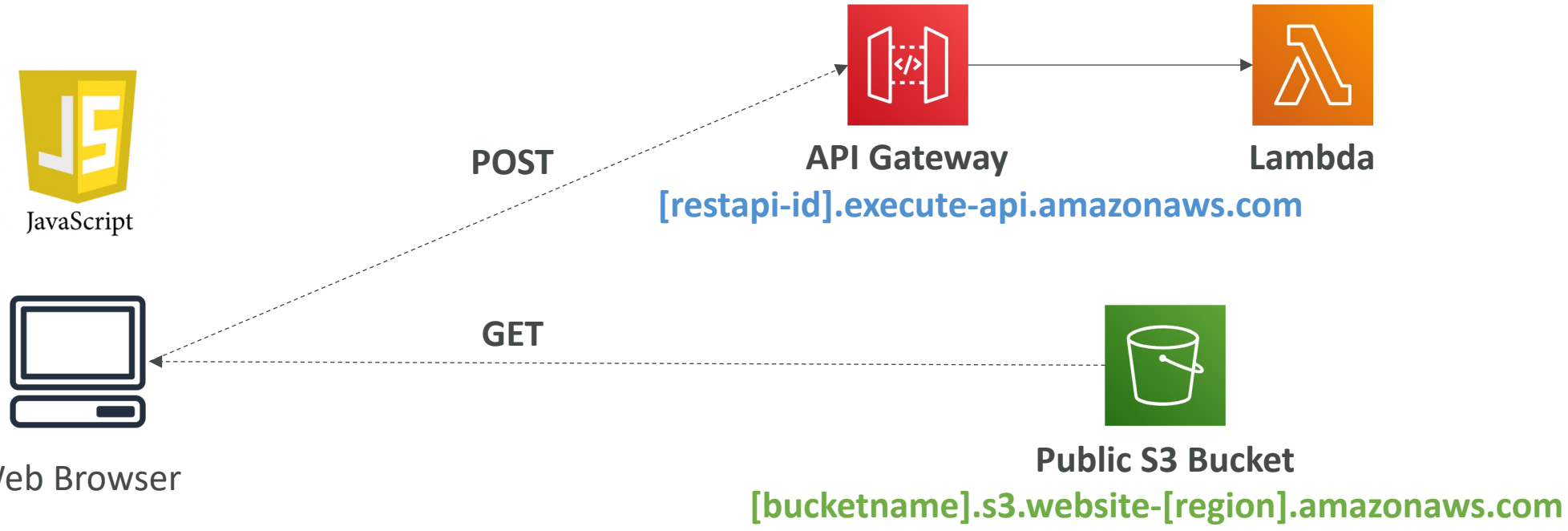


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Question 3 – Final Architecture
CORS is a Browser based security

CORS to allow calls with Origin **[bucketname].s3.website-[region].amazonaws.com**
Using the header **Access-Control-Allow-Origin**



Visits [bucketname].s3.website-[region].amazonaws.com
Makes API calls to [restapi-id].execute-api.amazonaws.com
With **Origin:** [bucketname].s3.website-[region].amazonaws.com

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

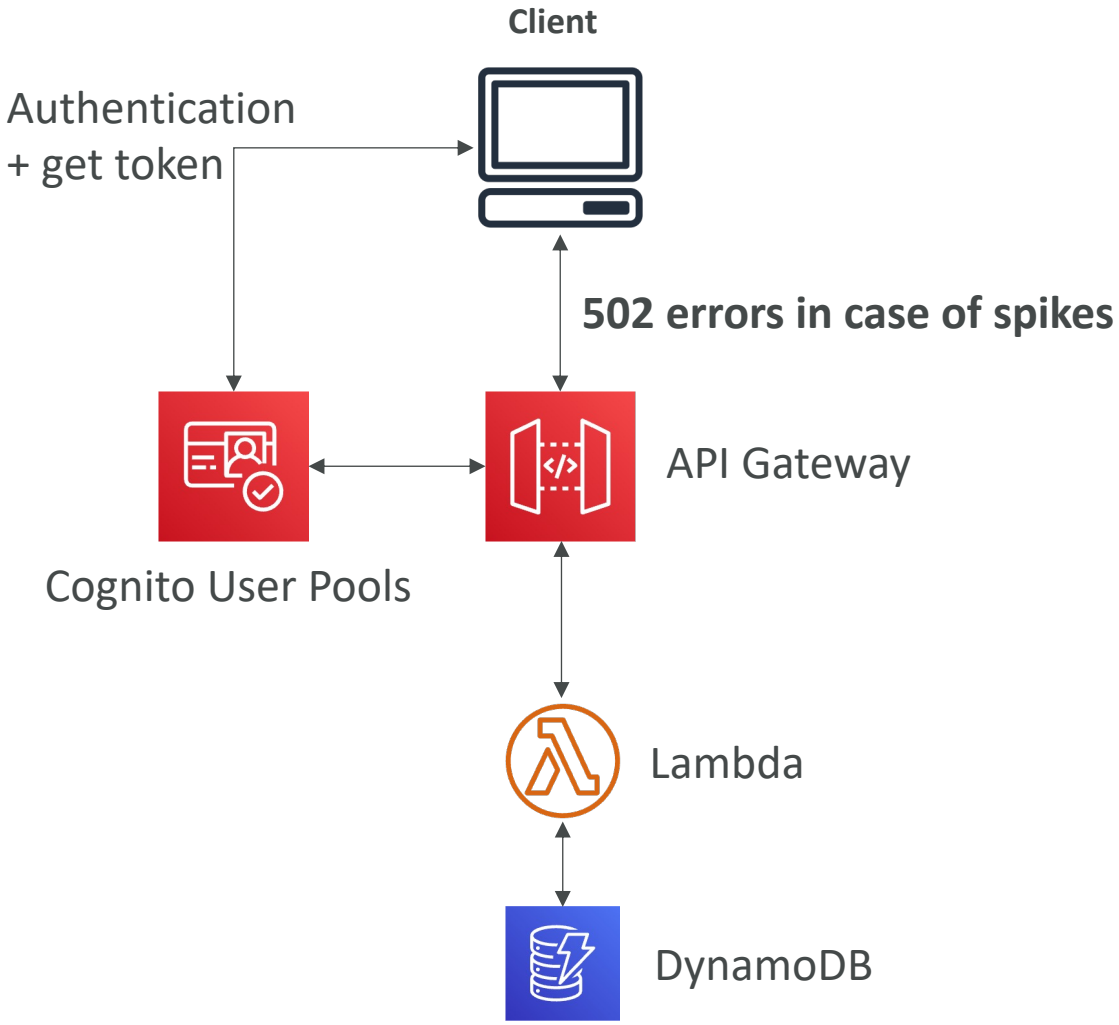
Question 4

- A retail company runs a serverless mobile app built on Amazon API Gateway, AWS Lambda, Amazon Cognito, and Amazon DynamoDB. During **heavy holiday traffic spikes**, the company receives complaints of **intermittent system failures**. Developers find that the API Gateway endpoint is returning **502 Bad Gateway errors to seemingly valid requests.**
- Which method should address this issue?

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Question 4 – Architecture



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

API Gateway - Errors

- 4xx means Client errors
  - 400: Bad Request
  - 403: Access Denied, WAF filtered
  - 429: Quota exceeded, Throttle

- 5xx means Server errors
  - **502: Bad Gateway Exception, usually for an incompatible output returned from a Lambda proxy integration backend and occasionally for out-of-order invocations due to heavy loads.**
  - 503: Service Unavailable Exception
  - 504: Integration Failure – ex Endpoint Request Timed-out Exception
    API Gateway requests time out after 29 second maximum

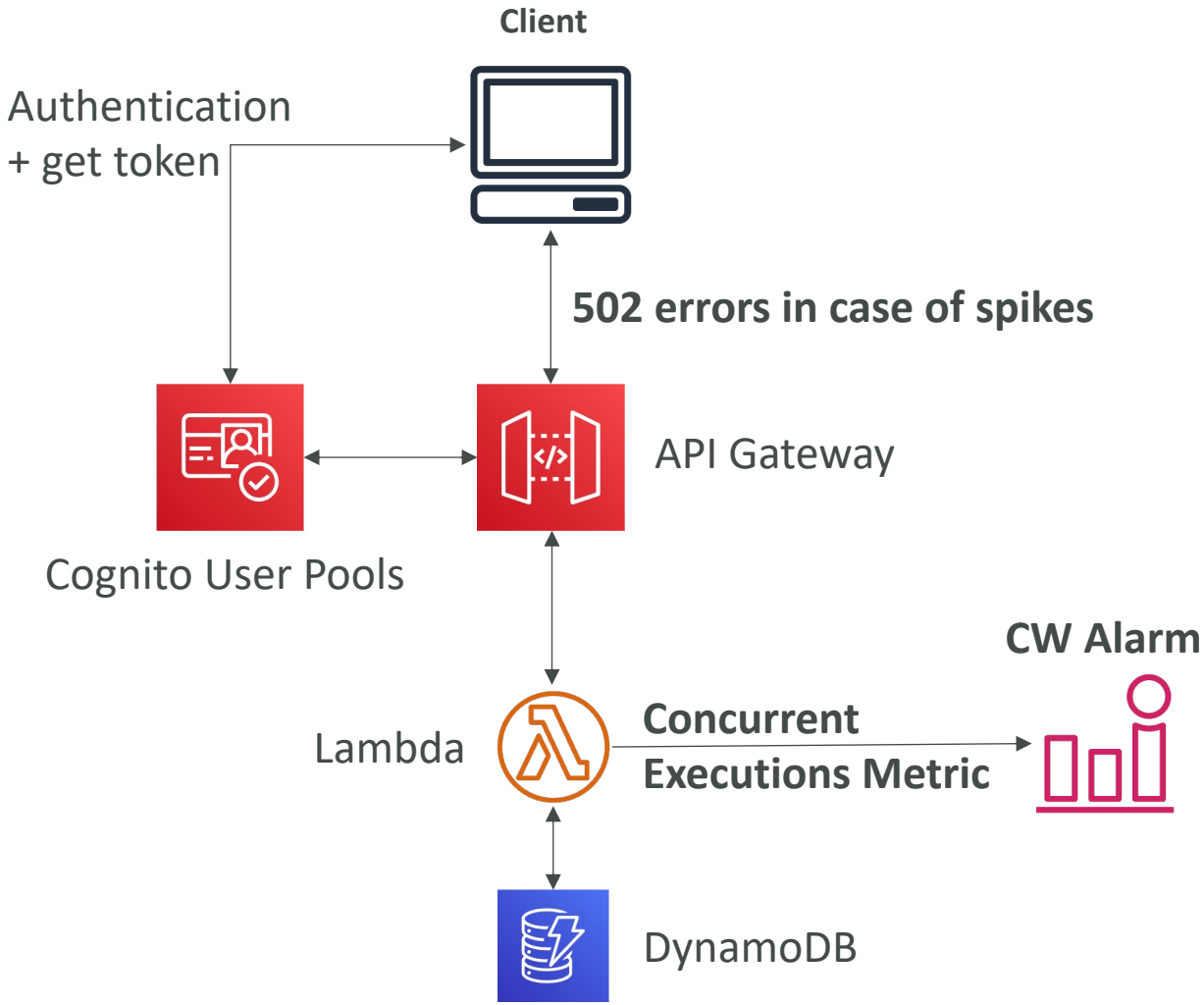
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option A

CORRECT ANSWER

- Increase the **concurrency limit for Lambda functions** and configure notification alerts to be sent by Amazon CloudWatch when the **ConcurrentExecutions** metric approaches the limit.



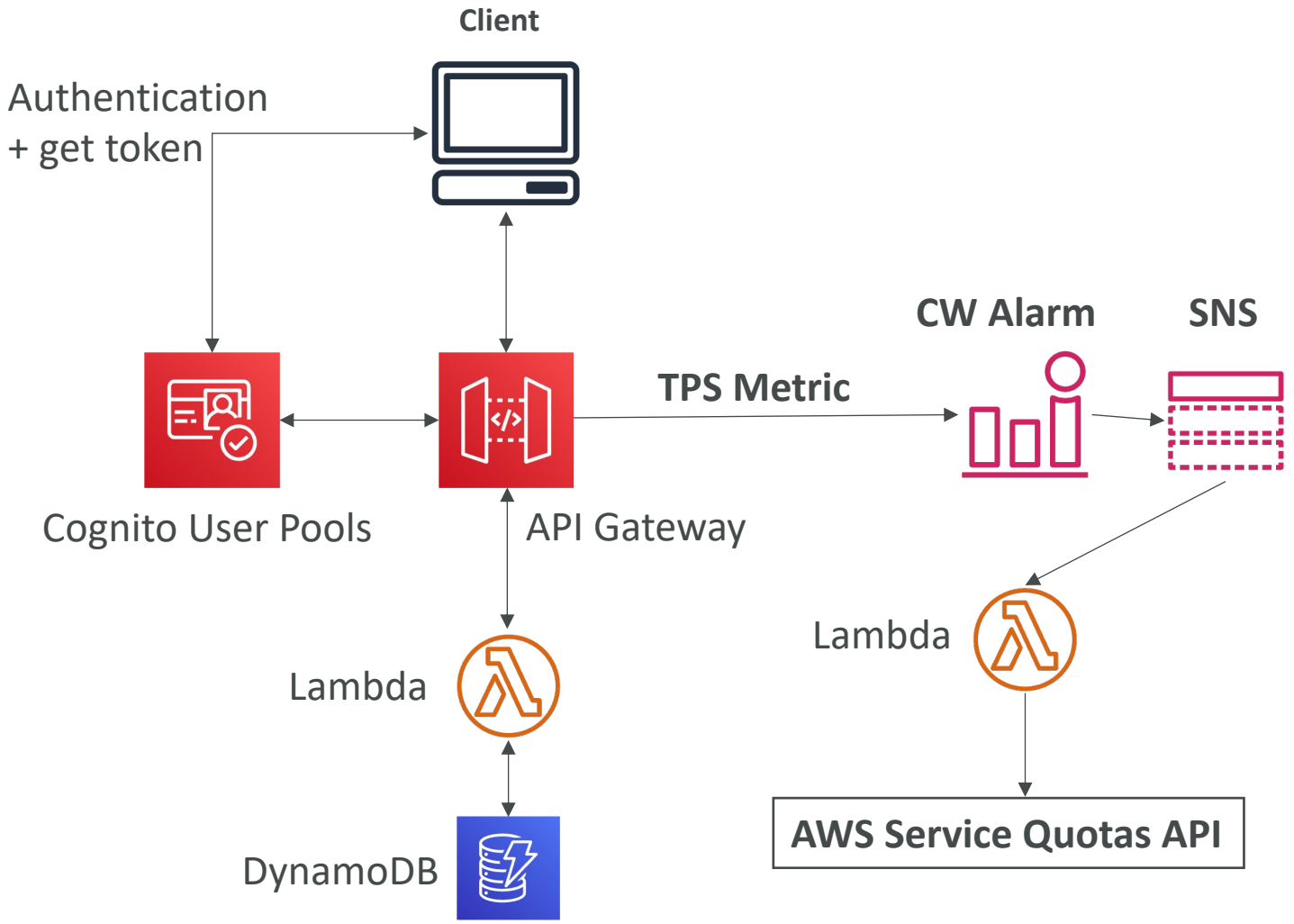
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option B

- Configure notification alerts for the limit of transactions per second on the API Gateway endpoint and create a Lambda function that will increase this limit, as needed.

Option B would work if we were receiving 429 errors
429: Quota exceeded, Throttle



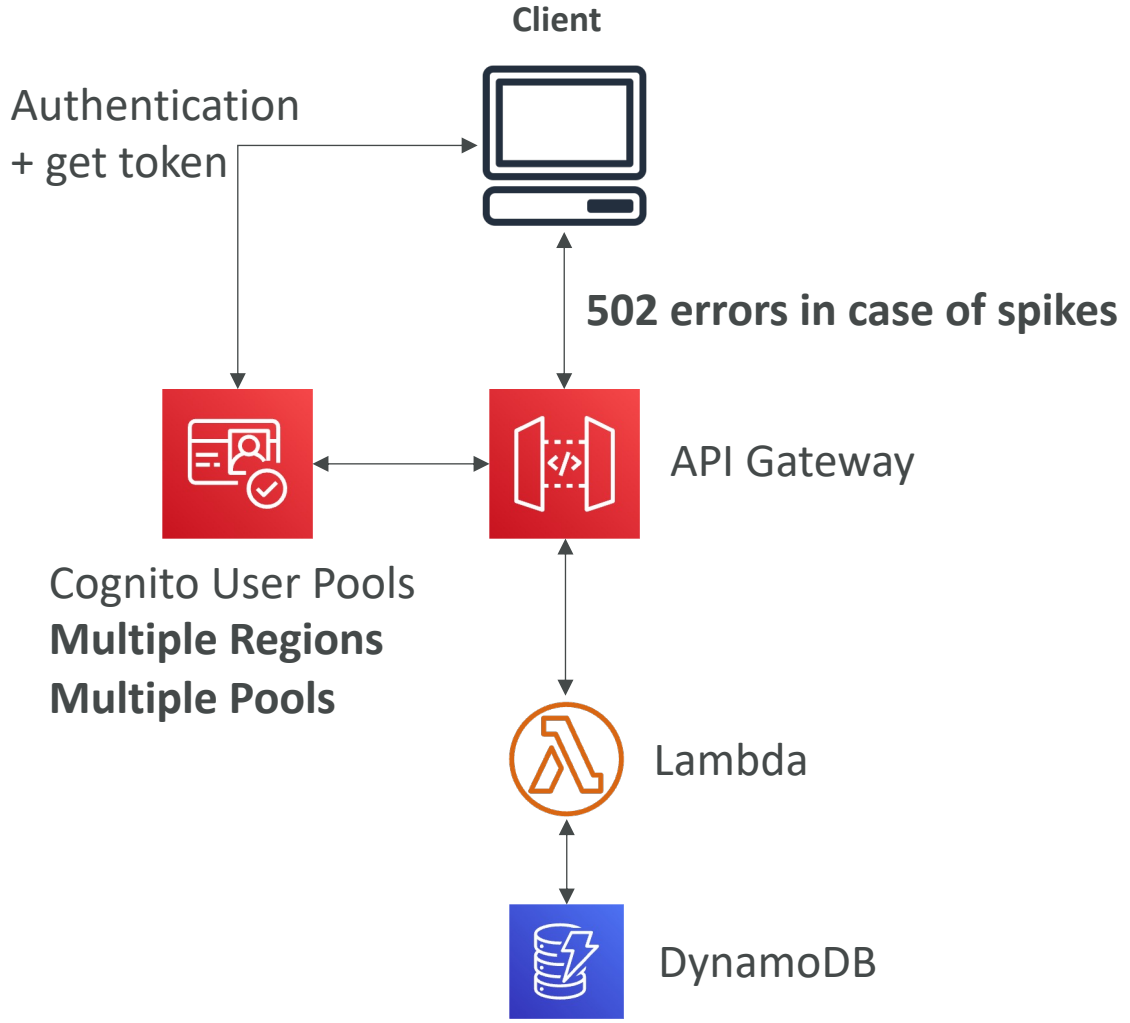
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option C

- Shard users to Amazon Cognito user pools in multiple regions to reduce user authentication latency.

Option C would be valid if we had performance issues At Cognito User Pools, but that's not the question

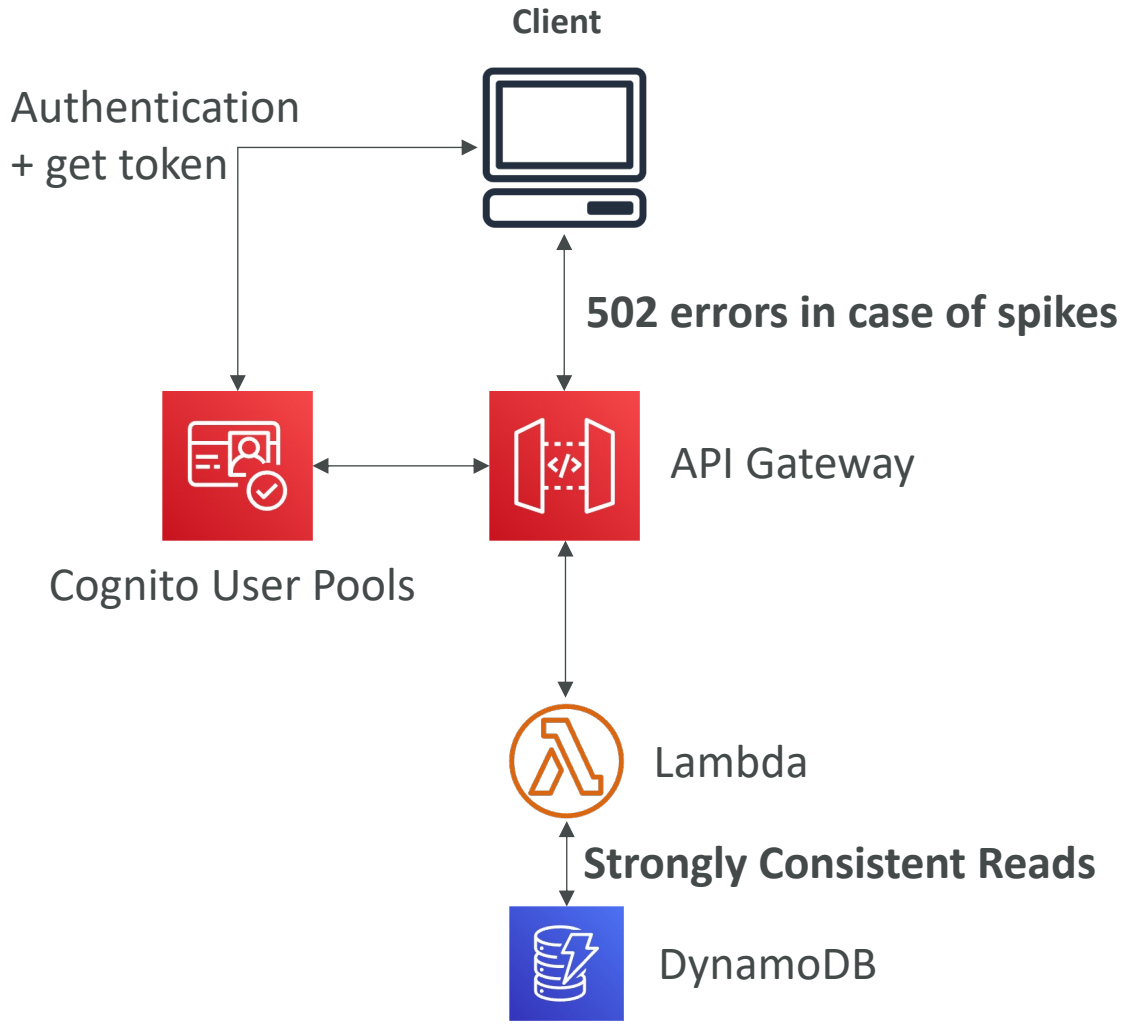


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option D

- Use DynamoDB strongly consistent reads to ensure the latest data is always returned to the client application.



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

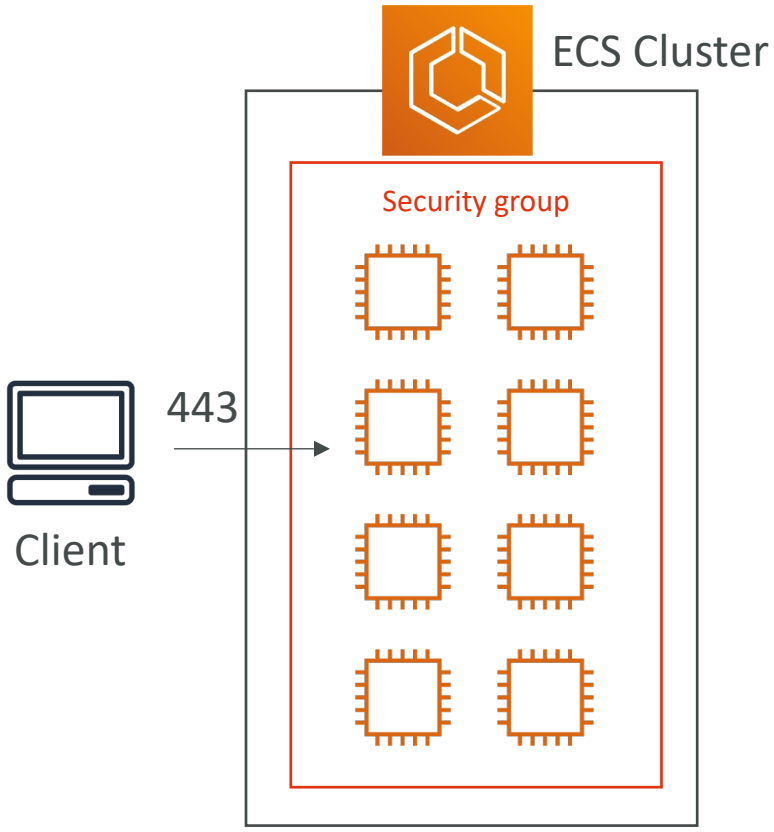
Question 5

- A company is launching a new web service on an Amazon ECS cluster. Company policy requires that the security group on the cluster instances block all inbound traffic but HTTPS (port 443). The cluster consists of Amazon 100 EC2 instances. Security engineers are responsible for managing and updating the cluster instances. The security engineering team is small, so any management efforts must be minimized.
- How can the service be designed to meet these operational requirements?

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Question 5 – Architecture

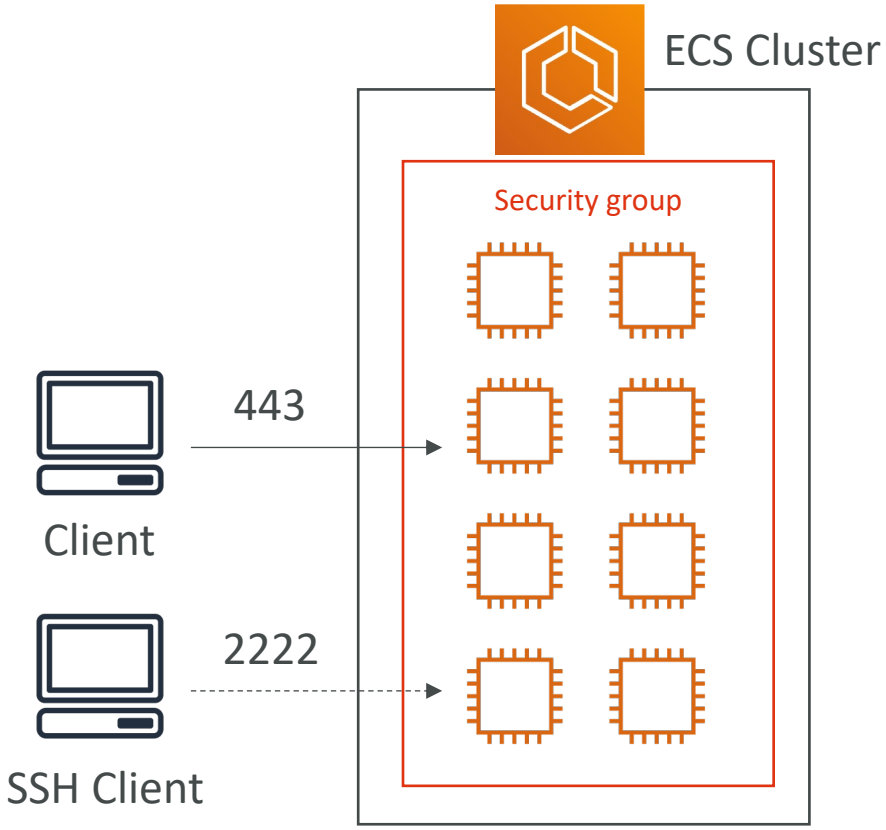


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option A

- Change the SSH port to 2222 on the cluster instances with a user data script. Log in to each instance using SSH over port 2222

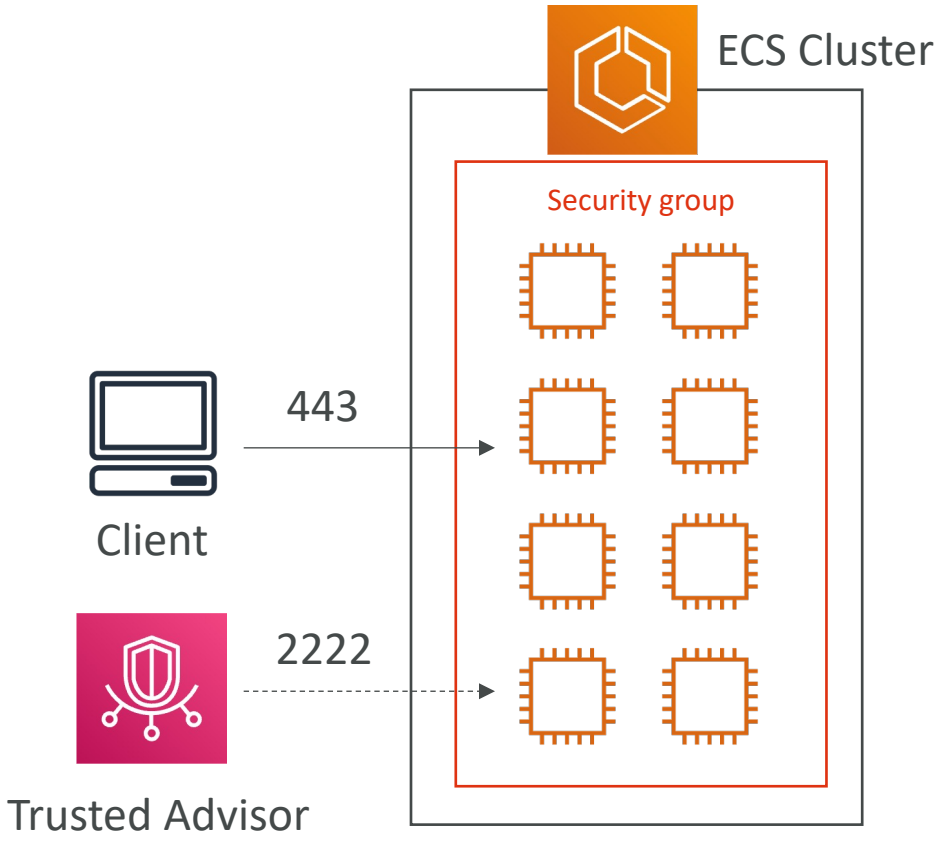


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option B

- Change the SSH port to 2222 on the cluster instances with a user data script. Use AWS Trusted Advisor to remotely manage the cluster instances over port 2222



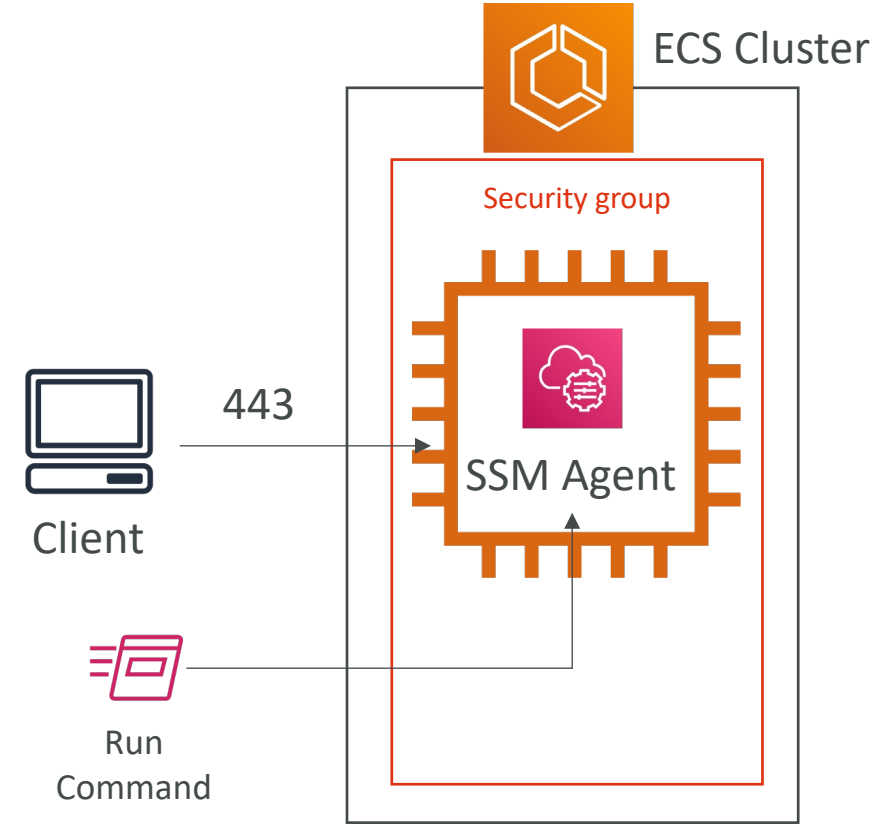
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option C

CORRECT ANSWER

- Launch the cluster instances with no SSH key pairs. Use the Amazon EC2 Systems Manager Run Command to remotely manage the cluster instances

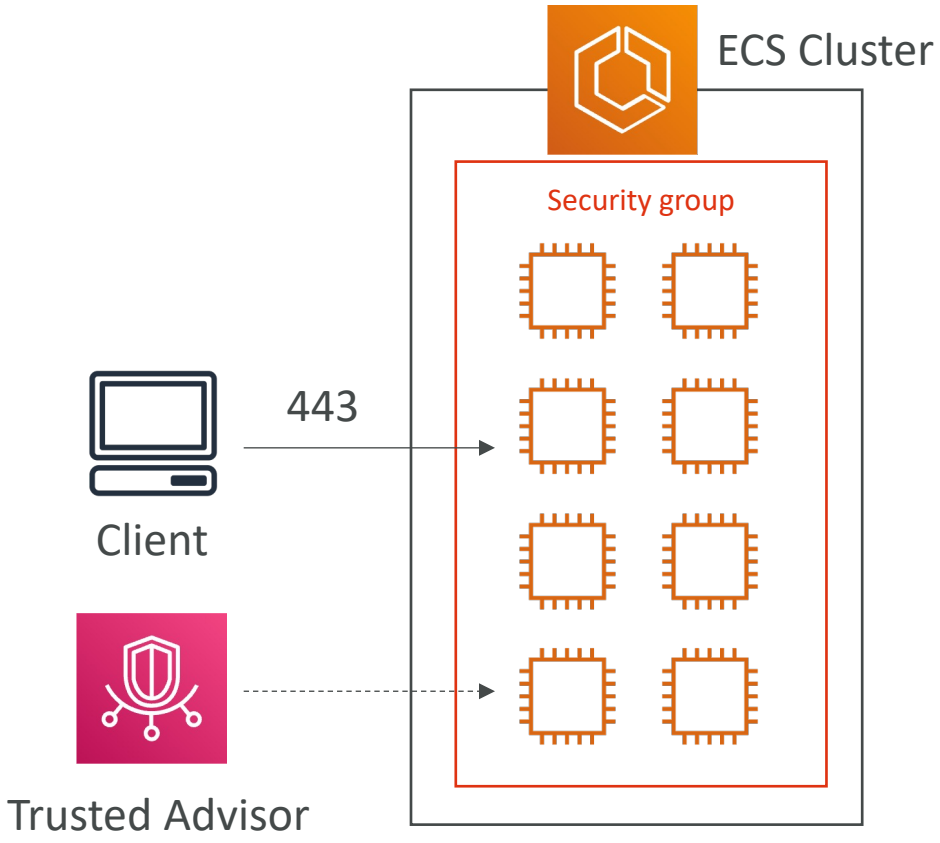


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option D

- Launch the cluster instances with no SSH key pairs. Use AWS Trusted Advisor to remotely manage the cluster instances.



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Question 6

- A company has two AWS accounts: one for production workloads and one for development workloads. Creating and managing these workloads are a development team and an operations team. The company needs a security strategy that meets the following requirements:
  - Developers need to create and delete development application infrastructure.
  - Operators need to create and delete both development and production application infrastructure.
  - Developers should have no access to production infrastructure.
  - All users should have a single set of AWS credentials.
- What strategy meets these requirements?

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option A

- In the development account:
  - Create a development IAM group with the ability to create and delete application infrastructure.
  - Create an IAM user for each operator and developer and assign them to the development group.
- In the production account:
  - Create an operations IAM group with the ability to create and delete application infrastructure.
  - Create an IAM user for each operator and assign them to the operations group.

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option B

- In the development account:
  - Create a development IAM group with the ability to create and delete application infrastructure.
  - Create an IAM user for each developer and assign them to the development group.
  - Create an IAM user for each operator and assign them to the development group and the operations group in the production account.
- In the production account:
  - Create an operations IAM group with the ability to create and delete application infrastructure.

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option C

- In the development account:
  - Create a shared IAM role with the ability to create and delete application infrastructure in the production account.
  - Create a development IAM group with the ability to create and delete application infrastructure.
  - Create an operations IAM group with the ability to assume the shared role.
  - Create an IAM user for each developer and assign them to the development group.
  - Create an IAM user for each operator and assign them to the development group and the operations group.

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option D

CORRECT ANSWER

- In the development account:
  - Create a development IAM group with the ability to create and delete application infrastructure.
  - Create an operations IAM group with the ability to assume the shared role in the production account.
  - Create an IAM user for each developer and assign them to the development group.
  - Create an IAM user for each operator and assign them to the development group and the operations group.
- In the production account:
  - Create a shared IAM role with the ability to create and delete application infrastructure.
  - Add the development account to the trust policy for the shared role.

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

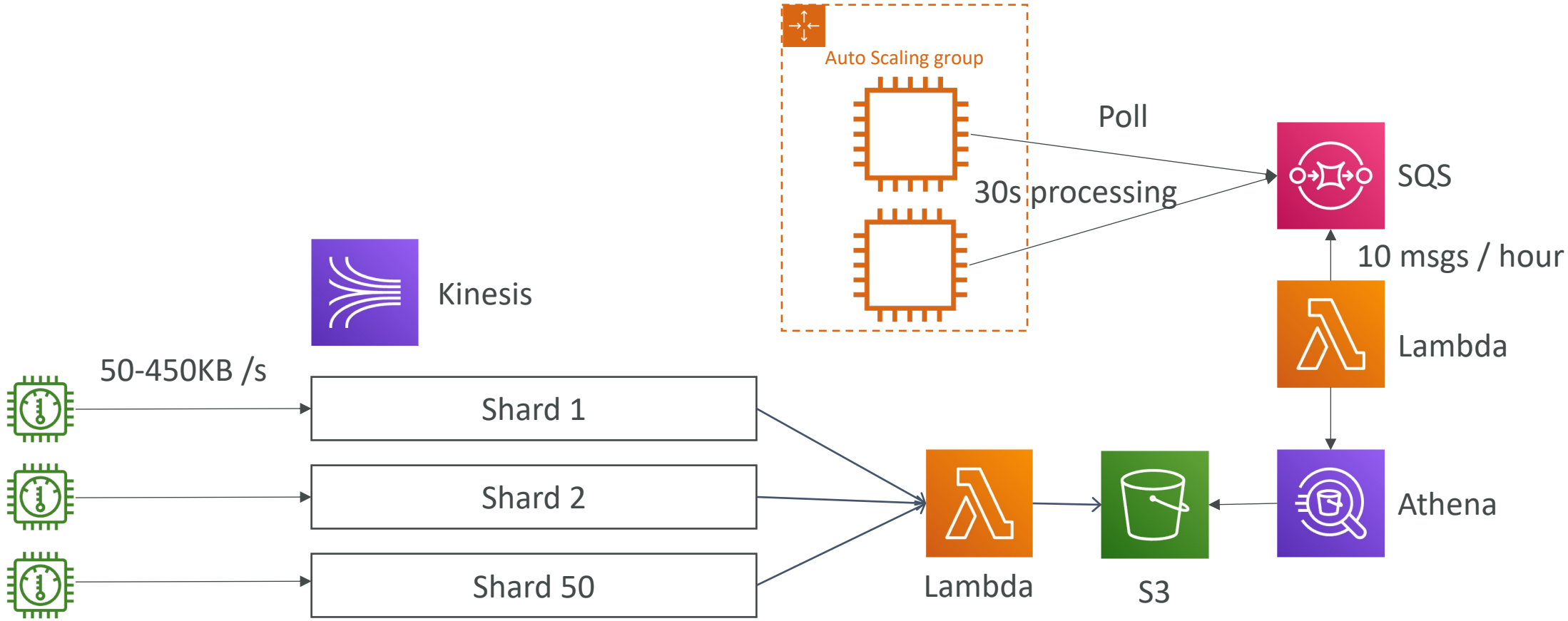
Question 7

- A solutions architect needs to reduce costs for a big data application. The application environment consists of hundreds of devices that send events to Amazon Kinesis Data Streams. The device ID is used as the partition key, so each device gets a separate shard. Each device sends between 50 KB and 450 KB of data per second. The shards are polled by an AWS Lambda function that processes the data and stores the result on Amazon S3.
- Every hour, an AWS Lambda function runs an Amazon Athena query against the result data that identifies any outliers and places them in an Amazon SQS queue. An Amazon EC2 Auto Scaling group of two EC2 instances monitors the queue and runs a short (approximately 30-second) process to address the outliers. The devices submit an average of 10 outlying values every hour.
- Which combination of changes to the application would MOST reduce costs? (Select TWO.)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Question 7 – Architecture



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Options (choose 2)

A) Change the Auto Scaling group launch configuration to use smaller instance types in the same instance family.

B) Replace the Auto Scaling group with an AWS Lambda function triggered by messages arriving in the Amazon SQS queue.

C) Reconfigure the devices and data stream to set a ratio of 10 devices to 1 data stream shard.

D) Reconfigure the devices and data stream to set a ratio of 2 devices to 1 data stream shard.

E) Change the desired capacity of the Auto Scaling group to a single EC2 instance.

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

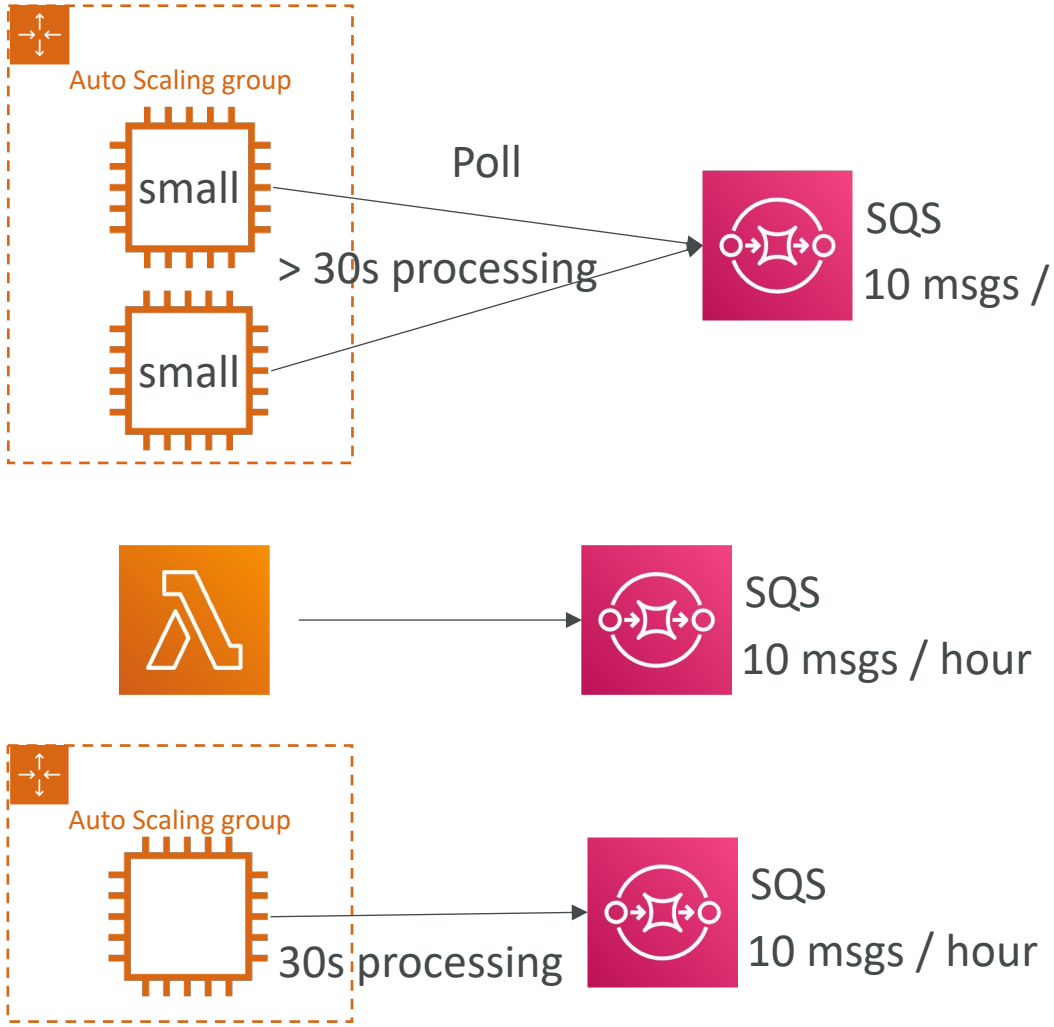
Option Group 1

CORRECT ANSWER B

A) Change the Auto Scaling group launch configuration to use smaller instance types in the same instance family.

B) Replace the Auto Scaling group with an AWS Lambda function triggered by messages arriving in the Amazon SQS queue.

E) Change the desired capacity of the Auto Scaling group to a single EC2 instance.



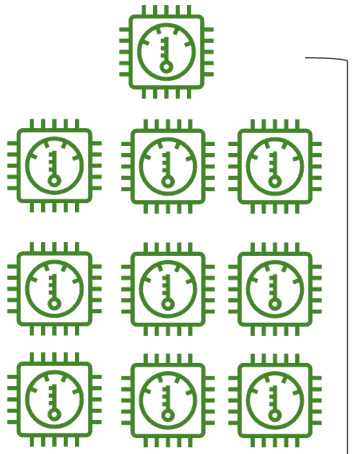
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option Group 2

CORRECT ANSWER D

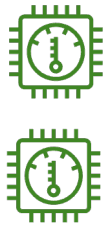
C) Reconfigure the devices and data stream to set a ratio of 10 devices to 1 data stream shard.



50-450KB /s x 10 = **500KB-4500KB / s**

Shard N

D) Reconfigure the devices and data stream to set a ratio of 2 devices to 1 data stream shard.



50-450KB /s x 2 = **100KB-900KB / s**

Shard N

Each shard has a limit of 1MB/s = 1000KB/s

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Question 8

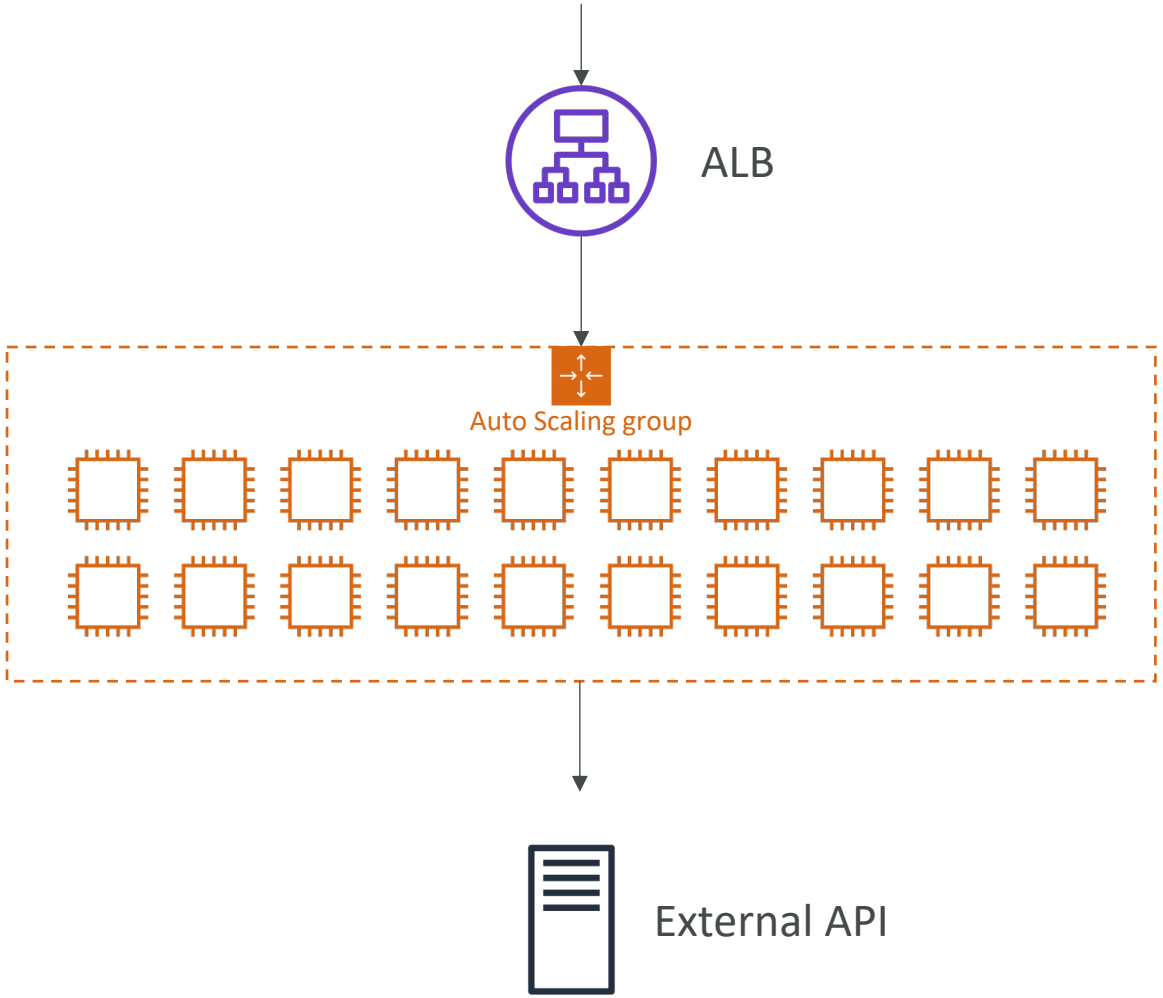
- A company operates an ecommerce application on Amazon EC2 instances behind an ELB Application Load Balancer. The instances run in an Amazon EC2 Auto Scaling group across multiple Availability Zones. After an order is successfully processed, the application immediately posts order data to an external third-party affiliate tracking system that pays sales commissions for order referrals. During a highly successful marketing promotion, the number of EC2 instances increased from 2 to 20. The application continued to work correctly, but the increased request rate overwhelmed the third-party affiliate and resulted in failed requests.

- Which combination of architectural changes could ensure that the entire process functions correctly under load? (Select TWO.)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Question 8 – Architecture



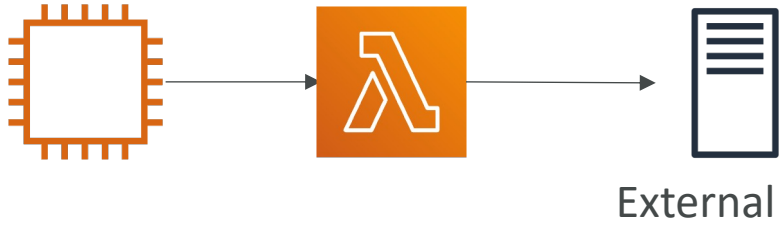
© Stephane Maarek

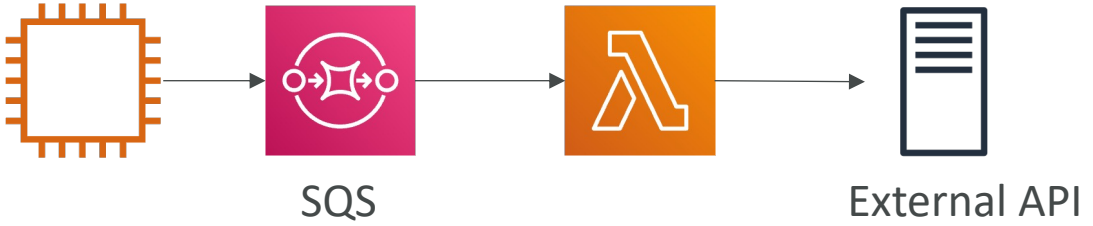
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option Group 1

CORRECT ANSWER B

- A) Move the code that calls the affiliate to a new AWS Lambda function. Modify the application to invoke the Lambda function asynchronously.
- B) Move the code that calls the affiliate to a new AWS Lambda function. Modify the application to place the order data in an Amazon SQS queue. Trigger the Lambda function from the queue.





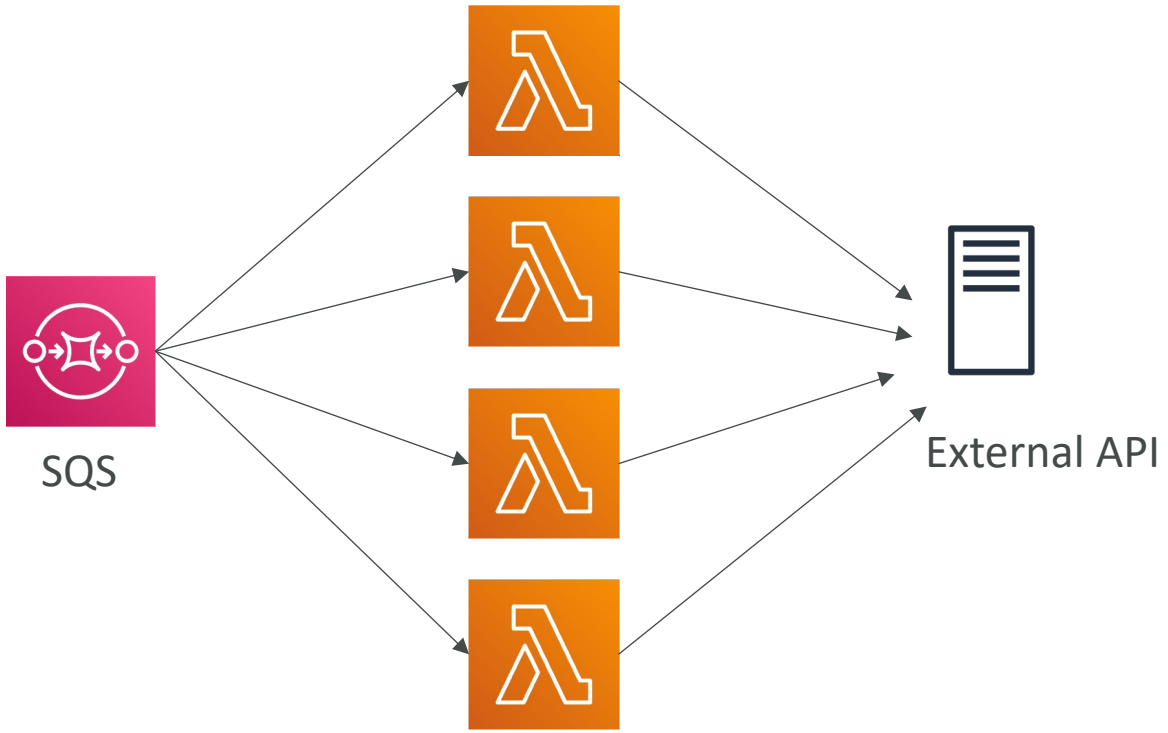
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option Group 2

CORRECT ANSWER D

- C) Increase the timeout of the new AWS Lambda function.
- D) Adjust the concurrency limit of the new AWS Lambda function.
- E) Increase the memory of the new AWS Lambda function.



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Question 10

- A company has deployed a multi-tier web application in the AWS Cloud. The application consists of the following tiers:
  - A Windows-based web tier that is hosted on Amazon EC2 instances with Elastic IP addresses
  - A Linux-based application tier that is hosted on EC2 instances that run behind an Application Load Balancer (ALB) that uses path-based routing
  - A MySQL database that runs on a Linux EC2 instance

- All the EC2 instances are using Intel-based x86 CPUs. A solutions architect needs to modernize the infrastructure to achieve better performance. The solution must minimize the operational overhead of the application.

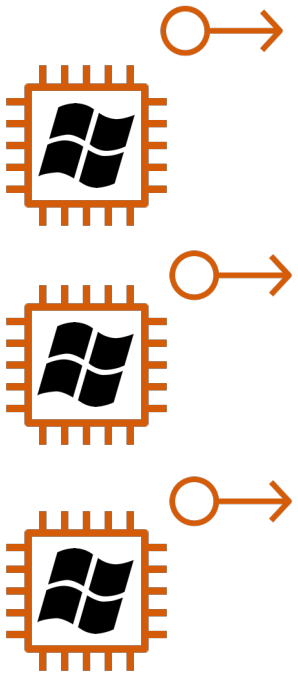
- Which combination of actions should the solutions architect take to meet these requirements? (Select TWO.)

© Stephane Maarek

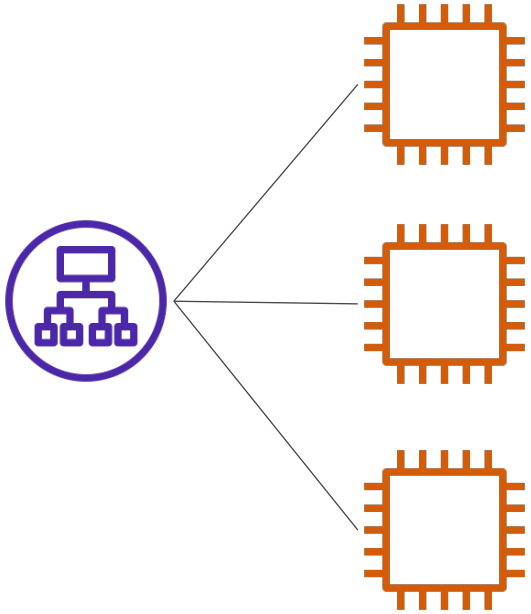
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Question 10 - Architecture

Web Tier



Application Tier



Database Tier

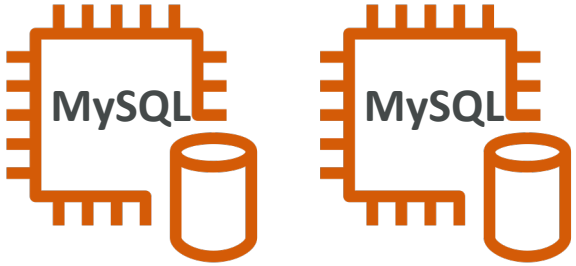
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Option Group 1

CORRECT ANSWER C

- A) Run the MySQL database on multiple EC2 instances.



- C) Migrate the MySQL database to Amazon Aurora Serverless.

- D) Migrate all EC2 instance types to Graviton2.
  - Not available for Windows Instances ✗

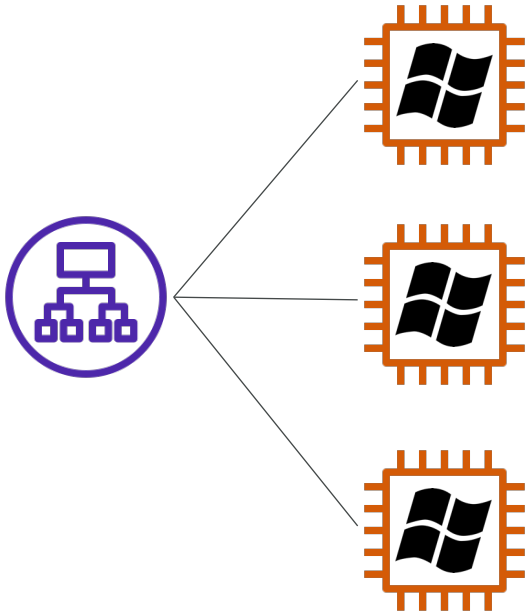
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

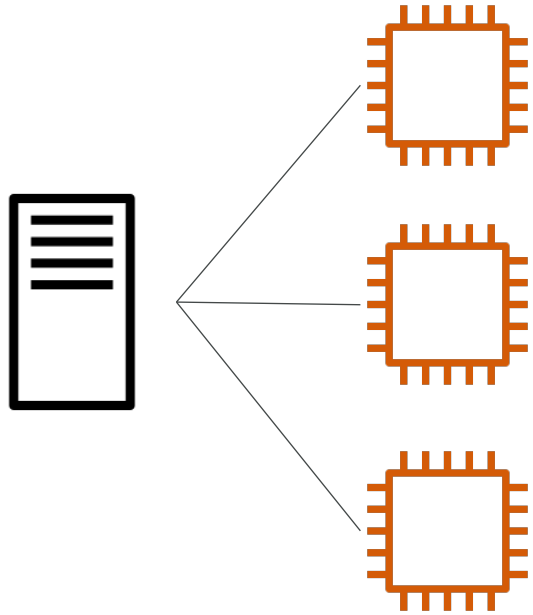
Option Group 2

CORRECT ANSWER B

- B) Place the web tier instances behind an ALB.



- E) Replace the ALB for the application tier instances with a company-managed load balancer.



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Your AWS Certification journey

Foundational

Knowledge-based certification for
foundational understanding of AWS Cloud.
No prior experience needed.

Associate

Role-based certifications that showcase your knowledge
and skills on AWS and build your credibility as an AWS Cloud professional.
Prior cloud and/or strong on-premises IT experience recommended.

Professional

Role-based certifications that validate advanced skills
and knowledge required to design secure, optimized,
and modernized applications and to automate processes on AWS.
2 years of prior AWS Cloud experience recommended.

Specialty

Dive deeper and position yourself as a trusted advisor to your
stakeholders and/or customers in these strategic areas.
Refer to the exam guides on the exam pages for recommended experience.

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Certification Paths – Architecture

Architecture

Solutions Architect

Design, develop, and manage cloud infrastructure and assets, work with DevOps to migrate applications to the cloud

optional for IT/ cloud professionals

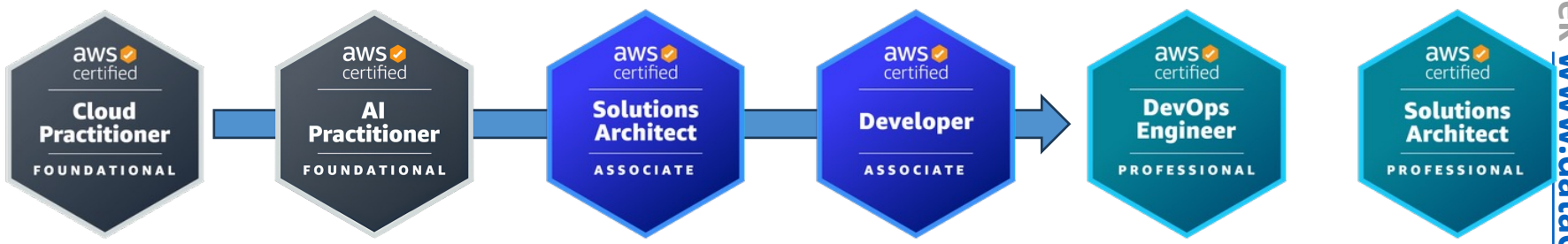
recommended for IT/cloud professionals to leverage AI

Dive Deep

Architecture

Application Architect

Design significant aspects of application architecture including user interface, middleware, and infrastructure, and ensure enterprise-wide scalable, reliable, and manageable systems



optional for IT/ cloud professionals

recommended for IT/cloud professionals to leverage AI

Dive Deep

https://d1.awsstatic.com/training-and-certification/docs/AWS_certification_paths.pdf

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Certification Paths – Operations

Operations
Systems Administrator
Install, upgrade, and maintain computer components and software, and integrate automation processes

optional for IT/ cloud professionals

Dive Deep

Operations
Cloud Engineer
Implement and operate an organization's networked computing infrastructure and Implement security systems to maintain data safety

optional for IT/ cloud professionals

Dive Deep

© Stephane Maarek

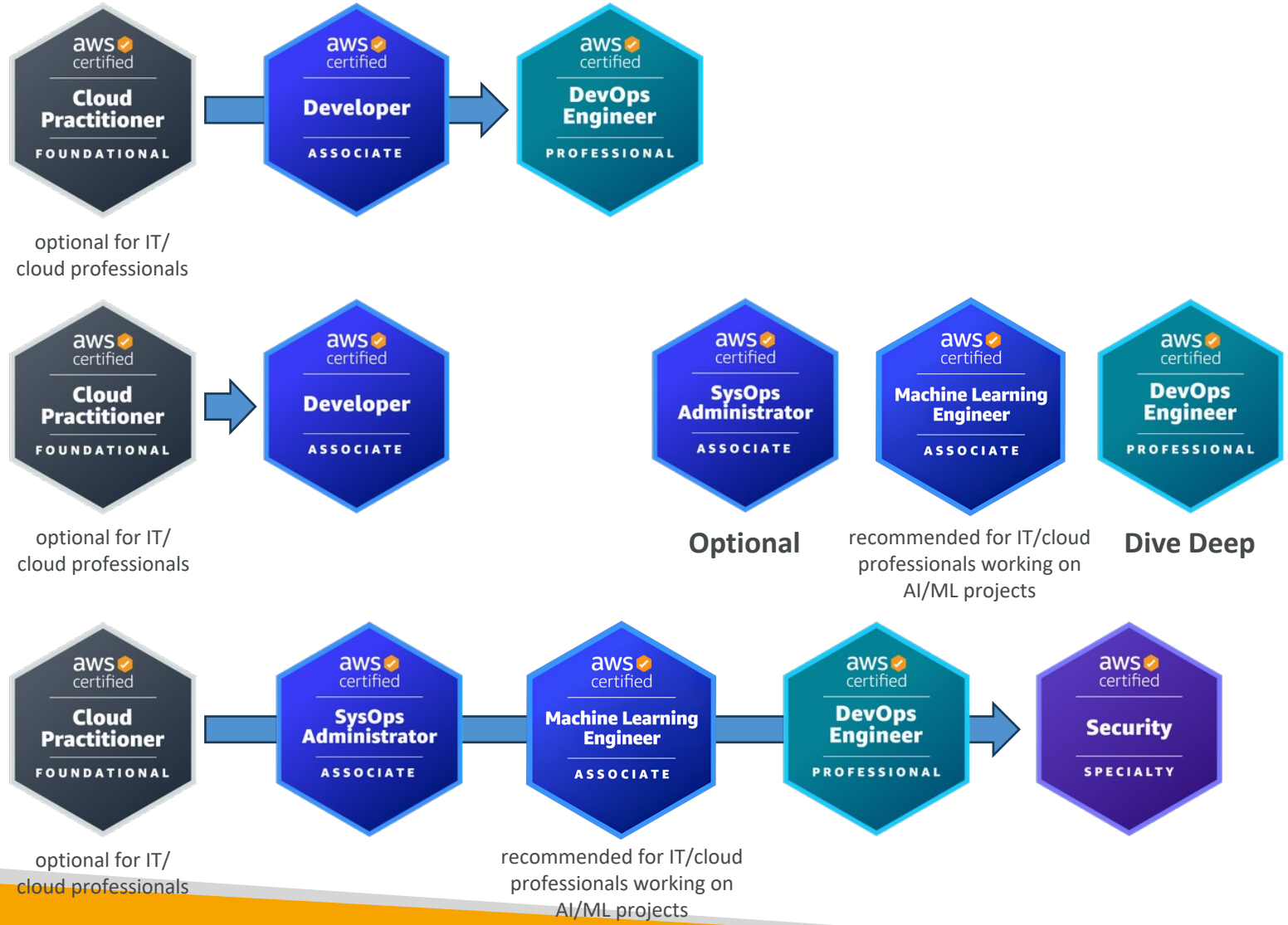
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Certification Paths – DevOps

DevOps
Test Engineer
Embed testing and quality best practices for software development from design to release, throughout the product life cycle

DevOps
Cloud DevOps Engineer
Design, deployment, and operations of large-scale global hybrid cloud computing environment, advocating for end-to-end automated CI/CD DevOps pipelines

DevOps
DevSecOps Engineer
Accelerate enterprise cloud adoption while enabling rapid and stable delivery of capabilities using CI/CD principles, methodologies, and technologies



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Certification Paths – Security

Security

Cloud Security Engineer

Design computer security architecture and develop detailed cyber security designs. Develop, execute, and track performance of security measures to protect information

optional for IT/ cloud professionals

recommended for IT/cloud professionals to secure AI/ML systems

Dive Deep

Security

Cloud Security Architect

Design and implement enterprise cloud solutions applying governance to identify, communicate, and minimize business and technical risks

optional for IT/ cloud professionals

recommended for IT/cloud professionals to secure AI/ML systems

Dive Deep

© Stephane Maarek

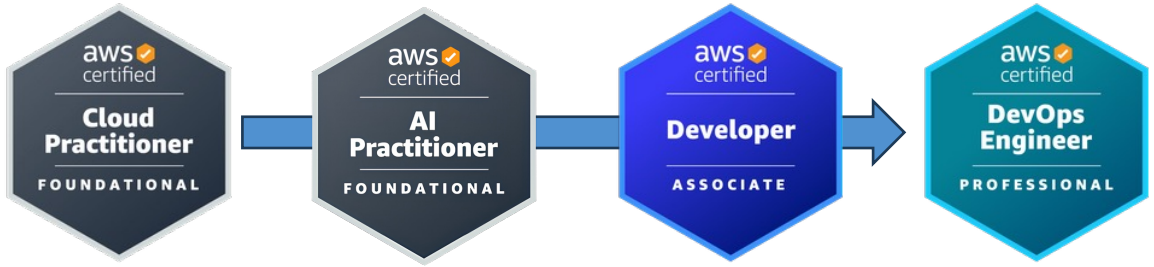
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Certification Paths – Development & Networking

Development

Software Development Engineer

Develop, construct, and maintain software across platforms and devices



optional for IT/ cloud professionals

recommended for IT/cloud professionals to leverage AI

Networking

Network Engineer

Design and implement computer and information networks, such as local area networks (LAN), wide area networks (WAN), intranets, extranets, etc.

optional for IT/ cloud professionals

Dive Deep

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Certification Paths – Data Analytics & AI/ML

Data Analytics

Cloud Data Engineer

Automate collection and processing of structured/semi-structured data and monitor data pipeline performance

optional for IT/ cloud professionals

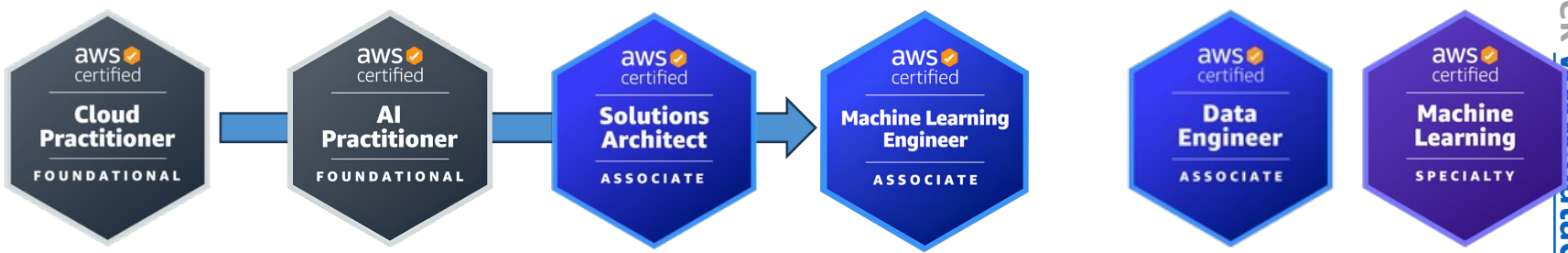
recommended for IT/cloud professionals working on AI/ML projects

Dive Deep

AI/ML

Machine Learning Engineer

Research, build, and design artificial intelligence (AI) systems to automate predictive models, and design machine learning systems, models, and schemes



optional for IT/ cloud professionals

optional for AI/ML professionals

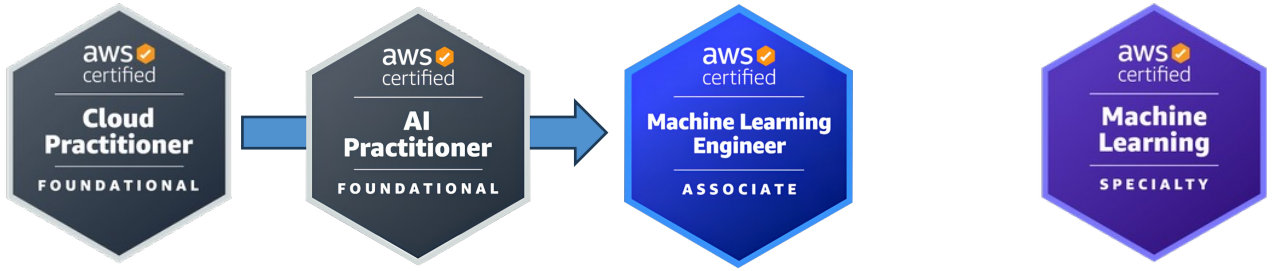
Dive Deep

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

AWS Certification Paths – AI/ML

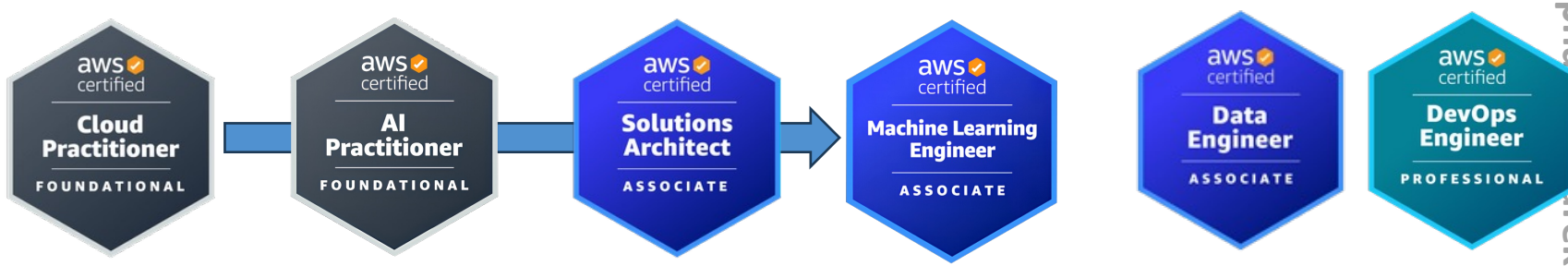
AI/ML
Prompt Engineer
Design, test, and refine text prompts to optimize the performance of AI language models



optional for IT/ cloud professionals

Dive Deep

AI/ML
Machine Learning Ops Engineer
Build and maintain AI and ML platforms and infrastructure. Design, implement, and operationally support AI/ML model activity and deployment infrastructure



optional for IT/ cloud professionals

optional for AI/ML professionals

AI/ML
Data Scientist
Develop and maintain AI/ML models to solve business problems. Train and fine tune models and evaluate their performance

optional for IT/ cloud professionals

optional for AI/ML professionals

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Congratulations

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Next steps

- Congratulations, you have covered all the domains!
- Make sure you revisit the lectures as much as possible

- The AWS Certified SA Pro exam is hard, and tests experience…
- Make sure you master every single concept outlined in this course

- For Hands On experience, have a look at the AWS Solutions Architect Associate, Developer Associate, SysOps Adminsitrator Associate, DevOps Professional courses

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com